Everaise Academy: Physics Mechanics

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Contents

Preface

> "I would rather have questions that can't be answered than answers that can't be questioned."
>
> Richard P. Feynman

This book was adapted from a series of handouts used during the 2020 session of Everaise Academy's introductory mechanics course. The ideas, theorems, and proofs presented in this book are the outgrowth of the effort of over a dozen people who have helped collaborate in making this project a reality.

In some sense, this book is a response to the vast array of introductory physics textbooks that currently flood the market. Too often, physics books are either rich in theory but lack challenging problems or consist of a vast array of problems and solutions without any exposition as to how a reader might come up with the solution in the first place. This book tries to remedy this: the theory is presented with an emphasis on challenging example problems and the motivation behind their solutions.

Each chapter also contains a vast array of practice problems ranging from easy exercises to challenging olympiad problems making this book especially suitable for students preparing for national and international physics competitions such as the $F = ma$ exam or the USAPhO. Solutions to the problems are also included, making the book apt for self-study.

Learning physics is not a passive task. Experience has shown that the only way to truly learn physics is by doing — solving challenging problems and actively paying attention to the examples is thus necessary.

You should be reading this book with a pencil and paper and actively thinking about the concepts as you read through the text. Get a notebook and write down your ideas, your thoughts, and your opinions about the content you are learning as you read through the text. You can not read physics the way you read a novel. Try solving the examples and problems yourself before reading the solution.

Of course, it's okay to give up sometimes, especially if this is your first brush with the subject, but it's important to spend a long time thinking about the examples and playing around with the problems to truly digest the material.

If you couldn't solve every problem, it's *okay* to look at the solution: if you could solve every single problem, that would be truly boring indeed. If you do give up, on a problem, try making little notes of problems you couldn't solve and try coming back to the problem a few days later and resolving the problem. By doing so, you can often latch on to the main ideas of the problem which is a crucial aspect of the learning process.

We are indebted to many people for making this project a reality. We greatly thank Ashmit Dutta for compiling examples and problems, writing explanations and solutions, and creating diagrams throughout the handouts. Furthermore, he served as the head editor and fixed most of the typos and errors found in the handouts. We thank Fiona Abney-McPeek for writing multiple handouts and solution manuals and Evan Kim for helping with writing examples, explanations, and solutions, and creating diagrams. We would also like to thank Jeffrey Hu, Nishant Bhakar, Abhay Bestrapalli, Atharva Mahajan, and Aniruddha Sharma for compiling exercises and problems, writing select handouts, and proofreading.

Finally, we make a small note regarding errors. Unfortunately, even with this extensive proofreading and copy-editing, it is a forlorn conclusion that the book will contain several typos and errors along the way. Your suggestions, comments, and corrections sent to team@everaise.org are most welcome.

Best wishes for success and enjoyment on your physics journey! Happy problem solving!

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I

Kinematics

1 Kinematics in One Dimension

We will begin our journey through physics by discussing the *way* in which things move. For now we'll be ignoring *why* things move but don't worry… we'll consider that very soon! Instead, we'll start with discovering the branch of physics that studies *how* objects move–the branch of physics called **kinematics**.

In this section, we will mainly be dealing with kinematics in one dimension. In the next handout, we will extend our understanding of kinematics to two dimensions. There will also be a brief review of vectors and scalars. The handout also comes with a number of exercises for you to work on at the end. Try your hardest to solve as many as you can!

You should read through this handout with a pencil in your hand and a piece of scratch paper at your side. We cannot emphasize this enough: **learning physics is not a passive activity.** You can try to follow along with the derivations we demonstrate here, but you'll learn and remember far more if you write things out, too.

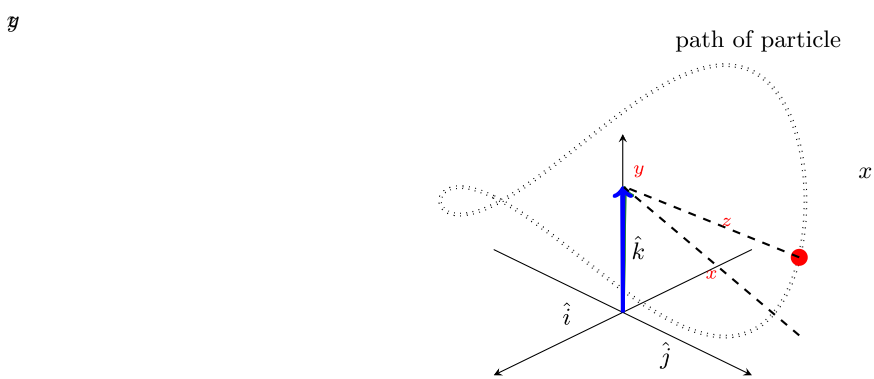
1.1 Vector and Scalar Quantities in Physics

We will preface our discussion of kinematics by briefly reviewing scalars and vectors.

A scalar is a quantity with magnitude only (such as temperature). A vector, in contrast, is a quantity having both a magnitude and a direction. In one dimension, the direction is indicated by the sign of the number. We will assume you know basic facts about vector addition, components, and the like.

There are several basic vector quantities that appear ubiquitously in kinematics. In kinematics, we describe the motion of a particle by specifying it's position, velocity, and acceleration with respect to time.

We will begin by introducing the following quantities which are quite intuitive to understand:



- **Position** (denoted by $\vec{r}$), which denotes the place in space a particle is located at a particular time.

- **Displacement** ($\Delta\vec{r}$ or $\vec{S}$), or the change in an object's position.

  Remark. The Δ symbol appears very often in physics, and means "change in." That is,

  $$\Delta\vec{x} = \vec{x}_f - \vec{x}_i,$$

  where $\vec{x}_f$ is the final position and $\vec{x}_i$ is the initial position.

We're also largely interested in the rate in which the particle's position changes with respect to time. The concept of **velocity** thus arises naturally.

Definition 1.1. The **average velocity** of a particle during any interval is defined to be the displacement divided over the time interval the displacement occurs. In other words,

$$\vec{v}_{\text{avg}} = \frac{\Delta\vec{r}}{\Delta t}$$

Notice that because Δt is a scalar quantity and $\Delta\vec{r}$ is a vector quantity that $\vec{v}_{avg}$ has the same direction as the displacement, $\Delta\vec{r}$. One thing to note about average velocity is that it does *not* depend on the motion between the two endpoints of the interval. That is, it doesn't matter if the particle speeds up or slows down or even reverses direction! All that matters is the endpoint of the motion! So though the average velocity may be useful in analyzing the *overall* motion of the particle, it would be more useful to have a function that could tell us the *instantaneous* velocity at any point in time. Fortunately, we have calculus to remedy this:

Definition 1.2. The **instantaneous velocity**, or **velocity** ($\vec{v}$) is the rate of change of position at a specific moment in time. That is,

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t}$$

We can similarly define the concepts of average acceleration and acceleration:

Definition 1.3. The **average acceleration** of a particle during any interval is defined to be the change in velocity over the change in time. In other words,

$$\vec{a}_{\text{avg}} = \frac{\Delta\vec{v}}{\Delta t}$$

Definition 1.4. The **instantaneous acceleration**, or simply **acceleration** is the rate of change of velocity. Mathematically, we have

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{x}}{dt^2}$$

Note that all these quantities are represented as vector functions of time. For example, we might say $\vec{x}(t) = t^2\hat{\imath}$, which means that the magnitude of the position of an object at time t is t^2 and the direction is in the $\hat{\imath}$ direction. Here's a quick example that highlights how these quantities might be used:

Video Example 1.5. A particle moves in the xy plane so that it's x and y positions vary with time according to $x = 3t^3 - 5t$, $y = t^2 - t$. Find the position, velocity, and acceleration of the particle at time $t = 3$, and find the average velocity and average acceleration of the particle from time $t = 0$ to time $t = 3$.

Solution. Watch https://youtu.be/BU05rmq17Os (timestamp 7:28) for the video solution to this problem! ■

Solution. The particle has position $\langle x(3), y(3)\rangle = [3(3)^3 - 5(3)]\hat{\imath} + [(3)^2 - (3)]\hat{\jmath} = 66\hat{\imath} + 6\hat{\jmath}$.

We note that the displacement of the particle is thus $\Delta x = x(3) - x(0) = 66\hat{\imath} + 6\hat{\jmath}$. The average velocity of the particle is thus $\Delta x/\Delta t = 22\hat{\imath} + 2\hat{\jmath}$.

The velocity of the particle as a function of time is

$$\vec{v} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \langle 9t^2 - 5, 2t - 1\rangle$$

so at time $t = 3$, the velocity is $\vec{v}(3) = [9(3)^2 - 5]\hat{\imath} + [2(3) - 1]\hat{\jmath} = 76\hat{\imath} + 5\hat{\jmath}$.

Noting that the velocity of the particle at $t = 0$ is $\langle -5, -1\rangle$, the average acceleration is

$$\vec{a}_{\text{avg}} = \frac{\Delta\vec{v}}{\Delta t} = \frac{76+5}{3}\hat{\imath} + \frac{5+1}{3}\hat{\jmath} = 27\hat{\imath} + 2\hat{\jmath}$$

The acceleration as a function of time is

$$\vec{a} = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \langle 18t, 2\rangle$$

so we have $\vec{a}(3) = 54\hat{\imath} + 2\hat{\jmath}$. ■

We will make a small note on language in physics: In every day speech, we use speed and velocity quite interchangeably. However, they are **not** the same in physics. **Speed** ($|\vec{v}|$) is a scalar quantity and the magnitude of velocity. This means that while speed is always nonnegative, velocity can be negative (or at least, the components of the velocity vector can be negative). The same distinction can be made for displacement (a vector) and distance (the corresponding scalar).

Here's a few quick examples that should clear up any misconceptions.

Example 1.6. If the speed of an object is increasing, what must be true about the directions of its velocity and acceleration?

Solution. If the speed of an object is increasing, then the velocity and a component of acceleration are in the same direction. This means they are either both positive or both negative. If the acceleration and velocity of an object are in different directions, the speed will decrease since acceleration is the rate of change of velocity. For example, if $v < 0$, and $a > 0$, then the velocity is becoming more positive, so the magnitude (speed) is decreasing. ■

Example 1.7. If an object's speed is constant, must the object's acceleration equal 0?

Solution. No. Remember that speed is a scalar, while velocity is a vector. This means an object can have constant speed but still experience a change in velocity. This cannot happen in one dimension, but as we will see later, this can happen in two-dimensions (in fact, it occurs when $\vec{a}$ and $\vec{v}$ are perpendicular).

However, one should note that if we talk about the object's velocity, the statement does hold; an object's velocity is indeed constant if and only if its acceleration is zero. ■

Let's do another quick example of how scalars and vectors differ.

Example 1.8. Professor Pylypovych leaves his home at exactly 4:00 PM and starts driving to the supermarket. After driving 1800 meters, he realizes he left his credit card at home, so he heads back to his house, getting there at exactly 4:03 PM.

(a) What is Professor Pylypovych's average velocity?

(b) What is Professor Pylypovych's average speed?

Solution.

(a) We see that Professor Pylypovych's displacement is $\Delta\vec{x} = x_f - x_i = 0$. Thus, his average velocity (displacement over time) is 0 as well.

(b) To calculate his speed, we see he traveled 3600 meters in 3 minutes, so his average speed is $3600/(3 \times 60) = 20$ m/s.

■

Lastly, We will do a simple example of calculating average acceleration from discrete data.

Example 1.9. Given the following table of data points, what is the average acceleration?

t (s)	$\vec{x}$ (m)
1	3
2	7
3	13
4	21

Solution. We can estimate a table of differences:

t (s)	$\Delta\vec{v}$ (m/s)
$[1,2]$	$7-3=4$
$[2,3]$	$13-7=6$
$[3,4]$	$21-3=8$

Taking any two of these points and computing $\frac{\Delta\vec{v}}{\Delta t}$ gives the desired answer of 2 m/s^2. ■

1.2 Graphical Relationships

We briefly went over the properties of derivatives in our calculus overview. We'll be using these properties in this section to better understand kinematics graphs. We'll mainly be focusing on the motion of objects in one-dimension for now, so we'll only be studying the variance of one particular scalar component with respect to time.

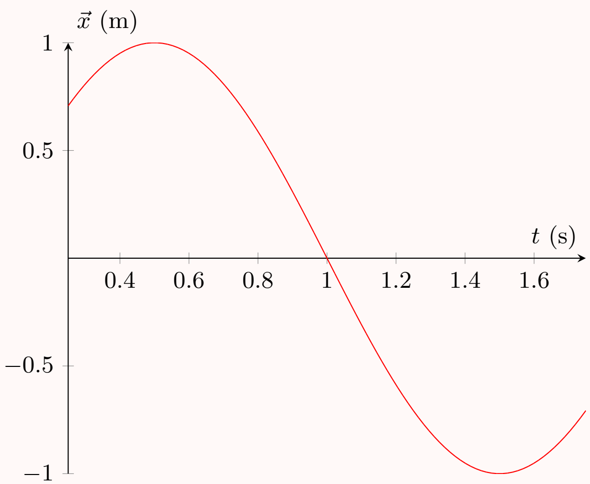
Consider a graph of the particle's position vs time (x vs t). Since the particle is moving only in one-dimension, we can treat x like a scalar function and graph it easily.

We'll begin by listing a few basic properties:

- Since $v = \frac{\mathrm{d}x}{dt}$, the slope of a x - t graph is given by v. Hence, if the graph has a positive slope, $\vec{v} > 0$, and likewise a negative slope indicates $\vec{v} < 0$. Remember that at any relative extrema, $\vec{v} = 0$.
- Likewise, $a = \frac{d^2x}{dt^2}$. Thus $\vec{a}$ can be determined by the concavity (or the "rate of change of the rate of change") of the graph. If the graph tends to curve upward as t increases, then the graph is concave up, or $\vec{a} > 0$. If the graph is concave down, then $\vec{a} < 0$. In the case of a straight line, $\vec{a} = 0$. Likewise, $\vec{a}$ is the slope of a velocity-time graph.

We'll now proceed with some examples that show these properties in action.

Example 1.10. A particle moves with position given by the following graph:



List the times at which the following events occur:

(a) The particle's position is at a maximum.

(b) The particle's speed is at a maximum.

(c) The particle turns around.

(d) The particle has zero acceleration.

(e) The particle's acceleration is at a maximum.

Solution.

(a) This is a position-time graph. We can see that maximum position occurs at $t = 0.5$.

(b) The particle's speed is at a maximum when the graph exhibits its steepest slope. This occurs at $t = 1$.

(c) The particle turns around when its velocity changes sign (recall velocity is the slope of the graph). This occurs at $t = 0.5$ and $t = 1.5$.

(d) The particle has zero acceleration when the slope of the graph is constant. Since the acceleration is negative for $t < 1$ (we can see this from the graph being concave down), and similarly, $\vec{a}$ is positive for $t > 1$, we see the graph must have passed through $\vec{a} = 0$ at some point; by symmetry it must be $t = 1$.

(e) This is when the velocity (slope of the graph) is changing the most, and we can see this occurs at $t = 0.5$ and 1.5.

■

If we can tell velocity from a position-time graph, what can we tell about position from a velocity-time graph? The inverse operation of differentiation is integration:

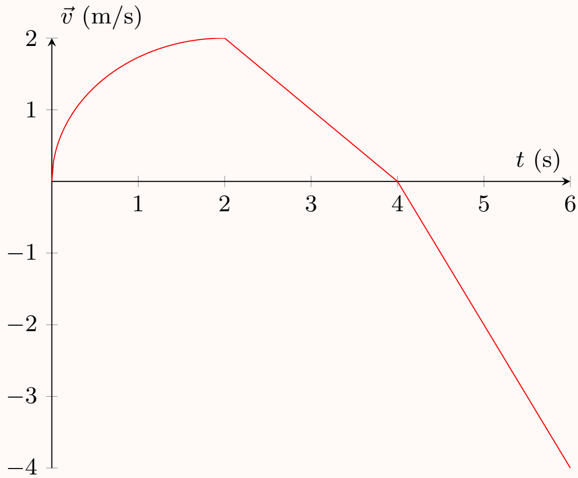
$$\int \vec{v}\ dt = \vec{x} + C.$$

where c is a constant determined by initial conditions. This relationship is also given by

$$\Delta\vec{x} = \vec{x}(t_f) - \vec{x}(t_i) = \int_{t_i}^{t_f} \vec{v}\ dt.$$

From a graphical perspective, the integral is the signed area under the curve. This means that for regions of the curve above the x-axis, the integral will simply be the area, and for regions of the curve beneath the x-axis, the integral will be the area but negative. Hence, the displacement of an object from time t_i to time t_f is given by the signed area under a $\vec{v}$ - t graph from t_i to t_f. Here's an example of what this looks like:

Example 1.11. A particle starts at position 2 meters at time $t = 0$, and moves with velocity given by the following graph:



You may assume the graph is a quarter circle during the first two seconds.

(a) On what intervals is the particle speeding up?

(b) What is the particle's maximum position?

(c) What is the particle's position at time $t = 6$?

Solution.

(a) The particle is speeding up when $\vec{v}$ and $\vec{a}$ have the same sign. Recalling $\vec{a}$ is the slope of the graph, we see they have the same sign (speeding up) on the interval $(0, 2) \cup (4, 6)$ whereas the signs are opposite (slowing down) on the interval $(2, 4)$.

(b) Recall that the displacement is equal to the area under the curve for a $\vec{v}$-t graph. After $t = 4$ seconds, the particle's position is decreasing since the velocity is negative, while before $t = 4$ seconds the position is increasing. Hence the maximum is exactly when $t = 4$. Recall that the displacement is equal to the area under the curve for a $v - t$ graph. We can split the first four seconds into two sections:

1. In the first two seconds, the graph is a quarter circle of radius two. The area is $\frac{1}{4}\pi r^2 = \pi$.
2. In the next two seconds, the graph is a triangle. The area formula is $\frac{1}{2}bh = 2$, where b is the base and h is the height.

The total displacement is $2 + \pi$, so the final position is $4 + \pi$.

(c) In the last two seconds, the graph is again a triangle. The area is 4 but the graph is negative, so we consider the displacement in this phase to be -4. We can subtract this from our answer from part (b) to get the final position is π meters.

■

That's really all there is to most kinematics graph problems. Just using the first two properties carefully and thinking about the situation well enough is more than enough to solve most of these kinds of problems.

1.3 Kinematics Equations for Constant Acceleration

In general, the equations of motion for a particle can be vastly complicated as they vary as functions of time. For the case of constant acceleration, however, the motion is quite a bit easier to analyze. There are a number of examples of motion with constant acceleration: objects falling near earth's surface or braking cars are typical examples.

The main reason why the constant acceleration case is so easy to analyze is because the instantaneous acceleration and the average acceleration become equal everywhere, that is,

$$a = a_{\text{avg}} = \frac{\Delta v}{\Delta t},$$

or equivalently,

$$\Delta v_x = a_x \Delta t.$$

From this equation, a number of other equations can be easily derived, either algebraically or through calculus.

In total, there are five kinematics equations that can be used to solve problems related to motion. Again, *note that these equations only hold if acceleration is constant.* Try deriving these equations on your own.

1. $\Delta \vec{x} = \frac{\vec{v}_i + \vec{v}_f}{2} \cdot t$. This follows from the graphical interpretation of displacement being area under a velocity vs. time graph, which is in this case a trapezoid with bases $\vec{v}_i$ and $\vec{v}_f$ and height t.
2. $\Delta x = \vec{v}_i t + \frac{1}{2}\vec{a}t^2$. This again follows from the graphical interpretation of displacement being area under a velocity vs. time graph, but incorporating Equation 2 to result in a trapezoid with bases $\vec{v}_i$ and $\vec{v}_i + at$, and height t, giving the appropriate area.
3. $\Delta x = \vec{v}_f t - \frac{1}{2}\vec{a}t^2$. This is very similar to the previous case, but now the bases of the trapezoid are $\vec{v}_f - at$ and $\vec{v}_f$.
4. $\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{a}\Delta\vec{x}$. This is perhaps the least intuitive of all the kinematics equations, making it the most important to memorize, as re-deriving it with intuition is difficult. Try to derive it on your own with algebra as an exercise. **Hint:** 22

   Exercise. For graphs where acceleration is not constant, show that this identity is generalized by the equation

   $$\vec{v}_f^2 = \vec{v}_i^2 + 2\int \vec{a}\, \mathrm{d}\vec{x}.$$

   Hint: 17

Note that $\Delta\vec{x}$ is also sometimes denoted by d or S. We'll now do some examples using these equations.

Video Example 1.12. A deer is running north at a constant speed of 8 m/s. A bear sees the deer when the deer is 20 meters to its north. The bear begins from rest and starts chasing the deer with a constant acceleration of 2 m/s^2. At what time does the bear catch the deer?

Solution. Watch `https://youtu.be/BU05rmq17Os` (timestamp 13:36) for the video solution to this problem! ■

Solution. We will use S instead of $\Delta\vec{x}$ here for convenience. We wish to solve

$$S_{\text{bear}} = 20 + S_{\text{deer}}.$$

By the kinematics equations,

$$\begin{aligned} S_{bear} &= v_i t + \frac{1}{2}at^2 = t^2 \\ S_{deer} &= vt = 8t. \end{aligned}$$

Substituting gives us

$$\begin{aligned} t^2 &= 20 + 8t \\ t^2 - 8t - 20 &= 0 \\ (t-10)(t+2) &= 0. \end{aligned}$$

The time cannot be negative, so we see that $t = 10$ seconds. ■

Remark. Many of the exercises and problems in this handout will deal with objects moving under the influence of gravity. For these problems, note that for an object near Earth's surface not in contact with any other objects, gravity will cause the object to accelerate downwards at $g = 10\ \text{m/s}^2$. Technically, the object might experience air resistance, but this is often assumed to be negligible.

The gravitational acceleration also decreases at higher altitudes, but the radius of the Earth is so great that the acceleration is assumed to be constant near the surface of the Earth. Often times, $10\ \text{m/s}^2$ is also used as an approximation for g because it is an easier number to work with.

1.4 Examples

We'll finish off with a few harder examples that show how we might use the equations we derived along with some clever intuition to solve problems. Try these examples by yourself first, but don't be afraid to look at the solution if you're struggling!

Example 1.13 (JEE Mock Test). Two students simultaneously start from the same place on a circular track and run for 2 minutes. In this time, one of them completes three and the other four revolutions with uniform speed. Due to thick vegetation in a circular area inside the track, either of the boys can see only a third of the track at a time (a sixth of the track both in front and behind them). How long during their run do they remain visible to each other?

Solution. It is obvious from the problem statement that the two students run with different speeds. We let t_1 be the time when one student first sees the other (of course, this starts at the time they have started running). Let student 'A' covering 3 round trips in the total time have a speed v_A, and the other v_B. Then since student B travels four times the distance of student A in the same time, we have

$$v_B = \frac{4}{3}v_A.$$

Now we consider when the students might be able to see each other. They will be able to see each other towards the beginning, since they initially start together, and towards the end, since they finish running together. Since student B does not 'lap' student A until the very end, these are the only two intervals where the see each other.

It takes student B 120 seconds to outrun student A by a full lap, so we conclude it takes 20 seconds for student B to outrun student A by a sixth of a lap. Therefore, the two students can see each other on the interval $0 \leq t \leq 20$ s.

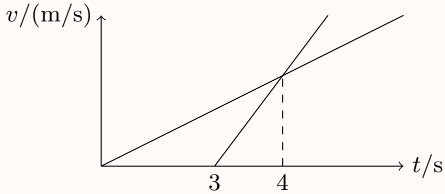
The second interval on which student A and student B can see each other is toward the end, when student B is catching up to student A. By symmetry, we have that it takes student B 20 seconds to "catch up" a sixth of a lap to student A, so the two students can also see each other on the interval $100 \leq t \leq 120$.

Thus, the answer would be $t = 20 + 20 = \boxed{40\text{ s}}$. Note that this is exactly a third of the time interval the students were running, which makes intuitive sense since exactly a third of the track is visible to them.

Remark. This problem is a great learning tool as it helps teach you the usage of speed, distance, and the displacement vector in a problem.

■

Example 1.14. Jack is chasing John and both of them are moving on the same path. After they pass a tree, their velocities v vary with time t as shown in the graph below. When will the chase end?



Solution. Let the velocity at which both lines meet be v_0. The accelerations of Jack and John respectively will be given as

$$a_{\text{Jack}} = \frac{v_0}{4}$$
$$a_{\text{John}} = \frac{v_0}{1}$$

The chase ends when Jack catches John or when the distance both people cover are the same. This means that

$$\frac{1}{2}a_{\text{John}}(t_0 - 3)^2 = \frac{1}{2}a_{\text{Jack}}t_0^2.$$

Substituting what we found for acceleration of each person gives us

$$\begin{aligned}\frac{1}{2}\left(\frac{v_0}{1}\right)(t_0 - 3)^2 &= \frac{1}{2}\left(\frac{v_0}{4}\right)t_0^2 \\ \frac{t_0}{2} &= t_0 - 3 \\ t_0 &= \boxed{6\text{ s}}.\end{aligned}$$

Remark. The main lesson of the problem is to use graphs to your advantage. You have to find the information the graph gives you, and take it to your advantage to manipulate variables and solve.

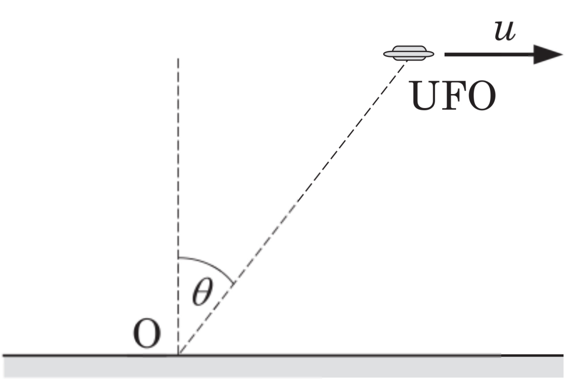
■

We'll finish off the handout with a final, harder example that also shows the power of calculus in physics problems:

Example 1.15 (Quantum Magazine)**.** In a particular scene of a science fiction movie, a UFO is flying horizontally at a very high altitude with a speed u that is $\eta < 1$ times the speed of light, c. The UFO is emitting sharp light pulses at regular and very small intervals. Find the speed of the UFO recorded by an observer on the group at point O when the UFO appears at an angle θ with the vertical. Ignore relativistic corrections.

A UFO (unidentified flying object) is believed to be a space ship used by aliens.

Solution. Let the time traversed between points A (directly above the observer) and B (at an angle θ to the observer) measured by the UFO be t, and let the altitude at point A to the observer be ℓ.



We then see that the total distance is

$$ut = \ell \tan\theta.$$

The observer measures the time t' it takes for the UFO to travel this distance by starting his stopwatch when he sees light from point A, then stopping it when he sees light from point B. Due to the finiteness of light, the observer will see the object in a time

$$\Delta t = \frac{\ell}{c\cos\theta} - \frac{\ell}{c} \implies t' = t + \frac{\ell}{c\cos\theta} - \frac{\ell}{c}.$$

The measured velocity of the UFO is

$$v' = \frac{\mathrm{d}x}{\mathrm{d}t'} = \frac{\mathrm{d}x}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}t'} = \frac{v}{\mathrm{d}t'/\mathrm{d}t}.$$

Differentiating our first equation tells us that

$$v = \frac{\ell}{\cos^2\theta}\frac{d\theta}{\mathrm{d}t} \implies \frac{d\theta}{\mathrm{d}t} = \frac{v\cos^2\theta}{\ell}.$$

Differentiating our second equation as well tells us

$$\frac{\mathrm{d}t'}{\mathrm{d}t} = 1 + \frac{\ell\sin\theta}{c\cos^2\theta}\frac{\mathrm{d}\theta}{\mathrm{d}x}.$$

Substituting $\mathrm{d}\theta/\mathrm{d}x$ into this gives us

$$\frac{\mathrm{d}t'}{\mathrm{d}t} = 1 + \frac{\ell\sin\theta}{c\cos^2\theta}\frac{v\cos^2\theta}{\ell} = 1 + \frac{v\sin\theta}{c} = 1 + \eta\sin\theta.$$

We plug into our first equation of v' to get that

$$v' = \boxed{\frac{v}{1+\eta\sin\theta}}$$

Remark. Note that using the chain rule along with calculus identities can help trivialize many kinematics problems.

■

1.5 Homework Problems and Solutions

Problem 1.1 (Morin)**.** [4] A ball is dropped, and then another ball is dropped from the same spot one second later. As time goes on while the balls are falling, the distance between them (ignoring air resistance, as usual):

(A) decreases

(B) remains the same

(C) increases and approaches a limiting value

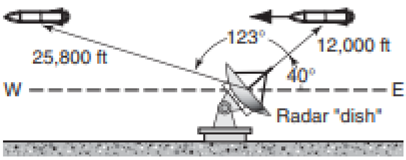
(D) increases steadily

Solution. Recall that $\Delta x = \frac{1}{2}gt^2$. If the two balls are released Δt seconds apart, then the distance between them can be expressed as

$$\frac{1}{2}g(t+\Delta t)^2 - \frac{1}{2}gt^2 = gt\Delta t + \frac{1}{2}g(\Delta t)^2.$$

We see this increases linearly as t goes to ∞, so our answer is (D). This conclusion is also intuitive: the velocity of the first ball is always greater than velocity of the second ball, so the gap in distance will continuously increase. ■

Problem 1.2 (HRK)**.** [5] A radar station detects a missile approaching from the east. At first contact, the range to the missile is 12,000 ft at 40.0° above the horizon. The missile is tracked for another 123° in the east-west plane, the range at final contact being 25, 800 ft; see Figure. Find the displacement of the missile during the period of radar contact.



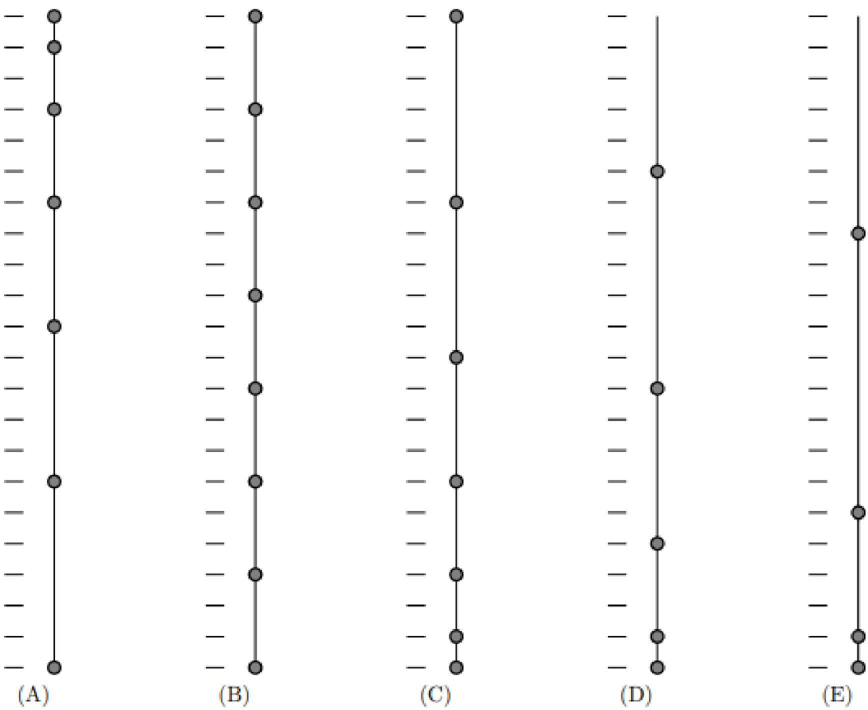
Solution. Finding the displacement is just law of cosines. We have

$$\Delta x^2 = 25,800^2 + 12,000^2 - 2 \cdot 25,000 \cdot 12,000 \cdot \cos 123^\circ.$$

So we can calculate that $\Delta x \approx \boxed{33900\,\text{ft}}$. However, a quicker method would have been to approximate it as horizontal.

Now to find the angle, we find the horizontal displacement and then use arccos to finish. The horizontal displacement is given by $25800 \cos 17^\circ + 12000 \cos 40^\circ$. So then we can find $\cos\theta \approx 0.99987$, so $\theta \approx \boxed{0.288^\circ}$. ■

Problem 1.3 ($F = ma$)**.** [5] You have 5 different strings with weights tied at various point, all hanging from the ceiling, and reaching down to the floor. The string is released at the top, allowing the weights to fall. Which one will create a regular, uniform beating sound as the weights hit the floor?



Solution. Choice (D) has the beads spaced out quadratically (at ticks 0, 1, 4, 9, etc.). Since we know Δx is a parabolic function of t, this will create uniform time intervals. ■

Problem 1.4 ($F = ma$). [6] A ball is bouncing vertically between a floor and ceiling, which are both horizontal and separated by 4 m. No speed is lost when the ball rebounds from hitting the floor, and when the ball hits the floor, it has a speed of 12 m/s. How long does a complete up-down cycle take?

Solution. Using the kinematics equations, the height of the ball is given by $h = 12t - \frac{1}{2}gt^2$, so substituting $h = 4$ and $g = 10$, we see $t = 0.4$ s for the ball to reach the ceiling. By symmetry, it takes 0.4 seconds to go back down, giving the final answer of $\boxed{0.8 \text{ s}}$. ■

Problem 1.5 ($F = ma$). [6] A car has a maximum acceleration of a_0 and a minimum acceleration of $-a_0$. The shortest possible time for the car to begin at rest, then arrive at rest at a point a distance d away is?

Solution. If we make a v-t graph of the motion, then we are restrained by $v(0) = v(t) = 0$, and the slope is limited between a_0 and $-a_0$. Since displacement is the area under the curve, we seek a graph that maximizes the area under the curve. This happens by accelerating at rate a_0 for the first half and accelerating at rate $-a_0$ for the second half. We see the time taken in the first half then satisfies

$$\frac{1}{2}a_0t^2 = \frac{1}{2}d \Rightarrow t = \sqrt{d/a_0}.$$

By symmetry, the second half will take equally long, so the total time taken is $\boxed{2\sqrt{d/a_0}}$. ■

Problem 1.6. [7] A basketball is thrown directly upwards with initial speed v_0. When it reaches its maximum height, a second basketball is thrown directly upwards with the same initial speed v_0. At what height do the two basketballs collide?

Solution. First, recall that $\Delta\vec{v} = \vec{a}t$. Hence, if we let $a = -g$ be the acceleration of gravity, the peak height occurs when $\vec{v} = 0$, or

$$t = \frac{v_0}{g}.$$

We also obtain the maximum height is

$$\frac{1}{2}g\left(\frac{v_0}{g}\right)^2 = \frac{v_0^2}{2g}.$$

We can now draw $\vec{x}$-t graphs for the paths of the two balls, and see by symmetry the intersection occurs at $t = \frac{3v_0}{2g}$. Since the movement of the first ball is a parabola, and this is half of the time on the way down, we see that it has only fallen down $\frac{1}{4}$ of the distance from its max height.

Thus the balls intersect at the height $\boxed{\dfrac{3v_0^2}{8g}}$.

Equivalently, we could solve for the time of intersection with the equation

$$v_0t - \frac{1}{2}gt^2 = v_0\left(t - \frac{v_0}{g}\right) - \frac{1}{2}\left(t - \frac{v_0}{g}\right)^2.$$

After we obtain $t = \frac{3v_0}{2g}$, as above, we can substitute into the left-hand side to obtain our answer. This solution is more computation heavy but also easier to see. ■

Problem 1.7. [12] Two tortoises, Alice and Bob, are spaced 20 m apart. They both begin to crawl toward each other at 0.5 m/s each. Trix the Rabbit starts next to Alice, and runs towards Bob. After reaching Bob, he runs back towards Alice, then back to Bob, and so on until Alice and Bob meet in the middle, at which point Trix stops.

(a) [4] If Trix runs at a constant speed of 2 m/s, what is the total distance Trix travels?

(b) [8] If Trix runs from Alice to Bob at rate 2 m/s, and runs from Bob to Alice at rate 8 m/s, what is the total distance Trix travels?

Solution.

(a) Note that Trix's speed is constant, and the question asks for distance, not displacement, so we can use the equation $x = vt = 2t$, where t is the time it takes for Alice and Bob to reach each other. Since the tortoises are moving towards each other at a combined speed of 1 m/s, this will take 20 seconds. Hence, the total distance will be $\boxed{40 \text{ m}}$.

(b) We can easily see that Trix reaches the middle for the first time in $10/2 = 5$ seconds. After Trix reaches the middle, there is a symmetry in the movement: she must travel equal distances in both directions since she stops in the middle. We can use this information to compute her average speed for the remaining time.

We wish to compute the average speed of traveling a distance $2d$ with speed v_1 for the first half of the distance, and speed v_2 for the second half of the distance. The time taken for the first half is then d/v_1, and the second half will take time d/v_2. Hence, total time is $d/v_1 + d/v_2$, so the average speed is given by

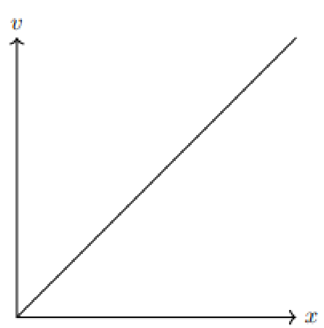
$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{2d}{d/v_1 + d/v_2} = \frac{2}{1/v_1 + 1/v_2} = \frac{2}{1/2 + 1/8} = 3.2 \text{ m/s}.$$

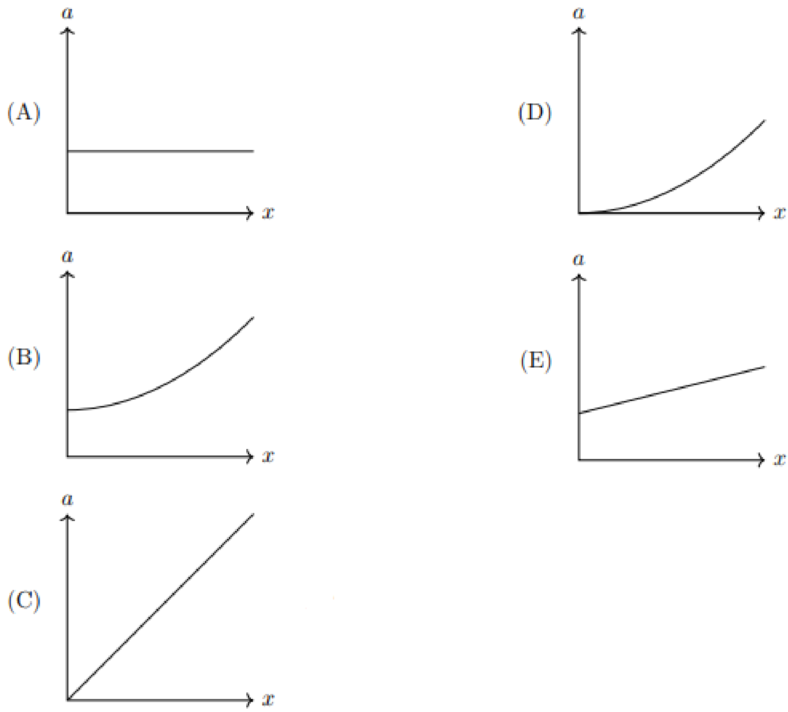
Recall that Trix travels for 20 seconds total. Therefore, the total distance is

$$5 \text{ s} \cdot 2 \text{ m/s} + 15 \text{ s} \cdot 3.2 \text{ m/s} = \boxed{58 \text{ m}}.$$

■

Problem 1.8 ($F = ma$). [7] The velocity versus position plot of a particle is shown below. Which following choices is the correct acceleration vs. position plot of the particle?





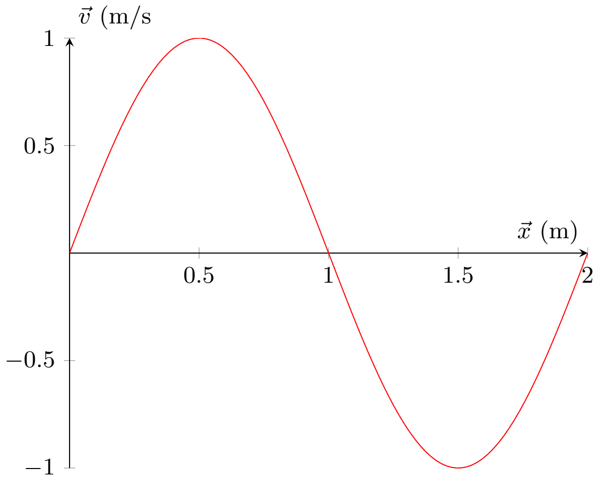
Solution. Since $dx = vdt$, we have that

$$a = v\frac{dv}{dx}.$$

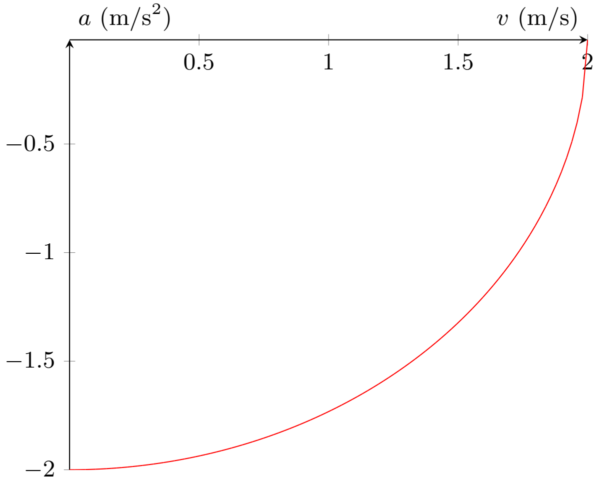
Since $\frac{dv}{dx}$ is constant, $a \propto v$. Since $v \propto x$, we conclude $a \propto x$, which gives graph (C). Alternatively, from $v = kx$, we can conclude $x = e^{kt}$, in which case $a = k^2e^{kt}$, again giving graph (C). ■

Problem 1.9. [16] We will now analyze some graphs where t is not the independent variable.

(a) [8] Consider the following sinusoidal velocity versus position plot. What is the acceleration of the object at $x = .25$ m?



(b) [8] Consider the following circular acceleration versus velocity plot. If the object has an initial velocity $v_0 = 2$ m/s and is given an infinitesimal negative acceleration, what is the displacement as the object comes to rest?



Solution.

(a) Note that the graph is the graph of $\sin(\pi x)$. Since we have $dx = v\ dt$, we can substitute to obtain

$$\frac{dv}{dx} = \frac{dv}{v\ dt} = \frac{a}{v}.$$

Rearranging gives that

$$a = v\frac{dv}{dx}.$$

We see that

$$\frac{dv}{dx}(.25) = \pi\cos(.25\pi) = \frac{\pi\sqrt{2}}{2},$$

and $v(.25) = \sqrt{2}/2$, so $a = \boxed{\pi/2}$.

(b) As with above, we have that
$$a = v\frac{dv}{dx}.$$
Rearranging, we see that
$$\begin{aligned} dx &= \frac{v}{a}dv \\ \implies \Delta x &= \int_{v_i}^{v_f} \frac{v}{a}dv. \end{aligned}$$
In this graph, we have $a = -\sqrt{4-v^2}$, so substituting gives
$$\Delta x = \int_2^0 -\frac{v}{\sqrt{4-v^2}}\,dv = \int_0^2 \frac{v}{\sqrt{4-v^2}}\,dv = \boxed{2},$$
where the integral was evaluated using the reverse chain rule.

■

Problem 1.10 ($F = ma$). [8] An object launched vertically upward from the ground with a speed of 50 m/s bounces off of the ground on the return trip with a coefficient of restitution given by $C_R = 0.9$, meaning that immediately after a bounce the upward speed is 90% of the previous downward speed. The ball continues to bounce like this; what is the total amount of time between when the ball is launched and when it finally comes to a rest? Assume the collision time is zero; the bounce is instantaneous. Treat the problem as ideally classical and ignore any quantum effects that might happen for very small bounces.

Solution. For an initial speed v_i, the time it takes for the ball to go up and down is given by $\Delta x = \frac{1}{2}at^2 + \vec{v}_i t = 0 \Rightarrow t = \frac{2v_i}{g}$.

Since the ball is initially launched upwards with speed 50 m/s, we see that the total time is
$$t_{\text{total}} = \frac{2}{g}(50 + 50\times .9 + 50 \times .9^2 + \cdots) = 10 \times (1 + .9 + .9^2 + \cdots).$$

We can evaluate the infinite product using the following method:
$$\begin{aligned} S &= 1 + .9 + .9^2 + \cdots \\ .9S &= \qquad .9 + .9^2 + \cdots \end{aligned}$$

Hence $(1 - .9)S = 1 \Rightarrow S = 10$. The total sum is therefore $\boxed{100 \text{ s}}$. ■

Problem 1.11 (Irodov). [12] Three points are located at the vertices of an equilateral triangle whose side equals a. They all start moving simultaneously with velocity v constant in modulus, with the first point heading continually for the second, the second for the third, and the third for the first. How soon will the points converge?

Solution. This problem may not seem like a one-dimensional motion problem, but it in fact is. Note that at any time, the three points are always the vertices of an equilateral triangle, and the center of the triangle does not change (by symmetry).

Let P denote one of the vertices of the triangle, and O the center of the triangle. Then the segment OP always forms a $30°$ angle with $v(t)$. Hence, the true rate at which P approaches the center is $v\cos 30^\circ = v\frac{\sqrt{3}}{2}$. The other "component" of the velocity goes towards rotating the vertex around the center.

The initial distance from P to O is given by $\frac{a}{\sqrt{3}}$, so the time taken is $\boxed{\dfrac{2a}{3v}}$. ■

1.5.1 Written Solutions

Problem 1.12 (Morin). [12] Suppose you watch a video backwards and slowed down by a factor α, so that if the video is one hour long when played normally, you watch it from end to beginning and it takes α hours to do so.

The video shows some physical motion. If the motion in the video has acceleration a, when played normally, what is the acceleration a' you would measure when watching the video backwards and in slow motion? Briefly explain your reasoning.

Solution 1. Note that when we watch an object move through a certain distance, $\Delta x = \frac{1}{2}at^2$ is invariant in. In the edited video, since t is increased by a factor of α, we see a has to be reduced by a factor of α^2. In other words, we have $dt' = -\alpha\, dt$, so we have

$$a' = \frac{d^2x}{dt'^2} = \frac{1}{\alpha^2}\frac{d^2x}{dt^2} = \boxed{\frac{a}{\alpha^2}}.$$

Note that even if it is backwards, the acceleration is still in the same direction, meaning there is no negative sign. ■

Solution 2. Let the position of an object in the video at a time t be $x(t)$. Then the object's position at time t' in the reversed video is $x(1 - t'/\alpha)$. Taking the second derivative of this with respect to t' gives

$$x''\left(1 - \frac{t'}{\alpha}\right) \cdot \frac{1}{\alpha^2} = \boxed{\frac{a}{\alpha^2}}.$$

■

2 Kinematics in Multiple Dimensions

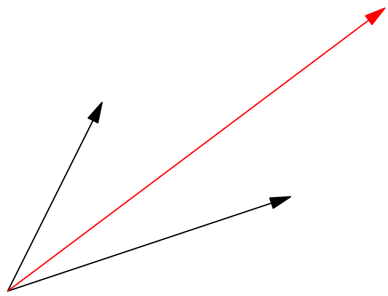
2.1 Adding and Decomposing Vectors

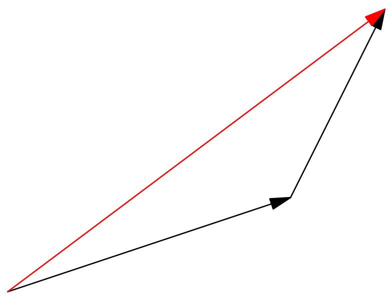
Many of you are probably already familiar with the addition and decomposition of vectors. We'll briefly review the topic in this section for those that are less familiar. In two-dimensions, we can write vectors in terms of their components. For example, $\vec{x} = \langle 0, 1\rangle$ indicates the object is located 1 m in the positive y-direction from the origin. Similarly, $\vec{x} = \langle t, t^2\rangle$ indicates that at any time t, an object has x-coordinate t and y-coordinate t^2.

To add vectors, we can simply add their components. Namely, if $\vec{x} = \langle a, b\rangle$ and $\vec{y} = \langle c, d\rangle$, then $\vec{x} + \vec{y} = \langle a + c, b + d\rangle$. As in one dimension, if we multiply vector $\vec{x}$ by a constant α, then it simply multiplies its components: $\alpha\vec{x} = \langle \alpha a, \alpha b\rangle$.

In two dimensions, the given magnitude of a vector is given by the equation $|\vec{x}| = \sqrt{a^2 + b^2}$.

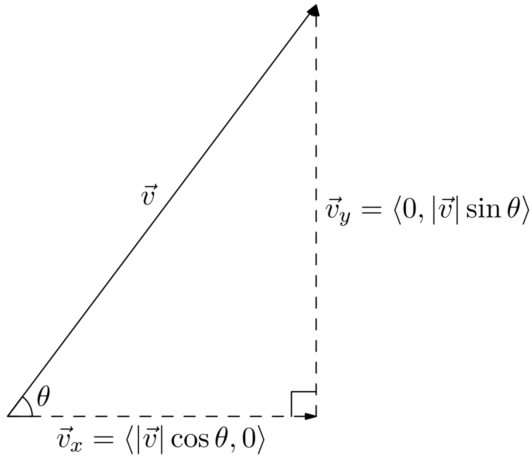
In practice, vector arrows can be drawn using the head-to-tail method. This means when adding vectors $\vec{x}$ and $\vec{y}$, draw $\vec{x}$ as you originally would, and place the tail of they $\vec{y}$ at the tip of $\vec{x}$. For example, adding the vectors $\langle 1, 1\rangle$ and $\langle 2, 1\rangle$ looks like this:





The resulting vector (shown in red) would be $\langle 3, 2\rangle$.

We can also decompose vectors, which is extremely important. Consider a vector $\vec{v}$ that forms an angle θ with the horizontal. We can decompose it into two perpendicular vectors:



That is, $\vec{v} = \vec{v}_x + \vec{v}_y$, and v_x and v_y are perpendicular. Those familiar with the unit circle will recall that if a vector $\vec{v}$ makes an angle θ with the positive x-axis, the following identity holds:

$$\vec{v} = \langle |\vec{v}| \cos\theta, |\vec{v}| \sin\theta \rangle$$

This provides a convenient way of decomposing a vector into perpendicular "components" from a known angle.

2.2 Kinematics in Two Dimensions

There are no really "new" kinematics equations in two-dimensions. Instead, we'll now treat our original equations more like *vector equations* rather than just scalar equations. We can also think of a vector equation as being akin to multiple different scalar equations, one for each component. For example, our equation

$$\vec{v} = \vec{v_0} + \vec{a}t$$

for constant acceleration systems, can be thought of as three scalar equations

$$v_x = v_{0x} + a_x t$$

$$v_y = v_{0y} + a_y t$$

$$v_z = v_{0z} + a_z t$$

Indeed, after decomposing vectors into their components, the two dimensional case largely becomes equivalent to the one-dimensional case! This is better explained with examples, so we'll go through a few quick examples here:

> **Video Example 2.1.** An object moves with acceleration vector $\langle 2, 3 \rangle$ and has initial velocity $\langle 1, 2 \rangle$. After 5 seconds, what is its displacement?

Solution. Watch `https://youtu.be/BeJFgx0Yxpo&t=0s` (timestamp 0:00) for the video solution to this problem! ■

Solution. Since we are given coordinates in this instance, it is easy to use components along the x and y axes. Using the kinematics equation $\Delta x = \frac{1}{2}at^2 + v_i t$, we can substitute a for the x-component of acceleration and v_i for the x-component of initial velocity. Doing the same for the y-component, we see that:

$$\Delta x = \frac{1}{2}(2)t^2 + (1)t$$
$$\Delta y = \frac{1}{2}(3)t^2 + (2)t$$

Plugging in $t = 5$ and simplifying gives $\vec{S} = \langle 30, 47.5 \rangle$. ■

Example 2.2. An object starts from rest with acceleration vector $\langle 3, 4 \rangle$. What is the object's velocity after it has traveled 10 meters?

Solution. This is actually a one-dimensional kinematics problem. The initial velocity is zero, so all motion occurs along a single line. In this case we can instead combine the components of acceleration: $|\vec{a}| = \sqrt{3^2 + 4^2} = 5$. Hence $v_f^2 = 2a\Delta x = 2 \times 5 \times 10 = 100 \Rightarrow v_f = 10$. ■

Example 2.3. A river flows from West to East with a constant current of 2 m/s. The river is 40 meters across, and Alice is located on the southern side. She can paddle her boat at a constant speed of 5 m/s.

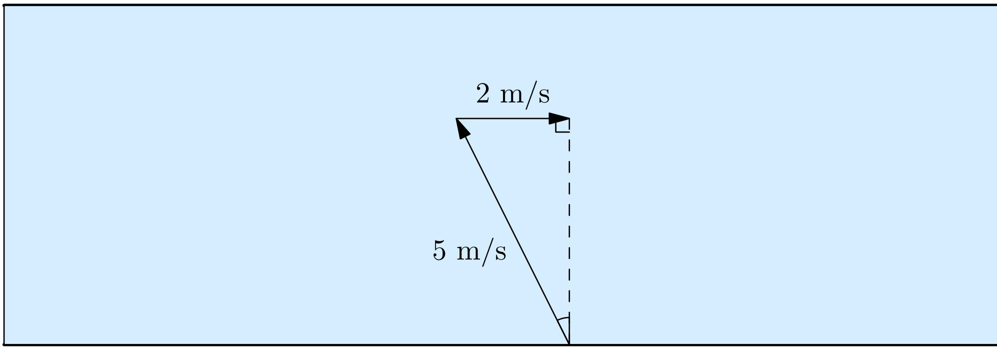
(a) What direction should Alice paddle if she wants to reach the north side of the river in the shortest time?

(b) What direction should Alice paddle if she wants to reach the north side of the river directly north from where he started? How much longer would it take for Alice to cross the shore compared to if she crossed the river as in part (a)?

Solution.

(a) The north-south component of Alice's velocity is greatest if she paddles directly north, so this is the optimal direction of paddling if she wants to reach the other side in the shortest time.

(b) We desire the Alice's net velocity to be completely in the north direction. Hence consider the following diagram:



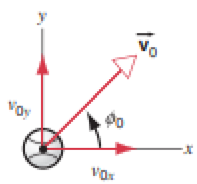
We see that our desired angle is in fact $\sin^{-1}(\frac{2}{5}) \approx 23.6°$ west of north.

In this instance we see the new net velocity is $\sqrt{5^2 - 2^2} = \sqrt{21}$. The time it takes to cross the river is $\frac{40}{\sqrt{21}} \approx 8.73$ seconds. In the optimal case, the time it would have taken would be $\frac{40}{5} = 8$ seconds. Hence it takes Alice 0.73 seconds longer.

■

2.3 Projectile Motion

One of the most common examples of two dimensional motion is that of a particle moving near Earth's surface under only the influence of gravity. As we saw in the first lesson, the acceleration of gravity can be treated as constant ($g = 9.8m/s^2$) near the surface of the Earth. We'll neglect all other forces that may act on the ball such as air resistance and/or friction. In this section, we'll try studying this motion in some more detail using our kinematics equations.



Consider a particle that is being launched with speed v_0 at an angle θ above the positive x-axis. We can consider the ball to be an object called a "projectile" as it is being "launched" off the ground. Let's try describing the path taken by the particle.

Recall we can decompose our acceleration and initial velocity vector to obtain:

$$\vec{a} = \langle 0, -g \rangle$$

$$\vec{v}_0 = \langle v_0 \cos\theta, v_0 \sin\theta \rangle.$$

Since $\vec{s} = \frac{1}{2}\vec{a}t^2 + \vec{v}_0 t$, we see that:

$$\vec{s} = \langle v_0 \cos(\theta)t, -\frac{1}{2}gt^2 + v_0 \sin(\theta)t \rangle$$

That is, we have horizontal and vertical components described by:

$$x = v_0 \cos(\theta) t$$
$$y = -\frac{1}{2}gt^2 + v_0 \sin(\theta) t$$

Projectile motion problems show up extremely often, and so it's important to be able to derive these equations for x and y. This derivation also makes sense physically, because intuitively, we'd think of the motion of a projectile to be a parabola. Indeed, relating the x and y coordinates, we see that the projectile does indeed follow parabolic motion.

Example 2.4. If a projectile is thrown with initial velocity v_i at an angle θ, what is the time it takes for the the projectile to reach the ground again?

Solution. We can set the displacement in the y-component to be zero. This gives:

$$-\frac{1}{2}gt^2 + v_i \sin(\theta) t = 0.$$

Solving gives

$$t = \boxed{\frac{2v_i \sin(\theta)}{g}}.$$

■

Video Example 2.5. How far does the projectile travel in the horizontal direction when it hits the ground? This is called the range of the projectile.

For a given velocity v_i, what is the optimal angle θ to maximize the range?

Solution. Watch `https://youtu.be/BeJFgx0Yxpo&t=312s` (timestamp 5:12) for the video solution to this problem! ■

Solution. From the previous example, we have that the time passed is

$$t = \frac{2v_i \sin(\theta)}{g}.$$

Then the horizontal distance traveled is

$$v_i \cos(\theta) t = \frac{2v_i^2 \sin(\theta)\cos(\theta)}{g}.$$

We can further simplify this using the identity $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$, which gives

$$R = \frac{v_i^2 \sin(2\theta)}{g}$$

We know the maximum value of sine occurs at 90 degrees, so $\theta = \boxed{45^\circ}$ is the optimal angle. ■

Example 2.6. What is the equation of trajectory of a projectile launched with speed v at an angle θ? That is, what is y as a function of x?

Solution. We look at our two parametric equations.

$$x = vt\cos\theta$$
$$y = vt\sin\theta - \frac{1}{2}gt^2.$$

We find that time is given by

$$t = \frac{x}{v\cos\theta}$$

which tells us the equation of motion is given by

$$v\sin\theta \cdot \left(\frac{x}{v\cos\theta}\right) - \frac{1}{2}g\left(\frac{x}{v\cos\theta}\right)^2.$$

Simplifying gives us

$$y = \boxed{x\tan\theta - \left(\frac{g\sec^2\theta}{2v^2}\right)x^2}.$$

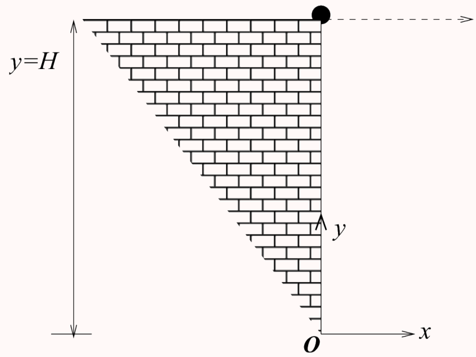
■

2.4 An Introduction to Air Resistance*

This section is a bit harder than others so feel free to skip it if you'd like. It's pretty interesting though so have a read through if you'd like a challenge!

All of the problems that we have encountered so far have factored in the fact that there is no **air resistance**. Air resistance is the resistance of air to a moving object. The air acts as a force that dissipates energy and makes the object slower. Usually, problems with air resistance simply just make the problem much less elegant and mathematically "bashy". But for curiosities sake, let us analyze an example from the 2008 Indian Physics Olympiad in which air resistance takes into play. Note that this problem is hard so if you don't know how to approach it right away, it's okay to look at the solution!

Example 2.7 (INPhO). Consider a ball which is projected horizontally with speed u from the edge of a cliff of height H as shown in the figure below. There is air resistance proportional to the velocity in both x and y direction i.e. the motion in the $x(y)$ direction has air resistance given by the $cv_x(cv_y)$ where c is the proportionality constant and $v_x(v_y)$ is the velocity in the $x(y)$ directions. Take the downward direction to be negative. The acceleration due to gravity is g. Take the origin of the system to be at the bottom of the cliff as shown in figure.



(a) Obtain expressions for $x(t)$ and $y(t)$.

(b) Obtain the expression for the equation of trajectory.

(c) Make a qualitative comparative sketch of the trajectories with/out air resistance.

(d) Given that the height of the cliff is 500 m and $c = 0.05\ \text{sec}^{-1}$. Obtain the approximate time in which the ball reaches the ground. Take $g = 10\ \text{m/s}^2$.

Solution. Note that when they mean, "the motion in the $x(y)$ direction has air resistance given by the $cv_x(cv_y)$ where c is the proportionality constant", they mean there is a negative factor of acceleration that decelerates the ball.

(a) Note that we can write that

$$a_x = \frac{dv}{dt} = -cv_x \implies \int_{v_0}^{v} \frac{dv}{v} = \int -cdt.$$

This implies that after integrating,

$$\ln \frac{v}{v_0} = -ct \implies v = v_0 e^{-ct}.$$

Remember that $v = dx/dt$ so

$$\frac{dx}{dt} = v_0 \exp(-ct) \implies \int_0^t dx = \int_0^t v_0 \exp(-ct).$$

After integrating, we find that

$$x(t) = \frac{-v_0}{c}(e^{-ct} - 1).$$

To find $y(t)$, we can essentially do the same thing. This time, $a_y = g - cv_y$ which means that

$$\frac{dv_y}{dt} = g - cv_y \implies \int dt = \int_0^{v_0} \frac{dv}{g - cv_y}.$$

By using a u substitution of $u = g - cv_y$, we find that

$$t = \frac{-1}{c} \ln \frac{g - cv_y}{g} \implies v_y = \frac{g}{c}(1 - e^{-ct}) = \frac{dy}{dt}.$$

Performing one last integral after seperating variables gives us

$$\int_y^H dy = \int_0^t \frac{g}{c}(1-e^{-ct})dt \implies H - y = \frac{g}{c}\left[t + \frac{1}{c}(e^{-ct}-1)\right].$$

By rearranging, we find that

$$y = H + \frac{g}{c^2} - \frac{g}{c}\left[\frac{1}{c}\exp(-ct) + t\right].$$

(b) Remember that our equation for $x(t)$ was

$$x = \frac{v_0}{c}(1-e^{-ct}) \implies e^{-ct} = 1 - \frac{cx}{v_0} \implies t = \frac{-1}{c}\ln\left(1 - \frac{cx}{v_0}\right).$$

Substituting our expression of time into $y(t)$ gives us

$$y = H + \frac{g}{c^2} - \frac{g}{c}\left[\frac{1}{c}e^{\ln\left(1-\frac{cx}{v_0}\right)} - \frac{1}{c}\ln\left(1-\frac{cx}{v_0}\right)\right].$$

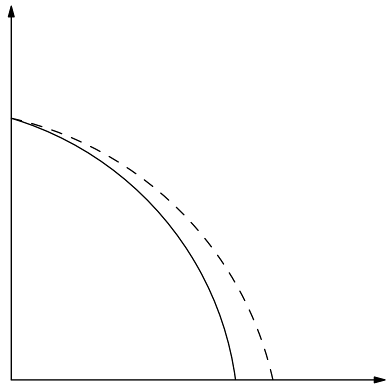
Simplifying gives us

$$\begin{aligned} y &= H + \frac{g}{c^2} - \frac{g}{c^2}\left(1 - \frac{cx}{v_0}\right) + \frac{g}{c^2}\ln\left(1 - \frac{cx}{v_0}\right) \\ y &= \boxed{H + \frac{gx}{cv_0} + \frac{g}{c^2}\ln\left(1 - \frac{cx}{v_0}\right)}. \end{aligned}$$

Remark. If c is small, then the range is limited and we can use a taylor expansion as shown below

$$\begin{aligned} y &= H + \frac{gx}{cv_0} + \frac{g}{c^2}\left[-\frac{cx}{v_0} - \frac{c^2x^2}{2v_0^2} - \frac{c^3x^3}{3v_0^3} + \dots\right] \\ y &\approx H - \frac{gx^2}{2v_0^2} - \frac{gcx^3}{3v_0^3}. \end{aligned}$$

(c) The trajectory would be smaller since there is air resistance which dissipates the total energy of the projectile. If the dashed line is the line where there is no air resistance, our sketch will look like this



(d) We remember that $y(t)$ is given by

$$y = H + \frac{g}{c^2} - \frac{g}{c}\left[\frac{1}{c}\exp(-ct) + t\right].$$

The projectile stops when $y = 0$ which means that

$$H + \frac{g}{c^2}(1 - e^{-ct} - ct) = 0.$$

Substituting

$$H = 500 \text{ m}$$
$$c = 0.05 \text{ s}^{-1}$$
$$g = 10 \text{ m/s}^2$$

we find that

$$t = \boxed{11.1 \text{ s}}.$$

■

2.5 Kinematics on Tilted Planes

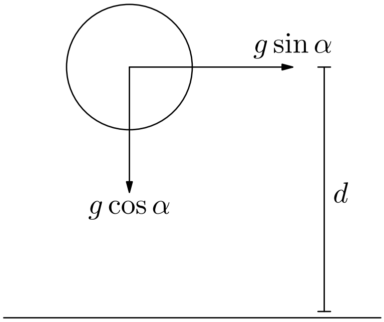
2.5.1 Projectile Motion

Sometimes in physics problems, you have problems relating projectile motion on a tilted incline. Let us work through some examples and see how the physics plays out.

Video Example 2.8. Josh stands at the base of an inclined plane that is slanted at an angle α. He throws the ball at an angle θ from the plane. What is the time of flight of the projectile?

Solution. Watch https://youtu.be/BeJFgx0Yxpo&t=701s (timestamp 11:41) for the video solution to this problem! ■

Solution. The main technique that is to be used in this problem is a **tilted coordinate system**. A tilted coordinate system is where you tilt the coordinates of a physical scenario such that the physics becomes much easier. In this case, when you tilt the *entire* plane by an angle α such that it becomes flat as shown below, you will get two components of acceleration.



Therefore, on tilted planes, the effective gravity $\vec{g}_{\text{eff}} = g\cos\alpha$. The equation for the time of flight can then be replaced with our new values to give

$$T = \frac{2v\sin\theta}{\vec{g}_{\text{eff}}} = \boxed{\frac{2v\sin\theta}{g\cos\alpha}}.$$

■

Concept. The free-fall acceleration on a ramp can be decomposed into two respective components:

$$\vec{g} = \vec{g}_x + \vec{g}_y$$

where $g_x = g\sin\alpha$ and $g_y = g\cos\alpha$.

Example 2.9 ($F = ma$)**.** A ball is released from release above an inclined plane and bounces elastically down the plane. As the ball progresses down the plane, the time and distance between each collision will:

(A) remain the same, and increase.

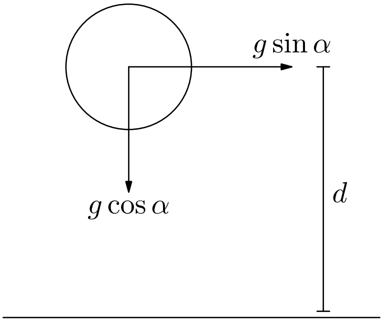
(B) increase, and remain the same.

(C) decrease, and increase.

(D) decrease, and remain the same.

(E) both remain the same.

Solution. We once again draw a diagram on a tilted coordinate system.

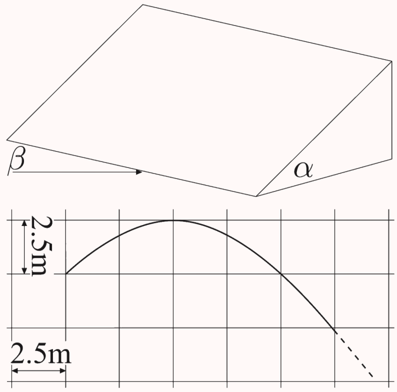


The ball can then be thought of as a ball that is bouncing up and down with a new effective acceleration of gravity of $\vec{g}_{\text{eff}} = g\cos\alpha$. Therefore, the time between each collision will remain the same as nothing is changing. In the x direction, the ball is acceleration with an acceleration of $g\sin\alpha$. This tells us that with each progressive collision with the ramp there will be more distance covered as the ball is accelerating. This means the answer is $\boxed{\text{(A)}}$. ■

2.5.2 Sliding

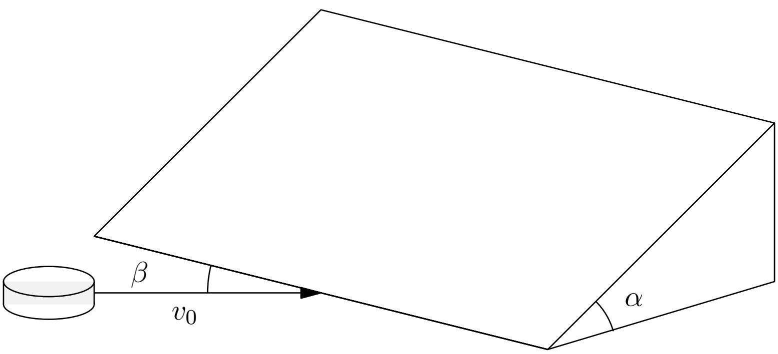
Usually, the same concepts required with projectile motion on tilted ramps apply to sliding motion on tilted planes as well. We can give an example of how this works.

Example 2.10 (Kalda). A puck slides onto an icy inclined plane with inclination angle α. The angle between the plane's edge and the puck's initial velocity $v_0 = 10$ m/s is $\beta = 60°$. The trace left by the puck on the plane is given in the figure (this is only a part of the trajectory). Find α under the assumption that friction can be neglected and that transition onto the slope was smooth.



Solution. When on the plane, the puck experiences no change in its x-velocity, which is

$$v_0 \cos\beta = 5 \text{ m/s}.$$



However, it experiences an acceleration parallel to the plane with magnitude

$$a = g \sin\alpha.$$

We note from the graph given that the puck drops 2.5 m below the apex of its trajectory while undergoing a horizontal displacement of $x = 5$ m.

The time it takes to complete its trajectory is

$$\begin{aligned} t &= \frac{x}{v_0 \cos\beta} \\ &= 1 \text{ s}. \end{aligned}$$

Therefore, we have that

$$\frac{gt^2 \sin\alpha}{2} = 2.5$$
$$\sin\alpha = \frac{5}{gt^2}$$
$$\alpha \approx \boxed{30^\circ}.$$

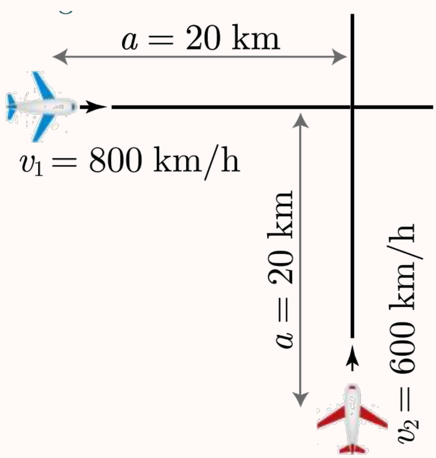
■

2.6 Reference Frames

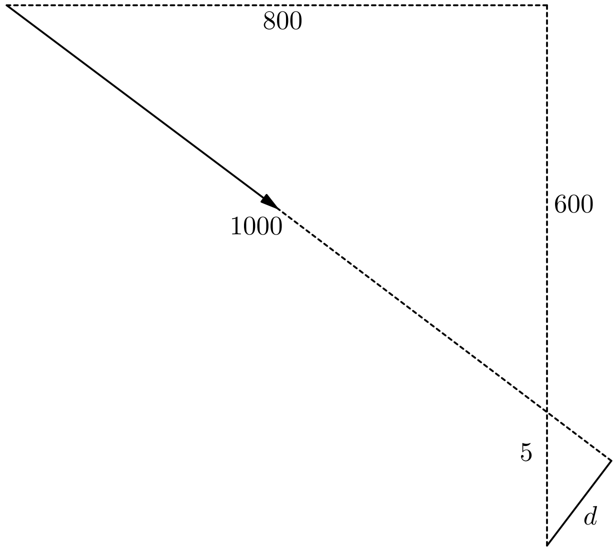
Consider the following scenario: you are sitting on a bench, and a car is driving towards you. From your perspective, you are stationary, and the driver of the car is approaching you with velocity $\vec{v}$. On the other hand, the driver does not see himself moving; he sees himself as stationary, and you are approaching him with velocity $-\vec{v}$.

If you want to determine the position of yourself relative to the car, both of these perspectives, called **reference frames**, are in fact valid. Depending on the problem, it may sometimes help to change reference frames. Consider the following example.

Example 2.11 (Kalda). Two planes fly at the same height with speeds $v_1 = 800$ km/h and $v_2 = 600$ km/h respectively. The planes approach each other; at a certain moment of time, the plane trajectories are perpendicular to each other and both planes are at the distance $a = 20$ km from the intersection points of their trajectories. Find the minimal distance between the planes during their flight assuming their velocities remain constant.



Solution. This is an excellent problem regarding several techniques that were presented before. We can view the problem from the reference frame of the red plane (i.e. what the pilot of the red plane would see). In this frame, the red plane is stationary, and we see the blue plane with a diagonally directed velocity.



The closest point is where a line from the red plane to the path of the blue plane is perpendicular. This turns out into a geometry problem where we have two similar right triangles.

We can break up the velocity of the blue plane into components (since the displacement is in the same direction as velocity, this is also the components of its displacement). The top triangle is a $3-4-5$ right triangle so the bottom right triangle must also be a $3-4-5$ right triangle.

Now all we need to know now is to determine how far away the blue plane is when it is directly overhead the red plane. The time it takes to reach this point is

$$\begin{aligned} t &= \frac{20 \text{ km}}{800 \text{ km/h}} \\ &= 0.025 \text{ h}. \end{aligned}$$

and the vertical distance it travels during this time is

$$\begin{aligned} \Delta y &= (600 \text{ km/h})(0.025 \text{ h}) \\ &= 15 \text{ km/h}. \end{aligned}$$

meaning the vertical separation is 5 km/h. Therefore

$$d = \boxed{4 \text{ km}}.$$

■

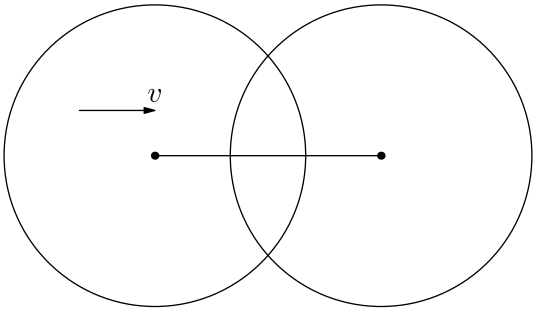
When using different reference frames, it helps to use the following two concepts.

Concept. Try to figure out some type of geometry from the system. For example, try to find similar triangles, angles, etc.

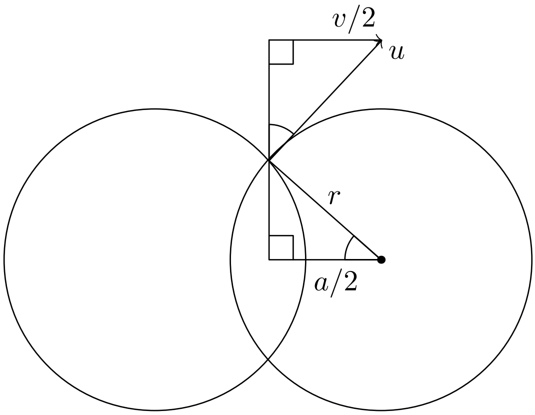
Concept. It is possible to figure out everything about a velocity or acceleration once we know one of its components and the direction of the vector.

Example 2.12. One of two rings with radius r is at rest and the other moves at a velocity v towards the first one. Find how the velocity of the upper point of intersection depends on a, the distance between the two rings' centres.

Solution. Consider the following diagram:



Moving into the reference frame moving leftward at velocity $v/2$ gives us the following diagram (where u is the velocity of intersection):



Using SAS similarity, we find that

$$\begin{aligned}
\frac{u}{v/2} &= \frac{r}{\sqrt{r^2-(a/2)^2}} \\
u &= \frac{v}{2r\sqrt{r^2-(a/2)^2}} \\
&= \boxed{\frac{v}{2\sqrt{1-(a/2r)^2}}}.
\end{aligned}$$

■

2.7 Homework Problems and Solutions

Problem 2.1 (Morin). [4] A wall has height h and is a distance l away. You wish to throw a ball over the wall with a trajectory such that the ball barely clears the wall at the top of its parabolic motion. What initial speed is required?

Solution. Recall that the time it takes for a projectile to reach the top is

$$t = \frac{v_i \sin\theta}{g}.$$

At this time, the x-distance traveled is

$$v_i \cos(\theta) t = l,$$

and the y-distance traveled is

$$v_i \sin(\theta) t - \frac{1}{2} g t^2 = h.$$

Substituting, we have that

$$\begin{aligned} v_i^2 \sin\theta\cos\theta &= gl \\ v_i^2 \frac{\sin^2\theta}{g} - v_i^2 \frac{\sin^2\theta}{2g} &= h \end{aligned}$$

Simplifying gives

$$\begin{aligned} v_i^2 \sin\theta\cos\theta &= gl \\ v_i^2 \sin^2\theta &= 2gh \end{aligned}$$

Dividing the equations gives $\tan\theta = 2\frac{h}{l}$. Therefore, some simple geometry gives

$$\sin\theta = \frac{2h}{\sqrt{4h^2 + l^2}}.$$

Thus

$$v_i = \frac{\sqrt{2gh}\sqrt{4h^2 + l^2}}{2h} = \boxed{\sqrt{\frac{g(4h^2 + l^2)}{2h}}}.$$

We now give a shorter solution. Let t be the time at which the ball reaches its maximum. Then we have the initial y-velocity $v_{y,i} = gt$, and the x-velocity is $v_x = l/t$.

Recall that

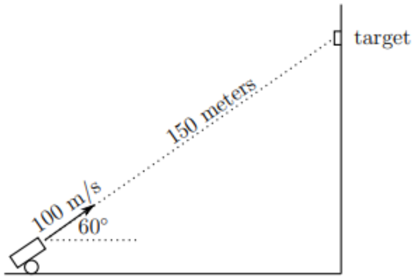
$$\begin{aligned} -\frac{1}{2} g t^2 + v_{y,i} t &= h \\ -\frac{1}{2} g t^2 + g t^2 &= h, \\ \Rightarrow t &= \sqrt{\frac{2h}{g}}. \end{aligned}$$

Thus, substituting for t gives

$$v_i = \sqrt{v_{y,i}^2 + v_x^2} = \boxed{\sqrt{\frac{g(4h^2 + l^2)}{2h}}}.$$

■

Problem 2.2 ($F = ma$)**.** [4] A 470 g lead ball is launched at a 60° angle above the horizontal with an initial speed of 100 m/s directly toward a target on a vertical cliff wall that is 150 m away as shown in the figure.



Ignoring air friction, by what distance does the lead ball miss the target when it hits the wall?

Solution. In the absence of gravity, the lead ball is situated to directly hit the target, and would do so in a time of $\frac{150\text{ m}}{100\text{ m/s}} = 1.5$ s. However, gravity is of course not absent. The displacement of the ball due to gravity will in fact give the distance by which the ball misses the target. Thus, we have

$$d = \frac{1}{2}gt^2 = 5\text{ m/s}^2 \cdot (1.5\text{ s})^2 = \boxed{11.25\text{ m}}$$

■

Problem 2.3 (HRK)**.** [5] In 1936, Jesse Owens set a new world long jump record of 8.09 m at the Olympics in Berlin, where $g = 9.8128\text{ m/s}^2$. Assuming his running speed and takeoff angle would be unchanged, how much different would the record have been if he had jumped in Melbourne, where $g = 9.7999\text{ m/s}^2$? (You can ignore air resistance during the jump.)

Solution. Recall that the time it takes for a projectile to reach its maximum height is

$$\frac{v_i \sin\theta}{g},$$

so we have the maximum height is

$$h = v_i t - \frac{1}{2}gt^2 = \frac{v_i^2 \sin^2\theta}{2g}.$$

We see that h and g are inversely proportional, so the record would instead have been $8.09\text{ m} \cdot 9.8128/9.7999 \approx 8.10$ m, or 0.0106 m longer. ■

Problem 2.4 (NBPhO)**.** [6] A boy is running on a large field of ice with velocity $v = 5$ m/s toward the north. The maximum acceleration that can be provided from friction is 1 m/s^2 Assume as a simplification that the reaction acceleration between the boy and the ice stays constant (in reality it varies with every push, but the assumption is justified by the fact that the value averaged over one step stays constant). What is the minimum time necessary for him to change his moving direction to point towards the east so that the final speed is also $v = 5$ m/s?

Solution. Let $\hat{\jmath}$ be directed North and let $\hat{\imath}$ be directed East. Then, the initial velocity is $v\,\hat{\jmath}$ and the final velocity is $v\,\hat{\imath}$. We then have that

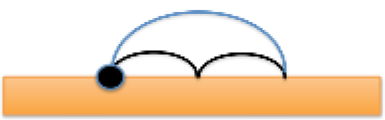
$$\begin{aligned}\Delta\vec{v} &= 5\,\hat{\imath} + 5\,\hat{\jmath}\\ |\Delta\vec{v}| &= 5\sqrt{2}\\ &= a\Delta t\end{aligned}$$

Since $a \leq 1$,

$$\Delta t \geq \boxed{5\sqrt{2}}.$$

■

Problem 2.5 (Morin). [6] A person throws a ball with speed v_0 at a 45° angle and hits a given target. At what angle should the ball be thrown with the same initial speed v_0 so that it makes the trajectory consisting of two identical bumps, as shown below? (Assume there is no loss of speed during the bounce.)



Solution. Recall that the range of a projectile is given by

$$\frac{v_0^2\sin(2\theta)}{g}.$$

Hence, the range of the first projectile is v_0^2/g, and for the second projectile we desire that

$$\frac{v_0^2\sin(2\theta)}{g} = \frac{v_0^2}{2g}.$$

In other words,

$$\sin(2\theta) = \frac{1}{2},$$

which we see occurs when $\boxed{\theta = 15^\circ}$. ■

Problem 2.6 (Morin). [7] A movie director wants a scene in which a car appears to fly off a cliff at 50 mph. To simulate this, the director will make a 1/100 scale model of a car and cliff, and have the model car fly off the model cliff.

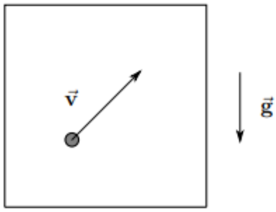
By what factor should the film be sped up or slowed down in order to look realistic, and what should the real speed of the toy car be as it approaches the cliff?

Solution. If the film of the model was played at normal speed, the person will fall unnaturally fast. Recall that g is constant in both reference frames. Hence, the model car will fall through a height h in a much faster amount of time in the model than a real car would fall through a height $100h$ in real life.

Since $h = \frac{1}{2}gt^2$, we see that we have to slow the video down by a factor of 10 for the film to seem realistic. ■

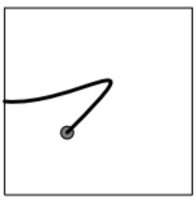
The following information applies to Problems 2.7 and 2.8.

Consider a particle in a box where the force of gravity is down as shown in the figure.



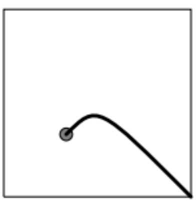
The particle has an initial velocity as shown, and the box has a constant acceleration to the right.

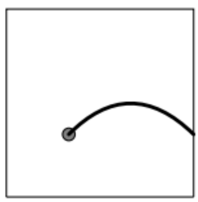
Problem 2.7 ($F = ma$)**.** [8] In the frame of the box, which of the following is a possible path followed by the particle?

(A) 

(B) 

(C) 

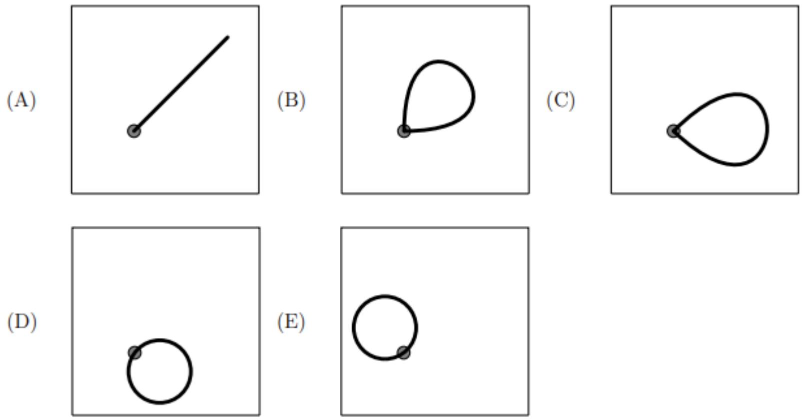
(D) 

(E) 

Solution. If the box has a constant acceleration to the right, then in *the frame of the box* (where the box is stationary), the particle is experiencing a constant acceleration to the left. Combined with the gravitational acceleration, this gives a net acceleration to the lower left.

A particle experiencing a constant net acceleration will under go parabolic motion, with the parabola's axis of symmetry having the same direction as the acceleration vector. Hence, we obtain (C) as our answer. ∎

Problem 2.8 ($F = ma$)**.** [7] If the magnitude of the acceleration of the box is chosen correctly, the launched particle will follow a path that returns to the point that it was launched. In the frame of the box, which path is followed by the particle?



Solution. In the previous problem, we saw that the motion is parabolic with acceleration toward the lower left. The only way the particle can return to its original position, then, is if it travels in a straight line parallel to its acceleration. Thus the answer is (A). Another approach could be to observe that a straight line is a very thin parabola, so (A) is the only one that matches the previous problem, while the other choices do not. ■

Problem 2.9. [10] A marble launcher that launches marbles at a fixed initial velocity v_0 is situated at a distance x away from a wall. The angle of the marble launcher can be varied, and let θ be the angle at which the marble hits the wall at the maximum possible height (ignoring air resistance). What is $\tan\theta$?

Solution. First we find the height h at which the marble hits the wall as a function of θ. Recall that we have

$$\begin{aligned} x &= v_0 \cos(\theta) t \\ t &= \frac{x}{v_0 \cos(\theta)}. \end{aligned}$$

By substituting, we have

$$\begin{aligned} h &= -\frac{g}{2} t^2 + v_0 t \sin(\theta) \\ h &= -\frac{g x^2}{2 v_0^2 \cos^2\theta} + x \tan(\theta). \end{aligned}$$

Now we find the critical point. We see that

$$\begin{aligned} \frac{dh}{d\theta} &= -\frac{g x^2}{v_0^2 \cos^2\theta} \tan\theta + \frac{x}{\cos^2\theta} && = 0 \\ -\frac{g\Delta x}{v_0^2}(\tan\theta) + 1 = 0 \\ \tan\theta &= \boxed{\frac{v_0^2}{g x}}. \end{aligned}$$

■

Problem 2.10 (Kalda)**.** [12] A cannon is situated in the origin of coordinate axes and can give initial velocity v_0 to a projectile, the shooting direction can be chosen at will. What is the region of space $\mathcal{R}$ that the projectile can reach?

Solution. We split up v_0 into it's vertical and horizontal components. From here we can see that each parameter x, y and z as a function of t is

$$z = v_0 t \sin\alpha - \frac{1}{2}gt^2 \tag{2.1}$$
$$x = v_0 \cos\alpha t \tag{2.2}$$
$$t = \frac{x}{v_0\cos\alpha} \tag{2.3}$$

From here, we substitute equation 3 into equation 1 to yield

$$z = v_0\left(\frac{x}{v_0\cos\alpha}\right)\sin\alpha - \frac{1}{2}g\left(\frac{x}{v_0\cos\alpha}\right)^2.$$

Simplifying with trigonometry yields

$$z = x\tan\alpha - \frac{gx^2}{2v_0^2}\sec^2\alpha$$
$$z = x\tan\alpha - \frac{gx^2}{2v_0^2}(\tan^2\alpha + 1)$$
$$0 = \frac{gx^2}{2v_0^2}\tan^2\alpha - x\tan\alpha + \frac{gx^2}{2v_0^2}$$

Here, we find a quadratic. For the region of space $\mathcal{R}$ to exist, the discriminant of this quadratic must be greater than zero. This tells us

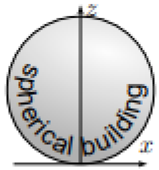
$$x^2 - 4\left(\frac{gx^2}{2v_0^2}\right)\left(z + \frac{gx^2}{2v_0^2}\right) \geq 0$$
$$x^2 - \frac{2gx^2}{v_0^2}z + \frac{g^2x^4}{v_0^4} \geq 0$$
$$\frac{2gx^2}{v_0^2}z \leq x^2 - \frac{g^2}{x^4}v_0^4$$

Simplifying this final expression gives us the answer. The region of space of $\mathcal{R}$ is

$$\boxed{z \leq \frac{v_0^2}{2g} - \frac{gx^2}{2v_0^2}}$$

■

Problem 2.11 (IPhO)**.** [14] A ball thrown with an initial speed v_0 moves in a homogeneous gravitational field of strength g; neglect the air drag. The throwing point can be freely selected on the ground level z = 0, and the launching angle can be adjusted as needed; the aim is to hit the topmost point of a spherical building of radius R (see fig.) with as small as possible initial speed v_0 (prior hitting the target, bouncing off the roof is not allowed). What is the minimal launching speed $v_{\min}$ needed to hit the topmost point of a spherical building of radius R?



Solution. Imagine the trajectory starting from the top of the building. This trajectory must be tangent to the building again because otherwise it can be brought closer and can be thrown with a smaller speed. We also know that the path must be contained within the region $\mathcal{R}$ mentioned in the previous problem. This was a parabola and we claim that this parabola is also tangent at the tangency point of the building and ball trajectory. This may be a bit tricky to think about, but you can imagine how it changes by increasing and decreasing the velocity of the ball. So we can use the parabola from the previous problem, and we solve for the tangency with the building. That tells us

$$z = R + \frac{v_t^2}{2g} - \frac{gx^2}{2v_t^2}$$
$$x^2 + z^2 = R^2.$$

(we shifted the coordinates). Then we substitute z into our equation for the circle. So we have

$$x^2 + R^2 + \frac{v_t^4}{4g^2} + \frac{g^2}{4v_t^4}x^4 + \frac{2Rv_t^2}{2g} - \frac{2Rg}{2v_t^2}x^2 - \frac{2v_t^2}{4v_t^2}x^2 = R^2.$$

This is a quadratic in x^2. Let's rearrange, and then we want to find a v_t so that the quadratic has only 1 solution. Why? Because we want tangency.

$$x^4\left(\frac{g^2}{4v_t^4}\right) + x^2\left(\frac{v_t^2 - 2Rg}{2v_t^2}\right) + v_t^2 \cdot \frac{v_t^2 + 4Rg}{4g^2} = 0.$$

We want the discriminant to be 0 so there is only one solution. So we have

$$\frac{(v_t^2 - 2gR)^2}{4v_t^4} - \frac{v_t^2(v_t^2 + 4gR)}{4v_t^4} = 0.$$

Solving this equation gives $8gRv_t^2 = 4g^2R^2$, so $v_t^2 = \frac{1}{2}gR$, so we apply conservation of energy and

$$\frac{1}{2}mv_0^2 - \frac{1}{2}mv_t^2 = 2mgR.$$

So $v_0 = \boxed{\sqrt{\frac{9}{2}gR}}$. ■

2.7.1 Written Solutions

Problem 2.12. [17] Farmer Danny's farm is located at the point $(0, -20)$, and the local market is located at the point $(80, 0)$. The line $y = 0$ is highway on which Danny can drive at a horizontal speed of $v = 25$. However, outside the highway Danny can only drive with a speed of $v = 15$.

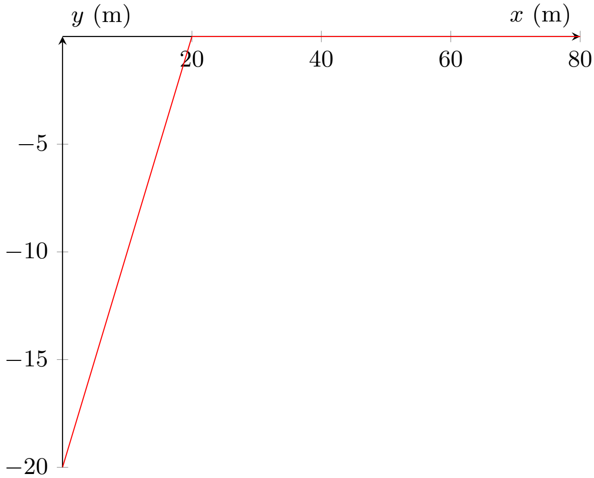
(a) [4] Draw a sketch of the general shape of the path Danny should follow to reach the market in the shortest possible time.

(b) [6] Let Danny's initial velocity be at an angle $\theta < 180°$ from the positive y-direction. Assuming Danny follows a path of the same shape sketched in the previous part, how much time does he take to reach the market (in terms of θ)?

(c) [7] At what angle θ does Danny reach the market in the shortest time?

Solution.

(a) The optimal path consists of a straight line to the x-axis, followed by a straight line to the right.



The angle/slope of the above graph need not be correct since the question just asks for a sketch of the general shape.

(b) First, we compute the position at which Danny reaches the x-axis. Some simple trigonometry gives this point to be $(20\tan\theta, 0)$. The time taken is given by

$$\frac{20}{15\cos\theta} = \frac{4}{3\cos\theta.}$$

The time it takes for Danny to get to the market after reaching the x-axis is then

$$\frac{80 - 20\tan\theta}{25} = \frac{16}{5} - \frac{4}{5}\tan\theta.$$

Thus, the total time it takes is

$$\frac{16}{5} - \frac{4}{5}\tan\theta + \frac{4}{3\cos\theta}.$$

(c) To find the optimal theta, we can find the critical point of the above expression.

Hence we have

$$\begin{aligned}
&\frac{d}{d\theta}\left(\frac{16}{5} - \frac{4}{5}\tan\theta + \frac{4}{3\cos\theta}\right) \\
&= -\frac{4}{5\cos^2\theta} + \frac{4\sin\theta}{3\cos^2\theta} = 0 \\
&= \sin\theta = \frac{3}{5} \Rightarrow \theta = \boxed{\sin^{-1}\left(\frac{3}{5}\right)}.
\end{aligned}$$

■

II

Dynamics

3 Forces

We'll now begin our first foray of what makes up the bulk of mechanics: dynamics. Today we'll look at *why* things move in addition to *how* they move.

3.1 Newton's First and Second Laws

To understand Newton's Laws, we first have to introduce the notion of a **force**. A force can be thought of as an interaction between two objects that causes an acceleration. Forces are vectors, and the units of their magnitude is the Newton ($\mathrm{N} = \mathrm{kg} \cdot \mathrm{m/s^2}$).

Last lesson, we briefly looked at reference frames. However, not all laws of physics are the same in all reference frames. Newton's First Law now defines an **inertial reference frame**, which is a reference frame in which Newton's laws hold and take their simplest form.

> **Theorem 3.1** (Newton's First Law). *In an inertial reference frame, an object's acceleration is zero if and only if the net external force acting on the object is zero.*

Unless otherwise stated, we will deal only with inertial reference frames.

An important distinction to make here is the term *net* force, which refers to the sum of all the forces acting on an object. The object can have forces acting on it, but as long as the sum of all those forces is zero, the velocity will not change. This brings us to Newton's Second Law.

> **Theorem 3.2** (Newton's Second Law). *In an inertial reference frame,*
>
> $$\vec{F}_{net} = m\vec{a},$$
>
> *where $\vec{F}_{net}$ is the net force acting upon and object, m is the mass of the object, and $\vec{a}$ is the acceleration of the object.*

If $\vec{a} = 0$ and $\vec{v} = 0$, where $\vec{v}$ is the velocity of the object, then the system is in static equilibrium. When $\vec{a} = 0$ and $\vec{v} \neq 0$, then the system is in dynamic equilibrium. *In either case, $\vec{a} = 0$ implies that a system is in equilibrium.* Note that we can treat forces like vectors, and the net force can be obtained simply from adding all the forces acting on an object. Here's a simple example.

> **Example 3.3.** A bowling ball of mass 5 kg is thrown vertically in the air. At a certain time, its acceleration is approximately 9 $\mathrm{m/s^2}$. What is the magnitude of the air resistance force acting on the ball at this instant? Use 10 $\mathrm{m/s^2}$ for the acceleration of gravity.

Solution. Recall that the gravitational force is $mg = 50$ N. By Newton's Second

Law, we have that

$$\vec{F}_g - \vec{F} = m\vec{a}$$
$$50 \text{ N} - F = 5 \text{ kg} \cdot 9 \text{ m/s}^2$$

Hence, the air resistance force has magnitude $F = 5$ N. ∎

Note that Newton's Second Law implies that the net force and acceleration vectors always point in the same direction.

3.2 Common Forces

In order to find the net force on an object, we have to be able to identify the forces acting on the object. *Outside of the gravitational force, most forces that appear in mechanics are contact forces, meaning you only need to consider what the object is in contact with. If you are ever stuck wondering what forces might act on an object, ask yourself: Have I accounted for the gravitational force? What is the object in contact with?* We now list some common forces:

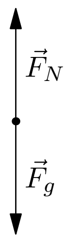
1. The **gravitational force** ($\vec{F}_g, mg$, or W) always exists between two objects with mass. We will only consider this force if one of those objects is massive, like a star or planet, as otherwise the interaction would be negligible. The gravitational force a person on Earth experiences always has magnitude mg, where m is equal to their mass. By Newton's second law, we see that the acceleration due to gravity is g, and is in fact constant.
2. The **normal force** ($\vec{F}_N$ or N) is a support force when two objects come into contact. For example, for a box lying on the ground, the ground is providing it with a normal force which prevents the box from accelerating downwards from the gravitational force it experiences. *The normal force always acts perpendicular to the surface.*
3. **Friction** ($\vec{F}_f$ or $\vec{f}$) is another extremely common force. Unlike normal forces, friction always acts parallel to the surface. A common misconception is that friction prevents motion; a more accurate statement is that friction prevents *sliding*. When you walk, the friction force is what lets you push forward and prevents your foot from sliding in the backwards direction. Friction also appears in the form of **air resistance**. Because friction is more complicated, we will discuss it in more in detail in the next lesson.
4. **Tension** ($\vec{F}_T$ or $\vec{T}$) is the force you feel when pulling on a rope (such as in tug of war). It is also the force you feel when stretching a rubber band. Tension forces always push away from the point of contact (that is, you cannot push an object using rope, you can only pull it).
5. **Compression** is the opposite of tension. This is the force that occurs when you squeeze two opposite ends of an object.
6. The **spring force** is applied by springs. Springs come up in a lot of problems, and we have a lot to talk about them, so we will discuss them more in detail in a later handout. In general though, ideal springs experience a force proportional to their displacement from rest length. This is called Hooke's Law:

$$\vec{F} = -k\vec{x}.$$

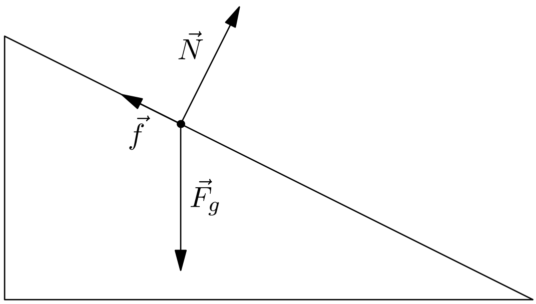
7. All the above forces, except for gravity, are **electromagnetic forces**. This is because of how electric charges in molecules interact when they get really close to each other. We will not be going in depth into the electromagnetic force in this course.
8. The **weak force** and **strong force** are the other two fundamental forces of nature, along with gravity and electromagnetism. The weak force determines radioactive decay. The strong force keeps a nucleus stable; it attracts protons together. Gravity is the weakest fundamental force, followed by the weak force, then the electromagnetic force, then the strong force, which is the strongest. Again, we will not be going in depth into the weak and strong forces in this course.

Applied forces ($\vec{F}$ or $\vec{F}_{\text{app}}$) describe the forces we typically think of. If you push a box, you are exerting a normal force on it, but we can also say you are applying a force or exerting an applied force.

When analyzing forces, the most common method is by drawing a free body diagram. This is essentially representing the object by a point, and drawing vectors to represent the forces it experiences. For example, the free body diagram for an ball lying on the ground might look like this:



$\vec{F}_N$ stands for normal force, while $\vec{F}_g$ stands for gravitational force. The free body diagram for a box lying on a ramp might look like this:

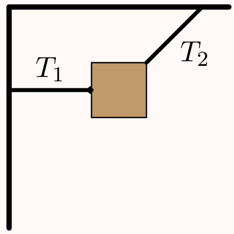


In general, free body diagrams are extremely helpful tools that help visualize the forces acting on an object.

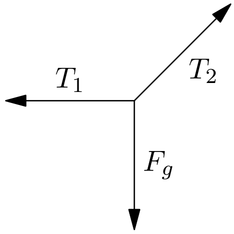
3.3 Static Equilibrium

Static equilibrium problems appear often, but the key realization is simply to apply Newton's First Law. Since there is no acceleration, the net force (and both its horizontal and vertical components) must equal zero. Here is a basic example.

Example 3.4. A box hangs from the ceiling suspended by two cords, one of which is horizontal, and one of which forms a 45 degree angle with the vertical, as shown below. If the box weights 40 N, what is the tension force in the cords?



Solution. Since the box is not accelerating, we can conclude the net force is zero (by Newton's Second Law). Now, we draw a free-body diagram:



Since, the vertical components of the forces must cancel, we see that the vertical component of T_2 is equal to mg, or 40 N. Since we are given T_2 forms a 45 degree angle with the vertical, we can use trigonometry to see that T_2 has a horizontal component of 40 N as well. This means T_2 has a magnitude of $40\sqrt{2}$ N $\implies \boxed{T_2 = 56.6 \text{ N}}$.

Likewise, the horizontal components of the forces must cancel, so $\boxed{T_1 = 40 \text{ N}}$. ■

Of course, these problems can be much more complicated, but the same strategy applies. Sometimes we will also see static equilibrium with more than one object. The forces can again be analyzed with Newton's First Law. However, we need to introduce a new rule as well.

3.4 Newton's Third Law

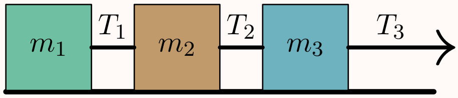
Theorem 3.5 (Newton's Third Law). *For every force between two objects, there is a force that is equal in magnitude and opposite in direction.*

Let's break down what this means. If an object experiences a "force," the force must be applied by another object. For example, the gravitational force you experience is the gravitational force the Earth exerts on you. When you stand, the floor exerts a normal force on you. When you sprint, the ground also exerts a friction force on you that accelerates you forward. A force is always an interaction between two objects.

Newton's Third Law states that this interaction is always symmetrical: just as the Earth exerts a gravitational force on you, you exert an *equally strong* gravitational force on it. If you push a wall, it will push you back with an equal amount of force.

Using Newton's second law combined with Newton's third law can be extremely helpful to analyze motion.

Example 3.6 (Walker). Three connected blocks are pulled to the right on a horizontal frictionless table by a force of magnitude $T_3 = 65$ N.



If $m_1 = 12$ kg, $m_2 = 24$ kg, and $m_3 = 31$ kg, calculate the following quantities:

(a) The acceleration of the blocks

(b) The tension T_1

(c) The tension T_2

Solution. Watch https://youtu.be/j0s0wsDz27o&t=427s (timestamp 7:07) for the video solution to this problem! ■

Solution. This is, quite literally, a classic textbook example using Newton's Laws.

We see that in the vertical direction, each block experiences a normal force and gravitational force, but these must cancel, so they can be ignored. We then need only consider the tension force. It is important to know that the tension force always "pulls" an object instead of "pushing" (you cannot push anything using string). This tells us the directions of the tension forces. We then note that both ends of a string will exert the same force, as given by Newton's third law.

As the blocks are pulled, they will also move together, meaning their accelerations will be equal. Using Newton's Second Law on each of the blocks individually, we can write the following equations:

$$\begin{aligned} T_1 &= 12 \text{ kg} \cdot a \\ T_2 - T_1 &= 24 \text{ kg} \cdot a \\ 65 \text{ N} - T_2 &= 31 \text{ kg} \cdot a \end{aligned}$$

Adding these equations, we see that $65 \text{ N} = 67 \text{ kg} \cdot a$, so $a = \boxed{0.970 \text{ m/s}^2}$. We actually see here another application of Newton's Third Law: we can view all three blocks as a single object, with only a single external force T_3; for those doing these problems for the first time, this solution might be harder to see, but it is a quicker method to compute the acceleration.

To solve part (b), we now look at block 1. We have $T_1 = 12 \text{ kg} \cdot a = \boxed{11.6 \text{ N}}$.

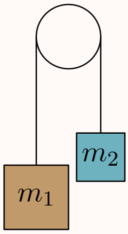
We can now solve part (c) by substituting into $T_2 - T_1 = 24 \text{ kg} \cdot a$, and we get $T_2 = \boxed{34.9 \text{ N}}$. ■

3.5 Systems in Physics

The previous problem briefly introduced us to the idea of a **system**. In physics, the term system is simply used to refer to a collection of objects that is being analyzed. Think of a system of equations — a system of equations is a group of equations being solved.

A basic example of a system is a massless pulley.

Example 3.7. Consider the system with a massless frictionless pulley shown below.



Given that $m_1 > m_2$, find

(a) the acceleration of the block with mass m_1 downward

(b) the acceleration of the block with mass m_2 upward

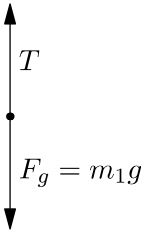
(c) in tension in the rope

This is a basic textbook example of analyzing pulley motion. We will first solve this using an application of Newton's laws.

Solution. Watch `https://youtu.be/j0s0wsDz27o&t=811s` (timestamp 13:31) for the video solution to this problem! ■

Solution. First, note that the equation describes a massless frictionless pulley. This means that the tension force on both sides of the pulley must be the same.

Let the magnitude of the acceleration of the block with mass m_1 be a_1, the block with mass m_2 be a_2, and the tension in the rope be T. Consider the free body diagram for the block of mass m_1:



From this, we can see that the net force on m_1 is $F_g - T$ downward (remember that $m_1 > m_2$). Thus, by Newton's Second Law,

$$m_1 g - T = m_1 a_1.$$

Similarly, we drawing the free body diagram for the block of mass m_2 gives

$$T - m_2 g = m_2 a_2.$$

Now, we have two equations and three unknowns. The third equation is a bit tricky. Note that because the rope is taut, its length will remain constant. This means that for every unit that unit length the left block goes down, the right block must also move up that same unit length. In other words, the acceleration of both blocks are of the same magnitude, just in opposite directions. This is our third equation:

$$a_1 = a_2.$$

Substituting the third equation into the second and adding that to the first gives

$$m_1 g - m_2 g = m_1 a_1 + m_2 a_1$$
$$a_1 = \boxed{\frac{m_1 g - m_2 g}{m_1 + m_2}}.$$

Thus, we must also have $a_2 = \boxed{\dfrac{m_1 g - m_2 g}{m_1 + m_2}}$.

From the second equation, we have $T = m_2(a_2 + g)$, so

$$\begin{aligned} T &= m_2 \left(\frac{m_1 g - m_2 g}{m_1 + m_2} + g \right) \\ &= m_2 \left(\frac{m_1 g - m_2 g + m_1 g + m_2 g}{m_1 + m_2} \right) \\ &= \boxed{\frac{2 m_1 m_2 g}{m_1 + m_2}}. \end{aligned}$$

■

This solution is a more canonical one. However, we can simplify the process of solving this equation by considering the pulley as a system.

Solution. Consider if we were to treat blocks m_1 and m_2 as a single entity (By Newton's Third Law, the net tension forces of the cord connecting them will sum to zero and can be ignored). Then the net external force on the system of block m_1 and m_2 is $(m_1 - m_2)g$.

You might wonder why the net external force is not $(m_1 + m_2)g$, since the gravitational force is pointing downward on both blocks. However, consider when the pulley accelerates: block m_1 moves down, and block m_2 moves up. In this case, which direction is acceleration?

We might better describe acceleration as in the counterclockwise direction, from which we get the right next external force $(m_1 - m_2)g$. An alternative explanation is that the gravitational force on block m_2 works opposing the direction of acceleration. Since net force and acceleration are always in the same direction, then $m_2 g$ should in fact be subtracted.

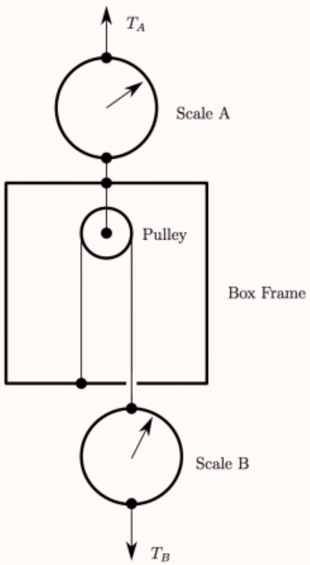
The total mass of the system is $m_1 + m_2$, so by Newton's Second Law we see that acceleration is

$$a = \frac{F_{net}}{m} = \boxed{\frac{m_1 g - m_2 g}{m_1 + m_2}}.$$

■

In the case of this pulley, this approach works because the acceleration is the same for objects. For other systems where the acceleration is the same, this approach will also work. However, in some systems objects will move with different accelerations. To analyze these systems, we will later introduce the idea of center of mass. First, we'll do another example of a system with a pulley.

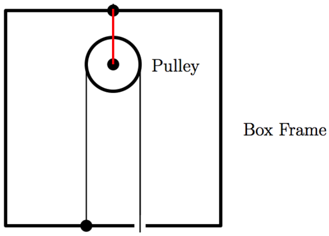
Example 3.8 ($F = ma$)**.** Consider the following diagram of a box and two weight scales. Scale A supports the box via a massless rope. A pulley is attached to the top of the box; a second massless rope passes over the pulley, one end is attached to the box and the other end to scale B. The two scales read indicate the tensions T_A and T_B in the ropes. Originally scale A reads 30 Newtons and scale B reads 20 Newtons.



If an additional force pulls down on scale B so that the reading increases to 30 Newtons, what will be the new reading on scale A?

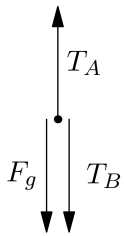
(A) 35 Newtons
(B) 40 Newtons
(C) 45 Newtons
(D) 50 Newtons
(E) 60 Newtons

Solution. Consider the system containing everything inside of the box frame.



Note that the pulley remains static with respect to the box frame, as the rope highlighted in red is taut and therefore has constant length. Similarly, the portion of rope that wraps around the pulley is also static with respect to the box frame, as it is attached to the box frame and is taut. This means that the acceleration of the objects inside the box frame is the same as the box frame itself, so we can consider the system as one object!

Let us analyze the free-body diagram for the box. We see that the only external forces are T_A, T_B, and gravity.



Thus,

$$T_A = T_B + F_g.$$

Now, consider what happens when the reading on scale B increases to 30 N; that is, T_B increases to 30 N. F_g does not change, as the mass of the stuff inside the box frame is constant. Therefore, T_A must change to oppose the increase in T_B.

The net change in T_B is 30 N − 20 N = 10 N, so T_A must increase by 10 N to counteract this change. The initial value of T_A is 30 N from the problem statement, so the answer is $30\text{ N} + 10\text{ N} = \boxed{40 \text{ Newtons}}$. ■

3.6 Center of Mass

The center of mass is an important property of a system. Here, we will define it and introduce some basic applications.

Definition 3.9. Given a system consisting of n points with masses $m_1, m_2, \ldots, m_n$ at positions $\vec{x}_1, \vec{x}_2, \ldots \vec{x}_n$ respectively, the system's **center of mass** is located at the weighted average of these points, or

$$\vec{x}_{\text{cm}} = \frac{\sum_{k=1}^{n} m_k \vec{x}_k}{\sum_{k=1}^{n} m_k}.$$

For a continuous mass distribution with total mass M, the center of mass is given by

$$\vec{x}_{\text{cm}} = \frac{1}{M} \int \vec{x}\, dm.$$

Here's a quick computational example.

Example 3.10. Find the center of mass of a 1 kg particle at $(0,0)$, a 1 kg particle at $(1,0)$, and a 2 kg particle at $(1,1)$.

Solution. The center of mass is at

$$\left(\frac{\sum_{k=1}^{3} m_k x_k}{\sum_{k=1}^{3} m_k}, \frac{\sum_{k=1}^{3} m_k y_k}{\sum_{k=1}^{3} m_k}\right) = \boxed{\left(\frac{3}{4}, \frac{1}{2}\right)}$$

■

You should also intuitively know centers of masses of many uniform objects. For instance, the center of mass of a uniform disk is at its center. The center of mass of a uniform solid triangle is at its centroid. However, some problems may ask you to compute the center of mass of a continuous object where the center of mass is not immediately obvious. This is when calculus is helpful.

Example 3.11. What is the center of mass of a uniform semicircular disk of radius r?

Solution. Let the diameter lie along the x-axis and let the axis of symmetry of the semicircular disk lie along the y-axis. Position the disk entirely above the x-axis. By symmetry, the x-coordinate of the center of mass is zero. However, we must compute the y-coordinate. By symmetry again, we can consider finding the y-coordinate of the center of mass of a quarter circle.

The key to solving these problems is finding what to substitute for $\mathrm{d}m$. Since a disk is two-dimensional, define the area mass density

$$\sigma = \frac{\mathrm{d}m}{\mathrm{d}A}.$$

Then we have

$$y_{\mathrm{cm}} = \frac{1}{M}\int y\sigma \,\mathrm{d}A.$$

Since σ is constant, we can take it outside the integral. Also, note that $dA = xdy$, so the equation simplifies to

$$\frac{\sigma}{M}\int xy \,\mathrm{d}y = \frac{1}{A}\int xy \,\mathrm{d}y.$$

For a quarter circle, we have $x = \sqrt{r^2 - y^2}$, so

$$y_{\mathrm{cm}} = \frac{1}{A}\int_0^r y\sqrt{r^2 - y^2}\,\mathrm{d}y = \frac{1}{A}\cdot\frac{r^3}{3}.$$

We have $A = \frac{1}{4}\pi r^2$, so $y_{\mathrm{cm}} = \frac{4r}{3\pi}$. Hence, the center of mass is at

$$\vec{x}_{\mathrm{cm}} = \boxed{\left(0, \frac{4r}{3\pi}\right)}$$

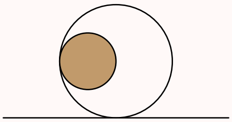
when we position the semicircular disk as above. ■

Theorem 3.12. *In a system with total mass M, the acceleration of the center of mass of the system is given by*

$$\vec{a}_{cm} = \frac{\vec{F}_{ext}}{M}$$

Corollary 3.13. *The velocity of the center of mass of a closed system (a system experiencing no net external force) will not change.*

Example 3.14. Consider a 2 kg hollow shell with radius 1 m and mass evenly distributed throughout the shell. A 1 kg solid ball of radius 0.5 m and mass evenly distributed is placed inside as shown.



When the solid ball is let go, it will roll back and forth inside the shell. Simultaneously, the shell will roll to the left, then to the right, then to the left and so on. Eventually, the ball will come to a stop at the bottom of the shell. Find the displacement of the shell from the original position at this time. Assume that no energy is lost due to friction.

Solution. Consider the system consisting of the ball and the shell. The external forces acting on this system are gravity and the normal force. Notice how both of these forces point in vertical directions. This means that the center of mass will only move vertically; it will not move horizontally.

We can use this fact to our advantage. Let us first calculate the x-position of the center of mass. Note that this will not change over time. Putting the shell on a coordinate plane where the origin is at the shell's center, we see that the center of the mass of the shell is at $(0, 0)$, and the center of mass of the solid ball is at $(-0.5, 0)$. Taking the weighted average of the two, we get that the center of mass of the entire system is at

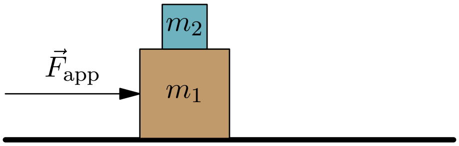
$$\left(\frac{0 \cdot 2 + (-0.5) \cdot 1}{3}, \frac{0 \cdot 2 + 0 \cdot 1}{3}\right) = \left(-\frac{1}{6}, 0\right).$$

Thus, the center of mass of the system at both the initial and final states has x-position $-\frac{1}{6}$. In the initial state, the center of the shell was at $(0, 0)$. In the final state, the solid ball rests at the bottom of the shell. Therefore, both the ball and the shell have the same x-coordinate, which is the same as the x-coordinate of the center of mass. Hence, the x-coordinate of the shell is $-\frac{1}{6}$ at the final state.

Our answer is $\boxed{\frac{1}{6} \text{ m to the left}}$. ■

3.7 Homework Problems and Solutions

Problem 3.1 (Morin). [3] You accelerate the two blocks in the figure below by pushing on the bottom block with a force of F. The top block moves along with the bottom block. What force directly causes the top block to accelerate?



(A) The normal force between the blocks
(B) The friction force between the blocks
(C) the gravitational force on the top block
(D) The force you apply to the bottom block

Solution. The friction force always acts to prevent sliding – if the top block did not accelerate then there would be sliding between the two blocks, so it is the frictional force that accelerates the top block. Thus the answer is $\boxed{(B)}$. ■

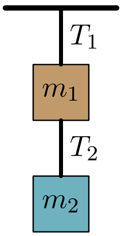
Problem 3.2. [4] The air resistance of a bowling ball of mass M falling is given by $\left|\vec{f}\right| = \alpha v^2$, where v is the speed of the bowling ball and α is a constant. Find the terminal velocity of a bowling ball near Earth's surface.

Solution. When the ball is at terminal velocity it is not accelerating so the net force is 0. Thus the force due to gravity and air resistance cancel out giving

$$\alpha v^2 = mg$$

and solving for v we get $v = \boxed{\sqrt{\frac{mg}{\alpha}}}$. ■

Problem 3.3. [4] The system of blocks below is in static equilibrium:



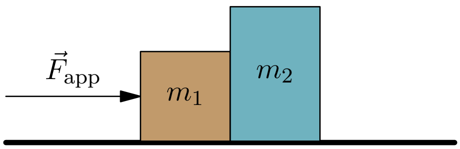
(a) What is the tension T_1?
(b) What is the tension T_2?

Solution. (a) Consider the system consisting of the two blocks m_1 and m_2 and the string connecting them. The external forces on this system are gravity, with magnitude $m_1 g + m_2 g$ and the force of tension from the top string, T_1. Since the system is in equilibrium, $T_1 = \boxed{m_1 g + m_2 g}$

(b) The forces on the bottom block are gravity, with magnitude $m_2 g$ and the force of tension T_2. Since it is in equilibrium $T_2 = \boxed{m_2 g}$.

■

Problem 3.4. [4] In the diagram below, a force $\vec{F}_{\text{app}} = 12$ N is applied to the block on the left. If $m_2 = 4$ kg and the normal force between the blocks is 8 N, what is m_1?

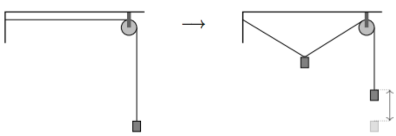


Solution. First we look at the forces on on the 4 kg block on the right. Gravity and the normal force with the surface balance out giving a zero vertical force. The only horizontal force is the normal force between the two blocks. Thus its acceleration is $8/4 = 2m/s^2$. The two blocks must have the same acceleration (if m_2 accelerates faster than m_1 then they will lose contact and so m_2 will necessarily stop accelerating). The net horizontal force on the leftmost block is $F_{appl} - f_{normal} =$ 12N - 8N = 4N the Thus we have

$$F_{app} = 4N = m_1 a = m_1(2\text{m/s}^2)$$

so $m_1 = 4/2 = \boxed{2\text{ kg}}$ ■

Problem 3.5 ($F = ma$)**.** [9] A block of mass m is attached to a massless string. The string is passed over a massless pulley and the end of the string is fixed in place. The horizontal part of the string has length L. Now a small mass m is hung from the horizontal part of the string, and the system comes to equilibrium. (Diagram not necessarily to scale.)



(a) [4] Neglecting friction everywhere, the tension at the end of the string is
  (A) $mg/2$
  (B) mg
  (C) $3mg/2$
  (D) $2mg$
  (E) $3mg$

(b) [5] During the process, the block has been raised by approximately a height
  (A) $0.15L$
  (B) $0.23L$
  (C) $0.31L$
  (D) $0.37L$

(E) $0.40L$

Solution.

(a) The only forces on the block at the end of the string are gravity with magnitude mg and tension T. Since the block is in equilibrium, the net force is 0 so these two forces must cancel giving $T = \boxed{mg}$, so the answer is (B).

(b) Since the string and pulley are massless, the tension is the same, mg, everywhere in the string. Now we look at the forces on the hanging block. Let θ be the angle the strings make with the horizontal. Recall that the tension points in the same direction as the string. Then the horizontal forces from the two ropes cancel, and the vertical component of the tension from each string is $mg\sin(\theta)$. Then the net vertical force on the block is

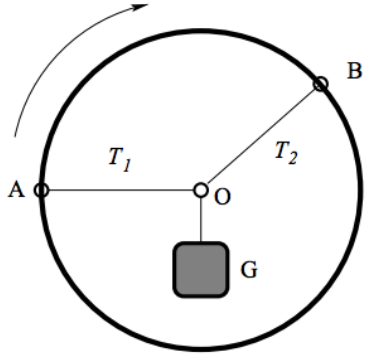
$$F_{net} = 2mg\sin(\theta) - mg = 0.$$

This gives $\theta = 30°$. Then the length of string between the wall and pulley is

$$L' = L/\cos(\theta) = \frac{2}{\sqrt{3}}L \approx 1.15L$$

This means the length of string changes by $\boxed{0.15L}$ giving choice (A). ■

Problem 3.6 ($F = ma$)**.** [6] A rigid hoop can rotate about the center. Two massless strings are attached to the hoop, one at A, the other at B. These strings are tied together at the center of the hoop at O, and a weight G is suspended from that point. The strings have a fixed length, regardless of the tension, and the weight G is only supported by the strings. Originally OA is horizontal.



Now, the outer hoop will start to slowly rotate 90° clockwise until OA will become vertical, while keeping the angle between the strings constant and keeping the object static. Which of the following statements about the tensions T_1 and T_2 in the two strings is correct?

(A) T_1 always decreases.

(B) T_1 always increases.

(C) T_2 always increases.

(D) T_2 will become zero at the end of the rotation.

(E) T_2 first increases and then decreases.

Solution. At the end of the rotation OA is vertical so the only horizontal force comes from OB. Since G is not accelerating horizontally, the tension in OB must be zero, so the answer is $\boxed{(D)}$.

Why are the other answer choices wrong? Initially T_1 is increasing. At some point, say t_1, OB will be horizontal and the diagram will essentially look like a reflection of the diagram in the problem. Then the value of T_1 will be equal to the initial value of T_2, which is greater than the initial value of T_1. This means T_1 increased and T_2 decreased. This eliminates choices (A) and (E) right away.

But when OA is vertical, T_2 will be zero, so the tension T_1 is just the weight of G. This means thatt T_2 increased and T_1 decreased since t_1. This eliminates choices (B) and (C). Thus we are left with choice D. ■

Problem 3.7 ($F = ma$)**.** [6] A massless rope passes over a frictionless pulley. Particles of mass M and $M + m$ are suspended from the two different ends of the rope. If $m = 0$, the tension T in the pulley rope is Mg. If instead the value m increases to infinity, the value of the tension

(A) stays constant

(B) decreases, approaching a nonzero constant

(C) decreases, approaching zero

(D) increases, approaching a finite constant

(E) increases to infinity

Solution. We will calculate the tension as a function of m. The force on the mass M is $Mg - F_t$ and the force on the mass $M + m$ is $(M + m)g - F_t$, so their accelerations are

$$\begin{aligned} a_1 &= \frac{(M+m)g - F_t}{M+m} \\ a_2 &= \frac{Mg - F_t}{M} \end{aligned}$$

and setting $a_1 = -a_2$ we get

$$\begin{aligned} \frac{(M+m)g - F_t}{M+m} &= \frac{Mg - F_t}{M} \\ M(M+m)g - MF_t &= F_t(M+m) - M(M+m)g \\ 2M(M+m)g &= F_t(2M+m) \end{aligned}$$

so solving this equation we finally get

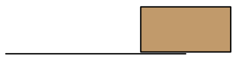
$$F_t = \frac{2M(M+m)g}{2M+m}$$

Interestingly, this is just g times the harmonic mean of the masses on the two sides of the pulley. As $m \to \infty$, this goes to $2Mg$ so the answer is (D). ■

Problem 3.8. [11] Preston has a bunch of 1 m long slabs, all with uniformly distributed mass, and a table. He first places a single slab down so that its edge lines up with the table's edge, then builds a bridge off the side of a table as follows.

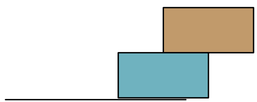
1. Preston pushes the slab structure currently on the table as far over the edge as possible without it collapsing.
2. Preston takes a new slab and aligns its edge with the edge of the table.
3. Preston takes this slab and moves it directly under the slab structure in step 1.
4. Go back to step 1.

For example, the bridge looks like this with one block:



Note how the structure will not fall over since its center of mass is on the edge of the table.

It looks like this with two blocks:



Again, note how the structure will not fall over.

(a) [2] Find how far from the edge of the table the rightmost part of a 1 block bridge is (after doing step 1).

(b) [4] Find how far from the edge of the table the rightmost part of a 2 block bridge is (after doing step 1).

(c) [5] Find how far from the edge of the table the rightmost part of an n block bridge is.

Solution. In general, the structure will topple if and only if its center of mass is hanging over the edge of the table.

(a) The center of mass of a one block structure is in the center of the slab, so half the slab will be hanging over the edge of the table. This means it will be $\boxed{0.5 \text{ m}}$ from the edge of the table.

(b) From our result for part (a), the top block will be slide over 0.5 m from the bottom block, so by symmetry the center of mass of the two blocks is 0.75 m from the edge of the structure. This means that it will be $\boxed{0.75 \text{ m}}$ from the edge of the table.

(c) Let l_k be the distance between the kth and $k + 1$st block from the edge of the configuration. For example the distance between the rightmost and second rightmost blocks is $l_1 = 0.5$m and the distance between the the second and third rightmost block is $l_2 = 0.25$m. Then if there are n blocks, the distance from the edge of the table to the rightmost block, what we want to find, is $L = l_1 + l_2 + \cdots + l_n$ where l_n is defined to be the distance between edge of the bottom block and the table. Note that l_k is a constant which does not depend on how many blocks are on the structure (assuming that there are at least k blocks of course).

For all n, we require that the center of mass of the structure, C_n is over the very edge of the table so it is about to but does not quite collapse. Let x_k be (the x-coordinate of) the center of mass of the kth block, letting $x = 0$ at the edge of the table. Then the distance from the table to the rightmost edge of the kth block is $l_n + l_{n-1} + \cdots + l_k$ so the center of mass of the kth block is at

$$x_k = \sum_{i=k}^{n} l_i - \frac{1}{2}.$$

The center of mass of the structure is the average of the center of masses of all the blocks, that is

$$\begin{aligned} 0 = C_n &= \frac{1}{n} \sum_{i-1}^{n} x_i \\ &= \frac{1}{n} \left(\sum_{i=1}^{n} l_i + \sum_{i=2}^{n} l_i + \cdots + (l_{n-1} + l_n) + l_n \right) - \frac{1}{2} \end{aligned}$$

Notice that for any given i, the term l_i shows up in exactly i of these sums, so this is just the same as saying,

$$0 = C_n = \frac{1}{n} \sum_{i=1}^{n} i l_i - \frac{1}{2}$$

so we get

$$\sum_{i=1}^{n} i l_i = \frac{n}{2}$$

for all n. In particular, for $n - 1$ we have

$$\sum_{i=1}^{n-1} i l_i = \frac{n-1}{2}$$

and so subtracting the two equations gives

$$n l_n = \frac{1}{2}$$

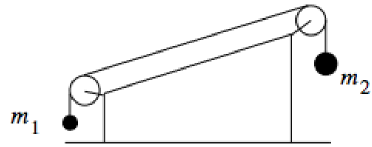
so we conclude that for any k, we have $l_k = \frac{1}{2k}$. Thus the edge of the rightmost block is a distance

$$\sum_{i=1}^{n} l_i = \boxed{\frac{1}{2} \sum_{i=1}^{n} \frac{1}{i}}$$

away from the edge of the table.

■

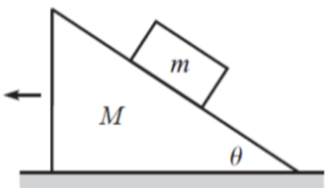
Problem 3.9 ($F = ma$)**.** [6] Two masses are attached with pulleys by a massless rope on an inclined plane as shown. All surfaces are frictionless. If the masses are released from rest, then the inclined plane



(A) accelerates to the left if $m_1 < m_2$
(B) accelerates to the right if $m_1 < m_2$
(C) accelerates to the left regardless of the masses
(D) accelerates to the right regardless of the masses
(E) does not move

Solution. The only forces acting in this system are the gravitational forces and tension forces all of which are vertical. Since the string is massless there cannot be any horizontal forces applied to it. Thus the inclined plane does not move and the answer is (E). ■

Problem 3.10 (Morin). [8] A block of mass m is held motionless on a frictionless plane of mass M and angle of inclination θ. The plane rests on a frictionless horizontal surface. The block is released. What is the horizontal acceleration of the plane?



Solution. We will do this problem by looking at the acceleration of the inclined plane and the block and solving the resulting system of equations. It gets a bit algebraic, but it is worth it! Let N be the normal force between the block and the inclined plane (note that since the plane is moving we can't just say $N = mg\cos(\theta)$ as we did before). Let a_p be the acceleration of the plane and a_{bx} and a_{by} the x and y components of the acceleration of the block (where the positive x-direction is pointing to the right). Then we get following the three equations from Newton's laws:

$$\begin{aligned} Ma_p &= -F_n\sin(\theta) \\ ma_{bx} &= \frac{F_n\sin(\theta)}{m} \\ ma_{by} &= \frac{mg - F_n\cos(\theta)}{m} \end{aligned}$$

But these accelerations cannot just be anything. The block has to remain in contact with the inclined plane throughout its motion. The block's acceleration to the right relative to the plane is $a_{bx} - a_p$, and its vertical acceleration is a_{by}. The total acceleration relative to the plane must be at an angle θ below the horizontal, so

$$\frac{a_{by}}{a_{bx} - a_p} = \tan(\theta)$$

We substitute our values for the acceleration that we found before into this equation:

$$\frac{g - N\cos(\theta)/m}{N\cos(\theta)\left(\frac{1}{m} + \frac{1}{M}\right)} = \tan(\theta)$$

Next we solve for the normal force N since it will be easy to find the accelerations from it, getting

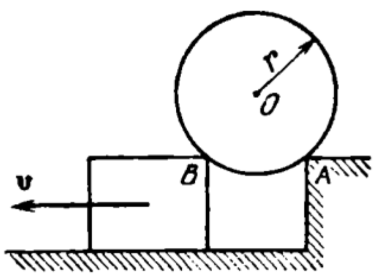
$$F_n = \frac{gmM}{M\cos(\theta) + \tan(\theta)\sin(\theta)(m + M)}$$

Then we substitute this in to find a_p, getting

$$a_p = -\frac{F_n \sin(\theta)}{M} = \boxed{-\frac{gm\sin(\theta)\cos(\theta)}{M + m\sin^2(\theta)}}$$

■

Problem 3.11 (Krotov). [10] A cylinder of mass m and radius r rests on two supports of the same height. One support is stationary, while the other slides from under the cylinder at a velocity v. Determine the force of normal pressure N exerted by the cylinder on the stationary support at the moment when the distance between points A and B of the supports is $AB = r\sqrt{2}$, assuming that the supports were very close to each other at the initial instant. The friction between the cylinder and the supports should be neglected.



Hint: 24

Solution. First let's figure out the horizontal velocity and acceleration of the cylinder. The x-coordinate of the center of mass of the cylinder is always at the midpoint of AB, so its velocity is the average of the velocities of points A and B which is just $v/2$. Thus the horizontal acceleration is $a_h = 0$.

Next we find the vertical velocity of the cylinder. Since we want to find the vertical acceleration, it is not enough to just find the velocity at the moment when the cylinders are $r\sqrt{2}$ apart but we want to find it as a function of the separation distance. Let θ be the angle of the radius drawn to the support points from the vertical. If we let w be the distance between the supports and h be the height that the center of mass is above the plane of the supports, then $h = r\cos(\theta)$ and $w = 2r\sin(\theta)$. At the instant in time that the problem is asking about, $\theta = \frac{\pi}{4}$ and $w = r\sqrt{2}$.

The vertical velocity v_v is given by

$$v_v = \frac{dh}{dt} = \frac{dh}{d\theta}\frac{d\theta}{dt}$$

In order to evaluate this we must calculate $\frac{dh}{d\theta}$ and $\frac{d\theta}{dt}$.

$$\frac{dh}{d\theta} = \frac{dr\cos(\theta)}{d\theta} = -r\sin(\theta)$$
$$\frac{d\theta}{dt} = \frac{d\theta}{dw}\frac{dw}{dt}$$

Since $w = 2r\sin(\theta)$, we have $\frac{dw}{d\theta} = 2r\cos(\theta)$ so this means $\frac{d\theta}{dw} = 1/(2r\cos(\theta))$ so substituting this into our expression we get

$$\begin{aligned}\frac{d\theta}{dt} &= \frac{d\theta}{dw}\frac{dw}{dt}\\ &= \frac{v}{2r\cos(\theta)}\end{aligned}$$

Finally we can put all this back into our equation for v_v to get

$$v_v = -r\sin(\theta)\frac{v}{2r\cos(\theta)} = -\frac{v\tan(\theta)}{2}$$

Now we will calculate the vertical acceleration a_v (the good news now is that some of the intermediate steps we did to find v_v will be helpful again here).

$$\begin{aligned}a_v = \frac{dv_v}{dt} &= -\frac{v}{2}\sec^2\theta\frac{d\theta}{dt}\\ &= -\frac{v^2}{4r}\sec^3(\theta)\end{aligned}$$

once we substitute in the value of $d\theta/dt$ that we found before. At the instant of time that we care about, $\theta = \frac{\pi}{4}$ so $\sec(\theta) = \sqrt{2}$, and substituting this in we have

$$a_v = -\frac{v^2}{r\sqrt{2}}$$

Now all we need to do is apply Newton's second law to get the force. The normal pressure force from the two supports is equal by symmetry, since the horizontal force is 0 and there is not friction. This means the net vertical force is just

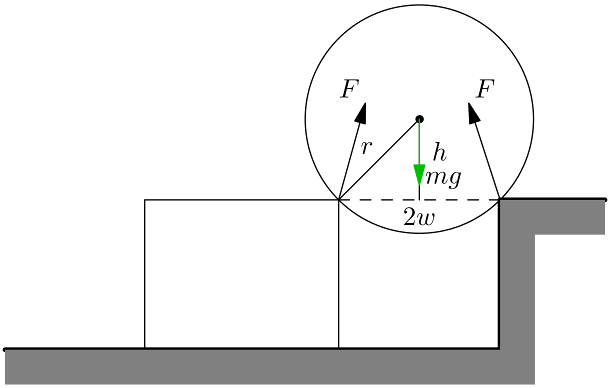
$$F_{net} = 2N\sin\theta - mg = \sqrt{2}N - mg = \sqrt{2}N - mg$$

Then by Newton's second law we have

$$\sqrt{2}N - mg = \frac{mv^2}{r\sqrt{2}}$$

and solving for N we get

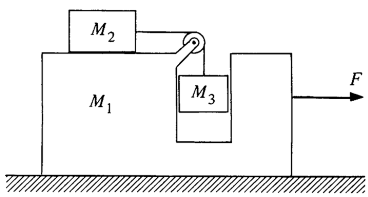
$$N = \boxed{\frac{mg}{\sqrt{2}} - \frac{mv^2}{2R}}$$



■

Problem 3.12 (Kleppner and Kolenkow). [14]

(a) [7] A "pedagogical machine" is illustrated in the sketch. All surfaces are frictionless. What force must be applied to M_1 to keep M_3 from rising or falling?



(b) [7] Consider the "pedagogical machine" in the case of where F is zero. Find the acceleration of M_1. **Hint:** 1

Solution.

(a) If F_t is the tension in the string, then in order to keep M_3 from rising or falling we must have $F_t = M_3 g$. Then the acceleration of M_2 is $a = F_t/M_2 = M_3 g/M_2$. Since M_3 is not rising or falling, the length of the string between M_2 and the pulley stays the same, so the acceleration of a_2 must be the same as the acceleration of M_1. Naively you might then just set $F = M_1 a$, but notice that the force F is not just accelerating block M_1 but is accelerating the entire system. It is the only external horizontal force on the system of the three blocks, and since all three blocks accelerate together, we have

$$F = (M_1 + M_2 + M_3)a = \boxed{\frac{(M_1 + M_2 + M_3)M_3 g}{M_2}}$$

(b) Let a_1, a_2 be the accelerations of blocks one and two and let a_3 be the vertical acceleration of block three – notice that its horizontal acceleration is just a_1 since it is carried along by the big block. We will get two equations in this problem, one for the condition that the length of the string stays the same and the other Newton's second law applied to block m_1.

First we will apply the condition that the length of the string stays the same throughout the motion. The acceleration of block two in the reference frame of the large block is $a_2 - a_1$, which tells us how the horizontal length of string changes, and a_3 tells us how the vertical length of the string changes. Then we have the condition

$$a_3 = a_2 - a_1$$

Using Newton's second law, we have $a_3 = g - F_t/m_3$ and $a_2 = F_t/m_2$, so substituting those in we have

$$g - \frac{F_t}{m_3} = \frac{F_t}{m_2} + a_1$$

solving for F_t (so we can substitute it into our next equation) we get

$$F_t = \frac{g + a_1}{\frac{1}{m_2} + \frac{1}{m_3}}$$

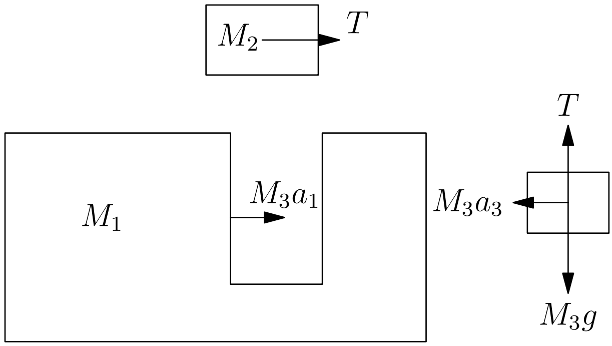
Next we will balance forces on the big block m_1. There is no net external force acting on the system consisting of all three blocks, so the only force acting on m_1 comes from blocks m_2 and m_3. Blocks 2 and 3 will have some horizontal acceleration – both because the tension force pulls m_2 to the right and because m_1 accelerates m_3 along with it and all this acceleration ultimately comes from their interactions with m_1. By Newton's third law, we can see that the net force acting on the system consisting of blocks two and three is the same as the net force acting on block one. This gives us the equation

$$-a_1 m_1 = a_1 m_3 + F_t$$

substituting in our value for F_t and solving for a_1, we get

$$\begin{aligned} -a_1 m_1 &= a_1 m_3 + \frac{g + a_1}{\frac{1}{m_2} + \frac{1}{m_3}} \\ -a_1(m_1 + m_3) &= \frac{g m_2 m_3 + a_1 m_2 m_3}{m_2 + m_3} \\ -a_1(m_1 + m_3)(m_2 + m_3) &= g m_2 m_3 + a_1 m_2 m_3 \\ -a_1[(m_1 + m_3)(m_2 + m_3) + m_2 m_3] &= g m_2 m_3 \\ a_1 &= \boxed{-\frac{g m_2 m_3}{(m_1 + m_3)(m_2 + m_3) + m_2 m_3}} \end{aligned}$$

This is a bit of a weird looking answer so we'll do a quick spot check to be sure this makes sense. First of all, it's the right units, surely a good sign. If m_1 goes to infinity then the acceleration goes to 0, which makes sense, and likewise if m_3 goes to infinity it also goes to 0. I'll stop here, but feel free to keep doing more of these until you are satisfied with your answer!



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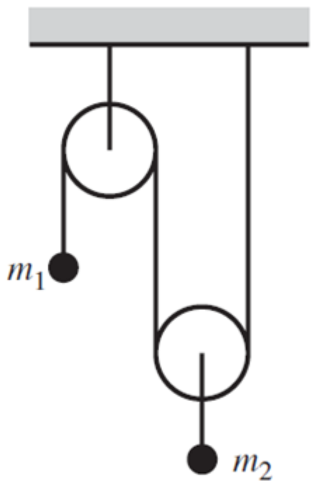
3.7.1 Written Solutions

Problem 3.13 (Morin). [15] Consider the double Atwood's machine shown below.



Compute the following quantities. **Hints:** 19

(a) [5] The acceleration of block m_1.

(b) [5] The acceleration of block m_2.

(c) [5] The tension force T in the string.

Solution. We will just solve all three parts at the same time here since solving one allows us to solve the others pretty easily. Let a_1 and a_2 be the acceleration of the masses (positive direction is taken to be upwards). Newton's second law gives us

$$a_1 = \frac{F_t}{m_1} - g$$
$$a_2 = \frac{F_t}{m_2} - g.$$

Next we will apply the conservation of string relation. The "acceleration" in the amount of string caused by m_1's motion is a_1 and by m_2's motion is $2a_2$, since it requires the length of the string to change on both sides of the pulley, so the conservation of string relation is

$$2a_2 + a_1 = 0.$$

Substituting in the values for a_1 and a_2, we get

$$2\left(\frac{F_t}{m_1} - g\right) + \frac{F_t}{m_2} - g = 0$$

so we can solve for F_t:

$$F_t\left(\frac{4}{m_2} + \frac{1}{m_1}\right) = 3g$$

so

$$\begin{aligned} F_t &= \frac{3g}{\frac{4}{m_2} + \frac{1}{m_1}} \\ &= \boxed{\frac{3gm_1m_2}{4m_1 + m_2}} \end{aligned}$$

Then we can plug this into our first two equations to get a_1 and a_2 as follows,

$$\begin{aligned} a_1 &= \frac{3m_2g}{4m_1 + m_2} - g = \boxed{\frac{g(2m_2 - 4m_1)}{4m_1 + m_2}} \\ a_2 &= \frac{3m_1g}{4m_1 + m_2} - g = \boxed{\frac{g(2m_1 - m_2)}{4m_1 + m_2}} \end{aligned}$$

Finally, let's do a quick check to make sure this all makes sense – yes our units our correct, and $2a_2 = -a_1$ as expected. If we let m_1 get large, then a_1 goes to $-g$ and a_2 goes to $g/2$, also as expected. This is looking good! ■

4 The Normal Force and Friction

4.1 Normal Force

We introduced the normal force in the previous handout, but in this handout we'll dive a bit deeper into the properties of the force while also pairing it with other common physical phenomena. Here we'll study the properties of friction and how they apply on inclines (and even moving inclines).

When you stand on the floor, you might "feel" a force acting on your feet. That is, you might "feel" your weight. What we actually "feel" as weight though is the force between the ground and our feet, or the normal force.

This is because for an object at rest experiencing just the gravitational and normal forces, since the net force is zero, we can conclude $N = mg$ which is a force of equivalent magnitude to that of our weight and is why we "feel" our weight on the floor. However, when an object experiences other forces or is accelerating, $N = mg$ is no longer necessarily true. The normal force N is also called the **apparent weight**, and is the reason why you feel slightly heavier when you ride an elevator that is accelerating up or when your on an airplane.

In real life, elevators move at constant velocity for the majority of the ride, only sharply accelerating and decelerating at the start and end of the ride, but sometimes problems will involve accelerating elevators. Here's an easy example.

> **Example 4.1.** You are standing on a scale in an elevator. When the elevator is at rest, the scale reads 50 kg. What does the scale read when the elevator is accelerating upward at 4 m/s^2?

Solution. We would like the acceleration to be 4 m/s^2 upward. Then we can write the following,

$$ma = N - F_g.$$

Thus, $N = ma + F_g = ma + mg = 200 \text{ N} + 490 \text{ N} = 690 \text{ N}$. ■

Problems will usually require using a strategy of this sort. Though obviously contests problems will generally be more difficult, by applying the same principles and being careful, you'll be able to solve almost all problems of this sort. We'll explore the other applications of the normal force in the following sections.

4.2 Static and Kinetic Friction

The normal force is also directly related to friction, another prominent force. Friction is the force that enables us to walk across surfaces: if there was no horizontal friction force that you applied on the ground (and by Newton's third law the ground applies on you), there would be no horizontal force that would push you forward. That's why its harder to move on ice; the friction is very small!

Friction is a force that opposes the relative motion between two objects and points parallel to the surfaces in contact. There are a couple important things to note about friction. All of the following laws have been verified experimentally and hold to a large degree of approximation.

1. For moving objects, the friction force is proportional to the normal force between them. (Amonton's First Law)
2. The friction force is independent of the contact area between two objects. (Amonton's Second Law)
3. The force of friction for moving objects is independent of the velocities of the objects in contact. (Coulomb's Law of Friction)
4. The friction force depends on the materials of the two surfaces in contact. (This is obvious, consider a tire on ice vs. a tire on asphalt)

There are two different kinds of friction: static and kinetic.

Definition 4.2. **Static friction** is friction that occurs between two objects that aren't moving relative to each other. The maximum value of static friction and the normal force are related by

$$f_s \leq \mu_s N,$$

where μ_s is a dimensionless constant of proportionality known as the coefficient of static friction. Note that F_s is **not** necessarily equal to $\mu_s N$; that is the maximum value. Instead, when applying a force on an object, the static friction will counteract the force applied on the object *until* the force is large enough to overcome the maximal value of the static friction ($\mu_s N$) in which case the static friction becomes kinetic friction. (Here's a relevant demo that might help you understand the concept better)

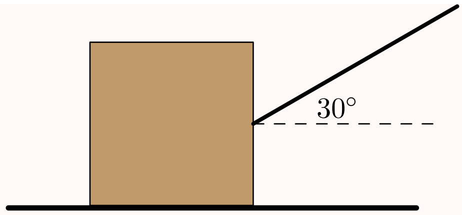
Definition 4.3. **Kinetic friction** is friction that occurs between two objects that are moving relative to each other. The value of kinetic friction is constant and independent of the relative velocities as stated above.

$$f_k = \mu_k N,$$

where μ_k is a dimensionless constant of proportionality known as the coefficient of static friction. The direction of the kinetic friction force on an object points in the opposite direction of its relative velocity with respect to the other object.

Generally, $\mu_k < \mu_s$, but some problems will state $\mu_k = \mu_s$ for simplicity.

Video Example 4.4. Professor Kim attaches a massless rope to a wooden crate of mass $m = 20\,\text{kg}$ and pulls at a 30° angle from the horizontal. What is the minimum force required to dislodge the crate? The coefficient of kinetic friction between the crate and the floor is $\mu_s = 0.8$. **Hint:** 18



Solution. Watch https://youtu.be/S4x0I-zgA_s&t=32s (timestamp 0:32) for the video solution to this problem! ■

Solution. Let F be the force with which he pulls. We require that the horizontal component of F, $F\cos(30^\circ)$ is greater than the maximum value of $f_s = \mu_s N$. The normal force also depends on the force applied. There is no vertical acceleration, so $N + F\sin(30^\circ) - mg = 0$, so $N = mg - F\sin(30^\circ)$. So

$$F\cos(30^\circ) > \mu_s(mg - F\sin(30^\circ)).$$

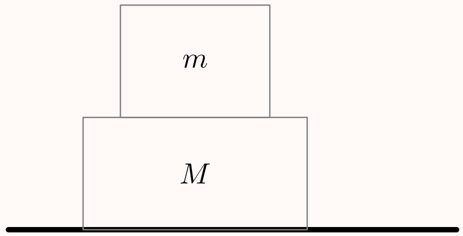
Solving for F, we have

$$F > \frac{\mu_s mg}{\cos(30^\circ) + \mu_s \sin(30^\circ)} = \boxed{124\text{ N}}.$$

■

Now we'll take a look at a pretty standard, but difficult example that gives some important insight into how some of these friction problems must be dealt with.

Video Example 4.5. Two blocks of mass m and M are stacked on top of each other as shown in the figure. You pull the bottom block with force F. The coefficient of friction between the two blocks is $\mu_k = \mu_s = \mu$. What are the accelerations of the blocks?



Solution. Watch https://youtu.be/S4x0I-zgA_s&t=275s (timestamp 4:35) for the video solution to this problem! ■

Solution. One of the most difficult parts of this example is intuitively understanding how the friction works. We have two cases depending on whether the top block slips with respect to the bottom block.

First assume that it doesn't slip. Then, the acceleration is simply $F/(m + M)$ as we can treat the blocks as one body. This case only happens if the frictional force is great enough. We can check this with whether or not the only horizontal force on the top block has enough force to make the acceleration $F/(m+M)$. We have

$$\mu mg \geq F_f = \frac{mF}{m+M},$$

so the acceleration is $F/(m+M)$ when $F \leq \mu(m+M)g$. The slight bit that may have been counter-intuitive is that the frictional force is pushing the top block forward. This seems weird since friction is making things move. However, this is correct. *Friction works note to reduce motion, but to reduce relative motion.*

Otherwise, the top block is slipping with respect to the bottom block. Then the forces on the bottom block are $Ma_M = F - F_f = F - \mu mg$ and for the top block, $ma_m = F_f = \mu mg$. So

$$a_M = \frac{F - \mu mg}{M},$$

and

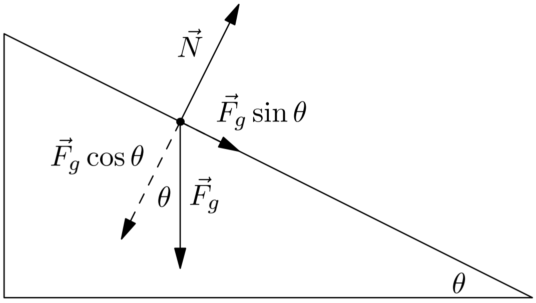
$$a_m = \mu g.$$

An interesting exercise you can think about is how the situation changes if you instead pulled the top block. ■

An important takeaway from that example is that one may have to consider both cases of slippage and non-slippage in problems. There will be a problem in the problems that requires you to consider those cases.

4.3 Inclined Planes

Inclined planes appear quite frequently in physics olympiads, and they are seen at all levels. We investigate the most basic case: we have a fixed inclined plane and we want to find the the accelerations/forces on a block on the plane.



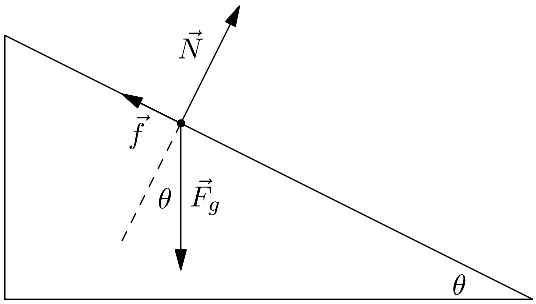
Clearly the block can't have any net force in the direction perpendicular to the plane as that would mean the block is either accelerating through the plane or flying out of it. So we can equate the component of gravity and the normal force, so

$$N = mg_{\perp} = mg\cos\theta.$$

Then we can calculate the net force, it is simply the component of gravity parallel to the plane. So

$$\Sigma F = mg_{||} = mg\sin\theta.$$

This is what would be derived using similar triangles in a standard physics classroom. In practice, these formulas appear so often that it is best to simply memorize them. From there, one can use the previous section to derive the net force when there is friction present.



All we need to apply is (assuming that the block is moving), is $F_f = N\mu_k = \mu_k mg\cos\theta$. So the net force is

$$\Sigma F = mg\sin\theta - \mu_k mg\cos\theta.$$

We also derive here the condition for the block to remain stationary. This is a simple task, we simply have

$$mg\sin\theta \leq \mu_s mg\cos\theta$$

for the net force to be 0. Thus, the condition is $\mu_s \geq \tan\theta$.

Theorem 4.6. *For a frictionless inclined plane, the net force on a free body points down the incline with magnitude*

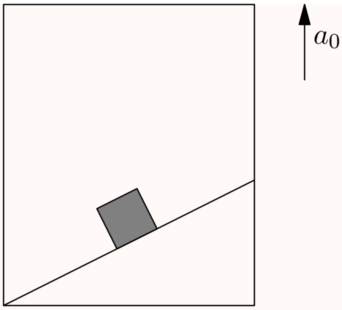
$$\Sigma F = mg\sin\theta.$$

For a rough inclined plane, if $\mu_s \geq \tan\theta$, the block remains stationary, otherwise, the net force points down the incline with magnitude

$$\Sigma F = mg\sin\theta - \mu_k mg\cos\theta.$$

Let us do a difficult example combining sections 4.1 and 4.3. Remember that the equations we derived previously were only for when the plane was completely still.

Video Example 4.7. A particle slides down a smooth inclined plane of elevation θ, fixed in an elevator going up with an acceleration a_0. The base of the incline has a length L. Find the time taken by the particle to reach the bottom.



Solution. Watch https://youtu.be/S4xOI-zgA_s&t=716s (timestamp 11:56) for the video solution to this problem! ■

Solution. The solution is quite simple if you use an accelerated reference frame and deal with fictitious forces (we'll deal with accelerated reference frames in week 19 of the course). However we'll work with the lab frame to show how one would solve these problems in the lab frame.

If the horizontal acceleration is a_x, we have that the upward acceleration is $a_0 - a_x \tan\theta$ (You can imagine a still elevator and then add the vertical acceleration). Then we have the following,

$$\begin{aligned} ma_x &= N\sin\theta \\ ma_y &= ma_0 - ma_x\tan\theta = N\cos\theta - mg. \end{aligned}$$

We can then solve for N in the second equation to obtain

$$N = \tan\theta(mg + ma_0 - ma_x\tan\theta).$$

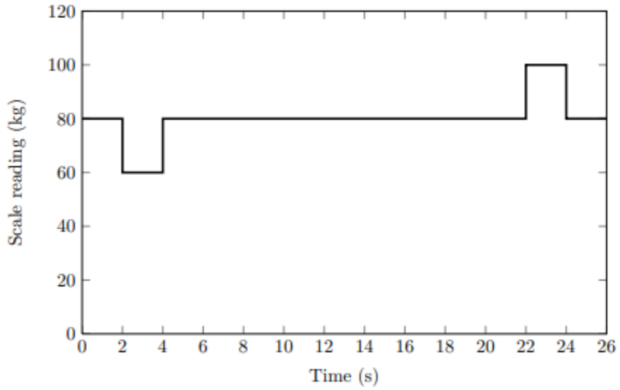
Substituting and solving for a_x gives us $a_x = \sin\theta\cos\theta(a_0 + g)$. So applying kinematics,

$$t = \boxed{\sqrt{\frac{2L}{\sin\theta\cos\theta(a_0+g)}}}.$$

■

4.4 Homework Problems and Solutions

Problem 4.1 ($F = ma$)**.** [8] A student steps onto a stationary elevator and stands on a bathroom scale. The elevator then travels from the top of the building to the bottom. The student records the reading on the scale as a function of time.



(a) [4] At what time(s) does the student have maximum downward velocity?
(b) [4] How tall is the building?

Solution. The choices for this problem were not given, but sometimes it is more instructive to not have choices.

(a) Let us think about what the scale reading means. If the scale reads M, then the net force downward on the student is given by $mg - Mg$ since a scale reads the normal force between the two surfaces.

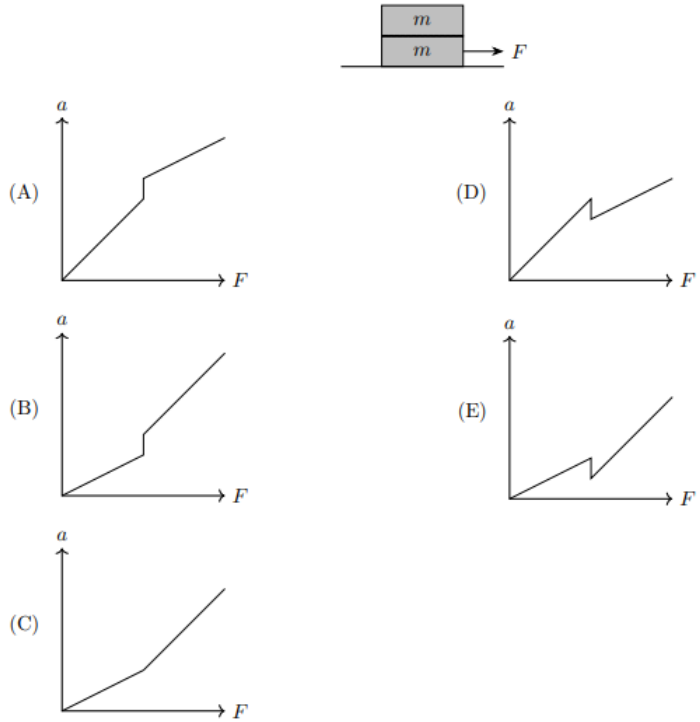
We know that initially the elevator is stationary, and thus 80 kg is the equilibrium reading, and it is the mass of the student. So only when the value M is less than 80 kg will the student begin accelerating downwards. Thus, the student only accelerates downwards between 2s and 4s, and then remains at 0 net force, until the student accelerates upwards again. So the maximum downward velocity is during **between 4 and 22 seconds**.

(b) We deduced earlier that the $F = mg - Mg$, so the acceleration downward at any point can be found using the scale reading, since $m = 80\,\mathrm{kg}$. We also know the only points of acceleration are at 2-4s and 22-24s. Also by symmetry, the student comes to rest right after 24s.

We can calculate the acceleration as $(20/80)g$ for 2-4s. So the distance covered here is $d_1 = \frac{1}{2} \cdot \frac{g}{4} \cdot 2^2 = 5\,\mathrm{m}$. Then for 4-22 secs, it will be at constant velocity of $v = 2 \cdot (20/80)g = 5\,\mathrm{m/s}$. So this is $d_2 18 \cdot 5 = 90\,\mathrm{m}$. Lastly, for 22-24 seconds, by symmetry $d_3 = 5\,\mathrm{m}$. So the final answer is $\boxed{100\,\mathrm{m}}$.

■

Problem 4.2 ($F = ma$). [4] Two blocks of mass m are placed on top of each other, and the bottom block is placed on the ground. The ground is frictionless. The static and kinetic coefficients of friction between the two blocks are μ_s and μ_k, with $\mu_k < \mu_s$. The blocks are at rest initially. When a constant horizontal force F is then applied to the bottom block, which of the following graphs could show its acceleration as a function of F?



Solution. Note that this is quite similar to an example we did, but we don't even need that to solve this problem. Let's look at 2 cases, the blocks slipping and not slipping.

For not slipping, we simply use $F = ma$ and the value for acceleration is given by $a_1 = F/2m$.

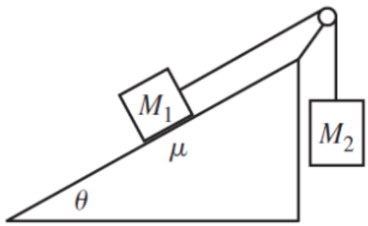
For slipping, then the friction between the two is given by $F_f = mg\mu_k$, so the acceleration of the bottom block is $a_2 = -g\mu_k + F/m$.

First, we notice that The slope of a_2 vs F is greater than the slope of a_1 vs F. So that eliminates choices A and D. Then let's examine the accelerations the two equations give at the boundary. This is when $\mu_s mg = F/2$ (see the example in the handout for an explanation). So when $F = 2\mu_s mg$, $a_1 = \mu_s g$, and $a_2 = (2\mu_s - \mu_k)g$, and since $\mu_s > \mu_k$, there should be a jump at the boundary, so the answer is (B). ■

Problem 4.3 (Morin). [10] Mass M_1 is held on a plane with angle of elevation θ, and mass M_2 hangs over the side. The two masses are connected by a massless string which runs over a massless pulley. The coefficient of kinetic friction between $M1$ and the plane is μ_k, and the coefficient of static friction is μ_s.

(a) [5] For a fixed mass M_1, what are the minimum and maximum possible values of M_2 such that the masses do not move?

(b) [5] Assuming M_2 is sufficiently large such that M_1 is pulled up the plane, what is the acceleration of the masses? What is the tension in the string?



Solution. This problem is applying equations from friction and forces.

(a) For the maximum, the forces on the M_1 block are friction, gravity, normal force and tension. The gravity and normal should cancel each other (perpendicular to the plane), leaving a force down the incline of $M_1 g \sin\theta$. Since this is the maximum, friction should also point down the plane, with magnitude $\mu M_1 g \cos\theta$. Thus, the maximum tension in the string can be $M_1 g \sin\theta + \mu_s M_1 g \cos\theta$. Thus, we divide by g to get the mass M_2 since the tension and gravity cancel on the 2nd block, and $M_{2_{max}} = \boxed{M_1 \sin\theta + \mu_s M_1 \cos\theta}$.

For the minimum, the gravity+normal force still points down, but tension and friction work together, both pointing up. Thus we can write

$$T + \mu M_1 g \cos\theta = M_1 g \sin\theta.$$

We have $T = M_2 g$, so $M_{2_{min}} = \boxed{M_1 \sin\theta - \mu_s M_1 \cos\theta}$.

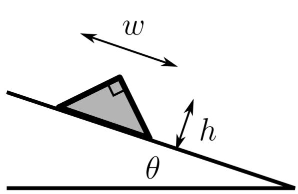
(b) We setup the F=ma equations for both blocks, and the magnitude of their accelerations must be equal since they are attached to one string.

$$\begin{aligned} M_1 a &= T - \mu_k M_1 g \cos\theta - M_1 g \sin\theta \\ M_2 a &= M_2 g - T. \end{aligned}$$

So we can eliminate and $\boxed{a = \dfrac{M_2 g - \mu_k M_1 g \cos\theta - M_1 g \sin\theta}{M_1 + M_2}.}$

■

Problem 4.4 ($F = ma$)**.** [5] A uniform solid right prism whose cross section is an isosceles right triangle with height h and width $w = 2h$ is placed on an incline that has a variable angle with the horizontal θ. What is the minimum coefficient of static friction so that the prism topples before it begins sliding as θ is slowly increased from zero?



(a) 0.71
(b) 1.41

(c) 1.50
(d) 1.73
(e) 3.00

Solution. We must determine the angle at which the prism topples and use that to find what the coefficient of friction must be. We wish to find when the centroid of the triangle is on the vertical.

But first, since this wasn't properly introduced in the section, we determine the minimum coefficient of friction at a given angle so that the block doesn't slip. We require that $mg \sin\theta \leq \mu mg \cos\theta$, and thus, the minimum is $\tan\theta$.

So we want to find $\tan\theta$ at the angle when the block topples. If you draw the diagram, you'll see that if the centroid is on the vertical, the angle of the incline is the angle between the altitude from the top corner and the median from the bottom right corner. So $\tan\theta$ is given by $h/(h/3) = 3$. Thus, the answer is E. ■

Problem 4.5 ($F = ma$)**.** [5] A box with weight W will slide down a 30° incline at constant speed under the influence of gravity and friction alone. If instead a horizontal force P is applied to the box, the box can be made to move *up* the ramp at a constant speed. What is the magnitude of P in terms of W?

Solution. First, we decipher what the first part of the problem tells us. It tells us that $\mu_k = \tan 30^\circ$. This will be useful later.

Now if a force P is applied horizontally, let's find the normal force between the ramp. This is given by $N = P\sin 30^\circ + mg\cos 30^\circ$ simply summing the components perpendicular to the ramp. From here we can use $F_f = N\mu_k$ to find friction, and then we must set the normal force along the ramp equal to 0. So,

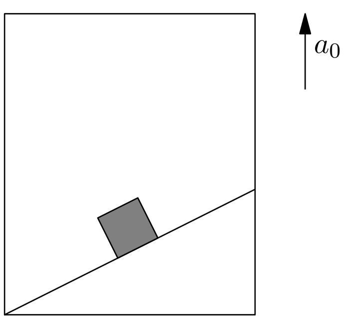
$$P\cos 30^\circ = mg\sin 30^\circ + N\mu_k = mg\sin 30^\circ + N\tan 30^\circ.$$

We now plug in N and then solve for P. We have

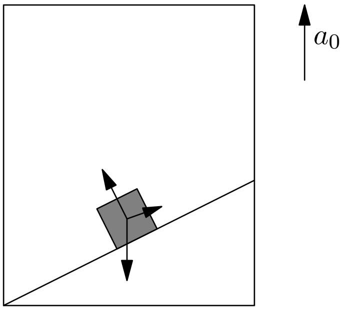
$$P(\sqrt{3}/2 - \sqrt{3}/6) = mg/2 + mg/2 = W.$$

Thus, $P = \boxed{\sqrt{3}W}$. ■

Problem 4.6. [6] A particle slides down a rough inclined plane of elevation θ, fixed in an elevator going up with an acceleration a_0. The coefficient of friction between the particle and the plane is μ. Find the minimum value of μ such that the block doesn't slip with respect to the plane.



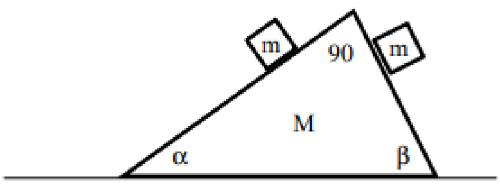
Solution. Let us draw the forces on the block in the elevator.



Now at this point, we can set equations and solve for everything. However, we can employ one really neat trick. We add the vectors of the normal force and frictional force. Let us denote it as $\vec{F} = \vec{N} + \vec{F_f}$. Now consider the direction of this vector. If this block isn't slipping in the elevator, then $\vec{F}$ must point directly upwards. Why? If it didn't, since gravity is the only other force on the block remaining, there would be a net horizontal force, implying that the block moves on the ramp and slips.

For $\vec{F}$ to point directly upwards, since $\vec{N} \perp \vec{F_f}$, $\tan\theta = F_f/N$. Note that the minimum value of μ implies that the block will be on the verge of slipping, or in other words, $F_f = \mu N$. So then we have $\boxed{\mu = \tan\theta}$, and we are done. If you did try to set equations and everything, do not be disappointed or annoyed by this simple solution, because the more bashy method is the natural approach. However, take note of this idea. ∎

Problem 4.7 ($F = ma$)**.** [7] A right-triangular wooden block of mass M is at rest on a table, as shown in figure. Two smaller wooden cubes, both with mass m, initially rest on the two sides of the larger block. As all contact surfaces are frictionless, the smaller cubes start sliding down the larger block while the block remains at rest. What is the normal force from the system to the table?



(a) $2mg$
(b) $2mg + Mg$
(c) $mg + Mg$
(d) $Mg + mg(\sin\alpha + \sin\beta)$
(e) $Mg + mg(\cos\alpha + \cos\beta)$

Solution. Consider the vertical forces on the triangular block. We have the vertical component of the normal force from the left block, which is given by $mg\cos\alpha \cdot \cos\alpha$.

Then we have the normal force from the right block, which is given by $mg\cos\beta\cdot\cos\beta = mg\sin^2\alpha$ analogously. Then we have the upward normal force from the floor, and gravity. So we have

$$mg\cos^2\alpha + mg\sin^2\alpha + Mg = N.$$

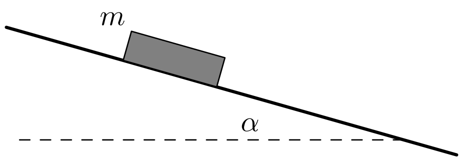
Thus, $N = Mg + mg = 1$. Our answer is (C). ■

Problem 4.8 (Kalda). [12] What is the minimum force needed to dislodge a block of mass m resting on an inclined plane of slope angle α, if the coefficient of friction is μ? Investigate the cases when:

(a) [6] $\alpha = 0$.

(b) [6] $0 < \alpha < \arctan\mu$.

Hint: It is helpful to realize that when the block is on the verge of slipping, the vector addition of the frictional force and the normal force make an angle $\arctan\mu$ with the plane. Also note that the force can point in any direction.



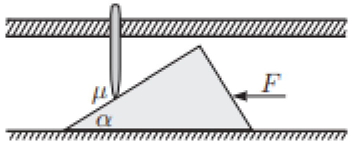
Solution. Let us first investigate part a, and then we will try to do part b. If you have not yet solved either, after reading part a, **do not** read part b, and try to solve it yourself. If you still can't, of course, read the solution.

(a) At this maximum point, the net force will be 0. Now let us consider the direction that the normal force + friction force points. This sum, call it $\vec{f} = \vec{F_f} + \vec{N}$ points at an angle $90 - \arctan\mu$ from the horizontal since at the point right before slippage, $F_f = N\mu$. We must have $\vec{f} + \vec{F_g} + \vec{F} = 0$. Using geometry (draw a diagram to see this), one can see that the minimum $\vec{F}$ occurs when $\vec{F} \perp \vec{f}$, meaning $F = F_g\sin(\arctan(\mu))$.

(b) We apply the same strategy, except that the angle at which the force $\vec{f}$ is at is different. The angle between the perpendicular from the plane to $\vec{f}$ is still the same $\arctan\mu$, but the gravitational force is at an angle α with the perpendicular, so the angle between the gravitational force vector and $\vec{f}$ is $\arctan\mu - \alpha$, so our answer is $F = mg\sin(\arctan(\mu) - \alpha)$.

■

Problem 4.9 (Kalda). [8] A wedge with the angle α at the tip is lying on the horizontal floor. There is a hole with smooth walls in the ceiling. A rod has been inserted snugly into that hole, and it can move up and down without friction, while its axis is fixed to be vertical. The rod is supported against the wedge; the only point with friction is the contact point of the wedge and the rod: the friction coefficient there is μ. For which values of μ is it possible to push the wedge through, behind the rod, by only applying a sufficiently large horizontal force?



Solution. The key insight is noting if the net vertical forces of the normal and friction forces were directed downwards then the stopper would be blocked. Let us then try to calculate the vertical components of forces that are involved. Let the normal force directed on the wedge be N. We then know that the vertical component of the normal force is clearly either $N\cos\alpha$ or $N\sin\alpha$. We can figure the component by chasing angles around, but an easier way is to imagine $\alpha \to 0$. In this case, the horizontal component of the normal force also goes to zero, which is the behavior of a sine function, so the horizontal component is $N\sin\alpha$. This in turn means that the vertical component of force involved is $N\cos\alpha$.

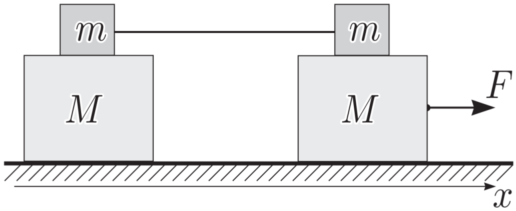
We now try to find the vertical component of friction involved. The friction force directed downwards on the direction of the wedge is μN (because N is already perpendicular, you do not have to manipulate it with trigonometric functions). This means that the vertical component of friction is $\mu N\sin\alpha$.

We now equate these, with an inequality where the vertical component of the normal force greater than the friction force for the wedge to pass through.

$$N\cos\alpha > \mu N\sin\alpha$$
$$\boxed{\mu < \cot\alpha}$$

A faster method would be to use the fact that at the maximum before slippage, the force of the normal force and friction force point in a constant direction, and we require that direction to be above the horizontal. The angle is $\arctan\mu$ below the normal. This means that we require $\arctan\mu < 90 - \alpha$ which means $\mu < \cot\alpha$. ■

Problem 4.10 (Kalda)**.** [10] A system of blocks sits on a smooth surface, as shown in the figure. The coefficient of friction between the blocks is μ, while that between the blocks and the surface is $\mu = 0$.



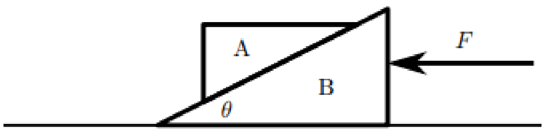
The bottom right block is being pulled by a force F. Find the accelerations of all blocks.

Solution. Thinking a little bit about the situation reveals that the only possibilities for motion is that everything moves together, or the big right block moves separately.

Suppose no slipping occurs, then clearly the acceleration is $a = F/(2m+2M)$. This occurs when $a(2m + M) \le \mu mg$, and plugging in a, when $F \le \dfrac{2\mu mg(m + M)}{2m + M}$.

Otherwise, suppose slipping does occur. Then, considering the forces on the right block, we have $a_1 = \dfrac{F - \mu mg}{M}$ for the right block, and for the three other ones, $a_2 = \dfrac{\mu mg}{2m + M}$. ■

Problem 4.11 (USAPhO). [10] A pair of wedges are located on a horizontal surface. The coefficient of friction (both sliding and static) between the wedges is μ, the coefficient of friction between the bottom wedge B and the horizontal surface is μ, and the angle of the wedge is θ. The mass of the top wedge A is m, and the mass of the bottom wedge B is $M = 2m$. A horizontal force F directed to the left is applied to the bottom wedge as shown in the figure.



Determine the range of values for F so that the the top wedge does not slip on the bottom wedge. Express your answer(s) using inequalities and in terms of any or all of m, g, θ, and μ.

Solution. Note the only external forces are friction between wedge B and the table, and the applied force F. Hence, we have

$$\sum F_{\text{ext}} = F - 3\mu mg = 3ma$$
$$F = 3ma + 3\mu mg.$$

We see that for small F (such as when $F = 0$), gravity will try to make block A to slide down the ramp, so friction will point up the ramp. Thus, we can apply Newton's second law for the net horizontal and vertical forces to obtain

$$\sum F_y = N\cos\theta + f\sin\theta - mg = 0$$
$$\sum F_x = N\sin\theta - f\cos\theta = ma.$$

The minimum force is applied when $f = \mu N$, so we can substitute and solve to obtain that

$$a_{\min} = g\frac{-\mu + \tan\theta}{1 + \mu\tan\theta}.$$
$$F_{\min} = 3mg\frac{(1 + \mu^2)\tan\theta}{1 + \mu\tan\theta}.$$

From our previous set of equations, we also see that once $N\cos\theta > mg$, friction will point down the ramp. so for sufficiently large F we have

$$\sum F_y = N\cos\theta - f\sin\theta - mg = 0$$
$$\sum F_x = N\sin\theta + f\cos\theta = ma.$$

Again, we have $f = \mu N$ at the extremal case, which gives us

$$F_{\max} = 3mg\frac{2\mu + (1-\mu^2)\tan\theta}{1-\mu\tan\theta}.$$

Therefore,

$$3mg\frac{(1+\mu^2)\tan\theta}{1+\mu\tan\theta} \le F \le 3mg\frac{2\mu + (1-\mu^2)\tan\theta}{1-\mu\tan\theta}.$$

However, we should also note some edge cases. For $\mu > \tan\theta$, no the block will not slide even if no force is applied (hence there is no minimum force). Similarly, for $\mu > \cot\theta$, the block will not slip upward either, so there is no maximum force. ■

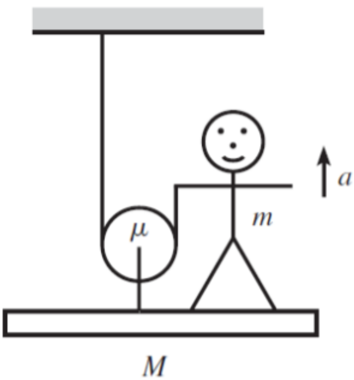
4.4.1 Written Solutions

Problem 4.12 (Morin). [15] A person stands on a platform-and-pulley system, as shown in the below figure. The masses of the platform, person, and pulley are M, m, and μ, respectively. The rope is massless. Let the person pull up on the rope so that she has acceleration a upward. (Assume that the platform is somehow constrained to stay level, perhaps by having the ends run along some rails.)

(a) [5] Find the tension in the rope.

(b) [5] Find the normal force between the person and platform.

(c) [5] Find the tension in the rod connecting the pulley to the platform.



Solution.

(a) Let us find the net external forces on the man + platform + pulley system. The only external forces are tension and gravity, with $2T$ upwards on the pulley, T downwards on the person, and gravity everywhere. So,

$$(\mu + m + M)a = T - (\mu + m + M)g.$$

So $T = (\mu + m + M)(a + g)$.

(b) We have the net force on the person is $N - T - mg = ma$. So,

$$N = ma + mg + (\mu + m + M)(a + g) = (\mu + 2m + M)(a + g).$$

(c) We have the net force on the pulley is $2T - T_2 - \mu g = \mu a$ where T_2 is the tension between the pulley and platform. So $T_2 = 2(\mu + m + M)(a + g) - \mu g - \mu a = (\mu + 2m + 2M)(a + g)$.

■

5 Momentum and Collisions

So far, we have only studied the dynamical quantity of force, which gave us a lot of information about the motion of an object. In theory, knowing the forces on an object at all time will be sufficient to analyze the entirety of the motion. In most cases however, it is cumbersome and often *impossible* to *always* use the force approach. In this chapter, we focus on another dynamical quantity, namely that of momentum, which will help us better analyze the motion of objects.

5.1 Linear Momentum

We'll preface our discussion of linear momentum by talking about a very special interaction - a **collision**.

We define a collision in physics to be slightly different from what you may currently think a collision is. When we have two objects with different initial motions that are originally so far apart that they do not apply a measurable force on the other, and they eventually collide such that the motion of one (or both) of the objects is drastically altered by an applied force of the other object, we say that the two objects have collided. Of course, this includes our common examples of a particle hitting another particle but it also includes a spaceship being "slingshot" by the gravity of a planet (unlike the particle, the spaceship never actually "touches" the planet).

How do we analyze the motion of these collisions? To do this, we define a quantity called the linear momentum of a body.

Definition 5.1. The **linear momentum** $\vec{p}$ of a particle is defined as the product of its mass and velocity:

$$\vec{p} = m\vec{v}.$$

The force acting on an object is then more generally described by the following:

> **Theorem 5.2** (Generalized Second Law). *In a closed system, the* ***force*** $\vec{F}$ *acting on a particle is defined as the rate of change of momentum:*
>
> $$\vec{F} = \frac{d\vec{p}}{dt}.$$

It is left as an exercise to see that for constant mass systems, the above definition does indeed yield $\vec{F} = m\vec{a}$.

So far, we have only defined the linear momentum for particles. How do we define the linear momentum for multi-particle systems? We will start this discussion by defining an object known as a rigid body which we will use as an approximation for all objects in this course henceforth (except in the chapter concerning Young's Modulus and elasticity):

Definition 5.3. A **rigid body** is defined as an object that can not be deformed and whose elasticity is infinite.

For these rigid body systems, just as we defined the net external force $\sum \vec{F}_{\text{ext}}$ on a system through the acceleration of its center of mass, it turns out we can define the momentum of an object through the velocity of its center of mass:

Definition 5.4. For a multi-particle system of N particles, we define the linear momentum $\vec{P}$ as the sum of the momentums of the N particles within the system:

$$\vec{P} = \sum_{n=1}^{N} \vec{p_n} = \sum_{n=1}^{N} m_n \vec{v_n} = M\vec{v_{\text{cm}}}.$$

Try to prove the last line by yourself using the definition of the center of mass.

Example 5.5. Professor Lee has three magic balls that have masses of $3, 4,$ and 5 kg respectively. The first ball has a velocity of $4\hat{\mathbf{i}}$, $5\hat{\mathbf{j}}$, and $3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ respectively. What is the linear momentum of the system?

Solution. Summing the momentum's of the individual particles, we get that the momentum is $\vec{P} = 27\hat{\mathbf{i}} + 40\hat{\mathbf{j}}$. ■

5.2 Impulse and Conservation Laws

We now define a quantity known as impulse for a force acting on an object. We like to think of one object applying an impulse on another object when it applies a force on it.

Definition 5.6. The **impulse** $\vec{J}$ of an arbitrary force $\vec{F}$ acting on a particle during a given time interval is defined as

$$\vec{J} = \int_{t_i}^{t_f} \vec{F} dt = \vec{F}_{\text{avg}} \Delta t.$$

For fixed forces, we have $F_{\text{avg}} = F$ so $J = Ft$.

We now prove a fundamental theorem which gives us some explanation for why we described this mysterious quantity and gives us some intuition as to what impulse truly represents.

Theorem 5.7 (Impulse-Momentum Theorem)**.** *The impulse of the net force acting on a particle during a given time interval is equal to the change in momentum of the particle during that interval. Mathematically, we have*

$$\vec{J}_{\text{net}} = \Delta\vec{p}.$$

The proof is left as an exercise to the reader (write $\vec{F} = m\vec{a}$ and integrate). The corresponding statement is also true for systems with multiple particles. This theorem gives us some intuition as to how the force(s) acting on an object affect the velocity of an object without actually knowing what happens at each moment in time!

With this in mind, we now show a type of law called a conservation law. Consider a system where the net force acting on the system is zero. Then, by the impulse-momentum theorem, we get that

$$\Delta\vec{P} = 0 \implies \vec{P_i} = \vec{P_f}.$$

That is, momentum is conserved for systems in an inertial reference frame. Thus, if we define our system appropriately, there will be no net force acting on the system so momentum will be conserved. We now give some examples:

Example 5.8. Professor Lee has two magic balls that lie on a plane. Ball 1 has a mass of 3 kg, and moves to the right with a velocity of 4 m/s, while ball 2 has a mass of 5kg and moves to the left with speed 2m/s. Given that after the two balls collide, ball 1 moves to the left with speed 3 m/s, what is the final speed of ball 2?

Solution. We choose the system including both balls. Note, because there are no net forces, momentum is conserved. Thus,

$$P_i = m_1 v_{1i} + m_2 v_{2i} = (3)(4) + (5)(-2) = (3)(-3) + (5)v - P_f \implies \boxed{v = 2.2 \text{ m/s}}.$$

■

Example 5.9 (USAPhO). Suppose you drop a block of mass m vertically onto a fixed ramp with angle θ with coefficient of static and kinetic friction μ. The block is dropped in such a way that it does not rotate after colliding with the ramp. Throughout this problem, assume the time of the collision is negligible.

(i) Suppose the block's speed just before it hits the ramp is v and the block slides down the ramp immediately after impact. What is the speed of the block right after the collision?

(ii) What is the minimum μ such that the speed of the block right after the collision is 0?

Solution.

(i) We let the direction of the incline be the x-axis and the direction normal to it be the y-axis. Remark that the block has a momentum of $mv\cos\theta$ in the y-direction. We first note that during the collision there should be an impulse on the block due to the normal force of the ramp, the friction, and gravity. Let J_n be the impulse due to the normal force and J_f be the impulse due to friction. Let v_f be the final velocity of the block. Because the collision is short, we assume the gravitational impulse to be negligible. Since the block does not go through the block, the momentum normal to the ramp after the collision should be 0. Then, by the impulse-momentum theorem,

$$J_n = -mv\cos\theta.$$

Similarly, note that we have

$$J_f = mv_f - mv\sin\theta.$$

Then, noting that $J_f = \mu J_n$ we thus get

$$-\mu mv\cos\theta = mv_f - mv\sin\theta \implies \boxed{v_f = v(\sin\theta - \mu\cos\theta)}.$$

(ii) Setting $v_f = 0$ we get $\boxed{\mu = \tan\theta}$.

Note how throughout the problem, we did not need to parameterize any forces with respect to time and what not. With the impulse momentum theorem, we were able to solve the problem knowing only what happened at the end-points of the motion. ■

5.3 Two-Body Collisions

As mentioned in the beginning of the handout, we are in large part interested in momentum because we are interested in analyzing the motion of collisions. In this section, we delve a bit into the specific types of collisions, specifically, we delve into the types of two-body collisions.

The key step in solving these problems is realizing that when two objects collide, they exert equal and opposite impulses on each other. *The total momentum of the system will not change, nor will the velocity of the center of mass.*

Definition 5.10. In an **elastic collision**, the total kinetic energy is conserved. We will learn kinetic energy next lesson, but there is an easier way to solve elastic collision problems then directly using kinetic energy. *In an elastic collision, the velocities of the objects in the center of mass reference frame are reversed.* Let's do a quick example to see what this looks like.

> **Example 5.11.** Two bowling balls collide head-on in an elastic collision. Bowling ball A has mass 1 kg and approaches at 2 m/s from the left, and bowling ball B has mass 2 kg and approaches at 4 m/s from the right. What are the velocities of the bowling balls after they collide?

Solution. To compute the velocity of the center of mass, we must first compute the total momentum and then divide it by the total mass. This gives:

$$\vec{v}_{\text{com}} = \frac{1 \times 2 + 2 \times (-4)}{1 + 2} = -2.$$

If we subtract -2 from our initial velocities, we see that in the center of mass reference frame, $\vec{v}_{A,\text{cm}} = 4$, and $\vec{v}_{B,\text{cm}} = -2$. After the collision, they will rebound will their initial speeds so $\vec{v}_{A,\text{cm}} = -4$, and $\vec{v}_{B,\text{cm}} = 2$. We now add back -2, to convert to our initial reference frame so we see that $\vec{v}_A = -6$ m/s and $\vec{v}_B = 0$ m/s. ■

Reference frames are extremely useful in momentum problems, and can greatly simplify a seemingly complicated problem.

> **Example 5.12.** Professor Lee hits a golf ball by swinging his golf club at 40 m/s. Assuming that the collision is elastic and does not significantly affect the path of the club, what is the approximate velocity of the golf ball?

Solution. Recall that momentum is conserved during a collision. If the path of the club is not significantly affected by the collision, we can assume that its momentum is much greater than the momentum of the ball. *That is, the center of mass reference frame is approximately the frame of the club.* In this frame, the golf club is stationary, and the ball is approaching it at 40 m/s. Since the collision is elastic, the ball with rebound with speed 40 m/s. We can convert back to our current reference frame by adding the velocity of the club to get that the golf ball moves at roughly 80 m/s. ■

Definition 5.13. An **inelastic collision** is any collision that is not elastic. If a collision is inelastic, more information must be given to find the final velocities of the colliding objects, i.e. the amount of kinetic energy lost or one of the final velocities).

Note. The amount of kinetic energy lost does not depend on the reference frame.

Definition 5.14. A **perfectly inelastic collision** is a type of inelastic collision where the most kinetic energy is lost. This is essentially a *hit-and-stick* collision, where objects end up moving together and have the same final velocities. The final velocity is given by the equation:

$$\vec{v} = \frac{\vec{P}}{M},$$

where $\vec{P}$ is the initial total momentum, and M is the total mass.

Example 5.15. A block is sitting on a frictionless surface, and a bullet is fired at it at speed v. There are three possible scenarios:

1. In scenario A, the bullet shoots through the block and keeps going, although at a slower speed.
2. In scenario B, the bullet sticks to the block and the two objects slide together.
3. In scenario C, the bullet rebounds off the block and moves in the opposite direction.

In which scenario does the block move the fastest? In which scenario does it move the slowest?

Solution. Note that initial momentum is the same in all three scenarios. By conservation of momentum, we have that

$$\vec{p}_{f,\text{block}} = \vec{P}_i - \vec{p}_{f,\text{bullet}}.$$

Since momentum is a vector, we see that $\vec{p}_{f,\text{block}}$ (and hence the final velocity of the block) is greatest when the bullet rebounds. On the other hand, it is the smallest when the bullet shoots through the block, since it retains some of its initial momentum. ■

5.4 Continuous Collisions

Another type of collision is where the collision is gradual and continuous. We have previously assumed that $\Delta\vec{p} = m\Delta\vec{v}$, but this only holds if mass is constant. When mass changes, we can instead use the equation $\vec{F} = \frac{d\vec{p}}{dt} = \vec{v}\frac{dm}{dt}$. Let's do two examples to see what this looks like.

Example 5.16. A bucket of mass m moves with velocity $\vec{v}$ along a frictionless surface of ice. At time $t = 0$, it starts to rain. The rain particles are hitting the train at a constant downward velocity $\vec{u}$ and at a constant rate c. The rain stops at time $t = \tau$, and none of the water evaporates. Assuming the bucket does not overflow, express the normal force of the ice on the bucket as a function of time.

Solution. Before it starts raining, the normal force is mg. When rain is falling, extra force is provided to decrease the momentum of the rain, equivalent to $\vec{u}\frac{dm}{dt} = \vec{u} \times c$. However, water also accumulates over time. The amount of water in the bucket at time $t, \in [0, \tau]$ is ct. Hence, while it is raining the total force is $mg + \vec{u} \times c + ctg$. Once it stops raining, the amount of water in the bucket is $c\tau$, and there is no longer a force from the rain drops hitting the bucket. Hence, the force would be $mg + c\tau g$ during this time period.

Hence, at any time the normal force is given by:

$$F_N(t) = \begin{cases} mg + \vec{u} \cdot c + ctg & \tau > t \geq 0 \\ mg + c\tau g & t \geq \tau \end{cases}$$

■

Example 5.17. A plate is placed on top of a force sensor, and the sensor is then zeroed. A uniform rope with mass M and length L is held vertically so that the bottom of the rope is just touching the plate. The rope is then dropped. What is the maximum force displayed on the force sensor?

Solution. Recall that $\frac{dp}{dt}$ for the rope is equal to $v\frac{dm}{dt}$. We can make the substitution

$$\frac{dm}{dt} = \frac{dm}{d\ell} \times \frac{d\ell}{dt} = \frac{M}{L} \times v.$$

Hence the force necessary to stop the falling rope is $\frac{Mv^2}{L}$. We can compute $v = gt$ to get $F = \frac{Mg^2t^2}{L}$. The mass of rope lying on the plate is

$$\Delta Mg = \frac{M}{L}(\Delta L)g = \frac{Mg^2t^2}{2L}.$$

Adding the two forces gives that the total force detected by the force sensor at any time is $\frac{3M}{2L}g^2t^2$. Then the maximum force occurs when the last bit of rope hits the force sensor. Since $\frac{g}{2}t_{\text{max}}^2 = L$, $t_{\text{max}} = \sqrt{2L/g}$. Substituting gives that the maximum force is $F_{\text{max}} = \boxed{3Mg}$. ■

5.5 Homework Problems and Solutions

Problem 5.1 ($F = ma$)**.** [3] A 3.0 kg mass moving at 30 m/s to the right collides elastically with a 2.0 kg mass traveling at 20 m/s to the left. After the collision, what is the speed of the center of mass of the system?

Solution. The center of mass of *the system* experiences no external forces so it's speed remains the same before *and* after the collision. Thus, we have

$$v_{cm,f} = v_{cm,0} = \frac{3(30) + 2(-20)}{3 + 2} = \boxed{10 \text{ m/s}}$$

■

Problem 5.2 ($F = ma$)**.** [6] Carts A, B, and C are on a long horizontal frictionless track. The masses of the carts are $m, 3m$, and $9m$. Originally cart B is at rest at the 1.0 meter mark and cart C is at rest on the 2.0 meter mark. Cart A is originally at the zero meter mark moving toward the cart B at a speed of v_0.

(a) [3] Assuming that all collisions are *completely inelastic*, what is the final speed of cart C?

(b) [3] Assuming that all collisions are completely elastic, what is the final speed of cart C?

Solution.

(a) All of the masses stick, so the final mass is $m + 3m + 9m = 13m$. Momentum is conserved, so

$$mv_0 = 13mv_f \implies v_f = \boxed{\frac{v_0}{13}}.$$

(b) Let's consider the first collision. The velocity fo the center of mass is $\frac{v_0}{4}$ so the speed at which the mass of $3m$ rebounds is $\frac{v_0}{4} + \frac{v_0}{4} = \frac{v_0}{2}$. The center of mass of the second collision has velocity $\frac{v_0}{8}$ so the velocity at which the $9m$ mass rebounds is $\frac{v_0}{8} + \frac{v_0}{8} = \boxed{\frac{v_0}{4}}$.

One could also use conservation of energy but we'll leave that solution as an exercise.

■

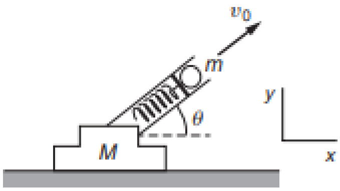
Problem 5.3 ($F = ma$)**.** [4] An object of mass m_1 initially moving at speed v_0 collides with an originally stationary object of mass $m_2 = \alpha m_1$, where $\alpha < 1$. The collision could be completely elastic, completely inelastic, or partially inelastic. After the collision the two objects move at speeds v_1 and v_2. Assume that the collision is one dimensional, and that object one cannot pass through object two. After the collision, what is the speed ratio $r_2 = v_2/v_0$ of object 2 is bounded by?

Solution. Let's think about this problem in the center of mass frame as all collisions are most easily seen *in* the com frame. It is clear that the velocity of the center of mass does not change as there are no external forces on the system. In the center of mass frame, the masses rebound with speeds less than or equal to their current speeds in the com frame so $0 \leq v_{2,com} \leq v_{cm} = \frac{1}{1+\alpha}v_0$. Translating to the lab frame gives

$$\frac{1}{1+\alpha}v_0 \leq v_2 \leq \frac{2}{1+\alpha}v_0$$

implying that the ratio is bounded between $\boxed{\frac{1}{1+\alpha} \leq r_2 \leq \frac{2}{1+\alpha}}$ ■

Problem 5.4 (Kleppner and Kolenkow)**.** [5] A loaded spring gun, initially at rest on a horizontal frictionless surface, fires a marble at angle of elevation θ. The mass of the gun is M, the mass of the marble is m, and the muzzle velocity of the marble (the speed at which the marble is ejected relative to the muzzle) is v_0. What is the final velocity of the gun?



Solution. The horizontal momentum is conserved. Thus,

$$0 = m(v_0 \cos\theta - v_f) - Mv_f = 0 \implies v_f = \boxed{\frac{m}{m+M} v_0 \cos\theta}$$

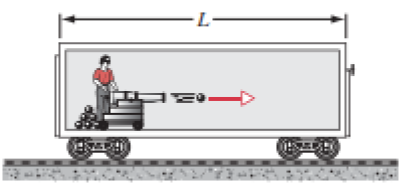
■

Problem 5.5 (HRK Altered)**.** [12] A cannon and a supply of cannonballs are inside a sealed railroad car of length L, as in the given figure. The cannon fires to the right; the car recoils to the left. The cannonballs remain in the car after hitting the far wall. Let m be the mass of the cannonballs and M be the mass of the car.

(a) [5] After all the cannnonballs have been fired, what is the distance the car can have moved from its original position?

(b) [3] What is the maximum possible distance the car has moved from its original position?

(c) [4] What is the speed of the car after all the cannonballs have been fired?



Solution.

(a) Let's take the position of the cannonballs to be $x = 0$. Then, since there are no external forces in the horizontal direction on the masses, the position of the center of mass remains the same. Assume the car shifts to the left a distance x. Then,

$$Mg\frac{L}{2} = Mg(\frac{L}{2} - x) + mg(L - x) \implies \boxed{x = \frac{m}{m+M} L}.$$

(b) This just occurs when the mass of the car is negligible compared to the mass of the cannonballs, or $m \ll M$. Then, $\boxed{x = L}$.

(c) Since there are no external forces on the system and the speed of the railroad car before firing was 0, the center of mass of the system initially has velocity 0. When the cannonballs are done firing, they move with the same velocity as the car. This in turn just implies the velocity of the car is $\boxed{0 \text{ m/s}}$ at the end of firing as well.

■

Problem 5.6 (Classical)**.** [10] A rocket in space propels itself forwards by eject mass backwards at a constant speed of 3000 m/s relative to the rocket. The initial mass of the rocket is 800, and the rocket loses mass at a rate of 5 kg/s. The rocket is at rest at time $t = 0$ s.

(a) [4] To two significant figures, what is thrust (in Newtons) provided by the rocket engine?

(b) [6] To three significant figures, what is the speed of the rocket (in m/s) at time $t = 100$ s? **Hint:**10

Solution.

(a) The rocket equation is

$$\sum F_{ext} = \frac{dm}{dt}v_e + m\frac{dv}{dt} = 0,$$

where v_e is the velocity of the rocket relative to the exhaust. The thrust $F_{thrust} = m\frac{dv}{dt}$ is then $F_{thrust} = \frac{dm}{dt}v_e = (5)(3000) = \boxed{15000 \text{ N}}$.

(b) Since $\sum F = 0$, we can manipulate the rocket equation to get the differential equation

$$-v_e\frac{dm}{m} = dv$$

Integrating, we get

$$-v_e\int_{m_0}^{m_f}\frac{dm}{m} = \int_{v_0}^{v_f} dv$$

which yields

$$v_f - v_0 = v_e \ln\frac{m_0}{m_f}$$

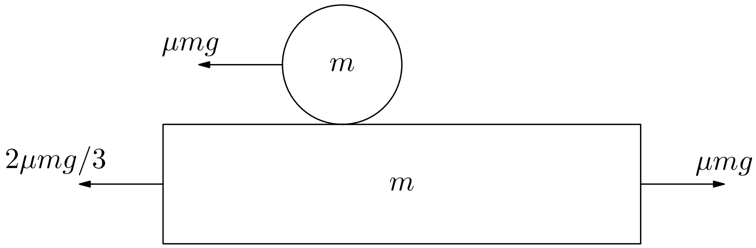
We thus get

$$v_f = (3000)\ln\frac{800}{800 - 5(100)} = \boxed{2940 \text{ m/s}}$$

■

Problem 5.7 ($F = ma$)**.** [6] A point mass m sits on a long block, also of mass m, which rests on the floor. The coefficient of static and kinetic friction between the mass and the block is μ, and the coefficient of static and kinetic friction between the block and the floor is $\mu/3$. An impulse gives a horizontal momentum p to the point mass. After a long time, how far has the point mass moved relative to the block? Assume the mass does not fall off the block. **Hint:** 16

Solution. After a long time, the point mass will not slip relative to the block so they will be moving with the same velocity. Thus, we just need to find the relative distance moved prior to slipping. Let's thus consider the forces that act on both objects. A free body diagram is shown below:



Let t be the time it takes to stop slipping. We then get the equation

$$p - \mu mgt = \frac{\mu mg}{3}t \implies t = \frac{3p}{4\mu mg}$$

The distance moved by the top block prior to slipping is thus

$$d_1 = -\frac{1}{2}\mu mgt^2 + \frac{p}{m}t$$

and the distance moved by the bottom block is

$$d_2 = \frac{\mu mg}{3}t^2$$

so the relative distance moved is

$$\delta d = d_1 - d_2 = \frac{2}{3}\mu mgt^2 + \frac{p}{m}t = \boxed{\frac{3p^2}{8m^2\mu g}}$$

■

Problem 5.8 (Kalda)**.** [6] A plank of length L and mass M is lying on a smooth horizontal surface; on its one end lies a small block of mass m. The coefficient of friction between the block and the plank is μ. What is the minimal velocity v that needs to be imparted to the plank with a quick shove such that during the subsequent motion the block would slide the whole length of the board and then would fall off the plank? The size of the block is negligible.

Solution. The force on the block is μmg forwards, so the force on the plank is μmg backwards. Thus, $a_b = \mu g$ and $a_p = -\mu\frac{m}{M}g$. Let t be the time it takes for the block to fall off. We thus have

$$d_b = \frac{1}{2}\mu gt^2$$

and

$$d_p = -\frac{1}{2}\mu\frac{m}{M}gt^2 + vt$$

Since the plank travels a distance L more than the block, we have $d_a - d_b = L$ or

$$\frac{-1}{2}\mu g(\frac{m}{M} + 1)t^2 + vt = L$$

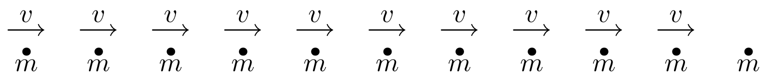
Since the discriminant must be positive, we get that

$$v^2 - 2L(\mu g(1 + \frac{m}{M}) \geq 0 \implies v \geq \sqrt{2L\mu g(1 + \frac{m}{M})}$$

This gives a minimum value of $\boxed{v = \sqrt{2L\mu g\left(1 + \frac{m}{M}\right)}}$.

(Alternatively, think of the motion in the frame of the block or the center of mass frame.) ■

Problem 5.9 (Morin)**.** [8] A stream of n clay balls with mass m move with speed v in a line across a frictionless table. The spacing between them is ℓ. An additional ball with mass m sits at rest in front of them, as shown in the figure. The front moving ball collides with the stationary ball and sticks to it and forms a blob of mass $2m$. Then the second ball collides with the blob and forms a blob of mass $3m$. And so on. How much time elapses between the instant shown below (when all the balls are separated by ℓ) and the last collision?



Solution. The key to this problem is to note that at the moment the $k-1$th ball collides with the clump of the $k-2$ previous balls and the stationary ball, the kth ball is still a distance l away from the $k-1$th ball. Thus, the kth ball just needs to move a distance l *relative* to the clump of the previous $k-1$ balls and the stationary ball. That is, if v_k is the velocity of the kth ball *relative* to the clump of the previous $k-1$ balls, then the time it takes for the kth ball to collide is $\frac{l}{v_k}$. To find v_k, let's first find the velocity of the clump of the $k-1$ previous balls. Since the total momentum of the $k-1$ previous balls is $(k-1)mv$, and since they amalgamate with each other and the stationary ball, we have

$$(k-1)mv = kmv_{clump} \implies v_{clump} = \frac{k-1}{k}v$$

Since the velocity of the kth ball is v, the relative velocity is

$$v_k - v - v_{clump} = \frac{1}{k}v$$

So the time it takes for the kth ball to collide is $t_k = \frac{k\ell}{v}$. The total time is then just

$$t = \sum_{k=1}^{n}\frac{k\ell}{v} = \boxed{\frac{\ell n(n+1)}{2v}}$$

■

Problem 5.10 (Irodov)**.** [8] A cannon of mass M starts sliding freely down a smooth inclined plane at an angle α to the horitzontal. After the cannon covered the distance l, a shot was fired, the shell leaving the cannon in the horizontal direction with a momentum p. As a consequence, the cannon stopped. Assuming the mass of the shell to be negligible, as compared to that of the cannon, determine the time it took to shoot the shell.

Solution. We can use either energy or kinematics to find that the velocity of the cannon after it has traveled a distance l is $v = \sqrt{2gl\sin\alpha}$. Note that the impulse imparted on the shell is $p\cos\alpha - m\sqrt{2gl\sin\alpha} \approx p\cos\alpha$ since the mass of the shell is negligible. This is equal and opposite to the impulse exerted on the cannon. Since the cannon stops, we have that this imparted impulse is equal to the sum of the initial momentum of the cannon plus the impulse imparted by gravity during the firing. We thus have

$$p\cos\alpha = M\sqrt{2gl\sin\alpha} + Mgt\sin\alpha \implies \boxed{t = \frac{p\cos\alpha - M\sqrt{2gl\sin\alpha}}{Mg\sin\alpha}}$$

■

Problem 5.11 (Irodov)**.** [8] A shell flying with velocity $v = 500$ m/s bursts into three identical fragments, so that the kinetic energy of the system increases by $\eta = 1.5$ times. What maximum velocity can one of the fragments obtain? (This problem requires concepts outside of what we've done so far. Skip this problem if you don't know kinetic energy yet)

Solution. Let v_1, v_2, v_3 be the horizontal velocities of the three particles. We have

$$v_1 + v_2 + v_3 = 3v$$

where v is the velocity of the original mass. By conservation of energy, we get

$$\frac{3}{4}mv^2 = \frac{1}{6}m({v_1}^2 + {v_2}^2 + {v_3}^2) \implies {v_1}^2 + {v_2}^2 + {v_3}^2 = \frac{9}{2}v^2$$

Since the problem is symmetric about v_1, v_2, v_3, finding the maximum of any one of the quantities will give the desired maximum. Let's try maximizing v_1. Substituting the first equation into the second and manipulating, we get

$$4{v_2}^2 + (4v_1 - 12v)v_2 + (4{v_1}^2 + 9v^2 - 12vv_1) = 0$$

Since the discriminant must be positive, we get

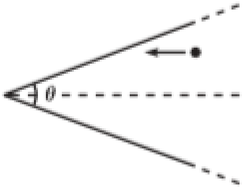
$$(4v_1 - 12v)^2 - 16(4{v_1}^2 + 9v^2 - 12vv_1) \geq 0$$

$$-48{v_1}^2 + 96vv_1 \geq 0 \implies -2v \leq v_1 \leq 2v$$

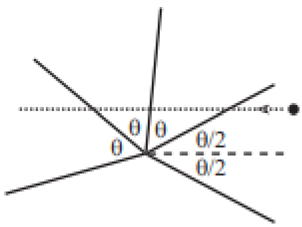
So the maximum velocity is just $v_1 = 2v = \boxed{1000 \text{ m/s}}$. ■

Problem 5.12 (Morin)**.** [10] A ball is thrown against a wall of a very long triangular room which has vertex angle θ. The initial direction of the ball is parallel to the angle bisector (see Figure). How many (elastic) bounces does the ball make? Assume the walls are frictionless.

Hint:6



Solution. This problem is *quite* difficult in that it requires a key bit of ingenuity: when the ball elastically hits the room it bounces off at the same angle it hits the wall. This is equivalent however, to if we reflect the *room* across the wall that it bounces off of. We can continue this process multiple times so that ultimately it is equivalent to the ball passing through the wall in a straight line similar to the figure:



So if the wall bounces n times, we can write

$$\frac{\theta}{2} + (n-1)\theta > 180^\circ$$

but

$$\frac{\theta}{2} + n\theta \leq 180^\circ$$

Putting these equations together gives

$$\boxed{\frac{180^\circ}{\theta} + \frac{1}{2} > n \geq \frac{180^\circ}{\theta} - \frac{1}{2}}$$

Thus n is the unique integer that lies between the above inequality. ■

5.5.1 Written Solutions

Problem 5.13 (NBPhO). [14] $n+1$ elastic balls are dropped so that they are exactly above each other, with a very small gap between each. The bottom ball has a mass of m_0, the one above has a mass of fm_0, the next has $f^2 m_0$, and so on, where the topmost ball has mass $f^n m_0$, where $f < 1$. At the moment when the bottom ball touches the ground, all the balls are moving with the speed v.

(i) [4] After the collision between the two bottommost balls, what is the speed v_1 of the second ball from the bottom?

(ii) [4] What is the speed of the topmost ball v_n after all collisions?

(iii) [6] How many times higher would that ball fly compared to the initial drop height h? Take $f = 0.5$ and $n = 10$.

Solution.

(a) The center of mass frame has velocity $\frac{1-f}{1+f}v$ upwards. Thus in the center of mass frame the second ball has velocity $v_1 = -v - \frac{1-f}{1+f}v = \frac{-2}{1+f}v$ so it rebounds to center of mass velocity $\frac{2}{1+f}v$. Thus in the lab frame, the velocity is $v_1 = \frac{2}{1+f}v + \frac{1-f}{1+f}v = \boxed{\frac{3-f}{1+f}v}$.

(b) Using the same method as above, we see in general that $v_k = \frac{2}{1+f} v_{k-1} + \frac{1-f}{1+f} v$. Applying this recursion n times gives $v_n = \left(2\left(\frac{2}{f+1}\right)^n - 1\right) v$.

(c) We know that the max height reached is $h' = \frac{{v_0}^2}{2g}$. The height the ball flys is $h = \frac{v^2}{2g}$. So the ratio of the heights is $\frac{h'}{h} = \left(\frac{v_0}{v}\right)^2 = \boxed{\left(2\left(\frac{2}{f+1}\right)^n - 1\right)^2}$. When $f = 0.5, n = 10$, this is about 1200.

■

6 Energy

You have no doubt heard of **energy** plenty of times, but describing energy is rather difficult. Textbooks often define energy as the capacity of an object to perform work, and will subsequently define work as the transfer of energy, which is not very helpful. In short, energy is the capacity of an object to create change.

What makes energy so useful is the law that the total energy of a closed system is conserved. However, many forms of energy, such as chemical, thermal, or sound energy are not useful in mechanics. We are interested specifically in **mechanical energy**, which is not always conserved.

The fact that energy is a *conserved* quantity allows us to analyze a vast array of difficult interactions as we can analyze merely the endpoints of the motion, similar to the way we did for linear momentum. Like with linear momentum, recognizing when we can use conservation of mechanical energy can help with solving a large range of problems, but unlike with linear momentum, mechanical energy can take different forms. In this section, we will begin by analyzing some of the various forms that energy can take and analyze how the conservation properties can be used to solve problems.

6.1 Kinetic Energy and Work

Kinetic energy is the energy of an object due to its motion while **potential energy** is the energy of an object due to its position. In any form, energy is a scalar and given in units of joules (J). There is only one form of kinetic energy but there are multiple types of potential energy: elastic, gravitational, chemical, electrostatic, etc.

We'll first define the kinetic energy for a point particle:

Definition 6.1. Consider a point particle of mass m that moves with speed v. We define kinetic energy (which we denote by K; it is also commonly denoted as E_k or KE) as

$$K = \frac{1}{2}mv^2$$

We now introduce a quantity called **work**:

Definition 6.2. For a force $\vec{F}$ exerted on an object as it is displaced, the work done can be calculated with the equation

$$W = \int \vec{F} \cdot d\vec{x} = \int F \cos\theta \, dx,$$

where θ is the angle between the vectors $\vec{F}$ and $\vec{dx}$.

Concept. You might be wondering where $\cos\theta$ comes from. In our integral definition of work, we used an operation called a *vector dot product.*

$$\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos\theta,$$

where θ is the angle between the directions of the two vectors $\vec{u}$ and $\vec{v}$. The reason the dot product is so useful is because if we have $\vec{u} = (u_1, u_2, \dots)$ and $\vec{v} = (v_1, v_2, \dots)$, then we have that

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + \cdots$$

While this form can be extremely useful, it rarely comes up. In general, it is sufficient to consider the dot product $\vec{u} \cdot \vec{v}$ as simply the product of the magnitudes u and v and the cosine of the angle between them.

For a constant force $\vec{F}$ applied on an object the experiencing a displacement vector $\vec{d}$, the work done on the object is given by

$$W = Fd\cos\theta.$$

Remark. Understanding the term $\cos\theta$ is extremely important to understanding work. When you walk on level ground, the gravitational force does no work on you, because the force and displacement are perpendicular (remember that $\cos 90^\circ = 0$). It might be helpful to view work in the form

$$W = F_{\parallel} x,$$

where $F_{\parallel}$ is the component of force parallel to the displacement vector. If the force and displacement are in the same direction, work is positive. If they are in opposing directions, then the work done is negative.

The quantity work is useful because it is directly related to change in energy.

Theorem 6.3 (Work-Energy Theorem). *The net work done on a closed system is equal to its change in kinetic energy. That is,*

$$W = \Delta K.$$

Proof. Recall that

$$W = \int \vec{F} \cdot d\vec{x}.$$

By Newton's Second Law, $\vec{F} = m\vec{a}$. Since mass is constant, we have that

$$W = \int m\vec{a} \cdot d\vec{x} = m \int \vec{a} \cdot d\vec{x}.$$

Now recall that $\vec{v} = \frac{d\vec{x}}{dt}$. The key insight is that we can substitute $d\vec{x} = \vec{v}\,dt$, giving

$$W = m \int \vec{a} \cdot \vec{v}\,dt$$

Since $\vec{a} = \frac{d\vec{v}}{dt}$, we use the reverse chain rule to obtain $W = \frac{1}{2}mv^2 + c$, and the result follows. ■

Note that when a force is parallel to the direction of motion, substituting $F = ma$ yields $W = mad$; an easy way to remember this form is "*work* makes you *mad*".

6.2 Conservation of Energy and Potential Energy

One of the most notable and well-known properties of energy is that it can neither be destroyed nor created, but instead converts forms. However, we are interested primarily in mechanical energy (i.e. energy in a usable form for motion), and the conversion of energy to sound or thermal energy is of course possible. This is called energy dissipation. In the context of mechanics, we will use the phrase "energy is conserved" to refer to "mechanical energy is conserved."

To understand when energy is conserved, we must introduce the notion of a conservative force.

Definition 6.4. A **conservative force** is a force such that the work done by the force on an object is path-independent. This also means the work done by the force when the object travels along a closed loop is zero.

Conservative forces will not dissipate mechanical energy. We will see that gravity and the spring force are conservative. The primary nonconservative force we will see is friction. However, forces such as air resistance, tension, compression, and the normal force are all nonconservative.

It is important to note that energy could be conserved when nonconservative forces are present. The key distinction here is whether the nonconservative forces do work.

Theorem 6.5. *The mechanical energy of a closed system is conserved if there is no net work done by nonconservative forces. More specifically, if the net work done by these forces is W_{nc}, then*

$$W_{\mathrm{nc}} = \Delta E_{\mathrm{sys}}.$$

All energy that is not kinetic energy is called potential energy, and is denoted U. Usually, a subscript is assigned based on the type of potential energy (i.e. gravitational potential energy is U_g). We now have the following corollary.

Corollary 6.6. *If the energy of a system is conserved, then*

$$\Delta U = -W.$$

Proof. From the previous theorem, we have that

$$\Delta E_{\mathrm{sys}} = 0$$

Since $E = K + U$, we have that $\Delta E = \Delta K + \Delta U$, which implies

$$\Delta U = -\Delta K.$$

The result follows from applying the work-kinetic energy theorem. ■

We now apply these results to gravitational potential energy.

Corollary 6.7. *Change in gravitational potential energy is given by*

$$\Delta U_g = mg\Delta h.$$

Proof. This result is relatively straight-forward, so we suggest you attempt this on your own before reading our solution.

Recall that at the surface of the Earth, the gravitational force is approximated as mg, which is constant. Then the work done by gravity in lifting an object a height Δh is

$$W_g = \vec{F} \cdot \vec{d} = -mg\Delta h.$$

Note the sign is negative because the gravitational force is points down where as displacement points up. If the object was lowered, then the work done would instead be positive.

We now recall that $\Delta U_g = -W_g$, and the result follows. ∎

You might have noticed that we only gave an equation for ΔU_g, which is change in U_g. If we don't have an initial value, however, how can we calculate gravitational potential energy?

In practice, we can set a reference height where $h = 0$. In this sense, energy is relative, and we can calculate gravitational potential energy with the equation

$$U = mgh,$$

where h is the altitude in relation to our reference height.

The gravitational force is only considered to be constant near the Earth's surface. For a system where gravity varies with distance, there is a different formula for gravitational energy that we will cover in a later handout, but it still builds on the same principle.

Conservation of energy is an extremely powerful tool. For example, the following problem could have been solved using kinematics, but using energy gives a much quicker solution.

Example 6.8. A block of mass m sits on a ramp with angle of elevation θ, and is initially at rest. Answer the following two questions:

(a) If the ramp is frictionless, what is the speed of the block after it descends through a vertical distance h?

(b) If the ramp and block have a coefficient of kinetic friction μ, and assuming the friction force is small enough for the block to slide, what is the speed of the block after it descends a vertical distance h?

Solution.

(a) By conservation of energy, we have that

$$\begin{aligned} \Delta K &= \Delta U_g \\ \frac{1}{2}mv^2 &= mgh \\ v &= \sqrt{2gh} \end{aligned}$$

(b) Recall that

$$W_{\text{nc}} = \Delta U_g + \Delta K.$$

The friction force is given by $\mu mg\cos\theta$, and the distance through which the block moves is $\frac{h}{\sin\theta}$. Hence, the work done by friction is $-\frac{\mu mgh}{\tan\theta}$ (work is negative since friction opposed direction of displacement). Since change in potential energy is still $-mgh$, we have that

$$mgh\left(1 - \frac{\mu}{\tan\theta}\right) = \frac{1}{2}mv^2.$$

Simplifying gives

$$v = \sqrt{2gh(1 - \frac{\mu}{\tan\theta})}.$$

■

Here is another quick conceptual example.

Example 6.9 ($F = ma$)**.** Liquid droplets store a given amount of potential energy per unit surface area, due to their surface tension. When two identical, nearly spherical liquid droplets coalesce on a certain type of surface, part of this energy can be converted into upward kinetic energy, causing the coalesced droplet to jump. Assuming the conversion is 100% efficient, how does the maximum height depend on the radius r of the initial droplets?

Solution. We have that the energy is proportional to surface area and thus proportional to r^2. On the other hand, mass m is proportional to r^3 since density is constant. Hence, since

$$\Delta E = mg\Delta h,$$

we have that Δh is proportional to r^{-1}. ■

6.3 Power

Definition 6.10. **Power** (P) is the rate at which work is done, and is measured in units of watts (W). That is,

$$P = \frac{\mathrm{d}W}{\mathrm{d}t}$$

Alternatively using, the Work-Energy Theorem,

$$P = \frac{\mathrm{d}K}{\mathrm{d}t} = \vec{F}\cdot\vec{v}.$$

These two equations are generally sufficient to solve any question asking about power. We'll now do a tricky example.

Example 6.11. A particle of mass $m = 4$ starts moving with initial position $x = 0$ and power P given by

$$P(x) = x^2 + 7.$$

If the particle has initial velocity $v_i = 3$, what is the velocity of the particle at

$x = 4$?

Solution. This example is rather odd because typically, we will see power as a function of time, in which case we have

$$W = \int P(t)\, dt.$$

However, in this case power is a function x. How is $\int P(x)\, dx$ related to $\vec{a}$ and $\vec{v}$? To get a better idea, we'll want to substitute $dx = v\, dt$. Hence, we see that

$$\begin{aligned} \int_{t_i}^{t_f} P(x) \cdot v\, dt &= \int_{t_i}^{t_f} Fv^2\, dt = m \int_{t_i}^{t_f} a \cdot v^2\, dt \\ &= \frac{1}{3} m v_f^3 - \frac{1}{3} m v_i^3. \end{aligned}$$

Therefore, we see that the velocity at $x = 4$ is given by

$$v_f = \sqrt[3]{v_i^3 + \frac{3}{m} \int_0^4 P(x)\, dx}.$$

We can evaluate

$$\int_0^4 x^2 + 7\, dx = \frac{4^3}{3} + 28 = \frac{148}{3},$$

and substituting $v_i = 3, m = 4$ gives

$$v_f = 4.$$

■

6.4 Potential Energy Graphs

By far the most common type of potential energy graph we will see is a graph of potential energy as a function of position. Recall our earlier equations that $W = \int \vec{F} \cdot d\vec{x}$ and $W = -\Delta U$. This gives the following relation:

$$F = -\frac{\mathrm{d}U}{\mathrm{d}x}$$

Thus, positions of equilibrium can be found by examining potential energy vs position graphs: wherever the derivative of the graph is zero, the net force is also zero and the object will be at equilibrium there. However, objects can be at stable and unstable equilibrium, and the distinction between the two, as the name implies, is how "stable" they are.

Definition 6.12. **Stable equilibrium** is a state of equilibrium where an object will tend to return to its original position after being disturbed.

Definition 6.13. **Unstable equilibrium** is a state of equilibrium where an object will tend to depart form its original position after being disturbed.

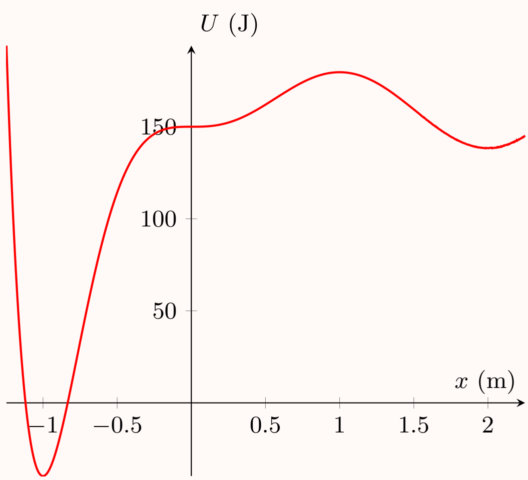
To evaluate whether equilibrium is stable, it is generally easiest to set $\vec{v} = 0$. In an ideal world, stability would not matter, since there would be no gust of wind or any sort of external force to push an object out of static equilibrium. In the real world, however, there do exist such forces that can push objects minutely out of equilibrium. In this case, some objects will return to their position of equilibrium, while others will move away from equilibrium after the slightest disturbance. For instance, a sphere placed perfectly on the top of another sphere could be at equilibrium, but it would be unstable since it would fall off upon the slightest force. In contrast, the same sphere placed at a bottom of a bowl would be at stable equilibrium since it could return to the same position even after a small push.

Graphically, positions of stable equilibrium are defined only where the derivative is zero and the second derivative is positive - all other positions of equilibrium are unstable. Intuitively, this makes sense if you imagine the potential energy graph as only a measure of gravitational potential energy, in which case the potential energy vs position graph would be a scaled version of an altitude map. This simplification would allow you to simply place the object at a position on the graph and visualize whether the object is at equilibrium and its stability (the "hills" would be unstable, where "valleys" would be stable).

Concept. The net force will always point towards regions of lower potential energy.

As such, a local maximum will always be a point of unstable equilibrium, and a local minimum will always be a point of stable equilibrium.

Example 6.14. Consider the below graph of potential energy of a particle in one-dimensional motion as a function of position. Answer the following questions.



(a) At which point(s), if any, is a particle in stable equilibrium?

(b) At which point(s), if any, is a particle in unstable equilibrium?

(c) At which point(s), if any, is equilibrium neither stable nor unstable (called semistable)?

(d) On the interval $-1 \leq t \leq 2$, give a half-second subinterval during which the magnitude of the force experienced by the object reaches a maximum.

(e) Assume the total energy of the system is 120 J and the particle has an initial position $x = -0.75$. Describe the motion of the particle.

Solution.

(a) Stable equilibrium occurs at $x = -1$ and $x = 2$.

(b) Unstable equilibrium occurs at $x = 1$.

(c) Semistable equilibrium occurs at $x = 0$.

(d) Recall that the magnitude of force is given by the magnitude of the slope of the graph. One such interval would then be $-1 \leq t \leq -0.5$.

(e) Note that the kinetic energy of the particle cannot be negative. Hence, the particle will be restricted on the right by roughly $x = -0.5$, and restrict on the left by an unlabeled point (roughly $x = -1.25$) where $U(x) = 150$ J. The particle will continually oscillate between these two points.

These two x-values are also called turning points.

■

Another type of potential energy graph that might appear is potential energy vs. time. Recall that $P = \frac{\mathrm{d}W}{\mathrm{d}t}$, and $W = -\Delta U$. Thus, we have that:

$$P = -\frac{\mathrm{d}U}{\mathrm{d}t}.$$

These graphs do not appear often, and often appear in conjunction with oscillations when they do. However, it is nonetheless helpful to remember the above relationship.

6.5 Internal Energy and Pseudowork

6.5.1 Internal Energy

Consider an ice skater who pushes herself off of a railing a the edge of a skating rink. Let's consider the conservation of energy equation.

Clearly as she pushes herself off the rink, her kinetic energy increases since she is now moving at a higher velocity than before. There are no net conservative forces so the change in potential energy must be 0. Also note that since the point of contact does not move during the push, the work that the wall does on the skater is 0. We thus have

$$0 = W = \Delta K + \Delta U > 0 \ldots$$

Huh? That's strange. Where does the missing kinetic energy come from? It appears that something must be wrong with our original equation.

Recall that our conservation of energy equation was derived only for *particles*. It is clear however that the ice skater is *not* just a particle since not all parts of the ice skater move in the same way. When she extends her arms, though her *body* move, her hands do not. So the ice skater has to be thought of as a *system of particles* that has an internal structure unlike that of single particle systems. Let's now try extending our idea of conservation of energy for these multi-particle systems.

So we can postulate that the system of particles can *store* energy in a form called *internal energy* E_{int} and then extend our equation for conservation of energy as follows:

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W_{\text{ext}}.$$

E_{int} may occur in the microscopic potential and kinetic energies of particles within the system and may arise within the system as things like heat or chemical energy. In the case of the ice skater, the ice skater is using her internal chemical energy to push herself off of the ice rink.

Example 6.15 (Classical). An ice skater of mass 50; kg pushes herself off of the edge of an ice rink and gains a speed of $3m/s$. What is the change in internal energy for the ice skater?

Solution. By our conservation of energy equation, we have

$$\Delta E_{\text{int}} = -\Delta K = -\frac{1}{2}(50)(3)^2 = \boxed{-225 \text{ J}}$$

That is, the ice skater uses $225J$ of her stored chemical potential energy (it could be through the food she ate or what not) to gain her speed. ■

6.5.2 Frictional Work

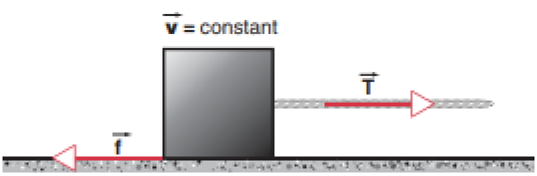
Consider a block sliding across a rough table and eventually coming to rest due to friction. Let's consider the system of the block and the table. Since there are no net forces on the system, we have

$$\Delta K + \Delta E_{int,system} = 0$$

As the kinetic energy of the block decreases, the internal energy of the system must increase. This increase in internal energy might occur as a slight increase in temperature from the surface of the block and the table (Try this yourself! For example, try rubbing your hands together. You'll see due to friction that your hands will heat up).

Now, we might be tempted to write that the magnitude of the work done by the frictional force on the block is just $|W| = fs$ where f is the magnitude of the friction force and s is the displacement of the block. As we'll see in this section however, this actually gives an *incorrect* value for the frictional work, because objects subject to sliding friction can *not* be treated as a particles from the standpoint of work and energy.

Let's consider an example where the block is pulled across a horizontal table with a string that exerts a constant tension force T such that the velocity remains constant. Since the velocity remains constant, the net force on the block must be 0.



This implies that $T = f$. Let's now try applying our conservation of energy equation to the block. Since we assumed the velocity remains constant, we have

$$\Delta E_{int,block} = W_T + W_f$$

where W_T is the work done by the string and W_f is the work done by friction. Now recall from our initial observation about friction, that when we drag the block across the surface, the block heats up so the internal energy $\Delta E_{int,block} > 0$. Lets say that the block moves through a distance s. We thus have $W_T = Ts$. Rearanging our equation, we thus get

$$W_f = -fs + \Delta E_{\text{int}}$$

However, recall that $E_{\text{int}} > 0$. This implies however that $|W_f| < fs$. So the energy that the friction transports *out* of the block system is actually *less* than fs because some of the energy remains as internal energy within the block.

That's just about the limit of what we can know with regard to the frictional work, unfortunately. How is it possible that a frictional force f, acting on an object that moves through a displacement s, does work that is in magnitude smaller than fs?

The frictional force acting on a sliding surface is not a single force acting at a single point, but rather many smaller forces acting at various surface points. The friction force can be regarded as the net effect of the forces at many microscopic welds, some occurring where protrusions from the table bond to the surface of the block, and others occurring where protrusions from the block encounter the surface of the table.

As the block moves through a displacement s, only those welds at the moving surface contribute to the work, so a portion of the frictional force does not contribute to the work, which is really why $W_f < fs$. Thus, without a microscopic model for the friction, it is very difficult to account for frictional work and can not directly calculate the quantity.

6.5.3 The Pseudowork Equation

This section is adapted from Bruce Sherwood's excellent article on Pseudowork that can be found here.

The past two subsections may have seemed somewhat gloomy. For complex systems, it appears that the conservation of energy equations are only able to offer us limited information. For example, in our figure skater example, the external force does not appear in the work equation because the point of contact doesn't move. Furthermore, the figure skater can not be treated like a particle, so there's an extra internal energy component that we have to deal with. We'll see however, in this section, that there *is* a remedy for this. Although we saw that the work energy equation does not hold in general for non-particle systems, we'll see that there is actually a **pseudowork** equation (or COM energy equation) for complex systems:

Theorem 6.16 (Pseudowork Equation). *Let's consider a system of particles. Say a net force F_{ext} acts on this system. Then, we have*

$$\int_{x_i}^{x_f} F_{ext} dx_{cm} = \frac{1}{2} M (v_{cm,f}^2 - v_{cm,i}^2)$$

Namely, we have

$$\int_{x_i}^{x_f} F_{ext} dx_{cm} = \Delta K_{cm}$$

Proof. Recall from Newton's second law we have

$$F_{\text{ext}} = M a_{\text{cm}}$$

Multiplying both sides by dx_{cm} we have

$$F_{\text{ext}} dx_{\text{cm}} = M a_{\text{cm}} dx_{\text{cm}}$$

$$= M \frac{dv_{\text{cm}}}{dt} dx_{\text{cm}} = M v_{\text{cm}} dv_{\text{cm}}$$

Integrating both sides, we have

$$\int_{x_i}^{x_f} F_{\text{ext}} dx_{\text{cm}} = \int_{v_{\text{cm},i}}^{v_{\text{cm},f}} M v_{\text{cm}} dv_{\text{cm}}$$

$$= \frac{1}{2} M (v_{\text{cm},f}^2 - v_{\text{cm},i}^2) = \Delta K_{\text{cm}}$$

■

Note this equation looks extremely similar to the work-energy theorem. However, it is important to note that the quantity on the left hand side is *not* work in the sense that we defined it, since dx_{cm} does not represent the displacement of the point of contact, but rather it represents the displacement of the center of mass of the system. Also note that the *translational* kinetic energy (of the center of mass) is the only type of energy that appears in the equation (not the rotational kinetic energy, potential energy, or other types of energy). We call this equation the *pseudowork* or *center of mass (COM) energy* equation. Note that the pseudowork equation is derived directly from Newton's second law and, although it is a useful formulation, it is not a new and independent principle (unlike the work, energy equation).

We'll see how this equation is used through a few examples:

Example 6.17. A block moves with velocity v across a rough surface with friction force f. Find the distance the block moves before coming to rest.

Solution. By the pseudowork equation, we have

$$F_{\text{ext}} s_{\text{cm}} = \Delta K_{\text{cm}},$$

so

$$fs = \frac{1}{2} M (v^2 - 0^2) \implies \boxed{s = \frac{Mv^2}{2f}}$$

■

Example 6.18. A ball of mass M rolls down an incline plane at angle θ with friction force f. Find the velocity of the center of mass of the ball after the ball moves down a distance d.

Solution. By the pseudowork equation, we have

$$(Mg\sin\theta - f)d = \frac{1}{2}M{v_{\mathrm{cm}}}^2$$

Thus

$$\boxed{v_{\mathrm{cm}} = \sqrt{2(Mg\sin\theta - f)d}}.$$

For those familiar with rotation, note that this equation is different from the conservation of energy equation that would have given

$$Mgd\sin\theta = \frac{1}{2}Mv_{\mathrm{cm}}^2 + \frac{1}{2}I\omega^2.$$

Both of these answers are technically correct and acceptable. ■

6.6 Homework Problems and Solutions

x

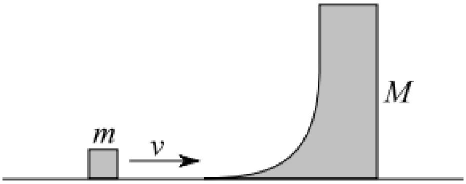
Problem 6.1 ($F = ma$)**.** [4] A ball is launched straight toward the ground from height h. When it bounces off the ground, it loses half of its kinetic energy. It reaches a maximum height of $2h$ before falling back to the ground again. What was the initial speed of the ball?

Solution. The potential energy of the of the ball at its second peak is $2mgh$, where m represents the mass of the ball. The total energy is also equal to $2mgh$ because kinetic energy is zero at this point. Since energy was conserved in the interval of time between the initial impact and the second peak, we know that the total energy following the initial impact is also equal to $2mgh$. Therefore, the total energy immediately before the initial impact was $4mgh$, of which mgh can be attributed to the potential energy when the ball was launched. Thus,

$$3mgh = \frac{mv^2}{2} \implies \boxed{v = \sqrt{6gh}}.$$

■

Problem 6.2 ($F = ma$)**.** [4] A block of mass m is launched horizontally onto a curved wedge of mass M at a velocity v. What is the maximum height reached by the block after it shoots off the vertical segment of the wedge? Assume all surfaces are frictionless; both the block and the curved wedge are free to move. The curved wedge does not tilt or topple.



Solution. Initially, all the energy of m is kinetic: $\frac{mv^2}{2}$, and its horizontal momentum is mv. Horizontal momentum and energy are both conserved. Note that the horizontal velocities of both masses must be equal, so $(m+M)v' = mv \implies v' = \frac{mv}{m+M}$ where v' represents this final velocity. Note then that of the original $\frac{mv^2}{2}$ energy, $\frac{(m+M)v'^2}{2} = \frac{m^2v^2}{2(m+M)}$ is being used to transport both masses horizontally. This leaves $\frac{mv^2}{2} - \frac{m^2v^2}{2(m+M)}$ energy that m can convert to gravitational potential energy. Thus,

$$mgh = \frac{mv^2}{2} - \frac{m^2v^2}{2\,(m+M)} \implies h = \frac{v^2(m+M) - mv^2}{2g\,(m+M)} = \boxed{\frac{Mv^2}{2g(m+M)}}.$$

■

Problem 6.3 ($F = ma$)**.** [4] A uniform block of 10 kg is released at rest from the top of an incline of length 10 m and inclination 30°. The coefficients of static and kinetic friction are are $\mu_s = 0.15$ and $\mu_k = 0.1$. The end of the incline is connected to a frictionless horizontal surface. After a long time, how much energy is dissipated due to friction?

Solution. Note that $W = F \cdot d$. Since F and d are parallel, $F \cdot d = Fd$. Now we only need to calculate F since $d = 10$ m is given:

$$F = mg\mu_k \cos\theta = 10 \text{ kg} \times 10 \text{ N/kg} \times 0.1 \times \cos 30^\circ \approx 8.66 \text{ N}.$$

Thus, $W = Fd \approx 10 \text{ m} \times 8.66 \text{ N} \approx \boxed{87 \text{ J}}$. Notice that coefficient of static friction was not used since it only takes place over a negligible distance. ■

Problem 6.4 (HRK)**.** [8] A stone of weight w is thrown vertically upward into the air with an initial speed v_0. Suppose that the air drag force f dissipates an amount fy of mechanical energy as the stone travels a distance y.

(a) [4] What is the maximum height reached by the stone?

(b) [4] What is the speed of the stone upon impact with the ground?

Solution.

(a) The initial energy of the stone is all kinetic: $\frac{mv_0^2}{2}$ where $m = \frac{w}{g}$. Suppose the stone reaches height h, then the energy dissipated due to air drag is fh and the remaining energy is stored as $mgh = wh$. Writing this information in an equation yields:

$$\begin{aligned} \frac{mv_0^2}{2} = \frac{wv_0^2}{2g} &= fh + wh \\ &= h(w+f), \\ \implies h &= \boxed{\frac{wv_0^2}{2g(w+f)}}. \end{aligned}$$

(b) The stone will lose another fh energy when returning to the ground. Combined with the fh energy lost during the upward part of the trajectory, the final energy of the stone upon impact will be $\frac{mv_0^2}{2} - 2fh$. Substituting known values into this expression gives:

$$\frac{mv_0^2}{2} - 2fh = \frac{wv_0^2}{2g} - \frac{fwv_0^2}{g(w+f)}.$$

Thus, if v_f is the final velocity, then

$$\begin{aligned}\frac{mv_f^2}{2} = \frac{wv_f^2}{2g} &= \frac{wv_0^2}{2g} - \frac{fwv_0^2}{g(w+f)},\\ \implies \frac{v_f^2}{2} &= \frac{v_0^2}{2} - \frac{fv_0^2}{w+f},\\ \implies v_f &= v_0\sqrt{1 - \frac{2f}{w+f}} = \boxed{v_0\sqrt{\frac{w-f}{w+f}}}.\end{aligned}$$

■

Problem 6.5 ($F = ma$)**.** [6] A car is driving against the wind at a constant speed v_0 relative to the ground. The wind direction is always opposite to the car's velocity, but its speed fluctuates about an average speed of v relative to the ground. The air drag force is Av_{rel}^2, where A is a constant and v_{rel} is the relative speed between the car and the wind. What is the average rate P of energy dissipation due to the air resistance?

(A) $\overline{P} = Av_0\left(v_0 + v\right)^2$
(B) $\overline{P} > Av_0\left(v_0 + v\right)^2$
(C) $\overline{P} < Av_0\left(v_0 + v\right)^2$
(D) Both (B) and (C) are possible depending on the situation
(E) Both (A) and (C) are possible depending on the situation

Solution. At an arbitrary instance of time, let the wind speed be $v + \delta v$ so that $v_{\text{rel}} = v_0 + v + \delta v$. The air drag force at this moment is $A\left(v_0 + v + \delta v\right)^2$ and since power is equal to force times velocity, the power at this moment is $Av_0\left(v_0 + v + \delta v\right)^2 = Av_0\left(v_0^2 + 2v_0\left(v + \delta v\right) + \left(v + \delta v\right)^2\right) = Av_0\left(v_0^2 + 2v_0\left(v + \delta v\right) + v^2 + 2v\delta v + \delta v^2\right)$. Note that on average, δv is zero, thus

$$\overline{P} = Av_0\left(v_0^2 + 2v_0v + v^2 + \delta v^2\right) > Av_0\left(v_0^2 + 2v_0v + v^2\right) = Av_0\left(v_0 + v\right)^2.$$

Thus, $\boxed{\overline{P} > Av_0\left(v_0 + v\right)^2}$. ■

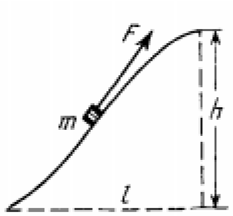
Problem 6.6 ($F = ma$)**.** [7] An escalator can carry passengers up a vertical distance of 10 m in 30 s. A mischievous person of mass 50 kg walks down the up-escalator so that they stay in place with respect to the building. If the child does this for 30 s, what is the total work the child performs on the escalator, in the frame of the building?

Solution. The average force that the child is exerting must obviously be $-mg = -500$. The velocity of the child is $\frac{10\text{ m}}{30\text{ s}} = \frac{1}{3}$ m/s. Since power is velocity times force, the average power the child is $-500 \times \frac{1}{3}$ W. Energy is power times time, so the desired answer is

$$\left(-500 \times \frac{1}{3}\text{ W}\right) \times (30\text{ s}) = \boxed{-5000\text{ J}}.$$

■

Problem 6.7 (Irodov)**.** [7] A body of mass m was slowly hauled up a hill by a force F which at each point was directed along a tangent to the trajectory. Find the work preformed by this force, if the height of the hill is h, the length of the base is l, and the coefficient of friction is k.



Solution. The net work done by F must equal the work done by all other forces acting on the mass, namely:

- gravitational force,
- normal force,
- frictional force.

The work done against gravitational force must clearly be mgh. Since normal force acts perpendicularly to the slope of the hill, it does zero work against F. To calculate the work that must be done to counter-act the frictional force, examine an arbitrary position on the hill where the angle of incline is θ. The frictional force at the position must be $mgk\cos\theta$ and the distance over which the frictional force is applied must be $\frac{\mathrm{d}l}{\cos\theta}$ where dl represents the horizontal displacement over the distance. Thus, the total frictional work that must be counter-acted is:

$$\int_0^l mgk\cos\theta\frac{\mathrm{d}l}{\cos\theta} = \int_0^l mgk\,\mathrm{d}l = mgkl.$$

The total work done by F is therefore $\boxed{mg(h+kl)}$. ■

Problem 6.8 ($F = ma$)**.** [7] A juggler juggles N identical balls, catching and tossing one ball at a time. Assuming that the juggler requires a minimum time T between ball tosses, the minimum possible power required for the juggler to continue juggling is proportional to N^k for some integer k. What is k?

Solution. Focusing only one one ball, the total time elapsed from the moment it leaves the juggler's hand to the moment it returns is NT. Thus, the juggler must toss it upwards with velocity of $\frac{NTg}{2}$ so that it returns NT time later. The energy applied on the ball is $\frac{mN^2T^2g^2}{8}$ where m is the mass of the ball. This energy is applied every T time for each of the N balls, so the juggler is applying an average power of $P = \frac{mN^2Tg^2}{8}$. Clearly P is proportional to N^2, meaning that $\boxed{k=2}$. ■

Problem 6.9 (Morin)**.** [15] Sand drops vertically (from a negligible height) at a rate σ kg/s onto a moving conveyor belt. The conveyor belt moves at a constant speed v.

(a) [4] What force must you apply to the belt in order to keep it moving at a constant speed v?

(b) [3] How much kinetic energy does the sand gain per unit time?

(c) [5] How much work do you do per unit time?

(d) [3] How much energy is lost to heat per unit time?

Solution.

(a) Recall the force is the change in momentum, and every second σ kg of sand is being forced to move at a velocity of v from an initial velocity of 0. Thus, the required force is $\boxed{\sigma v}$.

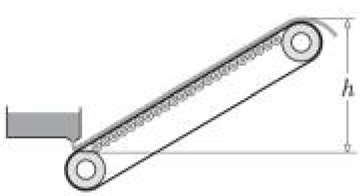
(b) Every second, σ kg of sand's velocity changes from 0 to v. Thus, the change in kinetic energy every second is $\boxed{\dfrac{\sigma v^2}{2}}$.

(c) Power is force times velocity, so in this case, power is $\boxed{\sigma v^2}$.

(d) The energy lost to heart per second is equal to the difference energy we put into the system (work) and the resulting change in kinetic energy of the system: $\sigma v^2 - \frac{\sigma v^2}{2} = \boxed{\dfrac{\sigma v^2}{2}}$.

■

Problem 6.10 (Pathfinder)**.** [15] A conveyor belt collects sand and transports it to a height h as shown in the figure. The angle of incline of the conveyor belt is θ. The sand falls on the belt with negligible speed at a constant mass rate μ. Friction between the belt and the sand particle is so high that the sand particles stop sliding almost instantaneously after they hit the belt. Acceleration due to gravity is g.



(a) [6] What should the speed of the belt be for the least possible driving force on the belt applied by the motor?

(b) [4] What is the power delivered by the motor to the belt, when the motor is applying the least possible driving force?

(c) [5] What is the power dissipated by the sand-belt system, when the motor is applying the least possible driving force?

Solution.

(a) Suppose the belt is moving at speed v. The driving forces must push the sand already on the belt and push the sand newly deposited on the belt up to speed.

Since the speed of the belt is v, at any given moment, there will be $m = \mu t = \frac{\mu h}{v \sin\theta}$ kg of sand on the conveyor belt. Since the acceleration of gravity down the incline is $g \sin\theta$, the driving force required to move this mass of sand is

$$F = \left(\frac{\mu h}{v \sin\theta}\right)(g \sin\theta) = \frac{\mu g h}{v}.$$

Furthermore, the newly deposited sand requires a force of μv to be pushed up to velocity v. Therefore, we have that

$$F(v) = \frac{\mu g h}{v} + \mu v.$$

By AM-GM, the minimum of driving force is $2\mu\sqrt{gh}$ which occurs at $v = \boxed{\sqrt{gh}}$.

(b) Power is force times velocity. Our prior work implies then that:

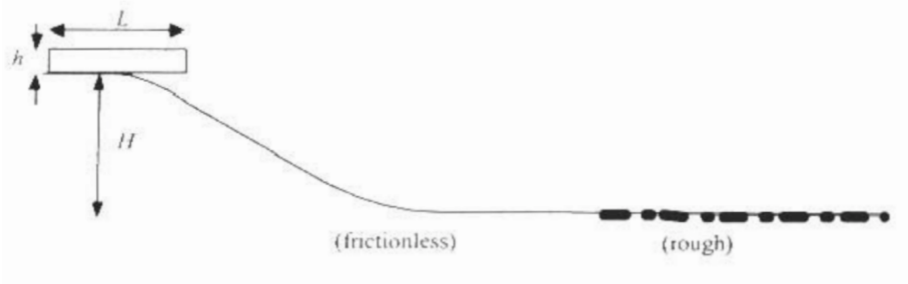
$$P = 2\mu\sqrt{gh} \times \sqrt{gh} = \boxed{2\mu g h}.$$

(c) Recall the total mass of sand on the conveyor belt is $\frac{\mu h}{v \sin\theta}$. The y-coordinate of the mass of sand ascends at a rate of $v \sin\theta$, so we have the potential energy changes at rate $\mu g h$. The newly added mass of sand gains mass at a rate of $\frac{1}{2}\mu v^2 = \frac{1}{2}\mu g h$, so we see the energy of the system increases at a rate of $\frac{3}{2}\mu g h$. Thus, the energy dissipated is $\boxed{\frac{1}{2}\mu g h}$.

■

6.6.1 Written Solutions

Problem 6.11 (USAPhO). [23] A block with uniform density, length L and height h, $h \ll L$, starts from rest at the top of a hill of height H as shown in the accompanying diagram. The hill is frictionless and there is a flat frictionless surface at the bottom of the hill that is longer than L. Then, there is a flat horizontal surface that has friction. The coefficient of friction between the block and the flat surface is μ. The leading edge of the block comes into contact with the surface with friction when the rest of the block is on a flat frictionless surface.



(a) If the hill has a certain critical height H_{crit}, the block comes to a stop at the instant that the entire block is on the surface with friction. (This is the instant at which the leading edge has traveled a distance L across the surface with friction). Express your answers to the following in terms of μ, L, and g.

(i) [3] Find the critical height H_{crit}.

(ii) [4] If the hill has critical height found in part (i), what is the time from the instant that the leading edge of the block encounters friction until the block comes to a stop?

(b) If the hill has a height $H < H_{\text{crit}}$, express your answers to the following in terms of H, μ, L, and g.

(i) [3] Find the distance that the leading edge of the block travels across the surface with friction before it comes to a stop.

(ii) [4] Find the time from the instant that the leading edge of the block encounters friction until the block comes to a stop.

(c) If the hill has a height $H > H_{\text{crit}}$, express your answers to the following in terms of H, μ, L, and g.

(i) [3] Find the distance the leading edge of the block travels across the surface with friction before it comes to a stop.

(ii) [6] Find the time from the instant that the leading edge of the block encounters friction until the block comes to a stop.

Solution.

(a) (i) Note that if a distance x of the block lies on the rough patch, the fraction of the block on the rough patch is x/L, so the friction force experienced is $\frac{\mu mgx}{L}$. Thus, setting the gravitational and frictional work equal, we have that

$$mgH_{\text{crit}} = \int_0^L \frac{\mu mgx}{L}\, dx = \frac{\mu mgL}{2}.$$

Thus we have $H_{\text{crit}} = \boxed{\frac{\mu L}{2}}$.

(ii) Recall that we have

$$v^2 = v_i^2 + 2\int a\, dx.$$

By conservation of energy $v_i^2 = 2gH_{\text{crit}} = \mu gL$. As with part (a)(i), we compute that

$$\int a\, dx = \int_0^x -\frac{\mu gx'}{L}\, dx' = -\frac{\mu gx^2}{2L}.$$

Thus we have

$$\frac{dx}{dt} = v = \sqrt{\mu gL - \frac{\mu gx^2}{L}},$$

and we can separate the differentials and integrate:

$$\int_{x_i}^{x_f} \frac{dx}{\sqrt{\mu gL - \frac{\mu gx^2}{L}}} = \int_{t_i}^{t_f} dt,$$

so

$$t = \int_0^L \frac{dx}{\sqrt{\mu gL - \frac{\mu gx^2}{L}}} = \sqrt{\frac{L}{\mu g}}\arcsin(1) = \boxed{\sqrt{\frac{L}{\mu g}}\frac{\pi}{2}}.$$

Remark. A much simpler solution to this problem (and to part (b)(ii)) uses simple harmonic motion (which we cover later in the course). Note that we can write

$$m\ddot{x} = -F_f = -\mu N = -\mu\left(m\frac{x}{L}g\right) \implies T = 2\pi\sqrt{\frac{L}{\mu g}}.$$

The motion is equivalent to a "quarter" of the period of a spring, which trivially gives us

$$\frac{\pi}{2}\sqrt{\frac{L}{\mu g}}.$$

We won't cover oscillations until much later in the course, but it might help to memorize the period for a spring is $2\pi\sqrt{\frac{k}{m}}$, where k is the spring constant.

(b) (i) Like with part (a)(i), we have

$$mgH = \int_0^d \frac{\mu mgx}{L}\,dx = \frac{\mu mgd^2}{2L}.$$

Thus we have $d = \boxed{\sqrt{\frac{2HL}{\mu}}}$.

(ii) Consider our integral from part (a)(ii). The only change is in the bounds of integration, and in v_i^2, which is now $2gH$. except for the bound:

$$t = \int_0^d \frac{dx}{\sqrt{2gH - \frac{\mu g x^2}{L}}} = \sqrt{\frac{L}{\mu g}}\arcsin\left(\sqrt{\frac{\mu}{2HL}}d\right) = \boxed{\sqrt{\frac{L}{\mu g}}\frac{\pi}{2}}.$$

(c) (i) Note that after the block has passed entirely onto the surface, the friction force is μmg, so we have

$$mgH = \frac{\mu mgL}{2} + \mu mg(d-L) = \mu mg(d - \frac{L}{2}).$$

Thus, $d = \boxed{H/\mu + L/2}$.

(ii) We first compute the time for the entire block to slide onto the rough surface. This essentially combines our integrals from parts (a)(ii) and (b)(ii):

$$t_1 = \int_0^L \frac{dx}{\sqrt{2gH - \frac{\mu g x^2}{L}}} = \sqrt{\frac{L}{\mu g}}\arcsin\left(\sqrt{\frac{\mu L}{2H}}\right).$$

Note the velocity is now $v_i = \sqrt{2gH - \mu gL}$, and $a = -\mu g$ is constant. Thus, the remaining time is

$$t_2 = \frac{\sqrt{2gH - \mu gL}}{\mu g}.$$

Thus, the total time in this case is

$$t = \boxed{\sqrt{\frac{L}{\mu g}}\arcsin\left(\sqrt{\frac{\mu L}{2H}}\right) + \frac{\sqrt{2gH - \mu gL}}{\mu g}}.$$

■

7 Central Forces

So far, we have dealt mainly with projectile motion with accelerations constant in both magnitude and direction. In this section however, we will deal with a very different kind of motion: one in which the magnitude of the velocity and acceleration is constant, but the direction of both the acceleration and velocity constantly changes. This situation is called **circular motion** and includes many examples such as the earth rotating around the sun, a ball being whirled around on a string, and so on.

7.1 Centripetal Acceleration and Centripetal Force

So far, we have only calculated acceleration in the same direction in motion. However, since velocity is a vector, an object with constant speed can still experience acceleration if its direction is changing. We will now look first at uniform circular motion as it is the most common and simplest example of this type of motion.

Definition 7.1. **Uniform circular motion** is the movement of an object along a circular trajectory with a constant speed.

Since the velocity (and hence kinetic energy) of the particle is constant, we have

$$P = \frac{dK}{dt} = Fv\cos\theta = 0.$$

We can then conclude that the net force and velocity vectors for the particle must be perpendicular. This leaves us with two possible directions for the acceleration vector, towards the center of the circle, or away from the center of the circle. Intuitively, since acceleration is the rate of change of velocity, we should conclude the acceleration vector is directed towards the center of the circle. We will call this inward direction the centripetal direction.

Definition 7.2. We define the **centripetal acceleration** (also called radial or normal acceleration) as the component of the acceleration vector in the centripetal direction. In the context of circular motion, an inward centripetal acceleration is positive, whereas a net outward acceleration is negative.

Note that during uniform circular motion, the particle will experience no net *tangential acceleration*, and hence acceleration and centripetal acceleration can be used interchangeably.

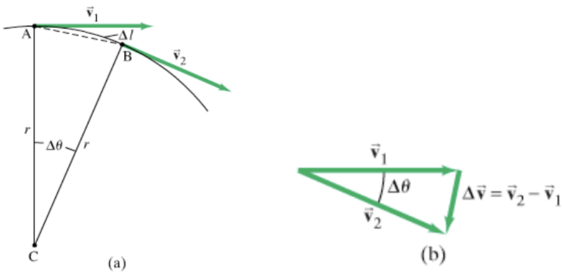
We now prove the following result which allows us to calculate centripetal acceleration.

Theorem 7.3. *For an object traveling in a circle of radius r at a constant speed v, the acceleration experienced by the object has magnitude*

$$a = \frac{v^2}{r},$$

and is always directed inwards towards the center of the circle.

Proof. Consider the following diagrams for a particle in circular motion. Diagram A illustrates two vectors at different instances while Δl is the distance along the arc from point A to B. Diagram B is a triangle of the two vectors $\vec{v}_1$ and $\vec{v}_2$ that is similar to the triangle ABC.



As the two triangles are similar, the following equation can be written.

$$\frac{d\vec{v}}{\vec{v}} = \frac{d\vec{r}}{\vec{r}} \Rightarrow \frac{d\vec{v}}{d\vec{r}} = \frac{\vec{v}}{\vec{r}}$$

By the chain rule, we have

$$\frac{d\vec{v}}{dt} = \frac{d\vec{v}}{d\vec{r}} \cdot \frac{d\vec{r}}{dt}.$$

Recall that $\vec{r}$ the position vector. Hence we can substitute to obtain:

$$\vec{a} = \left(\frac{\vec{v}}{\vec{r}}\right)\vec{v}$$

and the result follows. ■

Another proof can be done using a quantity called angular frequency and some basic calculus.

Definition 7.4. We define the **angular frequency** (ω) of a rotating object by the equation

$$\omega = \frac{d\theta}{dt}.$$

Equivalently, since the arc length s is given by $s = r\theta$, where r is the radius of rotation, we have that

$$\omega = \frac{v}{r}.$$

For an object undergoing uniform circular motion, we also have that the period $T = \frac{2\pi}{\omega}$, since ω is the rate of change of θ. Hence, we can also express ω as

$$\omega = \frac{2\pi}{T}.$$

Alternative Proof of Theorem 7.3. An important application of angular frequency is how it can parameterize circular motion. Since $\omega = \frac{2\pi}{T}$, we can express the displacement of a rotating object with the parameterization

$$\Delta\vec{r} = (r\cos\omega t, r\sin\omega t)$$

Some simple calculus can be used to find the formula for acceleration:

$$\vec{v} = \frac{d\vec{v}}{dt} = \omega(-r\sin\omega t, r\cos\omega t)$$

$$\vec{a} = \frac{d\vec{v}}{dt} = -\omega^2(r\cos\omega t, r\sin\omega t) = -\omega^2\vec{r}.$$

Since, $v = r\omega$, the result follows. This alternate form, $\vec{a}_c = -\omega^2\vec{r}$, is important as well. ■

Note that centripetal acceleration must result from a net force in the centripetal direction. We will call this the centripetal force.

Definition 7.5. The net force in the centripetal direction is called the **centripetal force**. By Newton's Second Law and Theorem 7.3, we have that the centripetal force satisfies

$$|\vec{F}_c| = \frac{mv^2}{r}.$$

Concept. It is important to understand that centripetal force is not a *type* of force like tension or gravity, and refers only to the direction of the force. Just as we can split a force into x and y components, we can split a force into centripetal and tangential components (since these two directions are always perpendicular). If an object experiences a net centripetal force, it is because it is experiencing forces that happen to act in the centripetal direction.

7.2 Common Examples of Circular Motion

Uniform circular motion can appear in many different forms. In this section, we will go over some relatively simple but common examples.

Example 7.6. A futuristic space station in the form of a large cylinder of radius 20 m rotates about its axis. An astronaut with a mass of 50 kg is standing on a scale within the space station. Any gravitational forces on the astronaut are negligible. If the scale reads 250 N, what is the angular frequency of the space station's rotation?

Solution. Recall that the measurement we see on a scale is in fact the normal force it is exerting on us. Hence, the net force in the centripetal direction is 250 N. We have that

$$\vec{F}_c = m\vec{\omega}^2 r,$$

$$\omega = \sqrt{\frac{F_c}{mr}} = \sqrt{\frac{250}{50 \times 20}} = \boxed{\frac{1}{2}\ \frac{\text{rad}}{\text{s}}}.$$

■

Example 7.7. An amusement park ride consists of a cylindrical-walled room of radius r that rotates at a fast enough pace such that tourists feel "pressed against the wall." Assuming the coefficient of static friction between the tourists and the wall is μ, what is the minimum angular velocity at which the tourists must rotate such that they do not slide down the wall?

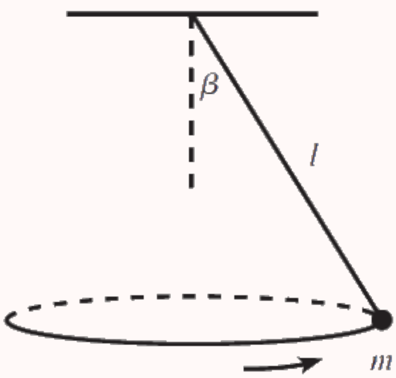
Solution. Note that the tourists are able to stick to the wall when it rotates at a high enough speed because the normal force increases to the degree that friction can counteract gravity (that is, $\mu N \geq mg$). The normal force is the only force in the centripetal direction, which impliesthe wall presses against the tourist with magnitude $N = mr\omega^2$.

Hence, we have

$$\mu mr\omega^2 \geq mg \implies \omega \geq \boxed{\sqrt{\frac{g}{\mu r}}}.$$

■

Example 7.8. A mass m hangs from a string of length ℓ and negligible mass. The mass swings around in a horizontal circle, with the string at a constant angle β from the vertical. What is the period (time it takes for a complete revolution) of the motion?



Solution. First, note that the circumference of the circle can be denoted as $2\pi\ell\sin(\beta)$. In order to find the period, we first find the net centripetal force. In this problem, the tension force, T, is equal to $\frac{mg}{\cos(\beta)}$ since the mass stays at a constant altitude. The centripetal force is given by $T\sin(\beta)$, and can thus be written as $mg\tan(\beta)$. Using the aforementioned formula for the centripetal force,

$$mg\tan(\beta) = m\frac{\vec{v}^2}{\ell\sin(\beta)}$$

$$\vec{v}^2 = g\ell\frac{(\sin(\beta))^2}{cos(\beta)}$$

$$\vec{v} = \sin(\beta)\sqrt{\frac{g\ell}{\cos(\beta)}}$$

Finally, we can calculate the period by dividing the distance of one full rotation by the velocity, giving the answer of

$$\frac{2\pi r}{\vec{v}} = \frac{2\pi\ell\sin(\beta)}{\sin(\beta)\sqrt{\frac{g\ell}{\cos(\beta)}}} = \boxed{2\pi\sqrt{\frac{\ell\cos(\beta)}{g}}}.$$

■

7.3 Nonuniform Circular Motion

Some problems may involve an object rotating at a changing speed. As you might expect, this is nonuniform circular motion.

Definition 7.9. **Nonuniform circular motion** is the movement of an object along a circular trajectory at a changing speed due to having nonzero tangential acceleration.

Just like with uniform circular motion, the centripetal component of $\vec{a}$ is still given by

$$\vec{a}_c = \frac{v^2}{r}.$$

We can also calculate the tangential component using the equation

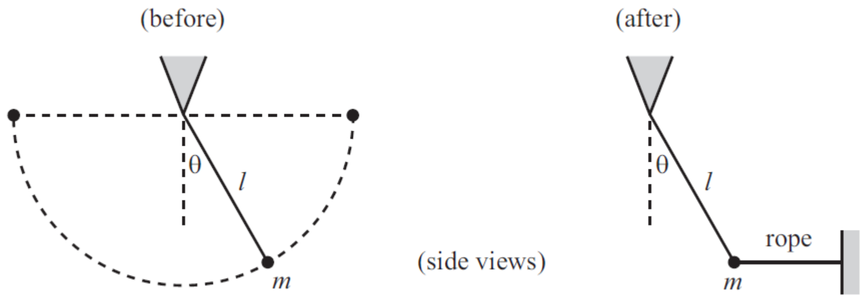
$$\vec{a}_T = \frac{d|\vec{v}|}{dt},$$

where $|\vec{v}|$ is the speed of the particle. This additional relation, along with conservation of energy, are the most common tools for problems with nonuniform circular motion. Since the two components are perpendicular, the total acceleration vector must also satisfy

$$\vec{a} = \sqrt{\vec{a}_c^2 + \vec{a}_T^2}.$$

We will now do some examples.

Example 7.10 (Morin). A pendulum with mass m and length ℓ swings back and forth between the two horizontal positions shown on the left in Fig. 5.13. Let the tension in the string as a function of θ be $T_1(\theta)$. The mass is then stopped at an angle θ and held in place with a horizontal rope, as shown below. Let the tension in the string (the pendulum's string, not the rope) as a function of θ be $T_2(\theta)$. For what θ does $T_1(\theta) = T_2(\theta)$?



Solution. Consider the first scenario. We compute the speed of the particle as a function of time using conservation of energy:

$$v = \sqrt{2g(\ell \cos(\theta))}.$$

We can now evaluate the net force in the centripetal direction:

$$\sum \vec{F}_c = T_1(\theta) - mg\cos(\theta) = \frac{mv^2}{\ell} = 2mg\cos(\theta).$$
$$T_1(\theta) = 3mg\cos(\theta).$$

In the second scenario, we have

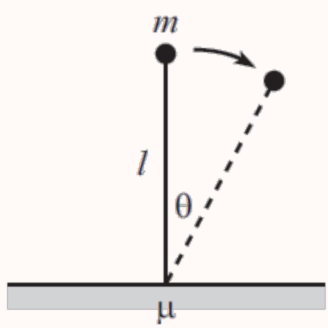
$$\sum \vec{F}_y = T_2(\theta)\cos(\theta) - mg = 0 \implies T_2(\theta) = \frac{mg}{\cos(\theta)}.$$

Therefore, we have

$$T_1(\theta) = T_2(\theta) \implies \theta = \boxed{\arccos\left(1/\sqrt{3}\right)}.$$

■

Example 7.11 (Morin). A massless stick with length ℓ stands at rest vertically on a table, and a mass m is attached to its top end, as shown below. The coefficient of static friction between the stick and the table is μ. The mass is given an infinitesimal kick, and the stick-plus-mass system starts to fall over. At what angle (measured between the stick and the vertical) does the stick start to slip on the table?



Solution. First, we can use conservation of energy to find the velocity v after the mass has swung through an angle θ. We have that

$$\begin{aligned}\frac{1}{2}mv^2 &= mg\ell - mg\ell\cos(\theta)\\ \Rightarrow v^2 &= 2g\ell(1-\cos(\theta)).\end{aligned}$$

Hence, we can find that the centripetal acceleration is

$$a_c = \frac{v^2}{\ell} = 2g(1-\cos(\theta)).$$

Let F be the sum of the friction and normal forces from the table on the stick. For the stick to not slip, F must be directed in the same direction as the stick. Since the stick is massless, we know that it will provide the mass with a force F as well. Since the centripetal component of the gravitational force is $mg\cos(\theta)$, we have that

$$\begin{aligned}F + mg\cos(\theta) &= 2mg(1-\cos(\theta))\\ \Rightarrow F &= mg(2-3\cos(\theta)).\end{aligned}$$

Hence F has vertical component $F\cos(\theta)$ and horizontal component $F\sin(\theta)$. Since the horizontal component is provided by static friction, it must hold that $F\sin(\theta) \le \mu F\cos(\theta)$, or $\mu \ge \tan\theta$.

As you might have noticed, though we have computed the force F, our solution has not yet used this computation. This is because even though $\mu \ge \tan\theta$ is a necessary condition, it is not sufficient. If $2 - 3\cos(\theta) < 0$, then $F < 0$, which means the ground provides the stick with a negative normal force! This is impossible, so we conclude for $\cos\theta > \frac{2}{3}$, the stick will *always* slip.

The second case applies when $\mu \ge \tan(\cos^{-1}(2/3))$, and some quick geometry can simplify this to $\mu \ge \sqrt{5}/2$. In other words, the critical angle ϑ is given by

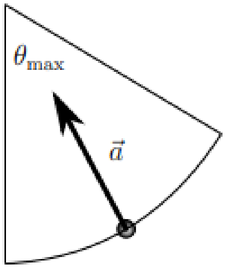
$$\vartheta = \boxed{\begin{cases}\tan^{-1}(\mu), & \mu \le \sqrt{5}/2\\ \tan^{-1}(\sqrt{5}/2), & \mu > \sqrt{5}/2\end{cases}}$$

■

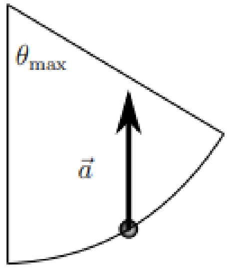
7.4 Homework Problems and Solutions

Problem 7.1 ($F = ma$)**.** [5] Consider the pendulum bob when it is at an angle $\theta = \frac{1}{2}\theta_{\text{max}}$ on the way up (moving toward θ_{max}). What is the direction of the acceleration vector?

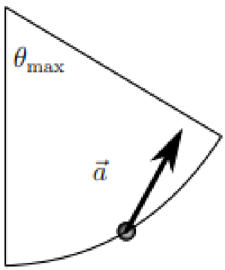
(A)



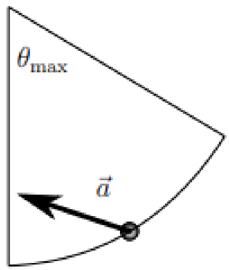
(B)



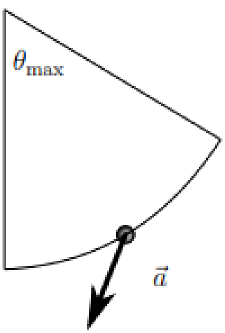
(C)



(D)



(E)



Solution. The pendulum bob experiences a centripetal acceleration akin to (A). However, note there is tangential acceleration slowing the pendulum bob, so we see the total acceleration vector must be $\boxed{(D)}$. ■

Problem 7.2 (Morin)**.** [5] A roller coaster car starts at rest and coasts down a frictionless track. It encounters a vertical loop of radius R. How much higher than the top of the loop must the car start if it is to remain in contact with the track at all times?

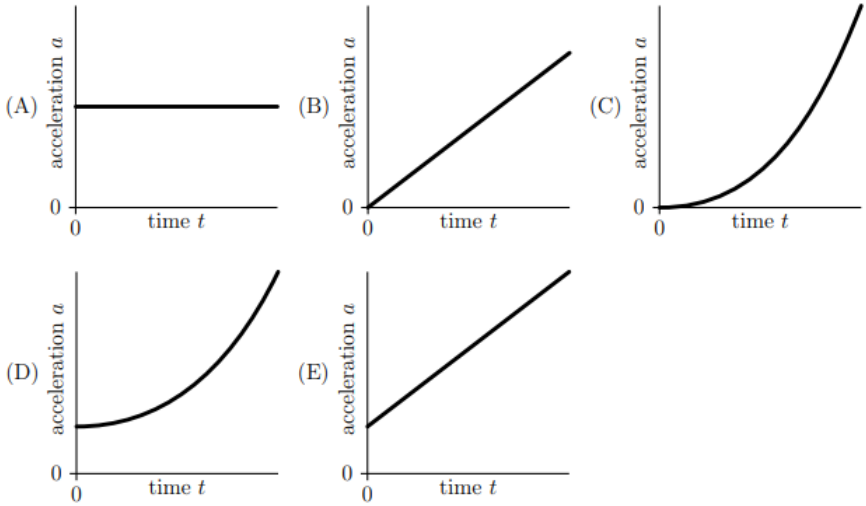
Solution. Let h be the height above the top of the loop at which the car starts. Then by conservation of energy the car will have speed $\sqrt{2gh}$ at the top of the track. Note the only two forces acting on the car are gravity and the normal force from the track (assuming v is high enough).

The boundary at which the track loses contact occurs when $N = 0$. Thus, we have

$$mg = \frac{mv^2}{r} = \frac{2mgh_{\text{crit}}}{R}.$$

Solving, we see that $h_{\text{crit}} = \boxed{\frac{R}{2}}$. ∎

Problem 7.3 ($F = ma$)**.** [5] A small bead slides from rest along a wire that is shaped like a vertical uniform helix (spring). Which graph below shows the magnitude of the acceleration a as a function of time?

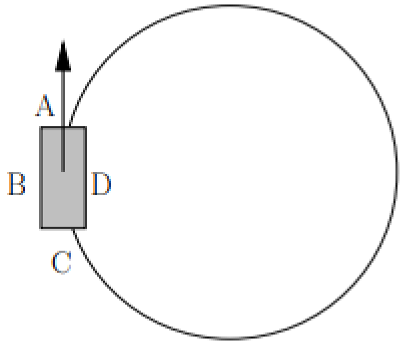


Solution. In this problem, we need to account for both tangential and centripetal acceleration. We note that tangential acceleration is constant ($g\sin(\theta)$, where θ is the angle of incline of the helix).

Recall that the tangential acceleration is the rate of change of speed. Hence, the speed of the particle is a function of t, and the centripetal acceleration is thus a function of t^2.

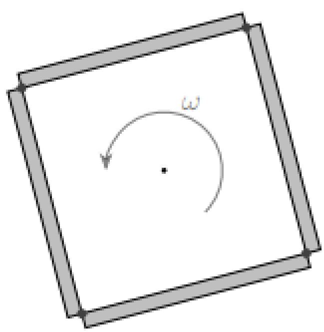
Adding these two components, we get the answer $\boxed{(D)}$. ∎

Problem 7.4. [5] [$F = ma$] A balloon filled with helium gas is tied by a light string to the floor of a car; the car is sealed so that the motion of the car does not cause air from outside to affect the balloon. If the car is traveling with constant speed along a circular path, in what direction will the balloon on the string lean towards?



Solution. The car is able to undergo circular motion due to the friction force acting in the centripetal direction. Hence, we might expect the balloon to lag behind the car, towards B (and leaning slightly towards C). However, because helium is less dense than air, the balloon will instead be pushed the opposite direction, mostly towards $\boxed{D}$. ■

Problem 7.5. [6] [$F = ma$] Four identical rods, each of mass m and length $2d$, are joined together to form a square. The square is then spun around its center, as shown in the figure, at an angular frequency of ω. What is the magnitude of the force that the joints between the rods (at the corners of the square) must bear?



Solution. This question can be a bit confusing because we must determine where we are analyzing the centripetal force. Consider one of the rods (i.e. the top rod). Let r be the distance from the center of the square to an arbitrary point on the top rod. Then the vertical component of the distance is d, and is constant invariant with r, so each point on the rod rotates with centripetal acceleration $a_c = \omega^2 d$.

Hence, the net vertical force on the rod is $m\omega^2 d$ downward – by symmetry the horizontal forces cancel. This force must be provided by the joints, and by symmetry again, we note the joints at the end of the rod form a $45°$ angle with the rod.

Thus, we have

$$2F_{\text{joint}}\cos(45^\circ) = m\omega^2 d \implies F_{\text{joint}} = \boxed{\frac{m\omega^2 d}{\sqrt{2}}}.$$

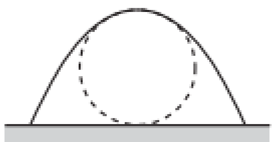
■

Problem 7.6 (Morin)**.** A projectile is fired with initial speed v and angle θ. What is the radius of curvature of the parabolic motion

(a) [5] at the top?

(b) [5] at the beginning?

(c) [5] At what angle should the projectile be fired so that the radius of curvature at the top equals half the maximum height, as shown in the below figure?



The radius of curvature is defined to be the radius of the circle that matches up with the path locally at a given point. It is also the same value of r used in Theorem 7.3.

Solution. Conveniently, the only force we have to deal with in this problem is gravity.

(a) The speed at the top of the motion is $v\cos(\theta)$. At the top, we see that the gravitational force is completely in the centripetal direction. Thus, we have

$$g = \frac{(v\cos(\theta))^2}{r} \implies r = \boxed{\frac{(v\cos(\theta))^2}{g}}.$$

(b) At the beginning, we see that the component of gravity perpendicular to the particle is $mg\cos(\theta)$. Thus, we have

$$g\cos(\theta) = \frac{v^2}{r} \implies r = \boxed{\frac{v^2}{g\cos(\theta)}}.$$

(c) Recall that the maximum height of a projectile is

$$h = \frac{v^2\sin^2\theta}{2g}.$$

Thus, we wish to solve

$$\frac{(v\cos(\theta))^2}{g} = \frac{v^2\sin^2\theta}{4g},$$

which occurs at $\theta = \boxed{\arctan(2)}$.

■

Problem 7.7 ($F = ma$)**.** [6] A unicyclist goes around a circular track of radius 30 m at a (amazingly fast!) constant speed of 10 m/s. At what angle to the left (or right) of vertical must the unicyclist lean to avoid falling? Assume that the height of the unicyclist is much smaller than the radius of the track.

Solution. Note that the normal force will slant with the unicycle. Let the unicycle slant an angle θ from the vertical. Using force analysis, we have

$$\sum \vec{F}_y = N\cos(\theta) - mg = 0,$$

so $N = \frac{mg}{\cos(\theta)}$. We now sum the forces in the centripetal direction. We have

$$\sum \vec{F}_c = N\sin(\theta) = \frac{mv^2}{r},$$

so we can conclude $\theta = \arctan\left(\frac{v^2}{gr}\right)$. Substituting, we obtain our answer of $\boxed{18.4^\circ}$. ■

Problem 7.8 ($F = ma$)**.** [6] A car drives anticlockwise (counterclockwise) around a flat, circular curve at constant speed, so that the left, front wheel traces out a circular path of radius R = 9.60 m. If the width of the car is 1.74 m, what is the ratio of the angular velocity about its axle of the left, front wheel to that of the right, front wheel, of the car as it moves through the curve? Assume the wheels roll without slipping.

Solution. Note that the two wheels will have the same angular velocity ω, which means their linear speed is proportional to their distance from the center. Note that the left front wheel travels in a path of radius 9.6 m, and the right front wheel travels in a path of radius $9.60 + 1.74 = 11.34$ m. Taking the ratio, our answer is $\boxed{0.847}$. ■

Problem 7.9. [7] A semicircular dome of radius r is glued to the table. A particle is initially at rest at the top of the dome, and is given an infinitesimal kick and starts to slide down the dome. At angle θ from the vertical, the particle loses contact with the dome. What is θ?

Solution. Note that at an angle θ from the vertical, the centripetal force from gravity is $mg\cos(\theta)$. The dome can provide an outward normal force, but not inward. Thus, the particle loses contact when we have

$$\frac{mv^2}{r} = mg\cos(\theta).$$

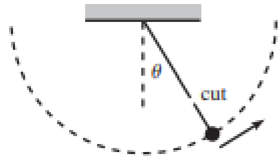
If the ball drops any further, the speed increases, but the right-hand side cannot increase. By conservation of energy, we have $v^2 = 2gr(1 - \cos(\theta))$, so substituting, we have

$$2mg(1 - \cos(\theta)) = mg\cos(\theta),$$

or $\theta = \boxed{\arccos(2/3)}$.

■

Problem 7.10 (Morin)**.** [8] A pendulum is held with its string horizontal and is then released. The mass swings down, and then on its way back up, the string is cut when it makes an angle θ with the vertical. What should θ be so that the mass travels the largest horizontal distance by the time it returns to the height it had when the string was cut?



Solution. Recall that the range of a projectile is

$$\frac{v^2\sin(2\theta)}{g}.$$

However, v depends on θ as well. Specifically, we have $v=\sqrt{2g\ell\cos(\theta)}$. Thus, we can rewrite the range as

$$\frac{v^2\sin(2\theta)}{g}=4\ell\cos(\theta)\sin(2\theta).$$

To optimize this, we take the derivative

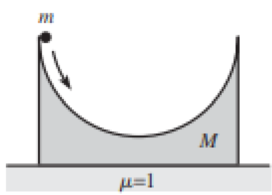
$$\frac{d}{d\theta}\cos(\theta)\sin(2\theta)=2\cos(\theta)\cos(2\theta)-\sin(\theta)\sin(2\theta)=0.$$

$$\tan(\theta)\tan(2\theta)=2\implies\frac{\tan^2(\theta)}{1-\tan^2(\theta)}=1$$

$$\implies\theta=\boxed{\arctan(\frac{\sqrt{2}}{2})}.$$

■

Problem 7.11 (Morin)**.** [10] A hemispherical bowl of mass M rests on a table. The inside surface of the bowl is frictionless, while the coefficient of friction between the bottom of the bowl and the table is $\mu=1$. A particle of mass m is released from rest at the top of the bowl and slides down into it. What is the largest value of m/M for which the bowl never slides on the table? **Hints:** 21



Solution. Consider an instant when the normal force N from the bowl on the particle is at an angle θ from the vertical. By conservation of energy, $v=\sqrt{2gr\cos(\theta)}$, where r is the radius of the bowl. We have that

$$\sum\vec{F}_c=N-mg\cos(\theta)=\frac{mv^2}{r}=2mg\cos(\theta),$$

or $N=3mg\cos(\theta)$. Note that the vertical component of N is $N\cos(\theta)$ and the horizontal component is $N\sin(\theta)$. Thus, analyzing the vertical forces, on the bowl, if the bowl does not slip, we have that

$$\sum\vec{F}_y=N_{\text{table}}-3mg\cos^2(\theta)-Mg=0.$$

$$\sum \vec{F}_x = F_f - 3mg\cos(\theta)\sin(\theta) = 0.$$

We have $f \le \mu N$, but since $\mu = 1$ this simplifies to $f \le N$. That is,

$$3mg\cos(\theta)\sin(\theta) \le 3mg\cos^2(\theta) + Mg.$$
$$\frac{m}{M} \le \frac{1}{3}\frac{1}{(\cos(\theta)\sin(\theta) - \cos^2(\theta))}.$$

We wish for this to be true for all θ, so we take the minimum of the right-hand side. We can optimize with calculus, but we'll instead give a neat algebraic solution for maximizing the denominator. Note that

$$\cos(\theta)\sin(\theta) - \cos^2\theta = \sin^2\theta + \cos(\theta)\sin(\theta) - 1.$$

Averaging both sides of the equation (since they are equal), we see this is equal to

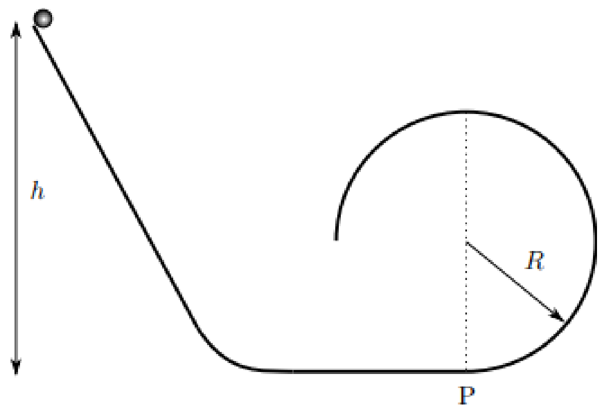
$$\frac{1}{2}\left(\sin(2\theta) - \cos(2\theta) - 1\right).$$

The maximum occurs at $\theta = 3\pi/8$, in which case $\sin(2\theta) - \cos(2\theta) = \sqrt{2}$. Thus, substituting, we have

$$\frac{m}{M} \le \boxed{\frac{2}{3}(\sqrt{2}+1)}.$$

■

Problem 7.12 (USAPhO, modified)**.** [10] A uniform solid spherical ball starts from rest on a loop-the-loop track. Assume the track is completely frictionless. However, it does not have enough speed to make it to the top of the loop. From what height h would the ball need to start in order to land at point P directly underneath the top of the loop? Express your answer in terms of R, the radius of the loop. Assume that the radius of the ball is very small compared to the radius of the loop.



Solution. Let P be the point $(0,0)$, and let the right and up directions be positive. Let the ball lose contact with the track at angle θ from the vertical measured from the center of the loop. The position of the ball at that time is given by

$$x = R\sin\theta, \quad y = R(1+\cos\theta).$$

Using conservation of energy, we have the speed when the ball leaves the loop is $v_0 = \sqrt{2g(h-y)}$, with horizontal velocity $-v_0\cos\theta$ and vertical velocity $v_0\sin\theta$. Hence, we can write equations for when the ball acts as a projectile:

$$y = -\frac{1}{2}gt^2 + v_0\sin(\theta)t.$$
$$x = -v_0\cos(\theta)t.$$

Note the horizontal distance traveled is $R\sin\theta$, so we have

$$t = \frac{R}{v_0}\tan(\theta).$$

Plugging in for t and y in our first equation, we get

$$\begin{aligned}
-R(1+\cos\theta) &= -\frac{1}{2}g\left(\frac{R}{v_0}\tan(\theta)\right)^2 + \frac{R\sin^2(\theta)}{\cos\theta}.\\
1+\cos\theta &= \frac{gR}{2v_0^2}\tan^2\theta - \frac{\sin^2(\theta)}{\cos(\theta)}.\\
1+\cos\theta &= \frac{\sin^2(\theta)}{\cos(\theta)}\left(\frac{gR}{2v_0^2\cos\theta} - 1\right)
\end{aligned}$$

The ball loses contact when the normal force is zero, which means

$$mg\cos\theta = \frac{mv_0^2}{R} \implies \frac{v_0^2}{gR} = \cos\theta.$$

Substituting, we get

$$1+\cos\theta = \frac{\sin^2(\theta)}{\cos\theta}\left(\frac{1}{2\cos^2(\theta)} - 1\right)$$

Note that $\cos\theta + 1 = 0$ is a solution, but this is clearly not meaningful. Hence, we can factor $\sin^2(\theta) = 1 - \cos^2(\theta) = (1+\cos(\theta))(1-\cos(\theta))$ and then divide by $1+\cos(\theta)$ on both sides. We then get

$$\begin{aligned}
1 &= \frac{1-\cos\theta}{\cos\theta}\left(\frac{1-2\cos^2(\theta)}{2\cos^2(\theta)}\right).\\
2\cos^3(\theta) &= 2\cos^3(\theta) - \cos^2(\theta) - \cos(\theta) + 1.
\end{aligned}$$

This simplifies to a quadratic, which we can then solve to get $\cos(\theta) = \frac{1}{2}$.

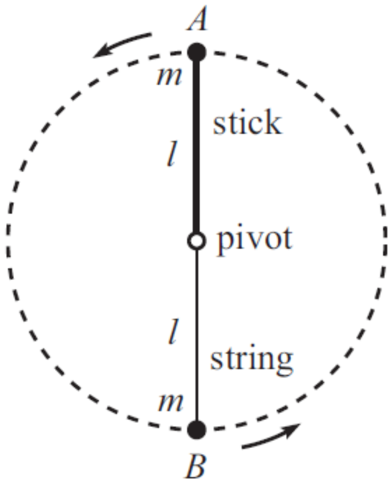
Thus, we have

$$h = \frac{v_0^2}{2g} + R(1+\cos\theta) = \frac{R\cos\theta}{2} + R(1+\cos\theta) = R(1+\frac{3}{2}\cos\theta) = \boxed{\frac{7}{4}R}.$$

■

7.4.1 Written Solutions

Problem 7.13 (Morin). [12] Ball A of mass m is connected to one end of a massless stick with length ℓ, the other end of which is connected to a pivot. The stick is held vertically above the pivot and is then given an infinitesimal kick (see figure). It swings down, and at the bottom of its motion it collides elastically with ball B, also of mass m, which hangs from a string with length ℓ and which is initially at rest. Ball B picks up whatever speed ball A had and then swings upward in circular motion.



(a) [7] At what point does ball B leave its circular motion and enter freefall projectile motion?

(b) [5] How high does ball B go in the resulting projectile motion? Assume the pivot is located at height $h = 0$.

Solution.

(a) Let θ be the angle from the vertical measured at the pivot. Then by conservation of energy, we have

$$v = \sqrt{2g\ell(1 - \cos\theta)}.$$

We can sum the forces in the centripetal direction to get

$$\sum \vec{F}_c = T + mg\cos\theta = \frac{mv^2}{\ell} = 2mg(1 - \cos\theta).$$

The string goes limp if $T = 0$, at which point we see that $\cos(\theta) = \frac{2}{3}$, or $\theta = \boxed{\arccos(\frac{2}{3})}$.

(b) Because the string is no longer "taut" it can no longer provide a tension force, and the ball will act exactly like a projectile. The max of a projectile is $\frac{v_0^2 \sin^2\theta}{2g}$. Recalling that $v_0 = \sqrt{2g\ell(1 - \cos\theta)}$, we see that

$$\Delta h = \frac{v_0^2 \sin^2\theta}{2g} = (\frac{5}{3})^2 \frac{\ell}{3} = \frac{5\ell}{27}.$$

However, this relative to the point of launch, which occurs at a height $\frac{2\ell}{3}$ above the pivot. Thus, the maximum height is $\boxed{\dfrac{23\ell}{27}}$.

■

8 Springs

8.1 The Spring Force

Definition 8.1. A **spring** is any form of elastic object that applies a restoring force (a force pointing back to equilibrium) when displaced. In most physics problems, springs are assumed to be ideal, meaning they follow *Hooke's Law*, which states that if the spring is displaced by x, the restoring force is given by

$$\vec{F} = -k\vec{x},$$

where k is a constant known as the *spring constant* which determines the stiffness of a spring. A spring's *rest length* is the length at which it is at equilibrium, relaxed.

Example 8.2. A mass of 5 kg is suspended from a ceiling using a massless spring with rest length 5 cm and a spring constant of 10 N/cm. What is the length of the spring when the system has come to rest?

Solution. Once the system has come to rest, the mass must have 0 net force on it, so we simply equate $mg - kx = 0$, so $x = \dfrac{mg}{k} = 4.9\,\text{cm}$, so the final length of the spring is $5 + 4.9 = 9.9$cm. ■

Remark. As we saw in this example, if the rest length of a spring is ℓ, then when we suspend a mass vertically from the spring (experiencing gravity), it will fall down a little bit before settling at equilibrium when the spring has length $\ell + \frac{mg}{k}$. The mass will be experiencing a net force $k(h - \ell) - mg = k(h - \ell - \frac{mg}{k})$, which, conveniently is just equivalent the force due to a spring of equilibrium length $\ell + \frac{mg}{k}$. This means that whenever you see a problem involving a mass hanging vertically from a spring you can simply treat it as being attached to a spring of the same spring constant with an adjusted length. Although in this course, we will primarily be dealing with gravity, this idea does, in fact, generalize to any constant force, not just gravity.

Now that we've covered the basics of how springs work, we can dive into some more interesting properties of them. What if we attach multiple springs to a block? How does it affect the motion and forces?

Consider the following setup. We have two springs of rest lengths ℓ_1 and ℓ_2 and spring constants k_1 and k_2 attached to the block. We wish to find the effective spring constant of the two springs, that is the spring constant of the spring that could replace the two springs and exhibit the exact same behavior.

Theorem 8.3. *Two springs of spring constants k_1 and k_2 in parallel are equivalent to a single spring of spring constant*

$$k_1 + k_2,$$

and the two springs in series are equivalent to a single spring of spring constant

$$\frac{1}{\frac{1}{k_1} + \frac{1}{k_2}}.$$

Proof. We will prove the case where the two springs have equal rest length ℓ. First we look at the configuration where the springs are attached in parallel, as shown below:

If the block is displaced a distance x from the rest length ℓ, then the springs will exert forces of $k_1 x$ and $k_2 x$, so the total force will be $(k_1 + k_2)x$. This means that the two springs in parallel are together equivalent to one spring rest length ℓ and equivalent spring constant $k_{equiv} = k_1 + k_2$.

Now what if they have different rest lengths ℓ_1 and ℓ_2? This case is slightly more complicated but still the same idea. The force due to the two springs is

$$\begin{aligned} F &= k_1(x - \ell_1) + k_2(x - \ell_2) \\ &= (k_1 + k_2)x - (k_1\ell_1 + k_2\ell_2) \end{aligned}$$

We want to get this to be an equation of the form $F = k_{equiv}(x - \ell_{equiv})$. Since the coefficient of x is $(k_1 + k_2)$, evidently we still have $k_{equiv} = k_1 + k_2$ for springs in parallel even when they have different rest lengths. Factoring out k_{equiv} we get

$$F = k_1 + k_2 \left(x - \frac{k_1\ell_1 + k_2\ell_2}{k_1 + k_2} \right)$$

This means that

$$\ell_{equiv} = \frac{k_1\ell_1 + k_2\ell_2}{k_1 + k_2},$$

that is, the weighted average of ℓ_1 and ℓ_2 (weighted by the spring constants).

Next we tackle the case where the springs are attached in series, shown below.

Suppose the block is displaced a distance x from equilibrium, changing the length of the first spring by x_1 and the second spring by x_2. Only the second spring exerts a force on the block and it has magnitude $k_2 x_2$. The effective spring constant k_{equiv} is defined such that $k_2 x_2 = k_{equiv} x = k_{equiv}(x_1 + x_2)$. Consider the point where the two springs are connected. The two springs are pulling in opposite directions with forces $k_1 x_1$ and $k_2 x_2$. But assuming the joint between springs is massless, the net force on it must be 0 so $k_1 x_1 = k_2 x_2$. Then using this to write $x_1 = x_2 \frac{k_2}{k_1}$, we get $k_2 x_2 = k_{equiv}\, x_2(\frac{k_2}{k_1} + 1)$ so solving for k_{equiv} we get

$$k_{equiv} = \frac{k_1 k_2}{k_1 + k_2} = \frac{1}{\frac{1}{k_1} + \frac{1}{k_2}},$$

that is, half the harmonic mean of the two spring constants.

■

Now that we've covered springs in series and parallel, here's a simpler application of those general cases.

Example 8.4. A spring of rest length $\ell = 2$ m and spring constant $k = 5\,\mathrm{N/m}$ is cut in half. What are the rest lengths and spring constants of the new springs now?

Solution. The rest lengths of the two springs will be 1m. Let k' be the spring constant of each of of the smaller springs. Then $k = \frac{1}{\frac{1}{k'}+\frac{1}{k'}} = \frac{k'}{2}$. Thus the new spring constant is 10 N/m.

Alternatively, we consider when the original spring is stretched one meter. Then each half of the spring is stretched half the distance, but it exerts the same spring force, so a spring of half the length will have double the spring constant. ∎

Thus we have the following takeaway:

Concept. If a spring of length ℓ has spring constant k, then a spring of length ℓ/α made of the same material will have spring constant αk.

8.2 Elastic Potential Energy

If a spring with a mass attached to it is displaced and let go, the mass will begin to move and thus will have some kinetic energy. This kinetic energy comes from the elastic potential energy stored in any compressed or stretched spring.

If the block is displaced from equilibrium by a distance A (we call A the **amplitude**) and released, it will initially accelerate towards the equilibrium position. But once it passes the equilibrium position, the force will now decelerate the block until it stops at a displacement of $-A$ and will accelerate it back in the opposite direction. Thus, we have the following:

- The block will oscillate between positions A and $-A$, where the block is at rest.
- The speed will reach a maximum at equilibrium.

This means we should expect the potential energy to be at a maximum when the mass is at its greatest displacement (A or $-A$), and zero when the displacement is 0.

Theorem 8.5. *The elastic potential energy of a spring displaced a distance x is*

$$U_{elastic} = \frac{1}{2}kx^2$$

Proof. Recall that we have

$$\Delta U_s = -W_s = -\int_0^x \vec{F} \cdot d\vec{x}.$$

Substituting $\vec{F} = -kx$, we have that $\Delta U_s = \int_0^x kx\, dx$. We define the elastic potential energy of an unstretched spring as zero, so we have

$$U_s = \frac{1}{2}kx^2.$$

A non-calculus explanation is the following: for every instance in which the spring is displaced a distance $d \leq x$, there is an instance in which the spring is displaced a distance $x - d$. Hence, on average, the force is $\frac{1}{2}(kd + k(x-d)) = \frac{1}{2}kx$. Since $W = \vec{F}_{\text{avg}} \cdot \vec{d}$, we have $\Delta U = \frac{1}{2}kx^2$. ■

Example 8.6. A 2 kg mass is attached to a spring with spring constant $k = 6.0$ N/m and displaced a distance $2m$ from equilibrium. When it is released what will its speed be a distance 1 m away from equilibrium?

Solution. By conservation of energy $\frac{1}{2}mv^2 + \frac{1}{2}kx_1^2 = \frac{1}{2}kx_0^2$ where x_1 and x_0 are 1 m and 2 m respectively. Solving for v we get

$$\begin{aligned} mv^2 &= k(a^2 - k^2) \\ v &= \sqrt{\frac{k}{m}(x_0^2 - x_1^2)} \\ &= 3 \text{ m/s.} \end{aligned}$$

■

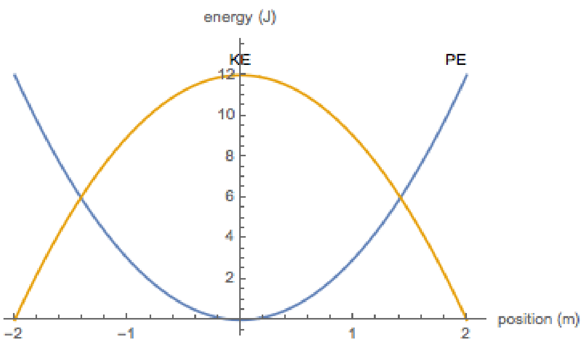
Example 8.7. For the above mass-spring situation, make a plot of the following:

(a) kinetic energy and potential energy vs. position

(b) velocity vs. position

Solution.

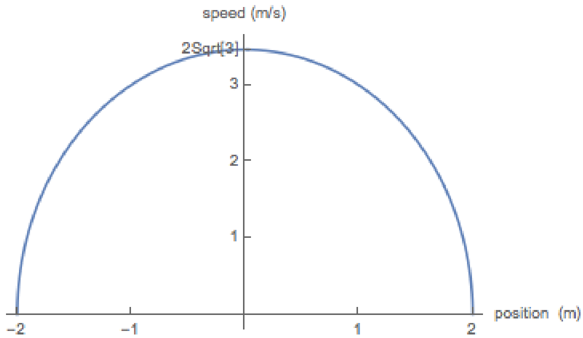
(a) The potential energy is $\frac{1}{2}kx^2 = 3x^2$ and so it reaches its maximum of 12 J at $x = \pm 2$m. The total energy is 12 J, so the kinetic energy is simply $12 - 3x^2$. We have the graph:



(b) Since the total energy is 12 J, we have

$$\begin{aligned} \frac{1}{2}(kx^2 + mv^2) &= 12 \\ \implies 3x^2 + v^2 &= 12 \end{aligned}$$

so the velocity vs position graph will be an ellipse with maximum velocity of $2\sqrt{3}$ m/s:

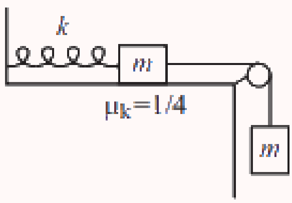


■

8.3 Applications of Elastic Potential Energy

In this section, we will deal with some relatively complicated examples, but which will be a good review of friction, energy, and what we've learned in dynamics in general.

Example 8.8 (Morin). Consider the system shown in the below figure, with two equal masses m and a spring with spring constant k. The coefficient of kinetic friction between the left mass and the table is $\mu = 1/4$, and the pulley is frictionless. The system is held with the spring at its relaxed length and then released.



(a) How far does the spring stretch before the masses come to rest?

(b) What is the minimum value of the coefficient of static friction for which the system remains at rest once it has stopped?

(c) If the string is then cut, what is the maximal compression of the spring during the resulting motion?

Solution.

(a) We use conservation of energy. Define the initial height of the suspended mass to be zero and let x be the distance the masses travel. Then the initial energy of the system is 0 and the final energy is $\frac{1}{2}kx^2 - mgx$. The only non-conservative force acting on the system is friction, which has a magnitude $\frac{1}{4}mg$ and acts over a distance x. Thus

$$\frac{1}{2}kx^2 - mgx = -\frac{1}{4}mgx$$

and solving for x we get $x = \boxed{\frac{3mg}{2k}}$.

(b) The mass experiences a force $mg + \mu_s mg$ to the right and a force kx to the left, so plugging in our value of x from part (a) we get the inequality

$$\frac{3mg}{2} - \mu_s mg \leq mg$$

so

$$\mu_s \geq \boxed{\frac{1}{2}}.$$

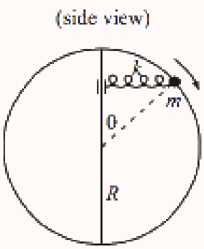
(c) We use conservation of energy. Let x' be the maximum distance that the spring is compressed from equilibrium. Immediately after the string is cut, the energy of the spring-block system is $\frac{1}{2}kx^2$ and the final energy will be $\frac{1}{2}kx'^2$. The work done by friction is $-\mu_k mg(x + x')$ so we get the equation

$$\frac{1}{2}k(x^2 - x'^2) = \mu_k mg(x + x')$$
$$\frac{1}{2}k(x - x') = \mu_k mg$$

and when we plug in our value of x from part (a) and solve for x', we get $x' = \boxed{\frac{mg}{k}}$.

■

Example 8.9 (Morin). A bead with mass m is free to slide on a fixed frictionless vertical hoop with radius R. A pole is located along the vertical diameter. A spring with spring constant k and relaxed length zero has one end attached to the mass, while the other end is free to slide without friction along the vertical pole. The bead is initially at rest at the top of the hoop and is given an infinitesimal kick. The situation at a later time is shown below. You can assume that the spring is always horizontal (this is a consequence of the facts that the spring is massless and the pole is frictionless).



Assume that before constructing this setup, you determine the value of k by noting that if you hold one end of the spring and let the bead on the other end hang at rest below, the length of the spring is R. This quickly tells you that $k = \frac{mg}{R}$. Given this value of k, at what angle θ (measured with respect to the vertical) is the normal force from the hoop on the bead equal to zero?

Solution. The force from the spring is $kR\sin(\theta)$ and the force from gravity is mg, so the components of these forces that point inwards are $kR\sin^2(\theta)$ and $mg\cos(\theta)$ respectively. If we let F_n be the normal force, then the net centripetal force felt by the bead is

$$\frac{mv^2}{R} = kR\sin^2(\theta) + mg\cos(\theta) + F_n \tag{8.1}$$

We want to solve for θ when the F_n is set to zero. In order to do this we will need to eliminate the velocity term, which we will do using conservation of energy. Defining the gravitational potential energy to be zero at the center of the hoop, we get that the initial energy of the system is mgR and the energy when the bead is at an angle θ is

$$E = \frac{1}{2}k(R\sin(\theta))^2 + mgR\cos(\theta) + \frac{1}{2}mv^2.$$

Then by conservation of energy, we can solve for mv^2 to get

$$mv^2 = 2mgR(1 - \cos(\theta) - k(R\sin(\theta)^2$$

Substituting into equation (8.1) and setting F_n to zero, we get

$$kR^2\sin^2(\theta) + mgR\cos(\theta) = 2mgR(1 - \cos(\theta) - k(R\sin(\theta))^2$$

Then we substitute in $k = \frac{mg}{R}$ and cancel mgR from all the terms to get

$$\begin{aligned}
\sin^2(\theta) + \cos(\theta) &= 2 - 2\cos(\theta) - \sin^2(\theta)\\
2\sin^2(\theta) + 3\cos(\theta) - 2 &= 0\\
2 - 2\cos^2(\theta) + 3\cos(\theta) - 2 &= 0\\
\cos(\theta)(3 - 2\cos(\theta)) &= 0
\end{aligned}$$

So we get $\cos(\theta) = \frac{3}{2}$, which is impossible, or $\cos(\theta) = 0$, so $\theta = \frac{\pi}{2}$ so the spring is horizontal when this occurs. ■

8.4 Elasticity

We now move onto the statics of deformable solids. We're already familiar with some solids that can deform (these are springs!). What we're going to do is figure out how to think of any sort of object as a type of spring.

We first introduce a series of ideas that will help reinforce our understanding of elasticity. But before that, what is elasticity? **Elasticity** is the ability of an object or material to resume its normal shape after being stretched or compressed; stretchiness. This definition goes back to when we were talking about deformable solids. In ideal scenarios all solids are elastic - meaning that they will resume back to their regular shape after being stretched. A non-elastic solid would be something like clay where clay would never go back to its original shape after being stretched.

Let us first consider a uniform density, elastic wire. If it is pulled with a force F, it will experience a change in length of Δl. If the wire was stiffer, Δl would be smaller and if it was less stiff, Δl would be greater. We define the stiffness of the wire as its **Young's modulus** and denote it by Y.

Now, the wire also has a cross sectional area of A. If A changes by a factor of γ we can then imagine γ^2 wires bundled together. This tells us that Δl will become smaller if A becomes bigger and Δl will become bigger if A becomes smaller.

Lastly, if the original length of the wire l increases, we can say that Δl will increase and vice versa. This is because we can imagine the wire as two separate wires that get stretched uniformly. This means that if both wires are stretched, the total change length will increase too.

All this information leads us to our final example for today.

Example 8.10. An elastic wire of uniform density, length l, cross-sectional area A, and Young's modulus Y is stretched by a force F. What is the stretch of the cable Δl?

Solution. Since $\Delta l \propto (F, l)$ and $\Delta l \propto (Y, A)^{-1}$, our final equation is

$$\Delta l = \frac{Fl}{YA}.$$

■

You might know that the force per unit area in fluids is given as pressure, in the case of elasticity, we define F/A as σ which is also known as the **stress** of the object. We also define $\Delta l/l$ as the relative stretch of the object. This is also called **strain** and is defined as ε.

Theorem 8.11. *The relation between tensile (compressive)stress σ and strain ε and is given by*

$$\frac{F}{A} = Y\frac{\Delta l}{l} \implies \sigma = Y\varepsilon$$

where Y is the Young's modulus of the deformable solid.

This formula might seem very similar to Hooke's Law:

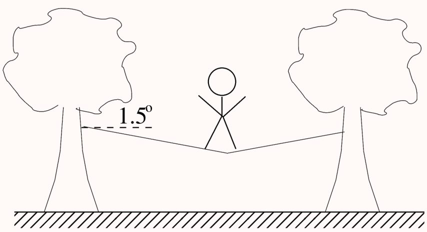
$$\sigma = Y\varepsilon$$
$$F = kx$$

and it is no mistake. Elastic solids act very similarly to springs!

Remark. There are also other types of strain such as the Bulk, Shear, and Lame Moduli. However, we won't be going into them that much in this handout.

With our newfound knowledge, let us tackle a problem from a recent $F = ma$ exam.

Example 8.12 ($F = ma$)**.** A massless cable of diameter 2.54 cm (1 inch) is tied horizontally between two trees 18.0 m apart. A tightrope walker stands at the center of the cable, giving it a tension of 7300 N. The cable stretches and makes an angle of 1.50° with the horizontal.



The Young's modulus is defined as the ratio of stress to strain, where stress is the force applied per unit area and strain is the fractional change in length $\Delta L/L$. The cable's Young's modulus is:

Solution. The change in length of the cable becomes

$$\frac{\Delta l}{l} = \frac{18/\cos 1.5° - 18}{18}.$$

The force is given as the tension T and we know the cross-sectional area A. Now plugging in our values to the formula gives us

$$\frac{F}{A} = Y\frac{\Delta l}{l} \implies Y = \frac{F}{A}\left(\frac{\Delta l}{l}\right)^{-1} = 4.2 \times 10^{10} \text{ N/m}.$$

Remark. Many problems define the Young's modulus for you since elasticity isn't actually in the IPhO syllabus.

■

Let us now look at a slightly harder example to finish this section.

Example 8.13 (Irodov)**.** A copper rod of length l is suspended from the ceiling by on of its ends. Find the elongation Δl of the rod due to its own weight.

Solution. We can take the average of the sums of all the stresses on the rod and treat this as the overall stress given to the rod. This is because the top of the rod is under the most stress while the bottom is under almost none. And since the stress doesn't increase exponentially, but rather linearly, the effective stress upon the cylinder at the center will be treated as the overall stress given upon the rod. Assuming the metal rod has a radius r and is uniform throughout, then the stress at the center will be given by (the force is given by the amount of weight the metal rod has to support from halfway)

$$\sigma = \frac{F}{A} = \frac{\pi r^2 \rho g \frac{l}{2}}{\pi r^2} = \frac{1}{2}\rho g l.$$

We know that the stress-strain relationship gives us

$$\epsilon = \frac{\Delta l}{l} = \frac{1}{Y}\sigma.$$

Since we know σ, replacing our expression for sigma into the relation gives us

$$\frac{\Delta l}{l} = \frac{1}{2Y}\rho g l$$

$$\Delta l = \frac{1}{2Y}\rho g l^2$$

■

Remark. Some other textbooks and sources denote Young's modulus with E.

8.5 Homework Problems and Solutions

Problem 8.1 ($F = ma$)**.** [4] Two springs of spring constants k_1 and k_2 are connected in series and stretched by a force F. What is the ratio of their potential energies, U_1/U_2?

(a) 1
(b) $\frac{k_1}{k_2}$
(c) $\frac{k_2}{k_1}$
(d) $\left(\frac{k_1}{k_2}\right)^2$
(e) $\left(\frac{k_2}{k_1}\right)^2$

Solution. The force acting on both springs is the same, so $\frac{x_1}{x_2} = \frac{k_2}{k_1}$. Hence, $\frac{U_1}{U_2} = \frac{k_1 x_1^2}{k_2 x_2^2} = \frac{k_2}{k_1}$ so the answer is (c). ■

Problem 8.2 (Morin)**.** [5] A mass m is attached to a spring with spring constant k and oscillates on a horizontal table with amplitude A. At an instant when the spring is stretched by $A/2$, a second mass m is dropped vertically onto the original mass and immediately sticks to it. If amplitude of the resulting motion is $A' = \alpha A$, then report α^2 in decimal?

Solution. Let v_0 be the speed of the mass right before the second mass lands on it and v' be the speed right after the collision. By conservation of energy we have

$$\frac{1}{2}mv_0^2 + \frac{1}{2}k\left(\frac{A}{2}\right)^2 = \frac{1}{2}kA^2$$

which gives $v_0 = \frac{A}{2}\sqrt{\frac{3k}{m}}$. During the collision, momentum is conserved so afterwards we have

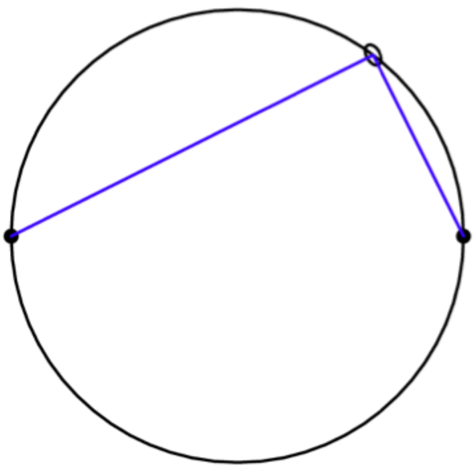
$$v' = \frac{v_0}{2} = \frac{A}{4}\sqrt{\frac{3k}{m}}.$$

Then we apply conservation of energy to figure out the new amplitude A':

$$\begin{aligned}\frac{1}{2}kA'^2 &= \frac{1}{2}(2m)v'^2 + \frac{1}{2}k\left(\frac{A}{2}\right)^2 \\ &= \frac{1}{2}kA^2\left(\frac{3}{8}+\frac{1}{4}\right)\end{aligned}$$

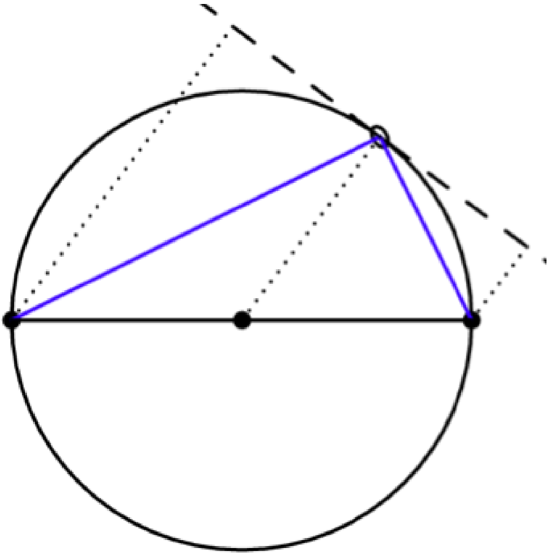
so $A' = \boxed{\sqrt{\frac{5}{8}}A}$. ■

Problem 8.3 (PhysicsWOOT)**.** [5] Suppose a ring is able to slide freely on a circle of radius r. Two zero-rest-length springs of spring constant k are connected to the circle 180 degrees apart. The springs are fixed where they are attached to the circle. The other side of each spring is attached to the ring. Find the acceleration of the ring.



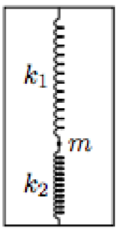
Solution. The ring does not move. Let the lengths of the springs be a and b. Since they have length 0, these are also the displacements. One way to think about this problem is as follows: Since the springs the potential energy is given by $\frac{1}{2}ka^2 + \frac{1}{2}kb^2$. Since the springs are attached on opposite ends of the circle, they will be at right angles to one another, so by the Pythagorean theorem $a^2 + b^2 = (2r)^2$ where r is the radius of the circle. This means that the potential energy will be the same no matter where the ring is on the circle so there will be no force.

We can also figure this out by looking at the forces. The force from a zero length spring is proportional to the length of the spring, so the force along the tangent to the circle is proportional to the projection of the length of the spring onto the tangent line. As we can see in the below diagram, these are the same.



Here we begin to see the very nice properties of zero length springs. ■

Problem 8.4 ($F = ma$). [5] A ball of negligible radius and mass m is connected to two ideal springs. Each spring has rest length ℓ_0. The springs are connected to the ball inside a box of height $2\ell_0$, and the ball is allowed to come to equilibrium, as shown. Under what condition is this equilibrium point stable with respect to small horizontal displacements?



(a) $k_1 > k_2$

(b) $k_2 > k_1$

(c) $k_1 - k_2 > \frac{mg}{\ell_0}$

(d) $\frac{k_1 k_2}{k_1 + k_2} > \frac{mg}{\ell_0}$

(e) $\frac{k_1 k_2}{k_1 - k_2} > \frac{mg}{\ell_0}$

AAPT Solution. Suppose we call z the height measured from the middle of the box and x the horizontal displacement, also measured from the middle of the box. When the ball is at equilibrium at some z_0 (which is negative), the springs obey

$$-k_1 z_0 - k_2 z_0 = mg$$

We can use this to find the equilibrium position,

$$z_0 = \frac{-mg}{k1 + k2}$$

.

Imagine a horizontal displacement of size x. If x is small, the tension in the top spring is not changed from the tension at equilibrium (to first order in x). So the restoring force is

$$F_1 = k_1 z_0 \sin\theta \approx k_1 z_0 \tan\theta = -k_1 z_0 \frac{x}{\ell_0 - z_0}$$

where θ is small is the angle the spring makes with the vertical. Similarly, the bottom spring exerts a force

$$F_2 = \frac{k_2 z_0 x}{\ell_0 + z_0}$$

pushing the ball away from equilibrium. For equilibrium to be stable, for positive x, we must have

$$F_1 + F_2 < 0$$

Putting in $F1$ and $F2$, we have

$$-\frac{k_1 x z_0}{\ell_0 - z_0} + \frac{k_2 x z_0}{\ell_0 + z_0} < 0$$

Dividing by x and z_0, which we recall is negative,

$$-\frac{k_1}{\ell_0 - z_0} + \frac{k_2}{\ell_0 + z_0}$$

Plugging in our expression for z_0 and rearranging this, we derive

$$k_1 - k_2 > \frac{mg}{\ell_0}$$

so the answer is (c). ■

Problem 8.5 (Physics StackExchange). [6] A copper rod of mass M, length ℓ, volume density ρ, and Young's modulus Y is suspended from the ceiling by on of its ends. A mass m is then attached to the bottom of the rod, and the rod will oscillate slightly. What is the effective spring constant of the rod?

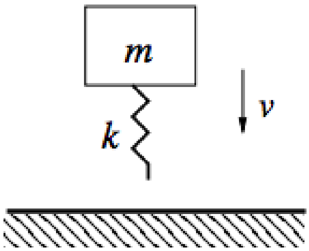
Solution. Note that $V = A\ell = \frac{M}{\rho}$. We have

$$\text{stress} = Y \cdot \text{strain}$$

$$F = \frac{YA}{\ell}\Delta\ell = \frac{YM}{\rho\ell^2}\Delta\ell$$

so the rod acts as a spring with effective spring constant $k = \boxed{\frac{YM}{\rho\ell^2}}$. ■

Problem 8.6 ($F = ma$). [6] A pogo stick is modeled as a massless spring of spring constant k attached to the bottom of a block of mass m. The pogo stick is dropped with the spring pointing downward and hits the ground with speed v. At the moment of the collision, the free end of the spring sticks permanently to the ground.



During the subsequent oscillations, the maximum speed of the block is

(A) v

(B) $v + \frac{2mg^2}{kv}$

(C) $v + \frac{mg^2}{kv}$

(D) $\sqrt{v^2 + \frac{2mg^2}{kv}}$

(E) $\sqrt{v^2 + \frac{mg^2}{kv}}$

Solution. We use conservation of energy. If we let ℓ be the rest length of the spring, the initial energy of the system is

$$E_i = mg\ell + \frac{1}{2}mv^2.$$

Then once the spring hits the ground, the block will oscillate and will have maximum speed at the equilibrium point. However since the block experiences gravity, the block's motion is equivalent to if it was attached to a spring of rest length $\ell - \frac{mg}{k}$. Letting v' be the maximum speed of the block, the energy when the block is at equilibrium position is

$$E_f = \frac{1}{2}mv_0^2 + mg(\ell - \frac{mg}{k} + \frac{1}{2}k(\frac{mg}{k})^2$$

Notice that we still have to include the last term, the elastic potential energy, since the effective equilibrium position is not the spring's true equilibrium length. Setting these two expressions equal, we get

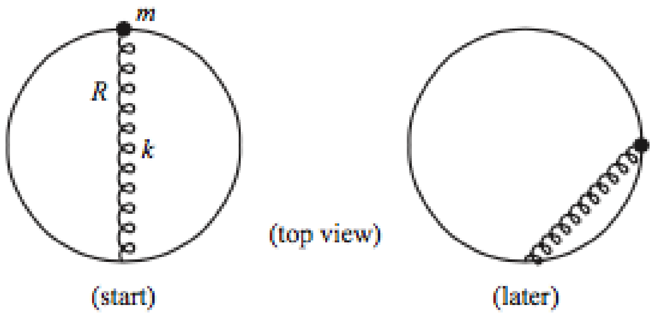
$$mg\ell + \frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + mg(\ell - \frac{mg}{k}) + \frac{1}{2}k(\frac{mg}{k})^2$$

and solving for v_0, we get

$$v_0 = \boxed{\sqrt{v^2 + \frac{mg^2}{k}}}$$

■

Problem 8.7 (Morin)**.** [6] A massless spring with spring constant k and relaxed length zero has one end attached to a given point on fixed frictionless horizontal hoop of radius R, while the other end is attached to a bead with mass m that is constrained to lie on the hoop. The spring is initially stretched across a diameter, with the bead at rest. The bead is then given an infinitesimal kick, and it gets pulled around the hoop by the spring, as shown below.



What is the normal force in the horizontal plane of the hoop (in other words, ignore gravity in this problem) that the hoop exerts on the bead at the moment the bead has gone a quarter of the way around the circle?

Solution (from Morin). When the bead has traveled a quarter of the way around the circle, the length of the spring has decreased from $2R$ to $\sqrt{2}R$, so conservation of energy gives

$$\frac{1}{2}k(2R)^2 = \frac{1}{2}k(\sqrt{2}R)^2 + \frac{1}{2}mv^2 \implies 2kR^2 = mv^2$$

After a quarter circle, the radial component of the spring force F_s is $F_s\cos(45°)$. The radial $F = ma$ equation is therefore

$$N + F_s\cos(45°) = \frac{mv^2}{R}$$

The spring force $F_s = (k\sqrt{2}R)$ so substituting in this and the value for v we get

$$N + (k\sqrt{2}R)\frac{1}{\sqrt{2}} = 2kR^2/R \implies \boxed{N = kR}.$$

■

Problem 8.8 ($F = ma$, 200 Puzzling Physics Problems)**.** [11] A uniform thin circular rubber band of mass M and spring constant k has an original radius R. Now it is tossed into the air. Assume it remains circular when stabilized in air and rotates at angular speed ω about its center uniformly.

(a) [6] What is the new radius of the rubber band?

(b) [5] Now repeat the same problem above but this time, the circular rubber band is now a circular band of copper with uniform density and Young's modulus Y.

Solution.

(a) The change in length of the rubber band is $\Delta x = 2\pi(R' - R)$. In order to calculate the force, consider a small sector of the rubber band of mass dm spanning an angle $d\theta$. The force pointing inwards exerted on this sector is, by the small angle approximation,

$$F_c = 2T\sin(d\theta/2) \approx T d\theta.$$

The centripetal acceleration is $\omega^2 R'$, so

$$\omega^2 R' dm = T d\theta$$
$$T = \omega^2 R' \frac{dm}{d\theta} = \omega^2 R' \frac{M}{2\pi}$$

Then applying $T = k\Delta x$ we get

$$\omega^2 R' \frac{M}{2\pi} = 2\pi k(R' - R)$$
$$R' = \boxed{\frac{4\pi^2 kR}{4\pi^2 k - \omega^2 M}}.$$

(b) A copper wire is equivalent to a spring with spring constant $\frac{YA}{\ell}$, where in this case, $\ell = 2\pi R$. Substituting this into the answer for problem 8.10, we have

$$R' = \frac{4\pi^2 \frac{YA}{2\pi R} R}{4\pi^2 \frac{YA}{2\pi R} - \omega^2 M}$$

Note that $M = \rho A \ell = 2\pi \rho A R$ since the band has uniform density. Plugging this in and simplifying, we get

$$R' = \frac{2\pi YA}{2\pi YA/R - \omega^2 2\pi \rho AR} = \boxed{\frac{YR}{Y - \omega^2 \rho R^2}}.$$

■

Problem 8.9. [7] A slightly conical wire of length L and end radii r_1 and r_2 is stretched by two forces F and F applied parallel to the length in opposite directions and normal to the end faces. If Y denotes Young's modulus, then what is the extension produced?

Solution. The total force on any horizontal cross-section of the wire is just F. So for a cross section of radius r, we have that the strain on this little piece of wire is

$$\epsilon(r) = \frac{F}{Y\pi r^2}$$

The total strain $\epsilon_{total} = \Delta L/L$ is the average of the strains $\epsilon(r)$ on each portion of the wire, and since the radius varies linearly down the wire, this just translates to

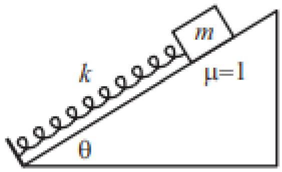
$$\begin{aligned}
\epsilon_{total} &= \frac{1}{r_2 - r_1} \int_{r_1}^{r_2} \epsilon(r) \mathrm{d}r \\
&= \frac{F}{Y} \frac{1}{r_2 - r_1} \int_{r_1}^{r_2} \frac{\mathrm{d}r}{\pi r^2} \\
&= \frac{F}{Y} \frac{1}{r_2 - r_1} \left(-\frac{1}{r}\right)\Bigg|_{r_1}^{r_2} \\
&= \frac{F}{Y} \frac{1}{r_2 - r_1} \left(-\frac{1}{r_2} + \frac{1}{r_1}\right) \\
&= \frac{F}{Y} \frac{1}{r_1 r_2}
\end{aligned}$$

Then since $\epsilon_{total} = \Delta L/L$ we get

$$\Delta L = \boxed{\frac{F}{Y}\frac{L}{r_1 r_2}}$$

■

Problem 8.10 (Morin). [14] A block with mass m lies on a plane inclined at angle θ. The coefficient of friction (both kinetic and static) between the block and the plane is $\mu = 1$. A massless spring with spring constant k is placed on the plane, with its lower end held fixed. The block is attached to the top end of the spring (see figure) and then moved down until the spring is compressed a distance ℓ relative to its relaxed length. The block is then released.



(a) [5] For what value of ℓ does the block rise back up exactly to its original position (where the spring is uncompressed)?

(b) [4] What is the condition on θ for which the block then starts to slide back down?

(c) [5] Assuming that the block does slide back down, how far down the plane does it go?

Solution.

(a) We use conservation of energy. The spring starts with an elastic potential energy $\frac{1}{2}k\ell^2$ which gets converted into gravitational potential energy and heat which is equal to the amount of work done by the frictional force. The magnitude of the potential energy is $mgh = mg\ell\sin(\theta)$ and the work done by friction is $\mu mg\cos(\theta)\cdot\ell = mg\ell\cos(\theta)$. Setting them equal we get

$$\frac{1}{2}k\ell^2 = mg\ell\cos(\theta) + mg\ell\sin(\theta)$$

$$\implies \ell = \boxed{\frac{2mg}{k}(\cos(\theta)+\sin(\theta))}$$

(b) At the moment, this spring is at equilibrium, so the only forces will be friction (at most $\mu mg\cos(\theta) = mg\cos(\theta)$) and gravity ($\mu mg\sin(\theta)$). The block will start to slide down if gravity is stronger than friction, that is

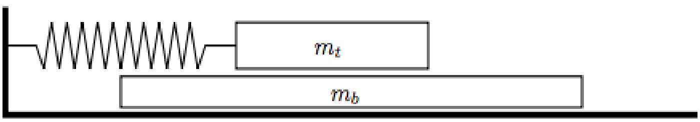
$$mg\sin(\theta) > mg\cos(\theta) \implies \tan(\theta) > 1 \implies \theta > \boxed{\frac{\pi}{4}}.$$

(c) This is similar to part (a). Let x be the distance it slides. In this case, the gravitational potential energy ($mgx\sin(\theta)$) is converted into elastic potential energy ($\frac{1}{2}kx^2$) and lost due to friction ($\mu mgx\cos(\theta)$), so we get

$$mgx\sin(\theta) = \frac{1}{2}kx^2 + \mu mgx\cos(\theta)$$
$$\implies x = \boxed{\frac{2mg}{k}(\sin(\theta) - \cos(\theta))}.$$

■

Problem 8.11 (USAPhO modified)**.** [14] A large block of mass m_b is located on a horizontal frictionless surface. A second block of mass m_t is located on top of the first block; the coefficient of friction (both static and kinetic) between the two blocks is given by μ. All surfaces are horizontal; all motion is effectively one dimensional. A spring with spring constant k is connected to the top block only; the spring obeys Hooke's Law equally in both extension and compression. Assume that the top block never falls off of the bottom block; you may assume that the bottom block is very, very long. The top block is moved a distance A away from the equilibrium position and then released from rest.



(a) [7] Depending on the value of A, the motion can be divided into two types: motion that experiences no frictional energy losses and motion that does. Find the value A_c that divides the two motion types. Write your answer in terms of any or all of μ, the acceleration of gravity g, the masses m_t and m_b, and the spring constant k.

(b) [7] Consider now the scenario $A \gg A_c$. In this scenario the amplitude of the oscillation of the top block as measured against the original equilibrium position will change with time. Determine the magnitude of the change in amplitude, ΔA, after one complete oscillation, as a function of any or all of A, μ, g, m_t, m_b, and k.

Solution.

(a) There will be no frictional energy losses if the bottom block moves with the top block at all times. This means at any instant in time, if the blocks have acceleration a, the frictional force $\mu m_t g$ is greater than $m_b a$, so

$$a \leq \frac{\mu m_t g}{m_b}$$

The maximal force exerted by the spring is kA so the maximal acceleration is

$$a_{max} = \frac{kA}{m_t + m_b}$$

Then,

$$a_{max} = \frac{kA}{m_t + m_b} \leq \frac{\mu m_t g}{m_b}$$

so solving for A we find that

$$A_c = \frac{\mu g m_t}{k}\left(\frac{m_t}{m_b} + 1\right)$$

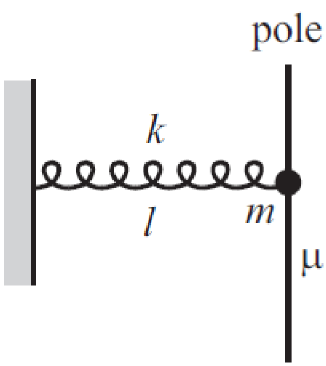
(b) Say that after half an oscillation, the new amplitude is A'. Since $A \gg A_c$, the bottom block is almost always lagging behind the top block, and when the speed bottom block does catch up to the top block, the top block will be near the amplitude of its oscillations and so it will be accelerating too much for the bottom block to stay caught up. Thus the frictional force is $\mu m_t g$ all the time, so the energy lost due to friction in half a cycle is $\mu m_t g(A + A')$. The change in energy over half a cycle is $\frac{1}{2}k(A'^2 - A^2)$. By conservation of energy,

$$\begin{aligned} -\mu m_t g(A + A') &= \frac{1}{2}k(A'^2 - A^2) \\ -\mu m_t g &= \frac{1}{2}k(A' - A) \\ A' - A &= -\frac{2\mu m_t g}{k}. \end{aligned}$$

Then the amplitude decrease over a full cycle is just twice this, or $\boxed{\dfrac{4\mu m_t g}{k}}$. ■

8.5.1 Written Solutions

Problem 8.12 (Morin). [17] A spring with spring constant k and relaxed length zero has one end attached to a wall and the other end attached to a ring with mass m. The ring is pulled to the side and is slipped over a vertical pole that is fixed at a distance ℓ from the wall, as shown in the figure. The coefficient of friction (both static and kinetic) between the ring and the pole is μ. The ring is held with the spring horizontal and is then released.



(a) [4] What is the normal force between the ring and the pole when the spring has fallen a distance y?

(b) [4] What is the cutoff value of μ, below which the ring will fall when it is released?

(c) [5] How far down the pole does the ring fall before bouncing back up? Assume μ is sufficiently small such that this occurs.

(d) [4] What is the cutoff value of μ, below which the ring will bounce back up at the bottom of its motion?

Solution.

(a) Suppose the spring has length s. Then the restoring force is ks and so the normal force is $ks\cos(\theta)$ while the vertical restoring force is $ks\sin(\theta)$ where θ is the angle of the spring relative to the horizontal.

$$F_n = ks\cos(\theta) = ks\frac{l}{k} = \boxed{kl}.$$

So the normal force is just kl no matter how far the ring has fallen, a nice property that results from the fact that the spring has rest length zero.

(b) When the spring is horizontal, the downward force is just mg and the upward force (due to friction) is $kl\mu$, so we have

$$kl\mu = mg \implies \mu = \boxed{\frac{mg}{kl}}.$$

(c) Say the ring falls a height y. The kinetic energy is zero at the start and end of the motion, so gravitational potential energy mgy is converted into elastic potential energy and lost due to work done by friction. Since, the normal force is kl throughout the motion, the work done by friction is just μkly. The initial elastic potential energy is $\frac{1}{2}kl^2$ and the final elastic potential energy is $\frac{1}{2}k(l^2+y^2)$, so conservation of energy gives the equation

$$mgy + \frac{1}{2}kl^2 = \frac{1}{2}k(l^2+y^2) + \mu kly$$
$$\frac{1}{2}ky^2 = mgy - \mu kly$$

so $y = \boxed{\frac{2}{k}(mg - \mu kl)}$.

(d) In order for the ring to bounce upwards, the vertical force due to the spring must be strong enough to counteract the gravitational force, and the frictional force pointing downward. As in part (a), the vertical force due to the spring is $ks\sin(\theta) = ksy/s = ky$ where θ is the angle of the spring relative to the horizontal. The frictional force is just μkl. Thus our equation is

$$ky > \mu kl + mg$$

Substituting in our value of (y) from part (c), we have

$$\mu kl < 2(mg - \mu kl) - mg$$
$$3\mu kl < mg$$
$$\mu < \boxed{\frac{mg}{3kl}}.$$

■

III

Rotational Motion

9 Rotational Kinematics

So far, other than dealing with circular motion, we have only dealt with translational motion. Today, we will start dealing with rotational motion.

9.1 Vector Cross-Product

Many of you will be quite familiar with the vector cross-product. However, we will review it here, as it is prominent throughout rotational mechanics. Recall the vector dot product:

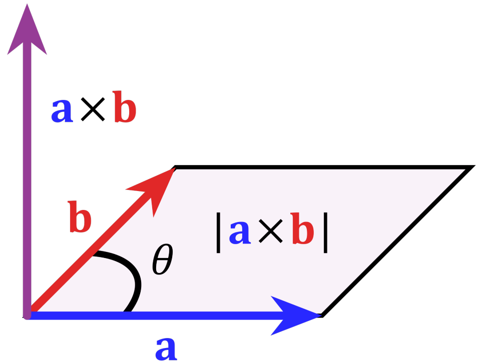
$$\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos\theta.$$

The **cross product** $\vec{u} \times \vec{v}$ is somewhat similar, and is given by

$$|\vec{u} \times \vec{v}| = |\vec{u}||\vec{v}| \sin\theta,$$

where θ is the angle between the two vectors. However, also note the extra absolute value: *unlike the dot product, the cross-product operation takes two vectors, and outputs another vector. This vector is also always perpendicular to both of the original vectors, as shown in the below diagram.* Oftentimes we will care only about the magnitude of the cross product, but in some cases the direction is extremely important, so it's something you should become familiar with doing.

The cross product also has magnitude equal to the area of the parallelogram that the two vectors span. That is, the vector $\vec{u} \times \vec{v}$ has magnitude equal to the area of the parallelogram below:



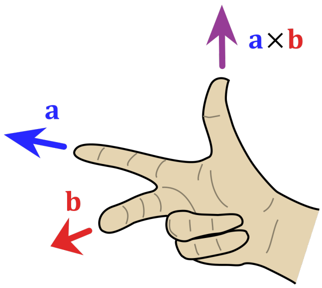
Concept. Recall that we can consider the magnitude of the dot product $\vec{u} \cdot \vec{v}$ as the product of a vector times the parallel component of another vector, or

$$|\vec{u} \cdot \vec{v}| = |\vec{u}_{\|}\vec{v}| = |\vec{u}\vec{v}_{\|}|$$

Similarly, the magnitude of the cross product of $\vec{u} \times \vec{v}$ can be considered as the product of a vector times the perpendicular component of another vector

$$|\vec{u} \times \vec{v}| = |\vec{u}_\perp \vec{v}| = |\vec{u}\vec{v}_\perp|$$

We know that the direction of $\vec{u} \times \vec{v}$ is perpendicular to both $\vec{u}$ and $\vec{v}$, but from this there are still two possible directions. The rule which we use to obtain the correct direction is called the **right hand rule**.



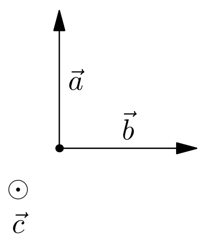
As the diagram shows, point your index finger in the direction of the first vector, your middle finger in the direction of the second. Then point your thumb straight up, and that should be the direction of the cross product. This is the right hand rule we like to use, however, there are several variants, and you can see them here, see which one works best for your hand. *Always make sure to use your right hand, and to not break your hand trying to use the right hand rule.*

Here are a couple facts that may be useful about the cross product.

1. The cross product is anticommutative, meaning $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$.
2. The cross product is distributive with addition, $\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$.
3. Scalar multiplication works, $(k\vec{u}) \times \vec{v} = \vec{u} \times (k\vec{v}) = k(\vec{u} \times \vec{v})$.

You may have noticed that the cross product gives 3D vectors, but the page is only 2D! To denote the direction of vectors in 3D, we say that a vector goes into the page or out of the page. If a 3D vector points into the page, it points away from reader. This is denoted by a $\otimes$. If it points out of the page, it points toward the reader, which is denoted by $\odot$. For example, the (direction of) the cross product of the two vectors shown here would be represented with $\odot$.

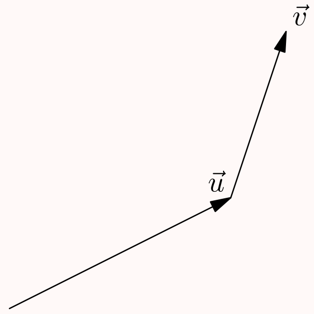
As an example, all of the vectors in the following diagram are perpendicular to each other. Can you see why?



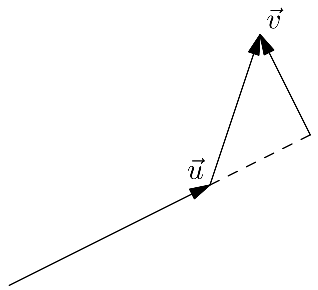
These symbols can be thought of kind of like an arrow. When the arrow points away from you, you see the fletching, hence the x. When the arrow points towards you, you see the point of the arrow, hence the dot.

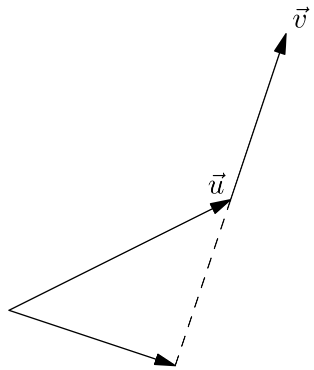
Now we look at a couple of simple examples, before we get on to the physics.

Example 9.1. Show, visually, the lengths that need to be multiplied to find the magnitude of $u \times v$ in the figure (this is a visual trick that's helpful in competitions). In what direction does the cross product point? Note the figure is 2D and in the plane of the paper, so indicate out of the page or into the page.



Solution. Here are the two ways visually that you can see the lengths being multiplied in the cross product. Both are perfectly fine!





The lengths are the new arrows that you see on the diagram. Can you notice how they're perpendicular to the other vector? This is a quick trick when you're looking at cross products in the future.

As for the direction, it is probably easier if you move the vector $\vec{v}$ back to the origin. Then applying the right hand rule, you should find that your thumb sticks out of the page. We will see the cross-product only briefly today, but it will appear frequently in angular dynamics.

■

9.2 Rotational Vectors

So what is rotation in the first place? Here we will deal usually with rotations of rigid bodies.

Definition 9.2. A **rigid body** is an object, which when rotated, all parts of the body move through the same angle.

You should be familiar with the idea of rotation from your everyday life. Rotation is anything that rotates really. You may not see it to often, but a lot of times the problems we see in olympiads will involve not something rotating about a fixed point (which is simple and boring). but things moving and rotating (think of a frisbee). Now that we've introduced this idea, let's get into some more rigorous concepts.

As we saw with circular motion, the direction of the velocity vector v was constantly changing. How can we describe uniform circular motion with a constant vector?

The answer is that we can position a vector, called the angular velocity ($\vec{\omega}$) along the axis of rotation, where the magnitude and direction of this vector indicate the direction of rotation.

The sign and of the vector are determined by the right hand rule. That is, using your right hand, curl your fingers in the direction of rotation. The direction in which your thumb points gives the direction of the angular momentum vector. For instance, for a counter-clockwise (CCW) rotation, if you let the fingers of your right hand curl in the direction of rotation, the thumb points out of the page.

Remark. By convention, the angular velocity is positive for a CCW rotation, and negative for a clockwise(CW) rotation. Additionally, it is imperative that the right hand is used; using the left hand will give the exact opposite sign.

We will now introduce several vectors that we will see replace translational position, velocity, and acceleration in rotation problems:

1. The **angular position** vector, $\vec{\theta}$, gives the angle between an object, and a specific reference line (typically the positive $\vec{x}$-axis). From geometry, if we let s denote arc length, then $s = r\theta$, where r is the radius of rotation. In other words,

$$\vec{\theta} = \frac{\vec{s}}{r}.$$

Note that we almost always use radians to measure angles because it is dimensionless and much easier to work with. The above relationship only holds if $\vec{\theta}$ is measured in radians. The next few also follow the same convention.

2. The **angular velocity** vector, $\vec{\omega}$, is the rate of change of angular position. That is,

$$\vec{\omega} = \frac{d\vec{\theta}}{dt}.$$

$\vec{\omega}$ also satisfies the equation

$$\vec{v}_T = \vec{\omega} \times \vec{r},$$

where v_T is the translational velocity.

Remark. $\vec{\omega}$ is also the same quantity referred to by the term **angular frequency**, and in uniform circular motion satisfies

$$\vec{\omega} = 2\pi f = \frac{2\pi}{T},$$

where f is the frequency of rotation, and T is the period of rotation.

3. Just as we have linear acceleration, we have **angular acceleration**, $\vec{\alpha}$, defined by

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt}.$$

If we let $\vec{a}_T$ denote tangential acceleration, then angular acceleration is given by

$$\vec{\alpha} = \frac{\vec{a}_T}{r}.$$

These vectors are analogous to vectors in regular kinematics. We have

Kinematics	Rotational
$\vec{x}$	$\vec{\theta}$
$\vec{v}$	$\vec{\omega}$
$\vec{a}$	$\vec{\alpha}$
t	t

These will be especially helpful in the next section, where you will see that it can be helpful for memorization.

The concept of expressing rotation with vectors can be difficult to get used to. To be fair, angular displacement is not a proper "vector." If we perform a series of 90 degrees rotations about the x, y, and z axes, the result is in fact dependent on the order of these operations, which is not true for a true vector quantity. *For simplicity, we will only examine rotation along a fixed axis, in which case these quantities can be treated as vectors.*

Example 9.3. If a object, say a stick, is rotating at angular velocity ω about its fixed end point, show that it is equivalent to (instantaneously) a stick moving with some center of mass velocity v (perpendicular to the stick) and rotating with some angular velocity ω'. Also, find the velocity v and ω' in terms of ω if the length of the stick is l.

Solution. If you think about the center of mass of the stick, if the end is fixed, we can say the "tangential" velocity is given by $\omega\frac{l}{2}$. So we can say the velocity $v = \frac{\omega l}{2}$. With that in mind, we can find ω'. We do this by noting the end of the stick is stationary. Then we have $v - \omega'\frac{l}{2} = 0$. Thus, $\omega' = \omega$. ■

In general, a lot of the simple rotational kinematic problems do not involve much more than applying the techniques from kinematics, just with rotation.

9.3 Rotational Kinematics

The translational kinematics equations hold with their rotational analogs as well. *Just like with the translational kinematics equations, these equations only hold if $\vec{\alpha}$ is constant.* To derive these, just use the same techniques as with regular kinematics. We leave this as an excercise to you.

1. $\Delta\vec{\omega} = \vec{\alpha}t$
2. $\Delta\vec{\theta} = \vec{\omega}_i t + \frac{1}{2}\vec{\alpha}t^2$
3. $\vec{\omega}_f^2 - \vec{\omega}_i^2 = 2\vec{\alpha}\Delta\vec{\theta}$
4. $\Delta\vec{\theta} = \frac{1}{2}(\omega_f + \omega_i)t$

These types of problems can be solved just like translational kinematics ones.

> **Example 9.4.** A drill bit is angularly accelerated from rest at $2\,\text{rad/s}^2$ until it reaches a maximum velocity of $6\,\text{rad/s}$. Then after 10 seconds of drilling, the drill bit slows down at $1\,\text{rad/s}^2$. If the drill bit drills 0.2 cm into the wall every revolution, how far does the drill bit drill?

Solution. We use the third equation and we find that the angular displacement is $\theta_1 = 9\,\text{rad}$ in the first stage. Then in the second stage, the displacement is simply $\theta_2 = 10 \cdot 6 = 60\,\text{rad}$. Lastly, the angular displacement in the last leg is $\theta_3 = \dfrac{-36}{-2} = 18\,\text{rad}$. So the total angular displacement is 87 rad. So the drill bit goes 2.77 cm into the wall (a bit unrealistic). ■

Let's get in one more exercise so that we get in the hang of it.

> **Example 9.5.** You are on a swing with radius 1.5 m, and as you reach the bottom, you are traveling with a horizontal velocity of 5 m/s. Assume (incorrectly) that you decelerate (angularly) at a constant rate of $2\,\text{rad/s}^2$ until you reach the top of your swing. What angle does the swing swing through?

Solution. First we must find the angular velocity of the swing at the bottom point. This is given by $\omega = \dfrac{v}{r} = 10/3\,\text{rad/s}$. Now we apply the 3rd equation from above. $0^2 - (10/3)^2 = 2(2)\Delta\theta$. So we have $\Delta\theta = 2.8 = 0.8\,\text{rad}$. ■

Recall that $\omega = \frac{d\theta}{dt} = v_T/R$. Using this definition, we see an object doesn't have to undergo circular motion to have an angular velocity ω about a point. Let's do an example:

> **Example 9.6.** Alice is sitting on a park bench. Bob, who is standing $d = 5$ meters away from Alice, throws a ball at an initial velocity $v = 5\sqrt{2}$ m/s and angle 45° from the vertical such that the ball lands on the ground right in front of Alice. What is the angular velocity of the ball with respect to Alice as a function of time for $0 \le t < 1$?

Solution. Let θ be the angle between Bob and the ball, measured at where Alice is sitting. Then we have

$$\theta = \arctan\left(\frac{\Delta y}{d - \Delta x}\right),$$

where where Δx and Δy are the vertical and horizontal displacements of the ball. Using the equations for projectile motion, we have

$$\theta = \arctan\left(\frac{-\frac{1}{2}gt^2 + \frac{vt}{\sqrt{2}}}{d - \frac{vt}{\sqrt{2}}}\right).$$

Substituting and simplifying, we have

$$\theta = \arctan\left(\frac{-t^2 + t}{1 - t}\right) = \arctan(t).$$

Thus, the angular velocity is given by

$$\omega = \frac{d\theta}{dt} = \frac{1}{1 + t^2}.$$

■

Remark. Typically, using calculus is overkill; for most problems, the relation $\omega = v_T/r$ is sufficient to find ω. However, this is still a useful tool to keep in mind for when v_T is hard to find (as shown by the above example).

9.4 Homework Problems and Solutions

Problem 9.1. [8] Your microwave starts from rest and begins accelerating an angular acceleration of 1 rad/s^2.

(a) [4] If its maximum angular speed is 2 rad/s, through what angle does the microwave turn while reaching maximum speed?

(b) [4] Describe the graph of angular displacement vs time and angular velocity vs time if you cook your hot pocket for 2 minutes.

Solution. This problem is just applying kinematic equations.

(a) We apply

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta.$$

So we solve, and

$$\Delta\theta = \frac{\omega_f^2 - \omega_i^2}{2\alpha} = \frac{4}{2} = \boxed{2\,\text{rad}}.$$

(b) You should have found for angular velocity vs time it was a line with positive slope for 2 seconds and then a flat line. As for angular displacement, you should have found part of a parabola that turns into a line with postive slope.

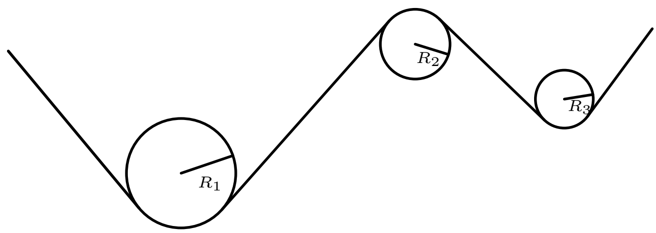
■

Problem 9.2. [12] In the diagram below, a rope is threaded around three pulleys of radii, $R_1 = 0.80$ m, $R_2 = 0.40$ m, and $R_3 = 0.20$ m. There is no slippage. For the following parts, give your answers in order of pulley 1, pulley 2, and pulley 3.

(a) [4]If the rope moves with constant speed 4.0 m/s, what is the angular velocity of pulley 1?

(b) [4]The rope accelerates from 5.0 m/s to 15.0 m/s in 5.0 s, what is the average angular velocity of pulley 2 for this time?

(c) [4] In the previous scenario, what is the average angular acceleration of pulley 3?



Solution.

(a) We know that the rim speed of a rotating object and it's angular velocity is related by $v = \omega R$. So for each we can calculate

$$\omega_1 = \frac{v}{R_1} = 5\,\mathrm{rad/s}$$

(b) A little bit of thought shows that the angular acceleration of the wheels will be constant, so we can simply calculate the final angular velocities and average, and this is also equivalent to just doing the calculations for $\bar{v} = 10\,\mathrm{m/s}$.

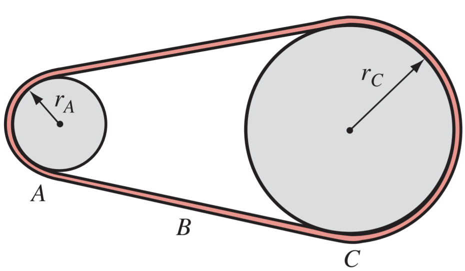
$$\bar{\omega}_2 = \frac{v_{avg}}{R_2} = 25\,\mathrm{rad/s}$$

(c) The average angular acceleration is simply $10/5 = 2\,\mathrm{m/s^2}$. We can relate angular acceleration and the acceleration by $\alpha = a/R$, so

$$\alpha_3 = \frac{a}{R_3} = 10\,\mathrm{rad/s^2}.$$

■

Problem 9.3 (HRK)**.** [6] Wheel A of radius $r_A = 10.0\,\mathrm{cm}$ is coupled by a belt B to wheel C of radius $r_C = 25.0\,\mathrm{cm}$ as shown. Wheel A increases its angular speed from rest at a uniform rate of $1.60\,\mathrm{rad/s^2}$. Determine the time for wheel C to reach a rotational speed of $100\,\mathrm{rev/min}$, assuming the belt does not slip. **Hint:** How are the linear speeds of the rims of the wheels related if the belt doesn't slip?



Solution. Let us first relate the angular speeds of A and C. We have the linear acceleration of the rims of the wheels are the same, so we can equate them as

$$\alpha_A r_A = \alpha_C r_C.$$

So we can calculate $\alpha_C = \dfrac{\alpha_A r_A}{r_C} = 0.64\,\text{rad/s}^2$. Now we just need to convert $100\,\text{rev/min}$ to rad/s and we have $100\,\text{rev/min} = 10.5\,\text{rad/s}$. So We divide to get $t = 10.5/0.64 = \boxed{16.4\,\text{s}}$. ■

Problem 9.4. [7] An object travels in a circle of radius r with angular velocity and angular acceleration given by

$$\omega = \frac{1}{t}$$

What is the magnitude of the translational acceleration of the object as a function of time?

Solution. We have to account for both centripetal acceleration and tangential acceleration. We see that

$$\alpha = \frac{d\omega}{dt} = -\frac{1}{t^2}.$$

Tangential acceleration is simply given by

$$a = \alpha r = -\frac{r}{t^2}.$$

Centripetal acceleration is given by

$$a_c = \omega^2 r = \frac{r}{t^2},$$

pointing toward the center. Thus, using 45-45-90 triangles, the magnitude of the total translational acceleration is given by $\boxed{\dfrac{r}{t^2}\sqrt{2}}$. ■

Problem 9.5 (HRK)**.** [8] A threaded rod with 12.0 turns/cm and diameter 1.18 cm is mounted horizontally. A bar with a threaded hole to match the rod is screwed onto the rod as shown. The bar spins at 237 rev/min. How long will it take for the bar to move 1.50 cm along the rod?

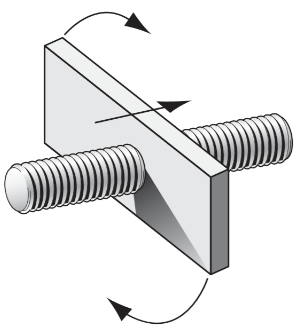


FIGURE 8-17. Exercise 24.

Solution. When one sees this problem, one could be quite confused, but using dimensional analysis to aid your intuition is quite helpful here. Notice that the 12.0 turns/cm is the same thing as 1/12.0 cm/turns. But we also know how many turns happen in a minute! That's just the angular speed!. So we have the rate at which the bar moves is 19.75 cm/min. Then we have $t = 1.50/19.75 = 4.56$ sec after converting from minutes to seconds. ■

Problem 9.6. [9] Jet sits on a park bench. A plane flies in at a constant speed v and at a constant altitude h above Jet. The plane flies in a straight line that eventually passes directly above Jet. At time t, Jet and the plane are a distance d away. At this instant, what is the angular velocity of the plane about the axis through Jet and perpendicular to the path of the plane (never intersecting the path)? **Hints:** 12

Solution. Let θ denote the angle between the line from Jet to the plane. We know the tangential velocity of the plane is given by

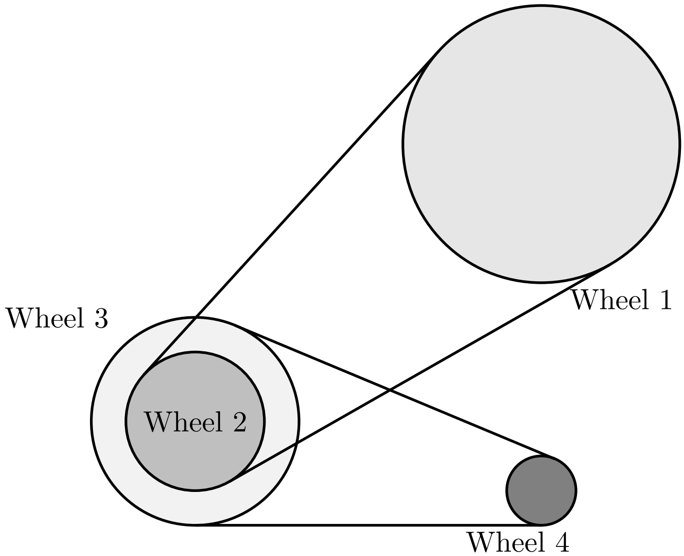
$$v = \omega d.$$

Thus, we just need to find the tangential velocity, the velocity perpendicular to the line between Jet and the plane. Drawing the diagram, we can see this component is $v\cos\theta$. So we can write,

$$\omega = \frac{v\cos\theta}{d}.$$

However, we need to answer without θ. So we can find $\cos\theta$ as h/d, so our answer is $\omega = \boxed{vh/d^2}$. ■

Problem 9.7. [9] The wheels in the diagram below have radii $R_1 = .80$ m, $R_2 = .40$ m, $R_3 = .60$ m, and $R_4 = .20$ m. If wheel 1 starts from rest and is given a constant angular acceleration of 6 rad/s^2, then find the angular displacement of wheel 4 after 9 seconds.



Solution. Let's relate the angular accelerations of Wheel 1 and wheel 4. First we know that the tangential acceleration (of the rope) for wheel 1 and wheel 2 are the same, meaning,

$$\alpha_1 R_1 = \alpha_2 R_2.$$

We also know the angular accelerations of wheel 3 and wheel 2 will be the same since they are attached together. Lastly, we have similarly,

$$\alpha_3 R_3 = \alpha_4 R_4.$$

So we can solve and we have

$$\alpha_4 = \frac{R_1 \cdot R_3}{R_2 \cdot R_4} \alpha_1.$$

So we can plug in our values, and we have $\alpha_4 = 6 \cdot 6 = 36\,\mathrm{rad/s^2}$. So we have the angular displacement is

$$\Delta\theta = \frac{1}{2}\alpha t^2 = 1458\,\mathrm{rad}.$$

■

Problem 9.8 (PhysicsWOOT)**.** [10] A circular disk lies between two horizontal plates. The bottom plate moves to the left at speed v and the top plate moves to the right at speed $2v$. Find all points on the disk moving with speed v.

Solution. First we find the speed of the speed of the COM of the disk. This is given by the average of the top and bottom plates, $v/2$. We also know, since relative to the COM the plates are moving at $3v/2$, the angular velocity of the disk is $3v/2r$.

Now to find the points where the speed is v, we can place the disk on the coordinate plane with the origin at the center of mass. Then we can write the velocity as

$$\langle v/2, 0\rangle + \omega\langle y, -x\rangle = \langle y \cdot 3v/2r + v/2, -x \cdot 3v/2r\rangle.$$

We want the magnitude of this vector to be v, and we can do that by dividing out the v and then setting the magnitude equal to 1. So we want to find the (x, y) that satisfy

$$(3y/2r + 1/2)^2 + (-3x/2r)^2 = 1.$$

We can multiply by $4r^2/9$, and we have

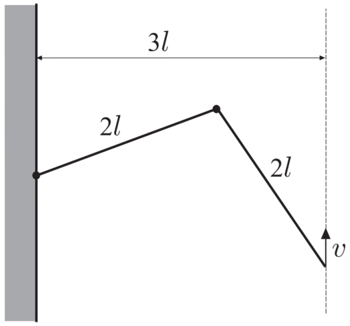
$$(y + r/3) + (-x)^2 = \left(\frac{2r}{3}\right)^2.$$

So the points moving with speed v make up a circle with radius $2r/3$ centered $r/3$ down from the center. This checks out since the bottom point, which moves left with speed v lies on this circle. ■

Problem 9.9 (Kalda)**.** [14] A hinged structure consists of two links of length $2l$. One of its ends is attached to a wall, the other is moving at distance $3l$ from the wall at constant vertical velocity v_0. Find the acceleration of the hinge connecting the links when

(a) [6] the link closer to the wall is horizontal.

(b) [8] the velocity of the connection point is zero.



Solution.

(a) First notice that the velocities of the hinge and endpoint are equal when the link is horizontal. We can see this because the velocity of the hinge must be vertical, and since the length of the rod is constant, it must match the vertical velocity of the end point (v_0).

Then the centripetal acceleration of the hinge is $\frac{v_0^2}{2l}$. Now consider the direction of the total acceleration. About the endpoint on the right, there can be no centripetal acceleration since at this point the rod is moving straight up with no spinning. So the acceleration must point perpendicularly to the right rod, and using some geometry,

$$a = \frac{v_0^2/2l}{\cos 30^\circ} = \frac{v_0^2}{\sqrt{3}l}.$$

(b) We can use a really sneaky trick for this part. Consider a reference frame moving upwards at v. Forces don't change in inertial reference frames, so this problem reduces to the first part, and we get $a = \frac{v_0^2}{\sqrt{3}l}$.

■

9.4.1 Written Solutions

Problem 9.10 (USAPhO)**.** [17] An object of mass m is sitting at the northernmost edge of a stationary merry-go-round of radius R. The merry-go-round begins rotating clockwise (as seen from above) with constant angular acceleration of α. The coefficient of static friction between the object and the merry-go-round is μ_s.

(a) [8] Derive an expression for the magnitude of the object's velocity at the instant when it slides off the merry-go-round in terms of μ_s, R, α and any necessary fundamental constants.

(b) [9] For this problem assume that $\mu_s = 0.5$, $\alpha = 0.2\ \mathrm{rad/s^2}$, and $R = 4$ m. At what angle, as measured clockwise from north, is the direction of the object's velocity at the instant when it slides off the merry-go-round? Report your answer to the nearest degree in the range 0 to 360°.

Solution. Though this is a USAPhO problem, don't be afraid of trying to do it; it is on the easier side.

(a) Now lets consider the moment just before the object slips. There will have to be a centripetal acceleration since it is in circular motion. But, there also has to be a tangent acceleration since the merry go round is accelerating angularly as well, and the frictional force has to account for both of these. We have the centripetal component of the force must have magnitude

$$F_c = \frac{mv^2}{R}.$$

As for the tangential component, we know that $a = \alpha R$, so

$$F_\perp = m\alpha R.$$

So the total force in the plane of the merry-go-round is

$$F = \sqrt{(mv^2/R)^2 + (m\alpha R)^2},$$

since the two forces are perpendicular. The maximum force in this plane is provided solely by the frictional force since all other forces are perpendicular, so we require that

$$mg\mu_s = \sqrt{(mv^2/R)^2 + (m\alpha R)^2}.$$

Solving for v,

$$v = \sqrt[4]{g^2\mu_s^2 R^2 - \alpha^2 R^4}.$$

(b) We have the equation

$$\omega_f^2 = \omega_i^2 + 2\alpha\theta.$$

Here we are interested in θ as it almost directly gives us the direction of the velocity. We have $\omega_f^2 = v^2/R^2$, so we can find

$$\theta = \frac{v^2}{2R^2\alpha} = 173\,\text{deg}.$$

(Make sure to convert to degrees). Then to get the velocity, since it is tangential, we just add 90°, giving us $\boxed{263^\circ}$.

■

10 Angular Dynamics

Rotation is perhaps the most challenging aspect of all of mechanics, so this section may be particularly challenging. As a result, we will be spending 2 days on it (and feel free to spend more time on the weekend).

10.1 Introduction

We discussed a number of different systems so far. Just to name a few:

1. Point Particles
2. Complex Systems (with many point particles)
3. Deformable bodies
4. Rigid Bodies

In this section we'll primarily focus on the motion of the latter kind of system, rigid bodies. This is because for rigid bodies, describing the rotation of the object is quite simple: the object merely translates about it's center of mass and rotates about its center of mass. This is not true in general, for other kinds of systems.

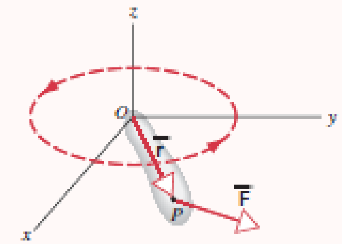
For example, consider two particles moving in opposite directions. It is definitely not true that the particles are rotating about the center of mass of the particles. However, consider some kind of moving *rigid* body, say, a pancake object in the plane. Intuitively, it is *clear* that the object is merely rotating about it's center of mass. It is thus easy to describe the equations of motion in this case for this kind of object.

One thing to note however, is that rotational motion is really nothing new. We won't be introducing any new physical laws: everything we derive in this section is a consequence of Newton's laws. Make note of the connections between the things we've already studied and the new things we derive in this section. We'll start with a preliminary example to give you the general gist of how everything is related. Try following along and if possible, try solving the examples yourself before reading the solution(s).

10.2 A Preliminary Example

Just as we have defined rotational analogues for the kinematic quantities of position,velocity, and acceleration we will be defining angular analogues for dynamical quantities such as force and momentum in this section. We preface our discussion by considering the following system:

> **Example 10.1.** A particle of mass m is attached to a massless rigid rod that is free to rotate about the z-axis at a position $\vec{r}$ relative to the origin. A force $\vec{F}$ is applied to the particle. Find the angular acceleration of the particle.



Let's try analyzing this problem in terms of the dynamical quantities we already know first. The force $\vec{F}$ can be split into both a component parallel to the position vector $\vec{r}$: $\vec{F}_{\|}$ and $\vec{F}_{\perp}$. We can thus consider the central forces and the tangential forces separately. Because the rod is rigid and is fixed to rotate about the origin, it will always provide a necessary force so that the centripetal force

$$\sum F_c = F_{rod} - F_{\|} = \frac{mv^2}{r}$$

That is, the mass retains circular motion.

The perpendicular component on the other hand provides a tangential acceleration to the mass. By newton's law, we get

$$\sum F_T = F_{\perp} = ma_T$$

We now find the angular acceleration in terms of the given quantities. Substituting $a_T = r\alpha$ we then get

$$F_{\perp} = F \sin\theta = mr\alpha \implies \alpha = \frac{F \sin\theta}{r}$$

where θ is the angle $\vec{r}$ makes with $\vec{F}$.

Indeed, this is one way to do the problem, but it seems it would be more natural to try finding the angular quantities directly. We thus define a quantity called a torque which will serve as our angular analogue for force.

Definition 10.2. Consider a force $\vec{F}$ that acts on a body at a position $\vec{r}$ relative to some chosen origin. We define the **torque**, $\vec{\tau}$ that the force acts upon the body relative to the origin to be

$$\vec{\tau} = \vec{r} \times \vec{F}$$

The direction of the resulting torque can of course be found with the right hand rule.

Why this mysterious definition? We would hope that just as the force is proportional to the acceleration of an object with $\vec{F} = m\vec{a}$, the rotational analogue force would be proportional to the angular acceleration of the object. Indeed, we find that it is.

Before we can proceed any further analyzing the rotation of the mass, we must first define a quantity called the *rotational inertia* of the mass relative to the origin. Just as we defined rotational analogs for position, velocity, and acceleration, the *rotational inertia* will serve as a rotational analogue for mass in this section. Mass can be seen intuitively as a quantative measure of an object's "resistance to acceleration" upon the action of a given force. More intuitively, we write

$$m = \frac{\vec{F}}{\vec{a}}$$

which clears this relationship somewhat. We will see that the rotational inertia serves an analogous role for rotation.

Definition 10.3. We define the **rotational inertia** or moment of inertia, I, of a particle with mass m at a distance r about a given axis by

$$I = mr^2.$$

Again, why this mysterious definition? To see this, we return to the original problem.

Remark in this instance, we have

$$\sum \tau = rF \sin\theta = rF_\perp = mra_T$$

Substituting in $a_T = r\alpha$ and $I = mr^2$, we get that

$$\sum \tau = rF_\perp = mr^2\alpha$$

$$\sum \tau = I\alpha.$$

Indeed, just as the net force was proportional to the acceleration of the object, we see that the net torque acting upon the object is equal to the *angular acceleration* of the object. The direction of the angular acceleration can then be found with the right hand rule (Point your thumb towards the direction of the angular acceleration. Then curl your fingers around your thumb; the direction your fingers curl is the direction of the angular acceleration).

Indeed we see that using the torque equation above we can directly arrive at the angular acceleration without splitting the force $\vec{F}$ into components and substituting $a_T = r\alpha$. Thus, in some sense it is *easier* to use the rotational quantities: it is important to reiterate however that no new physical laws are involved and that rotation is merely an extension of our current knowledge rather than new information in and of itself.

10.3 Inertia

We'll now try to derive the general equations for the rotational motion of rigid bodies rather than just singular particles. To do this however, we must first introduce the quantity of rotational inertia for rigid bodies in general.

Just as how mass is intuitively an object's "resistance to linear acceleration," upon the action of a given *force* the rotational inertia can be seen as a quantitative measure of an object's "resistance to *angular* acceleration," upon the action of a given *torque*, the rotational analogue for force.

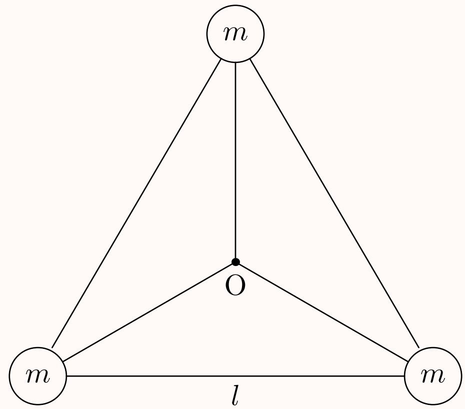
We have already introduced a definition for the rotational inertia for point particles. Let's now try defining the rotational inertia for rigid bodies that consist of discrete objects.

Definition 10.4 (Rotational Inertia is Additive)**.** Consider a collection of n particles that are attached in such a way that the overall system is rigid. Then, the rotational inertia I of the resulting system about the chosen axis is given by

$$I = \sum_{i=1}^{n} m_i r_i^2$$

where m_i is the mass of particle i and r_i is the distance of particle from the chosen axis.

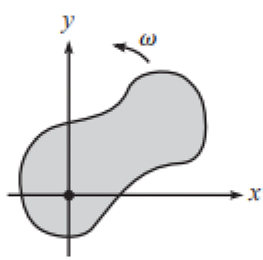
Example 10.5 (Classical)**.** Consider a system of three particles of mass m that are connected to each other by massless rigid rods of length l and to the origin which is located at the center of mass of the system. The three particles form an equilateral triangle. What is the rotational inertia of the system about an axis through the center of the triangle formed?



Solution. From the geometery of the problem we see that the distance from O to each of the masses is $\frac{l}{\sqrt{3}}$. Thus the rotational inertia of the system is $I = 3m(\frac{l}{\sqrt{3}}))^2 = ml^2$. ■

This is all well and good, but it can be cumbersome and practically impossible to calculate to sum the rotational inertia(s) of each particle in the system individually. It would be much easier if there was a way to sum the rotational inertias in a continuous manner. Fortunately, we have calculus to remedy this issue.

Let's consider some general rigid body in the x-y plane.



A pancake object in the x-y plane

We can imagine that our rigid object consists of a number of small mass elements with mass δm_n with each mass element being a distance r_n from the chosen axis. We then get from equation 10.4 that

$$I = \sum {r_n}^2 \delta m_n$$

Now if we assume our mass particles to be infinitesimally small (that is we take the limit $\delta m_n \to 0$), our sum becomes an integral in the usual way:

$$I = \lim_{\delta m_n \to 0} \sum {r_n}^2 \delta m_n = \int r^2 dm$$

We thus get the following result:

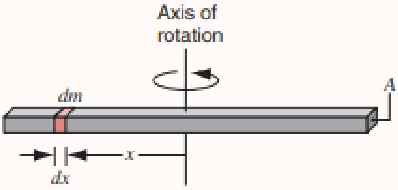
Theorem 10.6 (General Rotational Inertia Formula). *The rotational inertia of a rigid body about some axis is*

$$\int r^2 dm$$

where r is the distance from the infinitesimal mass elements dm.

We'll give a few examples to see how we can use this equation in practice. Make note of how we make calculus substitutions throughout as this is a fundamental skill:

Example 10.7. Find the rotational inertia of a thin, uniform rod of mass m and length l about its center of mass.



Solution. Let's consider an *arbitrary* element of mass dm a distance x away from the center. The mass of the element is equal to the linear mass density of the element times the length of the element. That is

$$dm = \lambda dx$$

where dx is the infinitesimal length of the element. Note that because the rod is uniform, we simply have $\lambda = m/l$. Summing these mass elements with equation 10.6, we thus get

$$I = \int r^2 dm = m/l \int x^2 dx$$

With $x = 0$ at the midpoint of the rod, the limits of integration are from $-l/2$ to $l/2$. We thus get

$$I = \frac{m}{l} \int_{-l/2}^{l/2} x^2 dx = \frac{m}{l} \frac{x^3}{3} \bigg|_{-l/2}^{l/2} = \boxed{\frac{1}{12} ml^2}$$

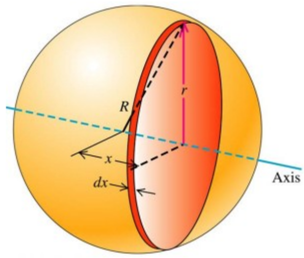
■

Now that you get the general gist of it, here's a few more examples. Try deriving the inertias first before looking at the solutions:

1. **Thin, circular Hoop**: Note $I = \int r^2 dm$ but since all points on the hoop lie the same distance r from the hoop, we can take out the r^2 so $I = r^2 \int dm = Mr^2$
2. **Plane or slab, about center:** Note that you can think of this as several rods stacked together, and by that thought process, you should arrive at $I = \frac{1}{12}Ma^2$. However, for a more rigorous calculation, we have

$$I = \int_0^b \frac{1}{12} \cdot \frac{M}{b} a^2 \, dx = \frac{1}{12} Ma^2.$$

3. **Thin spherical shell, about diameter:** We can consider the shell to be a series of thin rings like in the picture shown:



Let's consider the ring at angle θ relative to the axis. We have the radius of the hoop is $r = R\sin\theta$. The area of the ring is then $dA = 2\pi r(Rd\theta) = 2\pi(R\sin\theta)(Rd\theta) = 2\pi R^2 \sin\theta d\theta$. The mass of the ring is then just $\rho dA = \frac{M}{4\pi R^2}(2\pi R^2 \sin\theta d\theta) = M/2 \sin\theta d\theta$ We thus have that

$$I = \int r^2 dm = \int_0^\pi (R\sin\theta)^2 \frac{M}{2} \sin\theta d\theta$$

$$= \frac{MR^2}{2}[-\cos x + \frac{1}{3}cos^3 x\Big|_0^\pi] = \boxed{\frac{2}{3}MR^2}$$

4. **Solid sphere, about diameter:** We can consider a solid sphere to be made up of a ton of thin concentric shells. Each shell has mass has

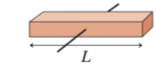
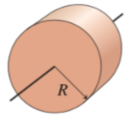
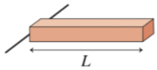
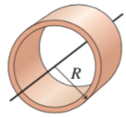
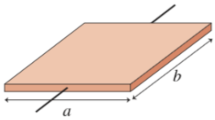
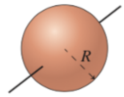
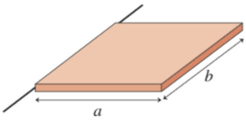
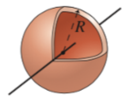
$$dm = \rho 4\pi r^2 dr = \left(\frac{M}{\frac{4}{3}\pi R^3}\right)(4\pi r^2)dr = \frac{3Mr^2}{R^3}dr$$

We thus have the moment of inertia of each shell is $dI = (2/3)dmr^2 = \frac{2Mr^4}{R^3}dr$ So the moment of inertia is

$$I = \int dI = \int_0^R \frac{2Mr^4}{R^3}dr = \boxed{\frac{2}{5}MR^2}$$

It is convenient to memorize the inertia of a few common objects. Make sure you know the rotational inertia constant (the c in $I = cmr^2$) for all of the following quantities:

TABLE 12.2 Moments of inertia of objects with uniform density

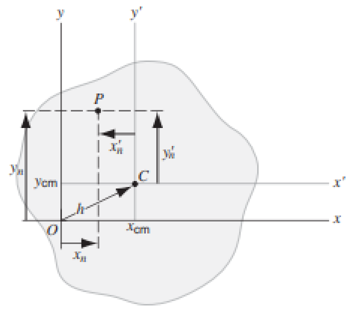
Object and axis	Picture	I	Object and axis	Picture	I
Thin rod, about center		$\frac{1}{12}ML^2$	Cylinder or disk, about center		$\frac{1}{2}MR^2$
Thin rod, about end		$\frac{1}{3}ML^2$	Cylindrical hoop, about center		MR^2
Plane or slab, about center		$\frac{1}{12}Ma^2$	Solid sphere, about diameter		$\frac{2}{5}MR^2$
Plane or slab, about edge		$\frac{1}{3}Ma^2$	Spherical shell, about diameter		$\frac{2}{3}MR^2$

So far we've generally only been finding the rotational inertias of objects about their center of mass. Perhaps we'd also like to find the rotational inertia about other points on the object. Of course, we could use equation 10.6 again to rederive the rotational inertia about some other point. Using equation 10.6 is quite tedious though, so it would be nice if there was an easier way to find the rotational inertia about some other point given that we know the rotational inertia about the center of mass. Fortunately, we do indeed have a formula that easily relates these two quantities: the **Parallel Axis Theorem**.

Theorem 10.8 (Parallel Axis Theorem)**.** *Let the inertia of an object of mass M around an axis through its center of mass be I_{com}. Then the inertia of the same object around a parallel axis a distance h away from the center of mass is given by*

$$I = I_{\text{com}} + Mh^2$$

Proof. Consider a thin, rigid pancake object in the x-y plane, which can be regarded as a collection of particles. We want to calculate the rotational inertia at a point O which lies a distance h away from the center of mass of the object at a point C.



A thin pancake object in the x-y plane

Let's represent each particle in the slab by its mass m_n and its coordinates x_n and y_n relative to the point O and it's coordinates x_n' and y_n' relative to the center of mass C. Relative to the point O, the center of mass has coordinates $(x_{\text{cm}}, y_{\text{cm}})$. Then, the rotational inertia about the point O is

$$I = \sum m_n r_n^2 = \sum m_n ({x_n}^2 + {y_n}^2)$$

Similarly, the rotational inertia about the center of mass C is

$$I_{\text{cm}} = \sum m_n {r_n'}^2 = \sum m_n ({x_n'}^2 + {y_n'}^2)$$

We note however, that by definition, we have $x_n = x_{\text{cm}} + x_n'$ and $y_n = y_{\text{cm}} + y_n'$. Substituting in these transformations, we then get

$$\begin{aligned}
I &= \sum m_n ({x_n}^2 + {y_n}^2) \\
&= \sum m_n (x_n' + x_{\text{cm}})^2 + (y_n' + y_{\text{cm}})^2 \\
&= \sum m_n ({x_n'}^2 + 2x_n' x_{\text{cm}} + {x_{\text{cm}}}^2 + {y_n'}^2 + 2y_n' y_{\text{cm}} + {y_{\text{cm}}}^2) \\
&= \sum m_n ({x_n'}^2 + {y_n'}^2) + 2x_{\text{cm}} \sum m_n x_n' + 2y_{\text{cm}} \sum m_n y_n' + (x_{\text{cm}}^2 + y_{\text{cm}}^2) \sum m_n .
\end{aligned}$$

Let's analyze the terms in the expression above. It is easy to see that the first term is just $I_{\text{cm}} = \sum m_n {r_n'}^2$. The next two terms appear to be calculating the coordinates of the center of mass *but in the center of mass system*. Thus, the second and third terms vanish. Finally, noting that ${x_{\text{cm}}}^2 + y_{\text{cm}}^2 = h^2$ (just the distance between the point O and C) and $\sum m_n = M$, we get that

$$I = I_{\text{cm}} + Mh^2$$

which proves the parallel axis theorem. ∎

Here's a simple example:

Example 10.9. Given that the rotational inertia of a uniform rod of mass M and length L rotating about an axis perpendicular to the rod and going through its center is $\frac{1}{12}ML^2$, find the rotational inertia of the same rod rotating about a perpendicular axis going through an endpoint of the rod.

Solution. This a simple application of the parallel axis theorem. In this case $h = \frac{L}{2}$, so

$$I = \frac{1}{12}ML^2 + M\left(\frac{L}{2}\right)^2 = \frac{1}{3}ML^2.$$

■

Another useful tool that you'll encounter in the $F = ma$ exam is the following theorem. Though you won't use it as often as the parallel axis theorem, the theorem has a few niche uses so make sure you know it. Note that this theorem *only* holds for flat, 2 dimensional pancake objects however, so don't use it where its not applicable.

Theorem 10.10 (Perpendicular Axis Theorem). *Consider a pancake object in the $x - y$ plane. Let the moment of inertia about some axis in the x-direction be I_x and define I_y and I_z analogously. We then have that*

$$I_z = I_x + I_y$$

Proof. Recall that

$$I_z = \int x^2 + y^2 dm$$

$$I_x = \int y^2 + z^2 dm$$

and

$$I_y = \int x^2 + z^2 dm$$

Thus summing we get

$$I_x + I_y = \int x^2 + y^2 dm + \int 2z^2 dm$$

But for flat pancake objects, note that $z = 0$. Thus,

$$I_x + I_y = \int x^2 + y^2 = I_z$$

as desired. ■ ■

Here's a quick example to see how the theorem is used in practice:

Example 10.11. Find the moment of inertia of a flat rectangle of mass and M side lengths a, b about the axis through the center of mass and perpendicular to the plane of the square.

Solution. We'll first calculate the moment of inertia in the x-direction. We have

$$I_x = \int r^2 dm = \int_{-a/2}^{a/2} x^2 \frac{M}{L} dx = \frac{1}{12}Ma^2$$

Similarly, we get

$$I_y = \frac{1}{12}Mb^2$$

So by the perpendicular axis theorem, we get that

$$I_z = I_x + I_y = \frac{1}{12}M(a^2 + b^2)$$

■

Corollary 10.12. *For a square of length L we thus get $I_z = \frac{1}{6}ML^2$. See also problem 10.2.*

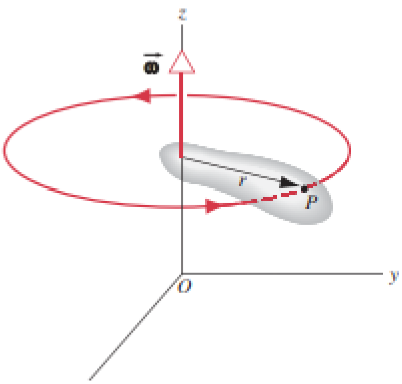
10.4 Angular Momentum and Kinetic Energy

Analogous to the momentum of a point mass, there is rotational analogue for momentum with respect to some origin:

Definition 10.13. We define the **angular momentum** of a point mass $\vec{l}$ by

$$\vec{l} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$$

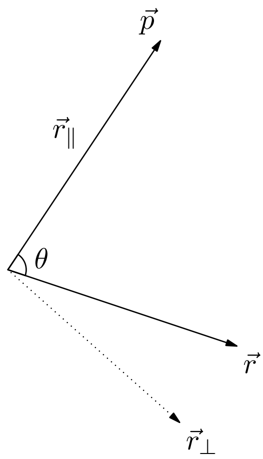
where $\vec{r}$ is the position of the particle with respect to the chosen origin.



We thus see the magnitude of the angular momentum can thus be given by

$$|\vec{l}| = rp\sin\theta$$

where θ is the angle the position vector makes with the momentum vector. Alternatively, we can break the $\vec{r}$ vector into the component *perpendicular* to the momentum vector and the component of the vector *parallel* to the momentum vector.



Splitting the r vector into its components

This is well and good for a single particle, but how do we find the angular momentum of a system? We first begin with what we may consider to be an obvious defintion for the angular momentum of a system of particles.

Definition 10.14. For a system of n particles, we define the angular momentum of the system to be the *sum* of the angular momenta of the n individual particles. That is, the angular momentum $\vec{L}$ is defined by

$$L = \sum_{i=1}^{n} \vec{l_i} = \sum_{i=1}^{n} \vec{r_i} \times \vec{p_i}$$

Of course, for continuous distributions, this sum becomes an integral.

This doesn't really give us any special insight as to how angular momentum is analogous to momentum however. We will see soon however, that just as the momentum $\vec{p}$ of a system is proportional to the velocity of the system v, we have that the angular momentum L is proportional to the angular velocity of the system ω.

There are some *slight* differences however, as rotation is quite a bit more complicated than the linear analogue. To simplify things, we begin our analysis by consider a special type of system: we consider a *rigid body* and we set our axis to be the center of mass of the object.

We'll also only be dealing with *flat* rigid objects in this section because both the $\mathbf{r}$ vector and the the $\mathbf{p}$ vectors always lie in the x-y plane so the $\mathbf{L}$ always lies in the $\hat{\mathbf{z}}$ direction. As we'll see, this simplifies the motion quite a bit because the angular velocity will always remain parallel to the angular momentum vector (this does not hold in general. See the optional section 10.6).

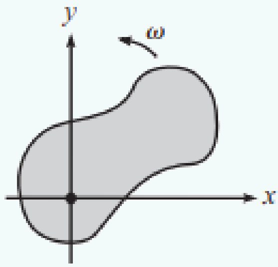
10.4.1 Rotation about the z-axis

We'll first consider the motion of a flat object that is pivoted at the origin and rotating with angular speed ω around the z-axis. We'll show (just as stated) that L truly is proportional to the angular velocity.

Theorem 10.15. *Consider a flat object in the $x-y$ plane that is pivoted at the origin and rotates with angular velocity $\vec{\omega}$. Then, the angular momentum L of the object is just*

$$\vec{L} = I\vec{\omega}$$

where I is the moment of inertia of the object about the origin.



Proof. Consider a small piece of the pancake of mass dm that is located a position (x, y). Notice that since the center of mass of the pancake is not translating, we get that the linear velocity of the particle is just $v = r\omega$ with the little piece traveling in a circle around the origin. The angular momentum dl of the piece is then

$$\vec{dl} = \vec{r} \times \vec{p} = dm(r^2\omega)\hat{z}$$

The $\hat{z}$ vector comes from the right hand rule. The angular momentum of the body is therefore

$$\vec{L} = \int \vec{dl} = \int r^2\omega\hat{z}dm = \omega\hat{z}\int r^2 dm$$

Note however that the moment of inertia about the z-axis is $I = \int r^2 dm$ so the last part of the equation simplifies to simply

$$L = I\omega\hat{z}$$

Since in this case, we have that $\vec{\omega} = \omega\hat{z}$, we get

$$\vec{L} = I\vec{\omega}$$

■

Note the surprising similarity to the equation $\vec{p} = m\vec{v}$! What is the kinetic energy T of the object? Not surprisingly, we'll see that the rotational kinetic energy formula is analogous to the transnational formula $K = \frac{1}{2}mv^2$. The derivation is largely similar to the derivation above (splitting the object into small pieces and then integrating) so try deriving the formula yourself before going into the proof:

Theorem 10.16. *The pancake object above has kinetic energy*

$$T = \frac{1}{2}I\omega^2$$

Proof. Each little piece dm has energy $dT = dm\, v^2/2$. Since $v = r\omega$ we have $dT = dm\, r^2\omega^2/2$. Integrating, we get

$$T = \int dT = \int r^2\omega^2/2dm = \frac{1}{2}\omega^2 \int r^2 dm = \frac{1}{2} I\omega^2.$$

∎

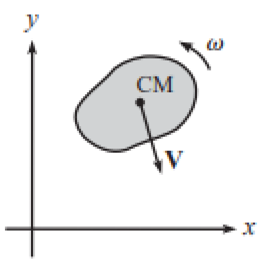
Note how similar the equation looks to $mv^2/2$!

10.4.2 General Motion in the x-y Plane

We'll now consider the general motion of the pancake object above. Instead of just rotating, we'll now consider a pancake object that both translates *and* rotates about its center of mass. For this motion (as shown in the figure), the various pieces don't just travel in a circle, so we can't just write $v = r\omega$ like we did in 10.15.

We'll end up seeing however, that the expressions for the angular momentum and the kinetic energy take on very nice forms if we work in the center of mass coordinates.

Expressed in words, the following theorem basically states that we can treat the rotation of a rigid body relative to the origin by treating the body like a point mass centered at the center of mass and finding the angular momentum and adding on the angular momentum due to the rotation of the body relative to the com.



A pancake object both
translating and rotating in the plane

Theorem 10.17 (Chasles' Theorem). *Consider a flat object of mass M in the $x-y$ plane that is rotating with angular velocity $\vec{\omega}$ about its center of mass and has a center of mass moving with velocity $\vec{V}$. Let the position of the center of mass (relative to the axis) be $\vec{R}$. Then, the angular momentum $\vec{L}$ of the object is*

$$\vec{L} = \vec{R} \times M\vec{V} + I_{\text{cm}}\vec{\omega}$$

where I_{cm} is the moment of inertia of the object relative to its center of mass. Equivalently, if $\vec{P}$ is the linear momentum of the object, we have

$$\vec{L} = \vec{R} \times \vec{P} + I_{\text{cm}}\vec{\omega}$$

Proof. Let's first try considering the motion about the center of mass frame. If we move *with* the center of mass, we'll note that every point on the particle is just in pure rotation about the center of mass; that is, we just get the situation in theorem 10.15. This motivates us to try considering the motion of the particles in the object about the center of mass frame *first*, before relating their motions to the origin.

Consider a small particle of mass dm on the flat object at a position $\vec{r'}$ relative to the center of mass of the pancake. The position of the particle relative to the origin is then

$$\vec{r} = \vec{R} + \vec{r'}.$$

Differentiating, we note that the velocity of the particle relative the origin is

$$\vec{v} = \vec{V} + \vec{v'}$$

where v' is the velocity of the particle relative to the center of mass. But remark that because in the center of mass frame the particle is just in pure rotation, we get that $v' = r'\omega$!

So the angular momentum of the particle is then

$$d\vec{l} = dm(\vec{r} \times \vec{v}) = dm((\vec{R} + \vec{r'}) \times (\vec{V} + \vec{v'}))$$

Integrating, we get

$$\vec{L} = \int d\vec{l} = \int (\vec{R} + \vec{r'}) \times (\vec{V} + \vec{v'}) dm$$

$$= \int \vec{R} \times \vec{V} dm + \int \vec{R} \times \vec{v'} dm + \int r' \vec{\times} \vec{V} dm + \int \vec{r'} \times \vec{v'} dm$$

Note however, that the second and third terms vanish because by defition we have that $\int r' dm = 0$ (that is, the sum of the positions of the particles relative to the center of mass is 0) and differentiating, we similarly have $\int v' dm = 0$. So we have

$$\vec{L} = \int \vec{R} \times \vec{V} dm + \int \vec{r'} \times \vec{v'} dm$$

$$= \vec{R} \times M\vec{V} + \int r^2 \vec{\omega} dm$$

$$= \vec{R} \times M\vec{V} + I_{\text{cm}}\vec{\omega}$$

as desired.

Alternatively, substituting $\vec{P} = M\vec{V}$, we get

$$\vec{L} = \vec{R} \times \vec{P} + I_{\text{cm}}\vec{\omega}$$

∎

In particular, note that if $\vec{V}_{\text{cm}} = 0$, then we just get the case of theorem 10.15!

Let's now try finding the kinetic energy T of the object.

Theorem 10.18. *The pancake object above has kinetic energy*

$$T = \frac{1}{2}MV^2 + \frac{1}{2}I_{\text{cm}}\omega^2$$

Proof. The kinetic energy of each of the small particles is $dT = \frac{1}{2}v^2 dm$. Integrating, we have

$$\begin{aligned} T = \int dT &= \int \frac{1}{2} v^2 dm \\ &= \int \frac{1}{2} \left| V + v' \right|^2 \\ &= \int \frac{1}{2} V^2 dm + \frac{1}{2} v'^2 dm \end{aligned}$$

where we go from the second to third line by noting that the $\int V v' dm$ term vanishes since $\int v' dm = 0$ by definition of the center of mass. So

$$\begin{aligned} T &= \frac{1}{2} V^2 \int dm + \frac{1}{2} r'^2 \omega^2 dm \\ &= \frac{1}{2} M V^2 + \frac{1}{2} I_{\mathrm{cm}} \omega^2 \end{aligned}$$

■

We now can deal with rotation in conservation of energy problems as well.

10.5 Torque

Just as we defined the rotational analogue for momentum, we'll now define a rotational analogue for force. We'll see that, in some specific situations, the rate of change of angular momentum is equal to the quantity called **torque** that we defined in the earlier section for point particles. That is, $\sum \vec{\tau} = \frac{d\vec{L}}{dt}$ which is the rotational analogue for Newton's Second Law $\sum \vec{F} = \frac{d\vec{p}}{dt}$. Indeed, we saw in section 10.2 that this equation *does* indeed, hold for point particles as

$$\frac{d\vec{L}}{dt} = \frac{d(I\vec{\omega})}{dt} = I\alpha$$

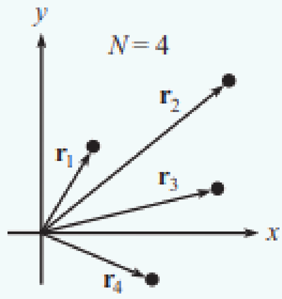
As in the previous section, we'll deal with the easier case of a fixed origin first, before exploring the more general case:

10.5.1 Rotation about the z-axis

We'll first show that the equation holds for objects that have a fixed origin.

Theorem 10.19 (Newton's Second Law for Rotation for Fixed Axes)**.** *Let's consider a system of N particles that are pivoted about the origin. Then,*

$$\frac{d\vec{L}}{dt} = \sum \tau_{ext}$$



Proof. Let's assume we have a collection of N discrete particles labeled with discrete indices i. In the continuous case, replace all the sums below with integrals. The angular momentum of the system is then

$$\vec{L} = \sum_{i=1}^{N} \vec{r_i} \times \vec{p_i}$$

The force acting on each particle can be seperated into the internal and external forces acting on each particle. That is, $\frac{d\vec{p_i}}{dt} = \sum F = F_i^{ext} + F_i^{int}$. Therefore, we can write

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}\sum_i \vec{r_i} \times \vec{p_i}$$

$$= \sum_i \frac{d\vec{r_i}}{dt} \times \vec{p_i} + \sum_i \vec{r_i} \times \frac{d\vec{p_i}}{dt}$$

$$= \sum_i \vec{v_i} \times (m\vec{v_i}) + \sum_i \vec{r_i} \times (F_i^{ext} + F_i^{int})$$

$$= 0 + \sum_i \vec{r_i} \times F_i^{ext} = \sum_i \tau_i^{ext}$$

■

It should be clear how we went from the third line to the fourth (the $\vec{v_i}$ vectors are parallel so the first term is equal to 0 and the internal forces cancel out. That is, the internal forces exert no net torque. This makes sense, because we wouldn't expect a rigid object with no external forces to start spontaneously spinning!

Note also we never assumed that the particles are rigidly connected to one another. The equation still holds even if the body is not a rigid body! Let's examine the rigid body case a little bit. Recall we showed that for flat, rigid body systems that $\vec{L} = I\vec{\omega}$. But then, we get $\frac{d\vec{L}}{dt} = \frac{d(I\vec{\omega})}{dt} = I\vec{\alpha}$. This gives us an equation similar to our more familiar $F = ma$ equation.

Corollary 10.20. *For rigid body systems about a fixed axis, we have*

$$\sum \tau_{ext} = I\vec{\alpha}$$

10.5.2 General motion in the $x - y$ plane

We'll move on to the more challenging case of general motion in the $x - y$ plane. In general, we'll see that we get a more complicated equation. We'll see however, that the equation simplies in a few specific situations.

Theorem 10.21 (Newton's Second Law for Nonfixed Origins). *Let the position of the origin be located at a position $\vec{r_0}$ (and the acceleration be $\vec{a_0}$) and let the position of the ith particle be $\vec{r_i}$. Let the total mass of the system be M and let the position of the center of mass of the system be $\vec{r_{\text{cm}}}$. Then,*

$$\frac{d\vec{L}}{dt} = \sum_i \tau_i^{ext} - M(\vec{r_{\text{cm}}} - \vec{r_0}) \times \vec{a_0}$$

Proof. The total angular momentum of the system relative to the (possibly accelerating?) origin is

$$\vec{L} = \sum_i (\vec{r_i} - \vec{r_0}) \times (\vec{v} - \vec{v_0})$$

Differentiating with respect to time, we get

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r_i} - \vec{r_0}) \times m_i(\vec{v_i} - \vec{v_0})$$

$$= \sum_i (\vec{v_i} - \vec{v_0}) \times m_i(\vec{v_i} - \vec{v_0}) + \sum_i (\vec{r_i} - \vec{r_0}) \times m_i(\vec{a_i} - \vec{a_0})$$

$$= 0 + \sum_i (\vec{r_i} - \vec{r_0}) \times (\vec{F_i^{ext}} + \vec{F_i^{int}}) - m_i(\vec{r_i} - \vec{r_0}) \times \vec{a_0}$$

$$= (\sum_i (\vec{r_i} - \vec{r_0}) \times \vec{F_i^{ext}}) - \sum_i m_i(\vec{r_i} - \vec{r_0}) \times \vec{a_0}$$

$$= (\sum_i (\vec{r_i} - \vec{r_0}) \times \vec{F_i^{ext}}) - \sum_i m_i(\vec{r_i} - \vec{r_0}) \times \vec{a_0}$$

You should check that the algebra does indeed work out from the 3rd line to the 4th line because of the distributivity of the cross product. Next, note that

$$(\sum_i (\vec{r_i} - \vec{r_0}) \times \vec{F_i^{ext}}) = \sum_i \vec{\tau_i^{ext}}$$

Also note that

$$\sum_i m_i \vec{r_i} = M\vec{r_{cm}}$$

by definition and that

$$\sum m_i \vec{r_0} = M\vec{r_0}$$

by $\sum_i m_i = M$. We thus get that

$$\frac{d\vec{L}}{dt} = \sum_i \tau_i^{ext} - M(\vec{r_{\text{cm}}} - \vec{r_0}) \times \vec{a_0}$$

as desired. ∎

The general equation appears to be quite a bit more nastier than in our fixed origin case because of the second term, which we dearly wish would go away. Fortunately, it appears that the second term *does* indeed disappear in the following two[1] cases:

- The origin is not accelerating. That is $\vec{a_0} = 0$.
- The origin is the center of mass of the system. That is, $\vec{r_{cm}} = \vec{r_0}$

This gives us an easy result that relates the rate of change of the angular momentum of the system to the total external torque. This is our $F = \frac{dp}{dt}$ analogue for rotation in general, though unlike the linear counterpart, *do* note that the rotational analogue only holds in the particular cases mentioned above:

> **Corollary 10.22.** *For systems about a non-accelerating axis or about the system's center of mass, we have*
>
> $$\sum \tau = \frac{d\vec{L}}{dt}$$

An immediate corollary of the above result is the following key result:

> **Corollary 10.23 (Conservation of Angular Momentum).** *If the total external torque on a system is zero, then its angular momentum is conserved. In particular, the angular momentum of an isolated system (one that is subject to no external forces) is conserved. That is, if $\sum \tau_{ext} = 0$, then*
>
> $$\vec{L_i} = \vec{L_f}$$

And for flat rigid body systems, as we discussed above, we can write that $\vec{L} = I\vec{\omega}$, which *finally* gives us our $F = ma$ analogue for rotation.

> **Corollary 10.24.** *For rigid body systems about a non-accelerating axis or about the system's center of mass, we have*
>
> $$\sum \tau_{ext} = I\vec{\alpha}$$

Again, make special note of for which origins the theorems and corollaries apply. Most of the time, you'll just be calculating the angular momentum and torque about the COM, a fixed point, or a point moving at constant velocity. These origins are our "safe" origins, so when you choose one of these origins, you can just use $\tau = I\alpha$ and $L = I\omega$ without much worry (given that the body is rigid, which most of the time it is).

10.5.3 Examples

We'll now work through a bunch of examples that will demonstrate how the theorems and corollaries derived in this chapter are used in practice.

We'll start by considering some examples of how conservation of angular momentum are used in practice:

[1]There is a third case where the $\vec{r_{cm}} - \vec{r_0}$ vector and the $\vec{a_0}$ vectors are parallel, but this case is seldom used, so we omit it

Example 10.25 ($F = ma$)**.** An ice skater can rotate about a vertical axis with an angular velocity ω_0 by holding her arms straight out. She can then pull in her arms close to her body so that her angular velocity changes to $2\omega_0$, without the application of any external torque. What is the ratio of her final rotational kinetic energy to her initial rotational kinetic energy?

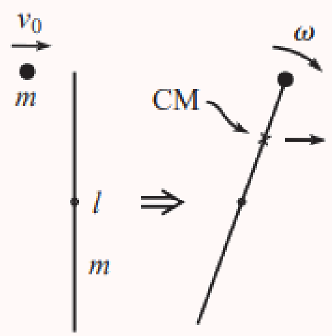
Solution. We have $T = \frac{1}{2}I\omega^2$. Let's think about what stays constant in this problem and what changes. Note that since there are no external torques on the system, the angular momentum $L = I\omega$ stays constant. Ah-hah! Since the kinetic energy is $T = \frac{1}{2}L\omega$, the ratio of the kinetic energy is

$$\frac{T_f}{T_i} = \frac{\frac{1}{2}L\omega_f}{\frac{1}{2}L\omega_i} = \frac{\omega_f}{\omega_i} = \frac{2\omega_0}{\omega_0} = \boxed{2}$$

■

We'll now consider a slightly more dynamic example:

Example 10.26 (Morin)**.** A mass m travels at speed v_0 perpendicular to a stick of mass m and length l, which is initially at rest. The mass collides completely inelastically with the stick at one of its ends and sticks to it. What is the resulting angular velocity of the system?



Solution. Let's consider the mass-rod system.Note that the center of mass of the system is located a distance $\frac{l}{4}$ above the center of the rod as shown in the picture. Let's try using conservation of angular momentum about our center of mass. We can use conservation of angular momentum since there are no *external* torques on our system. We then get

$$\frac{l}{4}mv_0 = L_f = I_{cm,sys}\omega$$

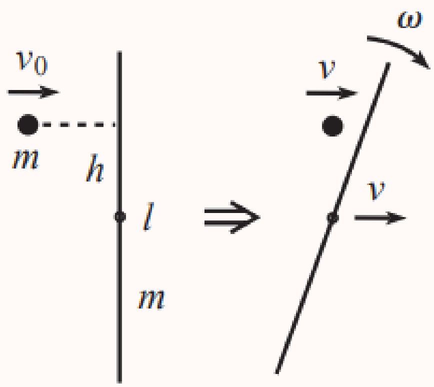
since the system becomes a rigid body after the mass sticks to the rod. We can find the moment of inertia of the system by adding the moment of inertia of the rod and the mass about the center of mass:

$$I_{sys,cm} = I_{cm,mass} + I_{cm,rod} = m(\frac{l}{4})^2 + (\frac{1}{12}ml^2 + m(\frac{l}{4})^2) = \frac{5}{24}ml^2$$

We thus get $\boxed{\omega = \dfrac{6v_0}{5l}}$ ■

Just as we can use conservation of momentum to solve a plethora of problems, we can also use conservation of energy for closed systems. Here's an example similar to the previous one for which conservation of energy works well.

Example 10.27 (Morin Adapted). A mass m travels perpendicular to a stick of mass m and length l, which is initially at rest. At what height h should the mass collide elastically with the stick, so that the mass and the center of the stick move with equal speeds after the collision?



Solution. Let's first find the common velocity v both the mass and the rod move with. Since horizontal momentum is conserved we have

$$mv_0 = mv + mv \implies v = \frac{v_0}{2}$$

Next, since this time the collision is *elastic* we can easily use conservation of energy to find the angular velocity of the rod:

$$\frac{1}{2}m{v_0}^2 = \frac{1}{2}mv^2 + (\frac{1}{2}mv^2 + \frac{1}{2}I_{rod}\omega^2)$$

Subsituting in $v = \frac{v_0}{2}$, we get

$$\omega = \frac{\sqrt{6}v_0}{l}$$

Note that conservation of energy does *not* give any information about the distance from the center of the rod that the mass collides with, because the distance r is absent in the conservation of energy equation. To find the variable h, we need to use one of our torque or momentum equations because they give us the variable (since $\tau = rF$ and $L = rp$).

Let's first let our system be the rod-mass system since it's a closed system. Let's try to find a suitable axis for which angular momentum is conserved. Consider the fixed axis through the point the perpendicular stick lied initially (rather than the center of mass of the system because the math is easier as you will see). Note that no torques are exerted on the system since there are no external forces. Since the axis is fixed, we are free to use conservation of angular momentum.

The initial angular momentum of the system is

$$L_i = mv_0$$

and the final angular momentum is

$$L = L_{mass,f} + L_{rod,f} = mv + (I_{rod}\omega + 0)$$

The equation of $L_{rod,f}$ follows from theorem 10.17 (Chasle's Theorem) where the 0 appears since the $\vec{R}$ vector and the $\vec{V}$ vector point in the same direction.

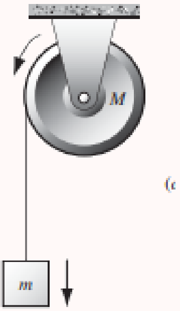
Equating the initial and final angular momentums and substituting $I = \frac{1}{2}ml^2$ and $\omega = \frac{\sqrt{6}v_0}{l}$, we get

$$\frac{1}{2}mv_0h = \left(\frac{ml^2}{12}\right)\left(\frac{\sqrt{6}v_0}{l}\right) \implies \boxed{h = \frac{l}{\sqrt{6}}}$$

■

Let's now get some more experience with torque. In general, torque can be used in a pretty similar manner to $F = ma$, after choosing an appropriate axis. Here's a simple example to start us off:

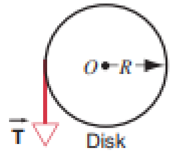
Example 10.28 (HRK Adapted). The figure below shows a pulley, which can be considered as a uniform disk of mass M and radius R, mounted on a fixed (frictionless) horizontal axle. A block of mass $m = M$ hangs from a light cord that is wrapped around the rim of the disk. Find the acceleration a of the falling block.



Solution. Let's first consider the forces on the mass m. Clearly, there is a tension force T acting upwards and the weight mg acting downwards. From Newton's second law, we get

$$\sum F = mg - T = ma$$

Let's now try considering the motion of the pulley.The pulley clearly will begin to rotate with some angular acceleration. To analyze the motion, let's consider an axis through the center of the disk. Since the pulley is fixed, our axis is fixed, and is thus a "safe" axis where we can use $\sum \tau = I\alpha$ to easily find the angular acceleration of the disk. Let's consider the torques on the disk:



Clearly, the only force that provides a torque on the disk is the tension T that acts on the edge of the disk. We thus get

$$\sum \tau = RT = I\alpha = \frac{1}{2}MR^2\alpha$$

Note that since the rope does not slip, a point on the rim of the pulley will have the same acceleration as that of the block. Thus,

$$a = R\alpha$$

Substituting, we get that $T = \frac{1}{2}Ma$ which gives

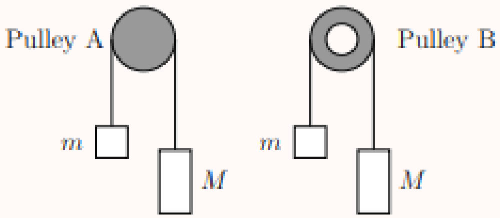
$$Ma = Mg - T = Mg - \frac{1}{2}Ma \implies a = \boxed{\frac{2}{3}g}$$

■

One thing you should be careful of when doing torque problems is to make sure your keeping correct track of which direction your torque is heading. If your net torque points *out* of the page, note that the angular acceleration will be counterclockwise, while if your net torque points *into* the page, the angular acceleration will point clockwise. Be *very* careful about which way you define to be positive, and the signs of your corresponding angular accelerations and torques.

We'll continue with a harder example from the 2014 $F = ma$ exam:

Example 10.29 ($F = ma$). Two pulleys (shown in the figure) are made of the same metal with density ρ. Pulley A is a uniform disk with radius R. Pulley B is identical except a circle of $R/2$ is removed from the center. When two boxes $M = \alpha m (\alpha > 1)$ are connected over the pulleys through a massless rope and move without slipping, what is the ratio between the accelerations in system A and B? The mass of pulley A is $M + m$.



Solution. Clearly this problem has to do with the difference in rotational inertias between the two pulleys. Let's try finding an expression for the ratio of the accelerations first to see just *how* it relates the two moment of inertias.

One thing to be *super* careful of in this problem is to note that the tensions on the two masses are *not* the same. This is because the rope does not slip, and since the pulley has mass, it must rotate, so there must be a difference in tension (and therefore a net torque) to compensate for the new angular acceleration.

Let's first write Newton's laws equations for the two masses. Since mass M is heavier, we predict that mass M will go down and mass m will go up. Let's let T_1 be the tension that acts on m and T_2 defined similarly for M. We then get

$$T_1 - mg = ma$$

$$Mg - T_2 = Ma$$

Let's now consider the torque about the center of the pulley (which is fixed). We then get

$$\sum \tau = RT_2 - RT_1 = I\alpha$$

Since the rope does not slip, we can write $\alpha = \frac{a}{R}$ and we thus get

$$R^2(T_2 - T_1) = Ia$$

Now, combining all our equations (try doing this yourself by summing the first two equations and substituting into our last equation), we get

$$Ia = R^2(M-m)g - R^2(M+m)a \implies a = \frac{R^2(M-m)g}{I + R^2(M+m)}$$

The ratio of the accelerations should then be

$$\frac{a_A}{a_B} = \frac{I_B + R^2(M+m)}{I_A + R^2(M+m)}$$

Ah-hah! The only thing left to do now is the laborous task of finding the moment of inertias of both objects. By the inertia formula for a disk we get

$$I_A = \frac{1}{2}(M+m)R^2$$

and through subtraction we get

$$I_B = \frac{1}{2}(M+m)R^2 - \frac{1}{2}(\frac{M+m}{4})(\frac{R}{2})^2 = \frac{15}{32}(M+m)R^2$$

Substituting into our expression for the ratio of accelerations, we get that

$$\frac{a_A}{a_B} = \frac{\frac{15}{32}+1}{\frac{1}{2}+1} = \boxed{\frac{47}{48}}.$$

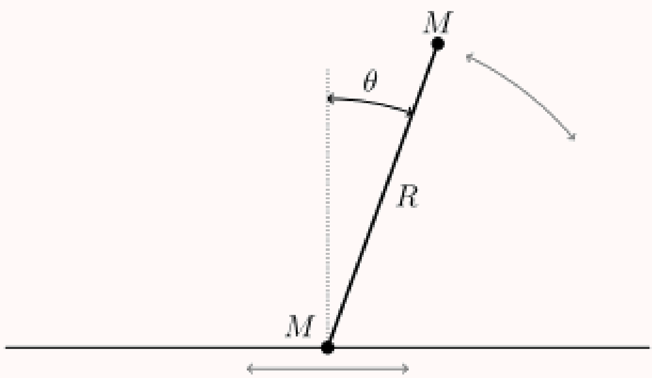
■

Though the algebra was a bit of a slog for this problem, the key steps for the solution should have been pretty clear.

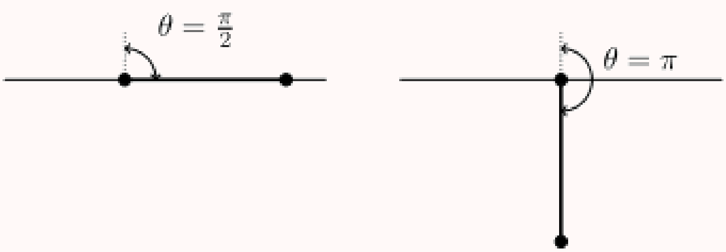
We're going to wrap up this section with an *extremely* challenging example from the 2019 USAPhO exam. This problem should bring together a lot of the previous things we've learned so far in the rotation unit and will serve as a "capstone" of sorts. Part b of the problem in particular is especially tricky, so keep a keen eye out for it. As always, try solving the problem first before looking at the solution. Good luck!

10.6 A Challenging Example

Example 10.30. A bead is placed on a horizontal rail, along which it can slide frictionlessly. It is attatched to the end of a rigid, massless rod of length R. A ball is attatched to the other end. Both the bead and the ball have mass M. The system is initially stationary, with the ball directly above the bead.The ball is then given an infinitesimal push, *parallel* to the rail. (Source: USAPhO)



Assume that the rod and ball are designed in such a way (not explicitly shown in the diagram) so that they can pass through the rail without hitting it. In other words, the rail only constrains the motion of the bead. Two subsequent states of the system are shown below:



(a) Derive an expression for the force in the rod when it is horizontal, as shown in the left above, and indicate whether the force is tension or compression.

(b) Derive an expression for the force when the ball is directly below the bead, as shown in the right above, and indicate whether the force is tension or compression.

(c) Let θ be the angle the rod makes with the vertical, so that the rod begins with $\theta = 0$. Find the angular velocity $\omega = d\theta/dt$ as a function of θ.

This problem is *quite* subtle, but it can be solved (without any calculus) if one is careful of the factors that restrain the motion.

Before starting the problem, we note that c is the most general part of the problem. Solving parts a and b should then give us some good physical intuition for part c so we go in order here. Some people prefer to do the general problem first, but we recommend trying some special cases (such as in part a or b) first.

Solution.

(a) We first think about what happens to the center of mass of the system. Because there are no net horizontal forces acting on our system, the center of mass must just drop vertically. We now do a more careful analysis. Let v_{cm} be the speed of the center of mass of the rod-ball system. We next focus our attention to the bead, as the bead being constrained to the rail is our main constraint. If the bead is constrained to the rail, it must mean that the bead does not have a vertical velocity (as otherwise it would fly off the rail). Since the vertical velocity of the bead is given by a superposition of the velocity of the center of mass and the tangential speed, we have

$$v_{\text{cm}} - \frac{R}{2}\omega = 0 \implies v_{\text{cm}} = \frac{R}{2}\omega$$

We now find ω through energy conservation:
By COE,

$$MgR = \frac{1}{2}(2M)(v_{\text{cm}})^2 + \frac{1}{2}I\omega^2$$

$$MgR = \frac{1}{2}(2M)\left(\frac{R}{2}\omega\right)^2 + \frac{1}{2}(2M)\left(\frac{R}{2}\right)^2\omega^2 = \frac{MR^2\omega^2}{2}$$

We thus get that

$$\omega = \sqrt{\frac{2g}{R}}$$

We thus get that the centripetal force on the ball is

$$F_c = (M)\left(\frac{R}{2}\right)(\omega)^2 = (M)\left(\frac{R}{2}\right)(\frac{2g}{R}) = Mg$$

Next, note that because the ball is horizontal, there is no additional tensile force on the rod due to gravity. So the net force on the rod is

$$T = \boxed{Mg}.$$

Because the radial force on the ball is directed towards the rod, the force on the rod by the ball must be directed outward, so the force must be a tension.

(b) We need to be *very* careful in this part as we now have to deal with the acceleration of the center of mass of the rod as well since the rod is now vertical. Let's first find the centripetal acceleration that the ball experiences. Since the bead experiences no vertical velocity (since the bead is constrained tot he rail) and since the velocity of the bead due to the angular velocity $r\omega$ is perpendicular to the rod at this point, we know that the velocity of the center of mass of the system has to be 0. Then, by conservation of energy, we can find the angular velocity:

$$2MgR = \frac{1}{2}I_{cm}\omega^2 = \frac{MR^2}{4}\omega^2 \implies \omega = \sqrt{8g}R$$

Then, the centripetal acceleration of the ball must be $F_c = M(\frac{R}{2})\omega^2 = 4Mg$. Since the mass of the ball is Mg, this then makes $F_{rod} = 5Mg$ a *very* tempting answer. Note however, because the rod is vertical, we must also account for the acceleration of the center of mass of the rod. Since the height of the center of mass is $y_{cm} = \frac{R}{2}\cos\theta$, we get $a_{cm} = \frac{-R}{2}(\omega^2\cos\theta + \alpha\sin\theta)$ which at $\theta = \pi$ is just $a_{cm} = 4g$. Now we *finally* get that

$$\sum F = F_c = F_{rod} - Mg - Ma_{cm} = 4Mg \implies F_{rod} = \boxed{9Mg}.$$

(c) We now try solving the general problem. We remark that because the bead must have no vertical velocity, we get the constraint

$$v_{\text{cm}} = \frac{R}{2}\omega\sin\theta$$

We next use energy conservation, akin to as we have done in part a and b. We have

$$Mg(R - R\cos\theta) = \frac{1}{2}(2M)v_{\text{cm}}^2 + \frac{1}{2}I\omega^2$$

Thus, we get

$$Mg(R - R\cos\theta) = Mv_{\text{cm}}^2 +$$

$$MgR(1-\cos\theta) = M(\frac{R}{2}\omega\sin\theta)^2 + \frac{MR^2\omega^2}{4} = \frac{MR^2\omega^2}{4}(\sin^2\theta + 1)$$

We thus get

$$\omega = \boxed{\sqrt{\frac{4g}{R}\frac{1-\cos\theta}{\sin^2\theta + 1}}}.$$

Checking the cases of $\theta = \frac{\pi}{2}$ gives us the value of ω in part a as does checking $\theta = \pi$ in part b, which gives us some satisfaction that our answer is indeed correct.

■

The above problem is *very* difficult and is about equivalent to what you would consider the *hardest* problem on the USAPhO exam to look like. Only about 2% of the students who took the exam got part b) correct, largely because they forgot to account for the acceleration of the center of mass so don't feel too bad if you couldn't solve the problem.

10.7 Homework Problems and Solutions

*This homework assignment is going across two days so it will be worth slightly more points at 150 points rather than the typical 100.

Problem 10.1. [10] A figure skater begins rotating with angular velocity $\omega = 30$ rad/s and rotational inertia $I = 2$ kg $\times$ m^2. To rotate faster, he moves his arms and legs inwards over the course of half a second so that his new rotational inertia is $3/4I$. His new angular velocity is ω'.

(a) [4] What is the ratio ω'/ω?

(b) [6] What is the skater's average power over this time interval?

Solution.

(a) The system is conservative so the angular momentum remains constant. So

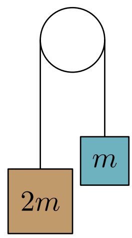
$$I\omega = I'\omega' \implies \frac{\omega'}{\omega} = \frac{I}{I'} = \boxed{\frac{4}{3}}.$$

(b) The average power is defined by $P = \frac{\Delta W}{\Delta t}$ So we have

$$\begin{aligned} P &= \frac{\frac{1}{2}I'\omega'^2 - \frac{1}{2}I\omega^2}{\Delta t} \\ &= \frac{\frac{1}{2}(2)(\frac{3}{4})(40)^2 - \frac{1}{2}(2)(30)^2}{0.5} = \boxed{600W}. \end{aligned}$$

■

Problem 10.2. [6] Consider the pulley system below consisting of two masses, one of mass m and the other of $2m$, attached to a pulley, which is a uniform disk of mass $4m$ and radius r.



Assuming the string does not slip, and using 10 m/s^2 for g, what is the magnitude of acceleration experienced by the block with mass m?

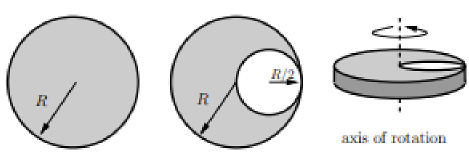
Hint: 11

Solution. There are multiple approaches to this problem. One approach would be to analyze the forces on the two masses, and the torque on the pulley, which is straightforward but perhaps longer. We will give a different solution. We can consider the mass $2m$ and m to have rotational inertias $2mr^2$ and mr^2 respectively. Note the net external torque is mgr, so we have

$$\sum \tau = I_{\text{total}}\alpha \implies mgr = 5mr^2\alpha,$$

or $\alpha = g/(5r)$. Then $a = \alpha r = g/5 = \boxed{2 \text{ m/s}^2}$. ■

Problem 10.3 ($F = ma$). [8]A uniform circular disk of radius R begins with a mass M; about an axis through the center of the disk and perpendicular to the plane of the disk the moment of inertia is $I_0 = \frac{1}{2}MR^2$. A hole is cut in the disk as shown in the diagram. In terms of the radius R and the mass M of the original disk, what is the moment of inertia of the resulting object about the axis shown?



Solution. Consider the inertia of the removed piece. Note that the mass has decreased by a factor of 4, and radius has decreased by a factor of 2. Hence, by the parallel axis theorem, we have

$$I_{\text{removed}} = (\frac{M}{4})(\frac{R}{2})^2 + \frac{1}{2} \cdot \frac{M}{4}(\frac{R}{2})^2 = \frac{3}{32}MR^2.$$

Subtracting this inertia, we see the new inertia is

$$I = I_0 - I_{\text{removed}} = \boxed{\frac{13}{16}MR^2}.$$

■

Problem 10.4 ($F = ma$)**.** [6] Consider a flat uniform square of mass M and side length L. Cut a circle out of the square that has a diameter equal to the length of the side of the square, with the same center as the square. Determine the moment of inertia of the remaining shape about an axis through the center and perpendicular to the plane of the sphere.

Solution. Recall the rotational inertia of the square is

$$I_0 = \frac{1}{6}ML^2.$$

If you do not remember the rotational inertia of a rectangle, you can also derive it via the perpendicular axis theorem. Consider the rotational inertia of the square about a line parallel to one of its sides and bisecting the square in half. Then the rotational inertia is the same as that of a rod, or $1/12ML^2$. By the perpendicular axis theorem, we see that the rotational inertia is double this value. The rotational inertia of the removed circle is

$$I_{\text{removed}} = \frac{1}{2}(\frac{\pi}{4}M)(\frac{L}{2})^2 = \frac{\pi}{32}ML^2.$$

Thus, we have $I = \boxed{(\frac{1}{6} - \frac{\pi}{32})ML^2}$.

■

Problem 10.5 ($F = ma$)**.** [8] The moment of inertia of a uniform equilateral triangle with mass m and side length a about an axis through one of its sides and parallel to that side is $(1/8)ma^2$. What is the moment of inertia of a uniform regular hexagon of mass m and side length a about an axis through two opposite vertices?

Solution. Subdivide the hexagon into six identical equilateral triangles. Then four of the equilateral triangles have an edge along the axis and have rotational inertia $\frac{1}{48}ma^2$.

The centers of mass of the other two equilateral triangles are a distance $\frac{a}{\sqrt{3}}$ from the axis of rotation. Note that the rotational inertia was $\frac{1}{8}(\frac{m}{6})(a^2)$ if they were a distance of $\frac{a}{2\sqrt{3}}$ away, so we must add an additional term $\frac{m}{6}(\frac{a}{2\sqrt{3}})^2$ to the inertia. Hence each of these equilateral triangles has an inertia of

$$\frac{ma^2}{48} + \frac{ma^2}{24} = \frac{3ma^2}{48}$$

Then we have

$$I = 4(\frac{ma^2}{48}) + 2(\frac{3ma^2}{48}) = \boxed{\frac{5}{24}ma^2}.$$

■

Problem 10.6 (HRK)**.** [10] A uniform disk of mass M can rotate without friction on a fixed axis. A string is wrapped around its circumference and is attached to a mass m. The string does not slip. What is the tension in the cord while the mass is falling?

Solution. Note that the forces acting on the mass. We have

$$T - mg = ma.$$

Now consider the torque on the disk about its center of mass. The only torque is from tension, so we have

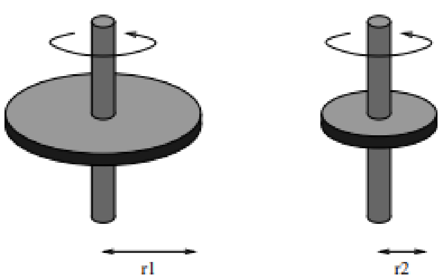
$$TR = \frac{MR^2}{2}\alpha = \frac{MR^2}{2} \times \frac{a}{R}.$$

Substituting using T from our force equation, we conclude

$$T = \boxed{\frac{mg}{1+2\frac{m}{M}}}.$$

■

Problem 10.7 ($F = ma$)**.** [10] Two disks are mounted on thin, lightweight rods oriented through their centers and normal to the disks. These axles are constrained to be vertical at all times, and the disks can pivot frictionlessly on the rods. The disks have identical thickness and are made of the same material, but have differing radii r_1 and r_2. The disks are given angular velocities of magnitudes ω_1 and ω_2 respectively, and are brought into contact at their edges. After the disks interact via friction, it is found that both discs come exactly to a halt. Which of the following must hold? Ignore effects associated with the vertical rods.



(A) $\omega_1^2 r_1 = \omega_2^2 r_2$
(B) $\omega_1 r_1 = \omega_2 r_2$
(C) $\omega_1 r_1^2 = \omega_2 r_2^2$
(D) $\omega_1 r_1^3 = \omega_2 r_2^3$
(E) $\omega_1 r_1^4 = \omega_2 r_2^4$

Solution. When the disks interact, by Newton's third law we have that disks experience forces of the same magnitude. Hence, the ratio of torques on the disks is proportional to their radii. Recall that

$$\tau = \frac{dL}{dt},$$

and we have

$$I\omega = mr^2\omega = \frac{\rho}{2}r^4\omega,$$

where ρ is the density of the disks. Hence, since $t = \frac{I\omega}{\tau}$ is equal for the disks, we have that $r^3\omega$ is the same for both disks. That is,

$$\boxed{\omega_1 r_1^3 = \omega_2 r_2^3}.$$

■

Problem 10.8 (Kalda)**.** [10] If one hits something rigid — e.g. a lamppost — with a bat, the hand holding the bat may get stung (hurt) as long as the impact misses the so-called centre of percussion of the bat (and hits either below or above such a centre). Determine the position of the centre of percussion for a bat of uniform density. You may assume that during an impact the bat is rotating around its holding hand.

Solution. Suppose that the blow occurs x away from the center of mass. Then we can write the following relation:

$$\Delta L = \int \tau \, dt = x \int F \, dt = x\Delta p.$$

So,

$$I\Delta\omega = \frac{mL^2}{12}\Delta\omega = x \cdot m\Delta v.$$

We want the change in speed of the end of the bat to be 0, so this means

$$\Delta v = \frac{\ell}{2}\Delta\omega.$$

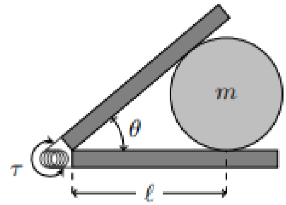
Plugging in the relation from above,

$$\frac{m\ell^2}{12}\Delta\omega = x \cdot m(\ell/2)\Delta\omega,$$

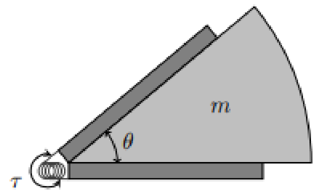
which gives $x = \ell/6$. So the distance from the hit must be $\ell/6$ from the center of the bat, or $\ell/2 + \ell/6 = \boxed{2\ell/3}$ from the hands. ■

Problem 10.9 ($F = ma$)**.** [17] A launcher is designed to shoot objects horizontally across an ice rink. It consists of two boards of negligible mass connected via a spring-loaded hinge, which exerts a constant torque τ on each board to keep them together. For both problems, neglect friction with either the ice or the launch boards.

(a) [7] A hard disc is pushed into the launcher between the boards until the boards make contact with it a distance ℓ from the hinge and are open to an angle θ, as shown in the figure. What is the minimum force necessary to hold the disc in this position?



(b) [10] The disc is removed and replaced with a pie-shaped wedge of the same mass m, so that the hinge is still initially held open at an angle θ, as shown in the following figure. If the wedge is released from rest, what is its speed after it exits the launcher?



Solution.

(a) Note that each board exerts a force $F = \frac{\tau}{L}$ towards the center of the hard disc. We can sum these two forces by summing the components along the perpendicular bisector of the launcher. Doing so gives that the boards exert a net force $F = \boxed{2\tau/\ell\sin(\theta/2)}$. This is also the force required to keep the disc in equilibrium, so this is our final answer.

(b) We have the energy stored by the launcher is

$$\int \tau \, d\theta,$$

but since τ is constant, using conservation of energy we see

$$\tau\theta = \frac{1}{2}mv^2 \implies v = \boxed{\sqrt{\frac{2\tau\theta}{m}}}.$$

■

Problem 10.10 (USAPhO)**.** [10] A car accelerates uniformly from rest. Initially, its door is slightly ajar. Calculate how far the car travels before the door slams shut. Assume the door has a frictionless hinge, a uniform mass distribution, and a length L from front to back. (This problem requires familiarity with oscillations so skip it if you haven't learned it yet)

Hint: 23

Solution. Let the force perpendicular to the rod that is acting at the hinge be F. Let the acceleration of the car be a_1, the acceleration of the hinge perpendicular to itself be a_2, and the angular acceleration about its center of mass be α. Now consider the acceleration of the hinge perpendicular to the stick. We have that it is $a_1 \sin\theta$, where θ is the angle at which the car is at. But we can then write

$$a_1 \sin\theta = -a_2 + \alpha\frac{L}{2}.$$

So $a_2 = \alpha L/2 - a_1 \sin\theta$. We have by $F = ma$, $F = ma_2$, and by $\tau = I\alpha$,

$$F \cdot \frac{L}{2} = -\frac{ML^2}{12}\alpha.$$

So we can combine our 3 equations and simply solve for α, giving

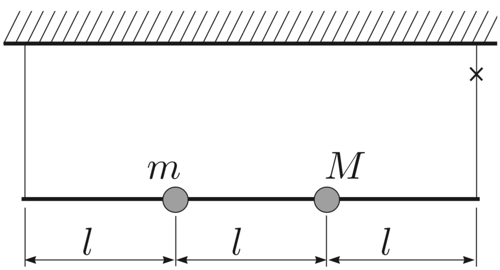
$$\alpha = \frac{3a\sin\theta}{2L} \approx \frac{3a}{2L}\theta.$$

This is simple harmonic motion, so the angular frequency is

$$\sqrt{\frac{\alpha}{\theta}} = \sqrt{\frac{3a}{2L}}.$$

We take 1/4th of the period to get the time to close, giving $t = \frac{\pi}{2}\sqrt{\frac{2L}{3a}}$, and using kinematics, $\frac{1}{2}at^2 = \boxed{\frac{\pi^2}{12}L}$. ∎

Problem 10.11 (Krotov)**.** [12] A light rod with length $3l$ is attached to the ceiling by two strings with equal lengths. Two balls with masses m and M are fixed to the rod, the distance between them and their distances from the ends of the rod are all equal to l. Find the tension in the second string right after the first has been cut.



Solution. We first find the angular acceleration of the rod about it's left end. The net torque (once the string is cut) is given by,

$$\tau = 2Mgl + mgl.$$

Now we find the moment of inertia, which is $(m+4M)l^2$. So the angular acceleration is

$$\alpha = \frac{2Mg + mg}{(m+4M)l}.$$

Then we can find the location of the center of mass to get the acceleration of the center of mass. The center of mass is a distance $\frac{m+2M}{m+M}l$ from the end point, so then the acceleration is

$$a = \frac{m+2M}{m+M} \cdot \frac{2Mg+mg}{m+4M}.$$

So then to get the tension, we use $F = (m+M)a$, and

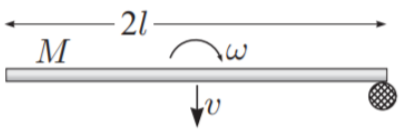
$$-T + (m+M)g = (m+2M)\frac{(m+2M)g}{m+4M}.$$

We simplify and solve for T, which gives us $T = \boxed{\frac{Mm}{4M+m}g}$. ■

Problem 10.12 (Kalda). [13] A rod of mass M and length $2l$ is sliding on ice. The velocity of the centre of mass of the rod is v, the rod's angular velocity is ω. At the instant when the centre of mass velocity is perpendicular to the rod itself, it hits a motionless post with an end. What is the velocity of the centre of mass of the rod after the impact if

(a) [6] the impact is perfectly inelastic (the end that hits the post stops moving);

(b) [7] the impact is perfectly elastic.



Solution.

(a) We conserve angular momentum about the post. The initial angular momentum about the post is given by

$$Mvl - \frac{Ml^2}{3}\omega.$$

The minus is there because of how the diagram was drawn. And the final angular momentum is given by

$$\frac{4Ml^2}{3}\omega_f.$$

Where ω_f is the angular velocity about the post. So we can just set these two equal. We also want to find the final velocity of the com of the rod, which is given by $l\omega_f$. So we solve for that, and we get

$$\omega_f l = \boxed{\frac{3}{4}v - \frac{l}{4}\omega}.$$

(b) We still conserve angular momentum about the post, but there's one more variable. We have

$$Mvl - \frac{Ml^2}{3}\omega = Mv_f l - \frac{Ml^2}{3}\omega_f.$$

We also conserve energy, and we have

$$\frac{1}{2}Mv^2 + \frac{1}{2}\frac{Ml^2}{3}\omega^2 = \frac{1}{2}Mv_f^2 + \frac{1}{2}\frac{Ml^2}{3}\omega_f^2.$$

We can solve these in whatever way we wish, but using difference of squares will probably be fastest. We have

$$\frac{1}{2}Mv^2 + \frac{1}{2}\frac{Ml^2}{3}\omega^2 = \frac{1}{2}Mv_f^2 + \frac{1}{2}\frac{Ml^2}{3}\omega_f^2 \implies -\frac{Ml^2}{3}(\omega_f^2 - \omega^2) = M(v_f^2 - v^2) \tag{10.1}$$

$$Mvl - \frac{Ml^2}{3}\omega = Mv_f l - \frac{Ml^2}{3}\omega_f \implies \frac{Ml^2}{3}(\omega_f - \omega) = Ml(v_f - v) \tag{10.2}$$

$$\implies l(\omega_f - \omega) = 3(v_f - v) \tag{10.3}$$

Dividing (10.1) and (10.3),

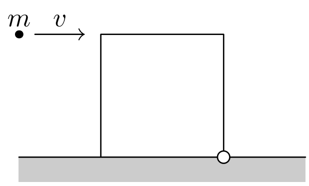
$$-l(\omega_f + \omega) = v_f + v$$

Adding (10.3) to this,

$$-2l\omega = 4v_f - 2v \implies v_f = \boxed{\frac{v - \omega l}{2}}.$$

■

Problem 10.13 (Morin)**.** [15] A square block with mass $m = 1$ kg and sidelength $\ell = 1$ m sits at rest on a table, with its bottom right corner attached to the table with a pivot, as shown in the figure. A ball, also with mass m, moves horizontally to the right with speed v and collides with the block and sticks to it at its upper left corner. The moment of inertia of the block around its center is $m\ell^2/6$. What is the value that v must be greater than in order for the block to tip over?



Solution. Note that once the center of mass of the system swings over the pivot, the box will naturally tip over. Hence, we will see if there is enough energy for the center of mass to swing over the pivot. To compute the initial kinetic energy, we must first compute the value of ω immediately after the collision. We'll first use the parallel-axis theorem to calculate the moment of inertia of the block:

$$I = I_{\text{block}} + I_{\text{ball}} = \left(\frac{m\ell^2}{6} + m(\frac{\ell}{\sqrt{2}})^2\right) + m(\ell\sqrt{2})^2 = \frac{2m\ell^2}{3} + \frac{8m\ell^2}{3}.$$

Note that the initial angular momentum is $mv\ell$, so by conservation of angular momentum, we see $\omega = \frac{3v}{8\ell}$.

Now consider the gravitational potential energy gain of the block when the center of mass is immediately above the pivot. We need only consider the height change of the center of mass of the system, and since the ball and block have the same mass, the center of mass is half way between the center of the block and the corner with the ball. Some geometry gives the height change is $\frac{3\ell}{4}(\sqrt{2}-1)$.

Thus, we have

$$\frac{1}{2}I\omega_i^2 = (2m)g \cdot \frac{3\ell}{4}(\sqrt{2}-1),$$

and substituting for I and ω_i gives

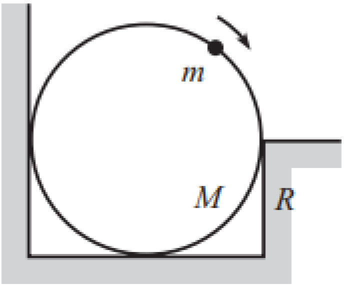
$$\frac{3}{16}mv^2 = \frac{3}{2}(\sqrt{2}-1)mg\ell,$$

$$\implies v = \boxed{\sqrt{8(\sqrt{2}-1)g\ell}}.$$

■

10.7.1 Written Solutions

Problem 10.14 (Morin). [15] A bead of mass m is positioned at the top of a frictionless hoop of mass M and radius R, which stands vertically on the ground. A wall touches the hoop on its left, and a short wall of height R touches the hoop on its right, as shown in the figure. All surfaces are frictionless. The bead is given a tiny kick, and it slides down the hoop, as shown. What is the largest value of m/M for which the hoop never rises up off the ground? (Note: It's possible to solve this problem by using only force, but solve it here by using torque.)



Solution. Let's use the point of contact of the hoop with the right wall. Let θ be the angle that the bead makes with respect to the line vertical through the center of the hoop. By conservation of energy, we have

$$\frac{1}{2}Mv^2 = MgR(1-\cos\theta) \implies v = \sqrt{2gR(1 = cos\theta)}$$

Now note that the centripetal force on the bead (directed radially) must be the *net* force on the bead in the centripetal direction, so the normal force that the hoop enacts on the bead is $3mg\cos\theta - 2mg$ radially. By Newton's third law, the same force is enacted on the hoop in the opposite direction. Note that the torque from this normal force must not counteract the weight of the hoop in order for the hoop to not leave the ground. We thus have

$$\tau_{hoop,bead} \leq \tau_{Mg}$$

so

$$R\cos\theta(2mg - 3mg\cos\theta) \leq RMg$$
$$\implies \frac{m}{M} \leq \frac{R}{2\cos\theta - 3\cos^2\theta}$$

To maximize the right hand side, we take the derivative of $2cos\theta - 3\cos^2\theta$ to minimize the quantity. Taking the derivative, we get $\cos\theta = \frac{1}{3}$ which gives us a result of

$$\frac{m}{M} \leq \boxed{3}.$$

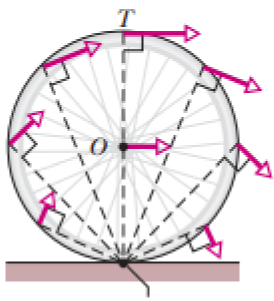
■

11 Rolling Motion

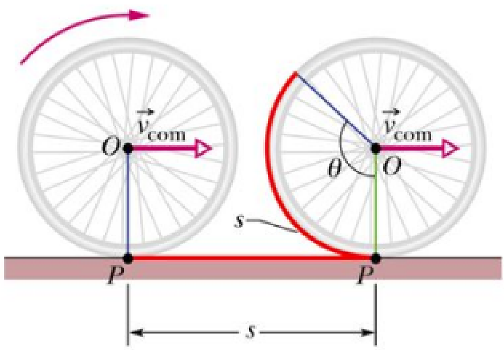
Today we will look at **rolling motion**, which is a combination of simultaneous rotational and translational motion.

11.1 Rolling without Slipping

Rolling without slipping will be our archetypal example of rolling motion. We can view a rolling object as a rigid body instantaneously rotating with angular velocity $\vec{\omega}$ about the point of contact between itself and floor.



However, because the point of contact is constantly changing, this perspective is only occasionally used. More commonly, we view a rolling object as an object that is undergoing simultaneous translation (with velocity $\vec{v}$) and rotation (with angular velocity $\vec{\omega}$) about its center of mass.



In particular, for an object rolling without slipping, the translational distance through which an object travels is equal to the arc length through which the object has rotated. Hence, we have that $s = r\Delta\theta$, or if we take the derivative with respect to time,

$$v = r\omega.$$

This in fact how we will define rolling without slipping.

Definition 11.1. We say an object is **rolling without slipping** if the object moves with center of mass velocity v and rotates about its center of mass with angular velocity ω such that $v = r\omega$. **An object undergoing rolling motion is experiencing simultaneous rotational and translational motion.**

Quite intuitively, the kinetic energy of a rolling object is the sum of its rotational and translational kinetic energies, or

$$K = K_T + K_R = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2.$$

Remark. The value of $\vec{\omega}$ is in fact the same in both views of rolling motion. When viewing rolling as pure rotation, we can use the parallel axis theorem to see that the kinetic energy is in fact

$$\begin{aligned} K &= \frac{1}{2}I\omega^2 = \frac{1}{2}(I_{\text{com}} + mr^2)\omega^2 \\ &= \frac{1}{2}I_{\text{com}}\omega^2 + \frac{1}{2}mv^2. \end{aligned}$$

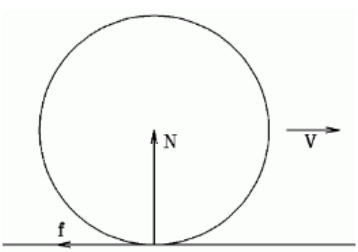
Hence it is evident that $\vec{\omega}$ is equal in both frames.

Also quite intuitively, different points in a rolling body experience different velocities. For example, the point of contact has velocity 0, whereas the top point has velocity $2v$. We use v to denote v_{com}, or the velocity of the center of mass of the body.

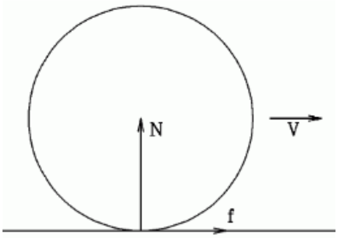
11.2 Friction in Rolling Motion

For an object rolling without slipping along a rough surface, you might expect the object might eventually come to rest due to the friction force. However, this is not exactly the case.

We might naively assume the following picture of friction:



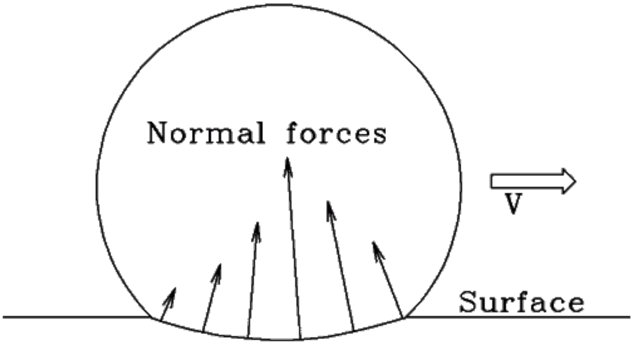
However, in this scenario we note that the friction force provides a clockwise torque that increases the rotational speed of the sphere! If the sphere is to continue rolling without friction, this would imply the sphere is gaining kinetic energy, which would not make sense. We might then naively assume our direction of friction is wrong, giving the following picture:



However, in this scenario the friction force provides a net force that increases the translational speed of the sphere, and we again run into the same problem.

Clearly, friction does not act on a rolling object the same way it does on a sliding object. Recall that friction prevents sliding/slipping. *As such, for an object that is already rolling without slipping, friction will act such that the object continues to roll without slipping.* In the case of an object rolling on level ground, this means friction will be absent altogether.

Remark. In the real world, it is clearly impossible for an object to continue rolling infinitely. So if it's rolling on level ground, why does the ball stop? The answer is that no rolling body is perfect *rigid.* That is, a rolling object will experience deformation, and will experience greater deformation on the forward side.



Ball rolling to the right, with surface deformation.
The deformation is greatly exaggerated.
Normal force components across the deformed region are not uniform in size.

This causes the normal forces to provide a net counterclockwise torque that slows the body. This is often called **rolling resistance** or sometimes **rolling friction** (though this term is quite misleading).

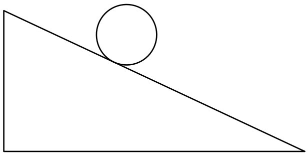
Friction is in fact what makes rolling motion possible. Let's do an example.

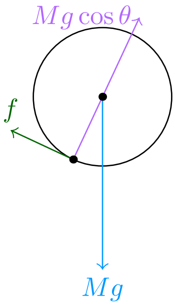
Example 11.2 ($F = ma$)**.** A hollow cylinder with a very thin wall (like a toilet paper tube) and a block are placed at rest at the top of a plane with inclination θ above the horizontal. The cylinder rolls down the plane without slipping and the block slides down the plane; it is found that both objects reach the bottom of the plane simultaneously. What is the coefficient of kinetic friction between the block and the plane?

Solution. This problem is not that hard if you consider the acceleration of both the block and the cylinder. For the block we have the $F = ma$ equation

$$ma_{\text{block}} = mg\sin\theta - \mu_k mg\cos\theta \implies a_{\text{block}} = g\sin\theta - \mu_k g\cos\theta$$

The acceleration of the hollow cylinder which will be a bit harder to find. First we draw a diagram.





We know that

$$mg\sin\theta - f = ma_{\text{cylinder}}$$

Newton's second law of rotation gives

$$-fr = I_{\text{cm}}\alpha \implies f = \frac{-I_{\text{cm}}\alpha}{R}$$

The rotational inertia of a hollow cylinder is mr^2. Substituting this result in for our equation gives us

$$f = \frac{-\left(mr^2\right)\left(-a_{\text{cm}}/r\right)}{r} = ma_{\text{cylinder}}$$

Taking this result back to our first equation for the hollow cylinder gives

$$mg\sin\theta - ma_{\text{cylinder}} = ma_{\text{cylinder}}$$

$$a_{\text{cylinder}} = \frac{1}{2}g\sin\theta$$

We now set $a_{\text{cylinder}} = a_{\text{block}}$ to get the equation

$$\frac{1}{2}g\sin\theta = g\sin\theta - \mu_k g\cos\theta$$

$$\mu_k = \frac{1}{3}\tan\theta$$

■

Another remarkable property of friction for an object rolling without slipping is the following.

Theorem 11.3. *For a rigid body that is continuously rolling without slipping, no net work will be done by friction.*

Proof. Because the body is "rigid," we can assume there is a single point of contact between the body and the surface on which it is rolling. Friction must act perpendicular to the surface of contact, so the torque provided is

$$\tau = fr.$$

Hence, the rotational work done by friction is equal to

$$W_R = \tau \cdot \theta = fr\theta.$$

On the other hand, the translational work done by friction is equal to

$$W_T = -f \cdot d.$$

In rolling motion, $d = r\theta$, so the total work done on the rolling body is zero. ∎

11.3 Analyzing Rolling Motion

Theorem 11.3 is important because it allows us to use conservation of energy! For a rigid body rolling down a ramp, energy is conserved. This provides us with a quick way of comparing different rigid bodies.

Corollary 11.4. *Given two bodies of mass m and radius r, that both roll without slipping down a ramp. If both objects start from rest and descend a height h, then:*

(a) The body with more rotational inertia will have the greater rotational kinetic energy.

(b) The body with more rotational inertia will take more time to descend a height h.

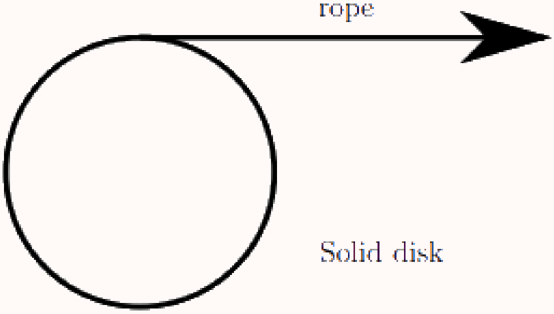
Proof. By conservation of energy, we have that

$$\begin{aligned} mgh &= \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2 \\ &= \left(\frac{I}{mr^2} + 1\right)(\frac{1}{2}mv^2). \end{aligned}$$

Hence, as I increases, the translational kinetic energy of the rotational object decreases. Thus we have that the body with greater rotational inertia will have more rotational energy. Likewise, this body will have less kinetic energy at any time, so it will take longer to descend the same height h. ∎

Students who just encounter rolling motion sometimes have a misconception that when applying a force F, the force is "split" between rotational and translational motion. However, this is not true; when rotational and translational motion happen simultaneously, Newton's laws should still be applied just as we have previously learned. The following example is *not* rolling motion, but is another example of combining translational and rotational motion, and is important nonetheless.

Example 11.5 ($F = ma$). A disk of moment of inertia I, mass M, and radius R has a cord wrapped around it tightly as shown in the diagram. The disk is free to slide on its side as shown in the top down view. A constant force of T is applied to the end of the cord and accelerates the disk along a frictionless surface.



After the disk has accelerated some distance, determine the ratio of the translational KE to total KE of the disk,

$$K_{\text{translational}}/K_{\text{total}}.$$

Solution. Note that the torque applied in this instant is

$$\tau = TR,$$

so the angular velocity of the disk at any time is

$$\omega = \alpha t = \frac{TR}{I}t.$$

Similarly, the velocity of the disk at any time is

$$v = at = \frac{T}{M}t.$$

Hence, the ratio of translational to total kinetic energy is

$$\begin{aligned}\frac{K_{\text{translational}}}{K_{\text{total}}} &= \frac{\frac{1}{2}M(\frac{T}{M})^2t^2}{\frac{1}{2}M(\frac{T}{M})^2t^2 + \frac{1}{2}I(\frac{TR}{I})^2t^2} \\ &= \frac{I}{I + MR^2}.\end{aligned}$$

■

The above result is evidently very different from rolling motion, since the ratio of translational to total kinetic energy now increases with rotational inertia. The reason for this is due to the fact that *the translational kinetic energy is in this scenario independent of inertia and rotational kinetic energy.* This occurs since there is no friction to oppose the sliding of the disk.

Remark. Compare this scenario to if the force T was applied perpendicular to the surface of the disk for an equal amount of time t. The kinetic energy change in this case would be less, since translational kinetic energy would be equal, but there would be no rotational kinetic energy.

This might be a bit confusing, as we have that

$$W = \int \vec{F} \cdot d\vec{x},$$

and since in both cases the disk has the same translational velocity, wouldn't distance (and hence work) be the same?

The answer is that we must also account for rotational work. In our initial example the cord wound around the disk will start to unravel, which makes rotational work possible. Recall that rotational work can be computed by

$$\Delta K_{\text{rot}} = \int \vec{\tau} \cdot d\vec{\theta}.$$

As rotational inertia increases, α (and hence θ) decreases. Hence, it makes sense that the ratio of translational to total kinetic energy will increase with rotational inertia.

11.4 Rolling with Slipping

Problems with **rolling with slipping** are not particularly complicated, but can be difficult for those who never encountered it before. Here is an example to give an idea of how to tackle these problems. *Remember that friction will act against the direction of motion until the body stops slipping.*

> **Example 11.6.** A uniform cue ball of mass m and radius r is struck by a large horizontal force F directed through the center of the ball. If the initial velocity of the ball after this collision is v_0, and the coefficient of friction between the ball and table is μ, for how long does the ball travel before it begins rolling without slipping?

Solution. In this problem, it does not necessarily hold that $v = r\omega$. Instead, friction will convert some of the translational kinetic energy to rotational kinetic energy such that the object rolls without slipping. Note the only external force on the ball is friction, where $f = \mu mg$. By Newton's Second Law for rotation, we have that

$$\begin{aligned}\vec{\alpha} = \frac{\vec{\tau}}{I} &= \frac{r \times \mu mg}{\frac{2}{5}mr^2} \\ &= \frac{5\mu g}{2r}.\end{aligned}$$

Since the ball initially does not rotate,

$$\vec{\omega} = \vec{\omega}_0 + \vec{\alpha}t = \frac{5\mu g}{2r}t.$$

We can write a similar equation for translational motion, since

$$\vec{a} = \frac{\vec{F}}{m} = -\mu g.$$

We have that the initial velocity is v_0, so $\vec{v} = v_0 - \mu g t$. We wish to find the time at which $\vec{v} = r\vec{\omega}$, so we have

$$v_0 - \mu g t = \frac{5\mu g}{2}t.$$

This happens at $t = \boxed{\dfrac{2v_0}{7\mu g}}$. ■

Just like with other rotational dynamics problems, we can use force and torque analysis in our usual way. Let's do another example.

Example 11.7. A uniform sphere of mass m is placed on a long board also of mass m, and is given an initial angular velocity ω_0 and initial horizontal velocity v_0. Friction between the board and ground is negligible, but friction between the board and sphere is not negligible. What is the critical velocity v_c such that the ball will eventually stop rotating with respect to the board?

Solution. By force and torque analysis, we see that

$$v_{\text{sphere}} = v_0 - \frac{F}{m}t = v - \mu g t,$$

$$\omega = \omega_0 - \frac{\tau}{m}t = \omega_0 - \frac{5\mu g}{2r}t.$$

We can similarly compute that $v_{\text{board}} = \mu g t$. In this case rolling without slipping occurs when

$$v_{\text{sphere}} = v_{\text{board}} = r\omega.$$

Plugging in, we see that

$$v - 2\mu g t = r\omega_0 - \frac{5}{2}\mu g t = 0.$$

Hence, we see that

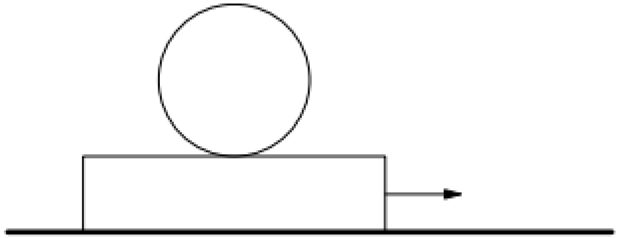
$$v = 2\mu g t = \boxed{\frac{4}{5}r\omega_0}.$$

■

11.5 Homework Problems and Solutions

For these problems, recall that the moment of inertia of a thin shell of radius r and mass m about the center of mass is $I = \frac{2}{3}mr^2$; the moment of inertia of a solid sphere of radius r and mass m about the center of mass is $I = \frac{2}{5}mr^2$.

Problem 11.1 ($F = ma$)**.** [5] A rectangular slab sits on a frictionless surface. A sphere sits on the slab. There is sufficient friction between the sphere and the slab such that the sphere will not slip relative to the slab. A force to the right is applied to the slab, with both the slab and the sphere initially at rest.

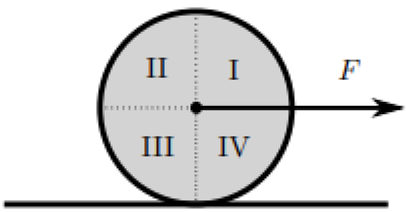


(a) In which direction will the sphere begin to spin?

(b) In which direction will the center of mass accelerate?

Solution. Since the sphere does not slip relative to the slab, it must experience a friction force to the right. The friction force provides the only torque on the sphere, and since it points right the sphere will accelerate counterclockwise. The frictional force is also the only horizontal force it experiences, so the center of mass will accelerate to the right. ■

Problem 11.2 ($F = ma$). [6] A uniform disk is being pulled by a force F through a string attached to its center of mass. Assume that the disk is rolling smoothly without slipping. At a certain instant of time, in which region of the disk (if any) is there a point moving with zero total acceleration?



(A) Region I

(B) Region II

(C) Region III

(D) Region IV

(E) All points on the disk have non-zero acceleration

Solution. Note that the force F provides a constant translational acceleration $\vec{a}$ to each point on the disk. If the total acceleration at a point is zero, the other components of acceleration must sum to point exactly to the left.

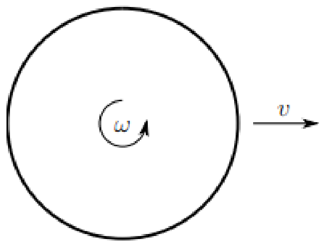
We now look at the acceleration caused by the rotational motion. Note that the centripetal acceleration $\vec{a}_c$ at any point will point towards the center of the disk. The tangential acceleration of the disk will point in the direction of rotation, since the disk is speeding up. We can do casework on the $\vec{a}_c$ and $\vec{a}_t$ vectors in each region of the disk:

1. In region I, $\vec{a}_c$ points to the bottom left and $\vec{a}_t$ points to the bottom right.

2. In region II, $\vec{a}_c$ points to the bottom right and $\vec{a}_t$ points to the top right.
3. In region III, $\vec{a}_c$ points to the top right and $\vec{a}_t$ points to the top left.
4. In region IV, $\vec{a}_c$ points top the top left and $\vec{a}_t$ points to the bottom left.

Thus we see that region IV is the only region where the net rotational acceleration $\vec{a}_c + \vec{a}_t$ can point completely to the left, which gives us our answer. ■

Problem 11.3 ($F = ma$)**.** [6] As shown in the figure, a ping-pong ball with mass m, radius R, with initial horizontal velocity v and angular velocity ω comes into contact with the ground. Friction is not negligible, so both the velocity and angular velocity of the ping-pong ball changes. What is the critical velocity v_c such that the ping-pong will stop and remain stopped? Treat the ping-pong ball as a hollow sphere.



Solution. If the ball is to come to rest, we can assume the total angular momentum to be zero. That is,

$$L = mv_cR + I\omega = 0,$$

so

$$mv_cR = \frac{2}{3}mR^2\omega \implies v_c = \boxed{\frac{2}{3}R\omega}.$$

■

Problem 11.4 (Morin)**.** [6] A uniform ball initially slides, without rotating, on the ground. Friction with the ground eventually causes the ball to roll without slipping. If the initial linear speed is v_0, what is the final linear speed?

Solution. The ball starts rolling when $v = r\omega$. Substituting $at = \frac{Ft}{m}$ and $\alpha t = \frac{\tau t}{I}$, we have

$$v_f = v_0 - \frac{ft}{m} = r\frac{frt}{\frac{2}{5}mr^2}.$$

Rearranging, the first equation becomes

$$\frac{ft}{m} = v_0 - v_f,$$

and the second equation becomes

$$\frac{ft}{m} = \frac{2}{5}v_f.$$

We can now set the right-hand sides equal to solve for v_f, and we obtain $v_f = \boxed{\frac{5}{7}v_0}$.

■

Problem 11.5 ($F = ma$). [6] A spherical shell of mass M and radius R is completely filled with a frictionless fluid, also of mass M. It is released from rest, and then it rolls without slipping down an incline that makes an angle θ with the horizontal. What will be the acceleration of the shell down the incline just after it is released?

Solution. The inertia of the shell about its center of mass is $I = \frac{2}{3}MR^2$. In this situation it is easier for us to find the angular acceleration by finding the gravitational torque relative to the contact point. Note that fluid is frictionless, which means it will not rotate within the shell and will have inertia equivalent to a point mass. Thus, by the parallel axis theorem, we have

$$I = I_c + 2MR^2 = \frac{8}{3}MR^2.$$

The torque given by the gravitational forces is

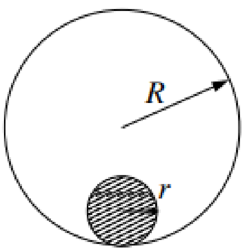
$$\tau = FR\sin\theta = 2MgR\sin\theta.$$

Thus, we can compute

$$\alpha = \frac{\tau}{I} = \frac{3g\sin\theta}{4R},$$

and in rolling without slipping $a = R\alpha$, so $a = \boxed{\frac{3}{4}g\sin\theta}$. ■

Problem 11.6 ($F = ma$). [7] A disk of radius r rolls uniformly without slipping around the inside of a fixed hoop of radius R. If the period of the disc's motion around the hoop is T, what is the instantaneous speed of the point on the disk opposite to the point of contact?

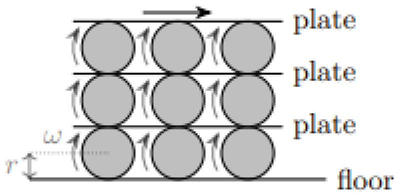


Solution. Recall the velocity at the top is twice the velocity of the center of mass. Now, we compute the distance the center of mass travels in a period T. It traverses the circumference of a circle of radius $R - r$, so we have

$$v_{\rm cm} = \frac{2\pi(R-r)}{T},$$

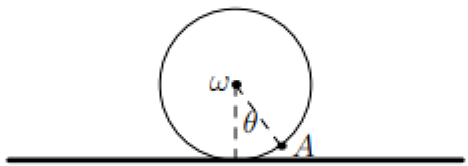
which we can double to get our answer $v = \boxed{4\pi(R-r)/T}$. ■

Problem 11.7 ($F = ma$). [7] A system of cylinders and plates is set up as shown. The cylinders all have radius r, and roll without slipping to the right with angular velocity ω. What is the speed of the top plate?



Solution. In this problem, we should view rotation as rotation at angular velocity ω about the contact point. In this case, we see the lowest plate moves at $2r\omega$ relative to the floor. The next plate up moves at $2r\omega$ relative to the lowest plate, or $4r\omega$ relative to the floor. Thus the top plate moves at speed $\boxed{6r\omega}$ relative to the floor. ■

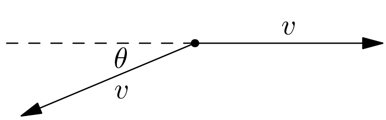
Problem 11.8 ($F = ma$)**.** [7] A wheel of radius R rolls without slipping at angular velocity ω.



For point A on the wheel at an angle θ with respect to the vertical, shown in the figure, what is the magnitude of its velocity with respect to the ground?

(A) ωR
(B) $\omega R \sin(|\theta|/2)$
(C) $\sqrt{2}\omega R \sin(|\theta|/2)$
(D) $2\omega R \sin(|\theta|)$
(E) $2\omega R \sin(|\theta|/2)$

Solution. Point A has translational velocity v to the right, and a rotational velocity v tangent to the wheel and directed to the lower left. Adding these two vectors, we have the following diagram:

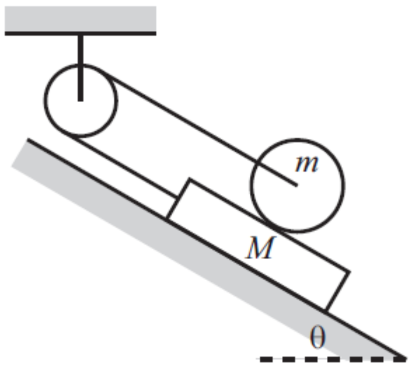


To sum these two vectors, consider the angle bisector between them. By symmetry, we can sum their components along the angle bisector. This is

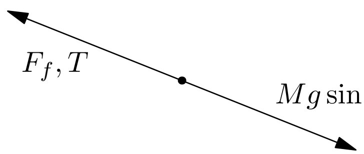
$$2v\cos(\frac{1}{2}(180-\theta)) = 2v\cos(90-\frac{\theta}{2}) = 2v\sin(\frac{\theta}{2}).$$

Substituting $v = R\omega$, we see the magnitude is given by $\boxed{2\omega R \sin(|\theta|/2)}$. ■

Problem 11.9 (Morin). [9] A uniform wheel with mass m and radius R lies on top of a board with mass M (you can assume $M > m$), which lies on a frictionless plane inclined at angle θ, as shown in the below figure. A massless string wraps around a massless pulley and has its ends tied to the board and the wheel's axle. Assume that the wheel rolls without slipping on the board, and assume that the two string segments shown are parallel to the plane. What is the acceleration of the board?



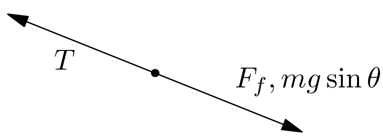
Solution. We are given $M > m$, so the wheel will move up the plane while the board slides downwards. Since the wheel rolls without slipping, it will roll counterclockwise. We can draw the following free body diagram for the block:



We subsequently use force analysis on the board:

$$\sum F = Mg\sin\theta - T - F_f = Ma.$$

Similarly, we can draw the following free-body diagram for the wheel:



We now use force and torque analysis on the wheel. We have

$$\sum F = T - F_f - mg\sin\theta = ma.$$

$$F_f R = \left(\frac{1}{2}mR^2\right)\alpha \implies F_f = \frac{mR\alpha}{2a}.$$

The next step is tricky. We might expect $\alpha = a/R$, but this is not correct. The wheel and board are accelerating in opposite directions, so relative to the board the wheel has acceleration $2a$. Thus, we have

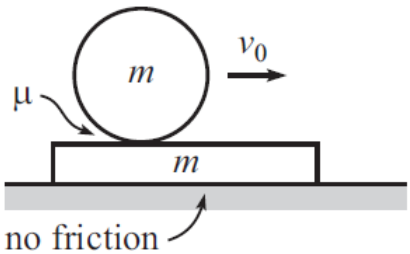
$$F_f = ma.$$

We now add our two equations for force analysis to get

$$(M-m)g\sin\theta - 2F_f = (M+m)a.$$

Substituting for F_f and solving for a, we get $\boxed{a = \dfrac{M-m}{M+3m} g\sin\theta}$. ■

Problem 11.10 (Morin). [9] A uniform solid cylinder with mass m and radius R lies on top of a long board which also has mass m, as shown in the below figure. The board is free to slide frictionlessly on a table, but there is kinetic friction between the cylinder and the board, with the coefficient of kinetic friction equal to μ. If the board is initially at rest, and if the cylinder is given an initial speed v_0 to the right, but without any initial rotation, what is the speed of the cylinder (with respect to the ground) when it finally rolls without slipping on the board?



Solution. First we right equations for the velocity of both the cylinder and board. The cylinder will have velocity given by

$$v_c = v_0 - \mu g t,$$

and the board will have velocity

$$v_b = \mu g t.$$

The torque experienced by the cylinder is $\mu m g R$, so the angular acceleration is

$$\alpha = \frac{\mu m g R}{\frac{1}{2}mR^2} = \frac{2\mu g}{R}.$$

Thus we have $\omega = \frac{2\mu g}{R}t$. in the reference frame of the board, the cylinder moves at speed $v_c - v_b$. Thus, when the cylinder is rolling without slipping we have $v_c - v_b = R\omega$. Substituting the appropriate functions, we have

$$v_0 - 2\mu g t = 2\mu g t \Rightarrow t = \frac{v_0}{4\mu g}.$$

Substituting for t now gives us $v_c = \boxed{\dfrac{3}{4}v_0}$. ■

Problem 11.11 (Kalda). [9] A ball is rolling along a horizontal floor in the region $x < 0$ with velocity $\vec{v_0} = (v_{x0}, v_{y0})$. In the region $x > 0$ there is a conveyor belt that moves with velocity $\vec{u} = (0, u)$ (parallel to its edge $x = 0$). Find the velocity $\vec{v} = (v_x, v_y)$ with respect to the belt after it has rolled onto the belt. The surface of the conveyor belt is rough (the ball does not slip) and is level with the floor.

Solution. Let us direct the z axis upward (this will fix the signs of the angular momenta). We first attempt to find the initial angular momentum. We note that

$$L = MvR + I\omega$$

Substituting in $I = \frac{2}{5}MR^2$ and $\omega = v/R$ gives us $L = \frac{7}{5}MvR$. In the x-axis, the sign of angular momentum is negative because of the right hand rule, and in the y-axis the sign of angular momentum is postive. This gives us,

$$L_x = -\frac{7}{5}Mv_{y0}R$$
$$L_y = \frac{7}{5}Mv_{x0}R$$

The ball will continue to move in the same velocity in the y-direction as no non-conservative forces are acting in the horizontal direction. In the x-axis, the ball will have a final velocity of u, which implies that the final angular momentum is

$$L_{xf} = -\frac{7}{5}Mv_yR - MuR$$

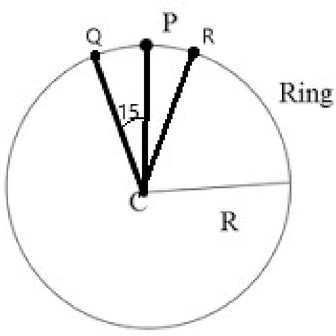
Setting this equal to the initial angular momentum because of conservation of angular momentum, we get

$$-\frac{7}{5}Mv_yR = -\frac{7}{5}Mv_yR - MuR$$
$$\frac{7}{5}v_{y0} = \frac{7}{5}v_y + u$$
$$v_y = v_{y0} - \frac{5}{7}u$$

This gives the final velocity to be $\boxed{\left(v_{x0}, v_{y0} - \frac{5}{7}u\right)}$.

■

Problem 11.12. [9] Three masses, O, P, and Q, are attached to the ring of mass m and radius r. Masses O and Q have mass $2m$ and the mass of P is m. The angle between 2 masses is 15 degrees as shown in the figure.



Find the maximum velocity the ring must roll with so that it doesn't hop while rolling, i.e. for the ring to roll normally (without slipping), not bouncing. Assume the ring is initially at rest and given a negligible push.

Solution. Since the maximum amount velocity in pure rolling will occur when P is on the bottom of the wheel (ask if you want more info), the change in potential energy compared to the original position is equal to

$$2mg(2R)\cos\theta + 2mg(2R)\cos\theta + mg(2R),$$

where $\theta = 15^\circ$. This is equal to

$$8mgR\cos\theta + 2mgR.$$

Factoring the above and applying law of conservation of energy gives

$$2mgR(1 + 4\cos\theta) = \frac{1}{2}I(\omega)^2 + \frac{1}{2}(6m)v^2.$$

Since $\omega^2 = \frac{v^2}{R^2}$, substituting gives

$$2mgR(1 + 4\cos\theta) = \frac{1}{2}v^2\left(\frac{I}{R^2} + 6m\right).$$

Solving for v, we have

$$v = \sqrt{\frac{4mgR(1 + 4\cos\theta)}{\frac{I}{R^2} + 6m}}.$$

We now find I. We are going to neglect the spokes mass and just assume that O, P and Q are just points on the rim. Since the entire mass distribution is a radius R from the center, we have

$$I = 6mR^2.$$

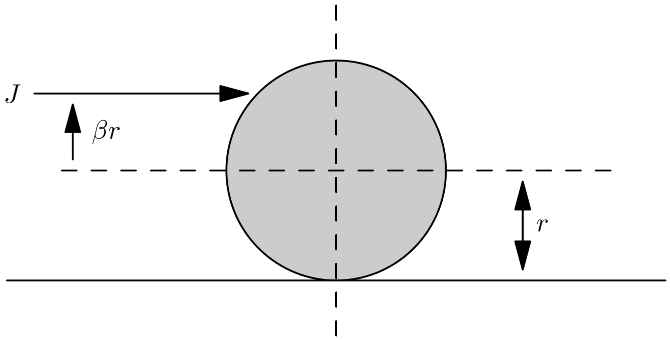
We can further simplify the radical to get

$$\boxed{v = \sqrt{\frac{1}{3}\cdot gR(1 + 4\cos 15^\circ)} = \sqrt{\frac{1}{3}\cdot gR(1 + \sqrt{2} + \sqrt{6})}}.$$

■

11.5.1 Written Solutions

Problem 11.13 (USAPhO). [14] A uniform pool ball of radius r and mass m begins at rest on a pool table. The ball is given a horizontal impulse J of fixed magnitude at a distance βr above its center, where $-1 \leq \beta \leq 1$. The coefficient of kinetic friction between the ball and the pool table is μ. You may assume the ball and the table are perfectly rigid. Ignore effects due to deformation. (The moment of inertia about the center of mass of a solid sphere of mass m and radius r is $I_{cm} = \frac{2}{5}mr^2$.)



(a) [6] Find an expression for the final speed of the ball as a function of J, m, and β.

(b) [8] For what value of β does the ball immediately begin to roll without slipping, regardless of the value of μ?

Solution.

(a) The translational impulse is J, and the rotational impulse relative to the center of mass is βrJ. After the ball starts rolling without slipping, we have $v = r\omega$, which means the total angular momentum is $L = I\omega + mvr = \frac{7}{5}mvr$. This must be equal to the sum of rotational and translational impulse, so we have

$$(\beta + 1)rJ = \frac{7}{5}mvr,$$

and rearranging gives

$$v = \boxed{\frac{5J}{7m}(\beta + 1)}.$$

(b) The cue ball will instantly start rolling with slipping if and only if $a = r\alpha$. We have that $a = \frac{F}{m}$, and the torque τ is given by

$$\tau = Fr\sin\theta = F\beta r.$$

The inertia of the ball is $I = \frac{2}{5}mr^2$, so substituting into $a = r\alpha$, we see

$$\frac{F}{m} = r \times \frac{F(h-r)}{\frac{2}{5}mr^2}$$

$$\frac{2}{5}r = \beta \implies \beta = \boxed{\frac{2}{5}}.$$

■

12 Statics

12.1 Linear and Rotational Equilibrium

In this section, we'll use some of the tools we learned in the forces and rotation handouts and apply them to a special subset of problems that are largely prevalent in Olympiads: **statics**.

Note that not much of the content in this handout is not necessarily *new*, rather, we are applying our current theory to a special type of problem. Thus, most of this handout will actually consist of solved examples rather than theory as usual.

We first provide some definitions:

Definition 12.1. A system is in **translational equilibrium** if

$$\sum \vec{F}_{\text{ext}} = 0$$

Definition 12.2. A system is in **rotational equilibrium** about some axis if

$$\sum \vec{\tau}_{\text{ext}} = 0$$

Definition 12.3. A system is in **static equilibrium** if the system is in both translational and rotational equilibrium. That is, we have that

$$\sum \vec{F}_{\text{ext}} = 0 \text{ and } \sum \vec{\tau}_{\text{ext}} = 0$$

But wait! Doesn't the net torque depend on the axis of rotation? Normally it does, but we will soon see that for statics problems they don't! In particular, we prove the following theorem:

Theorem 12.4 (Torques are equal in translational equilibrium). *If a system is in translational equilibrium, then the net torque is the same across all axes. That is, if the net torque about some axis located at a point O is $\sum \tau_O$ and the net torque about some axis located at a point P is $\sum \tau_P$, then*

$$\sum \vec{F}_{ext} = 0 \implies \sum \vec{\tau}_O = \sum \vec{\tau}_P$$

Proof. Assume that $\sum \vec{F}_{\text{ext}} = 0$ and let the net torque about axis O be $\vec{\tau}_0$. Relative to the origin O, force $\vec{F}_1$ is applied at the point located at $\vec{r}_1$, force $\vec{F}_2$ at $\vec{r}_2$ and so on. Then

$$\vec{\tau_O} = \sum_{n=1}^{N} \vec{\tau_n} = \sum_{n=1}^{N} \vec{r_n} \times \vec{F}_n$$

Let the displacement of point P with respect to point O be $\vec{r}_P$. Then, the net torque about point P is

$$\tau_P = \sum_{n=1}^{N} (\vec{r}_n - \vec{r}_P) \times \vec{F}_n = \sum_{n=1}^{N} \vec{r}_n \times \vec{F}_n - \sum_{n=1}^{N} \vec{r}_P \times \vec{F}_n$$

where the last line holds because of the distributivity of the cross product. However, note that since $\sum \vec{F}_{\text{ext}} = 0$, we have

$$\vec{\tau}_P = \sum_{n=1}^{N} \vec{r}_n \times \vec{F}_n - \sum_{n=1}^{N} \vec{r}_P \times \vec{F}_n = \sum_{n=1}^{N} \vec{r}_n \times \vec{F}_n - \vec{r}_P \times \left(\sum \vec{F}_{\text{ext}}\right) = \vec{\tau}_O - 0 = \vec{\tau}_O$$

This implies that if the system is in transitional equilibrium, then the net torque is the same across all all axes proving the theorem. ■

This theorem actually simplifies many things for us, because we can now actually choose *any* axis to get the extra torque condition! This is actually substantial, as often choosing the correct axis can simplify an extremely difficult problem into a rather simple one. This is because although theoretically, any axis you choose should give the same answer, choosing a particular axis through a point where multiple forces act greatly simplifies the problem as you can ignore those forces (since $\vec{r} = 0$). In the next section, we solve a few examples that highlight this technique:

12.2 Examples

We proceed with a few examples that highlight the basic structure of a statics problem. Try solving them first before you read the solution:

> **Example 12.5 (Classical).** A wooden plank of mass m of length l is propped up against a frictionless wall. The coefficient of static friction between the plank and the floor is μ. Find the angle θ the plank makes, as measured from the vertical.

Solution. We first consider the forces that act on the plank. There is going to be a gravitational force Mg acting at the center of mass of the plank, a normal force N between the floor and the plank, a frictional force $f = \mu N$ between the plank and the floor, and a force F between the wall and the plank.

Setting $\sum F_y = 0$ we get that

$$N - Mg = 0 \implies N = mg$$

Setting $\sum F_x = 0$, we get

$$F - \mu Mg = 0 \implies F = \mu Mg$$

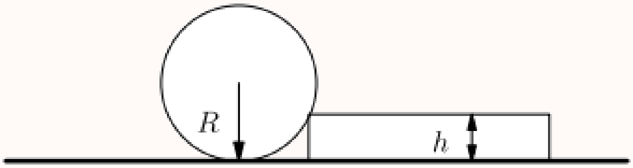
We now choose the point of contact between the floor and the plank to be our axis of rotation. This is because there are two forces acting on it, so we can eliminate two forces rather than just one in our torque equation. Setting $\sum \tau = 0$, we get

$$\frac{l}{2} \sin\theta Mg = l \cos\theta \mu Mg \implies \tan\theta = 2\mu \implies \boxed{\theta = \tan^{-1}(2\mu)}$$

In particular, note that the angle does not depend on the length of the board! This makes sense physically, as when you think about it, if you scale (dilate) the board, you would not expect it to slant at different angles. ■

We now delineate the way in which we can choose an appropriate axis to greatly simplify a problem. Using different axes should ultimately give the same result, but the choosing of an appropriate axis will often make problems much easier as we do not need to consider forces acting at the axis (because the moment arm will be zero). The following example clears up this method:

Example 12.6 (F=ma). A cylinder has radius R and weight G. You try to roll it over a step of height $h < R$. What is the minimum force needed to roll the cylinder over?



Solution. We first think a little bit about the constraints of the problem and what forces are to act on the cylinder. There is an applied force $\vec{F}$ that we can apply at any point on the cylinder and at any direction. There's going to be a contact force between the step and the cylinder as well. Finally, there's going to be a gravitational force $\vec{G}$ pointing down at the center of mass of the cylinder due to its weight. Let's think about what axis will be easiest to use to balance the torques and find F.

We first begin by asking the question: which force is the most complicated? We really have two places we can set our principal axis here: at the center of mass where the weight $\vec{G}$ acts or the point of contact between the cylinder and the step. Note that the weight G is a simple force that points verically down. In contrast, we don't really know what direction the contact force will face, only that it will cancel out the gravitational force and the applied force F. For this problem, we thus choose the point of contact as we can eliminate a much more complicated force (compared to the weight).

Note that the moment arm for the force $\vec{G}$ is the horizontal distance from the center of the cylinder to the step. By the Pythagorean Theorem, we have the torque from the weight force is

$$\vec{G}\sqrt{R^2 - (R-h)^2}.$$

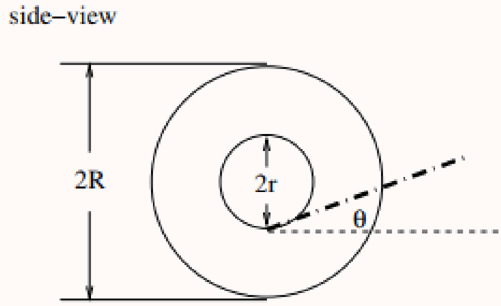
Next, note that we want to find the *minimum* force required to roll the cylinder over the step. To do this, we want the force to provide the maximum amount of torque with the smallest force possible. How do we do this? With a larger moment arm of course! The farthest point of contact between our axis and the cylinder is at the point on the opposite side of the cylinder, a distance $2R$ away. Letting F be the applied force, we then get that

$$\tau_{\text{net}} = 2RF - G\sqrt{R^2 - (R-h)^2} = 0 \implies \boxed{F = \frac{G\sqrt{2Rh - h^2}}{2R}}$$

■

The next example is a slightly trickier application of this method. Try solving it first before looking at the solution!

Example 12.7 (F=ma). A spool is made of a cylinder with a thin disc attached to either end of the cylinder, as shown. The cylinder has radius $r = 0.75$ cm and the discs each have radius $R = 1.00$ cm. A string is attached to the cylinder and wound around the cylinder a few times. At what angle above the horizontal can the string be pulled so that the spool will slip without rotating?

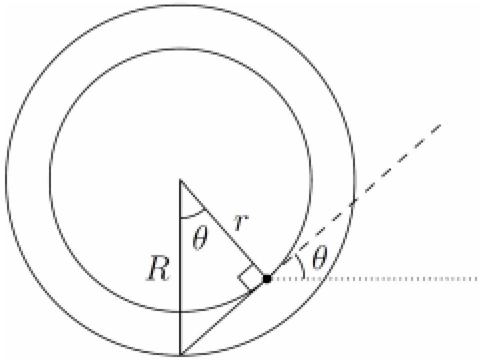


Solution. Let's first consider when we have slipping without rotating. This will begin to happen when there is no net torque and the net force just barely becomes nonzero. Basically, this situation begins to happen when the system is in static equilibrium! So above all, we first note that this problem is largely a statics problem.

Let's now try considering which forces act on the spool. This problem is quite tricky because we don't exactly know what the magnitudes of any of the forces are! At the center of mass of the spool, there will be the spool's weight W. At the point of contact between each disc there will be a normal force N (that may or may not be 0 depending on the magnitude of the applied force), a friction force f, and finally the applied force F by the string. Let's now consider a suitable axis so that we can find the angle θ the string makes with the horizontal.

Clearly the best candidate for an axis is the point of contact with the discs and the ground, because unlike all the other points, 2 forces act at that point, namely N and f.

We now consider when $\sum \tau = 0$. Notice that the line of action of the weight W passes through the point of contact with the disks and the ground (that is, our axis). Thus, the weight W contributes no torque relative to our axis. Then, the only force left is the applied force F. But because all the other forces contribute 0 torque, F must also contribute 0 torque! When does this happen? Well, F must *also* have a line of action through the point of contact.



Considering the geometry of the spool (as shown above), we thus get

$$\cos\theta = \frac{r}{R} = 0.75 \implies \boxed{\theta = 41.4^\circ}$$

■

This next example is quite tricky, as it requires a bit of ingenuity with the placement of the axis and one needs to be particularly careful about the forces that are acting. Again, try solving it on your own before looking at the solution. Good luck!

Example 12.8 (Morin). Two sticks, each of mass m and length l, are connected by a hinge at their top ends. They each make an angle θ with the vertical. A massless string connects the bottom of the left stick to the right stick, perpendicularly as shown in the figure. The whole setup stands on a frictionless table.

(a) What is the tension in the string?

(b) What force does the left stick exert on the right stick at the hinge? (Hint: No messy calculations required!)

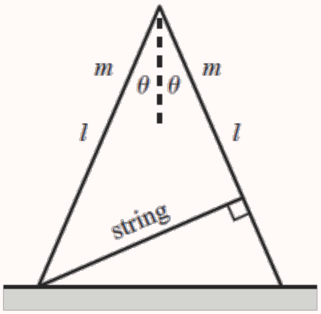
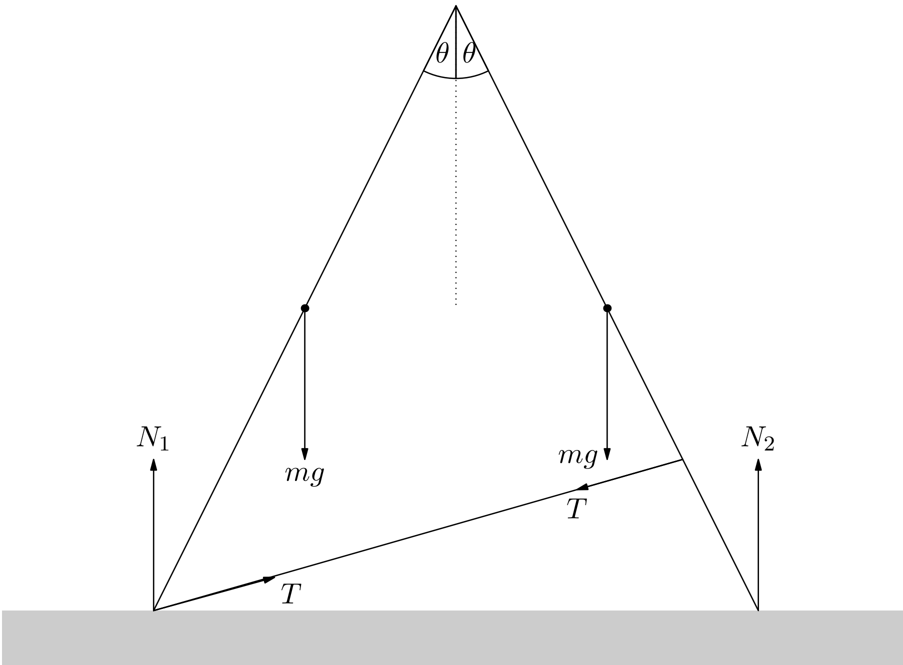


Fig. 2.37

Solution. (a) This problem is tricky in that we must define our systems precisely when finding the values of different forces that are pertinent to the problem. Let's first consider the forces acting on the system:



There are the weights of the rods mg acting at the center of masses of the sticks, a normal force N_1 acting on the left stick, a normal force N_2 acting on the right stick, and a tension T throughout the string. Since the system is in equilibrium, we get that

$$N_1 + N_2 = 2mg.$$

Let's get some more information about the normal forces. A clear target to use the torque equation is the axis perpendicular to the page through the point of contact of both sticks with each other, because we don't really know what this force is yet. By symmetry, the torques due to gravity on both sticks cancel out and since the line of action to the tension(s) on both sticks is the same, the torques due to the tension(s) also cancel out. Thus, we get that

$$\sum \tau = l \sin\theta N_1 - l \sin\theta N_2 = 0 \implies N_1 = N_2$$

So $N_1 = N_2 = mg$! So it turns out the string doesn't change the normal forces! Let's now define an appropriate system and axis to find the tension in the string. Because the string is massless, the tension T is the same throughout the string. We have a few appropriate choices to find the tension T. Let's choose our system to be the right stick, since the moment arm for the tension T in the right stick is easy to calculate. Again, we'll choose the pivot point as our choice of axis, because there are quite a few unknown forces acting on it. Balancing the torques on the right stick to be zero, we then get

$$\sum \tau = \frac{l}{2} mg \sin(\theta) + l \cos(2\theta) T - l \sin(\theta) N_2 = 0 \implies \boxed{T = \frac{mg \sin\theta}{2\cos 2\theta}}.$$

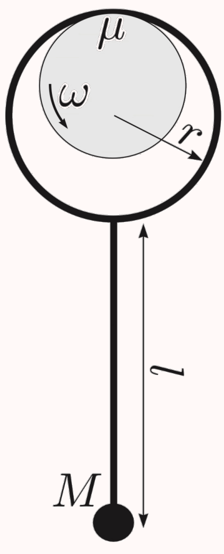
(b) Let's reconsider the forces on the right stick: remark that the normal force N_2 balances out the weight mg of the stick. The only other forces acting on the stick are the tension T of the rope on the stick along with the force the left stick exerts on the right stick. Ah-hah! Since each stick is in static equilibrium, these two forces must balance one another out, so the magnitude of the force F that the left stick exerts on the right stick must just be equal to the tension T and opposite in direction! That is

$$\boxed{F = \frac{mg \sin\theta}{2\cos 2\theta}}.$$

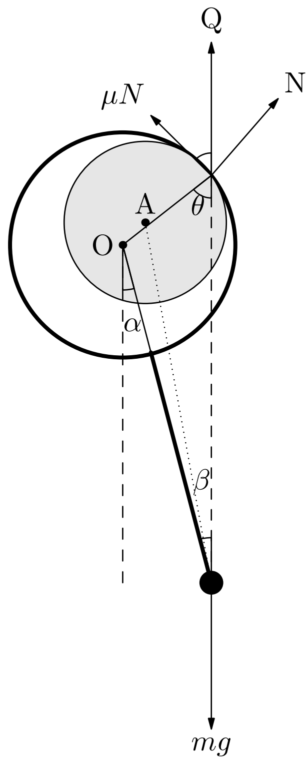
■

Our final example will show how helpful creating a diagram is in statics problems. Make sure to pay close attention to this problem as it is a very hard one.

Example 12.9 (Kalda). An end of a light wire rod is bent into a hoop of radius r. The straight part of the rod has length l; a ball of mass M is attached to the other end of the rod. The pendulum thus formed is hung by the hoop onto a revolving shaft. The coefficient of friction between the shaft and the hoop is μ. Find the equilibrium angle between the rod and the vertical.



Solution. The hardest thing about this problem is drawing a diagram. Here we provide a diagram for us to work with. Let O be the center of the hoop and A the center of the revolving shaft.



Let Q be the vector sum of the friction and normal forces[1],

$$Q = \sqrt{\mu^2 N^2 + N^2} = N\sqrt{\mu^2+1}$$

because the system is in equilibrium, then the frictional force, μN, must be equal to $mg\sin\theta$. We also know by simple trigonometry that $\mu N = Q\sin\theta$. Therefore, because the sum of forces are zero we have,

$$\mu N = mg\sin\theta = N\sqrt{\mu^2+1}\sin\theta.$$

We must now establish this relation in terms of β. One may look towards a torque analysis, however a more elegant mathematical approach is by the law of sines. We know by law of sines that

$$\frac{\sin\beta}{r} = \frac{\sin\theta}{r+\ell} \implies \sin\theta = \frac{(r+\ell)\sin\beta}{r}$$

Substituting this in for $\sin\theta$ we find

$$\mu N = N\sqrt{\mu^2+1}\,\frac{(r+\ell)\sin\beta}{r}$$

$$\sin\beta = \frac{r\mu}{(r+\ell)\sqrt{\mu^2+1}} \implies \boxed{\beta = \sin^{-1}\left(\frac{r\mu}{(r+\ell)\sqrt{\mu^2+1}}\right)}$$

[1] The frictional force is not constant throughout the entire process of slipping however it is maximum (or μN) when the shaft is at equilibrium angle.

Remark. There is in fact a smarter solution if you say that $\theta = \arctan \mu$. As an exercise, try to figure out why and try solving this problem again with that in mind.

■

12.3 Massive Ropes

We'll now consider a few other miscellaneous statics examples. These examples are pretty common and well known, so make sure you understand how we solve through them:

So far throughout the course, we've mainly been considering strings or massless ropes. What if the rope *does* have mass though? We'll consider the statically impacts this has in the following problem:

> **Example 12.10.** A rope of mass density $\lambda = 0.2$ kg/m and length $l = 4$ m is hung the top of the ceiling. Consider a point 3/4 of the way from the bottom of the rope. What is the tension in the rope at that point?

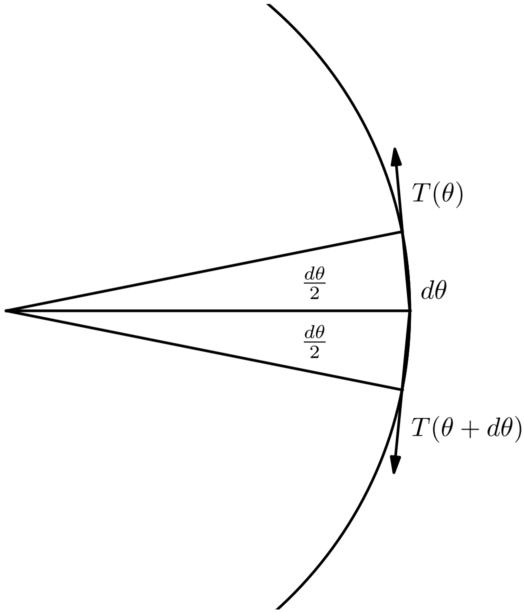
Solution. When a rope has mass, every point in the rope has to hold up the mass above it otherwise the rope wouldn't be in equilibrium. Therefore, the point 3/4 of the way from the bottom has to hold up the 1/4 of mass that is above it. Or in other words,

$$T = m_{\text{eff}} g = \lambda \frac{l}{4} g = 2 \text{ N}.$$

■

> **Example 12.11** (HRK). A massless rope is tossed over a wooden dowel of radius r in order to lift a heavy object of weight W off of the floor, as shown in Fig. The coefficient of sliding friction between the rope and the dowel is μ. Show that the minimum down- ward pull on the rope necessary to lift the object is $F_{down} = We^{\pi\mu}$.

Solution. Let the left of the dowel be at $\theta = 0$, and have θ increase as the rope wraps clockwise around the dowel.



Consider the forces on the $d\theta$ segment of rope. The net force is obviously 0.

$$N - T\left(\theta + d\theta\right)\sin\left(\frac{d\theta}{2}\right) - T(\theta)\sin\left(\frac{d\theta}{2}\right) = 0$$

$$N\mu - T\left(\theta + d\theta\right)\cos\left(\frac{d\theta}{2}\right) + T(\theta)\cos\left(\frac{d\theta}{2}\right) = 0$$

Since $\theta << 1$, $\cos(\frac{\theta}{2}) \approx 1$, and $\sin(\frac{\theta}{2}) \approx \frac{\theta}{2}$

$$2\mu T(\theta)\frac{d\theta}{2} = T(\theta + d\theta) - T(\theta)$$

$$\mu T(\theta) = T'(\theta) \implies T(\theta) = Ce^{\mu\theta}$$

$$T(0) = W \implies C = W$$

Thus, the minimum downward force is

$$F_{down} = T(\pi) = We^{\mu\pi}.$$

Remark. This is called the Capstan equation and is a classic problem found in almost any introductory mechanics textbook.

■

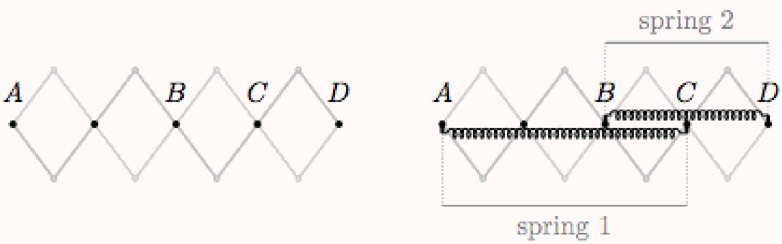
12.4 Virtual Work

So far, in the statics problems we have encountered thus far, we have mainly used free body diagrams and the like to solve problems. For some problems however, it is useful to consider energy in the statics problem. In statics problems however, because the system is, as the name suggests, static, we do not actually consider the dynamical changes in the system. Rather, we analyze *how* the energy were to be changed *had* the system been perturbed. *This* is the concept of **virtual work**. We'll go over a few examples in the section to highlight the core concept, but if your still struggling to understand the general procedure, refer to the pdf on the bottom of page 8.

Concept. Imagine that we are able to change the length in a string or rod. The length will increase by an infinitesimal amount δx. Equating the work $T\delta x$ (where T is the tension) by the change δK of the kinetic energy, gives us $T = \delta K/\delta x$. Since $\delta K + \delta U = 0$ in a conserved system, we thus get $T = -\delta U/\delta x$.

The best way to understand this concept is through a simple example.

Example 12.12 ($F = ma$)**.** An extendable arm is made from rigid beams free to pivot around the dots shown. Spring 1, with equilibrium length L_1, is attached between points A and C while spring 2, with equilibrium length L_2 is attached between B and D. The system is allowed to come to equilibrium. In equilibrium, what is the ratio of the tension in spring 1 to the tension in spring 2?



Solution. This problem can be solved very quickly with the principle of virtual work (in fact, you can probably solve it with a look of an eye!). Consider what happens when the setup is extended by a distance δx. The length of spring 1 would increase by $\frac{3}{4}\delta x$ and the length of spring 2 would increase by $\frac{1}{2}\delta x$ to compensate for length increase. This tells us that the virtual work done is given by

$$T_1 \frac{3}{4}\delta x + T_2 \frac{1}{2}\delta x.$$

Because the arm is in equilibrium, small extensions do not change the potential energy for very small δx, therefore by setting this equation of work to zero and then solving, we get the ratio of tensions to become $T_1/T_2 = -2/3$. ■

Let us look at one more example

Example 12.13. Four long and four half as long rods are hinged to each other forming three identical rhombi. One end of the contraption is hinged to a ceiling, the other one is attached to a weight of mass m. The hinge next to the weight is connected to the hinge above by a string. Find the tension force in the string.

Solution. We will use a virtual work approach.[2] In a static situation, the net force will be zero and as a result the potential energy will be at a minimum. Any slight displacement will create no change to the potential energy in first order.

Consider what happens when the mass is lowered by a distance dh. The potential energy would drop by $-mgdh$. The distance between hinges would each increase by $dh/3$ to compensate for the length increase. This means the string gets stretched by $dh/3$. The energy stored thus is:

$$Tdh/3$$

Setting these changes to zero gives:

$$-mgdh + Tdh/3 = 0 \implies \boxed{T = 3mg}.$$

■

12.5 Homework Problems and Solutions

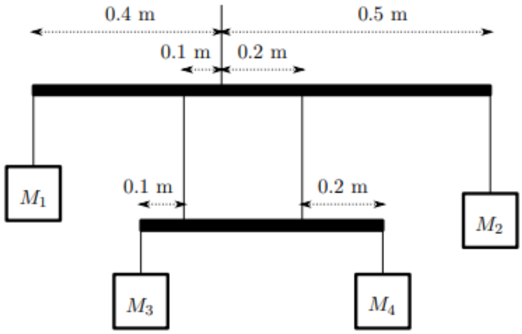
Problem 12.1. [5] A wooden equilateral triangle with uniform mass density is placed on an an inclined plane with angle of elevation θ. The surface is inclined with angle θ. What is the minimum θ in degrees such that the triangle will start to tip?

Solution. Consider the torques around the lowermost vertex of the equilateral triangle. Note that the normal force can only exert a clockwise torque about this point, so if the triangle is to remain balanced, the gravitational force must provide a counterclockwise torque. Note that when $\theta = 60^\circ$, the center of mass of the triangle is directly above the triangle, in which case gravity provides zero torque.

Any greater angle will cause gravity to provide a clockwise torque, so $\boxed{60^\circ}$ is the minimum angle at which the triangle will start to tip. ■

Problem 12.2 ($F = ma$)**.** [6] In the mobile below, the two cross beams and the seven supporting strings are all massless. The hanging objects are $M_1 = 400$ g, $M_2 = 200$ g, and $M_4 = 500$ g. What is the value of M_3 for the system to be in static equilibrium?

[2] If you are still unfamiliar with virtual work after this example, refer to this pdf for more examples: http://www.ce.siue.edu/examples/Worked_examples_Internet_text-only/Data_files-Worked_Exs-Word_&_pdf/Virtual_work.pdf



Solution. We first balance torques on the top rod about the point where the middle string touches the bar. This tells us that

$$0.4 \cdot M_1 g + 0.1 \cdot T_1 = 0.2 \cdot T_2 + 0.5 \cdot M_2 g.$$

where T_1 and T_2 are the tensions in the right and left strings (in the middle). Balancing the forces on the lower rod, we also see that

$$M_3 g + M_4 g = T_1 + T_2,$$

so we can substitute $T_2 = M_3 g + M_4 g - T_1$ to get

$$0.4 \cdot M_1 g + 0.3 \cdot T_1 = 0.2(M_3 g + M_4 g) + 0.5 \cdot M_2 g.$$

Then balancing torques on each of the lower right connection point, we see that

$$0.2 \cdot M_4 g + 0.3 \cdot T_1 = 0.4 \cdot M_3 g.$$

Rearranging, we obtain the system

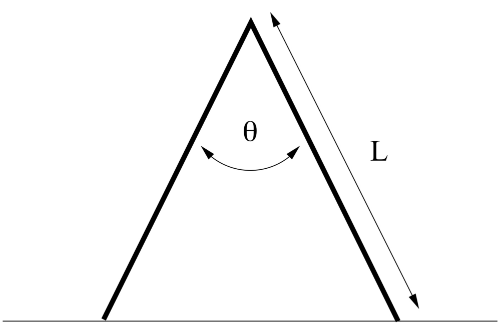
$$\begin{aligned} 0.3 \cdot T - 0.2 \cdot M_3 g &= 0.2 \cdot M_4 g + 0.5 \cdot M_2 g - 0.4 \cdot M_1 g \\ 0.3 \cdot T - 0.4 \cdot M_3 g &= -0.2 \cdot M_4 g \end{aligned}$$

and subtracting the equations gives

$$0.2 M_3 g = 0.4 \cdot M_4 g + 0.5 \cdot M_2 g - 0.4 \cdot M_1 g \implies M_3 = \boxed{700 \text{ g}}.$$

■

Problem 12.3 ($F = ma$)**.** [7] The sign shown below consists of two uniform legs attached by a frictionless hinge. The coefficient of friction between the ground and the legs is μ. What is the maximum value of θ such that the sign will not collapse?



Solution. First, let m be the mass of one half of a sign. Let's consider the torque on the sign about the point where the two halves meet (on one leg). Balancing torques,

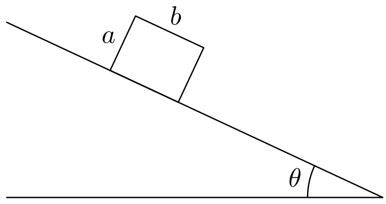
$$-mg \cdot \frac{L}{2}\sin\frac{\theta}{2} + N \cdot L\sin\frac{\theta}{2} = F_f L\cos\frac{\theta}{2}.$$

We have that N the normal force on the right endpoint of the leg is mg by symmetry. Furthermore, we have $F_f \le mg\mu$, so we must have

$$\frac{L}{2}mg\sin\theta/2 \le mg\mu \cdot L\cos\theta/2.$$

This yields $\boxed{\tan(\theta/2) \le 2\mu}$. ■

Problem 12.4 ($F = ma$)**.** [7] A uniform rectangular block with mass M, length b, and height a rests on an incline as shown. The incline and the block have a coefficient of static friction μ_s. The incline is moved upwards from an angle of zero through an angle θ. At some critical angle, the block will either tip over or slip down the plane. Determine the relationship between a, b, and μ_s such that the block will tip over (and not slip) at the critical angle.



Hint: This problem is very similar to the one above.

Solution. We know that a block begins slipping once $\theta = \arctan\mu$. As for tipping, it occurs when the center of mass of the block is past the vertical so there is a net torque. We can do a bit of angle chasing, and we see this happens when $\tan\theta = b/a$. So we want $\arctan(b/a)$ to be less than $\arctan\mu_s$, so the answer is $b/a < \boxed{\mu_s}$. ■

Problem 12.5 ($F = ma$)**.** [7] Three point masses m are attached together by identical springs. When placed at rest on a horizontal surface, the masses form a triangle with side length l. When the assembly is rotated about its center at angular velocity ω, the masses form a triangle with side length $2l$. What is the spring constant k of the springs?

Solution. We simply make sure that the net force is the centripetal force. The distance from the center, using geometry is given by $\frac{2l}{\sqrt{3}}$. So then,

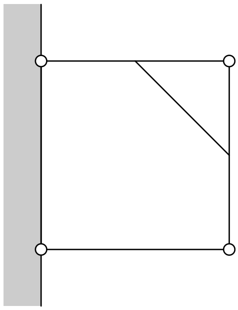
$$F_c = m\omega^2 \cdot \frac{2l}{\sqrt{3}}.$$

We can that the net force on the mass is given by $kl\sqrt{3}$ by simply adding vectors. Thus, we require

$$kl\sqrt{3} = m\omega^2 \cdot \frac{2l}{\sqrt{3}},$$

which means $k = \boxed{\frac{2}{3}m\omega^2}$. ■

Problem 12.6 (Morin). [8] Three sticks with mass m form three sides of a square, as shown in the figure. They are connected with pivots to each other and to a wall. The midpoints of the top and right sticks are connected by a massless string. What is the tension in the string?



Solution. There are many, many ways to solve this problem, some are faster some are slower, but they all involve mostly the same principles. But one of the ideas that we use here that significantly speeds up some part of the process is that we can balance torques on the whole entire system about one point. In this case we balance it about the upper left pivot. Most of the forces are internal or cancel out since they point toward the pivot, so the only one we worry about is the normal force from the wall on the lower stick and gravity. So if the length of the square is ℓ,

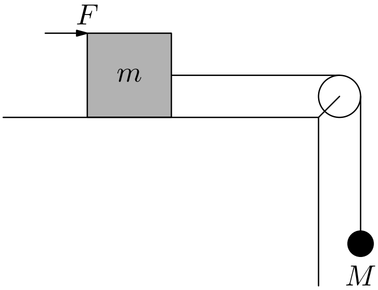
$$2mg\frac{\ell}{2} + mg\ell = N\ell.$$

So the normal force from the wall on the bottom stick is $2mg$. Then we balance forces on the bottom stick. We have the force from the right stick perpendicular to itself on the bottom stick is $2mg$, which in turn means the force perpendicular to it from the bottom stick is $2mg$. So we can balance torques on the right stick about the upper pivot. We have

$$2mg\ell = T\cos 45^\circ \ell/2,$$

meaning $T = \boxed{4\sqrt{2}mg}$. ■

Problem 12.7 (OPhO). [8] A solid cube of uniform density, mass m, and side-length ℓ rests on the ground. A horizontal force F is exerted on the top edge of one vertical face of the cube while a rope is strung from pulley to another mass of mass M. Assuming that the tension in the rope stays roughly the same, how far will the normal force move from the central axis of the block to stop the block from toppling over? Assume that there is sufficient friction such that the block does not slip and that the mass on the pulley does not accelerate. Note that $m > M$.



Solution. Consider the torques about the bottom right hand corner of the box. We have

$$F \cdot \ell + T \cdot \frac{\ell}{2} + N \cdot x = mg \cdot \frac{\ell}{2}.$$

Since $T = Mg$ and $N = mg$, we rearrange and we have

$$mgx = -F \cdot \ell - \frac{Mg\ell}{2} + \frac{mg\ell}{2}.$$

So we can find

$$x = \frac{\ell}{2} - \frac{F\ell}{mg} - \frac{M\ell}{2m}.$$

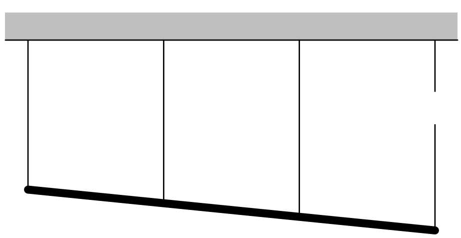
we subtract this value from $\ell/2$ to get the distance from the axis,

$$d = \boxed{\frac{F\ell}{mg} + \frac{M\ell}{2m}}.$$

Of course note that if x is negative, then the box will topple. ∎

Problem 12.8 (Kalda)**.** [9] A uniform bar with mass m and length l hangs on four identical light wires. The wires have been attached to the bar at distances $l/3$ from one another and are vertical, whereas the bar is horizontal. Initially, tensions are the same in all wires, $T_0 = mg/4$. Find tensions after one of the outermost wires has been cut.

Solution. We have to use an idea that one would not normally think of. We have to think of the wires as "stretchy". Then they all have the same "spring factor". Then once the outermost wire is cut, we have a situation like below: (though it is exaggerated).



So we can see that the average tension in these ropes is the tension in the middle left rope, so the tension in the middle left one is $mg/3$. Now we balance torques about the COM. So

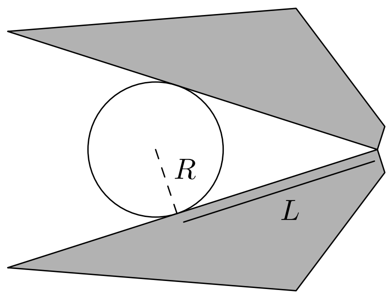
$$T \cdot \frac{l}{2} + \frac{mg}{3} \cdot \frac{l}{6} = \left(\frac{2mg}{3} - T\right)\frac{l}{6}.$$

Where T is the tension in the left rope. So we rearrange, and

$$T \cdot \frac{2l}{3} = \frac{mgl}{18},$$

so $T = \dfrac{mg}{12}$. So the tensions in the strings are $\dfrac{mg}{12}, \dfrac{mg}{3}, \dfrac{7mg}{12}$ (from left to right). ■

Problem 12.9. [9] A clamp is used to pick up a gumball with mass m and radius R. The contact point is a distance L from the pivot. Then the clamp is held parallel to the page. The following diagram is a top-down view.



Given that the coefficient of friction is μ, what is the minimum normal force each arm needs to exert on the gumball?

Solution. Let the normal force from the arms be N. Now we have to consider the component of the friction force perpendicular to the paper and also parallel. We have that the component perpendicular to the paper is $mg/2$, since the force of friction from each side has to lift the gumball. The force of friction parallel to the plane has to cancel the leftward component of the normal force, so

$$F_{f\|} \cos\theta = N \sin\theta,$$

where θ is half of the angle created by the clamps. So,

$$F_{f\|} = N \tan\theta = \frac{NR}{L}.$$

So then we have the total magnitude of the friction force is

$$F_f = \sqrt{(NR/L)^2 + (mg)^2}.$$

This needs to be equal to $N\mu$, so

$$N^2 \frac{R^2}{L^2} + (mg)^2 = N^2 \mu^2.$$

So we rearrange and solve for N, giving us $N_{min} = \boxed{mg\sqrt{\mu^2 - \frac{R^2}{L^2}}}$. ■

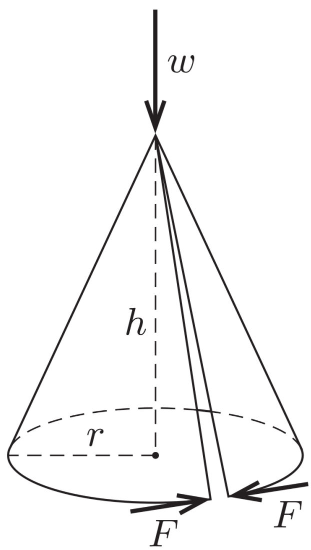
Problem 12.10 ($F = ma$)**.** [10] A circular table has radius R and $N > 2$ equally spaced legs of length h attached to its perimeter. Suppose the table has a uniform mass density with total mass m, and neglect the mass of the legs. Assuming the table does not slip, what is the minimum horizontal force needed to tip over the table?

Solution. Label the bottom of one leg X, and the bottom of an adjacent leg Y. The most effective way to tip the table is to apply a force opposite to the midpoint of XY. Hence, we use the midpoint of XY as our pivot point.

Note that about this axis, friction will provide no torque, and the normal force can only provide torque in one direction. When the table is on the edge of tipping, the torque from the applied force and the torque from the gravitational force will balance. The torque from the applied force is simply Fh. On the other hand, the gravitational force is a horizontal distance of $R\cos(\pi/N)$ from the pivot point, and will hence have torque $mgR\cos(\pi/N)$.

Equating the two torques gives us $F = \boxed{\frac{mgR}{h}\cos(\frac{\pi}{N})}$. ■

Problem 12.11 (200 More Puzzling Physics Problems)**.** [12] A cone with height h and a base circle of radius r is formed from a sector shaped sheet of paper. The sheet is of such a size and shape that its two straight edges almost touch on the sloping surface of the cone. In this state, the cone is stress-free.



The cone is placed on a horizontal, slippery table-top, and loaded at its apex with a vertical force of magnitude w, without collapsing. The splaying of the cone is opposed by a pair of forces of magnitude F acting tangentially at the join in the base circle (*see* figure). Ignoring any frictional or bending effects in the paper, find the value of F. **Hint:** Use the principle of virtual work!

Solution. We want to use virtual work. So suppose that the top moves by dh. Then we must figure out how much the circumference changes. We can first relate the radius and height,

$$r = \sqrt{l^2 - h^2},$$

where l is the slant height. So we have

$$\frac{dr}{dh} = \frac{-h}{\sqrt{l^2 - h^2}}.$$

So then we have

$$d(2\pi r) = \frac{-2\pi h}{\sqrt{l^2 - h^2}} dh = \frac{-2\pi h}{r}.$$

Now we want the total work to be 0, so we want

$$F \cdot d(2\pi r) + w \cdot dh = 0.$$

We plug in our values, and

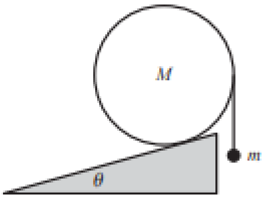
$$F \cdot \frac{2\pi h}{r} dh = w \cdot dh$$

$$\implies F = \boxed{\frac{wr}{2\pi h}}.$$

■

12.5.1 Written Solutions

Problem 12.12 (Morin). [12] A uniform cylinder of mass M sits on a fixed plane inclined at an angle θ. A string is tied to the cylinder's rightmost point, and a mass m hangs from the string, as shown in the figure. Assume that the coefficient of friction between the cylinder and the plane is sufficiently large to prevent slipping. What is m, in terms of M and θ, if the setup is static?



Solution. As always, we want both the net force and net torque on the cylinder to equal zero. Since friction is sufficient to prevent slipping, we can skip the step balancing forces. This is because if we choose the contact point as the center of rotation, both the normal force and friction will exert zero torque. Balancing the torques, we see that

$$Mg(R\sin\theta) = mg(R - R\sin\theta),$$

where the length $R - R\sin\theta$ comes from dropping an altitude from the pivot point to the string and applying trigonometry. Rearranging, we get

$$m = \boxed{M\left(\frac{\sin\theta}{1 - \sin\theta}\right)}.$$

■

IV

Harmonics

13 Oscillations

In the springs handout, we discussed how to determine the force that a spring exerts on an object, given by Hooke's law $F = -kx$. From this we got that the potential energy stored in a spring is given by $U = \frac{1}{2}kx^2$. From these two simple ideas we were able to solve a number of complex problems involving friction, gravity and various configurations of multiple springs.

However, while we were able to find the energy and motion of systems at different points in space, we were not able find them as a function of time. We saw that objects on a spring oscillate, but how long does each oscillation take? How can we write their position as a function of time?

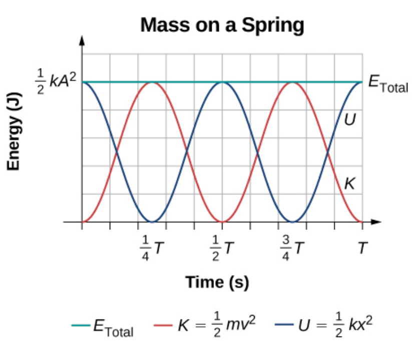
This handout will answer these questions, more precisely describing the motion of an object attached to a spring. Once we do this, we will also see striking similarities between springs and other seemingly unrelated systems such as pendulums, and we will hint at the broader generalizations of these principles in the differential equations section.

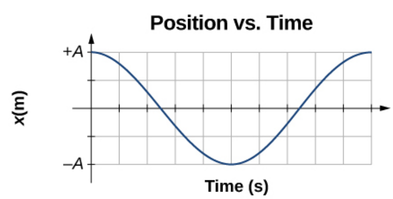
13.1 Simple Harmonic Motion

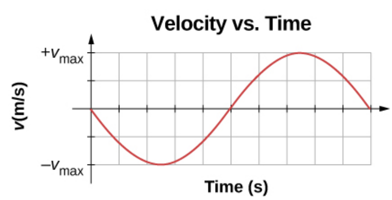
When you displace a spring with a mass on it and release it, in the absence of friction or any other driving forces, it will undergo what we call **simple harmonic motion**, oscillating back and forth. This motion is also known as simple harmonic motion, or SHM for short.

Simple harmonic motion is characterized by a restoring force pointing towards equilibrium. This restoring force has a magnitude proportional to the displacement from the equilibrium. For example, a mass on a spring is an example of simple harmonic motion since there is a restoring force on the mass of magnitude $|kx|$ that points towards the spring's equilibrium.

Plots of position, velocity and energy vs. time for a mass on a spring are shown below – be sure you have an intuitive understanding of these and how they compare with the graphs of the same quantities versus position that we examined in the springs handout.







These graphs look suspiciously like sine and cosine functions. We will explain this more in section 13.3 using calculus and basic properties of differential equations, but for solving problems it is sufficient to simply accept this is true. Using the assumption that the motion is sinusoidal, we prove the following theorem.

Theorem 13.1. *A mass m is attached to a spring of spring constant k and the equilibrium position is x_e. If the mass is displaced by a distance A and released at time $t = t_0$, then the position of the mass as a function of time is given by*

$$\begin{aligned} x(t) &= x_e + A\cos(\omega(t - t_0)) \\ v(t) &= -A\omega\sin(\omega(t - t_0)) \\ a(t) &= -A\omega^2\cos(\omega(t - t_0)), \end{aligned}$$

where $\omega = \sqrt{\frac{k}{m}}$.

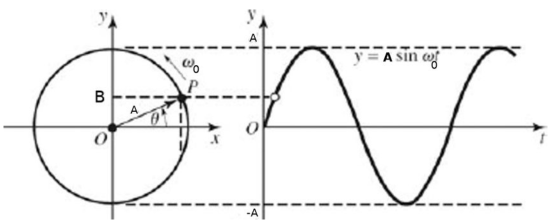
Proof. If we operate under the assumption that the motion is sinusoidal, all we need to do is to determine the value of ω. We will do this by comparing simple harmonic motion to a form of periodic motion which we are more familiar with – circular motion.

Consider a mass attached to the end of a string of radius A moving at a speed v in a circle around the origin. The mass is in uniform circular motion with angular velocity given by $\omega_0 = \frac{v}{A}$ and its position at time t (assuming $\theta = 0$ at $t = 0$) given by $(A\cos(\omega_0 t), A\sin(\omega_0 t))$. Thus, the motion of a mass oscillating on a spring with $\omega = \omega_0$ will simply be the same as the x-component for the motion of this mass attached to a string.

But how do we determine the angular velocity ω for a spring? To do this, we must account for the velocity of the mass as it oscillates on the spring, and match this with the velocity of the object undergoing circular motion. In a sense, we can use the oscillation of the x-component of circular motion as an analogy for the oscillation of a spring.

The speed in circular motion is given by $v = \omega A$ and the x-component of the velocity is $A\omega\sin(\theta) = A\omega\sin(\omega t)$. Thus, the maximum x-velocity occurs at $\theta = \frac{\pi}{2}$ and has magnitude ωA. The maximum velocity of the spring will be $\sqrt{\frac{kA^2}{m}} = \sqrt{\frac{k}{m}}A$. Thus,

$$\begin{aligned} \sqrt{\frac{k}{m}}A &= v_{max} = \omega A \\ \omega &= \sqrt{\frac{k}{m}}. \end{aligned}$$



Now that we finally know the angular frequency, we can plug this back into our expressions for the x-position and x-velocity of circular motion to get our equations of motion for the spring:

$$x(t) = A\cos\left(t\sqrt{\frac{k}{m}}\right)$$
$$v_x(t) = -A\omega\sin\left(t\sqrt{\frac{k}{m}}\right)$$

To find the acceleration of the mass on the spring, recall the magnitude of the centripetal acceleration is $\frac{v^2}{A} = \omega^2 A$, and so, as shown in the diagram, we see that the acceleration in the x-direction is simply this times the negative cosine of $\theta = \omega t$.

$$a(t) = -A\frac{k}{m}\cos\left(t\sqrt{\frac{k}{m}}\right).$$

■

As you can see, $a = -\frac{k}{m}x$, which is consistent with Hooke's Law. In fact, we have the following takeaway:

Concept. If a body experience acceleration $a = -Cx$ for a positive constant C, the body will experience simple harmonic motion with angular frequency $\omega = \sqrt{C}$. A calculus-based proof of this was in fact given in the solutions to Problem 6.11(a)(ii) and 6.11(b)(ii).

Thus, the motion of a mass attached to a spring is just some kind of sinusoidal motion with angular frequency $\omega = \sqrt{\frac{k}{m}}$.

We will now do a simple example to review what we've seen so far.

Example 13.2. A mass 2 kg hangs from a spring of spring constant $k = 3$ N/m. It is lifted a distance 1 m from equilibrium (note that this is the mass' equilibrium position, which is not the springs natural equlibrium length) and released at time $t = 0$. We define the equilibrium height to be $x = 0$.

(a) Using conservation of energy (discussed in the springs handout) find the velocity at each point x.

(b) Now find the $h(t)$ and $v(t)$, the position and velocity of the mass versus and evaluate them at $t = 1$s

(c) Verify that these values satisfy the condition you found in part (a).

Solution.

First recall that the force of gravity plus the force of the spring add to create a net force which is simply equivalent to the force of a spring whose equilibrium length is the position at which the mass is in equilibrium (we talked about this in the spring handout – verify it for yourself!) Let A be the amplitude 1m.

Then we use conservation of energy to get the velocity v a height h away from equilibrium:

$$\frac{1}{2}kh^2 + \frac{1}{2}mv^2 = \frac{1}{2}kA^2$$
$$mv^2 = k(A^2 - h^2)$$

Then we solve for v to get

$$v = \sqrt{\frac{k}{m}}\sqrt{A^2 - x^2} = \sqrt{\frac{3}{2}}\sqrt{1 - h^2} \tag{13.1}$$

(b) The motion is sinusoidal with $\omega = \sqrt{\frac{k}{m}} = 1.2247s^{-1}$ so

$$h(t) = A\cos(\omega t)$$
$$v(t) = -A\omega\sin(\omega t)$$

then we plug in values and $t = 1$s to get $h(1) = 0.3392$m and $v(1) = 1.1521$m/s.

(c) Indeed, $\sqrt{\frac{3}{2}}\sqrt{1 - 0.3392^2} = 1.1521$ as expected. In fact, if we plug in $h(t) = A\cos(\omega t)$ into equation 13.1 we get $v = A\omega\sin(\omega t)$ so this should work for any value of t, not just $t = 1$s.

■

So far we have been assuming that the equilibrium position of the spring is 0 and that the mass is released at time $t = 0$. But we can now generalize this to arbitrary equilibrium positions x_e and arbitrary start times t_0 by substituting in $t - t_0$ and $x - x_e$ to get the general theorem.

Note that

$$\begin{aligned} A\cos(\omega(t - t_0)) &= A(\cos(\omega t)\cos(\omega t_0) + \sin(\omega t)\sin(\omega t_0)) \\ &= B\cos(\omega t) + C\sin(\omega t) \end{aligned}$$

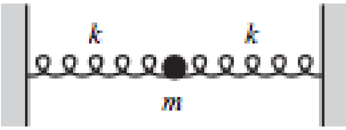
where $B = A\cos(\omega t_0)$ and $C = A\sin(\omega t_0)$. Then $\sqrt{B^2 + C^2}$. This is an equally valid way to express the motion of a mass attached to a spring. In fact in some cases, this equation is preferable because if does not require you to know the time t_0.

In some problems, rather than being given the time t_0 at which the spring begins its oscillations, you will be given the initial position and velocity at time 0. We could plug these conditions into Theorem 13.1 and solve these equations for t_0, however it is often easier to work from

$$x(t) = x_e + B\cos(\omega t) + C\sin(\omega t).$$

Plugging in $t = 0$, we see that the initial position x_0 of the particle is $x_0 = B + x_e$. If we take the derivative and plug in $t = 0$ then we see that the initial velocity is $v_0 = \omega C$. We also have that $A = \sqrt{B^2 + C^2}$. Thus, only knowing ω, x_e, x_0 and v_0, we can find the equation of the motion and the amplitude of the oscillations.

Example 13.3 (Morin). The springs in 13.1 are at their equilibrium length. The mass oscillates along the line of the springs with amplitude d. At the moment (let this be $t = 0$) when the mass is at position $x = \frac{d}{2}$ (and moving to the right), the right spring is removed. What is the resulting $x(t)$? What is the amplitude of the new oscillation?



Solution. We know the position at $t = 0$, and by conservation of energy we can also find the velocity at $t = 0$. Before the spring was removed, the effective spring constant was $2k$. Then the total energy was kd^2 and the potential energy at $\frac{d}{2}$ was $\frac{1}{4}kd^2$. Thus the kinetic energy at $\frac{d}{2}$ (which is $t = 0$) was $\frac{1}{2}mv_0^2 = \frac{3}{4}kd^2$, so $v_0 = \sqrt{\frac{3}{2}}\sqrt{\frac{k}{m}}d$. After the spring is removed we have $\omega = \sqrt{k/m}$. Thus our initial conditions are $x_0 = \frac{d}{2}$ and $v_0 = \sqrt{3/2}\omega d$.

From these we can get the equation of motion

$$\begin{aligned} x(t) &= x_0 \cos(\omega t) + \frac{v_0}{\omega}\sin(\omega t) \\ &= \frac{d}{2}\cos(t\sqrt{k/m}) + \sqrt{\frac{3}{2}}d\sin(t\sqrt{k/m}) \end{aligned}$$

and we calculate the amplitude

$$A = \sqrt{B^2 + C^2} = \sqrt{(d/2)^2 + (\sqrt{3}d/2)^2} = \frac{\sqrt{7}d}{2}$$

■

Other examples of simple harmonic motion include waves on string, sound, etc. Waves on a string are simple harmonic motion since there is a restoring tension force that brings the particles of the string back to equilibrium, which is when the string is straight. Sound is simple harmonic motion since there is a similar restoring force that brings, say, air particles back to their original position.

13.1.1 Period, Frequency and Angular Frequency

Often problems will ask you to compute the period, frequency or angular frequency/angular velocity of an oscillation. All these quantities contain the same information but it is important to be comfortable with them.

Recall that the angular velocity (or angular frequency) ω that we discussed above is the number of radians traveled per second. The **period** of an oscillation is the amount of time it takes to complete one oscillation and the **frequency** is the amount of oscillations that can be completed per unit time (commonly measured in $\text{Hz} = s^{-1}$). Thus we get that $\omega T = 2\pi$ and that $f = \frac{1}{T}$, so

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$
$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

For a transverse wave traveling on, say, a string, in one direction with velocity v, the period T is the amount of time it takes for one **wavelength** of a string to pass a certain point; a wavelength can be thought of as one oscillation. We usually denote the wavelength of a wave with λ (lambda). The frequency f is the amount of wavelengths that pass through a point per unit time.

Note that wavelengths per unit time multiplied by length per wavelength gives length per unit time, which is just velocity. This gives the equation

$$v = \lambda f.$$

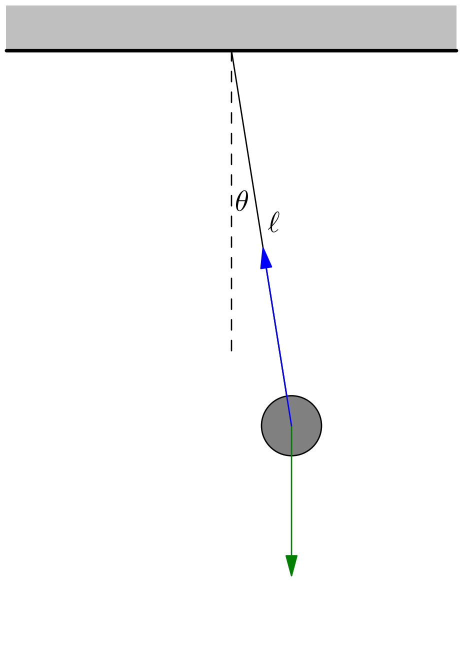
Another important equation for waves on a string is

$$v = \sqrt{\frac{F_T}{\mu}},$$

where v is the speed of a wave traveling along the string, F_T is the tension force in the string and μ is the linear mass density of the string.

13.2 Simple and Physical Pendulums

The same concepts of the previous section can be applied to simple pendulums. Simple pendulums consist of a mass and a massless rod pivoted so that the mass can swing. For small oscillations, a simple pendulum will undergo simple harmonic motion. Consider the following setup of the simple pendulum.



It's clear that this will oscillate since if it is moved right gravity will bring the mass back. Now let's analyze these oscillations using torque. If the mass is displaced by an angle θ the net torque on the mass is

$$\tau = mg\ell \sin\theta.$$

We apply $\tau = I\alpha$, and since $I = m\ell^2$, we have

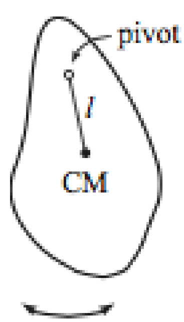
$$\alpha = \frac{g}{\ell}\sin\theta.$$

We learned in the previous section that if we have some equation of the form $a = \frac{k}{m}x$, then the resulting motion is oscillation with an angular frequency ω equal to $\omega = \sqrt{\frac{k}{m}}$. This should also hold for an equation of the form $\alpha = \frac{k}{m}\theta$. The only issue is that we have a $\sin\theta$ rather than a θ. This can be resolved by using the approximation $\sin\theta \approx \theta$ when θ is small. So we have

$$\alpha \approx \frac{g}{\ell}\theta,$$

meaning the angular frequency of oscillations is $\omega = \sqrt{\frac{g}{\ell}}$. Note that this formula only holds for small oscillations, and sometimes problems will ask you to get rid of that assumption.

Next, we apply a similar analysis to pendulums with an arbitrary shape. Say we have a pendulum who's center of mass is a distance ℓ away from its pivot and its moment of inertia about its pivot is I_p.



If the pendulum is displaced by an angle θ, then its torque about its pivot τ, is given by

$$\tau = mg\ell \sin(\theta)$$

so the angular acceleration is

$$\alpha = \frac{mg\ell}{I_p}\sin(\theta)$$

Be careful that you remember to use the moment of inertia about the *pivot* of the pendulum here (which we will often have to get by using the parallel axis theorem). If the displacement is small, we can again use the small angle approximation we get

$$\alpha \approx \frac{mg\ell}{I_p}\theta,$$

so the pendulum oscillates with angular frequency $\omega = \sqrt{\frac{mg\ell}{I_p}}$. We can summarize our results with the following statements.

Concept. Consider a pendulum of mass m, moment of inertia I_p and center of mass located a length ℓ from the pivot. At small values of θ, the pendulum will oscillate in simple harmonic motion with angular frequency

$$\omega = \sqrt{\frac{mg\ell}{I_p}}.$$

For a simple pendulum, this can be simplified to

$$\omega = \sqrt{\frac{g}{l}}.$$

We will now do a more complex example involving a physical pendulum.

Example 13.4 (Kleppner and Kolenkow).

(a) Find the period of a pendulum consisting of a uniform disk of mass M and radius R fixed to the end of a uniform rod of length l and mass m. The rod is attached to the center of the disk.

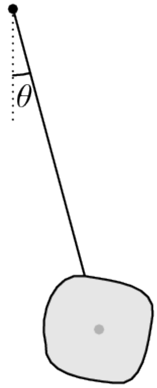
(b) How does the period change if the disk is mounted to the rod by a frictionless bearing so that it is perfectly free to spin?

Solution.

(a) Looking at the forces in the diagram below, we see that the torque about the pivot is given by

$$\tau = mg\frac{l}{2}\sin(\theta) + Mgl\sin(\theta) \approx (mg\frac{l}{2} + Mgl)\theta$$

using the small angle approximation.



Since $\tau = I_{pivot}\ddot{\theta}$, we get

$$\ddot{\theta} \approx \frac{(mg\frac{l}{2} + Mgl)}{I_{\text{pivot}}}\theta,$$

so ω, the angular frequency of the simple harmonic motion (not the actual angular frequency of the pendulum – well they are in fact equal when $\theta = 0$) is given by

$$\omega = \sqrt{\frac{(mg\frac{l}{2} + Mgl)}{I_{\text{pivot}}}}$$

and so the period is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I_{\text{pivot}}}{mg\frac{l}{2} + Mgl}}$$

The moment of inertia of the disk about its center of mass is $\frac{1}{2}MR^2$ so by the parallel axis theorem, its moment of inertia about the pivot is $\frac{1}{2}MR^2 + Ml^2$. The moment of inertia of the rod about the pivot is $\frac{1}{3}ml^2$.

$$I_{\text{pivot}} = \frac{1}{2}MR^2 + Ml^2 + \frac{1}{3}ml^2$$

Thus we have that

$$T = 2\pi\sqrt{\frac{\frac{1}{2}MR^2 + Ml^2 + \frac{1}{3}ml^2}{mg\frac{l}{2} + Mgl}}$$

(b) This part of the problem involves some slightly trickier rotational dynamics, though we apply simple harmonic motion in exactly the same. In order to understand how the rotating of the disk effects the system, we look at the total angular momentum of the system. When the disk is fixed in place we have $L = I_{\text{pivot}}\dot{\theta}$ which gives

$$L = \frac{1}{2}MR^2\dot{\theta} + Ml^2\dot{\theta} + \frac{1}{3}ml^2\dot{\theta}$$

But if the rod is free to rotate, it will naturally have angular frequency 0 rather than $\dot{\theta}$ so

$$L = Ml^2\dot{\theta} + \frac{1}{3}ml^2\dot{\theta}$$

$$\tau = \dot{L} = Ml^2\ddot{\theta} + \frac{1}{3}ml^2\ddot{\theta}$$

This simply amounts to removing the $\frac{1}{2}MR^2$ in the moment of inertia, so we get that the new period is simply

$$T = 2\pi\sqrt{\frac{Ml^2 + \frac{1}{3}ml^2}{mg\frac{l}{2} + Mgl}}$$

■

13.3 Differential Equations in Simple Harmonic Motion

This section is heavily calculus based and will look at differential equations.

Theorem 13.5. *If C is a positive constant, the solution to the differential equation*

$$\frac{d^2}{dx^2}f(x) = -Cf(x),$$

is $f(x) = A\sin(x\sqrt{C}) + B\cos(x\sqrt{C}) + C$ for some constants A, B, C.

Justification. Suppose we have a differential equation of the form

$$\frac{d^2}{dx^2}f(x) = -Cf(x)$$

for a positive constant C. What functions have a second derivative that is the negative of themselves? The first example that comes to mind is

$$\frac{d}{dx}\sin(ax) = a\cos(ax)$$
$$\frac{d}{dx}\cos(ax) = -a\sin(ax).$$

From this, we see can see immediately $\sin(x\sqrt{C})$ and $\cos(x\sqrt{C})$ are solutions to this differential equation. A second order differential equation has at most two independent solutions (which we state without proof), so the solutions to this equation are all the linear combinations of $\sin(x\sqrt{C})$ and $\cos(x\sqrt{C})$. ■

A mass on a spring, for example is described by the equation

$$\ddot{x} = -\frac{k}{m}x$$

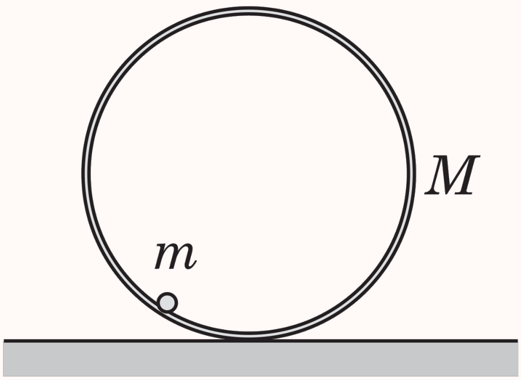
so its motion must be sinusoidal with angular frequency $\sqrt{\frac{k}{m}}$. Similarly, the motion of a simple pendulum undergoing small oscillations (where $\sin(\theta) \approx \theta$) is described by

$$\ddot{\theta} = -\frac{g}{\ell}\theta$$

so its motion is sinusoidal with angular frequency $\sqrt{\frac{g}{\ell}}$.

Let us look at an example where this helps us solve a problem.

Example 13.6 (Pathfinder). A thin rod of mass m is welded on the inner surface of a thin cylindrical shell of mass M ($M \gg m$) and radius R parallel to the axis of the cylinder. The composite body thus formed is placed on a horizontal floor. When disturbed slightly from equilibrium as shown in the figure, it undergoes small amplitude oscillations without sliding on the floor. Find the frequency of these oscillations. Acceleration of free fall is g.



Solution. Since the rod has deviated an angle θ from the center of the hoop, let us look at the torque on the hoop at that point. The moment of inertia about the contact point between hoop and ground is given by

$$I = (I_h + MR^2) + \underbrace{I_m}_{\text{very small}} \approx 2MR^2.$$

where $I_h = MR^2$ comes from the parallel axis theorem. Next we find torque on the hoop about the contact point. The force is mg and the distance between the contact point with the ground and with the rod is $2R\sin(\theta/2)$. Thus the torque is

$$\tau = mg\cos(\theta)2R\sin(\theta/2) \approx 2mgR\frac{theta}{2} \approx mgR\theta$$

using the small angle approximation $\sin(\theta/2) \approx \theta/2$ and $\cos(\theta) \approx 1$ We can now use the expression $\tau = I(-\alpha) = -I\ddot{\theta}$ (note that α is negative since the ball is rotating clockwise) to get

$$-2MR^2\ddot{\theta} \implies \ddot{\theta} = -\frac{mg}{2MR}\theta \implies \omega = \boxed{\sqrt{\frac{mg}{2MR}}}.$$

■

13.3.1 Energy

We have seen that an object undergoes simple harmonic motion if and only if it is governed by the equation of motion $\ddot{x} + Cx = 0$, meaning it experiences a restoring force of the form $F = -kx$ for some k. Then its potential energy would be $U =$

$-\int F \cdot dx = \frac{1}{2}Cx^2 + B$. The converse is in fact true as well. If an object is subjected to a potential energy function which is quadratic (with positive C), the restoring force will be linear, and it will experience simple harmonic motion.

This argument applies not only to springs but to any sort of system.

Example 13.7. Consider a simple pendulum consisting of a mass m attached to the end of a string of length ℓ. Find the potential energy as a function of θ and show that it will be quadratic for small angles.

Solution. When the mass is at an angle θ it will be at a height $\ell - \ell\cos(\theta)$ and so it will have potential energy

$$U = mg\ell(1 - \cos(\theta))$$

The small angle approximation says that $\cos(\theta) \approx 1 - \frac{\theta^2}{2}$ (these are simply the first two terms of the Taylor series expansion), so

$$U \approx mg\ell\frac{\theta^2}{2}$$

■

In the case of a pendulum, the torque is the derivative of the potential energy with respect to θ. Because, like a spring, the restoring torque is proportional to the displacement θ, the potential energy is also quadratic.

Concept. A system has potential energy of the form $U \propto z^2$ for a form of displacement z if and only if it experiences simple harmonic motion.

Remark. Taylor's theorem tells us that if we have a well behaved function we can locally approximate it as a power series. In this light, we can approximate an arbitrary potential energy function about a point x_0

$$U(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + a_3(x - x_0)^3 + \dots$$

where a_i is the ith derivative of U divided by $i!$. The terms with bigger powers of $(x - x_0)$ stay small for a longer time, so for sufficiently small oscillations (these may have to be very small depending on the function you are looking at) we can discard these terms to get

$$U(x) \approx a_0 + a_1(x - x_0) + a_2(x - x_0)^2$$

If we choose x_0 so that the system is in stable equilibrium, that is, the a_1 term is zero (and a_2 is positive because it is stable), then we get

$$U(x) \approx a_0 + a_2(x - x_0)^2$$

which is simply the potential energy for a spring of spring constant $2a_2$. So no matter what the potential energy function is, if a system is undergoing small oscillations, it can be approximated as simple harmonic motion! This approximation will be helpful in Problem 13.16.

We will now do an example which makes use of the energy formula for simple harmonic motion.

Example 13.8 ($F = ma$)**.** A particle moves in the xy plane with the potential energy

$$U(x) = 9kx^2 + 16ky^2$$

The particle can perform several different types of periodic motion. What is the ratio between the maximum and minimum periods?

Solution. The particle undergoes simple harmonic oscillations in the x and y directions independently. The period of simple harmonic motion is inversely proportional to the square root of k, so

$$\frac{T_x}{T_y} = \frac{\sqrt{16k}}{\sqrt{9k}} = \frac{4}{3}$$

If the particle is oscillating in both the x and y directions, then the total period T is the least common multiple of T_x and T_y because the particle must return to both its original x and y positions. Thus $T = 4T_y$. However it is also possible for the particle to only oscillate in the x direction or y direction in which case its period would be T_x or T_y respectively. The largest ratio is then $4t_y/T_y = 4$ ■

13.4 Homework Problems and Solutions

Problem 13.1 ($F = ma$)**.** [4] A light, uniform, ideal spring is fixed at one end. If a mass is attached to the other end, the system oscillates with angular frequency ω. Now suppose the spring is fixed at the other end, then cut in half. The mass is attached between the two half springs.

The new angular frequency of oscillations

(A) $\frac{\omega}{2}$

(B) ω

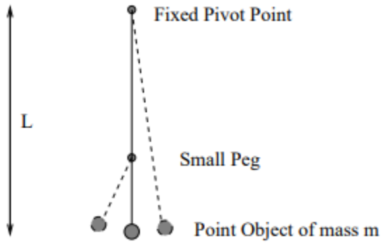
(C) $\sqrt{2}\omega$

(D) 2ω

(E) 4ω

Solution. As we saw in the springs handout, the spring constant of each of the smaller springs is $2k$ and since there are two of them, the effective spring constant of the second system is $4k$. Since $\omega = \sqrt{\frac{k}{m}}$ this means that ω is multiplied by 2 so we get (D) as the answer. ■

Problem 13.2 ($F = ma$)**.** [5] A simple pendulum of length L is constructed from a point object of mass m suspended by a massless string attached to a fixed pivot point. A small peg is placed a distance $2L/3$ directly below the fixed pivot point so that the pendulum would swing as shown in the figure below. The mass is displaced 5 degrees from the vertical and released. What is the period of this pendulum's oscillations?



(a) $\pi\sqrt{\frac{L}{g}}(1+\sqrt{\frac{2}{3}})$

(b) $\pi\sqrt{\frac{L}{g}}(2+\frac{2}{\sqrt{3}})$

(c) $\pi\sqrt{\frac{L}{g}}(1+\frac{1}{3})$

(d) $\pi\sqrt{\frac{L}{g}}(1+\sqrt{3})$

(e) $\pi\sqrt{\frac{L}{g}}(1+\frac{1}{\sqrt{3}})$

Solution. The period of a simple pendulum of length ℓ is $2\pi\sqrt{\ell/g}$, so the time it spends to the right of its starting position is $\pi\sqrt{L/g}$ and the time it spends to the left of its starting position is $\pi\sqrt{L/3g}$. Then the period is

$$T = \boxed{\pi\sqrt{\frac{L}{g}}(1+\frac{1}{\sqrt{3}})}.$$

■

Problem 13.3 ($F = ma$)**.** [10] A pendulum of length L oscillates inside a box.

(a) [5] A person picks up the box and gently shakes it horizontally with frequency ω and a fixed amplitude for a fixed time. To maximize the final amplitude of the pendulum, what is the value of ω?

(b) [5] A person picks up the box and gently shakes it vertically with frequency ω and a fixed amplitude for a fixed time. To maximize the final amplitude of the pendulum, what is the value of ω?

Solution.

(a) As with the previous part, we'll want the box and pendulum bob to be moving in the same direction. The pendulum bob experiences one left-right cycle in a period, so the angular frequency is just $\boxed{\sqrt{g/L}}$.

(b) Recall that a simple pendulum oscillates with angular frequency $\omega = \sqrt{g/L}$. To maximize the amplitude of the motion, we want the box and pendulum bob to be moving in the same direction. Since the pendulum bob experiences two up-down cycles in a period, this is double the angular frequency of the oscillation of the pendulum, or $\boxed{2\sqrt{g/L}}$.

■

Problem 13.4 ($F = ma$)**.** [5] A mass is attached to an ideal spring. At time $t = 0$ the spring is at its natural length and the mass is given an initial velocity; the period of the ensuing (one-dimensional) simple harmonic motion is T. At what time is the power delivered to the mass by the spring first a maximum?

(a) $t = 0$

(b) $t = T/8$

(c) $t = T/4$

(d) $t = 3T/8$

(e) $t = T/2$

Solution. The power delivered to the mass is $P = F \cdot v = -kxv$. We have

$$x(t) = A\sin\left(\frac{2\pi}{T}t\right)$$
$$v(t) = -v_{max}\cos\left(\frac{2\pi}{T}t\right)$$

Since we just want to find the time at which P is maximized we can just forget about the constants in the expressions for convenience. Then, plugging in these values of $x(t)$ and $v(t)$ we get

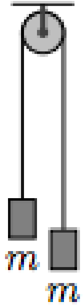
$$P = -C\sin\left(\frac{2\pi}{T}t\right)\cos\left(\frac{2\pi}{T}t\right) = -\frac{c}{2}\sin\left(\frac{4\pi}{T}t\right)$$

for some positive constant C that we don't really care about. This is maximized when $\sin = -1$, that is, $\frac{4\pi}{T}t = \frac{3\pi}{2}$ so

$$t = \boxed{\frac{3T}{8}}.$$

■

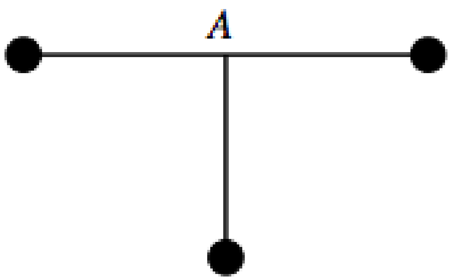
Problem 13.5 ($F = ma$)**.** [6] Two equal masses m are connected by an elastic string that acts like an ideal spring with spring constant k and unstretched length ℓ. The two masses are hung over a frictionless pulley. The two masses are both displaced downward by a small vertical distance x and simultaneously released from rest. What is the period of oscillation?



(A) $2\pi\sqrt{\frac{\ell}{g}}$
(B) $\pi\sqrt{\frac{2m}{k}}$
(C) $2\pi\sqrt{\frac{m}{k}}$
(D) $2\pi\sqrt{\frac{m}{k}+\frac{\ell}{g}}$
(E) There is not enough information to decide.

Solution. We look at the motion of the masses relative to the middle of the string. The spring constant of half of the string is $2k$, so the angular frequency for for one of the masses is $\omega = \sqrt{\frac{2k}{m}}$. Thus the period is $T = 2\pi\sqrt{\frac{m}{2k}} = \boxed{\pi\sqrt{\frac{2m}{k}}}$. ■

Problem 13.6 ($F = ma$)**.** [6] Three identical masses are connected with identical rigid rods and pivoted at point A. If the lowest mass receives a small horizontal push to the left, it oscillates with period T_1. If it instead receives a small push into the page, it oscillates with period T_2. What is the ratio T_1/T_2?



Solution. We have that for a pendulum $\omega = \sqrt{\frac{mg\ell}{I_{pivot}}}$ so the period is proportional to the square root of the moment of inertia I_{pivot} (for the same reason that for a spring the period is proportional to the square root of mass). When the mass receives a push to the left, the axis points straight out of the page so the moment of inertia is

$$I_1 = 3mR^2.$$

In contrast, when it receives a push into the page, the two masses on the ends lie of the axis, so the moment of inertia is

$$I_2 = mR^2.$$

Then $T_1/T_2 = \sqrt{I_1/I_2} = \boxed{\sqrt{3}}$.

■

Problem 13.7 ($F = ma$)**.** [7] A uniform bar of length L and mass M is supported by a fixed pivot a distance x from its center. The bar is released from rest from a horizontal position. The period of the resulting oscillations is minimal when

(A) $x = L/2$

(B) $x = L/2\sqrt{3}$
(C) $x = L/4$
(D) $x = L/4\sqrt{3}$
(E) $x = L/12$

Solution. The period is given by

$$T = f(\theta_0)\sqrt{\frac{I_{\text{pivot}}}{Mgx}}$$

for some function f that only depends on the initial angle from the vertical θ_0.

We use this function f instead of 2π since we deal with large initial angles. In this case, our pendulum starts from an initial angle of $\pi/2$ from the vertical. Note that for $\theta_0 << 1$, we have $f(\theta_0) \approx 2\pi$.

However, for this problem, the only term that matters is the $\sqrt{I_{\text{pivot}}/Mgx}$, since we release each bar from the same angle from the vertical; $f(\theta_0)$ can be treated like a constant.

By the parallel axis theorem, $I_{\text{pivot}} = \frac{1}{12}ML^2 + Mx^2$, so in order to minimize the period, we want to minimize the expression

$$\frac{\frac{1}{12}ML^2 + Mx^2}{Mgx} = \frac{\frac{1}{12}L^2 + x^2}{gx}.$$

This happens when $\frac{1}{12}L^2$ and x^2 satisfy the equality case of AM-GM, that is, when $x^2 = \frac{1}{12}L^2$, so $x = \boxed{L/2\sqrt{3}}$. ■

Problem 13.8. [7] A rope of length ℓ and mass per length λ is nestled inside a frictionless U-shaped tube so that the left end of the rope is a height h higher than the right end. When it is released what is the period of oscillations?

Solution. Let $h(t)$ be the height difference between the two sides. Then the gravitational force on the taller side will be stronger than the gravitational force on the shorter side by $g\Delta m = \lambda hg$ and so the rope will accelerate in the direction that evens out the height of the rope on the two sides. Thus we get

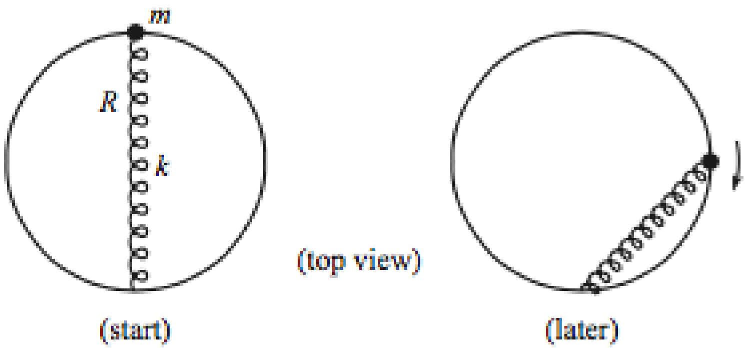
$$\ell\lambda a = F = -\lambda hg.$$

Notice that the acceleration of the rope $a = \ddot{h}/2$ because when the right side moves up by a distance dx then the left side moves down by dx so h changes by a total of $2dx$. Thus we have

$$\ddot{h} = 2a = -\frac{2hg}{\ell}.$$

Thus the angular frequency is $\sqrt{\frac{2g}{\ell}} \implies T = \boxed{\pi\sqrt{\frac{2\ell}{g}}}$. ■

Problem 13.9 (Morin). [8] A uniform solid cylinder with mass m and radius R is connected at its highest point to a spring (at its relaxed length) with spring constant k, as shown below. If the cylinder rolls without slipping on the ground, what is the angular frequency of small oscillations? Careful, the top of the cylinder moves more than the center!

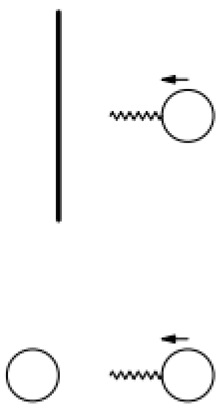


Solution. We will take torques about the point where the cylinder makes contact with the ground so that we do not have to take into account frictional force. Suppose the cylinder tilts by an angle θ about this point. Then the spring is stretched by a distance $2R\theta$, so it applies a torque $\tau = 4R^2k\theta$. To find the moment of inertia about this point we use the parallel axis theorem to get $I = \frac{1}{2}mR^2 + mR^2 = \frac{3}{2}mR^2$. Then the angular frequency of small rotations is

$$\omega = \sqrt{\frac{\tau/\theta}{I}} = \sqrt{\frac{4R^2k}{\frac{3}{2}mR^2}} = \boxed{\sqrt{\frac{8k}{3m}}}.$$

■

Problem 13.10 (PhysicsWOOT)**.** [16] Ignore gravity in this problem. A ball with a spring attached collides with a wall. The spring is massless. When the spring impacts the wall, it sticks to the wall. After the impact, the ball oscillates with frequency f and amplitude A. The initial velocity were the same but the spring instead collided with a stationary ball identical to the ball already attached to the other side of the spring.



(a) [8] What would the frequency of oscillations of a single ball be in terms of f, as viewed in the center-of-mass frame of the ball-spring-ball system?

(b) [8] What would the amplitude of oscillations of a single ball be in terms of A in the ball-spring-ball system?

Solution. We will look at the second situation in the reference frame of the center of mass of the two balls.

(a) In this reference frame the center of mass stays still, so we can pretend that it is fixed, say attached to a wall. Thus the oscillations of a ball in the second situation are equivalent to the oscillations of a ball attached to a spring which is half as long that is attached to a wall. The spring constant of a half-length spring would be $k' = 2k$. Then we have

$$f' = \frac{\omega'}{2\pi} = \frac{1}{2\pi}\frac{2k}{m} = \sqrt{2}\frac{\omega}{2\pi} = \boxed{f\sqrt{2}}.$$

(b) If the mass approaches with velocity v_0 in the first scenario, then in the second scenario it approaches with velocity $v_0/2$. Using conservation of energy, we see that in the first scenario

$$\frac{1}{2}mv_0^2 = \frac{1}{2}kA^2$$
$$\implies A = v_0\sqrt{\frac{m}{k}}.$$

Then in the second scenario we have

$$\begin{aligned} A' &= v_0'\sqrt{\frac{m}{k'}} \\ &= \frac{v_0}{2}\sqrt{\frac{m}{2k}} \\ &= \boxed{\frac{A}{2\sqrt{2}}}. \end{aligned}$$

■

Problem 13.11 (PhysicsWOOT). [8] A rod with uniform density, total mass m, and length l is pivoted about its center. The rod is horizontal and sits of a frictionless table. A spring of spring constant k is attached to the left side of the rod, and a second spring with the same spring constant is attached to the right side of the rod. What is the angular frequency of small-amplitude oscillations of the rod?

(a) $\sqrt{\frac{k}{m}}$

(b) $\sqrt{\frac{3k}{2m}}$

(c) $\sqrt{\frac{2k}{m}}$

(d) $\sqrt{\frac{3k}{m}}$

(e) $\sqrt{\frac{6k}{m}}$

Solution. We want to get an equation of the form $\ddot{\theta} = -C\theta$ in which case we will have that the frequency of oscillations is $\omega_1 = \sqrt{C}$ (we use ω_1 to distinguish it from rotational velocity).

First we calculate the torque on the rod about its center of mass when it is displaced by a small angle θ. The force from each spring is

$$F = kx = -\frac{kl\theta}{2}$$

for small angles. Then the torque is

$$\tau = -2F\frac{l}{2} = -\frac{kl^2\theta}{2}.$$

Next we calculate the moment of inertia of the rod about its center of mass:

$$I = \frac{ml^2}{12}$$

From this we find that the angular acceleration α is

$$\ddot{\theta} = \alpha = -\frac{\tau}{I} = -\frac{6k\theta}{m},$$

so we have that $\omega = \boxed{\sqrt{\frac{6k}{m}}}$ so the answer is (E). ■

13.4.1 Written Solutions

Problem 13.12 (Morin)**.** [18] A projectile of mass m is fired from the origin at speed v_0 and angle θ_0. It is attached to the origin by a spring with spring constant k and relaxed length zero.

(a) [6] Find $x(t)$ and $y(t)$.

(b) [8] Show that for small $\omega \equiv \sqrt{\frac{k}{m}}$, the trajectory reduces to normal projectile motion. And show that for large ω, the trajectory reduces to simple harmonic motion, that is, oscillatory motion along a line (at least before the projectile smashes back into the ground). What are the more meaningful statements that should replace "small ω" and "large ω"?

(c) [4] What value should ω take so that the projectile hits the ground when it is moving straight downward.

Solution. (a) The total force provided by the spring is $F = k\sqrt{x^2 + y^2}$. We will analyze the motion in the x and y direction separately. If θ is the angle of the particles position vector from the horizontal, then the force in the x and y directions are

$$F_x = -k\sqrt{x^2 + y^2}\cos(\theta) = -kx$$
$$F_y = k\sqrt{x^2 + y^2}\sin(\theta) - mg = -ky - mg$$

These surprisingly simple expressions are a nice property of zero length springs (see problem 13.16 for more fun with zero length springs). Then we have

$$\ddot{x} = -\frac{k}{m}x$$
$$\ddot{y} = -\frac{k}{m}y - g$$

Which means that both x and y satisfy Hooke's law with $\omega = \sqrt{\frac{k}{m}}$. Then all we have left to do in order to get the equations is work out the initial conditions. For x we want to figure out some constants A and B such that

$$x(t) = A\sin(\omega t) + B\cos(\omega t)$$

At $t = 0$ we have $x(0) = 0 \implies B = 0$ and $v_x(0) = v_0\cos(\theta_0) \implies A\omega = v_0\cos(\theta_0)$. Substituting these values back in we get our equation

$$\begin{aligned} x(t) &= \boxed{\frac{v_0\cos(\theta)}{\omega}\sin(\omega t)}. \\ &= \frac{v_0\cos(\theta)}{\sqrt{k/m}}\sin(t\sqrt{k/m}). \end{aligned}$$

The initial conditions for y are slightly trickier – the equation for y is $\ddot{y} = -\frac{k}{m}y - g$ and because of the $-g$ term, the equilibrium position is not at 0. Setting $\ddot{y} = 0$, then we get that the equilibrium position is at $y_0 = -g\frac{m}{k}$. This means we are looking for constants C and D such that

$$y(t) = C\sin(\omega t) + D\cos(\omega t) - g\frac{m}{k},$$

where $\omega = \sqrt{k/m}$ as before. At $t = 0$ we have $y(0) = 0 \implies D = gm/k$ and $v_y(0) = v_0\sin(\theta_0) \implies C\omega = v_0\sin(\theta)$. Substituting these value sback in we get our equation

$$\begin{aligned} y(t) &= \boxed{\frac{v_0\sin(\theta)}{\omega}\sin(\omega t) + \frac{gm}{k}(\cos(\omega t) - 1)} \\ &= \frac{v_0\sin(\theta)}{\sqrt{k/m}}\sin(t\sqrt{k/m}) + \frac{gm}{k}(\cos(t\sqrt{k/m}) - 1). \end{aligned}$$

(b) When ω is small we can approximate the sinusoidal functions to first and second order. We say

$$\sin(\omega t) \approx \omega t,$$

and using the Taylor series expansion of $\cos(\theta)$ to second order we have

$$\cos(\omega t) \approx 1 - \frac{(\omega t)^2}{2}.$$

Then substituting these into our expressions for $x(t)$ and $y(t)$ we get

$$x(t) \approx \frac{v_0\cos(\theta)}{\omega}\omega t = v_0\cos(\theta)t,$$

and

$$\begin{aligned} y(t) &\approx v_0 sin(\theta)t + \frac{g}{\omega^2}(1 - \frac{(\omega t)^2}{2} - 1) \\ &= v_0\sin(\theta)t - \frac{1}{2}gt^2, \end{aligned}$$

which is indeed the expression for v_x and v_y in normal projectile motion. Now what is the condition on ω that makes this approximation valid? We want to be able to say that $\omega t \ll 1$ at all points during the particles motion (before it hits the ground). The time at which the particle hits the ground is

$$t = \frac{2v_0 \sin(\theta)}{g},$$

so the condition that $\omega \ll 1/t$ is just

$$\omega \ll \frac{g}{2v_0 \sin(\theta)}.$$

Now suppose that ω is large. There isn't really a way to simplify the expression for $x(t)$ in this case, but notice that

$$y(t) = \frac{v_0 \sin(\theta)}{\omega} \sin(\omega t) + \frac{g}{\omega^2}(\cos(\omega t) - 1),$$

so when ω is large the second term is much smaller than the first term so

$$y(t) \approx \frac{v_0 \sin(\theta)}{\omega} \sin(\omega t),$$

and as before,

$$x(t) = \frac{v_0 \cos(\theta)}{\omega} \sin(\omega t).$$

Then at all times $y(t)/x(t) = \tan(\theta)$ and so the particle is oscillating in a straight line in the direction of v_0. What is the condition on ω that makes our approximation here valid? We want

$$\begin{aligned} \frac{g}{\omega^2} &\ll \frac{v_0 \sin(\theta)}{\omega} \\ \omega &\gg \frac{g}{v_0 \sin(\theta)} \end{aligned}$$

Essentially the motion of the particle depends on how long the timescale of the oscillations ($1/\omega$) is relative to the timescale of its projectile motion ($v_0 \sin(\theta)/g$ it doesn't really matter if there is a 2 or not). If the oscillations are much slower than the projectile motion, then all we can observe is the projectile motion, but if on the other hand they are much faster, then this means the oscillations due to the spring are much greater than the effects from gravity.

(c) We want $\dot{x}(t) = 0$, and during simple harmonic motion this occurs when the particle is at the amplitude of its oscillations, that is, $\sin(\omega t) = 1$ and $\cos(\omega t) = 0$. Substituting these values into the equation for $y(t)$ we have

$$0 = y(t) = \frac{v_0 \sin(\theta)}{\omega}(1) + \frac{g}{\omega^2}(0 - 1)$$

and solving for ω we have

$$\omega = \boxed{\frac{g}{v_0 \sin(\theta)}}.$$

Interestingly, this right between the limiting cases in part (b), when the oscillations and projectile motion have nearly equal effects, in a sense.

■

13.4.2 Extra Solutions

Problem 13.13 ($F = ma$)**.** A mass on a frictionless table is attached to the midpoint of an originally unstretched spring fixed at the ends. If the mass is displaced a distance A parallel to the table surface but perpendicular to the spring, it exhibits oscillations. The period T of the oscillations

(A) does not depend on A.

(B) increases as A increases, approaching a fixed value.

(C) decreases as A increases, approaching a fixed value.

(D) is approximately constant for small values of A, then increases without bound.

(E) is approximately constant for small values of A, then decreases without bound.

Solution. This a very hard problem that is difficult to get in a competition setting. Let ℓ be the length of half of the spring, and let each half of the spring have spring constant k.

Given a horizontal displacement A, we have the displacement of the spring is

$$\sqrt{\ell^2 + A^2} - \ell.$$

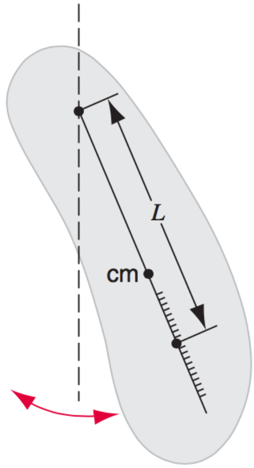
Summing the horizontal components of the spring force, we have

$$\begin{aligned} F &= k(\sqrt{\ell^2 + A^2} - \ell) \times 2\cos\theta \\ &= 2k(\sqrt{\ell^2 + A^2} - \ell) \times \frac{A}{\sqrt{\ell^2 + A^2}} \\ &= 2kA(1 - \frac{\ell}{\sqrt{\ell^2 + A^2}}). \end{aligned}$$

We note that $F = 2kA$ would be simple harmonic motion and the period would be independent of A. We note that as A increases, $\ell/\sqrt{\ell^2 + A^2}$ decreases. This means that as $A \to 0$, the F decreases faster than A. Likewise, as $A \to \infty$, F increases faster than A, eventually approaching $F = 2kA$, which has a fixed period (as it is SHM).

Thus, this indicates that the period decreases as A increases, eventually approaching a fixed value, giving us the answer (C). ■

Problem 13.14 (HRK)**.** A physical pendulum has two possible pivot points; one has a fixed position and the other is adjustable along the length of the pendulum, as shown in 13.14. The period of the pendulum when suspended from the fixed pivot is T. The pendulum is then reversed and suspended from the adjustable pivot. The position of this pivot is moved until, by trial and error, the pendulum has the same period as before, namely, T. Find the free-fall acceleration g in terms of T and L, the distance between two pivot points. Note that g can be measured in this way without needing to know the rotational inertia of the pendulum or any of its other dimensions except L.



Solution. Let l_1 and l_2 be the distances of the two pivot points from the center of mass. Then the equation of motion about pivot 1 is

$$I_{\text{pivot }1}\alpha = \tau = -Ml_1 g\theta,$$

where $I_{\text{pivot }1} = I_{\text{com}} + Ml_1^2$ by the parallel axis theorem. Then ω is given by

$$\omega = \sqrt{\frac{l_1 g}{I_{\text{com}} + Ml_1^2}},$$

which gives us

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I_{\text{com}} + Ml_1^2}{l_1 g}}.$$

By applying the same reasoning for pivot 2, we also have

$$T == 2\pi\sqrt{\frac{I_{\text{com}} + Ml_2^2}{l_2 g}}.$$

Our goal is to use these two equations to write g in terms of T, $L = l_1 + l_2$ and M. In order to do this, we need to get rid of I_{com} because we have pretty much no idea what the shape of the object is. Our strategy is to solve these two equations for I_{com}, set them equal and hopefully get something in terms of $l_1 + l_2$. Solving for I_{com} in the above equations we have

$$\left(\frac{T}{2\pi}\right)^2 Ml_1 g - Ml_1^2 = I_{\text{com}} = \left(\frac{T}{2\pi}\right)^2 Ml_2 g - Ml_2^2.$$

Simplifying, we get

$$\frac{T^2}{4\pi^2} g(l_1 - l_2) = l_1^2 - l_2^2,$$

and dividing both sides by $l_1 - l_2$, this reduces to

$$\frac{T^2 g}{4\pi^2} = l_1 + l_2 = L.$$

Now we can just solve for g to get

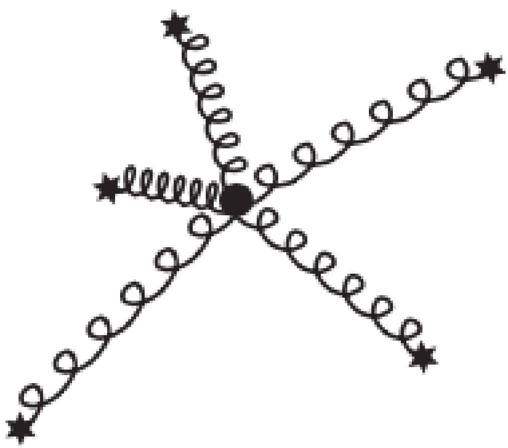
$$g = \boxed{\frac{4\pi^2 L}{T^2}}.$$

■

Problem 13.15 (Morin).

(a) A mass m is attached to two springs that have relaxed lengths of zero. The other ends of the springs are fixed at two points. The two spring constants are equal. The mass sits at its equilibrium position and is then given a kick in an arbitrary direction. Describe the resulting motion. How does the motion change (if it does) depending on the direction? What is the angular frequency? (Ignore gravity, although you actually don't need to.)

(b) A mass m is attached to n springs that have relaxed lengths of zero. The other ends of the springs are fixed at various points in space. The spring constants are $k_1, k_2, \ldots, k_n$. The mass sits at its equilibrium position and is then given a kick in an arbitrary direction. Describe the resulting motion. What is the angular frequency? (Again ignore gravity, although you actually don't need to)



Hints: 26

Solution.

(a) A nice property about zero length springs is that the force they exert is proportional to their length, (and like any spring, it is directed along their length).

Say we let $\vec{x_1}$ and $\vec{x_2}$ be the position vectors of the two springs, that is, the vector going from the fixed end of the spring to the end connected to mass m. For example, when the mass is at equilibrium position, $\vec{x_1} = l\hat{x}$ and $\vec{x_1} = -l\hat{x}$ if $2l$ is defined to be the separation between the two fixed points, and $\hat{x}$ denotes the unit vector in the horizontal direction. When the mass is displaced by some vector $\vec{r}$, we have $\vec{x_1} = l\hat{x} + \vec{r}$ and $\vec{x_1} = -l\hat{x} + \vec{r}$.

The key to this problem is realizing that because the springs have zero rest length, we conveniently get the simple relations

$$\vec{F_1} = k\vec{x_1}$$
$$\vec{F_2} = k\vec{x_2}$$

So if the mass is displaced by a vector $\vec{r}$ then the force it experiences is just

$$\begin{aligned}\vec{F} &= k(\vec{x_1} + \vec{x_2}) \\ &= k[(l\hat{x} + \vec{\Delta x}) + (-l\hat{x} + \vec{r})] \\ &= 2k\vec{r}\end{aligned}$$

Then we have

$$\vec{a} = \frac{2k}{m}\vec{r},$$

which miraculously has the same magnitude *no matter what direction we kick it in*. This gives us the angular frequency $\omega = \boxed{\sqrt{\frac{2k}{m}}}$

(b) This problem is very similar to the first part except generalized to n springs with different spring constants. Let $\vec{x_1}, \vec{x_2}, \ldots \vec{x_n}$ be the position vectors of the n springs. Then the key realization is that for any i, we have $\vec{F_i} = k_i\vec{x_i}$, so

$$F = k_1\vec{x_1} + k_2\vec{x_2} + k_3\vec{x_3} + \cdots + k_n\vec{x_n}. \tag{13.2}$$

Let $\vec{x}_{1\ \text{init}}, \vec{x}_{2\ \text{init}}, \ldots \vec{x}_{n\ \text{init}}$ be the initial position vectors of the springs. Then because the mass starts in equilibrium,

$$0 = \vec{F_{\text{init}}} = k_1\vec{x}_{1\ \text{init}} + k_2\vec{x}_{2\ \text{init}} + k_3\vec{x}_{3\ \text{init}} + \cdots + k_n\vec{x}_{n\ \text{init}} \tag{13.3}$$

When the mass is displaced by an arbitrary vector $\vec{r}$, then for all springs i we have $\vec{x_i} = \vec{x}_{i\ \text{init}} + \vec{r}$, so Eq. 13.2 becomes

$$\begin{aligned}\vec{F} &= k_1\vec{x_1} + k_2\vec{x_2} + k_3\vec{x_3} + \cdots + k_n\vec{x_n} \\ &= (k_1 x_1\vec{_{\text{init}}} + k_2 x_2\vec{_{\text{init}}} + k_3 x_3\vec{_{\text{init}}} + \cdots + k_n x_n\vec{_{\text{init}}}) + (k_1\vec{r} + k_2\vec{r} + \cdots + k_n\vec{r}).\end{aligned}$$

The first term is zero by Eq. 13.3, so this reduces to

$$\vec{F} = \vec{r}(k_1 + k_2 + \cdots + k_n),$$

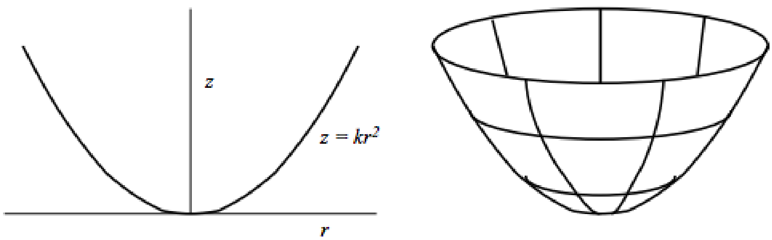
and so just like in part (a), the magnitude of the force amazingly does not depend on the direction that the mass was kicked. This gives us

$$\vec{a} = -\frac{1}{m}(k_1 + k_2 + \cdots + k_n)\vec{r}.$$

so $\omega = \boxed{\sqrt{\frac{k_1 + k_2 + \cdots + k_n}{m}}}$. Notice that in the case of two springs with equal spring constant, this reduces to our answer for part (a). Gotta love zero length springs!

■

Problem 13.16 (USAPhO(modified))**.** A particle is constrained to move on the inner surface of a frictionless parabolic bowl whose crosssection has equation $z = kr^2$. The particle begins at a height z_0 above the bottom of the bowl with a horizontal velocity v_0 along the surface of the bowl. The acceleration due to gravity is g.



(a) For a particular value of horizontal velocity v_0, which we will name v_h, the particle moves in a horizontal circle. What is v_h in terms of g, z_0, and/or k?

(b) Suppose that the particle now begins at a height z_0 above the bottom of the bowl with an initial velocity $v_0 = 0$.

(i) Assuming that z_0 is small enough so that the motion can be approximated as simple harmonic, find the period of the motion in terms any or all of the mass of the particle. m, g, z_0, and/or k. (See the remark in the section about approximations, but note that the approximation for this problem is pretty simple and does not actually involve Taylor expanding anything.)

(ii) Assuming that z_0 is not small, will the actual period of motion be greater than, less than, or equal to your simple harmonic approximation above? (You need not calculate the new value explicitly, but you should show some work to defend your answer.)

Solution.

(a) There are two forces acting on the ball, the normal force and the force of gravity, and the net force must be a centripetal (horizontal) force of magnitude $\frac{mv_h^2}{r}$. Thus the vertical component of the two forces cancel giving

$$mg = N\cos\theta.$$

where θ is the angle of the curve at r with respect to the horizontal. Newton's second law for the horizontal components gives

$$\frac{mv_h^2}{r} = N\sin\theta.$$

Dividing the two equations we get

$$\frac{mv_h^2}{r} = mg\tan(\theta).$$

Now $\tan(\theta)$ is just the slope of the curve at r, which is $\frac{d}{dr}kr^2 = 2kr = \frac{2z_0}{r}$ so substituting this in,

$$\frac{v_h^2}{r} = 2g\frac{z_0}{r}$$
$$\implies v_h = \sqrt{2gz_0}.$$

(b) We will give two different ways to think about this – one using forces and applying Hooke's law and one using energy so that we can see how the approximations play out in each of them. Let z be the height and r the radius at any point on the particle's path.

(i) Forces: We want to get an expression of the form $\ddot{r} = -Cr$ for some constant C. The force on the particle down the ramp is $F = mg\sin(\theta)$ where θ is the angle of the ramp relative to the horizontal, and the component of this force pointing inwards is $mg\sin(\theta)\cos(\theta)$. Thus,

$$\ddot{r} = -g\sin(\theta)\cos(\theta),$$

This may not seem too promising at first – we know that $\tan(\theta)$ is just the derivative of the curve, $\frac{d}{dr}kr^2 = 2kr$, but $\sin(\theta)$ and $\cos(\theta)$ are slightly uglier because they will involve using the Pythagorean theorem and dividing by a square root. Luckily, when z is very small, θ is very small and we can use the approximations $\sin(\theta) \approx \tan(\theta)$ and $\cos(\theta) \approx 1$ to get

$$\sin(\theta)\cos(\theta) \approx \tan(\theta) = 2kr.$$

a good approximation because it simplifies, but does not trivialize the problem. Thus we have

$$\ddot{r} = -2kgr,$$

giving $\omega = 2kg$ and a period

$$T = \boxed{2\pi\sqrt{\frac{1}{2kg}}}.$$

Energy: For simple harmonic motion, recall that we hope to have a quadratic potential energy function to get an equation of the form

$$E = \frac{1}{2}mCx^2 + \frac{1}{2}m\dot{x}^2$$

for a force $F = -Cmx$. In this problem, potential energy of the system is conventiently already in quadratic form,

$$U = mgz = mgkr^2.$$

The catch is that the kinetic energy is, in general, not just in the particle's changing radius but is also in the changing height:

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m(v_x^2 + v_y^2)$$
$$= \frac{1}{2}m\left(\left(\frac{dr}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2\right).$$

But since we can approximate z to be small, we ignore the vertical velocity, and we get

$$E = \frac{1}{2}m\left(\frac{dr}{dt}\right)^2 + \frac{1}{2}2gmkr^2.$$

giving $\omega = \sqrt{2gk}$ and

$$T = \boxed{2\pi\frac{1}{\sqrt{2gk}}}.$$

(ii) Forces: Our approximation was that

$$\ddot{r} = -g\sin(\theta)\cos(\theta) \approx -g\tan(\theta).$$

Since $\tan(\theta) > \sin(\theta)\cos(\theta)$, we overestimated the force, so the particle will actually oscillate more slowly than we expect, giving a greater period of motion than predicted.

Energy: Our approximation was that all the kinetic energy goes into the changing radius of the particle (we ignored the energy going into the changing height). In real life the radius will then change more slowly, so it will have a greater period of motion than predicted.

■

V

Celestial Mechanics

14 Gravitation

14.1 The Fundamental Forces

Throughout the course thus far, we have discussed the various properties of forces and the interactions that result because of them. However, we have yet to really discuss the fundamental *nature* of these forces and *why* they occur.

Recall from the Forces handout that in classical physics, all forces have been classified as belonging to one of the four **fundamental forces**. These forces include the **gravitational force**, **electromagnetic force**, **weak nuclear force**, and **strong nuclear force**. It has been theorized that there may be some other types of fundamental forces, but experimental evidence has yet to confirm such claims.

In this section, we will explore the fundamental nature of *one* of these four fundamental forces, namely *gravity*. The gravitational force arises due to one intrinsic property of all matter, namely *mass*. In the following section, we will delineate how mass affects the gravitational force among other things.

14.2 Gravitational Fields

Definition 14.1. We will see in this handout and the next handout that gravity is often key to the various planetary interactions that we collectively call **orbital mechanics**. The **gravitational force** is an attractive force between any two objects with mass. It can be experimentally shown that the magnitude of the gravitational force can be computed using the following relationship.

Theorem 14.2 (Newton's Law of Universal Gravitation). *The gravitational force between two point masses with masses M and m is given by the following law:*

$$F = \frac{GMm}{r^2},$$

where r is the distance between these two points and $G \approx 6.67 \times 10^{-11}$ is a constant called the ***gravitational constant****. The direction of this force is always attractive. In vector terms, the force on one particle can be written as*

$$\vec{F} = -\frac{GMm}{r^2}\hat{r}$$

where $\hat{r}$ is the unit vector pointing in the direction of the other particle.

Remark. As you might have noticed, so far in this course we have dealt with gravity between the Earth and another object. Because the gravitational constant G is so incredibly small (on the magnitude of 10^{-11}), we only need to consider gravitational forces between objects if at least one of them is rather large (i.e. planet, star, or other celestial body).

We have previously seen the gravitational force only in the form mg, where g was assumed to be constant with regard to altitude. However, $F_g = mg$ only holds for objects at the surface of the Earth, where a small change in altitude is negligible compared to the radius of the Earth. That is, g satisfies the relation

$$g \approx \frac{GM_e}{R_e^2},$$

where M_e is the mass of the Earth, and R_e is the radius of the Earth. However, when the change in radius is not negligible (namely in outer space), we must use Newton's Law of gravitation. Also, remember that the gravitational force is *conservative,* which means it will not dissipate mechanical energy.

Let us take an example of where the gravitational force is important. We consider the orbit of a mass around another mass (this is analogous to planetary orbits):

Example 14.3. Calculate the angular frequency of an object of mass m moving around a big mass M in radius r.

Solution. The gravitational force acts as the only centripetal force and thus we get the formula

$$\frac{GMm}{r^2} = m\omega^2 r.$$

By rearranging factors, we get

$$\omega = \boxed{\sqrt{\frac{GM}{r^3}}}.$$

■

Keep this result into mind for future problems as it is very important and comes up often.

Definition 14.4. We define the **gravitational potential energy** of an object (negative) to be the work per unit mass that would be needed to move an object from infinity to some location in space.

$$U_g = -\frac{GMm}{r}$$

Proof. Since the gravitational force is conservative, we can derive the potential energy of an object. Consider a particle of mass m that moves from two points, A to B on a radial path. A particle of mass M (which is assumed to be stationary and what m orbits around) is set at rest at the origin and exerts a gravitational force on m. As m moves from A to B, the work done on m by the gravitational force is

$$\begin{aligned} W_{AB} &= \int_A^B \vec{F} \cdot d\vec{r} = -\int_A^B F\ dr \\ &= -\int_{r_A}^{r_B} \frac{GMm}{r^2} dr = -GMm \int_{r_A}^{r_B} \frac{dr}{r^2} \\ &= -GMm \left(-\frac{1}{r}\right)\Bigg|_{r_A}^{r_B} = GMm\left(\frac{1}{r_B} - \frac{1}{r_A}\right). \end{aligned}$$

Next, we find that the change in potential energy of the system as m moves between points A and B is

$$\Delta U = U_B - U_A = -W_{AB} = GMm\left(\frac{1}{r_A} - \frac{1}{r_B}\right).$$

Instead of looking at the difference in potential energy, we can look at the value of the potential energy at a single point. We choose our reference frame point to be at an infinite separation between both particles. If we evaluate this at $r_B = \infty$ and $U_B = 0$ (since the potential energy far away will essentially be zero), and A represents an arbitrary point where the separation between the particles is r, the our equation becomes

$$U(\infty) - U(r) = GMm\left(\frac{1}{r} - 0\right)$$

or

$$U(r) = -\frac{GMm}{r}.$$

■

It might seem odd that gravitational potential energy is "negative," but remember that we are treating gravitational potential energy relatively. We are defining too infinitely far objects to have zero gravitational potential energy (since they will barely interact), and as such as objects get closer together, their potential energy decreases since we would have to perform work to separate them.

Concept. From the work-energy theorem, we have

$$F_g = -\frac{dU_g}{dr}.$$

This implies that the gravitational force always points in the direction of lower gravitational potential energy.

Definition 14.5. Additionally, we define the **gravitational field strength** ($\vec{g}$) by

$$\vec{g}(r) = -\frac{GM}{r^2}\hat{r} = \frac{\vec{F}_g}{m}.$$

This is another way of describing gravitational interactions, and some problems may use this term instead of defining the gravitational force directly since only one object is needed to define a gravitational field.

Given an object of mass m located at a position $\vec{r}$, the gravitational force is simply the gravitational field $\vec{g}(r)$ multiplied by m.

14.3 The Shell Theorems

The two shell theorems state the following:

1. **Shell Theorem 1:** A uniformly dense spherical shell attracts an external particle as if all the mass of the shell were concentrated at its center.
2. **Shell Theorem 2:** A uniformly dense spherical shell exerts no gravitational force on a particle located anywhere inside it.

Proving the shell theorems requires some extensive calculus (it took Newton 7 years to prove them). Therefore, we will not tackle them in this handout. If you are interested in the proof, we recommend looking at HRK (or the internet).

The shell theorems themselves don't seem that useful at first look. But in fact, they are extremely useful for solving many different types of problems: in the case of a planet orbiting a star, for instance, we can treat the bodies as point masses rather than need to account for their spherical mass distribution.

We will now do an example using this extremely amazing physics concept.

Example 14.6 (HRK). Suppose a tunnel could be dug through the Earth from one side to the other along a diameter. A particle of mass m is dropped into the tunnel from rest at the surface.

(a) What is the force on the particle when it is a distance r from the center?

(b) What is the speed of the particle when it is a distance r away from the center? Evaluate the speed at $r = 0$. Neglect all frictional forces and assume the Earth has a uniform density (apart from the tunnel that is).

Solution. Naively, we might expect the gravitational force to increase as we move toward the center of Earth. However, from the second shell theorem, we conclude that the gravitational force on the particle is only due to the portion of Earth that lies in a sphere of radius r where r is the distance of the particle from the center of the Earth. The first shell theorem allows us to conclude that the gravitational fore on the particle is all concentrated at the center of the Earth.

Let us define the variables M, R_E, and M_E where M is the total mass of the Earth enclosed in a sphere of r (where r is defined in the previous paragraph), R_E is the radius of the Earth, and M_E is the mass of the Earth. Assuming a constant density, we compare ratios of volumes

$$\frac{M}{M_E} = \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R_E^3} \implies M = M_E \frac{r^3}{R_E^3}.$$

Again, from the first shell theorem, we regard everything as mass concentrated at the center of the Earth which means that the gravitational force on the particle is

$$F = G\frac{mM}{r^2} = G\frac{m}{r^2} \cdot M_E \frac{r^3}{R_E^3} = G\frac{mM_E}{R_E^3} r.$$

We can treat r as the position vector $\vec{r}$ and note that the force $\vec{F}$ is moving in the opposite direction of $\vec{r}$ and therefore we can write

$$\vec{F} = -\frac{GmM_E}{R_E^3}\vec{r}.$$

Notice that this expression is very similar to Hooke's Law $F = -kx$. With this similarity, we can write the potential energy U of the system where $k = GmM_E/R_E^3$. Applying conservation of energy (and noting the potential energy is 0 at the center) we find that

$$\frac{1}{2}kR_E^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2.$$

Solving for v gives us

$$v = \sqrt{\frac{k}{m}(R_E^2 - r^2)} = \sqrt{\frac{GM_E}{R_E^3}(R_E^2 - r^2)}$$

■

From this example, we have the following takeaway:

Concept. Given a planet of radius R with uniform mass density, the gravitational force $\vec{F}_g$ experienced at a radius r satisfies the following:

- $\vec{F}_g$ is linear with r on the interval $0 \leq r < R$, and is equal to zero at $r = 0$.
- $\vec{F}_g$ is inversely proportional with r on the interval $r \geq R$.
- From the previous two relations, we have $\vec{F}_g$ reaches its maximum at $r = R$.

Exercise. Can you find the period of oscillations in the previous example?

14.4 Types of Orbital Velocities

The **escape velocity** is the minimum speed that initially needs to be given to an object such that it is able to completely escape the pull of gravity such that at $r \to \infty$ it has a speed of zero and a potential energy of zero. Conservation of energy tells us (the kinetic and potential energy will approximately be zero from far away):

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0$$

Solving for v gives:

$$\boxed{v = \sqrt{\frac{2GM}{R}}}$$

The **circular velocity** is similar to the escape velocity and is defined as the velocity required for a projectile to move in a circular orbit around the Earth. We can derive the circular velocity by using a force analysis.

$$F = \frac{mv^2}{r} \implies G\frac{mM}{r^2} = \frac{mv^2}{r} \implies v = \boxed{\sqrt{\frac{GM}{r}}}$$

Example 14.7 ($F = ma$)**.** The mass and radius of the Earth are M and R. If an object starting at a distance of R from the Earth's surface is moving at a velocity $v_0 = \sqrt{\frac{3GM}{2R}}$ tangentially, what is its trajectory?

Solution. At first, you may be slightly confused because the velocity the problem describes isn't anything that we have talked about before. However, you may realize that this problem really only tests common sense.

The velocity $v_0 > \sqrt{\frac{GM}{R}}$, which means that the object is moving too fast for a circular trajectory. This means that the velocity cannot be a circle but perhaps something elongated.

We also note that since $v_0 < \sqrt{\frac{2GM}{R}}$, so the velocity is too little for the object to escape and the object will continuously orbit the planet. This leads us to our answer that the trajectory of the object must intuitively be an $\boxed{\text{ellipse}}$. ■

Using this little example, we now define the velocity in different trajectories.

Concept. Based on the particle at a radius r from the center of a celestial body of mass M with velocity velocity v_0 directed tangentially, we have the following possibilities for the motion of the object:

- **Circle:** $v_0 = \sqrt{\frac{GM}{r}}$ on all parts of the trajectory.
- **Ellipse:** v_0 varies from $\sqrt{\frac{GM}{r}} < v_0 < \sqrt{\frac{2GM}{r}}$ on all parts of the trajectory.
- **Parabola:** $v_0 = \sqrt{\frac{2GM}{r}}$ on all parts of the trajectory.
- **Hyperbola:** $v_0 \geq \sqrt{\frac{2GM}{r}}$ on all parts of the trajectory.

The proofs of some of these results are rather complicated, so we will not cover them here. However, these trajectories are generally good to know.

14.5 Homework Problems

Problem 14.1 (HRK). [5] Show that on a hypothetical planet having half the diameter of the Earth but twice its density, what is the free fall acceleration in terms of g, where g is the free fall acceleration on the surface of the Earth?

Problem 14.2. [5] Assuming the Earth is a uniform sphere of radius R and that the gravitational free-fall acceleration at the surface is g_0, what is the approximate change in free-fall acceleration g at a height h above the surface? Use $g_0 = 10m/s^2$, $h = 100m$, $R = 6400km$ and report $10000\Delta g$ upto 3 significant figures. **Hint:** 8

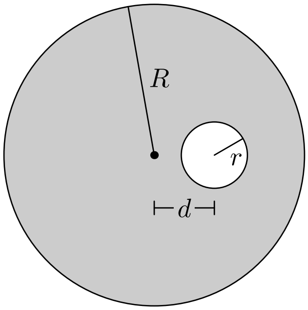
Problem 14.3 (Irodov). [5] A satellite orbits a planet with density ρ close to its surface. Find the period of its orbit. If it is of type $\alpha\rho^x G^y$, then find $\alpha + x + 3y$ upto 2 decimal places.

Problem 14.4 ($F = ma$). [6] A spherical cloud of dust in space has a uniform density ρ_0 and a radius R_0. The gravitational acceleration of free fall at the surface of the cloud due to the mass of the cloud is g_0.

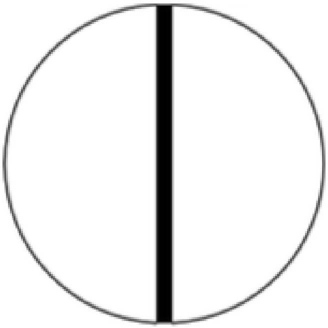
A process occurs (heat expansion) that causes the cloud to suddenly grow to a radius $2R_0$, while maintaining a uniform (but not constant) density. What is the gravitational acceleration of free fall at a point R_0 away from the center of the cloud due to the mass of the cloud now? Report g/g_0.

Problem 14.5 (Irodov). [6] A spherical planet has radius R and mass M. Then a spherical cavity of radius r is dug out. The cavity is centered at a distance $d > r$

from the center of the planet. Calculate the magnitude of the gravitational force on a mass m at the center of the planet. If answer is of format $GMm(r^x R^y d^z)$, then find $x + 2y + z$

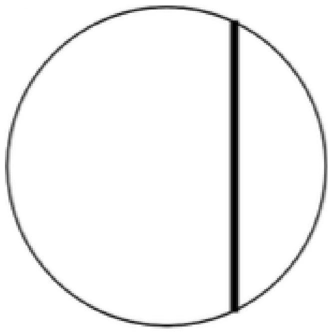


Problem 14.6 (PhysicsWOOT)**.** [10] Suppose a planet has uniform density ρ and radius R. The planet does not spin. You dig a hole straight through the planet from one side to the other, through the center. There is no air resistance in the hole. Use $G = 6.67 \times 10^{-11}$ and $\rho = 5510 kg/m^3$, report to closest minute.



(a) [5] Suppose you drop something in the hole. Find the time for it to return back to its starting point.

(b) [5] Suppose a hole is drilled straight through the planet, but off-center. Find the time to fall through such a hole.



Problem 14.7 (HRK)**.** [12] Nine small particles, each with mass M, are evenly arranged around a ring of radius R. Use $M = 2 \times 10^5$ and $R = 40m$

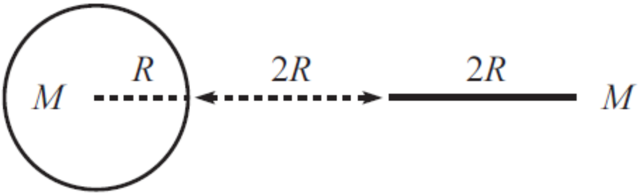
(a) [6] Calculate the net gravitational force on one of the particles due to the other eight particles in the ring. If it is represented in scientific format $a \times 10^b$, where a has 2 significant digits, then find a+b

(b) [6] Find the rotational period of the ring necessary to prevent the ring from collapsing under the mutual gravitational attraction of the particles. Report to the nearest day

Problem 14.8. [7] Find the gravitational potential energy of an object of mass m in the center of a uniform planet of mass M and radius R.. If it is of form $\alpha \frac{GMm}{r}$, then report α . **Hints:** 4 5

Problem 14.9 (Pathfinder)**.** [9] Let us consider the gravitational compression of an interstellar gas cloud. Assume that the gas has density $\rho = 10^{-15}$ kg/m^3 and fills a spherical space. Assume that the temperature is so low that the initial velocity of the gas particles is zero. How long does it take for the gas cloud to shrink? If it is represented in scientific format $a \times 10^b$, then find a+b.

Problem 14.10 (Morin)**.** [9] A uniform sphere with mass m and radius R is placed a distance $2R$ from a uniform stick with mass m and length $2R$, as shown below. The objects are attracted to each other due to gravity. If they are released from rest, what are their speeds right before they collide? If it is of format $v = \sqrt{\alpha Gm/2R}$, then report $\exp \alpha$

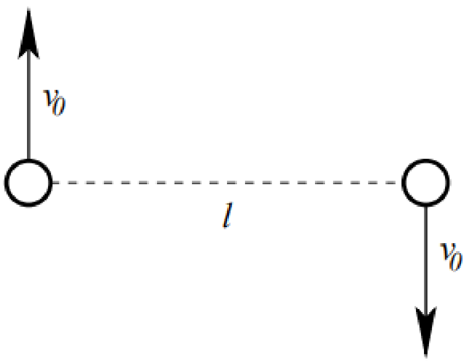


Hint: 27

Problem 14.11. [10] A spherical planet of mass M and radius R has density proportional to radius from the center. That is, $\rho(r) \propto r$ for $0 \leq r \leq R$. What is the gravitational potential energy of an object of mass m at a distance $r \leq R$ from the center of the planet? Use $R = 6400km$, $r = 3200km$, $M = 10^7 kg$, $m = 10kg$. If it is represented in scientific format $a \times 10^b$, where a has 2 significant digits, then find a+b **Hint:** 2

14.5.1 Written Problems

Problem 14.12 (USAPhO). [16] Two masses m separated by a distance l are given initial velocities v_0 as shown in the diagram. The masses interact only through universal gravitation.



(a) [3] Under what conditions will the masses eventually collide?
   (A) Energy= 0
   (B) angular momentum = 0
   (C) Momentum = 0
   (D) none of these

(b) [3] Under what conditions will the masses follow circular orbits of diameter l? Report $Gm/v_0^2 l$

(c) [4] Under what conditions will the masses follow closed orbits? Find minima and report $Gm/v_0^2 l$.

(d) [6] What is the minimum distance achieved between the masses along their path?
   (A) $x = l$ always
   (B) $x = l/(2Gm/lv_0^2 - 1)$ always
   (C) $x = l$ for $Gm < 2v_0^2 l$ and $x = l/(2Gm/lv_0^2 - 1)$ for $Gm > 2v_0^2 l$
   (D) $x = l$ for $Gm < 2v_0^2 l$ and $x = l/(Gm/lv_0^2 - 1)$ for $Gm > 2v_0^2 l$

14.6 Homework Solutions

Problem 14.1 (HRK). [5] Show that on a hypothetical planet having half the diameter of the Earth but twice its density, what is the free fall acceleration in terms of g, where g is the free fall acceleration on the surface of the Earth?

Solution. Let M denote the mass of Earth, and let M' denote the mass of the planet. Then since $V' = \frac{V}{8}$, and $\rho' = 2\rho$, we have $M' = \rho' V' = \frac{M}{4}$.

However, the gravitational force is proportional to $\frac{M}{R^2}$, and since R also shrinks by a factor of 4 here, we see that the new gravitational acceleration is also $\boxed{g}$. ■

Problem 14.2. [5] Assuming the Earth is a uniform sphere of radius R and that the gravitational free-fall acceleration at the surface is g_0, what is the approximate change in free-fall acceleration g at a height h above the surface?

Hint: 9

Solution. We have

$$g_0 = \frac{GM}{R^2},$$

and

$$g = \frac{GM}{(R+h)^2} = \frac{GM}{R^2 + 2Rh + h^2} \approx \frac{GM}{R^2 + 2Rh} = \frac{GM}{R^2} \times \frac{1}{1 + \frac{2h}{R}}.$$

Substituting g_0 and using the Taylor approximation $(1+x)^{-1} = 1 - x$, we see that

$$g = g_0\left(1 - \frac{2h}{R}\right), \implies \Delta g = \boxed{-g_0 \cdot \frac{2h}{R}}.$$

■

Problem 14.3 (Irodov). [5] A satellite orbits a planet with density ρ close to its surface. Find the period of its orbit.

Solution. The gravitational force is $F_g = \frac{GMm}{R^2} = \frac{4}{3}\pi R\rho Gm$. This is the only centripetal force, so it is equal to $m\omega^2 R$, so we have

$$\omega = \sqrt{\frac{4\pi\rho G}{3}} \implies T = \boxed{2\pi\sqrt{\frac{3}{4\pi\rho G}}}.$$

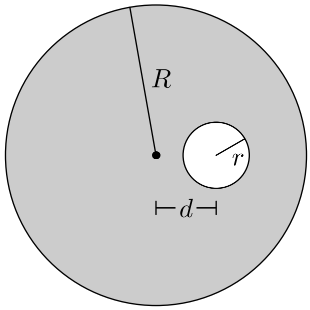
■

Problem 14.4 ($F = ma$). [6] A spherical cloud of dust in space has a uniform density ρ_0 and a radius R_0. The gravitational acceleration of free fall at the surface of the cloud due to the mass of the cloud is g_0.

A process occurs (heat expansion) that causes the cloud to suddenly grow to a radius $2R_0$, while maintaining a uniform (but not constant) density. What is the gravitational acceleration of free fall at a point R_0 away from the center of the cloud due to the mass of the cloud now?

Solution. Recall that by the shell theorem we care only about the sphere of radius R_0. As with the previous problem, we have $g = \frac{GM}{R^2} = \frac{4\pi R\rho G}{3}$, and since the radius is doubled, density is cut by a factor of 8. Hence we have $g = \boxed{1/8 g_0}$. ■

Problem 14.5 (Irodov). [6] A spherical planet has radius R and mass M. Then a spherical cavity of radius r is dug out. The cavity is centered at a distance $d > r$ from the center of the planet. Calculate the magnitude of the gravitational force on a mass m at the center of the planet.



Solution. Note that without the cavity, an object at the center of the planet would experience zero acceleration. Thus, the gravitational force has the same magnitude as if the planet was nonexistent, and the cavity were instead a sphere of mass. However, the force will point in the opposite attraction, being repulsive rather than attractive. First note that the mass of the cavity is proportional to radius, so we have

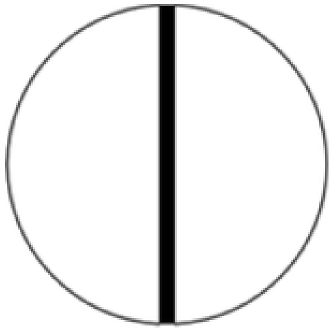
$$M_{\text{cavity}} = \frac{Mr^3}{R^3}.$$

Hence, the gravitational force is

$$\boxed{\frac{G(Mr^3)m}{d^2R^3}}.$$

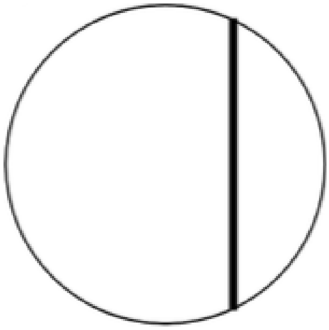
■

Problem 14.6 (PhysicsWOOT)**.** [10] Suppose a planet has uniform density ρ and radius R. The planet does not spin. You dig a hole straight through the planet from one side to the other, through the center. There is no air resistance in the hole.



(a) [5] Suppose you drop something in the hole. Find the time for it to return back to its starting point.

(b) [5] Suppose a hole is drilled straight through the planet, but off-center. Find the time to fall through such a hole.



Solution.

(a) Using the shell theorem, note that at a distance r from the center of the planet, we have

$$F = -\frac{4\pi G\rho m}{3} r.$$

This means the object will undergo simple harmonic motion with angular frequency $\omega = \sqrt{4\pi G\rho/3}$. Hence, we have

$$T = \boxed{\sqrt{\frac{3\pi}{G\rho}}}.$$

(b) Similar to above, we have that $F_y = -\frac{4\pi G\rho m}{3} r_y$, so we see our answer does not change.

$$T = \boxed{\sqrt{\frac{3\pi}{G\rho}}}.$$

■

Problem 14.7 (HRK). [12] Nine small particles, each with mass M, are evenly arranged around a ring of radius R.

(a) [6] Calculate the net gravitational force on one of the particles due to the other eight particles in the ring.

(b) [6] Find the rotational period of the ring necessary to prevent the ring from collapsing under the mutual gravitational attraction of the particles.

Solution.

(a) There is a 40° separation between any two masses as measured from their center of mass. Position one mass at the top. Note that if two masses are separated by a central angle $\theta < 180^\circ$, then their separation distance is $2R\sin(\theta/2)$. By symmetry, it is sufficient that we compute only the downward component of the gravitational force on the top mass exerted by the four masses on the left side and then multiply by 2. We see this is

$$\frac{2GM^2}{R^2}\left(\frac{\cos(70^\circ)}{4\sin^2(20^\circ)} + \frac{\cos(50^\circ)}{4\sin^2(40^\circ)} + \frac{\cos(30^\circ)}{4\sin^2(60^\circ)} + \frac{\cos(10^\circ)}{4\sin^2(80^\circ)}\right)$$

$$= \boxed{3.32\frac{GM^2}{R^2}}.$$

(b) Using part (a), we see that

$$F = 3.32GM^2/R^2 = M\omega^2 R,$$

and rearranging gives $\boxed{2\pi\sqrt{\dfrac{R^3}{3.32GM}}}$.

■

Problem 14.8. [7] Find the gravitational potential energy of an object of mass m in the center of a uniform planet of mass M and radius R. **Hints:** 3 13

Solution. Recall that at $r = R$ we can treat the planet's mass as a point mass, so we have

$$U(R) = -\frac{GMm}{R}.$$

For $r < R$, we have that $F_g = \frac{4}{3}\pi\rho Gmr$, so we can integrate this to obtain change in U_g. Namely we have

$$\Delta U = -\int_R^0 -\frac{4}{3}\pi\rho Gmr\, dr = \int_R^0 \frac{4}{3}\pi\rho Gmr\, dr = -\frac{2}{3}\pi\rho GmR^2.$$

Substituting $\rho = M/V = 3M/(4\pi R^3)$, we see that

$$\Delta U = -\frac{GMm}{2R},$$

so we see

$$U(0) = U(R) + \Delta U = \boxed{-\frac{3GMm}{2R}}.$$

■

Problem 14.9 (Pathfinder)**.** [9] Let us consider the gravitational compression of an interstellar gas cloud. Assume that the gas has density $\rho = 10^{-15}$ kg/m^3 and fills a spherical space. Assume that the temperature is so low that the initial velocity of the gas particles is zero. How long does it take for the gas cloud to shrink?

Solution. Note that for a specific particle, we need only consider the mass enclosed (by the Shell Theorem), and this does not change during the compression process. Hence, we can consider the path of the particle as a degenerate ellipse. If the particle starts from a distance x, the enclosed mass is

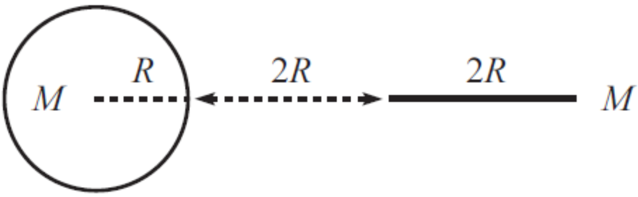
$$M = \frac{4}{3}\pi x^3\rho.$$

The semi-major axis of the ellipse is $x/2$, so we see half the period is

$$\frac{T_0}{2} = \frac{1}{2}\cdot 2\pi\sqrt{\frac{a^3}{GM}} = \pi\sqrt{\frac{3x^3}{32\pi Gx^3\rho}} = \sqrt{\frac{3\pi}{32G\rho}},$$

which is roughly $\boxed{2\times 10^{12}\text{ seconds}}$. ■

Problem 14.10 (Morin). [9] A uniform sphere with mass m and radius R is placed a distance $2R$ from a uniform stick with mass m and length $2R$, as shown below. The objects are attracted to each other due to gravity. If they are released from rest, what are their speeds right before they collide?



Solution. The easiest solution here is to use conservation of energy. Treating the sphere as a point mass and noting the stick has uniform density $m/2R$, we can integrate to obtain the initial potential energy, noting that dm' (an infinitesimal portion of the stick) is equal to $\frac{m}{2R}\,dr$:

$$U_i = -\int \frac{Gm\,dm'}{r} = -\frac{Gm^2}{2R}\int_{3R}^{5R}\frac{dr}{r} = -\frac{Gm^2}{2R}\ln(5/3).$$

Right before the objects collide, we again compute the gravitational potential energy:

$$U_f = -\int \frac{Gm\,dm'}{r} = -\frac{Gm^2}{2R}\int_{R}^{3R}\frac{dr}{r} = -\frac{Gm^2}{2R}\ln(3).$$

Hence, we have

$$U_f - U - i = -\frac{Gm^2}{2R}\ln(9/5).$$

This must be the change in potential energy, and since the masses are equal, they will have equal kinetic energy. Thus, we have

$$mv^2 = U_f - U - i \implies v = \boxed{\sqrt{\frac{Gm}{2R}\ln(9/5)}}.$$

■

Problem 14.11. [10] A spherical planet of mass M and radius R has density proportional to radius from the center. That is, $\rho(r) \propto r$ for $0 \le r \le R$. What is the gravitational potential energy of an object of mass m at a distance $r \le R$ from the center of the planet? 14

Solution. We can write $\rho(r) = cr$ for some constant c. In that case, the mass of a shell of radius r is

$$M(r) = \int_0^r cr(4\pi r^2)\,dr = c\pi r^4.$$

Hence, the force at a radius r is

$$F = \frac{GM(r)m}{r^2} = Gc\pi m r^2.$$

By integrating this force, we can find the gravitational potential energy as the radius of an object decreases. At radius R, the gravitational potential energy is

$$U(R) = -\frac{GMm}{R}.$$

Hence we have

$$\Delta U = -\int F\, dr = -\int_R^r -Gc\pi m r^2\, d\vec{r} = \int_R^r Gc\pi m r^2\, dr,$$

since $d\vec{r}$ and the force point in opposite directions. This gives

$$\Delta U = \int_R^r Gc\pi m r^2\, dr = \frac{1}{3}Gc\pi m(r^3 - R^3).$$

As a sanity check, we can check that the potential energy change is indeed negative (since the gravitational force points in the direction of lower gravitational potential energy). We now solve for c. We have that

$$M = \int \rho\, dV = \int cr\, dV = \int_0^R cr(4\pi r^2)\, dr = c\pi R^4 \implies c = \frac{M}{\pi R^4}.$$

Substituting gives

$$\Delta U = \frac{GMm}{3R^4}(r^3 - R^3).$$

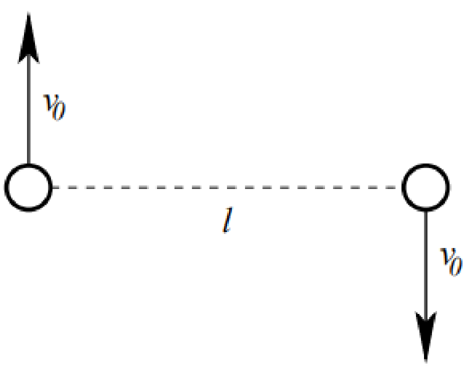
Then we have

$$U(r) = U(R) + \Delta U = \boxed{\frac{GMmr^3}{3R^4} - \frac{4GMm}{3R}}.$$

■

14.6.1 Written Solutions

Problem 14.12 (USAPhO)**.** [16] Two masses m separated by a distance l are given initial velocities v_0 as shown in the diagram. The masses interact only through universal gravitation.



(a) [3] Under what conditions will the masses eventually collide?

(b) [3] Under what conditions will the masses follow circular orbits of diameter l?

(c) [4] Under what conditions will the masses follow closed orbits?

(d) [6] What is the minimum distance achieved between the masses along their path?

Solution.

(a) Note that the angular momentum about their center of mass is $L = v_0 l$ and is constant, where l is the distance between the masses. Hence as the masses move closer together, v_0 increases proportional to $1/l^2$, which is a much faster rate than the potential energy increases. Hence from an energy perspective it is also impossible for the masses to collide if the angular momentum is nonzero.

(b) We can use the centripetal force equation:

$$\sum F_c = mv^2/r$$
$$\implies \frac{Gm^2}{l^2} = \frac{mv_0^2}{l/2}$$
$$\implies \boxed{Gm = 2v_0^2 l}.$$

(c) The masses will follow closed orbits if the total energy is negative. We see this happens when

$$mv_0^2 - \frac{gm^2}{l} < 0 \implies \boxed{Gm > v_0^2 l}.$$

(d) Because the masses will always move symmetrically about their center of mass, note that at the minimal separation, the velocity of each mass is perpendicular to the line joining the masses (this is true whether for a circle, ellipse, parabola, etc.). If we let x denote the minimum separation and let v denote the velocity at this minimum separation, then by conservation of momentum and conservation of energy, we have

$$vx = v_0 l$$
$$mv^2 - \frac{Gm^2}{x} = mv_0^2 - \frac{Gm^2}{l}.$$

Substituting $v = v_0 l/x$ gives

$$mv_0^2 \frac{l^2}{x^2} - \frac{Gm^2}{x} = mv_0^2 - \frac{Gm^2}{l}$$
$$\frac{l^2}{x^2} - \frac{Gm}{xv_0^2} = 1 - \frac{Gm}{lv_0^2}$$
$$\implies \frac{l^2}{x^2} - \frac{Gm}{lv_0^2}\left(\frac{l}{x}\right) + \left(\frac{Gm}{lv_0^2} - 1\right) = 0.$$

Solving gives solutions $x = l$ or $x = l/(\frac{Gm}{lv_0^2} - 1)$. we see the second solution is smaller for $Gm > 2v_0^2 l$, so we conclude the minimum separation is

$$x = \boxed{\begin{cases} l, & Gm \le 2v_0^2 l \\ l/(\frac{Gm}{lv_0^2} - 1), & Gm > 2v_0^2 l \end{cases}}.$$

■

15 Orbital Mechanics

We will now use our newfound knowledge about gravitation to study one of the core topics of astrophysics: the mechanics of planetary orbits, also known as **orbital mechanics**. Parts of this section will be somewhat more challenging and mathematically involved than others (for instances, the proofs of various laws) so feel free to skip over them.

15.1 Kepler's Laws

Kepler's three laws give some basic properties of planetary motion. Specifically, Kepler's laws state the following:

Theorem 15.1 (Kepler's Laws).

1. ***Law of Orbits:*** *The orbit of every planet is an ellipse with the Sun being one of the foci.*
2. ***Law of Areas:*** *As planets move, they sweep through elliptical arcs of equal area in equal amounts of time.*
3. ***Law of Periods:*** *As the radius of orbit changes, the period of orbit changes according to the proportion*

$$T^2 \propto a^3.$$

where a is the semi-major axis.

Note that Kepler's second law tells us is two things. First, consider the small area increment ΔA covered in a time interval Δt. The area of this approximately triangle wedge is given by its basic formula $A = \frac{1}{2}bh$ where the base is $r\Delta\theta$ and the height is r. The rate at which this area is swept out is

$$\frac{\Delta A}{\Delta t} = \frac{1}{2}\frac{(r\Delta\theta)(r)}{\Delta t}.$$

In the instantaneous limit, this in turn becomes

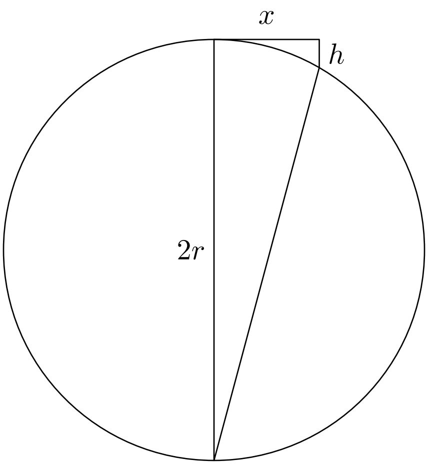
$$\frac{dA}{dt} = \lim_{\Delta t\to 0}\frac{\Delta A}{\Delta t} = \lim_{\Delta t\to 0}\frac{1}{2}r^2\frac{\Delta\theta}{\Delta t} = \frac{1}{2}r^2\omega$$

Second, assuming that the more massive body M is practically at rest. The angular momentum of the orbiting body m relative to the origin at the central body is, when choosing the z-axis to be perpendicular to the plane of origin $L_z = I\omega = mr^2\omega$. And therefore,

$$\frac{dA}{dt} = \frac{L_z}{2m}.$$

We will also now give a proof of Kepler's Third Law. The proof is rather long, and as such is optional to look over. However, the proof utilizes many nice things that we have been learning before and therefore, it is advised to look over it.

Proof. Let's first look at how to relate speed and acceleration (or gravitational forces). Let the body move in a circle for an infinitesimally short time so that the displacement in the horizontal direction (see figure) would be x and h in the vertical direction.



From the equality of boundary angles we get similar triangles

$$\frac{h}{x} = \frac{x}{2r-h} \approx \frac{x}{2r}, \quad h = \frac{x^2}{2r}.$$

Let this movement take place during the period Δt. Recall the formula for centripetal acceleration:

$$a_c = \frac{F}{m} = \frac{v^2}{r}.$$

Now move the object to the perigee of its elliptical orbit (closest point to a focus). The corresponding displacements are denoted by x' and h'. If we compress the ellipse in the direction of the semi-major axis by a coefficient $k = a/b$, we would get a circle with radius $r = b$. Doing that and combining it with our earlier result, we can see that:

$$\frac{h'}{k} = \frac{x'^2}{2b}, \qquad h' = \frac{x'^2 a}{2b^2}, \qquad \frac{F}{m} = \frac{v^2 a}{b^2},$$

where we have switched to acceleration and speed. The distance of the perigee from the focus is $a-c$, hence

$$F = \frac{GMm}{(a-c)^2}, \qquad v^2 = \frac{Fb^2}{ma} = \frac{GMmb^2}{ma(a-c)^2} = \frac{GM(a^2-c^2)}{a(a-c)^2} = \frac{GM(a+c)}{a(a-c)},$$

where we proceeded from the relation $a^2 = b^2 + c^2$. To determine the orbital period, we use Kepler's Second law (i.e. the time-area equality):

$$\frac{T}{S} = \frac{\Delta t}{\Delta S}, \qquad T = \pi ab \frac{\Delta t}{\frac{(a-c)v\Delta t}{2}} = \frac{2\pi ab}{(a-c)v},$$

where the area of the ellipse is $S = \pi ab$ and the area of the smaller section was from an isosceles triangle with base $v\Delta t$ and height $a-c$. Since we already have an expression for v^2, we can find that

$$T^2 = \frac{(2\pi)^2 a^2 b^2}{(a-c)^2 v^2} = \frac{(2\pi)^2 a^2 b^2 a(a-c)}{(a-c)^2 GM(a+c)} = \frac{(2\pi)^2 a^3 b^2}{GM(a^2-c^2)} = \frac{(2\pi)^2 a^3}{GM},$$

proving Kepler's Third Law (the square of the period is proportional to the cube of the semi-major axis). Note that the constant of proportionality is given by

$$\frac{T^2}{a^3} = \frac{4\pi^2}{GM}.$$

■

Kepler's Third Law is very important in many celestial mechanics problems.

Example 15.2. An object of mass m is separated a distance R away from a large mass M. Calculate the total amount of time required for m to reach M. Assume that $m \ll M$.

Solution. There are, in fact, two different ways we can solve this problem. One way is by conserving energy and then integrating velocity to find the total amount of time. This approach leads to some very nasty results but will lead to an elegant answer at the end. However, usually if there is an elegant answer, there will be an elegant solution. Note that the linear path of the $m \to M$ is like a degenerate ellipse. So we can use Kepler's laws!

As an ellipse gets very thin, its foci go towards the ends. So, the sun, which has to be at the focus of orbit, is at the very end of this elongated ellipse. From Kepler's third law,

$$T^2 \propto a^3 \implies T \propto a^{3/2}.$$

From the definition of the degenerate ellipse, the semi-major axis is just $a = R/2$. This means that the total period will be $\frac{T}{2^{3/2}}$ where T is the time taken for m to orbit M in a circular orbit. To find T, we again note from Kepler's laws that

$$\frac{T^2}{a^3} = \frac{4\pi^2}{GM} \implies T = 2\pi\sqrt{\frac{R^3}{GM}}.$$

Since the time taken to reach the large mass is just half of this period, the answer is

$$t = \frac{1}{2} \cdot \frac{1}{2^{3/2}} 2\pi\sqrt{\frac{R^3}{GM}} = \boxed{\frac{1}{2\sqrt{2}}\pi\sqrt{\frac{R^3}{GM}}}$$

■

There will be a similar problem using this idea of degenerate ellipses in the problems section of this handout. Make sure to try that problem to test your understanding of this concept.

Exercise. Show that in a binary star system, that two stars of masses M_1 and M_2 move in elliptical orbits one of whose foci (for each orbit) is at the center of mass of the system and show that the orbital period of the system is given by

$$T^2 = \frac{4\pi^2 \left(a_1 + a_2\right)^3}{G\left(M_1 + M_2\right)},$$

where a_1 and a_2 are the semi-major axes of the elliptical orbits.

15.2 Energy

Definition 15.3. The **total energy** of an object in orbital motion is given by

$$E = -\frac{GMm}{2a}.$$

Proof. Since the total energy of the object in orbit does not change, we write energy expressions at the apogee and perigee of the object's orbit:

$$E = \frac{mv_1^2}{2} - \frac{GMm}{r_1}, \quad E = \frac{mv_2^2}{2} - \frac{GMm}{r_2}.$$

We consider using conservation of angular momentum, $r_1 v_1 = r_2 v_2$. We multiply the first (energy) equation by r_1^2 and the second by r_2^2 and subtract, noting that $r_1 = a - c$ and $r_2 = a + c$:

$$E\left(r_1^2 - r_2^2\right) = -GMm(r_1 - r_2),$$

$$E = -GMm\frac{r_1 - r_2}{r_1^2 - r_2^2} = -\frac{GMm}{r_1 + r_2} = -\frac{GMm}{2a}.$$

■

If the total energy is negative, then the object is unable to escape the gravitational field of the star to the point at infinity since infinitely far away, the potential energy is zero and the total energy would be equal to the kinetic energy, which cannot be negative. Sometimes problems give an object enough speed to escape the gravitational field, in which case the total energy would be positive.

Example 15.4. An explosion takes place near a star, causing many small objects to fly outwards with speed v. The objects begin to move in elliptical orbits, with one of the foci at the star. Determine the locus (set of possible positions) of the second focus.

Solution. For this problem, one only needs to know the properties of an ellipse and the fact that the total energy, which is the same for all pieces, depends on the longer half-axis of the ellipse, which thus becomes the same for all pieces. Additionally, we know that the sum of the distances to the foci from each point on the ellipse is equal to twice the semi-major axis. We see that the total energy is given by

$$E = \frac{1}{2}mv^2 - \frac{GMm}{R} = -\frac{GMm}{2a}$$

where R is the distance from the star to the explosion and a is the semi-major axis of the resulting orbit. Since this value is constant, we can find that

$$a = \frac{GMR}{2GM - v^2R}.$$

We know that the sum of the distances to the foci is $2a$; one of these distances is already known to be R. Thus, the locus of the second foci of the ellipse is a circle (with center at the explosion) of radius

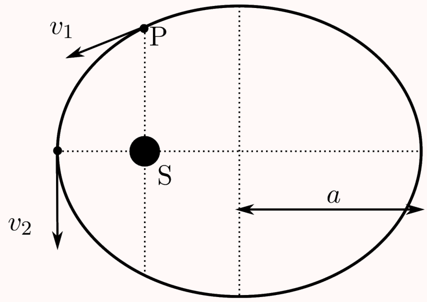
$$r = 2a - R = \frac{2GMR}{2GM - v^2R} - R = \boxed{\frac{v^2R^2}{2GM - v^2R}}.$$

■

15.2.1 The Vis-Viva Equation

The Vis-Viva equation is essentially a killer formula that can destroy several gravitation problems. We can first start with a small example.

Example 15.5 ($F = ma$). A planet orbits around a star S, as shown in the figure. The semi-major axis of the orbit is a. The perigee, namely the shortest distance between the planet and the star is $0.5a$. When the planet passes point P (on the line through the star and perpendicular to the major axis), its speed is v_1. What is its speed v_2 when it passes the perigee?



Solution. Because the sum of the distances from a point on the ellipse to the two foci is constant, we can solve for the length SP as

$$SP + \sqrt{SP^2 + a^2} = 2a \implies SP = \frac{3}{4}a.$$

Since we know the length of SP, the total energy at P is

$$E = \frac{1}{2}mv_1^2 - \frac{GMm}{3a/4}.$$

Since the length of the planet at periastron is $a/2$, the energy at periastron is

$$E = \frac{1}{2}mv_2^2 - \frac{GMm}{a/2}.$$

Lastly, we know that $E = -\frac{GMm}{2a}$ thus by replacing E into our conservation of energy equation at P, we have

$$E = \frac{1}{2}mv_1^2 + \frac{8}{3}E \implies v_1^2 = -\frac{10}{3}\frac{E}{m}.$$

The equation for energy at periastron is

$$v_2^2 = -6\frac{E}{m}$$

Finally, we have that

$$\frac{E}{m} = -\frac{3}{10}v_1^2 \implies v_2^2 = -6 \cdot \left(-\frac{3}{10}v_1^2\right) \implies v_2 = \frac{3}{\sqrt{5}}v_1$$

. ∎

What we have just done, is used the vis-viva equation without knowing it!

Theorem 15.6 (Vis-Viva Equation). *In an orbit,*

$$v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right).$$

Proof. Let a mass m be in orbit about a mass M. Let the velocity of m be v when it is a distance r from M and let a be the semi-major axis of the orbit. We then use conservation of energy to find that

$$E = \frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}$$
$$v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right).$$

■

We can now write a much shorter and quicker solution to this problem.

Solution. Because the sum of the distances from a point on the ellipse to the two foci is constant, we can solve for the length SP as

$$SP + \sqrt{SP^2 + a^2} = 2a \implies SP = \frac{3}{4}a.$$

The vis-viva equation then tells us

$$\frac{v_2^2}{v_1^2} = \frac{\frac{2}{a/2} - \frac{1}{a}}{\frac{2}{3a/4} - \frac{1}{a}} = \frac{4-1}{8/3-1} = \frac{9}{5}.$$

Thus, $v_2 = \frac{3}{\sqrt{5}}v_1$. ■

15.2.2 The Virial Theorem

Theorem 15.7 (Virial Theorem). *The relationship between the average kinetic energy $\langle T \rangle$ and potential energy $\langle U \rangle$ in an orbit is given by*

$$\langle T \rangle = -\frac{1}{2}\langle U \rangle.$$

While this theorem is not as useful as the vis-viva equation, it is still very important as we can express the total energy soley in terms of kinetic or potential energy.

Example 15.8 ($F = ma$). A satellite is in a circular orbit about the Earth. Over a long period of time, the effects of air resistance decrease the satellite's total energy by 1 J. What happens to the kinetic energy of the satellite?

Solution. By the virial theorem, the total energy in an orbit is given by

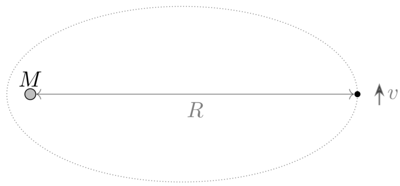
$$E = \langle T \rangle + \langle U \rangle = -\langle T \rangle$$

Because the satellites total energy decreases by 1 J, this means the total kinetic energy increases by 1 J. ■

15.3 Homework Problems

Problem 15.1 (NAO). [4] Imagine that our Sun was suddenly replaced by an M-dwarf with a mass half that of the Sun (By Ashmit of course). If our Earth kept the same semi-major axis during this change, what would Earth's new orbital period be around the M-dwarf? Report $(T'/T)^2$

Problem 15.2 ($F = ma$). [4] A mote of dust is initially located at distance R from the sun, which has mass M. At this point, the mote has a small tangential velocity v. Which of the following is a good approximation for the distance of closest subsequent approach between the mote and the sun? If it is of the format $\alpha R^a v^b G^c M^d$, then report $\alpha + a + 2b + c + d$



Problem 15.3 ($F = ma$). [5] An astronaut standing on the exterior of the international space station wants to dispose of three pieces of trash. They face the station's direction of travel with the Earth to their left. From the astronaut's perspective, the three pieces are thrown (I) left, (II) right, and (III) up. To the astronaut's frustration, some of the pieces of trash return to the space station after several hours. In which ways, (I), (II), and/or (III) can the astronaut do this? Report sum of numbers of pieces which come back.

Problem 15.4 (Mock $F = ma$). [5] Rick Astley is on a new planet with a grandfather clock calibrated on Earth.A person from earth asks him a question about time. Now, Rick has promised to never give you up and never let you down, so he has to answer, but the time scale on the new planet is different. This new solar system is an exact copy of our solar system except all distances have been doubled. For example, the distance from the planet to the star will be doubled, and the radius will be doubled as well, among other lengths. The density stays the same. Find out Earth years will Rick Astley measure to be one year on this new planet?

Problem 15.5 (Irodov). [6] A planet moves around the Sun (mass M) along an ellipse so that its minimum distance to the Sun is r_1, and its maximum distance to the sun is r_2. Find the period of its revolution. Use $M = 2 \times 10^{30} kg$, $r_1 = 1.47 \times 10^{11} m$, $r_2 = 1.52 \times 10^{11} m$ and $\pi = \pi$ (What did you even think I was gonna give lmao). Report in days rounded to nearest day.

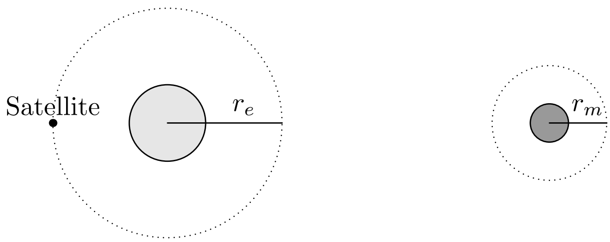
Problem 15.6 ($F = ma$). [7] A body of mass M and a body of mass $m \ll M$ are in circular orbits about their center of mass under the influence of their mutual gravitational attraction to each other. The distance between the bodies is R, which is much larger than the size of either body.

A small amount of matter $\delta m \ll m$ is removed from the body of mass m and transferred to the body of mass M. The transfer is done in such a way so that the orbits of the two bodies remain circular, and remain separated by a distance R. Which of the following statements is correct?

(A) The gravitational force between the two bodies increases.
(B) The gravitational force between the two bodies remains constant.
(C) The total angular momentum of the system increases.
(D) The total angular momentum of the system remains constant.
(E) The period of the orbit of two bodies remains constant.

Problem 15.7 (Pathfinder). [8] A small moon of radius r and mass m is orbiting around a planet of mass M in a circular orbit of radius R ($R \gg r$) always keeping the same face towards the planet. If an object on the moon closest to the planet is in weightlessness, find a suitable expression for the radius of the orbit. Use $r = 1m$, $m = 1kg$ (yes, the moon has 1kg mass, problem?) , and $M = 109503kg$

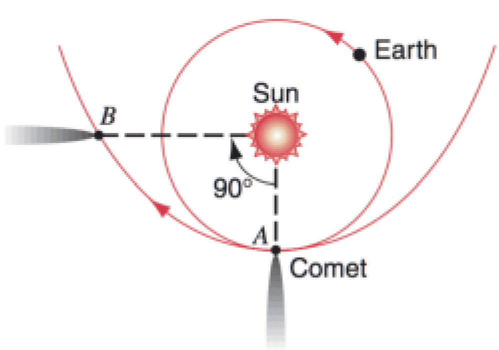
Problem 15.8 (Mock $F = ma$). [8] NASA wishes to bring a small rocket of mass m to the moon. First, it starts off in a circular orbit at a semi-major axis r_e around Earth. The engines fire in a negligible amount of time and bring the rocket to the moon, a distance d away. You may assume that $d \gg r_e$. Once the rocket reaches the moon, the engines fire once again and bring the rocket into a circular orbit at a semi-major axis r_m. The mass of the Earth and the moon are M_e and M_m respectively.



What is the total impulse the engines must impart on the rocket? Use $m = 419kg$, $r_e = 10m$, $M = 10^1 1kg$, ignore relativistic effects due to unrealistic values please and thank you. Report 3 significant digits

Problem 15.9 (Kalda). [9] An object is thrown vertically from the ground and reaches a distance $R = R_M$ above the surface of the Earth before returning. If R_M is the radius of the earth, determine the time of flight of the object. Use $R = 10m$, $M = 10^1 1kg$, report to nearest second

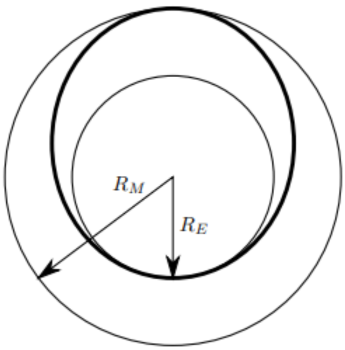
Problem 15.10 (HRK). [11] A comet passes by the Sun as shown, in a parabolic path.



How long, in years, does the comet take to get from point A to point B? Report only 1 significant digit. **Hint:** 7

Problem 15.11 (USAPhO). [15] A ship starts out in a circular orbit around the sun very near the Earth and has a goal of moving to a circular orbit around the Sun that is very close to Mars. It will make this transfer in an elliptical orbit as shown in bold in the diagram below. This is accomplished with an initial velocity boost near the Earth Δv_1 and then a second velocity boost near Mars Δv_2.

Assume that both of these boosts are from instantaneous impulses, and ignore mass changes in the rocket as well as gravitational attraction to either Earth or Mars. Don't ignore the Sun! Assume that the Earth and Mars are both in circular orbits around the Sun of radii R_E and $R_M = R_E/\alpha$ respectively. The orbital speeds are v_E and v_M respectively. Use $\alpha = 0.25$



(a) [5] Derive an expression for the velocity boost Δv_1 to change the orbit from circular to elliptical. Express your answer in terms of v_E and α.Report $\Delta v_1/v_E$ in 3 significant figures.

(b) [5] Derive an expression for the velocity boost Δv_2 to change the orbit from elliptical to circular. Express your answer in terms of v_E and α. Report $\Delta v_2/v_M$ in 3 significant figures.

(c) [5] What is the angular separation between Earth and Mars, as measured from the Sun, at the time of launch so that the rocket will start from Earth and arrive at Mars when it reaches the orbit of Mars? Express your answer in terms of α.Report θ/π

15.3.1 Written Solutions

Problem 15.12 (HRK). [18] A pair of stars revolve about their common center of mass. On of the stars has a mass M that is twice the mass M of the other; that is, $M = 2m$. Their centers are a distance d apart, d being large compared to the size of either star.

(a) [6] Derive an expression for the period of revolution of the stars about their common center of mass in terms of d, m and G. Report $T = \alpha d^a G^b M^c$, then report $\alpha + a + b + c$ to 3 significant figures

(b) [6] Compare the angular momenta of the two stars about their common center of mass by calculating the ratio L_m/L_M.

(c) [6] Compare the kinetic energies of the two stars by calculating the ratio K_m/K_M.

15.4 Homework Solutions

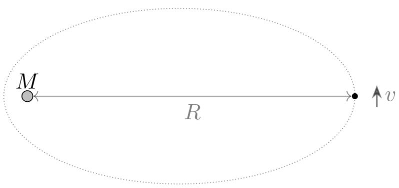
Problem 15.1 (NAO). [4] Imagine that our Sun was suddenly replaced by an M-dwarf with a mass half that of the Sun. If our Earth kept the same semi-major axis during this change, what would Earth's new orbital period be around the M-dwarf?

Solution. Recall that Kepler's Third Law gives the formula

$$\frac{T^2}{a^3} = \frac{4\pi^2}{GM}.$$

If M halves, and a remains constant, then T increases by a factor of $\sqrt{2}$, so the new orbital period is $\boxed{T\sqrt{2}}$. ■

Problem 15.2 ($F = ma$). [4] A mote of dust is initially located at distance R from the sun, which has mass M. At this point, the mote has a small tangential velocity v. Which of the following is a good approximation for the distance of closest subsequent approach between the mote and the sun?



Solution. Let the radius and velocity at the perigree be R' and v'. Then we have by conservation of angular momentum that

$$mv'R' = mvR.$$

We note that we can approximate both the initial kinetic energy and initial potential energy as 0, since v is small and R is rather large. Then by conservation of energy we have

$$\frac{-GMm}{R'} = \frac{1}{2}mv'^2 = 0,$$

and substituting $v' = vR/R'$ and solving for R' gives

$$R' = \boxed{\frac{R^2v^2}{2GM}}.$$

■

Problem 15.3 ($F = ma$). [5] An astronaut standing on the exterior of the international space station wants to dispose of three pieces of trash. They face the station's direction of travel with the Earth to their left. From the astronaut's perspective, the three pieces are thrown (I) left, (II) right, and (III) up. To the astronaut's frustration, some of the pieces of trash return to the space station after several hours. In which ways, (I), (II), and/or (III) can the astronaut do this?

Solution. Recall by Kepler's third law that T depends only on semi-major axis of rotation a. Since the energy change in the trash is negligible, a will not change significantly, so all three pieces of trash will return after one full orbit. ■

Problem 15.4 (Mock $F = ma$)**.** [5] The Little Prince is on a new planet with a grandfather clock calibrated on Earth. This new solar system is an exact copy of our solar system except all distances have been doubled. For example, the distance from the planet to the star will be doubled, and the radius will be doubled as well, among other lengths. The density stays the same. How many Earth years will the Little Prince measure to be one year on this new planet?

Solution. Kepler's 3rd law says $T^2 = \frac{4\pi^2}{GM} r^3$. When the distance doubles, mass of sun goes from M to $8M$ and radius of orbit goes from r to $2r$. Doing the calculations tell us that the time period does not change by doubling distance.

Alternatively, we can go by the more mathematical route by substituing $M = 4/3\pi R_e^3$ in kepler's law to get $T' = T(\frac{r'/r}{R_e'/R_e})^{3/2}$. Now, as all distances scale equally, we can conclude that $T = T'$ ■

Problem 15.5 (Irodov)**.** [6] A planet moves around the Sun (mass M) along an ellipse so that its minimum distance to the Sun is r_1, and its maximum distance to the sun is r_2. Find the period of its revolution.

Solution. Note that the semi-major axis is given by $a = (r_1 + r_2)/2$. By Kepler's Third Law, we have

$$T = 2\pi\sqrt{\frac{a^3}{GM}} = 2\pi\sqrt{\frac{(r_1+r_2)^3}{8GM}} = \boxed{\pi\sqrt{\frac{(r_1+r_2)^3}{2GM}}}.$$

■

Problem 15.6 ($F = ma$)**.** [7] A body of mass M and a body of mass $m \ll M$ are in circular orbits about their center of mass under the influence of their mutual gravitational attraction to each other. The distance between the bodies is R, which is much larger than the size of either body.

A small amount of matter $\delta m \ll m$ is removed from the body of mass m and transferred to the body of mass M. The transfer is done in such a way so that the orbits of the two bodies remain circular, and remain separated by a distance R. Which of the following statements is correct?

(A) The gravitational force between the two bodies increases.

(B) The gravitational force between the two bodies remains constant.

(C) The total angular momentum of the system increases.

(D) The total angular momentum of the system remains constant.

(E) The period of the orbit of two bodies remains constant.

Solution. Recall the exercise we gave earlier in the chapter. It stated that the period of a binary star system is

$$T = 2\pi\sqrt{\frac{(a_1+a_2)^3}{G(M_1+M_2)}}.$$

In this case we see the sum of the masses does not change, so the period remains constant, giving an answer of (E). We'll now prove the exercise as well. Note the distances from the center of mass are $r_1 = \frac{mR}{M+m}$ and $r_2 = \frac{MR}{M+m}$. The centripetal force is gravity for both of the binary stars, so we see

$$Mr_1\omega^2 = G\frac{Mm}{R^2} \implies \omega = \sqrt{\frac{Gm}{r_1R^2}}.$$

Substituting for r_1, we obtain that

$$\omega = \sqrt{\frac{G(M+m)}{R^3}},$$

and the result follows. ■

Problem 15.7 (Pathfinder)**.** [8] A small moon of radius r and mass m is orbiting around a planet of mass M in a circular orbit of radius R ($R \gg r$) always keeping the same face towards the planet. If an object on the moon closest to the planet is in weightlessness, find a suitable expression for the radius of the orbit.

Solution. Let us analyze a point that is in between both objects a distance R from M. It has a mass m_0. This mass will experience a centrifugal force of

$$F_{\text{centrifugal}} = m_0\omega_0^2(R-r).$$

We can find ω_0^2 from the circular velocity or

$$v_{\text{circular}}^2 = \frac{GM}{R} \implies \omega_0^2 = \frac{GM}{R^3}.$$

From a force balance, we have that

$$\frac{Gmm_0}{r^2} + \frac{GMm_0}{R^3}(R-r) = \frac{GMm_0}{(R-r)^2}.$$

Using a taylor expansion, we can approximate

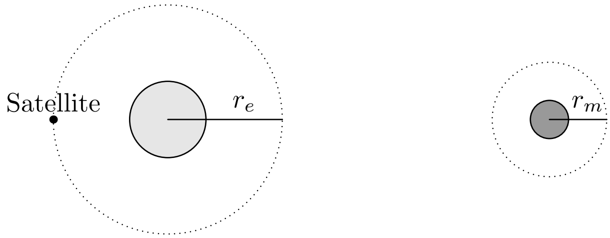
$$\frac{GMm_0}{(R-r)^2} \approx \frac{GMm_0}{R^2}\left(1+\frac{2r}{R}\right).$$

Now by simplifying, we find that

$$\frac{m}{r^2} = \frac{3Mr}{R^3} \implies R = \boxed{r\left(\frac{3M}{m}\right)^{1/3}}.$$

This point described in the problem is similar to the Lagrange point, 5 such points exist for 2 body systems ■

Problem 15.8 (Mock $F = ma$)**.** [8] NASA wishes to bring a small rocket of mass m to the moon. First, it starts off in a circular orbit at a semi-major axis r_e around Earth. The engines fire in a negligible amount of time and bring the rocket to the moon, a distance d away. You may assume that $d \gg r_e$. Once the rocket reaches the moon, the engines fire once again and bring the rocket into a circular orbit at a semi-major axis r_m. The mass of the Earth and the moon are M_e and M_m respectively.



What is the total impulse the engines must impart on the rocket?

Solution. Initially the rocket is in a circular orbit around Earth. It has an orbital speed of

$$v_1 = \sqrt{\frac{GM_e}{r_e}}$$

Let d be the distance between Earth and the moon. After performing a burn, the rocket should not have an elliptical orbit where the furthest point is roughly at an altitude d. Realistically, it would be $d + r_e + r_m$ but we can assume $r_e \ll d$ and $r_m \ll d$. Thus the semi-major axis will now be $d/2$. Per the vis-visa equation, the speed will now be:

$$v_2 = \sqrt{GM_e\left(\frac{2}{r_e} - \frac{2}{d}\right)}$$

Substituting in values, we can calculate $\Delta v_1 = v_2 - v_1$ Now let us determine the speed of the spacecraft once it has reached the moon. Conservation of energy gives:

$$\begin{aligned}\frac{1}{2}v_3^2 - \frac{GM_m}{r_m} - \frac{GM_e}{d} &= \frac{1}{2}v_2^2 - \frac{GM_e}{r_e}\\ v_3^2 &= GM_e\left(\frac{2}{r_e} - \frac{2}{d}\right) - \frac{2GM_e}{r_e} + \frac{2GM_m}{r_m} + \frac{2GM_e}{d}\\ &= \frac{2GM_m}{r_m}\end{aligned}$$

so we have

$$v_3 = \sqrt{\frac{2GM_m}{r_m}}$$

Now we need to circularize the orbit. After circularizing, the speed will then be:

$$v_4 = \sqrt{\frac{GM_m}{r_m}}$$

so the change in speed will be:

$$\Delta v_2 = v_3 - v_4 = \left(\sqrt{2} - 1\right)\sqrt{\frac{GM_m}{r_m}}$$

so the total change in velocity the engines need to impart (impulse/mass) is:

$$\Delta v_1 + \Delta v_2 = \sqrt{GM_e\left(\frac{2}{r_e} - \frac{2}{d}\right)} - \sqrt{\frac{GM_e}{r_e}} + \left(\sqrt{2} - 1\right)\sqrt{\frac{GM_m}{r_m}}$$

We can make this easier if we let them assume d is very big such that:

$$\Delta v_T = \left(\sqrt{2}-1\right)\left(\sqrt{\frac{GM_m}{r_m}}+\sqrt{\frac{GM_e}{r_e}}\right)$$

Note: this solution shows the method to solve this in the general case. However, since the problem stated that d is very large, we can say that as the rocket performs the burn to intercept with the moon, it creates an elliptical orbit where the speed at aphelion is zero and the potential energy is also zero. Thus we can use energy conservation:

$$-\frac{GM_e m}{r_e^2}+\frac{1}{2}mv^2 = 0 \implies v = \boxed{\sqrt{\frac{2GM_e}{r_e^2}}}.$$

■

Problem 15.9 (Kalda). [9] An object is thrown vertically from the ground and reaches a distance $R = R_M$ above the surface of the Earth before returning. If R_M is the radius of the earth, determine the time of flight of the object.

Solution. The flight path of the object is along an infinitely narrow ellipse with semi-major axis R (one focus is at the center of the earth and the other is at the tip of the trajectory of the object). The period of orbit depends solely on the semi-major axis (by Kepler's third law). Furthermore, the area swept through by the radius vector is proportional to the time of flight. Here, the radius vector covers an area consisting of half the area of an ellipse and an isosceles triangle (of base $2b$ and height a):

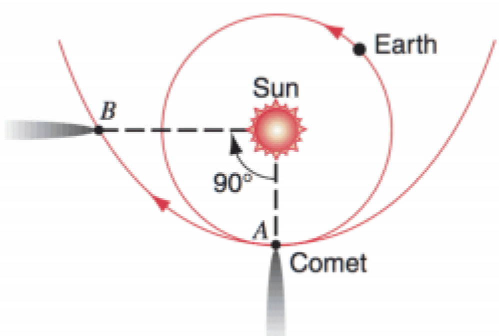
$$\frac{\tau}{S'} = \frac{T_0}{S_0}, \quad \text{where} \quad S_0 = \pi ab \quad \text{and} \quad S' = \frac{1}{2}\pi ab + ba.$$

So we have that

$$\tau = \frac{\frac{1}{2}\pi+1}{\pi}T_0 = \frac{\frac{1}{2}\pi+1}{\pi}\cdot 2\pi\sqrt{\frac{a^3}{GM}} = \boxed{(\pi+2)\sqrt{\frac{R^3}{GM}}}.$$

■

Problem 15.10 (HRK). [11] A comet passes by the Sun as shown, in a parabolic path.



How long, in years, does the comet take to get from point A to point B? **Hint:** 15

Solution. Let R be the closest distance from the Sun to the Comet and let r be the distance between the Sun and the Comet when the comet has gone through an angle θ.

The first thing to note is that the angular momentum of the system remains constant. Since the Sun is approximately static, we have $L = Rmv$ where v is the speed of the commet at the point of closest contact. Since for a parabola we have $E = 0$, we thus have

$$\frac{1}{2}mv^2 = \frac{GMm}{R} \implies v = \sqrt{\frac{2GM}{R}}$$

The total angular momentum of the system is then $L = \sqrt{2GMR}$. Since $L = I\omega$, we furthermore have that

$$L = \sqrt{2GMR} = mr^2\frac{d\theta}{dt}$$

It should be clear that by integrating both sides we can find the period T.

Let's now try to find r in terms of R and θ in order to integrate our expression. Recall that the distance from a point on the parabola to the focus is equal to the distance of the point on the parabola to the directrix of the parabola. Since the distance from the focus to the parabola is $2R$, we thus have

$$2R = r\cos\theta + r \implies r = \frac{2R}{1+\cos\theta}.$$

Substituting into our original expression we have:

$$\sqrt{2GMR} = \frac{4R^2}{(1+\cos\theta)^2}\frac{d\theta}{dt}.$$

Rearranging our expression, we obtain

$$dt = \frac{4R^2}{\sqrt{2GMR}}\frac{1}{(1+\cos\theta)^2}d\theta.$$

Integrating our expression, we get

$$\int_0^T dt = \frac{4R^2}{\sqrt{2GMR}}\int_0^{\frac{\pi}{2}}\frac{1}{(1+\cos\theta)^2}d\theta.$$

The right hand integral is $\frac{2}{3}$ (we will skip the computation since it is straightforward). We thus get that

$$T = \frac{8R^{\frac{3}{2}}}{3\sqrt{2GM}}.$$

Let's now calculate this quantity in terms of years. Recall from Kepler's Third Law, we have that

$$1 \text{ year} = \frac{2\pi R^{\frac{3}{2}}}{\sqrt{GM}}.$$

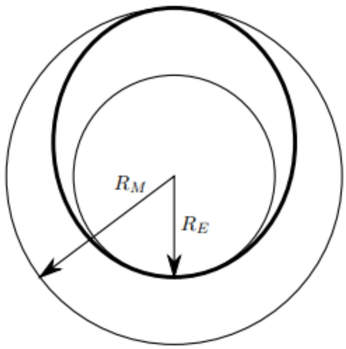
We thus have

$$T = \frac{4}{3\pi\sqrt{2}} \text{ years} \approx \boxed{0.300} \text{ years}.$$

■

Problem 15.11 (USAPhO). [15] A ship starts out in a circular orbit around the sun very near the Earth and has a goal of moving to a circular orbit around the Sun that is very close to Mars. It will make this transfer in an elliptical orbit as shown in bold in the diagram below. This is accomplished with an initial velocity boost near the Earth Δv_1 and then a second velocity boost near Mars Δv_2.

Assume that both of these boosts are from instantaneous impulses, and ignore mass changes in the rocket as well as gravitational attraction to either Earth or Mars. Don't ignore the Sun! Assume that the Earth and Mars are both in circular orbits around the Sun of radii R_E and $R_M = R_E/\alpha$ respectively. The orbital speeds are v_E and v_M respectively.



(a) [5] Derive an expression for the velocity boost Δv_1 to change the orbit from circular to elliptical. Express your answer in terms of v_E and α.

(b) [5] Derive an expression for the velocity boost Δv_2 to change the orbit from elliptical to circular. Express your answer in terms of v_E and α.

(c) [5] What is the angular separation between Earth and Mars, as measured from the Sun, at the time of launch so that the rocket will start from Earth and arrive at Mars when it reaches the orbit of Mars? Express your answer in terms of α.

Solution.

(a) We will find the velocity v' of the rocket right after the boost using conservation of energy. The energy of an elliptical orbit is

$$E = -\frac{GMm}{2a}.$$

and in this case $a = (R_E + R_M)/2$. Then at the moment that the velocity is boosted, the kinetic plus potential energy is the total energy, giving

$$\frac{1}{2}mv'^2 - \frac{GMm}{R_E} = -\frac{GMm}{R_E + R_M}.$$

Then we can solve for v':

$$v'^2 = -2GM\left(\frac{1}{R_E + R_M} - \frac{1}{R_E}\right) \tag{15.1}$$

$$= 2GM\frac{R_M}{R_E(R_E + R_M)}. \tag{15.2}$$

Since we want to get this in terms of V_E and α, we need to express v_E in terms of R_M and R_E as well. The orbit is circular so we set centripetal acceleration equal to the force due to gravity to get the equation,

$$\frac{v_E^2}{R_E} = \frac{GM}{R_E^2} \implies v_E = \sqrt{\frac{GM}{R_E}}.$$

Then we can write v' in terms of v_E so equation (15.1) becomes

$$v' = v_E\sqrt{\frac{2R_M}{R_E + R_M}} = v_E\sqrt{\frac{2}{1+\alpha}}.$$

Then subtracting v_E and v' we get our final answer:

$$\Delta v_1 = v' - v_E = \boxed{v_E\left(\sqrt{\frac{2}{1+\alpha}} - 1\right)}.$$

(b) This part uses pretty much the same ideas as part (a). Like before, we use $K + U = E$ for the ellipse to calculate the speed v_a, this time at the apogee of the ellipse:

$$\frac{1}{2}mv_a^2 - \frac{GMm}{R_M} = -\frac{GMm}{2a}.$$

The calculation works out just much the same as before, except for R_M and R_E are essentially switched, so when we solve for v_p we get

$$v_a = \sqrt{\frac{2GMR_E}{R_M(R_M + R_E)}} \tag{15.3}$$

Now we find v_M in terms of R_E and R_M so that we can find the difference:

$$\frac{v_M^2}{R_M} = \frac{GM}{R_M^2} \implies v_M = \sqrt{\frac{GM}{R_M}}.$$

Substituting this into equation (15.3) we get

$$\begin{aligned} v_a &= v_M\sqrt{\frac{2R_E}{R_E + R_M}}, \\ &= v_M\sqrt{\frac{2}{1+\alpha}} \end{aligned}$$

so the difference is

$$\Delta v_2 = v_M - v_a = v_M\left(1 - \sqrt{\frac{2}{1+\frac{1}{\alpha}}}\right).$$

But we want to write this in terms of V_E, not V_M. Luckily, v_M can be easily expressed in terms of V_E:

$$v_M = \sqrt{\frac{GM}{R_M}} = v_E\sqrt{\frac{R_E}{R_M}} = v_E\sqrt{\alpha},$$

and substituting this into the expression for Δv_2 we get the final answer:

$$\Delta v_2 = \boxed{v_M\sqrt{\alpha}\left(1 - \sqrt{\frac{2}{1+\frac{1}{\alpha}}}\right)}.$$

(c) We want to find the angle that Mars travels through while the rocket moves an angle π through its own orbit (half its orbital period). If T_R and T_M are the orbital periods of the rocket on its elliptical trajectory and Mars on its circular trajectory, then

$$\frac{\theta_M}{\theta_R} = \frac{\omega_M}{\omega_R} = \frac{T_R}{T_M}$$

Thus we are looking for the ratio of the periods of the two orbits which we get through Kepler's third law:

$$\frac{T_R}{T_M} = \left(\frac{(R_E + R_M)/2}{R_M}\right)^{3/2} = \left(\frac{1+\alpha}{2}\right)^{3/2}$$

When the rocket travels an angle $\theta_R = \pi$, then Mars travels an angle

$$\theta_M = \pi \frac{T_R}{T_M} = \pi \left(\frac{1+\alpha}{2}\right)^{3/2}$$

Then the initial separation between Earth and Mars is

$$\theta = \pi - \theta_M = \boxed{\pi \left(1 - \frac{1+\alpha}{2}\right)^{3/2}}.$$

■

15.4.1 Written Solutions

Problem 15.12 (HRK)**.** [18] A pair of stars revolve about their common center of mass. On of the stars has a mass M that is twice the mass M of the other; that is, $M = 2m$. Their centers are a distance d apart, d being large compared to the size of either star.

(a) [6] Derive an expression for the period of revolution of the stars about their common center of mass in terms of d, m and G.

(b) [6] Compare the angular momenta of the two stars about their common center of mass by calculating the ratio L_m/L_M.

(c) [6] Compare the kinetic energies of the two stars by calculating the ratio K_m/K_M.

Solution.

(a) We use Newton's law of gravitation, while assuming a circular orbit. Since the masses are in ratio 1 to 2, we can find the distance from the smaller star to the center is $2d/3$. So we have that the net force is

$$m\omega^2 \cdot (2d/3).$$

The only force is gravitational force which has magnitude

$$G\frac{2m^2}{d^2}.$$

So we can set these two equal, and we solve for ω, which gives

$$\omega = \sqrt{\frac{3Gm}{d^3}}.$$

So the period is

$$T = \boxed{2\pi\sqrt{\frac{d^3}{3Gm}}}.$$

(b) The ratio between the angular momenta can be written as

$$\frac{L_m}{L_M} = \frac{mr(\omega r)}{MR(\omega R)} = \frac{r^2}{2R^2}.$$

The lengths are $2d/3$ and $d/3$, so

$$\frac{L_m}{L_M} = \boxed{2}.$$

(c) We have

$$\frac{K_m}{K_M} = \frac{m(\omega r)^2}{M(\omega R)^2}.$$

Since ω cancels, this is the same as the previous case, so our answer is $\boxed{2}$.

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VI

Fluids

16 Fluid Statics

We begin our study of fluids by defining what exactly a fluid is.

Definition 16.1. We define a **fluid** to be a substance that deforms under external pressure or forces. Typically, when we refer to a fluid, we refer to either a liquid or a gas. Solids, on the other hand, are *not* fluids as they are *not* malleable under external forces.

Another important property of any substance is its density, defined as mass per unit volume. We now refer to a particular type of fluid that we will use throughout the handout.

Definition 16.2. A fluid is said to be **homogeneous** if it has the same density throughout. That is, if ρ is the density of the fluid, m is the mass, and v is the volume, then

$$\rho = \frac{m}{v}$$

To study fluids, however, we first make precise the notion of **pressure**.

16.1 Pressure

A fluid exerts a force perpendicular to any surface in contact with it such as a container. Even when a fluid is at rest, its particles move around, and apply a force on, like you in a swimming pool.

Now imagine the surface of a fluid at rest. For it remain at rest the fluid must exert equal forces on it from both sides. In this context that we introduce pressure. Consider a small surface of area ΔA and let the normal force exerted on this surface on each side be ΔF.

Definition 16.3. We define the pressure to be

$$p = \frac{\Delta F}{\Delta A}$$

If the pressure is the same at all points on a surface (which will usually be the case in this course), then

$$p = \frac{F}{A}$$

where F is the net normal force on one side of the surface.

At first, this may seem arbitrary or not particularly useful, but the truth is the quite the opposite. We shall be using pressure throughout.

The SI unit of pressure is called the pascal, where

$$\text{pascal} = \text{ Pa} = \frac{\text{Newton}}{\text{metre}^2}.$$

Check that this is in accordance with our definition of pressure. Two more important units are the bar, defined by

$$1 \text{ bar} = 10^5 \text{ Pa}$$

and the atmosphere (atm). Atmospheric pressure p_a is defined to be the pressure exerted by by earth's atmosphere, at sea level. p_a changes due to weather and elevation. 1 atm is an average value and defined to be $101,325$ Pa. We will assume this value to be the pressure at sea level from here.

16.2 Pressure and Density

We intuitively know that pressure increases with depth (when you go down deep during diving, for example, your ears start to hurt because of the pressure). This is because an increased amount of mass on top exerts a higher amount of force on the same area. But what is the exact relation?

Consider two surfaces of area A at a distance h from each other. Our key observation is that the the force one lower surface is equal to the force exerted on the higher surface and the force exerted by the volume of fluid between them. We then get

$$F_1 = F_2 + mg, \tag{16.1}$$

where F_1 and F_2 are the forces on the surfaces and m the mass in the volume between them. This is not a very useful relation because m is arbitrary. To get over this, we apply some clever manipulations.

Note that F_1 can be written as $p_1 A$ where p_1 is the pressure at the lower surface, and similarly for the other surface. For m we try to represent it in terms of the density of the fluid (ρ), a more "global" quantity.

$$m = V\rho = Ah\rho$$

where V is the volume, which can be written as Area $\times$ Height. Substituting all this in (16.1) we get

$$p_1 A - p_2 A = Ahg\rho,$$

Remarkably the area, an undesirable quantity gets cancelled leaving us with the following relation.

Concept. The change in pressure between two points is given by the equation

$$p_1 - p_2 = \rho g h.$$

At this point it is worth noting a few facts. Note the usefulness of pressure. It is a quantity whose value only depends on the difference in vertical length. Also, never forget that for this to work, the density ρ and free-fall acceleration g need to be constant throughout. Finally, the pressure at the surface of a liquid exposed to the air is considered to be the atmospheric pressure.

Example 16.4. Water stands 12 m deep in a storage tank whose top is open to the surface the atmosphere. The tank is cylindrical with radius 7 m. What is the pressure and force exerted on the bottom surface?

Solution. The pressure at the surface of the water must be equal to the pressure of the air so we have

$$\begin{aligned} P_{\text{bottom}} &= \rho g h + P_{\text{atm}} \\ &= 1000\ \text{kg/m}^3 \times 9.8\ \text{m/s}^2 \times 12\ \text{m} + 1\ \text{atm} \\ &= 2.2 \cdot 10^5\ \text{Pa}. \end{aligned}$$

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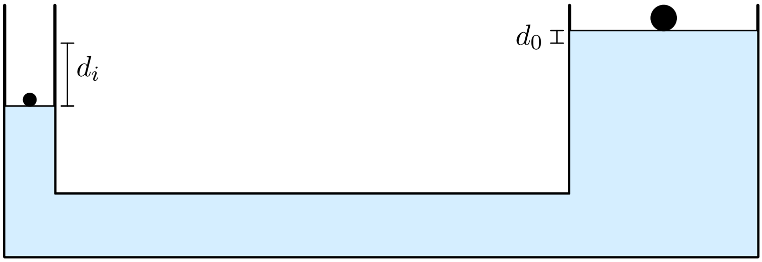
16.3 Pascal's Principle

If we let p_0 be the pressure at the surface, we can restate (16.2) as

$$p = p_0 + \rho g h,$$

where p now represents the depth at at depth h. We notice that if we increase p_0 by a certain amount, regardless of the value of h, the value of p will increase by the same amount (convince yourself this is true). This is the crux of Pascal's Principle. Being more specific, if we have an enclosed fluid and apply some pressure with say, a piston, this difference in pressure will transmit throughout.



This can have some very helpful practical applications when we realize the following.

In a hydrualic lift, a piston with small cross sectional area A_1 exerts a force F_1 on the surface of a liquid such as oil. The applied pressure

$$p = \frac{F_1}{A_1}$$

is transmitted throughout a connecting pipe, to a larger piston of area A_2 . The applied pressure is the same, so

$$p = \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

or

$$F_2 = \frac{A_2 F_1}{A_1}$$

The hydraulic lift is thus, a force multiplying device, with a multiplication factor being the ratio of the areas of the pistons. Dentists chairs, car lifts and jacks use this principle. Pascal's principle is also often formulated as follows.

> **Concept** (Pascal's Principle). A pressure change at any point of an enclosed incompressible fluid is transmitted undiminished throughout the fluid and to the walls of the container.

> **Example 16.5.** A 1 ton car is being held on a hydraulic platform of area 8 m^2. This is connected to a piston of area 2 cm^2. Find the force that needs to be applied on the piston to keep the car in place. (Note 1 ton is 907 kg)

Solution.

$$F_2 = \frac{A_2 F_1}{A_1} = \frac{907\text{ kg} \times 9.8\text{ m/s}^2 \times (2 \cdot 10^{-4})\text{ m}^2}{8\text{ m}^2} = 0.222\text{ N}$$

■

16.4 Archimedes' Principle

The Archimedes' Principle states the following:

Theorem 16.6. *When a body is completely or partially submerged in a fluid, the fluid exerts an upward force on the body equal to the weight of the fluid displaced by the body.*

We will not go over the proof of Archimedes' Principle in this course. Read the theorem a couple of times and understand it well, as it is *very* important and can be applied very tricky ways.

Note the following consequences of Archimedes' Principle. While doing so please do not proceed before being thoroughly convinced. Do ask if you have questions. Again, going over each of these (and trying to prove them) is very important.

(a) If a body is completely submerged, then whatever the depth, the fluid applies the same force.

(b) A denser fluid applies a larger force than a less dense fluid for a fully submerged body.

(c) Shape doesn't matter (for the body); only the volume does.

(d) In a partially submerged body we only take into account the volume inside the fluid for calculating the buoyant force.

Note that while calculating the buoyant force, we use the density of the fluid but the volume of the body.

Example 16.7. Calculate the buoyant force on a fully submerged body of density $3\ \mathrm{kg} \times \mathrm{m}^{-3}$ and volume $1\ \mathrm{m}^3$ in a fluid of density $1.5\ \mathrm{kg} \cdot \mathrm{m}^{-3}$. Also calculate the net force.

Solution. The buoyant force is equal to the weight of the displaced fluid, namely,

$$F_b = \rho g V = 14.7\ \mathrm{N}$$

Then the total force is

$$F_b - F_{\mathrm{grav}} = (\rho_1 - \rho_2) g V = -14.7\ \mathrm{N}$$

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Now we shall consider what will happen to a body of density ρ and Volume V released in a fluid of density σ.

(a) Case 1: $\rho \geq \sigma$

Intuitively, the body should sink. Math will give the same result; we have 2 forces to consider. The buoyant force is upwards and has a value of $V\sigma g$ (why?). The weight of the body has a magnitude of $mg = V\rho g$. and acts downwards. Thus, the net force is downwards with a magnitude of $V(\rho - \sigma)g$. Therefore, the body sinks.

Of course, when $\rho = \sigma$ the net force is zero and one can consider the body to be in equilibrium with the liquid.

(b) Case 2: $\rho < \sigma$

Intuitively, the body should float, but by how much? We have the same net force as before: $V(\sigma - \rho)g$ upwards. Thus, it moves upwards. But how will it attain equilibrium? This happens when you observe that while $V\rho g$ remains fixed, the V in $V\sigma g$ can change if the body is not fully submerged (why?). To highlight the difference, we rename the volume of the body submerged as V' at equilibrium. Now we can write

$$V'\sigma g = V\rho g$$

or

$$\frac{V'}{V} = \frac{\rho}{\sigma}.$$

In other words, the fraction of a light body below the surface is equal to the ratio of the densities of body to the fluid.

Example 16.8. We have a body of density 5 kg/m^3 and volume 10 m^3. When we release the body into a certain fluid X we find 8 m^3 of the body below the surface. What is the density of X?

Solution. When the body is in equilibrium, we have

$$\begin{aligned} F_b &= \rho_{\text{fluid}} V_{\text{submerged}} g = F_{\text{grav}} = \rho_{\text{body}} V g \\ \implies \rho_{\text{fluid}} &= \frac{\rho_{\text{body}} V}{V_{\text{submerged}}} = 6.25 \text{ kg/m}^3. \end{aligned}$$

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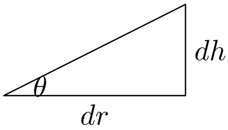
16.5 Examples

We will now do some slightly more complex examples:

Example 16.9 (Kalda). A vertical cylindrical vessel with radius R is rotating around its axis with the angular velocity ω. By how much does the water surface height at the axis differ from the height next to the vessel's edges?

Solution. Consider the following diagram.



The diagram shows the cross-section of the small ring of water which is a distance r away from the center of the cylinder. Letting P be the hydrostatic pressure, we see the vertical force is

$$0 = P(r)2\pi r\, dr - g\, dm.$$

The centripetal force on the water is

$$\omega^2 r\, dm = P(r) 2\pi r\, dh = P(r) 2\pi r \tan(\theta)\, dr$$

Then substituting in the first equation, we get that $\omega^2 r dm = g \tan(\theta) dm$ so

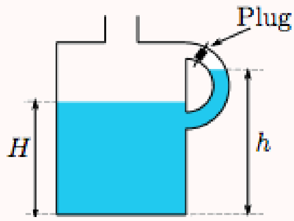
$$\frac{dh}{dr} = \tan(\theta) = \frac{\omega^2 r}{g}.$$

To get the height difference we integrate this from $r = 0$ to $r = R$

$$h = \int_0^R \frac{\omega^2 r}{g}\, dr = \boxed{\frac{\omega^2 R^2}{2g}}.$$

■

Example 16.10 ($F = ma$)**.** The handle of a gallon of milk is plugged by a manufacturing defect. After removing the cap and pouring out some milk, the level of milk in the main part of the jug is lower than in the handle, as shown in the figure. If ρ is the density of the milk, what is the gauge pressure at the bottom of the milk jug?



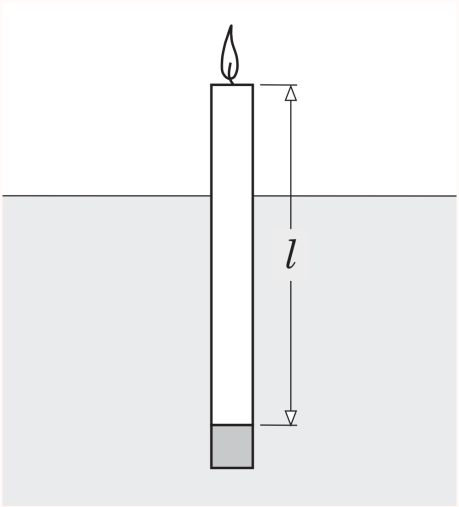
Solution. This problem may initially seem confusing since there is no definite height of the milk, so it is not immediately clear how we would apply our formula $P = \rho g h$. But the problem is asking for the gauge pressure, meaning, the pressure of the bottom of the milk jug minus the ambient atmospheric pressure, and since the air pressure in the handle is less than ambient air pressure, the height of the milk in the handle is irrelevant for our purposes. We have $P = P_{atm} + \rho g H$ (balancing forces in the center of the jug) so then the gauge pressure is $\rho g H$. ■

Example 16.11 (PhysicsWOOT)**.** A beaker of water sits on a scale. You put your finger in the water without touching the beaker or the scale. Does the reading on the scale change?

Solution. When you put your finger in, the water level goes up, so the pressure at the bottom of the beaker goes up and hence the scale reading goes up. Another way to think about it is that the water exerts a buoyant force up on your finger, so by Newton's third law your finger exerts a downward force on the water which increases the scale reading. ■

Next we will do a trickier example dealing with stable equilibrium in fluid statics. This concept does not come up very much and will not be needed on the problem set but will be useful if you want to attempt the Extra problems.

Example 16.12 (BelPhO). To make a candle float stably in water in vertical orientation, an aluminum cylinder of radius equal to that of the candle is glued at the bottom of the candle. Length of this cylinder is 1.0 cm and densities of wax, aluminum, and water are $0.8\ \text{g/cm}^3$, $2.7\ \text{g/cm}^3$, and $1.0\ \text{g/cm}^3$ respectively.



(a) Find the maximum and minimum length ℓ of the candle so that it can float in water stably in vertical orientation. The radius of the candle is small enough to ignore horizontal shifting of the point of application of the buoyant force.

(b) When the candle is lit, it consumes wax. If consumption of wax reduces the length of the candle at a constant rate of 1.0 mm/min, how long can the candle be used for lighting?

Solution.

(a) To float, the depth of immersion cannot exceed the total length of the candle-cylinder structure and to float stably in the vertical orientation, the center of the buoyancy must be vertically above the center pf mass of the candle-cylinder structure. If S is the cross-sectional area, then the depth of immersion can be measured by the formula

$$\rho_w g S\ell + \rho_a g S h = \rho_0 g S d \implies d = \frac{\rho_w \ell + \rho_a h}{\rho_0}.$$

With what we said before, $d \le \ell + h$ which means

$$\frac{\rho_w \ell + \rho_a h}{\rho_0} \le \ell + h.$$

Solving this tells us that 8.5 cm $\leq \ell$. Let us now use a coordinate system . The coordinate of the center of candle is given by

$$y_d = \frac{d}{2} = \frac{1}{2}\frac{\rho_w \ell + \rho_a h}{\rho_0}$$

and the coordinate of the center of gravity is given by

$$y_g = \frac{\rho_a h \frac{h}{2} + \rho_w \ell \left(h + \frac{\ell}{2}\right)}{\rho_a h + \rho_w \ell}.$$

Taking into account that $y_d \geq y_g$, we have

$$\frac{1}{2}\frac{\rho_w \ell + \rho_a h}{\rho_0} \geq \frac{\rho_a h \frac{h}{2} + \rho_w \ell \left(h + \frac{\ell}{2}\right)}{\rho_a h + \rho_w \ell}.$$

Solving this inequality and plugging in gives us $\ell \leq 18.54$ cm.

(b) The candle will continue burning until it is as long as $\ell_{\text{min}} = 8.5$ cm. This means that

$$t = \frac{\ell - \ell_{\text{min}}}{v} = \frac{3.5}{0.1} = 35 \text{ min}.$$

■

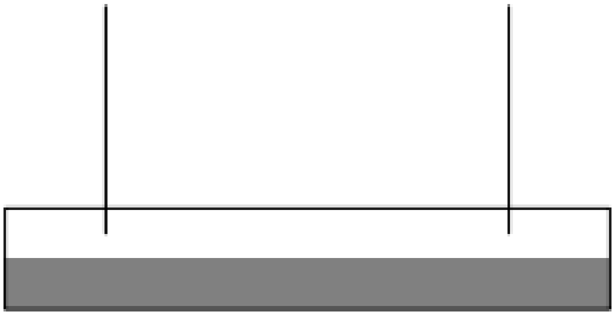
16.6 Homework Problems and Solutions

Problem 16.1 (PhysicsWOOT, $F = ma$)**.** [11] The following are several interesting applications of Archimedes' principle.

(a) [3] You have a glass with some ice water in it. Assuming you don't drink any of the water, when the ice cube melts will the water level change?

(b) [4] You're in a canoe in a lake. In the canoe is a rock. You toss the rock overboard and it sinks to the bottom of the lake. Does the water level in the lake go up or down? (careful!)

(c) [4] A trough half-filled with water is suspended from wires, as shown. The tension is initially the same in each wire.



A boat is placed in the trough directly under the left wire. It floats without touching the sides of the trough or overflowing the water. How does the tension in each of the wires change as a result?

Solution.

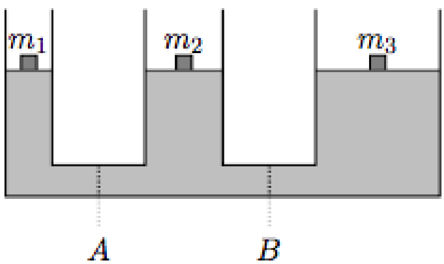
(a) No, the water level will not change. By Archimedes' principle, the water that the ice displaces has the same mass as the ice. Once the ice melts, the added water also has the same mass as the ice (it was the ice). Thus the amount of water that was displaced by the ice is the same as the amount of water that the melted ice cube adds.

(b) The water level goes down. The mass of the water displaced by the canoe is the same as the mass of all the objects in the canoe combined. The rock originally displaces a mass of water equal to its weight, which is a volume of water greater than its original volume (since the rock is denser than water). Hence, the water level will go down as the volume of water displaced decreases.

(c) The water will distribute itself so that the pressure is even across the bottom of the trough regardless of where you place the boat (otherwise the pressure difference would cause horizontal water movement). Thus the tension increases in both wires by the same amount so that it is enough to support the weight added by the boat.

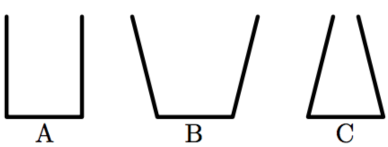
■

Problem 16.2 ($F = ma$)**.** [5] In the device shown, blocks of various masses are placed on pistons so that the device is in equilibrium. (The fluid in the drawing is to scale.) There are valves that are both initially open at locations A and B. One of the valves is closed, and the system is allowed to come to equilibrium. How will this affect the height of the mass m_2?



Solution. Since they are already at equal heights, the pressure at the bottom is the same everywhere, so putting a valve doesn't change anything, so the height of m_2 will stay the same. ■

Problem 16.3 ($F = ma$)**.** [5] Flasks A, B, and C each have a circular base with a radius of 2 cm. An equal volume of water is poured into each flask, and none overflow. Rank the force of water F on the base of the flask from greatest to least (if possible)



Solution. The force F at the bottom of the flasks depends directly on the pressure there, which means we want to rank the heights of the fluids in the flasks. C is the narrowest, so the water will rise the highest, so the force will be greatest for C, so we can do the same, and the order is C, A, B. This may seem like a contradiction since it is the same amount of water. However, notice the slope of the containers and in what direction they provide a force on the water. ■

Problem 16.4. [5] The density of ice is 92% that of water. What percentage of the iceberg is above the water (i.e. the tip of the iceberg)?

Solution. We require that the net force on the iceberg is 0, that is

$$\rho_{ice} V g = \rho_w V f g,$$

where f is the fraction underwater. Then we can clearly see $f = 92\%$. So 8% of the iceberg is above the water. ■

Problem 16.5. [10] Find the angular frequency of oscillation of each of the following scenarios:

(a) [5] The liquid in a U-shaped tube of liquid with a total length of liquid ℓ oscillates vertically.

(b) [5] A vertical stick of length ℓ and density ρ floating in liquid with density ρ_0 oscillates vertically.

Solution.

(a) Note this problem is very similar to Problem 13.8. Let h denote the height difference between the two sides. Then if the liquid has density ρ, the pressure difference is $\rho g h$. Then since both ends of the U-shaped tube will experience an acceleration, we see that

$$\ddot{h} = -2\frac{F}{m} = -2\frac{\Delta P \cdot A}{\rho V} = -2\frac{\Delta P}{\rho \ell} = -2\frac{g \Delta h}{\ell},$$

which implies

$$\omega = \boxed{\sqrt{\frac{2g}{\ell}}}.$$

(b) Consider the change in buoyant force caused by the stick sinking by a length $\Delta\ell$. As with the previous part, we have

$$\ddot{\ell} = -\frac{F}{m} = -\frac{\rho_0 \cdot A\Delta\ell \cdot g}{\rho \cdot A\ell} = -\frac{\rho_0 g \Delta\ell}{\rho \ell}$$

$$\implies \omega = \boxed{\sqrt{\frac{\rho_0 g}{\rho \ell}}}.$$

■

Problem 16.6. [6] A balloon filled with air submerged in water at a depth h experiences a buoyant force B_0. The balloon is moved to a depth of $2h$, where it experiences a buoyant force B. Assuming the water is incompressible [density cannot change] and the balloon and air are compressible [density can change], the buoyant force B satisfies

(A) $B \geq 2B_0$
(B) $B_0 < B < 2B_0$
(C) $B = B_0$
(D) $B < B_0$
(E) it depends on the compressibility of the balloon and the air

Solution. The volume of the balloon will decrease, and thus the buoyant force decreases. ∎

Problem 16.7 ($F = ma$)**.** [6] When a block of wood with a weight of 30 N is completely submerged under water the buoyant force on the block of wood from the water is 50 N. When the block is released it floats at the surface. What fraction of the block will then be visible above the surface of the water when the block is floating?

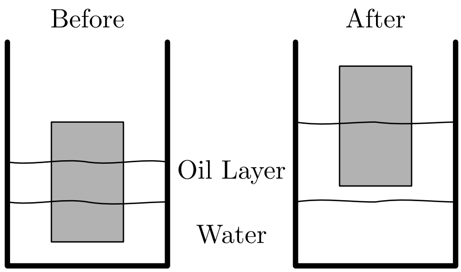
Solution. When the block is floating, there is zero net force on it. Therefore, the buoyancy force must be exactly opposite of the force from gravity. Therefore, the buoyancy force when the block is floating is 30 N.

If V is the volume of the whole block and V_{float} is the volume of the block that is submerged while the block is floating, we have

$$\frac{30\text{ N}}{50\text{ N}} = \frac{\rho g V_{\text{float}}}{\rho g V} = \frac{V_{\text{float}}}{V},$$

which means that 3/5 of the block is submerged while the block is floating. Therefore $\boxed{2/5}$ of the block is visible when the block is floating. ∎

Problem 16.8 ($F = ma$)**.** [6] A 3.0 cm thick layer of oil with density $\rho_o = 800\text{ kg/m}^3$ is floating above water that has density $\rho_w = 1000\text{ kg/m}^3$. A solid cylinder is floating so that 1/3 is in the water, 1/3 is in the oil, and 1/3 is in the air. Additional oil is added until the cylinder is floating only in oil. What fraction of the cylinder is in the oil?



Solution. Let the mass of the cylinder be m and the volume be V. Since the cylinder is in equilibrium before we add the oil, the buoyancy force from both liquids balance out the force from gravity. This gives

$$mg = \rho_o g V/3 + \rho_w g V/3.$$

Suppose that after oil is added, the fraction of the cylinder in the oil is α. Again, the buoyancy force from the oil balances out the force from gravity, so

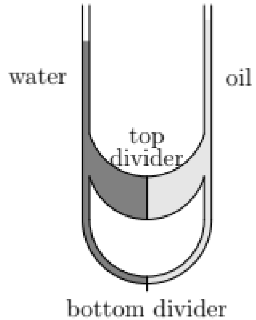
$$mg = \rho_o g \alpha V.$$

Equating these two equations gives

$$\rho_o gV/3 + \rho_w gV/3 = \rho_o g\alpha V \implies \alpha = \frac{\rho_o + \rho_w}{3\rho_0} = \boxed{3/4}.$$

■

Problem 16.9 (PhysicsWOOT). [7] A U-shaped tube has a connector between its two sides. There are massless, movable dividers in the middle of the connector and at the bottom of the tube. The dividers are initially held stationary. Water is poured in the left side of the tube and oil is poured in the right side. Water is more dense than oil. The heights of the water and oil columns are such that the pressure is equalized between the water and oil at the bottom divider.



Then the dividers are then allowed to slide left or right slowly. The dividers settle into a final position such that

(A) neither divider has moved.

(B) the top divider has moved left and the bottom divider has not moved.

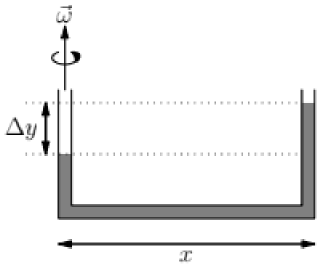
(C) the top divider has moved left and the bottom divider has moved right.

(D) the top divider has moved right and the bottom divider has not moved.

(E) the top divider has moved right and the bottom divider has moved left.

Solution. When the system is initially held in equilibrium, we are told the pressure at the bottom divider is equalized. Since water is more dense than oil, this means that the pressure from oil is greater at the top divider (since there is a smaller change in pressure). Hence, we see that the top divider will move left. This increases the pressure of the water at the bottom divider, so the bottom divider will move right, giving our answer $\boxed{\text{(C)}}$. ■

Problem 16.10 (PhysicsWOOT). [7] A pipe consists of two vertical sections connected by a horizontal section of length x. The width of the pipe is small compared to the length of the sections of pipe. The pipe spins around one of the vertical sections at angular frequency ω. What is the difference in heights of the fluid Δy between the vertical segments of pipe in terms of ω, x, and gravitational acceleration g?



Solution. Consider the pressure change between the two columns. We have

$$\Delta P = \rho g \Delta y \implies F = \rho g A \Delta y.$$

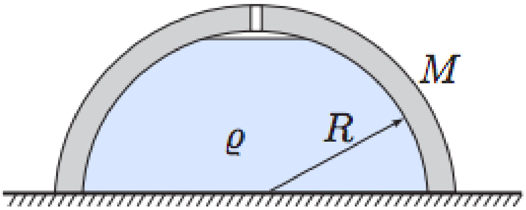
Note that we have

$$\rho g A \Delta y = \int \omega^2 r \, dm = \int_0^x \rho A \omega^2 r \, dr = \frac{\rho}{2} A \omega^2 x^2$$

$$\implies \Delta y = \boxed{\frac{\omega^2 x^2}{2g}}.$$

■

Problem 16.11 (Kalda)**.** [8] A hemispherical container is placed upside down on a smooth horizontal surface. Through a small hole at the bottom of the container, water is then poured in. Exactly when the container gets full, water starts leaking from between the table and the edge of the container. Find the mass of the container if water has density ρ and radius of the hemisphere is R.



Solution. Water starts leaking between the table and edge of the container when the normal force from the ground on the container is 0, that is the water pressure is just enough to support the container. To find the force exerted by the water on the container we could integrate over the surface of the container and do some trigonometry. However a more clever way to go about it is to find the total force of the ground on the water and set this equal to the weight of the water plus the bowl.

Since the top of the container is open to the air, the gauge pressure at the bottom of the water is just

$$P_{bot} = \rho g R$$

so the force of the ground up on the water is $P_{bot}A = \rho g \pi R^3$. This force supports the weight of the both bowl and water, so we have

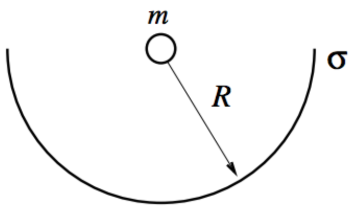
$$F = \rho g \pi R^3 = Mg + \rho V g = Mg + \rho g \frac{2}{3} \pi R^3.$$

Solving for M we get

$$M = \boxed{\rho g R^3 \frac{\pi}{3}}$$

■

Problem 16.12 ($F = ma$). [9] A particle of mass m is placed at the center of a hemispherical shell of radius R and mass density σ, where σ has dimensions of kg/m^2.



The gravitational force of the shell on the particle is

(A) $(1/3)(\pi G m \sigma)$

(B) $(2/3)(\pi G m \sigma)$

(C) $(1/\sqrt{2})(\pi G m \sigma)$

(D) $(3/4)(\pi G m \sigma)$

(E) $\pi G m \sigma$

Hints: 25 20

Solution. The force on a small amount of mass dm in the shell is

$$dF = \frac{Gmdm}{R^2}.$$

One way to solve this would be to use trigonometry and integrate over the entire shell (we would have to take into account the fact that the force is pointing in a different way at each point on the shell...). However the hint suggests a much nicer way to approach this – since the force on each part of the shell is the same it is analogous to if the shell was filled with a fluid of constant pressure,

$$P = \frac{dF}{dA} = \frac{Gmdm/dA}{R^2} = \frac{Gm\sigma}{R^2}.$$

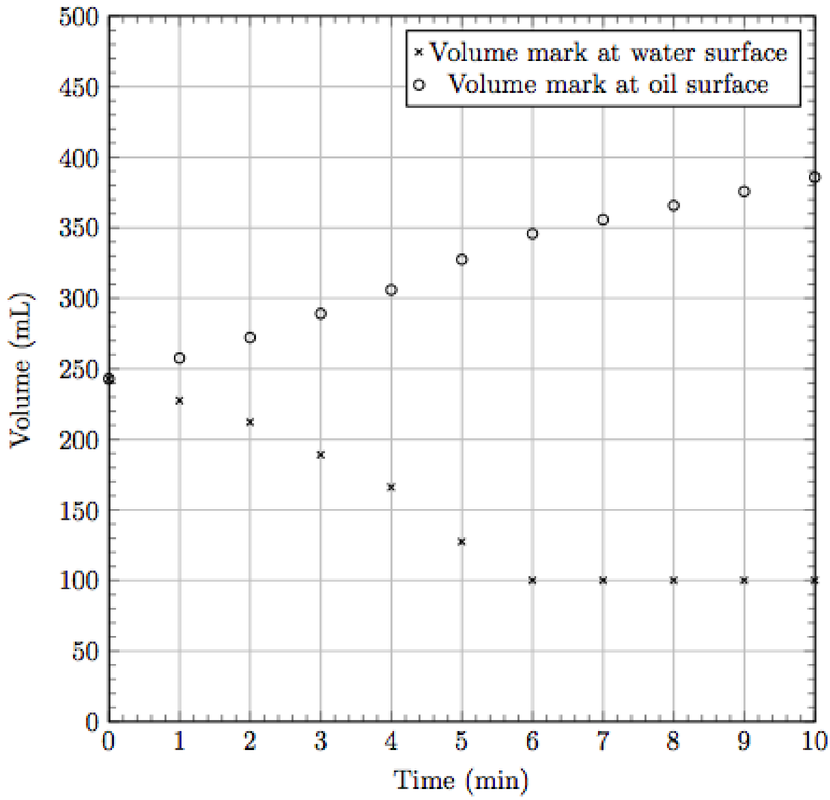
Of course the force of gravity will be in the opposite direction as the force exerted by this imaginary fluid, but otherwise everything should work out exactly the same. Imagine capping off the end of the hemisphere with a circle. Then the system would be in equilibrium so the force of pressure on the circle would have to be the same as the force of pressure on the sphere.

$$F = PA = \frac{Gm\sigma}{R^2}\pi R^2 = \boxed{\pi G m \sigma}.$$

■

16.6.1 Written Solutions

Problem 16.13 (USAPhO). [15] A graduated cylinder is partially filled with water; a rubber duck floats at the surface. Oil is poured into the graduated cylinder at a slow, constant rate, and the volume marks corresponding to the surface of the water and the surface of the oil are recorded as a function of time.



Water has a density of 1.00 g/mL; the density of air is negligible, as are surface effects. Find the density of the oil.

Solution. First lets understand what this graph is telling us since there is a lot going on here. As oil is added, the duck starts to rise. It displaces less and less water until it is completely immersed in the oil and displaces no water at all. This occurs at approximately $t = 6$min, the point at which the bottom line flattens out. When the duck is partially in the water and partially in the oil, the amount of oil it displaces may be changing with time (in fact it will be increasing – why is this?). However the rate of change of the top line after $t = 6$min will tell us exactly the rate at which oil is added to the container.

Now let's convert our intuition to some equations. Let V_{oil} and V_{water} be the true volumes of oil and water in the container (not equal to the readings) and let $V_{\text{oil, disp}}$ and $V_{\text{water, disp}}$ be the volumes of oil and water displaced by the duck. Since the duck has constant mass we have

$$\begin{aligned}
\rho_w V_{\text{water, disp}} + \rho_{\text{oil}} V_{\text{oil, disp}} &= \text{const.} \\
\rho_w \Delta V_{\text{water, disp}} + \rho_{\text{oil}} \Delta V_{\text{oil, disp}} &= 0 \\
\implies \rho_{\text{oil}} &= -\rho_w \frac{\Delta V_{\text{water, disp}}}{\Delta V_{\text{oil, disp}}}
\end{aligned}$$

The change in the volume of water displaced by the duck $\Delta v_{\text{water, disp}}$ is just the initial water level minus the final water level, $243 - 100 = 143$mL. Finding the change in the displaced oil is a bit more complicated since the amount of oil is increasing throughout the process. The rate at which you add oil to the container is given by the slope of the top line after 6 min,

$$\frac{dV_{\text{oil}}}{dt} = \frac{380 - 340}{4} = 10 \text{ mL/min}.$$

so the final volume of oil is 10min $\cdot$10mL/min = 100mL. At the end, we have that the difference between the two lines is

$$280 = V_{\text{oil}} + V_{\text{oil, disp}} = 100 + V_{\text{oil, disp}}$$

the amount of oil displaced by the duck at the end is $\Delta V_{\text{oil, disp}} = 180$mL. Thus we have

$$\rho_{\text{oil}} = -\rho_w \frac{\Delta V_{\text{water, disp}}}{\Delta V_{\text{oil, disp}}} = -1.00 \text{ g/mL} \cdot \frac{143}{180} = \boxed{0.79 \text{ g/mL}}.$$

Of course answers will vary because of differences in measurement with the slopes. The official AAPT answer is 0.77 g/mL, but don't worry if you get something slightly off.

This is not the only way to solve this problem – there are several different equivalent approaches. All of them somehow make use of the fact that the mass of the displaced water is constant throughout and use the change in slope of the two lines – the value of the bottom line is

$$y_1 = V_{\text{water}} + V_{\text{water, disp}}$$

and the value of the top line is

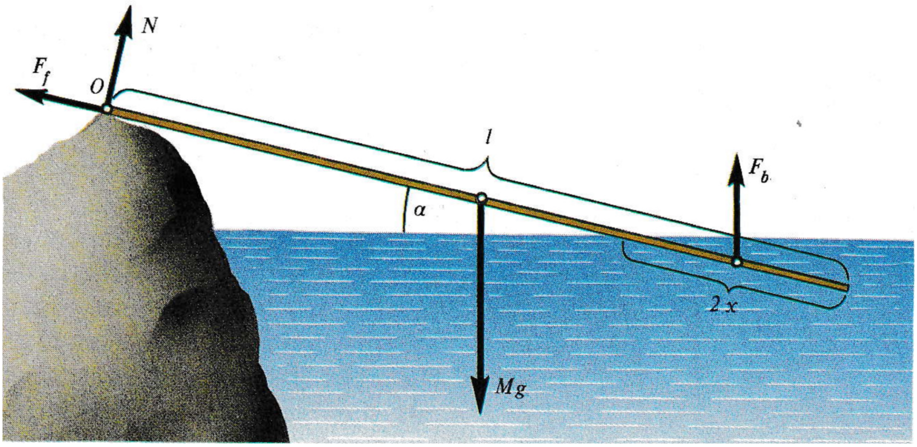
$$y_2 = V_{\text{oil}} + V_{\text{water}} + V_{\text{oil, disp}} + V_{\text{water, disp}}.$$

■

16.6.2 Extra Solutions

Problem 16.14 (Quantum Magazine)**.** A thin plank of length $\ell = 5$ m is propped up at its higher end by a stone that emerges above the water's surface to a height $H = 1$ m. What is the minimal coefficient of friction between the stone and the plank that is needed for the plank to remain at rest? Let the density of the plank be $\rho = 200$ kg/m^3 and the density of water be $\rho_0 = 1000$ kg/m^3.

Solution. Let θ be the angle between the water and the plank, and let A be the cross sectional area of the plank. Then the length of the plank above water is $\frac{H}{\sin\theta}$, so the length of the rod below water is $\ell - \frac{H}{\sin\theta}$.



Therefore, we have

$$F_g = \rho A \ell g$$
$$F_b = \rho_0 A \left(\ell - \frac{H}{\sin\theta}\right) g.$$

Since the rod is at equilibrium, we know that the torque about any point is zero. In this situation, we have three unknown variables, μ, N, and θ. To minimize the number of variables in our equation, we'll pick the origin to be the contact point of the plank with the rod. Then, the distance to the force of gravity is $\ell/2$, and the distance to the force of buoyancy is the average of ℓ and $\frac{H}{\sin\theta}$, i.e. $\frac{1}{2}\left(\ell + \frac{H}{\sin\theta}\right)$. This gives us the equation for torque:

$$0 = F_g(\ell/2)\cos\theta - F_b \cdot \frac{1}{2}\left(\ell + \frac{H}{\sin\theta}\right)\cos\theta \implies F_g \ell = F_b\left(\ell + \frac{H}{\sin\theta}\right).$$

This implies that

$$\rho A \ell^2 g = \rho_0 A \left(\ell - \frac{H}{\sin\theta}\right)\left(\ell + \frac{H}{\sin\theta}\right) g$$
$$\implies \rho \ell^2 = \rho_0 \left(\ell^2 - \frac{H^2}{\sin^2\theta}\right)$$
$$\implies \sin^2\theta = \frac{H^2}{\ell^2(1-\rho/\rho_0)}.$$

We also know that the net force on the plank is zero. Taking the components of force parallel and perpendicular to the plank,

$$\mu N + F_b \sin\theta = F_g \sin\theta \implies \mu N = \sin\theta(F_g - F_b)$$
$$N + F_b\cos\theta = F_g\cos\theta \implies N = \cos\theta(F_g - F_b).$$

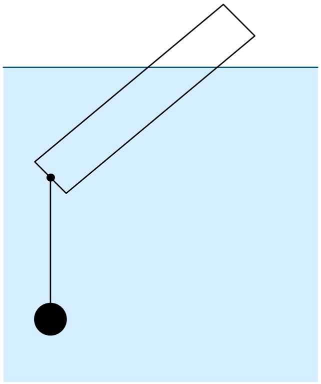
Dividing these equations, we have $\mu = \tan\theta$.

We know that $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta} = \frac{\sin^2\theta}{1-\sin^2\theta}$, so

$$\begin{aligned}\mu^2 &= \left[\frac{H^2}{\ell^2(1-\rho/\rho_0)}\right] \Big/ \left[\frac{\ell^2(1-\rho/\rho_0)-H^2}{\ell^2(1-\rho/\rho_0)}\right] \\ &= \frac{H^2}{\ell^2(1-\rho/\rho_0)-H^2} \\ \mu &= \frac{H}{\sqrt{\ell^2(1-\rho/\rho_0)-H^2}} \approx \boxed{0.229}.\end{aligned}$$

■

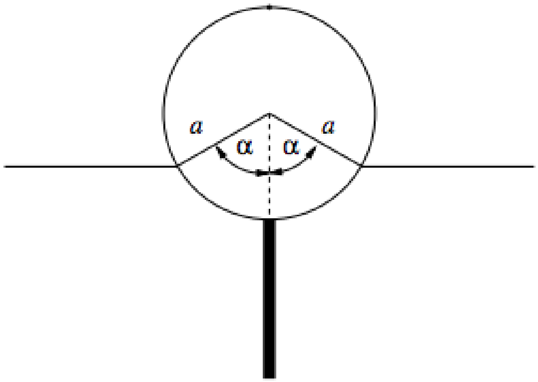
Problem 16.15 (Mock $F = ma$)**.** A thin homogeneous cylindrical float is made out of a light substance with a density 0.5 g/cm^3. A lead sinker of density 204.53 g/cm^3 is tied with fishing line to the bottom of the float. Let the mass of the float be m, and the mass of the sinker be M. What conditions must the ratio of the masses of the sinker and the float satisfy for the float to rest vertically in the water? (Neglect the forces of surface tension. The density of water is 1 g/cm^3).



Solution. ■

Problem 16.16 (IPhO)**.** A buoy consists of a solid cylinder, radius a, length ℓ, made of lightweight material of uniform density d with a uniform rigid rod protruding directly outwards from the bottom halfway along the length. The mass of the rod is equal to that of the cylinder, its length is the same as the diameter of the cylinder and the density of the rod is greater than that of seawater. This buoy is floating in sea-water of density ρ.

(a) In equilibrium derive an expression relating the floating angle α, as drawn, to d/ρ (don't bother solving for α) Neglect the volume of the rod.



(b) In the approximation that the cylinder swings about its horizontal central axis, determine the frequency of swing again in terms of g and a. Neglect the dynamics and viscosity of the water in this case. The angle of swing is assumed to be small.

Solution.

(a) The mass of the cylinder is $m = d\pi a^2 l$ so the total mass of the bouy is

$$2m = 2\pi a^2 ld.$$

Next we find the volume of the water displaced by subtracting the area of the two triangles with right angle α from the area of the sector spanned by angle 2α. The area of a sector of a circle is proportional to the angle, and since the area of a sector of angle 2π radians is just πa^2, then we can deduce that the area spanned by angle 2α is $a^2\alpha$. The are aof the two right triangles is $a^2 \sin(\alpha)\cos(\alpha)$ using some trigonometry. This gives us

$$V_{disp} = \alpha a^2 l - a^2 l \cos(\alpha)\sin(\alpha).$$

Archimedes' principle tells us $\rho V_{disp} = 2m$ so substituting in the values for V_{disp} and $2m$ we have

$$\rho\alpha a^2 l - a^2 l\rho\cos(\alpha)\sin(\alpha) = 2\pi a^2 ld,$$

giving us our simplified relation

$$\alpha - \cos(\alpha)\sin(\alpha) = \frac{2\pi d}{\rho}.$$

(b) Since we assumed that the rod is infinitely thin, it does not displace any water, so when calculating the buoyant force we can ignore it and treat the buoy like it is just a cyliner. If we rotate the cylinder by an angle θ about its center, the amount of water displaced does not actually change. This means the buoyant force stays the same and since it acts directly under the cylinder's center of mass, it actually does not exert any torque on the cylinder. This means that the only torque on the buoy is due to gravity. Also notice that when we rotate the cylinder by some θ, since the buoyant force does not change, the buoy will neither rise nor sink, so the center of the cylinder really is the center of rotation as the problem assumes.

Our first step is to calculate the torque (due to gravity) about the center of the cylinder. The center of mass of the system is at the part where the rod connects to the cylinder and there is a force $2mg$ where m is the mass of the cylinder, so the torque is

$$\tau = 2mga\sin(\theta).$$

Next we find the moment of inertia about the center of the cylinder. We have

$$I_{cyl} = \frac{1}{2}Ma^2,$$

and since the length of the rod is $2a$ we have

$$I_{rod} = \frac{1}{12}M(2a)^2,$$

so using the parallel axis theorem

$$\begin{aligned} I &= \frac{1}{2}Ma^2 + (\frac{1}{12}M(2a)^2 + M(2a)^2) \\ &= \frac{29Ma^2}{6}. \end{aligned}$$

Then for a displacement angle θ we get

$$\alpha = \frac{\tau}{I} = \frac{12g\sin(\theta)}{29a} \approx \frac{12g\theta}{29a}.$$

Thus we get

$$\omega = \sqrt{\frac{12g}{29a}},$$

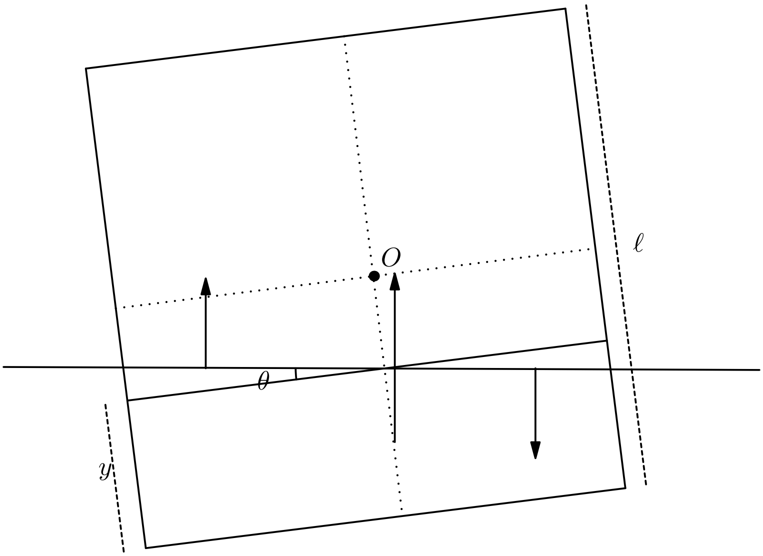
so the frequency is

$$f = \boxed{\frac{1}{2\pi}\sqrt{\frac{12g}{29a}}}.$$

■

Problem 16.17 (Kalda)**.** If a beam with square cross-section and very low density is placed in water, it will turn one pair of its long opposite faces horizontal. This orientation, however, becomes unstable as we increase its density. Find the critical density when this transition occurs. The density of water is $\rho_w = 1000\ \mathrm{kg/m^3}$.

Solution. Conceptually, what would happen is that if the block is extremely light and the square cross section is given a tiny push, there will have a restoring torque causing it to be in stable equilibrium. However, at a certain density, the equilibrium position will not be when the sides of the square are parallel to the water. In fact, the new equilibrium position will be rotated a tiny angle θ where $\theta \ll 1$.



We can represent the submerged portion as three separate masses. The long horizontal line that extends past the square is the water level. Therefore, we can recognize that the submerged part represents a trapezoid. This can be perfectly represented as a rectangle that has the same area as the trapezoid. However, if we try to balance torques with this setup, we will fail because there are certain edge effects that are not covered. Therefore, we need to add a triangle of density ρ an identical triangle with density $-\rho_o$ to make it resemble its original shape.

Let the width of the square be ℓ and the height of the rectangle be y. Balancing forces we have:

$$\rho_o \ell^2 g = \rho_w \ell y g \implies \frac{y}{\ell} = \frac{\rho_o}{\rho_w}$$

Let us now balance torques around the center of mass at O. In an equilibrium position, the torques will sum to zero. The torque from the buoyant force from the rectangle is:

$$\tau_1 = \rho_w g\,(\ell y)\left(\frac{1}{2}(\ell - y)\right)\sin\theta$$

where θ is the angle the bottom of the beam makes with the horizontal. The triangular parts will also provide a torque from the buoyant force. Note that the buoyant force caused by the negative mass triangle will be negative and point in the other direction. The torque of each is:

$$\tau_2 = \rho_w g\left(\frac{1}{2}(\ell/2)^2\sin\theta\right)\left(\frac{2}{3}(\ell/2)\right)$$

where $\frac{2}{3}(\ell/2)$ is the perpendicular distance from the center of mass of the triangle to the center of mass of the square. Notice that since $\theta \ll 1$ we can sum torques and set it to zero:

$$\begin{aligned} 0 &= \tau_1 - 2\tau_2 \\ \frac{\ell^3}{12} &= \frac{\ell y(\ell - y)}{2} \\ \frac{\ell^2}{6} &= y(\ell - y) \end{aligned}$$

From earlier, let's substitute $\frac{y}{\ell} = \frac{\rho_o}{\rho_w} \equiv f$ and we'll get:

$$\frac{\ell^2}{6} = \ell^2 f(1-f) \implies f^2 - f + 1/6 = 0$$

Using the quadratic formula we get:

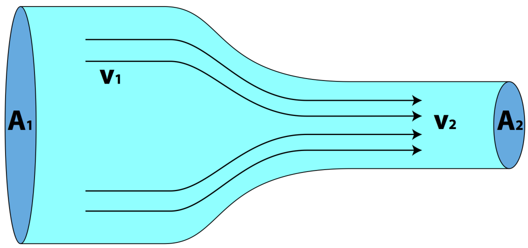
$$f = \frac{\rho_o}{\rho_w} = \frac{1}{2}\left(1 - 3^{-1/2}\right).$$

■

17 Fluid Dynamics

17.1 The Equation of Continuity

When a fluid is in motion, it must move in such a way that mass is conserved (otherwise it would break the law of conservation of mass). Let us consider, in detail, the flow of fluid through a pipe.



Let us say the pipe contracts and then expands as shown below. Fluid enters at point A where the cross sectional area is A_1, squeezes through point B with cross-sectional area A_2 and then leaves the pipe out of point C with cross sectional area A_1. Once again, the fluid must move in a way such that mass is conserved. This means that the velocity of the fluid cannot be the same at all points when traveling through this pipe. Similarly, under non-ideal situations, the density of the fluid cannot be the same. Generally, most problems require a steady flow rate which means that the density of the fluid does *not* change with respect to time.

Let the speed of the fluid at point A be v_1. In a *small* time interval dt, the fluid will travel an approximate distance of $v_1 \cdot dt$. This tells us that the volume dV_1 will be approximately $A_1 v_1 dt$. If the density of the fluid at that point is ρ_1, then the mass ($dm \approx \rho_1 dV_1$) of the fluid that crosses through A_1 is approximately

$$dm = \rho_1 A_1 v_1 dt.$$

This result that we have obtained is interesting and allows us to bring up a certain definition.

Definition 17.1. The **mass flux** is the rate of mass flow per unit area. Its units are defined as $[ML^{-2}]$. It is also generally defined as $\dot{m}$ or μ. In this course, we will define the mass flux as μ.

Going back and applying our definition to the equation up above tells us that

$$\frac{dm}{dt} = \mu_1 = \rho_1 A_1 v_1.$$

We can apply a similar equation at point B where the fluid density is ρ_2, velocity is v_2, and the cross-sectional area is A_2.

$$\mu_2 = \rho_2 A_2 v_1.$$

Since there are no external forces and the flow is steady, then we can equate the flux at points A and B to result in the equation

$$A_1 v_1 = A_2 v_2.$$

Theorem 17.2. *By conservation of mass, we have that under ideal conditions, the fluid movement in a pipe abides by the equation*

$$A_1 v_1 = A_2 v_2$$

where A represents the area and v represents the velocity at different points.

The continuity equation has many applications.

Example 17.3 (USAPhO). When a faucet is turned on, a stream of water flows down with initial speed v_0 at the spout. For this problem, we define y to be the vertical coordinate with its positive direction pointing up. (a) Assuming the water speed is only affected by gravity as the water falls, find the speed of water $v(y)$ at height y. Define the zero of y such that the equation for v^2 has only one term and find y_0, the height of the spout. (b) In this case, the radius r of the stream of water is a function of vertical position y. Let the radius at the faucet be r_0. Find $r(y)$.

Solution. Note that the first part of the problem does not require the use of the continuity equation but is needed for part b which does use the continuity equation.

(a) We apply energy conservation at two different points on the spout. One at the initial location with velocity v_0 and height h and another when the water streams falls a distance y and gains a velocity v. From energy conservation to get

$$\frac{1}{2}mv_0^2 + mgy_0 = \frac{1}{2}mv^2 + mgy \implies v = \sqrt{v_0^2 + 2g(y_0 - y)}.$$

The equation for v^2 has three terms, but we were asked to choose the zero of y such that there is only one. Evidently, two of the terms must cancel, and these must be the two constant terms, since the final term varies with y. That means we need

$$v_0^2 + 2gy_0 = 0 \implies y_0 = -\frac{v_0^2}{2g}.$$

Substituting this result back into our equation for v tells us

$$v = \boxed{\sqrt{-2gy}}.$$

(b) The same volume of water must fall through any horizontal cross-section of the stream each second because water doesn't disappear during its fall, and its density if constant. This means that we can use the continuity equation

$$A_1 v_1 = A_2 v_2.$$

If the initial radius is r_0, then we have

$$\pi r_0^2 \cdot v_0 = \pi r(y)^2 v \implies r(y) = r_0 \sqrt{\frac{v_0}{v}}.$$

From part a we know what v is, therefore we can plug back in to get

$$r(y) = \boxed{r_0 \sqrt[4]{\frac{v_0^2}{-2gy}}}$$

Exercise 17.4. Fluid flows into a pipe with initial cross-sectional area of 5 cm^2 at a velocity of 3 m/s. The pipe then contracts till the cross-sectional area is 3 cm^2. (a) Calculate the velocity of the fluid, (b) Conceptually describe why the fluid would become faster/slower based on your previous answer.

■

17.2 Bernoulli's Equation

We now introduce Bernoulli's Equation. This equation is essentially the equation of conservation of energy applied to fluid dynamics.

Theorem 17.5. *Bernoulli's equation states that for a steady, incompressible, nonviscous, and irroational flow of a constrained fluid, the following holds true:*

$$P + \frac{1}{2}\rho v^2 + \rho g h = const.$$

where P is the pressure, ρ is the density, v is the velocity, and h is the height.

Proof. Consider fluid flowing in through a pipe as shown below.

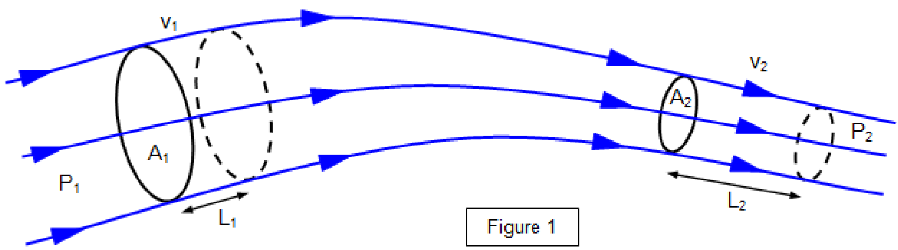


Figure 1

If the flow is steady (and there is no viscosity), then the energy of any given volume of fluid should be constant. We examine the volume of fluid initially contained between the solid areas A_1 and A_2. After some time dt the volume of fluid has moved further down the pipe (so it is enclosed between the dashed areas). The pressures at A_1 and A_2 do some work on this volume of fluid,

$$W = F \cdot d = P_1 A_1 L_1 - P_2 A_2 L_2$$

but $A_1L_1 = A_2L_1 = dV$ since the volumetric flow rate is a constant, so

$$W = (P_1 - P_2)dV$$

As the fluid moves, its velocity and thus kinetic energy increases slightly. After time dt, effectively, the fluid that was contained in the space L_1 has now relocated to the space L_2. Thus the change in kinetic energy is

$$\begin{aligned}\Delta K &= \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \frac{1}{2}\rho A_1 L_1 v_2^2 - \frac{1}{2}\rho A_2 L_2 v_1^2 \\ &= \frac{1}{2}\rho(v_2^2 - v_1^2)dV\end{aligned}$$

To be more thorough, we include the change in energy due to gravitational potential energy as well. If A_1 is located at a height h_1 and A_2 is at a height h_2, then the change in gravitational potential energy is

$$\Delta U = \rho g(h_2 - h_1)dV.$$

We have $W = \Delta K + \Delta U$, so

$$(P_1 - P_2)dV = \frac{1}{2}\rho(v_2^2 - v_1^2)dV + \rho g(h_2 - h_1)dV$$

Canceling dV, we get our final equation,

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2.$$

The left hand side is the initial energy per unit volume and the right hand side is the final energy per unit volume. ■

Although in our proof we were looking at fluid in a pipe, this argument holds for any steady streamline of fluid flow. Be aware though that it only holds for steady flows, meaning the velocity at each point stays the same through time, otherwise the energy in a given volume of fluid may not be conserved.

When the velocity is constant, Bernoulli's equation simply reduces to

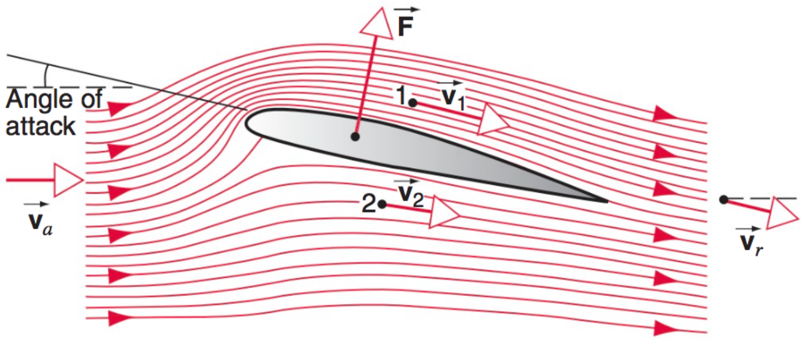
$$P + \rho g h = const.$$

which is familiar from our lesson on fluid statics. When the pressure is constant it reduces to

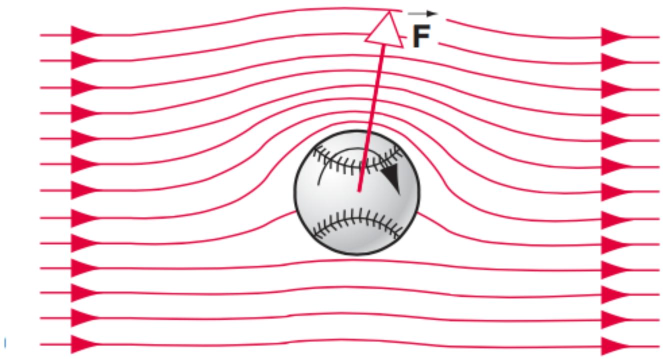
$$\frac{1}{2}\rho v^2 + \rho g h = const.$$

which is just the familiar work-energy principle. What is surprising is the relationship between pressure and kinetic energy – in general the pressure will be higher when the fluid is moving more slowly and lower when the fluid is moving more quickly.

Bernoulli's principle has a number of interesting examples in real life. One of the more famous examples is in the flight of airplanes – airplane wings are designed so that the air moves faster and thus has lower pressure above the wing than below it, as shown in 17.2. This pressure difference exerts a lift force on the wing. Another explanation for the lift on an airplane wing is that the air is deflected downward, exerting an upward force on the wing by Newton's third law. Both of these explanations are valid.

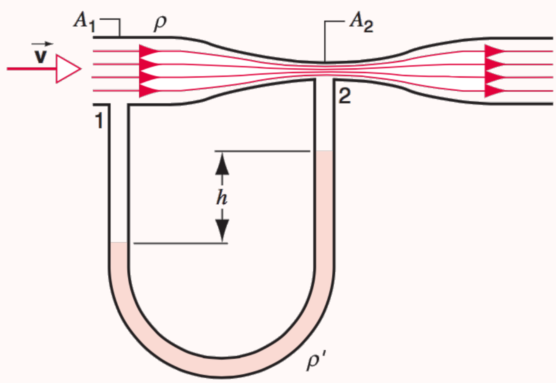


Bernoulli's principle an also be used to understand the flight of a spinning baseball. Because the air has some viscosity, the air close to the baseball will be carried along with the ball's spinning motion as show in 17.2, so in the frame of the baseball, the air will be moving faster on one side than on the other. By Bernoulli's principle, this will create a pressure difference, causing the baseball to turn relative to its expected path.



With Bernoulli's Principle and the continuity equation, you can solve almost any fluid dynamics problem.

Example 17.6 (HRK). The following is an image of a Venturi meter, a device used to measure the flow speed of fluid in a pipe. A fluid of density ρ flows through a pipe of cross-sectional area A_1. At the throat the area is reduced to A_2, and a manometer tube is attached, as shown. Let the manometer liquid, such as mercury, have a density ρ'. Given h, find the flow speed v.



Solution. The force of gravity on the mercury in the manometer is $\rho' gha$ where a is the width of the manometer tube, and the counter force due to the pressure difference is $(P_1 - P_2)a$. Since the liquid is in equilibrium, we see that the pressure difference is

$$P_1 - P_2 = \rho' gh. \tag{17.1}$$

Our next step is to relate the pressures and velocities, which we do using Bernoulli's law. But first we need to figure out the speed of the fluid at point 2. The continuity equation tells us

$$v_2 = v\frac{A_1}{A_2}.$$

Finally can apply Bernoulli's equation to points 1 and 2,.

$$P_1 + \frac{1}{2}\rho v^2 = P_2 + \frac{1}{2}\rho\left(v\frac{A_1}{A_2}\right)^2$$

$$P_1 - P_2 = \frac{1}{2}\rho v^2\frac{A_1^2 - A_2^2}{A_2^2}$$

Setting this equal to (Eq. 17.1) we have

$$\rho' gh = \frac{1}{2}\rho v^2\frac{A_1^2 - A_2^2}{A_2^2}$$

Now all we have to do is solve for v to get

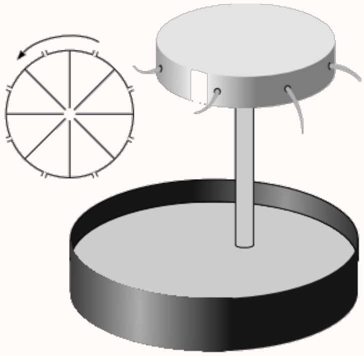
$$v = \boxed{A_2\sqrt{\frac{2\rho' gh}{\rho(A_1^2 - A_2^2)}}}.$$

■

Now we will do a more complicated example.

Example 17.7 (Est-Fin Physics Olympiad)**.** Consider the following construction of a water pump. A vertical tube of cross-sectional area S_1 and height h leads from an open water reservoir to a cylindrical rotating tank of radius R and negligible height. An engine rotates the tank at angular velocity ω The rotating tank is filled with water and has holes along its perimeter of net cross-sectional area S_2.

The water density is ρ, the air pressure p_0 and the pressure at which water boils (saturated vapor pressure) is p_k.



(a) Find the pressure at the edge of the tank if the holes are closed.

(b) From now on, all the holes are open. Find the velocity of the water jets with respect to the ground.

Solution.

(a) The centrifugal force at a radius r is $\omega^2 r\ dm$. Using the substituting $dm = \rho\ dV = \rho A\ dr$, we see that

$$F_c = \int_0^R \omega^2 r\ dm = \int_0^R \rho A \omega^2 r\ dr = \rho A \frac{\omega^2 R^2}{2},$$

so the pressure difference between the middle and the edge of the centrifuge is $\frac{\rho\omega^2 R^2}{2}$. Thus the pressure at the middle of the centrifuge where it connects to the pipe bringing water from below is $P_0 - \rho g h = P_2 - \frac{\rho\omega^2 R^2}{2}$, so

$$P_2 = P_0 + \frac{\rho\omega^2 R^2}{2} - \rho g h.$$

(b) Let v_0 be the velocity of the water jets in the rotating frame. The water at the edge of the centrifuge should have the same energy whether or not the holes are open or closed. Bernoulli's equation tells us the energy per unit volume of a fluid, so we can equate $P + \rho v^2 + \rho g h$ for the case when the holes are closed ($P = P_2$, $v = 0$) and the holes are open ($P = P_0$, $v = v = v_0$) to get

$$P_2 = P_0 + \frac{1}{2}\rho v_0^2$$
$$v_0 = \sqrt{\frac{(P_2 - P_0)}{\rho}} = \sqrt{\omega^2 R^2 - 2gh}$$

However in the ground reference frame, the velocity has an extra component in the tangential direction ωR so the total velocity is $\sqrt{2(\omega^2 R^2 - gh)}$

■

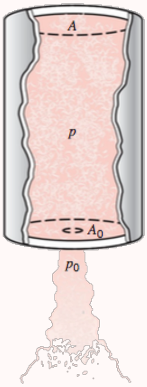
17.3 Momentum

If Bernoulli's law is conservation of energy for fluids, then what happens to conservation of momentum? Conservation of momentum is generally less useful for fluids as it is for Newtonian dynamics. When a fluid is flowing in a pipe, the pipe will often exert forces on the fluid and it is not immediately clear what these forces are, so the fluid's momentum is not generally conserved. By contrast, a *stationary* pipe cannot do any work on the fluid, so Bernoulli's principle will always hold. However, there are some instances where conservation of momentum is helpful.

While the momentum of a fluid may not be conserved, the momentum of the fluid pipe system is conserved and thus by figuring out the change in velocity of a fluid using Bernoulli's principle, we can figure out the force it exerts on the pipe containing it.

Let's first do a simple example to illustrate this approach.

Example 17.8. A rocket has a chamber of cross-sectional area A with incompressible gas of density ρ at a pressure P. The gas shoots out through an opening of width A_0. Letting P_0 be the ambient atmospheric pressure, compute the thrust on the rocket.



Solution. Using Bernoulli's equation we get the velocity v_0 that the gas leaves the rocket in the reference frame of the rocket,

$$P - P_0 = \frac{1}{2}\rho(v_0^2 - v^2)$$

where v is the velocity of the gas inside the rocket. The continuity equation tells us $Av = A_0 v_0$, so if $A \gg A_0$ then $v_0 \gg v$ and we can approximate v to be zero. Then,

$$P - P_0 = \frac{1}{2}\rho v_0^2$$

so the speed that gas exits the rocket is

$$v_0 = \sqrt{\frac{2(P - P_0)}{\rho}}$$

The thrust on the rocket is $dp/dt = v_0\, dm/dt$. The mass of gas flowing out in a time dt is just $\rho A_0 v_0$, so

$$F = v_0\, dm/dt = \rho A_0 v_0^2$$

substituting in our value for v_0, the thrust is

$$F = 2A_0(P - P_0)$$

■

Remark. It appears that the force on the exhaust gas is due to a pressure difference $(P - P_0)$ over an area A_0, so we might naively expect that the force should be $A_0(P - P_0)$. However, as we saw, the force is actually twice this. The reason is that the pressure difference $P - P_0$ does not actually occur right at the opening – the gas does not instantaneously have a higher velocity right when it leaves the rocket. Rather, as the gas gets closer to the opening it speeds up slightly causing the pressure to decrease slightly. This means that the full pressure P is not applied over A_0 but is actually applied over a wider area. Now, we might expect that since P is applied over an area A and P_0 over an are A_0, then the force should be $PA - P_0 A_0$. But it turns out this isn't exactly true either since the bottom surface of the rocket provides a force that slows down the exhaust gas. This gives us

$$PA - P_0 A_0 > 2A_0(P - P_0),$$

which is definitely true since we assumed $A \gg A_0$.

Now let's go back to the example pipe we used to come up with Bernoulli's law to get an idea of how momentum conservation plays out more generally.

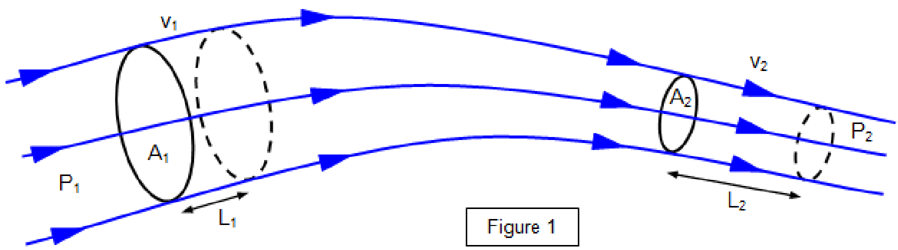


Figure 1

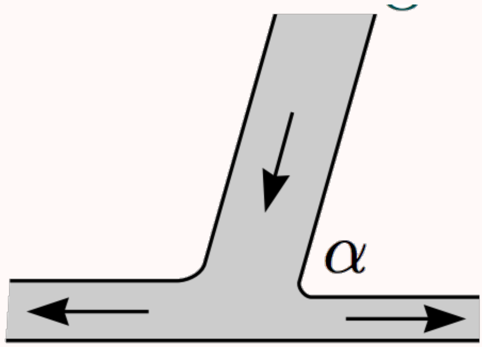
Suppose we wished to find the force that the fluid exerts on the pipe in the region between A_1 and A_2. Newton's second law tells us

$$F = \frac{dp}{dt} = (v_2 - v_1)\frac{dm}{dt} = (v_2 - v_1)\rho A_1 v_1 = (v_2 - v_1)\rho A_2 v_2$$

If we know the pressure difference and initial flow rate, we can calculate this with Bernoulli's law

In the next example, the fluid is not flowing in a pipe at all, so we get conservation of momentum in one component

Example 17.9 (Kalda). A stream of water falls against a trough's bottom with velocity v and splits into smaller streams going to the left and to the right. Find the velocities of both streams if the incoming stream was inclined at an angle α to the trough (and the resultant streams). What is the ratio of amounts of water carried per unit time in the two outgoing streams?



Solution. We can assume that the streams of water have neglibable depth so their hydrostatic pressure is just atmospheric pressure P_0. Then by Bernoulli's law, since the pressure and potential energy of the two outgoing streams is the same, the velocity must be the same, say v. Let μ_1 and μ_2 be the mass per unit time (amount of water) in the left and right-flowing streams respectively. Then the mass per unit time of the incoming stream is $\mu_1 + \mu_2$ In order to figure out their ratio, we use conservation of momentum. Since the trough's bottom is horizontal (and frictionless) it exerts no horizontal force on the water. Thus the horizontal momentum going into the junction per unit time must be the same as the momentum going out of the junction per unit time, that is,

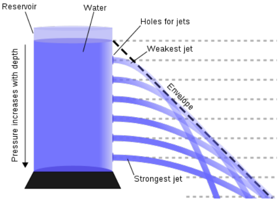
$$\begin{aligned}
(\mu_1 + \mu_2) v \cos(\alpha) &= \mu_1 v - \mu_2 v \\
\mu_2 (1 + \cos(\alpha)) &= \mu_1 (1 - \cos(\alpha)) \\
\implies \frac{\mu_2}{\mu_1} &= \frac{1 - \cos(\alpha)}{1 + \cos(\alpha)} = \tan^2(\frac{\alpha}{2}).
\end{aligned}$$

■

17.4 Applications and Further Theorems

17.4.1 Toricelli's Law

Toricelli's law is something we all have observed in our day to day lives.



The above diagram should make it clear as to what we are talking about. Torricelli himself made a mathematical law about this through observation, but we now show that this a special case of Bernoulli's law-

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

Now we take, for convenience, one point at the surface of the cylinder, because there we can assume the velocity of fluid flow to be 0 (why?). Further, at the opening, the only pressure that acts on the fluid is atmospheric , and hence $P_1 = P_2$.

Exercise. Under all these assumptions show that the velocity of flow at a point P is

$$v_P = \sqrt{2g(h - h_P)}$$

where h is the height of the tower and h_p the height of the point P.

Can you now explain why the fluid jets take parabolic trajectories (Hint: use your knowledge of kinematics). Can you also see why the higher you go, the nearer to the base of the tank the jet 'splashes'?

17.4.2 Viscosity and Turbulence

Viscosity is internal friction in a fluid. It opposes the motion of one part of a fluid relative to the other. Like 'regular' friction, it can be called a necessary evil. For example, it makes using a paddle on a canoe hard, but it is also the reason the paddle works. Intuitively, 'thin' liquids like gasoline and water have lesser viscosities than 'thicker' liquids like honey. Viscosities are temperature dependent, increasing for gases and decreasing for liquids as temperature increases.

Also, due to viscosity it turns out that the pressure required to sustain a given volume flow rate through a cylindrical pipe of length L and Radius R is proportional to L/R^4 . You can now see why even a slight thinning of arteries (due to deposition of Fat) can lead to much bigger amount of pressure needing to be exerted.

Turbulence is the state of (non-laminar) irregular chaotic flow. Bernoulli's equation is not applicable to turbulent flow. As a matter of fact, the more the viscosity, the less the turbulence. Can you see why? Also flow speed is an important factor if we consider fluids of the same viscosity. Near some critical speeds, flow speed tends to be less turbulent. If we lessen the speed, or increase it, flow becomes unstable.

17.5 Homework Problems and Solutions

Problem 17.1. [4] Consider a fluid flowing through a pipe that makes an upward turn, so that now it is flowing horizontally with a vertical shift upward from where it was flowing before. By Bernoulli's principle we would expect the velocity to decrease. However, suppose we make the pipe so that its area is constant. By continuity equation we would expect an increase in its area, but we forced the pipe to have constant area. How is this apparent contradiction resolved?

Solution. We need to also take into account the change in pressure of the water. Bernoulli's equation in its full form is

$$P_1 + \rho g h_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g h_2 + \frac{1}{2}\rho v_2^2$$

If the height increases but the pressure drops sufficiently, then the speed does not need to change; the pressure difference pushing the water forwards and up does the work required to increase its potential energy. ■

Problem 17.2 (Cutnell & Johnson). [5] Water flows straight down from an open faucet. The cross-sectional area of the faucet is $1.8 \cdot 10^{-4}\text{m}^2$, and the speed of the water is 0.85 m/s as it leaves the faucet. Ignoring air resistance, find the cross-sectional area of the water stream at a point 0.10 m below the faucet.

Express your answer in mm^2 and round to the nearest mm^2; note that the problem gives dimensions in meters. However, do **not** include "mm^2" in your answer.

Solution. When the water flows out of the faucet, it is in free fall, so by conservation of energy (also a corollary of Bernoulli's equation) we have

$$\frac{1}{2}\rho v_0^2 + \rho g h = \frac{1}{2}\rho v^2$$

so

$$v = \sqrt{v_0^2 + 2gh}.$$

Then we can use the continuity equation

$$vA = v_0 A_0$$

and plugging in the value of v from conservation of energy,

$$\begin{aligned} A &= \frac{v_0 A_0}{\sqrt{v_0^2 + 2gh}} \\ &= \boxed{93 \text{ mm}^2} \end{aligned}$$

■

Problem 17.3 (Cutnell & Johnson). [5] An airplane wing is designed so that the speed of the air across the top of the wing is 251 m/s when the speed of the war below the wing is 225 m/s. The density of air is 1.29kg/m^3. What is the lifting force on a wing of area 24.0m^2?

Round your answer to the nearest 1000's.

Solution. We will use Bernoulli's equation. Since the air over the top of the wing is a negligible height above the air under the wing, we will ignore the $\mu g h$ term. We have

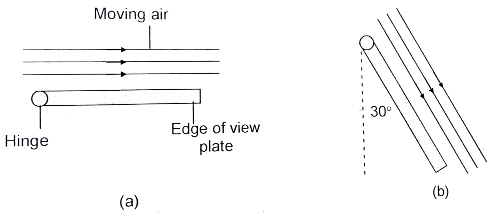
$$\begin{aligned} P_{top} + \frac{1}{2}\mu v_{top}^2 &= P_{bot} + \frac{1}{2}\mu v_{bot}^2 \\ P_{bot} - P_{top} &= \frac{1}{2}\mu v_{top}^2 - \frac{1}{2}\mu v_{bot}^2. \end{aligned}$$

Plugging in values into the RHS, we have

$$\begin{aligned} P_{bot} - P_{top} &= \frac{1}{2} \cdot 1.29 \text{ kg/m}^3 (251 \text{ m/s})^2 - \frac{1}{2} \cdot 1.29 \text{ kg/m}^3 (225 \text{ m/s})^2 \\ &= 7983 \text{ N/m}^2. \end{aligned}$$

Note that $P_{bot} - P_{top}$ is the net pressure downwards. When we multiply the pressure by the area it affects, in this case, the size of the wing, it will give us the net force. Multiplying our above answer by 24.0 m^2, we get an answer of $\boxed{192000 \text{ N}}$. ■

Problem 17.4 (Cutnell & Johnson). [6] A uniform rectangular plate is hanging vertically downward from a hinge that passes along its left edge. By blowing air at 11.0 m/s *over the top of the plate only,* it is possible to keep the plate in a horizontal position, as illustrated in part *a* of the drawing. To what value should the air speed be reduced so that the plate is kept at a 30.0° angle with respect to the vertical, as in part *b* of the drawing?



Round your answer to the nearest 0.01.

Solution. Like in problem 17.3 we use Bernoulli's equation and we can ignore the ρgh term since the height difference is negligible. We get

$$P_{bot} - P_{top} = \frac{1}{2}\rho v_{top}^2 - \frac{1}{2}\rho v_{bot}^2$$

Then the force on the the plate is

$$F = \Delta P A = \frac{A\Delta P}{2} v^2$$

always pointing perpendicular to the plane. When the plate is horizontal the torque required to keep it from falling is

$$\tau_1 = mg\frac{\ell}{2}$$

if ℓ is the length of the plate. In the second scenario the torque is

$$\tau_2 = mg\sin(\theta)\frac{\ell}{2} = \frac{mg\ell}{4}.$$

Then

$$2 = \frac{\tau_1}{\tau_2} = \frac{F_1}{F_2} = \frac{v_1^2}{v_2^2}$$

so $v_2 = v_1/\sqrt{2} = \boxed{7.78 \text{ m/s}}$. ■

Problem 17.5 ($F = ma$). [6] Two streams of water flow through the U-shaped tubes shown. The tube on the left has cross-sectional area A, and the speed of the water flowing through it is v; the tube on the right has cross-sectional area $A' = 1/2A$. If the net force on the tube assembly is zero, what must be the speed v' of the water flowing through the tube on the right? Neglect gravity, and assume that the speed of the water in each tube is the same upon entry and exit.

Solution. The force on the tube is proportional to the change in momentum of the water flowing through it per unit time (Newton's second law).

$$F = \frac{dP}{dt} = (v_in - v_out)\frac{dm}{dt}$$

where dm/dt is the amount of mass that enters the tube per unit time, that is, ρ times the volumetric flow rate Av so

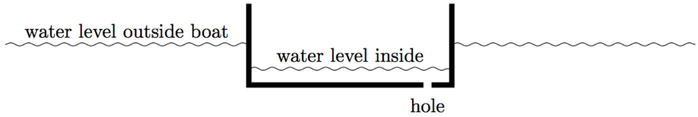
$$F = 2v^2 \rho A.$$

Since the force on the assembly is 0 we have

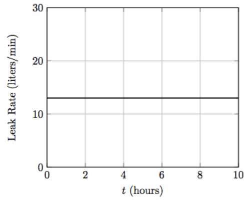
$$v^2 A = v'^2 \frac{A}{2}$$

giving $v' = \boxed{v\sqrt{2}}$ ■

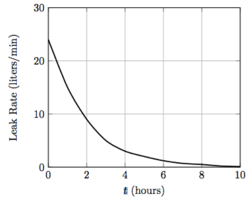
Problem 17.6 ($F = ma$)**.** [6] A small hole is punched into the bottom of a rectangular boat, allowing water to enter the boat. As the boat sinks into the water, which of the following graphs best shows how the rate water flows through the hole varies with time? Assume that the boat remains horizontal as it sinks.



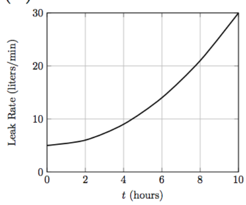
(A)



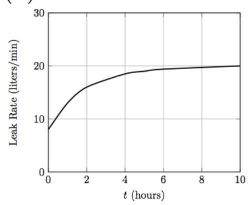
(B)



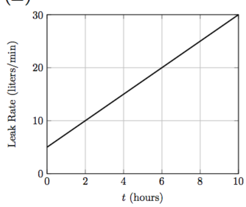
(C)



(D)



(E)



Solution. The difference between the pressure inside and outside the hole is just determined by the mass of the boat and the area it covers, so it is constant no matter how far the boat has sank. Then by Bernoulli's principle if the pressure difference is constant, the speed must be constant so graph (A) is the answer. ■

Problem 17.7 ($F = ma$)**.** [6] A car is turning left along a circular track of radius r at a constant speed v. A cylindrical beaker is placed vertically inside the car. The beaker has a small hole on its right side. If the water's highest point in the beaker is a height h above the hole, at what instantaneous speed does the water escape the hole from a passenger's perspective?
Hint: It's simpler than it looks!

Solution. When the car turns the water will seemingly feel a centrifugal force pointing horizontally outwards and so the surface will be slanted (as we saw in the example in fluid statics). However it turns out we actually do not really care about this because conveniently we are already given the height of the water directly above the hole. Look at the vertical forces on the water above the hole – they are unaffected by the centripetal acceleration, and so balancing gravity with pressure, we get that the pressure by the hole is

$$P_h = \rho g h$$

just as we would expect. (Note that the pressure does vary horizontally across the beaker, we just don't care about that effect.) Then by Bernoulli's principle we have

$$P_h = \frac{1}{2}\rho v^2$$

since the speed of the water in the container is 0, and the gauge pressure outside the container is 0. Thus,

$$v = \boxed{\sqrt{2gh}}$$

■

Problem 17.8 ($F = ma$)**.** [7] A 1500 Watt motor is used to pump water a vertical height of 2.0 meters out of a flooded basement through a cylindrical pipe. The water is ejected through the end of the pipe at a speed of 2.5 m/s. Ignoring friction and assuming that all of the energy of the motor goes to the water, which of the following is the closest to the radius of the pipe? The density of water is $\rho = 1000\text{kg/m}^3$.

(A) 1/3 cm
(B) 1 cm
(C) 3 cm
(D) 10 cm
(E) 30 cm

Solution. We want to set the power equal to the amount of energy transferred to the water. During a small increment of time dt, effectively a small mass dm is lifted up by 2.0m and accelerated from rest to 2.5m/s. This means conservation of energy gives

$$P = \frac{dK}{dt} = \frac{1}{2}\frac{dm}{dt}v^2 + \frac{dm}{dt}gh$$

where $\frac{dm}{dt}$ is the rate at which mass flows through the pipe, ie. ρ times the volumetric flow rate Av. Substituting this it we have

$$P = \rho A v\left(\frac{1}{2}v^2 + gh\right)$$

so solving for $A = \pi r^2$ this becomes

$$\pi r^2 = \frac{2P}{\rho(v^3 + 2ghv}$$

so

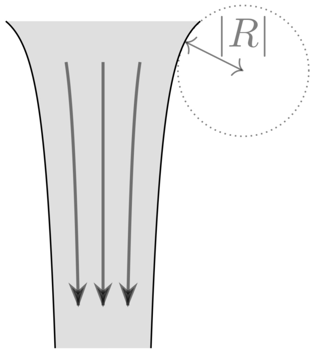
$$r = \sqrt{\frac{2P}{\rho\pi(v^3 + 2ghv)}} = 10\text{ cm}$$

so the answer is (D). ■

Problem 17.9 (USAPhO). [8] This problem continues from the USAPhO problem example. The water-air interface has some surface tension, σ. The effect of surface tension is to change the pressure in the stream according to the *Young-Laplace equation*,

$$\Delta P = \sigma\left(\frac{1}{r} + \frac{1}{R}\right)$$

where ΔP is the difference in pressure between the stream and the atmosphere and R is the radius of curvature of the vertical profile of the stream, visualized below. ($R < 0$ for the stream of water; the radius of curvature would be positive only if the stream profile curved inwards.)



For this part of the problem, we assume that $|R| \gg |r|$, so that the curvature of the vertical profile of the stream can be ignored. Also assume that water is incompressible. Accounting for the pressure in the stream, find a new equation relating for $r(y)$ in terms of σ, r_0, v_0, and ρ, the density of water. You do not need to solve the equation for r.

Hint: This problem really isn't as hard as you think it is. In fact, you don't need to know anything about surface tension to solve this problem! Because you have an expression for pressure from the Young-Laplace equation, you can use a certain equation for the fluid flow in the stream.

Solution. We will use Bernoulli's equation and the continuity equation. Continuity gives

$$v(y) = v_0\frac{r_0^2}{r^2}$$

Since $|R| \gg |r|$, the Young-Laplace equation just becomes

$$\Delta P(y) = \frac{\sigma}{r(y)}$$

Then Bernoulli's equation is

$$\frac{1}{2}\rho v_0^2 + \frac{\sigma}{r_0} + \rho g y_0 = \frac{1}{2}\rho v(y)^2 + \frac{\sigma}{r} + \rho g y$$

In the example part (a) we set $y_0 = -\frac{v_0^2}{2g}$. Substituting this, and $v(y)$ into Bernoulli gives

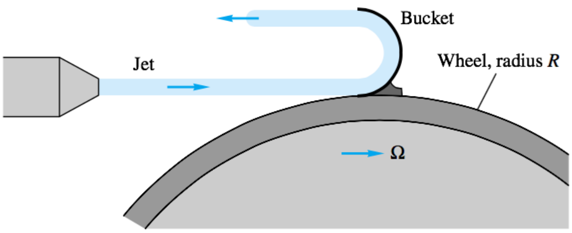
$$\frac{\sigma}{r_0} = \frac{1}{2}\rho\frac{v^2 r_0^4}{f^4} + \frac{\sigma}{r} + \rho g y$$

which simplifies to

$$\frac{1}{2}\rho v_0^2 \frac{r_0^4}{r^4} + \rho g y = \sigma \left(\frac{1}{r_0} - \frac{1}{r} \right).$$

■

Problem 17.10 (White). [10] A liquid jet of velocity v_j and area A_j strikes a single 180° bucket on a turbine wheel of radius R rotating at angular velocity ω as in the below figure. Derive an expression for the power P delivered to this wheel at this instant as a function of the system parameters. At what angular velocity is the maximum power delivered? How would your analysis differ if there were many, many buckets on the wheel, so that the jet was continually striking at least one bucket?



Hint: Yes it does differ if there are many buckets. Try looking at this in the reference frame of the bucket and finding the force exerted by the water jet.

Solution. Let's look at the situation in the reference frame of the bucket. Say the water comes in with velocity v_1 and bounces off with velocity v_2 in the fixed frame. Then the bucket sees a jet approaching it with velocity $v_j = v_1 - \omega R$ and so the water bounces off with velocity $v_j = \omega R - v_1$. This means that the change in momentum of the water jet is

$$F = \frac{dp}{dt} = (v_2 - v_1)\frac{dm}{dt} = 2(\omega R - v_1)\frac{dm}{dt}$$

The mass flow rate is just ρ times the volume flow rate $A_j v_j$. Careful here, we can't just plug v_1 into this equation – in the reference frame of the bucket the water is approaching with velocity $v_j = v_1 - \omega R$ so there is actually less mass per unit time hitting the bucket than if the bucket had been stationary. Plugging this in, we have

$$F = \frac{dp}{dt} = 2\rho A_j (\omega R - v_1)^2.$$

Then the power is

$$P = F\omega R = 2\omega R \rho A_j (v - \omega R)^2$$

Note that we can get this same result looking at the change in energy of the water stream, although this way is a bit more confusing in my opinion. Since $\Delta v = 2(\omega R - v_1)$ have $v_2 = 2\omega R - v_1$ so

$$\begin{aligned} \frac{dK}{dt} &= \frac{1}{2}\frac{dm}{dt}(v_2^2 - v_1^2) \\ &= \frac{1}{2}\rho A_j (v - \omega R)(4\omega^2 R^2 - 4\omega R v_1) \\ &= 2\rho A_j \omega R(\omega R - v)^2. \end{aligned}$$

Now we want to find the value of ω which maximizes the power, that is, maximizes the function

$$f = v^2\omega R - 2v\omega^2 R^2 + \omega^3 R^3$$

We take the derivative with respect to ω and set it equal to 0 to get the quadratic

$$0 = 3R^2\omega^2 - 4vR^2\omega + v^2 R$$

and solving for ω using the quadratic formula we have

$$\omega = \frac{v}{3R} \quad \text{or} \quad \omega = \frac{v}{R}.$$

The second case is impossible, so $\omega R = \frac{v}{3}$ giving

$$P_{max} = \frac{8}{27}\rho A_j v^3$$

Now suppose there are enough buckets on the wheel so that the jet was continually striking at least one bucket. Previously, when calculating the momentum we said that the mass flow rate is $\rho A_j(v - \omega R)$ since this is indeed the amount of mass striking a given bucket per unit time. But now we are told that all the mass in the jet strikes the wheel, so that the total amount of mass striking the wheel per unit time is $\rho A_j v_1$. Now we get that the force is

$$\begin{aligned} F' = \frac{dp}{dt} &= 2(\omega R - v_1)\frac{dm}{dt} \\ &= 2\rho A_j v_1(\omega - v_1) \end{aligned}$$

so now the power is

$$P' = 2\omega v_1 \rho A_j (v - \omega R).$$

To maximize the power we want to maximize the quadratic

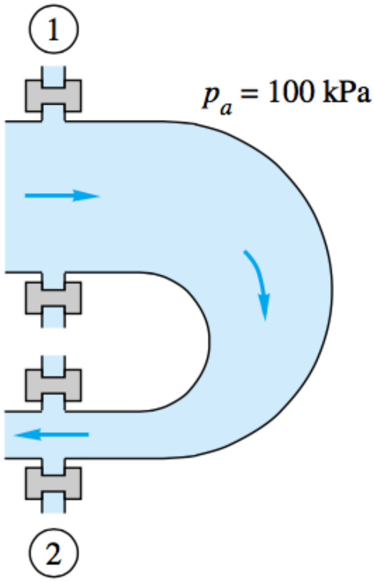
$$f = \omega R(v_1 - \omega R)$$

which gives $\omega R = v_1/2$ so

$$P'_{max} = \frac{1}{2}\rho A_j v^3$$

■

Problem 17.11 (White – modified)**.** [11] Water flows steadily through a reducing pipe bend, as shown below with diameters $D_1 = 25$ cm and $D_2 = 8$ cm. The water coming in has static pressure $P_1 = 350$ kPa and velocity $v_1 = 0.2$ m/s, and the atmospheric pressure is $P_a = 100$ kPa. Neglecting bend and water weight, estimate the total force which must be resisted by the flange bolts.



Solution. Our plan is to sum all the forces on the water-joint system and set it equal to the change in momentum per time of the water flow. The total force exerted on the system is

$$F_{net} = (P_1 - P_{atm})A_1 + (P_2 - P_{atm})A_2 - F_{bolts}$$

since we have to take into account the additional atmospheric pressure pushing from the right side. Then Newton's second law gives

$$F_{net} = P_1 A_1 + P_2 A_2 - F_{\text{bolts}} = \frac{dP}{dt} = v_2 \frac{dm}{dt} - v_1 \frac{dm}{dt} \tag{17.2}$$

We see that in order to find F_{bolts} we still need to figure out v_2, P_2, and the mass flow rate dm/dt.

Let's start with the simplest – finding v_2, the velocity of the water coming out of the pipe bend. We apply the continuity equation $-v_1 A_1 = v_2 A_2$ to find

$$v_2 = -\frac{v_1 A_1}{A_2} = -21.48\text{m/s}.$$

Now we can use Bernoulli's equation to find P_2 the static pressure of the water coming out of the bend,

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

so

$$P_2 = P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2) = 119.3\text{kPa}$$

Finally we have

$$\frac{dm}{dt} = \rho A_1 v_1 = \rho A_2 v_2$$

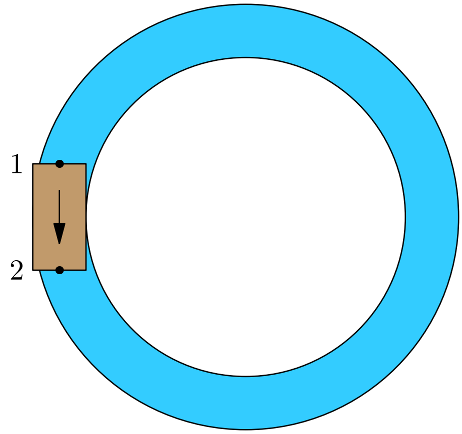
Now we can finally return to equation 17.2 and solve for the force exerted by the bolts:

$$
\begin{aligned}
F_{\text{bolts}} &= (P_1 - P_{atm})A_1 + (P_2 - P_{atm})A_2 + \frac{dm}{dt}(v_1 - v_2) \\
&= (P_1 - P_{atm})A_1 + (P_2 - P_{atm})A_2 + \rho A_1 v_1 (v_1 - v_2) \\
&= (350,000 - 100,000)\left(\frac{\pi}{4}0.25^2\right)\text{N} + (119,300 - 100,000)\left(\frac{\pi}{4}0.08^2\right)\text{N} \\
&+ 1000\left(\frac{\pi}{4}0.25^2\right)0.2(0.2 + 21.48)\text{N} \\
&= 14900\text{N}
\end{aligned}
$$

■

Problem 17.12 (Poiseuille's Law and Hydraulic Analogy)**.** [15] So far, we have mostly been ignoring the effects of viscosity and friction. In this problem, we will deal with what happens when there is a resisting force in a pipe.

Consider a loop of pipe with radius R, cross sectional area $A = \pi R^2$ and length L, along with a pump pushing water in one direction, as shown below. The pressure at points 1 and 2 are P_1 and P_2, respectively.



(a) [2] Find the force on the water that results because of the pressure difference from points 1 and 2.

(b) [3] Treat the pipe as if it were cylindrical. The resisting force as a function of the distance away from the center of a pipe, caused by the friction of the pipe and viscosity of the water, is given by $F_f(r) = -\mu(2\pi r L)\frac{\mathrm{d}v}{\mathrm{d}r}$, where μ is a constant and $\frac{\mathrm{d}v}{\mathrm{d}r}$ is the change in velocity of the water with respect to the radius r away from the center of the pipe. Remember that there is friction, so the velocity of the water is not the same everywhere in the pipe. Is F_f greater than, less than, or equal to the force given in part (a)? Why or why not?

(c) [5] Assuming that the velocity of the water at the edge of a pipe is $v(R) = 0$, solve for v using integration.

(d) [2] Find the average value of v over a cross section by integrating v from 0 to R, then dividing by πR^2.

(e) [2] Solve for $P_2 - P_1 = \Delta P$. Now write your equation in terms of Av instead of v – what is this equivalent to? This equation is known as Poisseuille's Law.

Exercise. Notice that there are a bunch of constants in the equation, like π, μ, or L. Identify these constants. We will call this product of constants the resistivity of the pipe. Now, rename pressure difference to voltage difference, and rename volumetric flow rate to current. If you know a little bit about electronic circuits, does this sound familiar? For more information, search up "hydraulic analogy".

Solution.

(a) The force exerted on the fluid a point 2 is P_2A pointing counterclockwise and the force exerted on the fluid at point 1 is P_1A pointing clockwise, so the force is $A(P_2 - P_1)$. (Note that we only have a notion of 'force' in this case because the radius of the pipe is the same everywhere so all the fluid accelerates together.)

(b) The force caused by the pressure difference is the same as the resistive force. If they were not equal then the fluid would accelerate in some direction indefinitely – either the forces would at some point become equal or it would keep on accelerating infinitely, which is physically impossible.

(c) We treat the pipe as if it were cylindrical, that is, ignoring the fact that it loops around. Look at the mass of the fluid dm that is at a certain radius r from the center of the pipe. This fluid forms a cylindrical shell of thickness dr and mass $dm = \rho 2\pi r L dr$.

The water closer to the center moves faster than the fluid closer to the edges. This means the frictional force from fluid inside the cylindrical shell (a radius $r - dr/2$) points forwards as the faster moving fluid will intuitively try to speed up the fluid around it. Similarly, the friction from the fluid outside the shell (a radius $r + dr/2$) points backwards. Then Newton's second law gives

$$0 = F_{net} = F_f(r - dr/2) - F_f(r + dr/2) + (P_1 - P_2)dA$$

where dA is the cross sectional area of this cylinder, $2\pi r dr$. Thus we get the equation

$$(P_1 - P_2)2\pi r dr = F_f(r + dr/2) - F_f(r - dr/2)$$

Dividing both sides by dr and taking the limit as $dr \to 0$, this becomes

$$2\pi r(P_1 - P_2) = \frac{dF_f}{dr}$$

If we integrate with respect to r, we see that

$$\begin{aligned} \pi r^2(P_1 - P_2) &= F_f \\ &= -\mu(2\pi r L)\frac{dv}{dr}. \end{aligned}$$

So rearranging and integrating once again, we have that

$$\begin{aligned} dv &= \frac{(P_2 - P_1)r}{2\mu L}dr \\ \int dv &= \int \frac{(P_2 - P_1)r}{2\mu L}dr \\ v &= \frac{(P_2 - P_1)r^2}{4\mu L} + C. \end{aligned}$$

Since v has to be 0 when $r = R$, we determine that the constant C must be $-R^2(P_2 - P_1)/(4\mu L)$ so this gives

$$v = \boxed{\frac{(P_2 - P_1)}{4\mu L}(R^2 - r^2)}.$$

(d) At a radius r the area element is $2\pi r dr$ because intuitively the amount of fluid at a given radius is proportional to $2\pi r$. Then we can just perform the integral as follows:

$$\begin{aligned}
v_{\text{avg}} &= \frac{1}{\pi R^2}\int_0^R 2\pi r v(r) dr \\
&= \frac{1}{\pi R^2}\int_0^R \frac{(P_2 - P_1)}{4\mu L}(R^2 - r^2) 2\pi r dr \\
&= \frac{(P_2 - P_1)}{2\mu L R^2}\int_0^R r R^2 - r^3 dr \\
&= \frac{(P_2 - P_1)}{2\mu L R^2}\left(\frac{R^2 r^2}{2} - \frac{r^2}{4}\right)\Big|_0^R \\
&= \frac{(P_2 - P_1)}{2\mu L R^2}\frac{R^2}{4} \\
&= \boxed{\frac{(P_2 - P_1)R^2}{8\mu L}}.
\end{aligned}$$

(e) In the previous part we got the equation

$$v_{\text{avg}} = \frac{(P_2 - P_1)R^2}{8\mu L}$$

so all we have to do is solve this equation for $\Delta P = P_2 - P_1$ to get

$$\Delta P = \frac{8\mu L v_{\text{avg}}}{R^2}$$

We want to get this in terms of the volumetric flow rate $Av = \pi R^2 v_{\text{avg}}$ so we multiply the numerator and denominator of the fraction by πR^2 to get

$$\Delta P = \boxed{\frac{8\mu L A v_{\text{avg}}}{A R^2}}.$$

Note that Av_{avg} is the volumetric flow rate.

The resistivity of the pipe will be $\rho = 8\mu\pi$ so the equation simplifies to

$$\Delta P = \frac{\rho L(\text{flow rate})}{A^2}.$$

Note that if we were to rename ΔP to V and flow rate to I we get

$$V = \frac{\rho L I}{A^2}$$

or if we let $R = \rho L/A^2$ then

$$V = RI$$

which is a statement equivalent to Ohm's law! Here we still have resistance proportional to L, but the only difference is that in Poisseuille's law we have resistance is inversely proportional to A^2 while in Ohm's law resistance is inversely proportional to A. ■

17.5.1 Written Solutions

Problem 17.13 (Kalda, modified). [12]

In this problem we find the velocity u of the propagation of small waves in shallow water.

This may sound intimidating but don't worry, we will walk you through it. Learning how to attack an open ended problem like this is an important skill in physics, but here we try to be clear about what you can and cannot assume.

1. The water is considered shallow if the wavelength is considerably larger than the depth of the water H. Thanks to this we can assume that along a vertical cross-section the horizontal velocity of all particles is the same.
2. The smallness of the waves means that their height is significantly smaller than the depth of the water, allowing us to assume that the horizontal velocity of the water particles v is significantly smaller than the wave velocity, u. In this light, we can get rid of terms that are second order in smallness, (eg. h^2, hv, v^2).

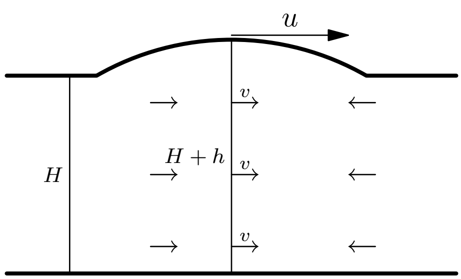


Diagram of the wave in fixed reference frame

We will analyze this problem from the reference frame which is moving at the speed of the wave u so that the flow is steady (we don't really know how to think about flow in a pipe that is changing shape).

(a) [6] Letting h be the height of the wave, find the velocity of the water at the wave's peak.

(b) [6] Now apply Bernoulli's theorem to an imaginary 'tube' of water at the surface of the wave, and use appropriate approximations to find the wave velocity u.

Solution.

(a) The reference frame of the wave is moving at a speed u so the velocity of the water far away from the wave is $-u$ and the velocity of the water at the peak is $v - u$. Then we can use the continuity equation

$$(u - v)(h + H) = Hu$$

and so solving for v we have

$$v = \boxed{\frac{uh}{h + H}}.$$

Alternately if we expand the continuity equation we get

$$uh + uH - vh - vH = uH$$

and since the vh term is second order in smallness we can get rid of it and so solving for v we obtain

$$v \approx \boxed{\frac{uh}{H}}.$$

(b) Consider a strip of water along the surface of the wave. The pressure is just atmospheric pressure everywhere. In the reference frame moving with the wave, the speed of the water at the top of the wave is $u - v$ and the speed far away is u. Then we apply Bernoulli's theorem to get

$$\frac{1}{2}\rho(u-v)^2 + \rho g h = \frac{1}{2}\rho u^2$$

The continuity equation from part (a) tells us that

$$u - v = \frac{Hu}{h+H}.$$

Substituting this into Bernoulli's equation and solving for u,

$$\frac{1}{2}\rho u^2 \frac{H^2}{(h+H)^2} + \rho g h = \frac{1}{2}\rho u^2$$
$$\frac{1}{2}\rho u^2 H^2 + \rho g h(H^2 + 2hH + h^2) = \frac{1}{2}\rho u^2(H^2 + 2hH + h^2)$$

We discard the h^2 terms since they are second order in smallness and obtain

$$\frac{1}{2}\rho u^2 H^2 + \rho g h(H^2 + 2hH) = \frac{1}{2}\rho u^2(H^2 + 2hH)$$
$$\rho u^2 hH = \rho g h H(H + 2h)$$
$$u = \sqrt{g(H+2h)} \approx \boxed{\sqrt{gH}}.$$

■

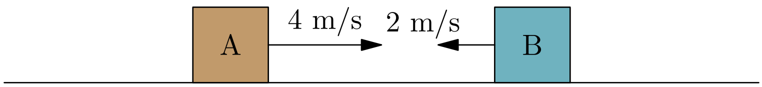
VII

Other Topics

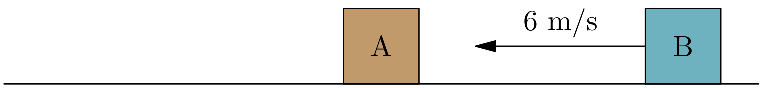
18 Fictitious Forces

18.1 Introduction

So far throughout the course, we have mainly been looking at problems from one specific reference frame: the lab frame. This frame of reference, also known as the environmental frame, is like a third person view. To an observer in the lab, they might see, say block A moving at 4 m/s and block B moving at 2 m/s towards each other.



We have dealt with other references frames as well – namely inertial reference frames as defined by Newton's First Law. Consider what block A experiences. From the frame of reference of block A, block B is moving toward A at 6 m/s. The observer appears to be traveling 2 m/s towards A. Finally, block A does not move relative to itself, so A sees that A has a velocity of 0 m/s.



Problems may arise, however, when we acceleration. Consider a scenario in which block A experiences a constant force of 2 N right. In the lab frame, this works out; a force is applied, so A accelerates to the right.

However, what happens in A's reference frame? From A's perspective, a force is applied onto A. But A does not accelerate relative to itself; A's change in velocity is always 0 relative to itself! This scenario appears to break Newton's first law, so it appears that our reference frame is *not* inertial (Recall that an inertial frame of reference is a frame in which Newton's laws hold).

This presents problems for calculating what happens in a non-inertial reference frame: how do we deal with them if Newton's Laws don't work? Fortunately, there is a solution: it appears that we can use Newton's Laws and pretend that a reference frame is inertial if we introduce our own fake or fictitious force to "counteract" the acceleration of the reference frame. We'll now go over the types of fictitious forces and see how they change non-inertial reference frames to inertial ones.

18.2 Noninertial Frames of Reference

We'll start by defining what exactly a fictitious force is:

Definition 18.1. A **fictitious force** (also called a pseudo-force), is a force acting on a mass whose motion is within a non-inertial frame of reference, such as an accelerating or rotating reference frame that allows us to apply Newton's Laws.

There are **four** different kinds of fictitious forces in mechanics as we will soon see:

- The **translational force**, which is caused by relative acceleration in a straight line.
- The **centrifugal force**, caused by rotation around a central axis.
- The **Coriolis force**, which acts on objects that are in linear motion inside of a frame of reference that rotates about an axis.
- The **Azimuthal force** (also called the Euler force), which occurs when an angular acceleration occurs. It is essentially the rotation equivalent of the translational force.

The Azimuthal force is not seen all that often so we will not be covering it. Feel free to check out Chapter 10 of David Morin's Introduction to Classical Mechanics if you want to learn more about the force.

Mathematically, the above statement means that the following holds:

Theorem 18.2. *Suppose $\vec{a}_{cm}$ is the acceleration of the COM of an object of mass M in some noninertial reference. Let F_{net} be the net external forces acting on the object and $\vec{F_{fict}}$ be the net fictitious forces acting on the object. Then*

$$Ma = \vec{F}_{net} + \vec{F}_{fict}$$

Since $\vec{F}_{fict} = \vec{F}_{trans} + \vec{F}_{cent} + F_{cor} + \vec{F}_{azm}$, we have

$$Ma = \vec{F}_{net} + \vec{F}_{trans} + \vec{F}_{cent} + F_{cor} + \vec{F}_{azm}$$

18.2.1 The Translational Force

We'll now go through each of the individual fictitious forces. Let's begin with perhaps the most intuitive of all the fictitious forces: the translational force.

We'll state the following theorem without proof:

Theorem 18.3. *Suppose a particle is moving in a non-inertial, accelerating reference frame that accelerates with acceleration $\vec{a}$. Then,*

$$\vec{F}_{trans} = -M\vec{a}$$

So transposing this ficititious force will allow us to use Newton's Second Law.

The above theorem is more easily understood through examples.

Example 18.4 (Classical). Imagine that you are standing on a train that is accelerating to the right with acceleration a. Analyze the equations of motion in the non-inertial frame of the train.

Solution. Let a_{train} be your acceleration relative to the train. Note that in the reference frame of the train, you are at rest (because relative to the train you do not move). Since in the lab frame, the only force acting on you is the fictitious force we see that $F_{net} = F_f = ma$. Then, in our non-inertial frame, we have $F_{trans} = m(-a)$ so we see that

$$0 = ma_{train} = F_f + F_{trans} = ma + m(-a) = 0$$

so it seems our equations *do* indeed work out, at least in this example. ■

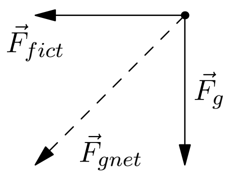
The above example may have been somewhat tautological and may not have been very insightful as to *why* fictitious forces are useful. In general, any problem that can be solved with fictitious forces can be solved *without* fictitious forces as well in an inertial frame of reference. It is often times however, more convenient to think of the problem in terms of a non-inertial frame.

Non-inertial frames are present throughout our lives. For example, when you are in an accelerating car, you are in a non-inertial frame.

Example 18.5. A helium balloon is put inside a car. If the car accelerates forward, in what direction will the balloon go relative to a driver inside the car?

Solution. The car accelerating forward is equivalent to a gravitational force backward for the balloon. We will now look at the gravities acting on the balloon. $\vec{F}_{fict}$ is our fictitious force, and $\vec{F}_{gnet}$ is the net gravitational force from our two "gravities".

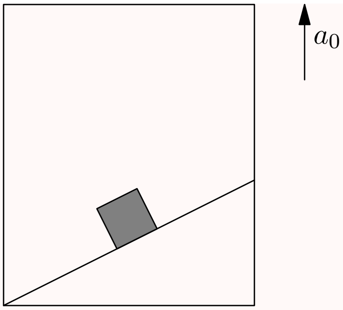
Side view of car



We can see that our net "gravity" points down and to the left, which is down and backward from the driver's perspective.

Under normal conditions, helium balloons rise since their buoyant force is greater than their weight. Thus, in the car, the balloon's buoyant force will point in the opposite direction of the net "gravity" and be stronger than it. The balloon's net acceleration will be in the opposite direction of our "gravity", which is $\boxed{\text{up and to the front}}$ of the driver, though it may only go forward since it will hit the ceiling. ■

Example 18.6. A particle slides down a smooth inclined plane of elevation θ, fixed in an elevator going up with an acceleration a_0. The base of the incline has a length L. Find the time taken by the particle to reach the bottom.



Solution. This was an example in the inclined planes handout. It was mentioned that this problem is trivialized by fictitious forces; we will now show this.

The acceleration of the elevator a_0 upwards is equivalent to a gravitational force of ma_0 downward, where m is the acceleration of the particle. The net gravitational force downward is therefore $mg + ma_0$, so the gravitational acceleration downward is $g+a_0$. Our problem reduces to a inclined plane problem with gravitational acceleration $g + a_0$. Finding the component of acceleration of the block parallel to the plane and using kinematics yields an answer of $t = \sqrt{\frac{2L}{\sin\theta\cos\theta(a_0+g)}}$. ■

18.2.2 The Centrifugal Force

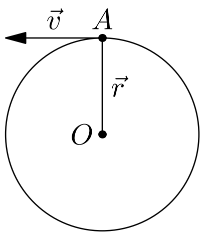
We'll now consider the centrifugal force which goes in hand with the centripetal force as viewed in the inertial frame. The centrifugal force acts on rotating frames as follows:

Theorem 18.7. *Consider an object moving in a noninertial frame of reference rotating with angular velocity ω with respect to some origin. Then, the centripetal force on the object is*

$$F_{cent} = mr\omega^2 = \frac{mv^2}{r}$$

where r is the distance from the origin and the direction of F_{cent} is pointed radially outward from the origin.

We'll now consider a simple example that will illuminate how this works in practice. Consider uniform circular motion:



In the lab frame, we see that the net force on A is a centripetal force such that the object remains in circular motion. The net force on the object therefore has magnitude $\frac{mv^2}{r}$ and points toward the center.

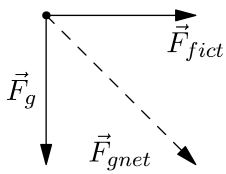
Now we switch to the reference frame of A. In this frame, the object is completely still! Just like with the previous examples, this breaks Newton's second law since this reference frame is not inertial.

The centripetal force still points in the same direction and magnitude, so it provides a force of $\frac{mv^2}{r}$. Since the object is not moving relative to itself, we must introduce a fictitious force to cancel this centripetal force so our object remains still, meaning that we can treat the reference frame as inertial. This centrifugal force must have the same magnitude $\frac{mv^2}{r}$, but point outward.

> **Example 18.8.** A helium balloon is put inside a car. If the car turns left, in what direction will the balloon go relative to a driver inside the car?

Solution. Like in the example for the translational force, we can make a fictitious centrifugal force that points radially outward. Since the car turns left, our force should point to the right.

View of car from the back



We can see that $\vec{F}_{fict}$ points right and $\vec{F}_g$ points down, so $\vec{F}_{gnet}$ points down and to the right. Therefore, the helium balloon will go $\boxed{\text{up and to the left}}$ of the driver, though it may only go left since it will hit the ceiling. ■

> **Example 18.9.** A toy car drives inside a cylindrical pipe with radius r, as shown by the blue path below.
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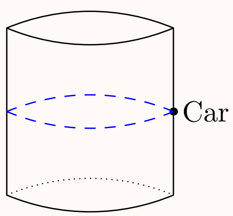
> 
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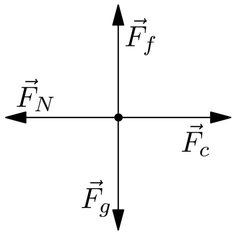
>
> If the cylinder has a static friction coefficient of μ, what is the minimal speed the car must drive with so that it does not slide down the cylinder?

Solution. Consider the free body diagram of the car in the car's frame.



$\vec{F}_c$ is our fictitious centrifugal force.

In order for the car to not slide downwards, we must have $F_f = \mu F_N \geq F_g$. The car does not accelerate relative to itself, so its horizontal components must add to 0. As such, we have $F_N = F_c = \frac{mv^2}{r}$, where m is the mass of the car.

Thus,

$$\begin{aligned} \mu F_N &\geq F_g \\ \mu \cdot \frac{mv^2}{r} &\geq mg \\ v &\geq \boxed{\sqrt{\frac{gr}{\mu}}}. \end{aligned}$$

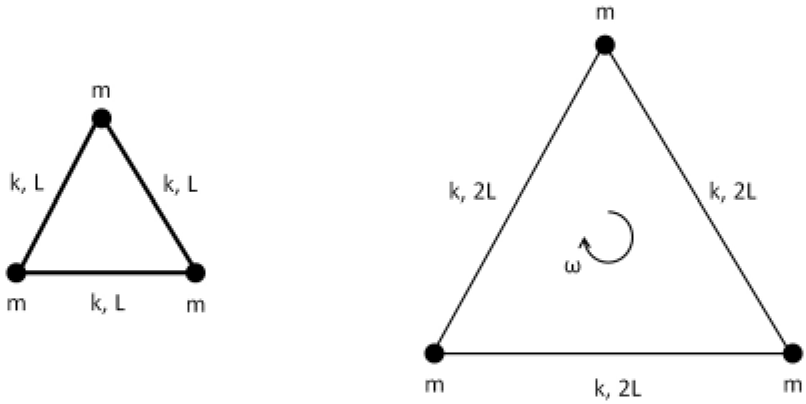
■

Remark. You may have noticed that the example could be done by noting that the centripetal acceleration is $\frac{v^2}{r}$, then noting that the normal force must be equal to this acceleration times the mass of the car. This is the case for this problem, but there are many problems that are greatly simplified by introducing a centrifugal force.

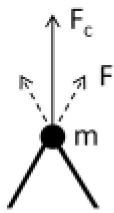
We'll now do a few harder examples that show how we might use the centrifugal force in contest problems:

> **Example 18.10** ($F = ma$). Three point masses m are attached together by identical springs. When placed at rest on a horizontal surface the masses form a triangle with side length l. When the assembly is rotated about its center at angular velocity ω, the masses form a triangle with side length $2l$. What is the spring constant k of the springs?

Solution. Let's try thinking about the problem in the frame of reference such that moves with the same angular velocity ω as the assembly. It's easier to understand *why* the springs extend further in this frame of reference because the point masses are simply at rest rather than rotating. Since we are in a rotating (noninertial) frame, we must add centrifugal forces on the three point masses that are directed radially outward.



Let's see why exactly the springs extend by considering the forces on a particle:



Since each point mass remains at rest in the rotating frame, the spring force must now counteract the centrifugal force. Note that since at rest, the spring has length l, the spring is extended by a length $2l - l = l$, so

$$F_c = 2k\Delta x \cos 30^\circ = 2k(l)\frac{\sqrt{3}}{2} = \sqrt{3}kl = mr\omega^2$$

Since in our case $r = \frac{2}{\sqrt{3}}l$, we have

$$\frac{2m}{\sqrt{3}}l\omega^2 = kl\sqrt{3} \implies \boxed{k = \frac{2m}{3}\omega^2}$$

. ■

For this last example, it is *much* more practical to use the centrifugal force because the angular momentum vector actually turns out to *not* to be parallel to the angular velocity in the lab frame so more technical knowledge (about inertia as a tensor) is needed. The usage of fictitious forces however, simplifies the problem quite a bit.

Example 18.11 (USAPhO). A unicyclist of total height h goes around a circular track of radius R while leaning inward at an angle θ to the vertical. The acceleration due to gravity is g.

1. Suppose $h << R$. What angular velocity ω must the unicyclist sustain?
2. Now model the unicyclist as a uniform rod of length h, where h is less

than R but not negligible. This refined model introduces a correction to the previous result. What is the new expression for the angular velocity ω? Assume that the rod remains in the plane formed by the vertical and radial directions, and that R is measured from the center of the circle to the point of contact at the ground.

Solution.

1. Let's work in the frame rotating with angular velocity ω, where the unicyclist is static. This is a much easier frame to work in than the lab frame because we can choose a fixed axis (perhaps through the point of contact between the unicyclist and the ground) while in the lab frame, the axis is also rotating/moving in a circle.

   There are four forces acting on the body: the normal force and friction force at the point of contact, gravity at the center of mass, and the centrifugal force. Let's try analyzing the torque about an axis through the point of contact (this is of course, because there are 2 forces acting at this point so it simplies the problem quite a bit). Since the unicyclist is static, we have $\sum \tau = 0$.

   If $h << R$, all parts of the unicyclist can be treated as being a distance R away from the center of the circle, so we can treat the centrifugal force as being a single force acting at the center of mass of the unicyclist (this is analogous to how if we treat gravity as uniform we can treat it as acting on the center of mass of the object).

   If the center of mass of the object is a distance l from the point of contact, we have

   $$\sum \tau = m\omega^2 Rl\cos\theta - mgl\sin\theta = 0 \implies \boxed{\omega = \frac{g}{R}\tan\theta}.$$

2. The centripetal acceleration now varies across the body of the unicyclist. Let's calculate the torque due to the centripetal force τ_c by calculating the small centripetal forces $d\tau_c$ acting at a point a distance z from the point of contact and then integrating. Note the centripetal force at any point on the unicycle is

   $$dF_c = dmr\omega^2$$

   directed radially outwards, where r is the distance from the center of the disk. So the torque is

   $$d\tau = s\cos\theta dF_c = \omega^2 rsdm$$

   where s is the distance of the point from the point of contact. Through some geometry, one can easily find that the distance r of any point on the unicycle is given by $r = R - s\sin\theta$. Also note that since the mass of the rod is uniform, we have $dm = \frac{m}{h}ds$ So we get that

   $$d\tau = \omega^2(R - s\sin\theta)s\frac{m}{h}ds$$

   We thus have

   $$\tau_c = \int d\tau = \int_0^h \omega^2(R - s\sin\theta)s\frac{m}{h}ds$$

$$= m\omega^2 h\cos\theta\left(\frac{R}{2} - \frac{h}{3}\sin\theta\right)$$

We also have that gravity acts at the center of mass of the rod and in the opposite direction so $\tau_g = -mg\frac{h}{2}\sin\theta$, so the net torque is

$$\sum \tau = \tau_g + \tau_c = 0$$

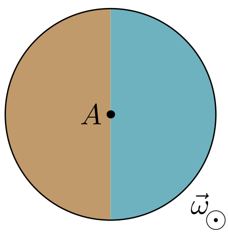
which gives us

$$\tau = \boxed{\sqrt{\left(\frac{g}{R}\tan\theta\right)\left(1 - \frac{2h}{3R}\sin\theta\right)^{-1}}}$$

■

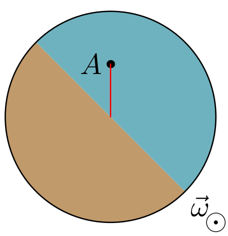
18.2.3 The Coriolis Force

The Coriolis force is not tested too often on the $F = ma$ exam, but it's useful to know just in case it *does* show up. The Coriolis force is the fictitious force that an object experiences when it *moves* in a rotating reference frame. Consider an object on a rotating disk, as shown below.

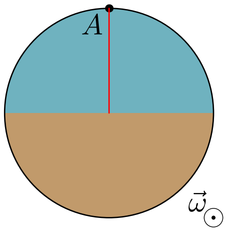


In this case, the object A starts at the center, and the disk rotates at ω counter-clockwise about its center. The disk is colored to show its rotation later on.

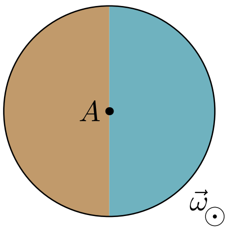
Let us say that A travels upward at some velocity. Now, consider what the path of A looks like from the lab frame. After some time t_1, the path of A may look like this.



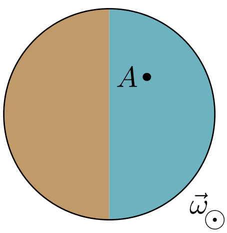
We see that A has moved up because of its velocity, and that the disk has rotated a little because of its angular velocity. Eventually, A will reach the edge of the disk, and the path looks like a straight line segment.



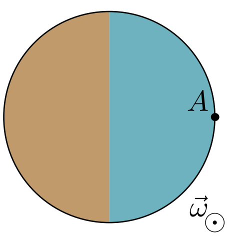
Now, consider this from the rotating frame of the disk. From the rotating frame, the disk always remains upright. The starting position is the same as in the lab frame.



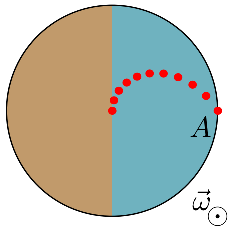
We know where A is at after time t_1 from the lab frame; see the second diagram. However, from the rotating frame, A will be at the following position after t_1.



We can see this if we rotate the second diagram clockwise a little. When A reaches the edge of the disk, it will be at the following position.



By plotting points, we can see the curved path of A.

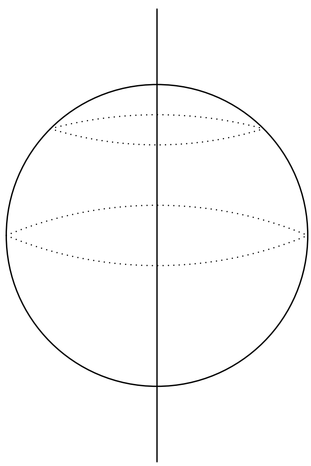


We may wonder how this path can be curved if the path was straight in the lab frame. The answer is that this curvature comes from a fictitious force: the Coriolis force.

Essentially, the Coriolis force comes from the fact that the inside of the disk is spinning slower than the outside. Thus, the further out A travels, the more A "lags" behind the disk's rotation.

The Coriolis Force is also the reason storms in the North spin counterclockwise and storms in the south spin clockwise. In this case, the Coriolis Force acts in 3D.

Consider the northern hemisphere. In this hemisphere, as one goes more north, the rotation of the Earth covers a smaller distance. We can see below that the top path is shorter than the path along the equator.



However, both of these paths complete in 24 hours! This means that an object on the equator will be moving faster than an object closer to the north.

The result is that air moving towards the equator enters a faster moving environment, so it "lags" behind the counterclockwise rotation of the Earth and gets deflected westward. Similarly, air from the equator moving towards the north pole enters a slower environment, so it moves counterclockwise faster than the surrounding air. As such, it gets deflected eastward.

If there is a region of low pressure, we may get winds in the Northern Hemisphere that move like so:

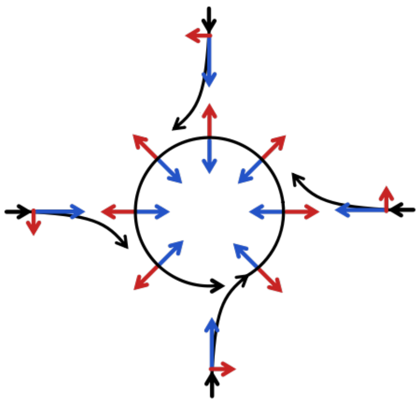


Image source

The blue forces are from a pressure difference. The higher pressure air outside experiences a force towards the region of lower pressure.

The red forces are from the Coriolis force. We can see that the air moving towards the equator (down) is deflected westward, and the air moving towards the north pole (up) is deflected eastward.

As we can see, this results in hurricanes having a counterclockwise motion in the Northern hemisphere. In the Southern hemisphere, the red arrows are reversed, so the opposite occurs; hurricanes will spin clockwise.

Like before, we'll state but not prove that the Coriolis Force can be represented algebraically as follows:

Theorem 18.12. *Suppose an object is moving with a velocity v relative to a reference frame rotating with angular velоctiy ω. Then,*

$$F_{coriolis} = -2\vec{\omega} \times \vec{v}$$

We can then calculate the acceleration of an object from the Coriolis force as

$$\vec{a} = -2\vec{\omega} \times \vec{v},$$

where $\vec{a}$ is the acceleration of object, $\vec{\omega}$ is the angular velocity of the environment, and $\vec{v}$ is the velocity of the object.

We will now see how this might apply in practice:

Example 18.13 ($F = ma$)**.** A man standing at 30° latitude fires a bullet northward at a speed of 200 m/s. The radius of the Earth is 6371 km. What is the sideway deflection of the bullet after traveling 100 m?

Solution. The Coriolis acceleration is given by

$$\vec{a} = -2\vec{\omega} \times \vec{v}.$$

$\vec{\omega}$ is the angular velocity of the Earth and $\vec{v}$ is the velocity of the bullet. Evaluating this cross product gives an eastward acceleration with magnitude $a = 2\omega v \sin\theta$.

We are given that the bullet is fired at 30° latitude, so $\theta = 30°$, and $a = \omega v$. We also know that $\omega = \frac{2\pi}{\text{day}}$.

The deflection of the bullet is given by $d = \frac{1}{2}at^2$, where $t = \frac{100 \text{ m}}{v}$. This means that

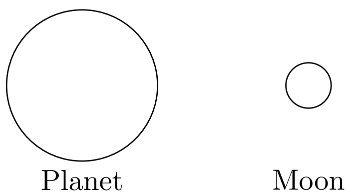
$$\begin{aligned} d &= \frac{1}{2}(\omega v)\left(\frac{100 \text{ m}}{v}\right)^2 \\ &= \frac{\omega(100 \text{ m})^2}{2v} \\ &\approx 1.8 \text{ mm}. \end{aligned}$$

Thus, we have a deflection of $\boxed{1.8 \text{ mm east}}$. ■

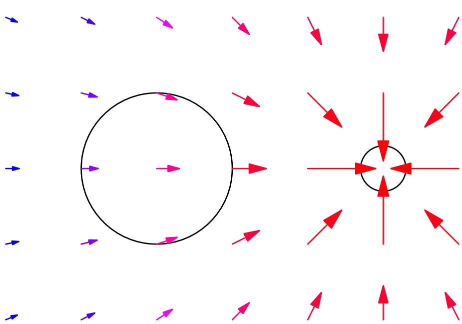
18.3 Tidal Forces

A fictitious **tidal force** can occur when there is a gradient, or change in gravitational field.

Consider a plane and one of its moons, as shown below.



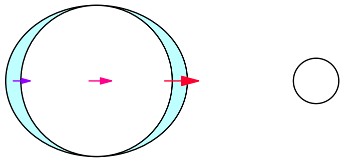
The gravitational field due to the moon looks something like the following.



Note how the gravitational force on the part of the planet closer to the satellite is stronger. Similarly, the force on the part of the planet farther from the satellite is weaker. Finally, the force on the center of the planet is in between these two.

The net result is that parts of the planet closer to the satellite are pulled away from the center, while the center is pulled away from the parts farther from the satellite. If the planet were covered in a fluid, say water, then the water closer to the satellite is pulled away from from the center, and the center is pulled away from the water farther away.

As such, we get tides.



On Earth, the satellite that causes tides is a natural one; it is the Moon. As the Moon rotates around the Earth, the high and low tides change positions. There are always two high tides on opposite sides of the Earth and two low tides on opposite sides of the Earth. Equivalently, any place on Earth will experience two high tides and two low tides in a day.

We'll now derive an algebraic expression for the tidal force. We'll first state the definition of the tidal force slightly more rigorously so we have a base to work with algebraically.

Definition 18.14. The **tidal force** is the force on a particle by the moon at a point on the surface of a planet relative to the force on the particle at the center of the planet by the moon.

We'll now show that the tidal force can be calculated as follows:

Theorem 18.15 (Tidal Force). *Suppose we have a planet of radius R and a moon a distance r away. Suppose the moon has mass M. Consider a particle of mass m on the surface of the planet. Then, the tidal force on the particle can be expressed as*

$$F_{tidal} = \frac{2GMmR}{r^3}$$

Proof. The proof of this is just a straight application of the definition. The force of the particle at the surface is

$$F_{mM} = \frac{GMm}{(r-R)^2}$$

and the force of the particle if it was at the center is

$$F = \frac{GmM}{r^2}$$

So we have

$$F_{tidal} = \frac{GMm}{(r-R)^2} - \frac{GMm}{r^2} = GMm(\frac{(2r+R)R}{r^2(r-R)^2}) \approx \frac{2GmMrR}{r^4} = \frac{2GmMR}{r^3}.$$

■

In extreme gravitational environments, such as near a black hole, this difference in gravitational strength can be so great that an object in the environment will get stretched apart. This is called spaghettification.

18.4 Homework Problems and Solutions

Problem 18.1 (Morin)**.** [5] A block with mass m sits on a frictionless plane inclined at an angle θ, as shown in Fig. 4.7. If the plane is accelerated to the right with the proper acceleration that causes the block to remain at the same position with respect to the plane, what is the normal force between the block and the plane?

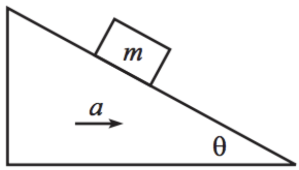


Figure 4.7

Solution. We balance forces vertically, as the vertical acceleration of the block is zero. Then we see that

$$\sum F_y = N\cos\theta + mg = 0 \implies N = \boxed{mg/\cos\theta}.$$

■

Problem 18.2 (Morin)**.** [5] It is possible to experience "zero gravity" on board an airplane if the plane travels up and down with the appropriate $v(t)$ along a path that looks roughly like the one shown below. Which region of the path causes the passengers to experience a feeling of "weightless-ness"?

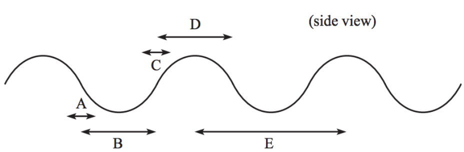
(A) A: where the path is approximately a straight line with negative slope

(B) B: the bottom part of the dips

(C) C: where the path is approximately a straight line with positive slope

(D) D: the top part of the bumps

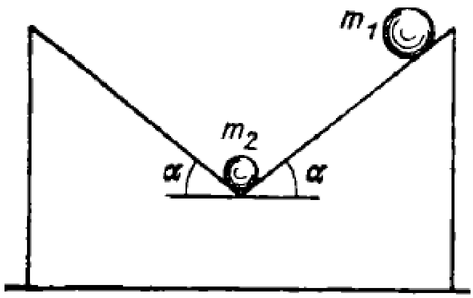
(E) E: the entire path



Solution. We consider the reference frame of the plane. If zero gravity is to be experienced on the plane, the net force in this frame must be 0. Now let's consider A. The fictitious translational force will point up and to the left (in the diagram, not in the plane). A quick drawing shows that this can't create 0 net force when added with gravity. This also eliminates E because A is part of E and also eliminates C since it is analogous.

Now let's consider B. We can think of it as circular motion, so there will be a centrifugal fictitious force downward, which means that the net force can't be 0. So it must be $\boxed{\text{D}}$, which you can see has a fictitious force upward, and can cancel gravity. ■

Problem 18.3 (Krotov)**.** [6] Two balls are placed as shown in Fig. 17 on a "weightless" support formed by two smooth inclined planes each of which forms an angle a with the horizontal. The support can slide without friction along a horizontal plane. The upper ball of mass m_1 is released. Determine the condition under which the lower ball of mass m_2 starts "climbing" up the support.



Solution. Consider the reference frame of the wedge. Now we will work with the boundary case. Consider the forces on the small ball. Since they must cancel, we require $a/g > \tan\alpha$. So now we want to find a different expression for $\tan\alpha$ so that we can get an inequality.

For this we now consider the forces on the big ball. The vectoral sum of these forces must point down the plane, so we can take the ratio of the componenets. If N is the normal force on the big ball, we have

$$\frac{Mg - N\cos\theta}{a + N\sin\theta} = \tan\alpha.$$

Now notice that N is the same normal force as the one on the small ball, you can see this by balancing forces on the *weightless* wedge. So we can substitute $N\cos\theta = mg$ and $N\sin\theta < ma$. Note we also had $a > g\tan\alpha$ So we can substitute all of these in carefully and turn our equation into an inequality. We have

$$\frac{Mg - mg}{Mg\tan\alpha + mg\tan\alpha} > \tan\alpha.$$

So we can cancel g and rearrange and we get

$$M(1 - \tan^2\alpha) > m(1 + \tan^2\alpha).$$

Dividing and using the trig identity for $\cos 2\alpha$, we get

$$\boxed{m < M\cos 2\alpha}.$$

■

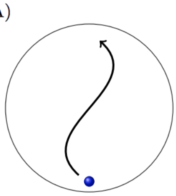
Problem 18.4 (PhysicsWOOT)**.** [6] Earth rotates with angular frequency vector $\vec{\omega}$. Objects moving on Earth experience a Coriolis force $\vec{F}_c = -2\vec{\omega} \times \vec{v}$. When a ball is dropped from a tower, it falls almost straight down, but is deflected by the Coriolis force a distance x to the East. If the ball is thrown vertically from the bottom of the tower so that it reaches the top of the tower at the apex of its flight, how much and in what direction is it deflected by the Coriolis force before it hits the ground?

Solution. When the ball is dropped, its velocity-time plot is linear, beginning at zero. Since $a_x \propto F_{coriolis} \propto v$, we see that the horizontal acceleration is also linear with time, so the horizontal velocity is quadratic, and the horizontal displacement is cubic. That is, we see that the distance displaced is $\frac{1}{3}v_h t$, where v_h is the final velocity.

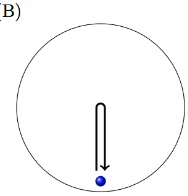
When the ball is thrown up, the Coriolis force will point in the opposite direction, and this time the acceleration starts out large and approaches zero. This is the opposite of the graph from the previous problem, and we see the distance displaced is $v_h t - \frac{1}{3}v_h t = \frac{2}{3}v_h t = 2x$. On the way back down, the displacement is the same, so we our answer is $\boxed{4x \text{ to the West}}$. ■

Problem 18.5 ($F = ma$)**.** [6] A child in a circular, rotating space station tosses a ball in such a way so that once the station has rotated through one half rotation, the child catches the ball. From the child's point of view, which plot shows the trajectory of the ball? The child is at the bottom of the space station in the diagrams below, but only the initial location of the ball is shown.

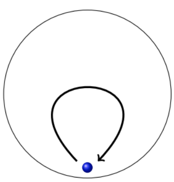
(A)



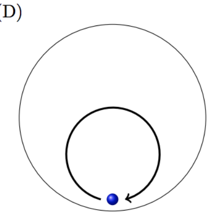
(B)



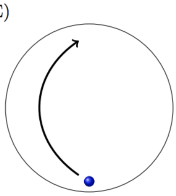
(C)



(D)

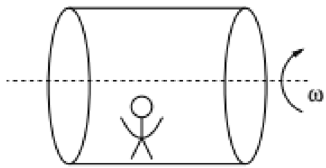


(E)



Solution. We can rule out choices (A) and (E) since we know the ball returns to the child. Now we know the coriolis force is going to make the ball deflect a little, so it can't be (B) either. Lastly, choice (D) doesn't make sense because that would mean he threw the ball horizontally, so our answer is $\boxed{\text{(C)}}$. ■

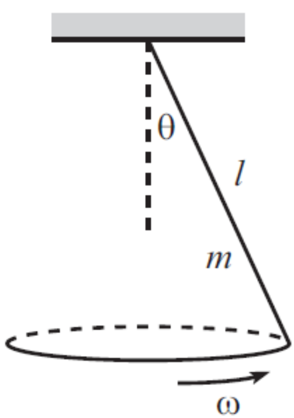
Problem 18.6 ($F = ma$)**.** [7] A cylindrical space station produces 'artificial gravity' by rotating with angular frequency ω. Consider working in the reference frame rotating with the space station. In this frame, an astronaut is initially at rest standing on the floor, facing in the direction that the space station is rotating. The astronaut jumps up vertically relative to the floor of the space station, with an initial speed less than that of the speed of the floor. Just after leaving the floor, the motion of the astronaut, relative to the space station floor,



(A) always has a component of acceleration directed toward the floor, and they land at the same point they jumped from.
(B) always has a component of acceleration directed toward the floor, and they land in front of the point they jumped from.
(C) always has a component of acceleration directed toward the floor, and they land behind the point they jumped from.
(D) has a component of acceleration directed away from the floor, and they land behind the point they jumped from.
(E) has a zero acceleration relative to the floor, and the astronaut never reaches the floor again.

Solution. Finding the direction of the acceleration is quite easy, the centrifugal force points toward the floor. Now for where they land we consider the coriolis force. This is just an exercise in using the right hand rule, and this gives us that the force points in the direction the astronaut is facing, and this means he will land in front of his original location. ■

Problem 18.7 (Morin). [7] A uniform stick with mass m and length ℓ is attached by a pivot to a ceiling. Initial conditions have been set up so that the stick rotates around the vertical axis, making a constant angle θ with respect to the vertical, as shown. Find the angular speed ω by considering the setup in the reference frame that rotates along with the stick (at frequency ω) around the vertical axis.



Solution. Consider the torques on the rod relative to the pivot point. There is a torque provided by gravity ($mg\frac{\ell}{2}\sin\theta$), and there is a torque provided by the centrifugal force. Note the centrifugal force on a tiny piece of mass is $\omega^2 r\ dm$, where r is the radius of rotation. Then to find the total torque from the centrifugal force, consider an infinitesimal piece of mass a distance x from the pivot. We see that the centrifugal torque experienced is

$$d\tau = d\vec{F} \times \vec{r} = \omega^2(x\sin\theta)dm(x\cos\theta) = \omega^2 x^2 \sin\theta\cos\theta\ dm,$$

and we can substitute $dm = m/\ell \cdot dx$. Integrating, we see that

$$\tau_{\text{centrifugal}} = \frac{m\omega^2\sin\theta\cos\theta}{\ell}\int_0^\ell x^2 dx = \frac{1}{3}m\omega^2\ell^2\sin\theta\cos\theta.$$

We have that $\tau_{\text{grav}} = \tau_{\text{centrifugal}}$, so we can substitute to solve for ω :

$$\frac{1}{2}mg\ell\sin\theta = \frac{1}{3}m\omega^2\ell^2\sin\theta\cos\theta \implies \omega = \boxed{\sqrt{\frac{3g}{2\ell\cos\theta}}}.$$

■

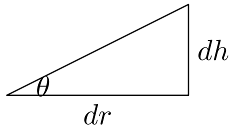
Problem 18.8 ($F = ma$)**.** [8] A circle of rope is spinning in outer space with an angular velocity ω_0. Transverse waves on the rope have speed v_0, as measured in a rotating reference frame where the rope is at rest. If the angular velocity of the rope is doubled, what is the new speed of transverse waves, as measured in a rotating reference frame where the rope is at rest?

Hint: Recall that the speed of a wave on a string is $v = \sqrt{\frac{F_T}{\mu}}$ from the Oscillations handout.

Solution. In the frame where the rope is at rest, we have a centrifugal force $F = mr\omega^2$. This is balanced by the tension, so we see that $T \propto \omega^2$. Since the speed of a wave on a string is given by $v = \sqrt{T/\mu}$, we see that $v \propto \omega$, so the new speed is $\boxed{2v_0}$. ■

Problem 18.9 (Kalda)**.** [9] A vertical cylindrical vessel with radius R is rotating around its axis with the angular velocity ω. By how much does the water surface height at the axis differ from the height next to the vessel's edges?

Solution 1. This is a repeat with an example from the fluids handout. First, consider the following diagram.



The diagram shows the cross-section of the small ring of water which is a distance r away from the center of the cylinder. Letting P be the hydrostatic pressure, we see the vertical force is

$$0 = P(r)2\pi r\, dr - g\, dm.$$

The centrifugal force on the water is

$$\omega^2 r\, dm = P(r) 2\pi r\, dh = P(r) 2\pi r \tan(\theta)\, dr$$

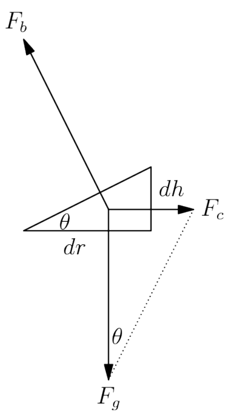
Then substituting in the first equation, we get that $\omega^2 r dm = g \tan(\theta) dm$ so

$$\frac{dh}{dr} = \tan(\theta) = \frac{\omega^2 r}{g}.$$

To get the height difference we integrate this from $r = 0$ to $r = R$

$$h = \int_0^R \frac{\omega^2 r}{g}\, dr = \boxed{\frac{\omega^2 R^2}{2g}}.$$

■



Solution 2.

Consider a small "chunk" of the water surface with radial length dr, height dh, mass dm, and angle of elevation θ. Taking the rotating reference frame of the chunk (in which the net force on the chunk is zero), gravity, with magnitude $F_g = g\, dm$ and the centrifugal force, with magnitude $F_c = r\omega^2\, dm$, sum to the negation of the buoyant force F_b, which is perpendicular to the water surface. Thus

$$dh = \tan\theta\, dr = \frac{F_c}{F_g} dr = \frac{r\omega^2}{g} dr,$$

from which it follows that

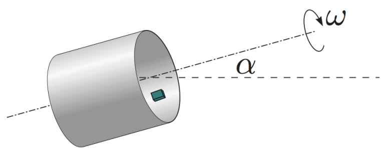
$$h = \int dh = \int_0^R \frac{r\omega^2}{g} dr = \boxed{\frac{\omega^2 R^2}{2g}}$$

■

Problem 18.10 (Kalda)**.** [14] A cylinder with radius R spins around its axis with an angular speed ω. On its inner surface there lies a small block; the coefficient of friction between the block and the inner surface of the cylinder is μ. Find the values of ω for which the block does not slip (stays still with respect to the cylinder). Consider the cases where

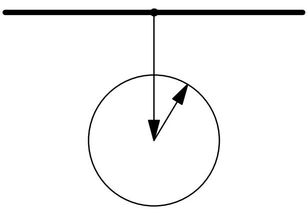
(a) [8] the axis of the cylinder is horizontal.

(b) [6] the axis is inclined by angle α with respect to the horizon.



Solution.

(a) We take the rotating reference frame of the cylinder. Now we draw the block in this rotating frame, but we'll mark it as a dot for simplicity. We can think of the endpoint of the gravitational force being on a circle, since it rotates, and then for the centrifugal force it shifts this circle entirely downward. So here is our free body diagram:



We haven't drawn the friction or normal force yet, but the circle represents all the locations in which the sum of the gravitational force and the normal force points. Now we use an idea that is frequently used in statics problems. The sum of the normal and friction force makes an angle less than or equal to $\arctan \mu$ with the normal. That means this circle of possiblities must be entirely contained within that range, so we can draw lines tangent, and we see that $\omega^2 R \geq mg/\sin(\arctan \mu) = \boxed{g\sqrt{\mu^2+1}/\mu}$.

(b) With this problem we can do nearly the same trick by choosing the plane with the three vectors that we are concerned with, $\vec{F}_c$, $\vec{F}_g$, and $\vec{F}_N + \vec{F}_f$. Now the circle from the first part is actually only a half circle this time, and it goes from α with the centrifugal force to $180+\alpha$ with the normal force. Now since this is only an arc, we have two cases, if the endpoints touch the tangents or if you have to go closer to touch the tangents, and it isn't tangent when it does.

For the first case we draw the tangents again, and this gives us $\omega^2 R \geq \dfrac{g\sqrt{\mu^2+1}}{\mu}$ if the angle $\alpha < 90 - \arctan \mu = \cot \alpha$.

Now for the other case, we use law of sines. We essentially have a triangle with angle $\arctan \mu$, α and a side of length mg. By law of sines,

$$m\omega^2 R \geq \frac{mg}{\sin \arctan \mu} \sin(180 - \alpha - \arctan \mu)$$

Using $180 - \arctan \mu = \arctan \mu$ and some identities,

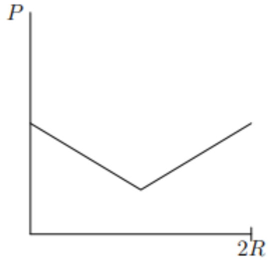
$$m\omega^2 R \geq \frac{mg}{\sin \arctan \mu} (\sin \arctan \mu \cos \alpha - \cos \arctan \mu \sin \alpha) = mg(\cos \alpha - \mu^{-1} \cos \alpha).$$

So for this case $\alpha > \cot \alpha$, the condition is $\omega^2 R \geq \boxed{g(\cos \alpha - \mu^{-1} \cos \alpha)}$.

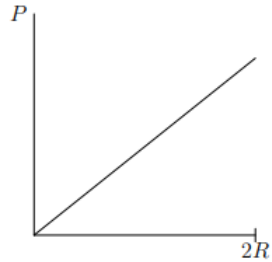
■

Problem 18.11 (Mock $F = ma$). [9] In deep space, a large cylindrical spaceship with radius R is rotating with an angular velocity ω. A drone takes off from the ground and in the frame of the spacecraft, flies directly upwards with a constant velocity to the other side a distance $2R$ away. In the frame of the rotating spaceship, which graph shows the power P of the drone as a function of distance?

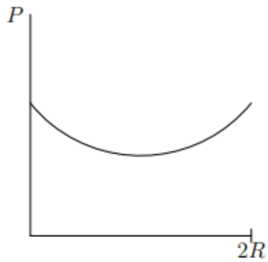
(A)



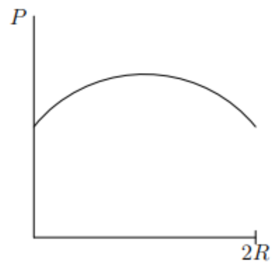
(B)



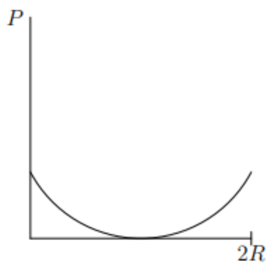
(C)



(D)



(E)



Solution. The force required to move at constant velocity is given by:

$$F = \sqrt{F_{\text{coriolis}}^2 + F_{\text{centrifugal}}^2} = \sqrt{4m^2\omega^2v^2 + m^2\omega^4R^2}$$

thus the power would be:

$$P = m\omega v\sqrt{4v^2 + \omega^2R^2}$$

Hence, we see that the answer of (C), as R decreases in the middle.

Remark. Some of you may not understand why the graph never touches zero. This is because you thought the Coriolis force does no work since it is perpendicular to the Centrifugal force. However, this reasoning is invalid because while technically the Coriolis force doesn't do any work; an actual, real flying drone would still need to spend energy to counter it, e.g. by pushing air in the opposite direction.

■

18.4.1 Written Solutions

Problem 18.12 (Morin, modified)**.** [18] You may have heard that toilets swirl in opposite directions in opposite hemispheres and the same with drains. In this problem we estimate the affect that the Coriolis force has on the water swirling down the drain.

(a) [9] If the water travels a distance d during a time t straight toward the drain, what is the sideways deflection of the water (up to a dimensionless constant)? Let the earth have angular speed ω.

(b) [9] Plug in some reasonable values, for the equation you got in part (a). Is the Coriolis force responsible for your water swirling?

Solution.

(a) [9] The magnitude of the Coriolis acceleration is given by $2\omega v \sin\theta$, where ω is the angular speed of the Earth, v is the speed of the water, and θ is the angle between $\vec{\omega}, \vec{v}$. Since the water travels a distance d in a time t, the average acceleration is about $a = 2\omega \cdot \frac{d}{t} \sin\theta$. Note that $\sin\theta$ is a small dimensionless constant, so we ignore it to get an acceleration of

$$a \approx 2\omega \cdot \frac{d}{t}.$$

The Coriolis force acts perpendicular to the velocity of the water (since a cross product is taken), so the deflection of the water is about

$$\begin{aligned} d &\approx \frac{1}{2}at^2 \\ &\approx \frac{1}{2} \cdot 2\omega \cdot \frac{d}{t} \cdot t^2 \\ &= \boxed{\omega\, dt}. \end{aligned}$$

(b) [9] The earth travels 2π radians in $24{\cdot}60{\cdot}60$ seconds, so we take $\omega = \frac{2\pi}{24\cdot 60\cdot 60}\ \mathrm{s}^{-1} \approx 7.3 \cdot 10^{-5}\ \mathrm{s}^{-1}$.

The water in a toilet bowl is about 20 cm wide, so we will take $d \approx 0.1$ m considering the swirl of the water.

A toilet bowl takes around 5 s to flush, so we will take $t \approx 5$ s.

Plugging these numbers in, we get that the Coriolis force causes a deflection of around $7.3 \cdot 10^{-5} \cdot 0.1 \cdot 5 \approx 3.7 \cdot 10^{-5}$ m, or around 0.037 mm. This is clearly negligible to and washed out by even small initial velocities. Thus, the Coriolis force is not responsible for water swirl.

■

19 Dimensional and Error Analysis

19.1 Dimensional Analysis

This section is adapted from Section 1.3 of Morin's Introduction to Classical Mechanics

The dimensions or units of a quantity are the powers of mass, length, and time associated it with. For example, the dimensions of speed are length per unit time.

There are 2 main reasons for always checking your units: for one, they give you a surefire way of knowing that your answer is *wrong* although it can never tell you if your answer is indeed correct. For example, if you are solving for the velocity of say, some particle, and you get an answer with units of $\mathrm{m}^2/\mathrm{s}^3$, you *know* your answer must be wrong because velocities have units of m/s! Thus, it is always important to check your units (or dimensions) after solving a problem as you have one way of checking your work. The other main benefit of dimensional analysis is that you get a sense of what your final answer *should* look like. We'll see this in the following (classical) example. Of course, this problem can be solved with the results from our previous chapters, but lets try seeing how far we can get with *only* dimensional analysis.

We'll be using the notation "[ ]" for dimensions of some quantity in this section, and M, L, T for the mass, length, and time dimensions respectively. For example, $[v] = L/T$, $[a] = L/T^2$ and so on.

Example 19.1 (Classical). A mass m hangs from a massless string of length l and swings back and forth in the plane of the paper. The acceleration due to gravity is g. What can we say about the frequency of oscillation?

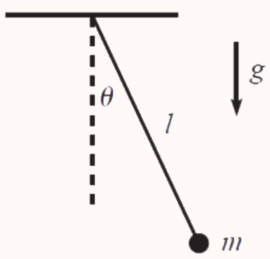


Fig. 1.1

Solution. Lets consider the dimensional quantities we are given in this problem. We have the mass which has a dimension $[m] = M$. We have the gravitational field g with dimension $[g] = L/T^2$, which should surely affect the frequency of the oscillation. We have the length of the string l which has dimensions $[l] = L$. Finally, we have the angle θ_0 that the mass initially makes, but this is a dimensionless quantity. We know that frequencies have dimensions of $[\omega] = 1/T$. How can we use our given quantities to give units of $1/T$?

The only combination of our dimensional quantities that have units of $1/T$ is $\sqrt{g/l}$ (try checking this for yourself). But we can't rule out any dependence based on the dimensionless variable θ_0, so the most general possible form of the frequency of the pendulum is

$$\omega = f(\theta_0)\sqrt{\frac{g}{l}},$$

where f is a dimensionless function of the dimensionless variable θ_0. ■

Remark. Indeed, it just so happens that for small oscillations, $f(\theta_0) \approx 1$, which is where we get our equation $\omega = \sqrt{g/l}$ in the SHM handout. However, there is no way to show this through dimensional analysis alone. For this, we need to go through the full physical derivation. Dimensional analysis can only show us *how* certain quantities must relate and gives us an *idea* of the final answer, but can never give us the full answer.

Remark. It actually turns out for larger θ_0,

$$f(\theta_0) = \int_0^{\theta_0} \frac{\sqrt{8}}{\sqrt{\cos\theta - \cos\theta_0}} d\theta = 2\pi(1 + \theta_0^2/16 + ...)$$

Quite the complicated function indeed!

We give a few more classic examples and then give a few examples from contests. Dimensional analysis is pretty intuitive and can be a really helpful tool in your physics intuition. Rather than thinking about problems *specifically about* dimensional analysis, it is good to think about dimensional analysis *all the time* in *every* problem.

Example 19.2 (Morin). How does the speed of waves on a string depend on its mass M, length L, and tension (that is, force) T?

Solution. We haven't really discussed waves so far, so many of you may not even know what a wave is! We note however, that with dimensional analysis, *we don't even have to know what a wave is.* Let's let v denote the speed of the wave. Then $[v] = L/T$. We have $[M] = M$, $[L] = L$, and $[T] = ML/T^2$. Let's let

$$[v] = [M]^\alpha [L]^\beta [T]^\gamma$$

Then

$$LT^{-1} = M^{\alpha+\gamma} L^{\beta+\gamma} T^{-2\gamma}$$

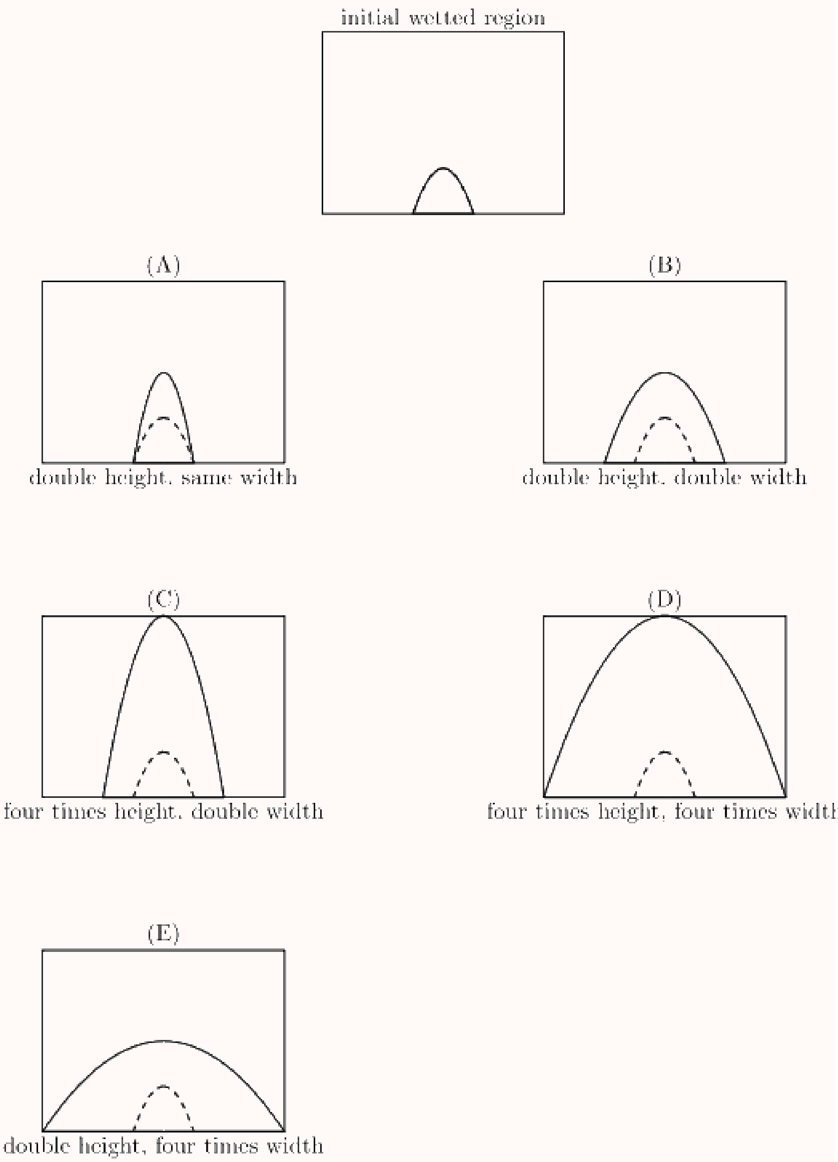
We thus get

$$\begin{aligned} \alpha + \gamma &= 0 \\ \beta + \gamma &= 1 \\ -2\gamma &= -1 \end{aligned}$$

Solving this system, we get $(\alpha, \beta, \gamma) = (-\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ Thus we have that $\boxed{v \propto \sqrt{\frac{LT}{M}}}$ ■

Remark. Indeed, when one does a mechanical analysis, we actually get that the proportionality constant is 1, that is: $v = \sqrt{\frac{LT}{M}}$.

Example 19.3 ($F = ma$)**.** A water hose is at ground level against the base of a large wall. By aiming the hose at some angle, and squirting the water at some speed v, one wets a region of the wall, as shown below. If the speed of the water is doubled, what is the new region that can be wetted? Ignore the effects of the water splashing beyond the point of contact. In the the answer choices, the dotted line marks the initial wetted region.



Solution. This problem is quite tricky, because a lot of information is omitted. We can't exactly do a mechanical analysis, because we don't exactly know how the mechanisms of the hose work (for example, clearly the hose doesn't shoot fluid in a straight line, so the flow is clearly turbulent, not laminar). In fact, we don't really know *anything* other than the fact that the wetted region depends on the initial speed v of the water and probably the gravitational acceleration g. We want to find two length quantities that both have dimensions of $[L]$. Then, the only combination of our quantities that gives $[L]$ is

$$[L] = [v]^2[g]^{-1}$$

Thus, we see that the distance in both the horizontal directions depend on the *square* of the velocity of the fluid. So when the speed of the fluid is doubled the wetted distance in both the horizontal and vertical direction should quadruple. So the answer is $\boxed{\text{(D)}}$. ■

19.2 Error Analysis

Suppose you measure some quantities X, Y, Z with uncertainties $\delta X, \delta Y, \delta Z$ (these are usually determined by the precision of the measurement device). What if we want to find the **uncertainty** of some combination of these quantities? Here are some of the rules that are used in the F=ma. Those of you that have taken a statistics course should be quite familiar with these laws. The proofs of these laws are long and tedious; they are not really involved so they will be omitted.

Theorem 19.4. *If Q is some combination of sums and differences of independent quantities such that*

$$Q = a + b + \cdots + c + \cdots - (x + y + \cdots + z)$$

then

$$\delta Q = \sqrt{(\delta a)^2 + (\delta b)^2 + \cdots + (\delta c)^2 + (\delta x)^2 + (\delta y)^2 \cdots + (\delta z)^2}$$

That is, uncertainties add in quadrature

Example 19.5. Professor Lee wishes to measure the length of a pencil. Unfortunately, he only has a 4 cm long ruler! His ruler has an uncertainty of 10%. He makes three measurements of 4 cm, 4 cm, and 2 cm and sums them up to get the uncertainty of his pencil. What does he get for the measurement of the ruler and what is the uncertainty?

Solution. We get that he measures the ruler to be $4 + 4 + 2 = 10$cm long. We have that the uncertainty of the first measurement is $4(0.1) = 0.4$, second measurement is $4(0.1) = 0.4$, and the third is $2(0.1) = 0.2$. We thus get that the uncertainty of the measurement is

$$\sqrt{0.4^2 + 0.4^2 + 0.2^2} = 0.3$$

So he measures the ruler to be 10 ± 0.3 cm long. ■

Okay so that's addition, but what about multiplication and subtraction. Our second law deals with precisely these operations. Our second law is similar in nature to the first one, but is slightly different in that it deals with **relative uncertainties** rather than absolute ones. The relative uncertainty of a quantity x is given by $\delta x/x$, and is the "percentage uncertainty." Depending on the context, it is often easier for us to work with relative uncertainties.

Theorem 19.6. *If Q is some combination of products such that*

$$Q = \frac{a \times b \times c \times \cdots}{x \times y \times z \times \cdots},$$

then

$$\frac{\delta Q}{Q} = \sqrt{(\frac{\delta a}{a})^2 + (\frac{\delta b}{b})^2 + (\frac{\delta c}{c})^2 + \cdots + (\frac{\delta x}{x})^2 + (\frac{\delta y}{y})^2 + (\frac{\delta z}{z})^2 + \cdots}$$

We note one extra corollary here, which follows from the laws seen in the handout:

Corollary 19.7. *Let X be a variable with uncertainty δX. Assume all uncertainties are Gaussian. Then, the uncertainty of mean of n measurements of the variable X (which we denote by $\bar{X}$) has uncertainty*

$$\delta \bar{X} = \frac{\delta X}{\sqrt{n}}$$

Proof. Remark that since

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X$$

we have that

$$\delta \bar{X} = \frac{1}{n}\sqrt{n(\delta X)^2} = \frac{\delta X}{\sqrt{n}}$$

as desired (since uncertainties add in quadrature). ■

Problems with applications of these laws appear frequently in the $F = ma$. For contests, we recommend just memorizing these laws by heart and applying them to the problems. These problems tend to be pretty straightforward so make sure to pick them off! Let's do an example:

Example 19.8. A student runs a lap of 100 ± 6 m at a speed of 5 ± 0.4 m/s. What is the uncertainty in the computed time t for which the student runs?

Solution. From our formula, we have $\frac{\delta t}{t} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta v}{v}\right)^2} = \sqrt{(0.06)^2 + (0.08)^2} = 0.1$. Since we have $t = 100/5 = 20$, we obtain that $\delta t = 2$ s. ■

We will also talk briefly about weighted error. When different independent measurements of a certain variable have different errors, it is best not to average them, but to take a **weighted average**:

$$t_w = \alpha t_1 + (1-\alpha)t_2.$$

Thus far, we have dealt with adding and multiplying errors of independent variables. However, there is also another important formula that allows us to compute the error of a single-variable function.

Theorem 19.9. *If $z = f(x)$, then the uncertainty δz is given by*

$$\delta z = |f'(x)|\, \delta x.$$

Note the similarity between the error terms and differentials; the intuition behind this formula is that for comparatively small errors, the ratio $\delta z/\delta x$ becomes equivalent to the derivative:

$$\frac{\delta z}{\delta x} = \frac{dz}{dx.}$$

As a corollary of this result, we have the power rule:

Corollary 19.10. *If $z = x^k$, then the uncertainty δz is given by*

$$\delta z = |kx^{k-1}|\, \delta x.$$

Most problems will not ask for δz for a complicated function of x, so this is the most commonly used result of the above theorem. Note that as a result of this corollary, we have that if $y = x^k$, then they have the relative error is given by

$$\frac{\delta y}{y} = |k| \frac{\delta x}{x}.$$

Let's do another simple example:

Example 19.11. The radius of a sphere is $2.0 \pm .1$ m. What is the uncertainty in the volume of the sphere?

Solution. We have that

$$V = \frac{4\pi}{3} r^3,$$

and using the power rule gives that

$$\delta V = 4\pi r^2 \delta r$$

$$\implies \delta V = 4\pi \cdot 4 \cdot .1 \approx 5.0 \text{ m}^3.$$

■

Error analysis problems are comparatively straightforward; just apply the appropriate relationships to calculate the error. Good luck!

19.3 Homework Problems and Solutions

Problem 19.1 (PhysicsWOOT). [4] A wire is made of a material with density known to very good accuracy. The length of the wire is also known to very good accuracy. The radius of the wire is uncertain by about 2%. What is the uncertainty in the mass of the wire?

(A) about 2%
(B) about 4%
(C) about 6%
(D) about 8%
(E) about 10%

Solution. We have $m = \rho l \pi r^2$, so $\delta m = 2\rho l \pi r \cdot \delta r$, which means

$$\frac{\delta m}{m} = \frac{2\rho l \pi r \cdot \delta r}{\rho l \pi r^2} = \frac{2\delta r}{r} = \boxed{4\%}.$$

■

Problem 19.2 ($F = ma$). [5] In an experiment to determine the speed of sound, a student measured the distance that a sound wave traveled to be 75.0 ± 2.0 cm, and found the time it took the sound wave to travel this distance to be 2.15 ± 0.10 ms. Assume the uncertainties are Gaussian. The computed speed of sound should be recorded as

(A) 348.8 ± 0.5 m/s
(B) 348.8 ± 0.8 m/s
(C) 349 ± 8 m/s
(D) 349 ± 15 m/s
(E) 349 ± 19 m/s

Solution. Since we have $v = x/t$, it holds that

$$\frac{\delta v}{v} = \sqrt{(\frac{\delta x}{x})^2 + (\frac{\delta t}{t})^2}.$$

Plugging in the appropriate value gives that $\delta v = 5.4\% \cdot v$. Hence, we obtain our answer

$$v = \boxed{349 \pm 19 \text{ m/s}}.$$

■

Problem 19.3 ($F = ma$). [6] A group of students wish to measure the acceleration of gravity with a simple pendulum. They take one length measurement of the pendulum to be $L = 1.00 \pm 0.05$ m. They then measure the period of a single swing to be $T = 2.00 \pm 0.10$ s. Assume that all uncertainties are Gaussian. The computed acceleration of gravity from this experiment illustrating the range of possible values should be recorded as

(A) 9.87 ± 0.10 m/s^2
(B) 9.87 ± 0.15 m/s^2

(C) $9.9 \pm 0.25 \text{ m/s}^2$
(D) $9.9 \pm 1.1 \text{ m/s}^2$
(E) $9.9 \pm 1.5 \text{ m/s}^2$

Solution. Recall the period formula for a simple pendulum:

$$T = 2\pi\sqrt{\frac{L}{g}} \implies g = 4\pi^2 L/T^2.$$

We have

$$\frac{\delta g}{g} = \sqrt{\left(\frac{\delta L}{L}\right)^2 + \left(\frac{\delta T^2}{T^2}\right)^2},$$

and we see that the relative uncertainty of L is 5% and the relative uncertainty of T^{-2} is 10% (using the power rule). This gives us a relative uncertainty of g of 11%, so we compute

$$g = \boxed{9.9 \pm 1.1 \text{ m/s}^2}.$$

■

Problem 19.4 ($F = ma$). [6] Steve determines the spring constant k of a spring by applying a force F to it and measuring the change in length Δx. The tools he uses to measure F and Δx both have a constant absolute uncertainty, leading to an uncertainty in k of δk_S. If Tiffany measures the same spring constant with the same tools but by using a force that is five times larger, what will her uncertainty in k be in terms of δk_S?

Solution. By $F = kx$ we have

$$\frac{\delta k}{k} = \sqrt{\left(\frac{\delta F}{F}\right)^2 + \left(\frac{\delta x}{x}\right)^2},$$

and since δF and δx are fixed, we see that $\frac{\delta k}{k}$ decreases by a factor of 5 when x and F are increased by a factor of 5. Since k is fixed, we see $\boxed{\delta k_T = 0.2\delta k_S}$. ■

Problem 19.5 ($F = ma$). [6] Three students make measurements of the length of a 1.50 meter rod. Each student reports an uncertainty estimate representing an independent random error applicable to the measurement.

- Alice: A single measurement using a 2.0 meter long tape measure, to within ± 2 mm.
- Bob: Two measurements using a wooden meter stick, each to within ± 2 mm, which he adds together.
- Christina: Two measurements using a machinist's meter ruler, each to within ± 1 mm, which she adds together.

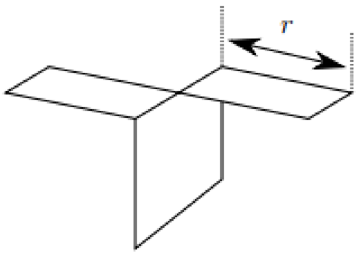
The students' teacher prefers measurements that are likely to have less error. What is the correct order of preference?

Solution. Alice's error is simply 2.0 mm. Recall that when summing independent quantities, the uncertainty adds in quadrature, which implies Bob's error is $2\sqrt{2}$ mm, and Christina's error is $\sqrt{2}$ mm. Hence we see that the correct order of preference is Christina, followed by Alice, who is followed by Bob. ■

Problem 19.6 ($F = ma$)**.** [6] Assume that the drag force for a fish in water depends only on the typical length scale of the fish R, its velocity v, and the density of water ρ. A pufferfish is about 10 cm in length and swims at about 5 m/s. How fast does a clown fish, about 1 cm in length, need to swim such that it experiences the same drag force as the pufferfish?

Solution. Using dimensional analysis, note that $F \propto R^2 v^2 \rho$. Since ρ is constant, we have R^2v^2 is constant for both the pufferfish and clown fish. Hence, $10 \cdot 5 = 1 \cdot v$, which gives $v = \boxed{50 \text{ m/s}}$. ■

Problem 19.7 ($F = ma$)**.** [12] A paper helicopter with rotor radius r and weight W is dropped from a height h in air with a density of ρ.



Assuming that the helicopter quickly reaches terminal velocity, a function for the time of flight T can be found in the form

$$T = kh^\alpha r^\beta \rho^\delta W^\omega,$$

where k is an unknown dimensionless constant (actually, 1.164). α, β, δ, and ω are constant exponents to be determined.

(a) [6] Determine α.

(b) [6] Determine β.

Solution.

(a) At terminal velocity, the speed is constant so $h = vT$, implying $T \propto h$, or $\boxed{\alpha = 1}$.

(b) Using dimensional analysis, we can conclude that since W is the only quantity with seconds $\omega = -1/2$. To cancel the unit of kilograms, we must also have $\delta = 1/2$.

Now to compute β, we can consider the unit of meters. We have

$$\alpha + \beta - 3\delta + \omega = 0,$$

which gives us $\boxed{\beta = 1}$.

■

Problem 19.8 ($F = ma$)**.** [8] The depth of a well d, is measured by dropping a stone into it and measuring the time t until the splash is heard at the bottom. What is the maximum value of d for which ignoring the time for the sound to travel gives less than 5% error in the depth measurement? The speed of sound in air is 330 m/s.

Solution. By kinematics we have $d = \frac{1}{2}gt^2 \implies t = \sqrt{2d/g}$, and we have $\delta t = d/330$. Using the power rule we see that

$$\frac{\delta d}{d} = \frac{2\delta t}{t} = \frac{2 \cdot d}{330\sqrt{2d/g}} = \frac{\sqrt{2gd}}{330} \leq 5\%,$$

$$\implies d \leq \boxed{14 \text{ m}}.$$

■

Problem 19.9 ($F = ma$)**.** [9] The power output from a certain experimental car design to be shaped like a cube is proportional to the mass m of the car. The force of air friction on the car is proportional to Av^2 , where v is the speed of the car and A the cross sectional area. On a level surface the car has a maximum speed v_{max}. Assuming that all versions of this design have the same density, then how is v_{max} related to m?

Solution. We have that the area of the car is proportional to $m^{2/3}$. Hence, since we have

$$P = Fv \propto Av^3 \propto m^{2/3}v^3,$$

and the power P is proportional to the mass m of the car, we see that $v_{\text{max}}^3 \propto m^{1/3}$, or $\boxed{v_{\text{max}} \propto m^{1/9}}$. ■

Problem 19.10 (PhysicsWOOT)**.** [20] A simple pendulum consists of a string of length $\ell \pm \delta\ell$ connected to a mass m. We wish to find the local gravitational acceleration to a good deal of accuracy.

(a) [5] Estimate the condition on acceptable relative error in the period that will negligibly affect the results (negligible compared to other sources of errors).

(b) [5] Estimate the condition on amplitude such that the formula for the small-amplitude period can be used while negligibly affecting the results. You may need to know $\sin\theta \approx \theta - \dfrac{\theta^3}{6}$.

(c) [5] Estimate the condition on mass of the string such that it needs to be accounted for in the calculations.

(d) [5] If the mass is a sphere connected to the string, estimate the condition on the radius of the sphere such that it needs to be accounted for.

Solution.

(a) The simple pendulum equation of $T = 2\pi\sqrt{\dfrac{l}{g}}$ can be written as $T^2 = 4\pi^2(\dfrac{l}{g})$ where $g = \dfrac{4l\pi^2}{T^2}$. Using the product rule and taking the absolute value,

$$\left(\frac{\delta g}{g}\right)^2 = \left(\frac{\delta \ell}{\ell}\right)^2 + \left(\frac{2\delta T}{T}\right)^2.$$

So the error condition on the relative error is that

$$\boxed{\frac{\delta T}{T} \ll \frac{\delta \ell}{2\ell}}.$$

(b) Since torque is $F_g \ell \sin\theta$, Note that the error in angular acceleration is roughly $\delta\ell/\ell$. Since $\theta - \sin\theta \approx \theta^3/6$, and relative uncertainties add in quadrature when to quantities are multiplied, we see that this error is negligible if

$$\frac{\theta^3/6}{\theta} \ll \frac{\delta\ell}{\ell} \implies \boxed{\theta \ll \sqrt{6}\frac{\delta\ell}{\ell}}.$$

(c) Recall that torque is proportional to mass times distance from the pivot. Note that if $m_s\ell \ll m\delta\ell$, then torque from the string is negligible compared to error in the torque from the pendulum bob. Rearranging, we see that

$$\boxed{m_s \ll m\frac{\delta\ell}{\ell}}.$$

(d) Recall that the relative error in torque is roughly $\delta\ell/\ell$. Then if $\delta I/I \ll \delta\tau/\tau$, then the the error from inertia is roughly insignificant (since $\alpha = \tau/I$ and relative uncertainties are added in quadrature). Hence, we need not take into account $\delta I/I$ if

$$\frac{2/5mr^2}{m\ell^2} = \frac{\delta I}{I}\ell\frac{\delta\tau}{\tau} = \frac{\delta\ell}{\ell}.$$

Rearranging, we see that $\boxed{r \ll \frac{5}{2}\ell\,\delta\ell}$.

■

19.3.1 Written Solutions

Problem 19.11 (PhysicsWOOT). [18] The power P dissipated in a resistance R when there is a voltage V applied across it is given by

$$P = \frac{V^2}{R}.$$

The resistance is

$$R = 2.00\ \Omega,$$

and is known very precisely. The voltage is

$$V = 10.4 \pm 0.4\ \text{V}.$$

(a) [9] Find the value of P and estimate its associated error δP using error propagation formulas.

(b) [9] Find the error in P a second way. In this method, take the lowest estimated value of the voltage, 10.4 V − 0.4 V, and use this voltage to estimate P. Then repeat with the highest-estimated voltage. Use the range in P obtained from these estimates to find the uncertainty in P. How do your results from this method compare to those in part (a)?

Solution.

(a) Substituting into the formula, we get $P = \dfrac{10.4^2}{2} = 54.08$. Since $P = \frac{V^2}{R}$,

$$\frac{\delta P}{P} = \sqrt{(\frac{\delta V^2}{V^2})^2 + (\frac{\delta R}{R})^2},$$

and as the resistance is known very precisely,

$$\frac{\delta P}{P} = \sqrt{(\frac{\delta V^2}{V^2})^2} = \frac{\delta V^2}{V^2}.$$

Now, finding δV^2 using product rule, we see $\delta V^2 = 2V \cdot \delta V = 20.4 \cdot 0.4 = 8.32$. We can then find the value through substitution:

$$\frac{\delta P}{P} = \frac{\delta V^2}{V^2} = 0.0769$$

$\delta P = 54.08 \cdot 0.0769 = 4.158$ and $P = 54.08$, so our final answer is

$$P = \boxed{54.08 \pm 4.158 \text{ W}}.$$

(b) First considering when $V = 10.4 - 0.4 = 10$, $P = 100/2 = 50$. On the other hand, when $V = 10.4 + 0.4 = 10.8$, $P = \dfrac{10.8^2}{2} = 58.32$. Averaging these results, we see that

$$\boxed{P = 54.16 \pm 4.16 \text{ W}}.$$

The results from part (b) are almost identical to that of part (a).

■

VIII

Appendix

20 Calculus Overview

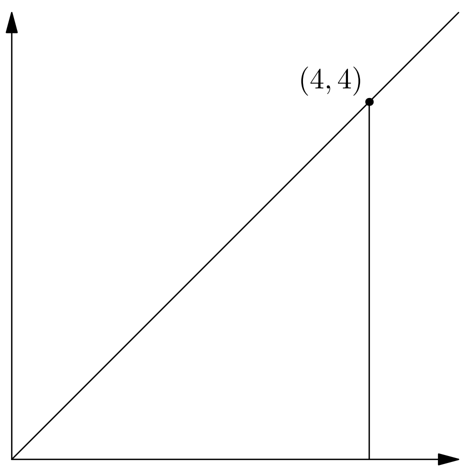
In this handout, we'll quickly outline the basics of calculus that will be necessary for this course. For those already familiar with calculus, feel free to skim through or skip the handout.

20.1 Limits and Continuity

Limits are really important tools in physics as they allow us to find the behavior of various physical systems at their endpoints (or their "limits") which are often more intuitive than the general behavior of the system. We won't really be going into the technical mathematical definition of what a limit is here, but we will instead focus on attaining a good intuitive understanding of what a limit is. A limit is basically a value that a function approaches at a certain point. For example, if $f(x) = x$, then

$$\lim_{x \to 4} f(x) = 4.$$

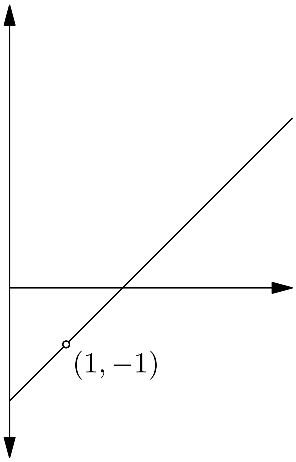
We can graphically represent this as well:



As we just saw, for a value a in the domain of f, $\lim_{x \to a} f(x) = f(a)$. However, limits can be defined for values not in the domain of f. For example, consider the function

$$f(x) = \frac{x^2 - 3x + 2}{x - 1}.$$

Note that $x^2 - 3x + 2 = (x - 1)(x - 2)$, so the graph of $f(x)$ is the same as $x - 2$, except with a "hole" located at $x = 1$, where $f(x)$ is not defined.



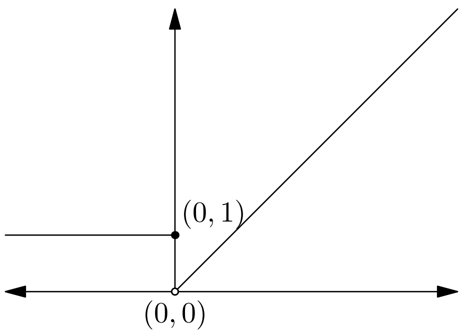
Hence, even though $f(1)$ is undefined, we have that

$$\lim_{x\to 1} f(x) = -1,$$

since f is defined near and still approaches $(1,-1)$, even though it is not defined at that point specifically.

Note that $\lim_{x\to a}$ implies that the limit is two-sided, and equal on both sides. For some functions, this might not be the case. consider the function

$$f(x) = \begin{cases} x, & x > 0 \\ 1, & x \leq 0. \end{cases}$$



We denote $\lim_{x\to a^+}$ as the limit from the positive side, and likewise $\lim_{x\to a^-}$ as the limit from the negative side. Thus, we have

$$\lim_{x\to 0^+} f(x) = 0,$$
$$\lim_{x\to 0^-} f(x) = 1,$$
$$f(0) = 1$$

from the graph above. Since $\lim_{x\to 0^+} f(x) \neq \lim_{x\to 0^-} f(x)$, we have that $\lim_{x\to 0} f(x)$ is undefined. From this, we can see that limits give us a precise way to define continuity.

Definition 20.1. $f(x)$ is **continuous** at $x = a$ if:

1. $\lim_{x \to a} f(x)$ exists.
2. $f(a)$ is defined.
3. $\lim_{x \to a} f(x) = f(a)$.

Essentially, this means that you can take a pen, approach $(a, f(a))$ from either the positive or negative side, and go through it without picking up the pen.

Many functions are continuous at all points in their domain, such as polynomials, exponential functions, or sine and cosine functions. Some notable exceptions include functions such as trigonometric functions $\tan, \sec, \csc, \cot$; and rational functions such as $\frac{1}{x}$. Many of the courses we will be dealing with in this class will be continuous so make sure to be able to differentiate a continuous function from one that is not.

20.2 Derivatives

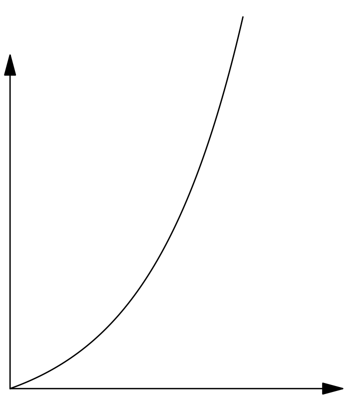
20.2.1 Definition

For a linear function $y = mx + b$, we know that the slope, m, gives the rate of change of the function. But how can we generalize this notion of rate of change to x^2, $\sin(x)$ or other functions, where the rate of change is not constant?

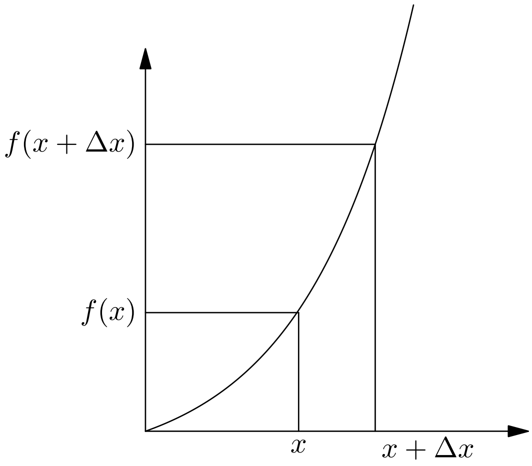
To find the instantaneous rate of change at a point $x = a$ is equivalent to finding the slope of the tangent line to the graph at $x = a$. Why does this work? The tangent line intersects the graph locally at one point only, but it can be thought of as the line through two infinitesimally close points on f. This means the tangent line's slope matches the slope through these two infinitesimally close points, which is really just the slope locally at that one point.

Since the value of this slope might be constantly changing, the rate of change of $f(x)$ would be a function of x as well. This is what we call the derivative of $f(x)$ with respect to x.

Let's say we are given a function $f(x)$. Mathematically, how should we define the derivative of $f(x)$? Consider the graph of an arbitrary function:



Let's consider how we calculate the average slope over an interval:



The slope of any line is given as $\frac{\text{rise}}{\text{run}}$, so thus the average slope over the this interval $(x, x+\Delta x)$ is given by $\frac{f(x+\Delta x)-f(x)}{\Delta x}$. However, this is an *average* and not very accurate. To find the slope at a specific instant, we take the limit as the interval approaches zero. In other words, we take the limit as $\Delta x \to 0$. Over this "infinitesimally" small interval, the function acts like it is linear. This is what motivates the following definition of a derivative.

Definition 20.2. For a function $f(x)$, the **first derivative** of $f(x)$ with respect to x is given by

$$\frac{\mathrm{d}}{\mathrm{d}x} f(x) = \lim_{\Delta x \to 0} \left(\frac{f(x+\Delta x) - f(x)}{\Delta x} \right)$$

There are many ways of denoting the derivative of a function. The derivative of a function $f(x)$ can also be denoted by any of the following: $f'(x)$, f', $\dot{f}$, or $\frac{df}{dx}$.

> **Concept.** The key takeaway here is that the derivative of a function $f(x)$ gives the rate of change of $f(x)$ as x changes.

20.2.2 Differentiation Rules

We will now give basic rules to evaluate the derivative of various functions. We will not give proofs, since they are rather computational and come from evaluating the limit definition of a derivative, but they are easily accessible online. For the following identities, let f and g be functions of x. Then:

1. Differentiation is linear. This means that for a constant c,

$$(c \cdot f)' = c \cdot f',$$

and

$$(f+g)' = f' + g'.$$

2. The Power Rule:

$$(f^n)' = n \cdot f^{n-1} \cdot f'.$$

When $f(x) = x$, we have the simplified form

$$(x^n)' = nx^{n-1}.$$

3. The Product Rule:

$$(fg)' = fg' + f'g.$$

4. The Chain Rule:

$$(f(g(x)))' = f'(g(x))g'(x).$$

The chain rule also has an alternate form,

$$\frac{df}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx}.$$

This has an important application, as we will now see.

Concept (Implicit Differentiation). So far, we have only taken the derivative of a function $f(x)$. However, sometimes it is burdensome to express a function in the form $f(x) = \cdots$. In this case, we can simply take the derivative with respect to x using the chain rule.

To illustrate this, consider the following example.

Example 20.3. For the graph of

$$x^3 + y^2 = 64,$$

give $\frac{dy}{dx}$ as a function of x and y.

Solution. In this case, y is not a function of x. However, we can still find the slope through differentiation. Taking the derivative of both sides gives

$$\frac{d}{dx}\left(x^3 + y^2\right) = \frac{d}{dx}(64),$$

and applying the chain rule gives

$$3x^2 + 2y\frac{dy}{dx} = 0.$$

Now rearranging gives that

$$\frac{dy}{dx} = -\frac{3x^2}{2y}.$$

■

The derivatives of sine and cosine will also show up in physics, and as such are rather important. The differentiation rules are as follows. For a constant a,

$$\frac{d}{dx}\sin(ax) = a\cos(ax)$$
$$\frac{d}{dx}\cos(ax) = -a\sin(ax).$$

The differentiation rules for logarithms and exponentials don't appear particularly often in physics, but we will list them here as well.

$$\frac{d}{dx}\left(\log_a(f(x))\right) = \frac{f'(x)}{f(x)\ln a}$$
$$\frac{d}{dx}\left(a^{f(x)}\right) = f'(x)a^{f(x)}\ln a.$$

This concludes our summary of basic differentiation rules. In a typical calculus class, a lot of time is spent memorizing and applying these rules. For our purposes, however, this is generally unnecessary. The most important differentiation rules to know are the power rule and the rules for sine and cosine. The chain rule is also helpful, though mainly for implicit differentiation.

20.3 Optimization

As you might have noticed, in our definition of derivative we used the term *first derivative*. It is typically implied that the term "derivative" refers to the first derivative. However, higher-order derivatives do exist and are important as well.

Definition 20.4. For a function $f(x)$, define the ***k*-th derivative** of $f(x)$ with respect to x (denoted by $f^{(k)}(x)$) as the first derivative of the $(k-1)$th derivative of $f(x)$:

$$f^{(k)}(x) = \frac{d}{dx}\left(f^{(k-1)}(x)\right)$$

For example, the second derivative of a function $f(x)$ is the derivative of $f'(x)$.

Second-order derivatives show up frequently in physics, but derivatives of even higher orders are rarely useful. The second derivative of $f(x)$ is often denoted by $f''(x)$, f'', $\ddot{f}$, or $\frac{d^2f}{dx^2}$.

Recall that $f'(x)$ is the rate of change of $f(x)$, and that $f''(x)$ is the rate of change of $f'(x)$. This will intuitively give us the following results.

Concept.

1. A function $f(x)$ is increasing when $f'(x) > 0$, decreasing when $f'(x) < 0$, and constant when $f'(x) = 0$. *For x in the domain of f, points $(x, f(x))$ where $f'(x) = 0$ or where $f'(x)$ is undefined are called* ***critical points****.*
2. A function $f(x)$ is convex when $f''(x) > 0$, concave when $f''(x) < 0$, and linear when $f''(x) = 0$. *Points where $f''(k) = 0$ are called* ***inflection points****.*

For those unfamiliar with convex and concave functions, here is a simplified explanation. The term convex (also called concave up) refers to a function whose graph is "bowl-shaped", while the term concave (also called concave down) refers to a function whose graph is "dome-shaped."

The reason critical points are so interesting is that all local minimums or local maximums of a function must occur at a critical point (though not all critical points must be local extrema). Note that if $f'(x)$ changes from positive to negative, then $f(x)$ has changed from increasing to decreasing. There must be a peak at some point; this is in fact the point at which $f'(x) = 0$. Hence, we have the following result:

Concept.

1. A function $f(x)$ has a local minimum at $x = c$ if $f'(x)$ changes from negative to positive at $x = c$.
2. A function $f(x)$ has a local maximum at $x = c$ if $f'(x)$ changes from positive to negative at $x = c$.

We'll do a quick example to see how this helps with optimization problems.

Example 20.5. Olenna has 1000 feet of fence with which she wants to make a rectangular garden. One side of the fence will lean against a castle wall and will not need to be fenced. Find the dimensions of the garden that will maximize its area.

Solution. Let the dimensions of the garden by $w \times \ell$, where (without loss of generality), we let ℓ be the length of the unfenced side. Then we have the following equation:

$$\begin{aligned} 2w + \ell &= 1000 \\ \Rightarrow \ell &= 1000 - 2w. \end{aligned}$$

Hence, the area of the garden can be expressed as a function of w:

$$A = w\ell = w(1000 - 2w).$$

Note that when $A'(w) = 0$, this is a critical point, which will give us the value of w for our desired relative maximum. Setting A' equal to zero gives

$$\begin{aligned} A' &= 1000 - 4w = 0 \\ w &= 250, \ell = 500. \end{aligned}$$

■

Similarly, any relative maxima or minima of $f'(x)$ must occur at an inflection point of $f(x)$. However, inflection points are also important because any change in the concavity of $f(x)$ must occur at an inflection point.

Maximizing and minimizing quantities are crucial in many physics problems, so make sure you know how. This sums up the basics of using calculus for optimization problems. However, differentiation has another important application, as we will see in the following section.

20.4 Related Rates

In problems where there are multiple variables which are functions of a single variable, typically time(t), the relationship involving their rates of change can be obtained by differentiating with respect to that common variable. We'll now do some examples to show what we mean.

Example 20.6. Loras is baking a spherical tart, the volume of which is expanding at a rate of 24 cubic centimeters per minute. At what rate is the radius r changing when $r = 4$ ft?

Solution. Note that the problem tells us $\frac{dV}{dt} = 24$, and wants us to evaluate $\frac{dr}{dt}$ when $r = 4$. The volume of the tart as a function of radius is given by

$$V = \frac{4}{3}\pi r^3,$$

as with any sphere. Taking the derivative with respect to time, we have that

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

Hence, substituting gives us that

$$\frac{dr}{dt} = \frac{3}{8\pi} \text{ ft/min.}$$

■

The key here is to differentiate with respect to t, and substitute the rate we are already given.

We'll now do another example that might seem initially like a physics problem, but is again solved easily using calculus.

Example 20.7. A 13 foot ladder is leaning against a castle wall, and the top of the ladder is sliding downward at a rate of 0.5 m/s. How quickly is the base of the ladder sliding when the base is 12 m away from the castle wall?

Solution. Let x denote the horizontal distance from the wall to the base of the ladder, and let y denote the vertical distance from the floor to the top of the ladder. The length of the ladder is 13 feet, so by the Pythagorean Theorem, we have that

$$x^2 + y^2 = 169.$$

Taking the derivative with respect to t, we have that

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0.$$

Now recall the conditions give $x = 12$ and $\frac{dy}{dt} = -.5$. By the Pythagorean Theorem again, we see that at this instant $y = 5$. Hence, substituting gives that

$$\frac{dx}{dt} = \frac{5}{24} \text{ m/s.}$$

■

20.5 Integrals

As we have seen, we can obtain the derivative of a function through differentiation. However, given the derivative of a function, can we determine the original function?

The answer is not quite. Remember that the derivative is just the slope of a graph. If we shift the graph vertically, the derivative will not change (the derivative of a constant is zero). However, given a specific point on the original graph, we can determine the original function. This process is called integration.

Definition 20.8. For a function $f : [a, b] \to \mathbb{R}$, we call a differentiable function $F : [a, b] \to \mathbb{R}$ an anti-derivative of f if $F'(x) = f(x)$ for all $x \in (a, b)$.

Definition 20.9. Define the **indefinite integral**

$$\int f(x)\, dx$$

as the set of all anti-derivatives of f. Since all anti-derivatives of f differ only by a constant, this is generally notated as

$$\int f(x)\, dx = F(x) + c,$$

where F is an anti-derivative of f, and c represents an arbitrary constant. You may be asking, why do we have c there? The main thing is, that when you differentiate $F(x) + c$ back, you'll notice that

$$\frac{d}{dx} F(x) + c = \frac{d}{dx} \int f(x) dx = f(x)$$

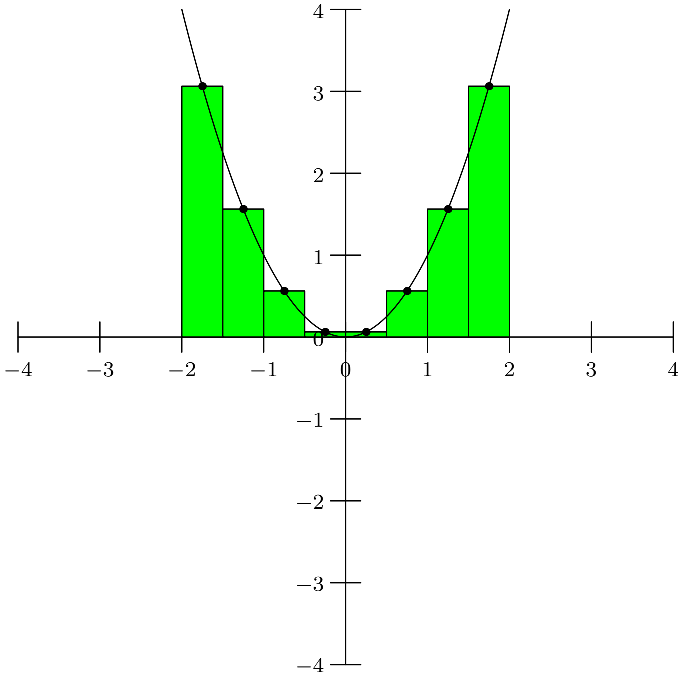
since the derivative of a constant c is simply the slope of the line $y = c$ which is just zero. If we have initial values, we can find the value of c which can help us in real life models.

For example, we have that the indefinite integral of x is

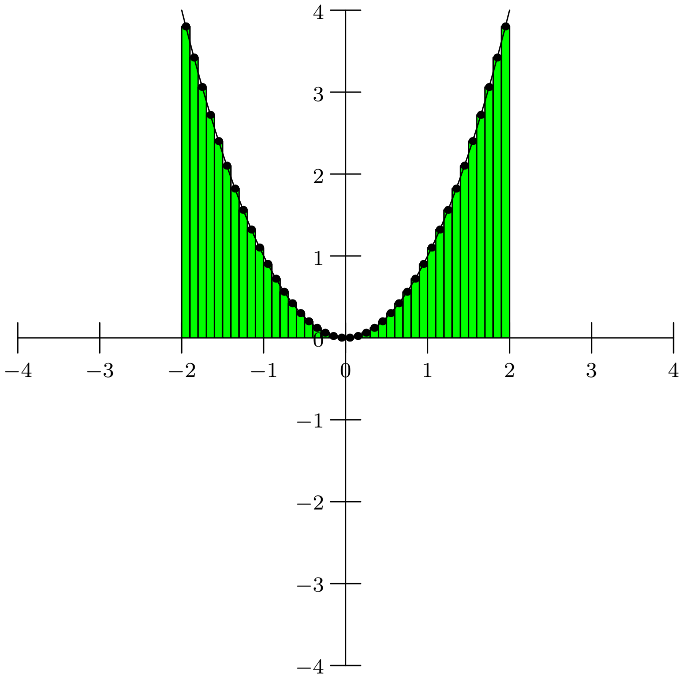
$$\int x\, dx = \frac{1}{2} x^2 + C.$$

To determine a function from its derivative, we can evaluate the indefinite integral, and determine c by substituting an arbitrary point satisfied by the function.

Another useful form of integration is called the **definite integral**. The motivation for the definite integral is to take the signed area under the curve. One way of approximating the area is using rectangles:



This is not very accurate, but by shrinking the intervals of each rectangle, we can improve the approximation. In fact, by shrinking the time intervals to be infinitesmally small, we can obtain the exact area.



Definition 20.10. Given a function $f(x)$ continuous on the interval $[a, b]$, divide $[a, b]$ into n equal subintervals of width Δx and from each interval choose a point x_i. Then we define the definite integral of $f(x)$ from a to b as

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x.$$

Concept. The key takeaway is that the definite integral of $f(x)$ evaluates the signed area under the curve of the graph of $f(x)$. Areas under the x-axis are evaluated with a negative sign.

Let us look at a table of important integral identities in calculus now:

$\int du = u + c$	$\int \frac{du}{u} = \ln\lvert u\rvert + c$
$\int a^u du = \left(\frac{1}{\ln a} a^u\right) + c$	$\int u^n du = \frac{u^{n+1}}{n+1} + c, \qquad n \neq -1$
$\int \cos u \quad du = \sin u + c$	$\int \sin u \quad du = -\cos u + c$
$\int \tan u \quad du = -\ln\lvert\cos u\rvert + c$	$\int \sec u \quad du = \ln\lvert\sec u + \tan u\rvert + c$
$\int e^u du = e^u + c$	$\int \sec^2 u \quad du = \tan u + c$
$\int \csc^2 u \quad du = -\cot u + c$	$\int \csc u \cot u \quad du = -\cot u + c$

Let us look at a quick example of where integrating will be beneficial.

Example 20.11 (2019 Excellence in Mathematics Individual Test). Suppose $\frac{dy}{dt} = 20t - 0.1t^2$, where y represents the quantity of bolts produced, in thousands, after t hours of manufacturing. If $y(0) = 100$, determine the total number of bolts, in thousands, produced over the interval $2 \leq t \leq 5$. Round your answer to the nearest hundredth.

Solution. We simply take the definite integral

$$dy = \int_2^5 (20t - 0.1t^2)dt \implies y = 206.1$$

to see 206.1 bolts were produced on the time interval. ■

We now state without proof the following important theorem which allows us to compute definite integrals.

Theorem 20.12 (Fundamental Theorem of Calculus). *This theorem consists of two parts:*

(a) *Let f be a continuous real-valued function on $[a, b]$, and let F be a function on $[a, b]$ defined by*

$$F(x) = \int_a^x f(t)\,dt.$$

> *Then it holds that F is an anti-derivative of f on $[a, b]$.*
>
> *(b) Let f be a real-valued function on $[a, b]$, and F an antiderivative of f on $[a, b]$. Then it holds that*
>
> $$\int_a^b f(x)\,dx = F(b) - F(a).$$

Hence, just as we can calculate indefinite integrals with anti-derivatives, we can also calculate definite integrals using anti-derivatives.

20.6 Differential Substitutions

In physics we sometimes will be given integrals that are difficult to integrate, so it is important to know which substitutions we can make. For instance, if we have

$$A = \int y\,dx,$$

we can write this as

$$dA = y\,dx.$$

We will now give a list of substitutions for differentials that are often useful in physics:

1. $$dA = x\,dy.$$
   This substitution comes as a symmetry of $dA = y\,dx$ by switching x and y.
2. For a circular shape, it might help that we use the substitution
   $$dA = 2\pi r\,dr,$$
   where r is distance from the origin. This substitution comes from considering an infinitesimal change in area change as adding a circular ring of radius r and infinitesimal thickness dr (which will hence have an area of $2\pi r \cdot dr$).
3. In general, we have the volume differential is equal to
   $$dV = A\,dz,$$
   where A is the cross-sectional area and dz is an infinitesmal thickness.
4. For a cylindrical shape of radius R length ℓ, we also have the volume differential
   $$dA = 2\pi r\ell\,dr = \pi R^2 dz.$$
   The first substitution is a generalization of the circular volume differential (integrating from r), whereas the second substitution comes from $dV = A\,dz$ (integrating along the long axis of the cylinder).
5. Similarly, for spherical shapes have that the volume differential is equal to
   $$dV = 4\pi r^2\,dr.$$
   This comes from considering the change in volume as adding a spherical shell of radius r and infinitesimal thickness dr.

We will now do an example.

Example 20.13. A circular disk of radius R has its center located at the origin $(0,0)$. The the circular disk has density given by the function x^2+y^2. What is the mass of the disk?

Solution. We wish to compute

$$M = \int dm.$$

Note that in this case, density is dm/dA. Hence, we have

$$\frac{dm}{dA} = x^2+y^2 \implies dm = (x^2+y^2)\,dA.$$

It is easier for us to now convert to polar coordinates. We have

$$\int (x^2+y^2)\,dA = \int r^2 dA = \int_0^R r^2 \cdot 2\pi r\,dr = \frac{\pi}{2}R^4.$$

■

20.7 Differential Equations

Throughout the course, we will have to solve a number of **differential equations** to find the time-dependence of physical quantities. A differential equation is one that involves derivatives (which in our case, will mostly be with respect to time) of the variable we're solving for. Here are a few examples of differential equations:

1. $\frac{dy}{dx} = 3$
2. $\frac{dy}{dx} = x\sin x$
3. $x\frac{dy^2}{dx^2} = \frac{1}{\ln y}$

There are many types of differential equations, with some more complicated than others. *Solving* a differential equation usually entails finding the dependence of one of the variables on another. This is more easily said than done however, as often times solving a differential equation can become *extremely hard* and at times even impossible. In this course however, we'll only be dealing with a very small subset of differential equations. Namely, we'll only be dealing with **ordinary differential equations** (ODE), a differential equation with one or more functions of a *single* variable, with all derivatives relative to that same variable. All the differential equations in the list above are ordinary differential equations (Note that we do not have any partial derivatives and all derivatives are with respect to x).

Now, in a typical differential equations course, you'll general learn a number of tips and tricks to solving various types of ODEs. In this course however, you'll only really have to be able to solve differential equations that are *separable*. The solutions to all other types of differential equations will be given.

Sometimes the differentials in the differential equations can be *separated* and then integrated which results in a solution: for example, consider the differential equation

$$\frac{dy}{dx} = -\frac{x}{ye^{x^2}}$$

Note that we can separate the xs and ys to their own side of the equation so that we get

$$ydy = \frac{x}{e^{x^2}}dx$$

We now note that we can integrate both sides which gives

$$\int ydy + C_1 = \int \frac{x}{e^{x^2}}dx$$

$$\frac{1}{2}y^2 = \frac{1}{2}e^{-x^2} + C_2$$

for some constants C_1, C_2. Note however, that this gives us a solution to the differential equation: namely by multiplying both sides by 2 and then squaring, we get that solutions of the form

$$y = \pm\sqrt{e^{-x^2} + C}$$

where C is some constant forms the solution set to the differential equation.

For more information on separable differential equations see An Introduction to Separable Differential Equations.

We'll encounter other separable equations often in the future so make sure you know how to solve them when presented with one.

20.8 Taylor Approximations and Series

The final application of calculus we present is in approximations. Consider the following theorem, which we state without proof.

Theorem 20.14. *An infinitely differentiable function $f(x)$ can be represented by the sum*

$$f(x) = \frac{f(a)}{0!}(x-a)^0 + \frac{f'(a)}{1!}(x-a)^1 + \frac{f''(a)}{2!}(x-a)^2 + \cdots.$$

The motivation behind this theorem is that substituting the nth derivative of $f(x)$ (denoted $f^{(n)}(x)$) for $f(x)$ is equivalent to differentiating the right-hand side n times. Since this must hold for any $n \geq 1$, we obtain this series as the sum. This sum is called the **Taylor series** centered at $x = a$, and when centered at $x = 0$, it is also called the **Maclaurin series**.

The Maclaurin series of $(1-x)^{-1}$, for example, is given by

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \cdots$$

The reason these series expansions are so useful is that under certain constraints, they allow us to approximate more complicated functions. For instance, if $x \ll 1$, then we can approximate that $1/(1-x) \approx 1+x$, since the other terms are all exponentially smaller in magnitude. Similarly, we have the binomial expansion:

$$(1+x)^n = 1 + nx + \binom{n}{2}x^2 + \cdots + nx^{n-1} + x^n.$$

As with the previous Maclaurin series, for small values of $x \ll 1$, we can approximate $(1+x)^n \approx 1 + nx$.

Two other Maclaurin series that give useful approximations in physics are

$$\sin(x) = \frac{1}{1!}x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots$$
$$\cos(x) = \frac{1}{0!} - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots$$

Hence for small values of $x \ll 1$, we can approximate that $\sin(x) \approx x - \frac{1}{6}x^3$ and $\cos(x) = 1 - \frac{1}{2}x^2$. Sometimes, it might even be necessary to approximate $\sin(x) \approx x$ and $\cos(x) \approx 1$. In this case, tangent can also be approximated by $\tan(x) \approx x$. This specific approximation is referred to as the *small-angle approximation* which roughly holds when $x < 10^{\text{deg}}$.

In summary, for small values of $x \ll 1$, we have the following:

$$\frac{1}{1-x} \approx 1 + x$$
$$(1+x)^n \approx 1 + nx$$
$$\sin x \approx x$$
$$\cos x \approx 1 - \frac{x^2}{2}.$$

All four of these approximations will be useful in physics problems and allow us to approximate otherwise chaotic calculations, particularly in the presence of nasty integrals that would be otherwise be difficult (or at times impossible) to solve. They will also make a number of our formulas “nicer” although we still need to make sure we uphold the conditions for which these approximations hold.

Hints

1. Taking the answer from part (a) and dividing by $M_1 + M_2 + M_3$ is actually wrong. Note that this asks for the acceleration of M_1, not the entire system.
2. This problem is similar to problem 14.8. Note that when integrating, you may need the differential $dV = 4\pi r^2\ dr$.
3. Recall that we can obtain the change in gravitational energy by integrating the gravitational force with respect to distance.
4. Recall that we can obtain the change in gravitational energy by integrating the gravitational force with respect to distance.
5. By the Shell Theorem, the gravitational force is linear for $r < R$.
6. You do not need to solve for an explicit answer, just write down an inequality that gives the answer n.
7. If you apply Kepler's laws and properties of conics, this problem can be done with almost no computation.
8. You may need to use some Taylor Approximations, or more specifically: $\frac{1}{1+x} = 1 - x$.
9. You may need to use some Taylor Approximations, or more specifically: $\frac{1}{1+x} = 1 - x$.
10. Consider a infinitesmally small interval of time. What is the increase in speed after a small amount of mass is ejected? You should get an equation in terms of dm and dv.
11. Note that this pulley system differs from massless pulleys we have seen in the past because the pulley has mass. This means the tension on both sides is not necessarily the same, and instead, the pulley will experience an angular acceleration and will rotate just like the rest of the system.
12. What is θ in this case? How is ω related to θ?
13. By the Shell Theorem, we also have that the gravitational force is linear for $r < R$.
14. This problem is similar to problem 14.8. Note that when integrating, you may need the differential $dV = 4\pi r^2\ dr$.
15. If you apply Kepler's laws and properties of conics, this problem can be done with almost no computation.
16. When does the point mass stop slipping on the block?
17. It is difficult to integrate with respect to x since a is usually a function of t, not x. What can we substitute for dx instead?
18. $N \neq mg$
19. What is the conservation of string relation here?
20. The particle exerts a pressure on the shell. Another way to exert pressure would be to close up the shell and add a gas.
21. The angle you're concerned with is not $45°$.
22. Try moving the squares to one side and using difference of squares, as well as appropriate identities from above.
23. Use the approximation $\sin\theta \approx \theta$, which holds for small angles.
24. What are the horizontal velocity and acceleration of the cylinder? Can you use this to figure out the vertical component of acceleration?
25. This problem is about gravity, but trying to use gravity as a force to solve the problem is hard. Could you perhaps find a way to change this from gravity problem into a pressure problem?

26. The springs have zero rest length . . . why might this be important?
27. How can we use conservation of energy here?

Index